# THE JOY OF COMPUTING USING PYTHON

Prof. Sudarshan Iyengar
IIT Ropar

# INDEX

# Introduction To Programming

Welcome to the NPTEL course, ‘The joy of computing”. I plan to start with the very informal session with the couple of my friends sitting here, ravikiran and sowjanya, maybe we should say hi to everyone. So I have handpicked a couple of my students who know little about programming and I would like to see how I can teach them the joy of computing. So initially this session will be more of asking questions and then answering, mostly asking questions from your side and I will try to answer at my best ok? So why are you people here firstly? Basically I want to learn programming. Right, have you tried learning programming before any time? I just tried but it seem to be really difficult sort of. So difficult hasn’t you have tried programming and you found it extremely harder task to program is it? Or is it that you find it disinteresting? It is interesting but there is a notion that programming is difficult so there is that clear in us that programming is difficult so. That’s true many people find programming is very difficult any particular reason why you think programming is difficult? Because the teacher who taught me was directly started that “rules should be followed” and the actual procedure there was no joy or there was no fun in it. Right, so have you used programming anywhere, let’s say you had a real life situation and u felt the need to write a piece of code and then validate it and verify it or have you used programming anywhere outside your curriculum, let’s say apart from course on programming huh first course on programming I am sure these days high school have, high scholars learn programming right very basic stuff and given that you people have mathematics background off course you would have learned some packages software and even programming for that matter so outside your curriculum have u tried learning something? Not really. Have you felt the need to oh here is a place where I think I should get to my desk try to write a piece of code and then to see what’s the output, has that happened ever? No I have never felt the need or urge to. So the course will actually be full of that in fact we will be giving you, ok we will clear ting occasion to feel the in need to go write a piece of coded and then see what would be the output. Right? And most of the course will be simply trying to mimic a game, stimulate a game and then try to play with each other on the computer right. Heard of rock, paper scissors. Yeah so we have seen where couple of people will be playing, rock paper scissors and then person feels the need to write the computer program for it. And there are many more instances where we create that desire in you to see, if this can be programmed and seen right? Ok.

# Why Programming?

One of these questions that many people have is why do we even programmed? Right?  What is the need for it? U seems the digital world most of what you see on your cell phone on your computers or be it on the internet webpage all of these are the in fact some programs. Someone should have written a piece of code for all these things tats it if I can tell you in one sentence what is programming? It’s about getting things done really really fast right? You can always ask me this question can we replace? Can we do without computers? Can we do without any of these programming? Off course yes we can. The point is it makes us life really really really simple, it makes it really fast and nice for you to execute a piece of talk for example If you where to go to your huh train reservation system right? And see when exactly you had travelled from this location to this location. It’s very easy for you to put a search query inside your reservation website and see where all you have travelled. Otherwise you should maintain a big log of where you go what you have done and then you may have to search through it manually. Let’s say for instance I am travelling from Delhi to bang lore ok, what do I do? I go to the railway station ensure that the my name is there on the chart board the train and my friend is checking whether the train has started on time or not so that he knows it reaches on time and he can even trace where the exactly my train is. And has I am having my journey I open YouTube and then I can watch a good video may be good movie, right?  And then I can send a couple of emails to my friends and then I reach Bangalore after reaching Bangalore, I open my cell phone and then try booking a taxi through the popular application. Right? And then I reach back home and pay the taxi person the money that my phone tells me to pay. You see the entire thing from boarding my train until reaching my home I have been using applications all of them not even excluding one all of them  are bits and pieces of programming. A whole lot of programming the whole of digital world is of been made of these programming only and the atomic blocks of it how exactly do we  go about getting programming done is what we will be initiating you all with. So you mean to say every step of our life every day every minute some where we are using programs? Almost every where we are using it today I can say because you see, you got to a shop and you do online transfer pay it though your favourite application to the shop keeper in fact every shop keeper these days they have one application or the other through which you can transfer money to them right? So we can hardly imagine a life today without digital way of handling things let’s say we pay money or we getting Amazon getting your favourite product delivered through Amazon it all involves the digital world and by digital world that’s the system what happens in this world is the whole lot of code, whole lot of programs where in you precisely tell a computer what to do what not to do how to handle exceptions and things like that.

# PROGRAMMING FOR EVERYBODY

What all you said all of those really amazing which fascinates me that by using programming so much but I know little of it so. Do u think anybody can learn it? So that's a good question, this is the frequently asked questions that can I learn programming? I would say programming is the actually quiet common seneschal. But there is one problem that a student faces majorly that is a whole lot of discouragement so it's like if you come and tell me you want to learn French and I start of giving you literature in French, that you obviously will be discouraged. First point, second point is when you have someone around you who is doing really well with programming that's the second form of discouragement, That's the huge form of discouragement that you should be there is a popular tip for chess players when you want to learn chess you should not play with a champion, you should play with someone who is like you may be slightly better but not a lot better right? So second discouraging factor for someone to learn programming is that they end a felling extremely discourage by looking at someone who does really well. First one is whole lot of details we start with some complicated stage of programming we don't start from scratch, secondly people around you sort of discourage you and put you off. A one word answer to your question is YES in fact any one can program and it's fairly easy process. So can you ball juggle? If I give you three balls can u juggle? It's quite difficult. It's quite difficult. Right? But you see in about a week or two weeks time anyone can learn ball juggling it's all a matter of practice so programming is also that it is in fact much much easier then learning a new language and its very very common seneschal so in this course what we will be doing is we will be ensuring that we start from something that is not programming but it is a lot of fun to do and then slowly introduce you all to python programming right? Again not through whole lot of details and the do's and don'ts for programmers but we what we will be doing is will be making you solve a problem and then learn a concept in programming. I think this is the fun way to learn programming than learning the old school way of reading a entire book on a programming language and trying to do everything parallel and not knowing why is this even used and how to go about doing this and what not exercise problems that you don't understand things like that, so the whole of this course will mostly be focused on, you pick a question, you try to answer it and while you are answering it you will be learning a concept.

# Any Prerequisites

Do I need to know something else pre-requisites? Ok. There is absolutely no pre-requisites assumed for the course, all you need to know is you must understand what I talk tats the only pre-requisite, some amount of computer handling, let’s say if u know how to handle your key board and if you know how to click on something and open what is called a python terminal that should be more than enough. Right? If you just follow our instructions you should be able to do that.

# Where To Start?

I have heard people telling that computer has some languages? People understand only certain set of rules and like also heard that something called C, C++. But I don’t really know where to start because some say for this you know, you have to know this. And some say for this u have to know something else so I don’t really know where to start programming? Yes there is a huge confusion for a beginner. Where to start? What to do and things like that. You see talk about programming language people talk to you about python, C, C++, what not right? So let me tell you something, most of these things have one thing in common and that is to understand what programming logic is. Right? There are some three to four most important skills that one must develop. That should be more than enough. Once you develop that you in fact can converse in its like the art of articulation, if you know how to talk well after learning a language I am sure you can do well in that language. Ok? Learning many languages is different from the art of articulation so the art of articulation analogously translates to understanding what is called the programming logic. You will see more of it during the run offer course let me tell you that it doesn’t really matter which programming language you learn. What matters is to understand how to think like a computer scientist by that I mean how to ensure that your brain can actually think how to solve piece of problem using programming. For that I would strongly suggest that you people go by the course, the course is going to assume nothing but we will be teaching python. Python is considered to be a very easy and quick language to learn at the same time extremely powerful. Right!

# Why Do We Have So Many Languages?

I have one more question, why do you have so many languages? Why can't you have just one universal language? Or is it necessary to have so many languages? So this questions answer is slightly tricky that even I don't think even I know the right answer for this question, but if I ready to ask u the same question why do you have so many languages in the world people speak right? You have look at our country like India we have so many languages and let alone that dialects 100's of them, right. So the reason is some people find so laze and comfort in getting a few tasks done using a particular type of programming language and it develops with time. Then they realize, ok if this kind of programming language is used for, let's say for scientific purposes only it is not helping us in building any business application and then they switch to a customize programming language and say, so this we will use for business oriented applications, for example Pascal was a general purpose programming language, Fulton, I am talking about 30-40 years ago 'Fulton' its stands for formula translator ok? That was mainly used for scientific reasons and then came 'COBOL', 'COBOL' was used for common business oriented language that's what It stands for, it was used for business oriented applications. And then they thought ok fine we need something really rob stand nice and then they came out with C and then they realize in C there was lot of problem s when a programmer was programming, it was not easy on his mind so they created this notion of objects and that was the birth of object oriented programming. Right, there is very good difference between C and C++. C++ helps a person program better by keeping his mind free from any sort of ambiguity and confusions when he is coding. And then came python because python was known to be sort of they kept in mind the physiology of program. C to C++ are big leap then the python was like everyone should program, how do we go about doing that and then since its open for lot of people contributed to it and today its stand really tall as a fantastic programming language. So it is all evolution of mans thinking based on the need for something, that's the good question by the way. Yeah was slightly thinking as you said it was evolved to us that what I can generalize from what you have told now, so do you think it will further evolve will be further move ahead or.. that's again a very interesting question again I am not so sure whether I know the answer off course yes I personally believe that next generation will one thing is for sure I have been observing the programming is getting easier and easier with time. Firstly because of the available online resources, secondly because of the inherent ease with which you can code with the kind of tools and techniques available today. For example I am not sound little technical don't mind there is something called integrated development environments IDES where it makes your life really simple to write a piece of, it's like you are talking in some language if you make a mistake you have a earphone small earphone where you hear you made a mistake here you correct it. Or you are typing something and you make a typing mistake and your word processor says this is the right spelling not just that it corrects it automatically or says there is a possible grammar error here I am just giving you analogies right, even in programming there are such beautiful what is called IDEs which helps you program really really better to the future of programming would be any one can program any amount of it and with the whole lot of confidence and the skill set require for one to program back in 80’s is very different from what it is today in fact there are software’s called scratch a very popular one which I am going to teach you people to begin with you will see the actually fun to program it’s a whole lot of fun to program in fact you can develop a games of your choice by using tool called scratch. Will see more of it.

# How To Go About Programming

So you have told us that programming has evolved to be very easier these days, so can you tell us how we will be leaning or how we go about programming in this course? So in this course, we are going to start from scratch. By scratch I mean absolutely nothing. We only assume that you know how to use your mouse and your keyboard that's it ok and from there we are going to go ahead and teach you almost everything of what one is supposed to know in programming right? At the end of the course you will be ready to call yourself a programmer and you can actually start programming some non tribunal things. So you mean to say after twelve weeks of the course I will be really ah knowing all the requirements of requirements to write a piece of code. See this I should, I should probably take a pause and then I should make you all realise that programming is a whole lot of practice. It is probably, it has two steps, one is you need some motivation to programme, that the course is giving you in abundance. No doubts about it, secondly you need to do a whole lot of practice, practice comes with, you should be inspired, you should be some sort of self motivation, you should sit with your friends and then talk to them and then see if there is something that you can take up as a challenge and then write a piece of code and then see the output that is something that we should gradually cultivate right? With time. This is more like singing, you see I mean you cannot become a fantastic famous singer over a night; it involves a whole lot of practice, a whole lot of thinking, a whole lot of culturing of your voice and things like that. Programming is exactly that. In this connection I must tell you something that there are many people who give up on programming thinking that it is not for them and I must tell you all something that to the best of what I have seen to the best of what is known already, programming is quite straight forward and comes with just few hours of practice but a huge deterrent for a programming is the de motivation which generally emanates from your surroundings or your inability to solve something and things like that. So what you must do is go in baby steps, don't try to do something very complicated in the beginning of your leaning of the programming, go in baby steps and go as much as possible and talk to a whole lot of people, and off course as in always keep trying. Right? Nothing like it. Ahh you said about programming and how we go about it, how can mathematician make use of it? Even I am from math background. So you are from mathematics background, be it math, chemistry, economics, physics you name it. Today there is a requirement for all of us to calculate things to analyze things, right? Something as non trivial as how much water you intake every day? Right? What is your, what is the change in weight from the past three four years of an individual ok? What is the amount of carbohydrates that you take? Protein that you take? Can can  have something to say about your health. To analyze this kind of data and to infer something you need programming so you see an nutrition specialist requires programming, see all though you have packages which can be used to get all these things done, you must keep in your mind that this packages have a programme inbuilt, because of which the package works and it sometimes helps for you to understand how to programme so that you can customize whatever you need to do as per your requirements. So i would say programming is definitely not a must for everyone but is a great add on to your skill no matter what you are doing and no matter what background you hale from.

# Why To Learn Programming?

Now a day you have applications apps for everything. You have websites for everything, you name it you have it, so why do you think we have to learn something about that because now a day we have for all the utility you get something, you have applications. Again a good question. So you see the fact that there are so many buildings isn't a reason why you cannot have a new building? Right! The fact that you have so many buildings doesn't stop you from understanding how buildings are made. May be you may want to go one floor up of an existing building. Right! So if you understand how one can programme? There are a whole lot of situations where you will feel something is missing here and I can go on and do add on to it, right? That can happen if you know the background programming that they have used. See it's not an absolute must for everyone to know programming. But you to have an edge in your respective subject programming is a must. Be it you said you are from mathematics background, physics background, economics, assume you are doing business a small scale business still knowing how to write a piece of code and get things done comes a long way in helping you to have that extra edge over others. Right! Forget others to do something that is out of the box and out of the knob, you need to understand how to programming, programming is one skill that helps you get things done really really fast, I am doing to illustrate this very soon with you people by using test cases, where you will realise that without programming you cannot imagine getting o things done.

# What Is Programming?

So I will give you a quick illustration on what is programming? Ok. I am going to programme you people right now, ok. Look at this, I am going to show you my hand, if I show you in this hand, if I say this it means coffee, if I say this two it means tea ,this is water. Can we play this game now? Yeah yeah. Coffee! Water, Tea, water, coffee, tea, water. Ok. Let me add a two more items to it. Coffee, tea, water, juice, cookies. Can you remember? Yeah. Tea, tea, water, juice, cookies, Water, cookies, juice, tea, he is copying. Ah! Juice, cookies, tea, ok. So let's add a small variant to this. You see what happened? I told them to make these assignments for these symbols and they did the assignment. As I was showing them this numbers they could easily say what they represent. How about this? I will show the item using these fingers, whatever coffee, tea, water, juice, cookie and in this hand I will show the number of coffee that I want, number of coffee cups basically. If I say this, it means three cups of coffee, If I do this, quick! Five cookies perfect! Shall we go ahead? Three cops of water, three cups of water. I think he has not quoted right, let him do it, you remain silent ok? Oye! I can't see from this side. Two cups of juice, yeah! Two cups of juice where you took some time for me to figure out ok! What is this? Four cups of tea, Very nice. One cup of tea, one cup of coffee, four cups of juice, two cups of!!Two cookies, two cookies and so on. So what I just now did is like I programmed their minds to follow a particular code line of thought and then they could figure out what I mean by this. What I meant by this and things like that. So whenever I show this to sowjanya it means help me with two cups of tea, I am sorry four cups of tea. If I say this it means help me with two cookies. Correct! So on and so for so symbolically we can represent an item with something and I just programmed their minds to follow a particular rule. Right? Ok. Probably you wondering what has this to do with the programming? I am just trying to get your minds started on what one means by giving instructions. Programming is all about that.

# How To Give Instructions

So let's do one thing, so you all know the age old way of doing a paper boat, observe carefully. How is a paper boat done? We take a sheet of paper ok? A very clear to you people from this angle but i am sure most of u already knows how to make a paper boat. Now imagine these two people are robots! They know nothing. Do you know how to make a paper boat? Probably not! No! Correct good. So I will teach you step by step, you follow exactly what I do. Firstly this is a rectangle, this is a rectangle right? I wish to make a square out of it, how do i do that? Just follow whatever I am doing, we can keep it here. May be you can tear this, all that we did was we made this rectangle, a square. Right? By crossing it like this and that's the way you make a square. Right? And simply do whatever I am gonna tell you people. Collect all the papers here, look at this, fold it once like this and fold it again like this, simple ok. Show it to everyone so that they can see what we are doing and then you see what's happening here, off course I am not going to teach you how to make a boat in this course but I am trying to tell you people that we give instructions and people follow and we get things done. Ok let's go ahead now fold this like this take one leaf of this and fold it like this and the rest three leafs will be folded on the other side. This reminds me of my school days! So then u knew how to make a boat is it? No they never taught you, rarely I remember ok! Ok. Good. So what you do is you can you should fold it like this basically you should put your finger inside and fold it like this and you will get something like form of a kite and then pluck this out, pluck this out like this and you will get a boat. Tada! Done? all of you, what is it that I did just now? I instructed a couple of people here on how to make a boat and here we have three boats. Correct? so similarly we probably can now make a plane out of paper or any origami art, I do it, I give instructions step by step and you also do it. If you fail to understand my steps you will not be able to get the boat as simple as that. So! Not the exact boat, u may get the some other shape or u may not end up u might end up u might end up not getting a boat finally! Right? So what if we get stuck in middle and it's not opening it at all, you force it open it will tear ha ha ha, you will be tearing a sheet right? So the point I am trying to make is by following a set of instructions he accomplishes something that I ask him to accomplish, similarly your computer can actually understand your instructions, when you instruct the right way. Imagine i instruct! I tried giving this instruction in Telugu or Kannada or French or Italian or some other language which you don't understand, you will not able to accomplish it or if there are any ambiguity in the instruction you will not be able to accomplish it. Right? So what is important here is to talk and give instruction in a way that whether it's a computer or a human being you are instructing in a way that the other end the computer understand what you are instructing. Right! Any questions? We didn't discuss anything to have questions oh go ahead! I will just sum up so just told that whatever instructions we give for the other person it must be very clear, crystal clear. Clear, ambiguous, is it to follow and he must be able to produce whatever you are trying to ask him to produce him or her? Right? Ok so this way we actually make a computer listen to you, take your instructions and then accomplish it and computers have a capability of accomplishing something really really fast right.

# Introduction To Scratch

So trust me on this, most of your questions can be solved once we start programming. Let us start from their basics and try to build up one step at a time; most of your questions cannot be answered in plain English although I have tried my level best to do that. You will realize that the best way to answer it is to experience it all by yourself. Ok. So let us not get going with our programming journey with the very nice online package called scratch. This actually a programming language and trust me the pre requisites, the age from where one can start using this programming language is actually just eight. An eight year old can understanding this programming language really well and trust me it’s actually very addictive. Look at this little cat here and look at all the commands over here. You know these commands give this cat a whole lot of power and let’s see what exactly are the powers that you can enable the cat to perform something right? When you say move ten steps I think it’s obvious for you all. The moment you say move ten steps she must moves ten steps ahead so let me say move ten steps! Move ten steps, she is not moving! Probably there something else I must do. What is that? So it is double clicking on the “move 10 steps” that is what is going to make her move and if you change the 10 steps to 50 steps, I type 50 here; it’s going to make her move 50 steps. How? Not by just clicking on this once but click on this twice. A good thing to do is possibly open this scratch window by going to scratch.mit.ed it is there in the description of this lesson and then you can try it all by yourself as I am trying it. So move “50 steps” she moves. In another time move 50 steps, she moves. Let me say move 100 times, 100 steps she moves 100 steps. I am going to pull her by the tale and bring her back, ok so again move 100 steps, she moves. Move move perfect! And as you can see there are self explanatory stuff here. “Turn 15 degrees” says a command, when I keep it here and double click on it look she is going to move 15 degrees. Correct? Let me make her move completely right? She move she move she moves and finally she comes back to her original state ok? I can also move her in the opposite direction, I made her move clockwise, I can make her move antI clockwise, look at this double click, double click, double click she is moving and double click this she moves in the clock wise direction. Go ahead 100 steps, turn by 15 degrees go ahead 100 steps turn 15 degrees, go ahead 100 steps turn by 15 degrees go ahead turn turn turn turn turn turn turn and then go ahead, you see I can command her using these commands. She went back let me pull her back let me turn her she comes back to her original position. Ok. What have you learnt so far? You have learnt the following. Here is the cat and you can instruct her to move the way you want, the way you want here is basically the set of command given here, some of them sound complicated but don’t worry with time you will be able to understand almost everything of what is given here. As of now we just learnt how to make her move ahead and how to make her turn the way we want. Let us end this lesson by making her turn by let’s say 90 degrees so this will make her turn by 90 degrees, right? Again let’s say 70 degrees and I am just typing it here 70 and then again let’s say 80 degrees, 80 degrees and so on right? So now it’s time for you all not to go ahead with the next lesson but to open the scratch window and try playing around with her and make her move around, turn around, and do whatever you want. Sky is the limit as you can see and its whole lot of fun making her do stuff ok? You can in fact explore other things here also, although we will be covering all these things it will be good if you can go ahead and look at this mini laboratory of how to programme a small unit graphical figure? We just now made the cat rotate; move around this was just the beginning. You can do a lot more with the scratch then this let us see more of it in the forth coming videos.

We saw how we can make this cat move, rotate and things like that. Now what I am gonna tell you is what is called grouping these instructions? Some I am gonna group them you easily guess what that mean? Basically I just zoomed the commands but it's easily visible for you. What I am going to do is, I am going to make her move by 100 steps and again move back by -100 steps, what is this mean? Clicking on this will make her move ahead, clicking on this will make her move back to her position it's like go ahead 100 steps and come back by 100 steps. Right? Ok. And then let me say 'I want her to turn by 90 degrees', so when I click on this she turns by 90, again turns by 90, again turns by 90, and again turns by 90. So what I want her to do is let's say first move 100 steps and then turn by 90 and again move by 100 steps so I should bring back move once again keep it here make it 100 steps and then turn by 90 degrees. What will this do? Let me keep this aside. So when I double click on this she will move back by -100 but when I double click on this she will perform all this 4 things continuously, simultaneously ok, so let me just remove this, this is not required I double click on it and click on delete. ok? And it's gone. So when I double click on this, the first one she will move 100 steps, rotate by 90 degrees and again move by 100 steps ahead and then rotate by 90 degrees, let us see whether this happens or not? Boom!! She comes here and then she settles here, we couldn't we couldn't recognise what exactly happened right? Now if I click on this once again she will move ahead by 100 rotate move ahead by100 rotate and stop. See that happened now. I click, she comes here, I click, She goes back the movement is so sudden but we are unable to observe what's happening? Right! So there is a way in which we can actually make her wait between the commands that is called if you click on control you will see what is called wait here, you can insert wait she will move ahead wait for one second and again rotate by 90 degrees, I want her to move, I wanted to wait her for one second once again and then move 100 steps and then again wait, ok now what will this do? Very simple see most of scratch is self explanatory, I would say all these video lessons are not so important. What is important is you should experiment with what stand for what? Now let me double click on this, let's see what happens? Self explanatory isn't it? Move by 100 steps moved, waiting for one second move, rotate and stop! I repeat, move by 100 steps, rotate, move by 100 steps, rotate. Even that I have given one second time this is happened. Correct? What do I want? I want her to move ahead let's say by 200 steps, what will happen right now? She will move a whole lot, rotate, move just a little 100 rotate and stop. So I want her to move 200 steps always and go back to her original position. She went away somewhere! Let me pull her back. So I wanted to start here move by 200 steps, she is somewhere here, rotate and then she will move, rotate and then she will move, rotate and then she will move, she will complete a loop you see, she should get back to her original position. So what do I do for that, self explanatory I should again make her to move by 200 steps, go to motion click on it put move here move by 200 steps I want her to wait for a second, so go to control and then say wait for a second move by 200 steps and then again rotate right? You must again make her rotate by 90 degrees again I want a wait in between as you can see otherwise you cannot see her move, computer is very fast, the computer will make all these commands get executed in a fraction of a second you will not be able to see her move properly ok and again rotate by 90 degrees, lets revise. 200 she comes ahead, 90 degrees-she rotates, 200 steps, she comes down, 90 degrees-she rotates, 200 steps-she goes here, 90degrees-she rotates and again I must say move by 200 steps isn't that right? She should move four times, once she goes straight, rotate and move she comes down, rotate and move she goes to the left, rotate and move she goes up once again control wait for a second tada! There we are just double click on this go, rotate, go, rotate, go rotate, go rotate. Ahh! She is not rotating because I am not rotating here so what I should do that the end I must make her rotate once again, so rotate by 90 degrees hip hip hurray!! We are done. Wait for one second and this will make her do the trick so but I will I will I am rotate her by 90 degrees right now otherwise she will not start from there, so I will create a new command just to rotate her by 90 now we are all set, bring her here and then double click on this, this is a separate command we can double click and she will rotate like this once twice thrice four times as when double click on this, this will happen but when you double click here move, rotate, move, rotate, move rotate, move and then rotate is what's happening right? These were the list of commands which made you write your first programme, a set of instructions for this cat to move and complete a square. Remember the example of coffee, tea, 5 of them, 3 of them and things like that correct! I gave you people instructions and you followed it remember? 2 coffee, 3 tea, cookies! 4 cookies and things like that right? Similarly I give him instructions, I give the cat instructions to rotate to move ahead few steps and he does that ok. We just saw how we can group a bunch of commands! And execute them in an instance. This is one of the most powerful concepts in computers and especially one of the aspects of programming that is worth appreciates it. You can write a long piece of code running through several consaflaince which will execute something very quickly, let's say you click on your icon on your computer and the programme opens this particular thing is actually a piece of code, a long piece of code but it acts so instantly that you will not realise that it is so longer code that we double click and it gets executed in fraction of a second. Fine, now we will go ahead and understand what are called the looping structures. If you want to get a bunch of things done repeatedly there is a nice way of doing it in programming and let's see more of it in forth coming sessions.

# Introduction To Loops

We are on to discussing something very significant, important and a novel concept for a beginner ok. What is our ultimate aim? To make our cat recite five tables ok? And now I am going to introduce you people on what is called a variable? Ok. That's what is the important significant thing for a programmer I was talking about. Go to data click on it, click on make a variable, it will ask you for a name put any name you want now here is a confusing part you can put your name or you can simply put a letter you can simply enter whatever you want here, a word also let me simply enter "S" to begin with and click on ok, you will get a S here, you saw the difference right? Correct? Now pull this aside and keep it here double click on it you will get a 0, a variable is always 0 unless otherwise initiated, what do i mean by that? I will tell you next, pull this this side it says set S to i will put 15 here which means when you double click on this S actually becomes 15, double click on this S we are getting 15 right? Also for your reference this shows, the variable value which is 15 here correct? Ok. Now when I, let me tell you what is change. When I say change S by 1 you will see what happens, double click on it S changes by 1 means S gets incremented by 1 ok? Double click on S you will see 16 now, you also see 16 here right? If you say change S by let's say 20 what will happen? The existing value 16 will get added by 20, which makes it 36. You see you must practice a whole lot of handling variables, exercises; you should try playing around just this thing, just this much. Make a variable and try to see what exactly happens? Let us do a small exercises now let's go to data create a variable, let me call it X. Create another variable, yes you can do as many as you want call it Y and you are done, you have X and Y correct? And let me set the values of X and Y, how do I set it? You go here, you will get all the current variables that you have defined X, I will set X to something, Y to something ok? Let me do that 10 and Y will be let's say some 12 ok? You still see that X is 0, Y is 0 correct? Let me also keep them here as you know, you can keep these here double click and then see what their values are? It is 0. That is because I am not double clicked on this the moment I double click on this, this two get executed so double click now X will be 10 Y will be 12 as is evident here and also double click 10, 12 perfect! Now what I want to do is add these two variables, how do i do that? Go to operators you have an addition operator here, bring that here and then go back to data, bring set here create another variable namely Z you see what I am doing Z gets created. What is Z? Double click 0 ok? I am going to set Z to be X+Y how do I do that? First put X here then put Y here and then say set Z to X+Y and bingo!! You see what's going to happen now? Double click on this Z is now 0 but double clicking on it will make X 10, Y 12, which already the case by the way and then will make Z become equal to Z+Y which is that should become 22 here, should become 22, let's see what happens? Come here and then double click boom! See Z became 22, Z now is 22 correct? So you can do a whole lot of such operations here ok? Let's do more for the sake of explaining things to you. I want to subtract also ok? Subtract what from what? Go to data subtract let's say X-Y ok? Double click on it you will get the answer but then if you want to set it on a new variable then you should use set let me call it a new variable, make a variable, let's say diff. Diff gets created please note variables need not necessarily be a letters it can even be words like this strings like this. So I am going to say set diff to be the difference so bring it here and then double click on it, now my difference will be X-Y whatever X was whatever Y was the difference will be diff perfect it showing -2 and you want this to be sequentially executed simply keep it here, now tell me what happens if I make X as 35 and Y as 15 what will happen to Z? It will be X+Y, Z should become 50. X is 35, Y is 15 why is it showing 10 and 12 that is because I am yet to execute these things. Ok? Double click here and all these values will change. X should become 35, Y should become 15, Z should become sum of X and Y which is 50 and diff should become the difference which is 35-15 what is it? 20. Correct? We must see 35, 15, 50 and 20 let's see if this is visible? Double click boom! Yes perfect 35, 15 50 and 20. So now you have understand how to set a variable to a value, perform a few operations assign the values to a new variable. Ok? Go back to operators explore all these things look at data you can create variable and use them to do whatever you want. So what was our mission? We wanted our cat to recite the five tables, how do we go about doing that? Let's see. First let me go to data and create a variable let me call it't''t' for tables multiplication tables ok? And then I want to set't' to 5 and then I want t to change by 5 every time ok? And I want my cat to say these, how do I make her say this? Say hello so say go to data take t and out that here so you are making t=5 you change t by 5 no! I think before that we should make her say t and then you can change t by 5 and then make her say t once again how do i do that? Simply copy paste say t once again ok? Double click 't' set to 5 and she says 't' 5 and then changes 't' by +5 and then says 't' but this happen so fast you cannot see a five come here, let me try it once again. Oh see you can't even see because it gets executed so quick so what will I do? I will go to control and I will say wait one second, with this you will see that a 5 comes here, you wait for a second so that you see it and then t changes by +5 and then't' is again displayed which is 10 let me see this, double click huh see 5 comes and then 10 comes after a seconds time, double click 5 and then 10 so what should I do? I want this to repeat, what to repeat? Wait for one second change't' by 5 and say't'. Right click on this and say duplicate you will get the same thing keep it down; this will now be 5, 10, 15 if you know what i mean. Correct? Let me see double click 5, 10, 15 good my cat can say 5 tables up to 5*3 is 15. Let me make her say up to 50 5*10 is 50 how do i do that? Keep copy pasting this below, right click duplicate, means you will get a duplicate of that keep pasting this below like this duplicate down, duplicate down, duplicate down, I have not kept a count of how many times I am saying let me double click and check. 5, 10, 15, 20, 25, 30, 35, 40 this is a dumb cat as I told you she will only do what is being instructed, she is not intelligent enough to understand that we want up to 50 and she must go up to 50. Unless and otherwise specified she will not do it. So what need to be done now? Yes i must duplicate this just a couple of times and there I am, this makes it 10 says of't' which changes by 5 every time and this must make our cat recite the 5 tables from 5 times 1 is 5 up to 5 times 10 is 50 so basically you see 5, 10, 15, 20 up to 50 let see if she does it! Double click 5, 10, 15, 20, 25, 30, 35, 40, 45, 50 hip hip hurray!!! There you are so we made our cat recite the 5 table. So we made her recite the 5 table from 5, 10, 15 up to 50 I have a feeling there is a better way to do this and what is that better way? Can you think? Can you try before proceeding any further? It will be a good exercise for you all. Now if not I am always there to help you, maybe we want to pause here and then try it to do it all by yourself, if not lets continue. Go to control and then look at this repeat, keep it here and then simply take, let me duplicate this I don't want anything beyond this ok? Let me delete this I only want I want a set‘t’ to 5 to begin with and then repeat this 9 times, repeat what? I want to say‘t’ first 5 is being said wait for about one second ok? And then again let me delete this i want the cat to say t after changing t by 5 ok? So see what i am going to do now change t by 5 and then say‘t’ and wait for one second and I am done! Let me delete this and try explaining what I just did. ‘t’ will be 5, cat will say 5 we wait for one second and I then repeat this nine times ok? Which means i change t by 5 which means ‘t’ become 10 I say 10, I wait for one second and this happens nine times which means i will go on and recite the entire thing, correct? The same thing the same thing can be done in such a small piece of code, do you see the power of this repeat loop here but let’s see whether it does or not? Double click and there we go 5, 10, 15, 20, 25, 30, 35, 40, 45, 50 yes we did exactly this bigger code equivalent by writing a small code correct? Ah is there any way I can even better this code let me try bettering this code. All of you observe, i duplicate this I go here I simply want to repeat something ten times repeat what ten times? I want the cat to say five first so for that t=5 and then I want this to be done inside, I don’t want any of these things, let me delete these things see what this code does first sets‘t’ to 5 and then repeats say t change by 5 and then waits for a second let me see whether it does what it is suppose to do. Double click 5, 10 so on let see whether its stops at 50 yes 35, 40, 45, 50. You are just wondering why exactly i did this, this looks exactly similar to this even does what this does but this will give me enormous amount of flexibility, and what is that. See what I am going to do I am gonna do small magic for you all. Click on make a variable and create ‘any’ I will type any here, ‘any’ is now a variable ok? And now I will set t to not 5 but any and I will change ‘t’ by not 5 but any and i will now set any to let’s say 7 pause the video observe the code, what exactly this code does? And then un pause play the video and see if you are right or not? Ok. I suppose you all pause the video and gave your guess now double clicking on this will simply give me 7 tables isn’t it? How? Let’s see whether it gives me 7 tables or not. Double click 7, 14, 21, 28, 35, 42, 49, 63 and 70 this gave me 7 tables that because whatever was given here I tried replacing 5 by another variable which I could flexibly change, what do I mean by that? If you make this eight or nine let’s say it gives me the table for nine look at this double click and i will get tables for 9 right? 27, 36, 45, 54, 63, 72, 81 and 90 that is basically 9 table in fact you can even make it recite 19$^{th}$ table which by the way I never learnt as a kid I knew tables only up to 9, I dint even know 12 table but let me see whether it can recite 19 table very unfortunately I will not be able to verify given that I don’t know by heart so this is going on I know we should stop on 190 let me see. 171, 190 perfect! So I now can make the cat recite any table which even I dint learn as a school kid.

# MORE ABOUT LOOPS

We know performed one of the major tasks in programming called the looping, we told you how we can make our cat reside the multiplication tables, well we made our cat do some mental exercise you see it is time for us to make her do some physical exercise. We will make her move back and forth by using our programming skills. Ok, friends see I have written a very long code, very patiently I have tried writing this code what exactly is this code? Look at this, initially I am going to move twenty steps take a half a second break there and then come back twenty steps, half a second pause go ahead forty steps see it's like first day cat moves twenty steps and comes back next it goes forty steps and comes back sixty steps comes back minus sixty steps, eighty steps and then comes back and so on as you can see so on so on, so on, move hundred, minus hundred move one twenty and then minus one twenty which means come back move one forty and then come back one sixty, one eighty, two hundred and then come back as  you would have guessed, if you have guessed this cat will now keep moving ahead like this incrementally will get bolder and bolder and move away from her house, she will try going as far as possible, let's see how this works by double clicking on this. Double click; ahead back, back, ahead back go ahead back, go bolder and bolder go twenty steps ahead twenty steps ahead and finally stop once you reach two hundred steps. Correct? Let's do this once again and then watch one, two, three, four, five, six, seven, eight, nine and ten so twenty steps, two hundred steps which is ten perfect, now I have a question for you all, she is doing basically ten ahead's and ten backs in succession right, is there any way I can use a repeat loop and then put something so that the same thing can be achieved by making her by do something ten times without going through this ordeal of writing this big of code, now what am I trying to say, is there any way you can use this and reduce this to a smaller code, because its looks like we are repeating the same stuff but these values are changing correct? How do we go about it? Can you now figure out by looking at all the options here, if you can do this all by yourself, we strongly suggest that you spend at least fifteen to twenty minutes time figuring out, how you can achieve the same thing using the repeat loop, if you cannot solve it get back and continue watching the videos.

# SOLUTION TO LOOPING PROBLEM

So I hope you people tried doing what I asked you to do, Right? If you dint? Don't worry we are there to help you. How do we go about this? Here we are going to help you out with understanding of this really really involved computer science concept. The question that I asked in a previous video clip is not a very straight forward one, if you are not into programming already you should have found it very difficult to even understand what I was asking you to do. So don't worry listen to me careful, let's go step by step.

Firstly let me teach you what is this data thing here, click on it you will get these two things, click on make a variable and you will get, you will ask you will be asked to type in a variable name, let me now type a variable name as a num click on ok. So you see a num gets created here it is assigned zero and also num sits here, pull it and keep it here double click on it you will gets it value zero. So num is zero so I can set num to some number put this here, set num to ten when you do this and then execute this number is now ten you see its updated to ten here and when you go here and double click on this it will say ten, it will say ten correct? Now when you say change num by one and execute that num initially was set to ten and then you changed it by one which means by one means what you incremented ten by one and you get eleven double click on num you will see eleven correct? Let us double click on this once again this is just like these are all blocks of instruction you are giving right? Num is equal to eleven why don't you double click on this? I double clicked go back you will see twelve here come back double click it becomes thirteen, double click fourteen, double click fifteen so what do I do? Look at this I will take go to this control and take a repeat loop and put this here can you guess what's going to happen now? I repeat num was fifteen and I am going to change num by one ten times when I double click on this you see the computer is going to very quickly execute this piece of instruction ten times no no that is very certain thing we want to pause the video and think what's happening here? One even parallelly try doing what exactly I am trying to do here right? It's not so easy same time not very difficult it takes times to this to sink it, I double click on this and you will see the magic happen, I double click num is fifteen now I am going to double click this, double clicking yeah I did it now immediately this became twenty five, why? That's because num got incremented by one ten times, if I make this fifty then num gets incremented by one fifty times and so many double click on this I mean to double click by the way but if I double on this num will get incremented by fifty which means this becomes seventy five let me see double click boom! This became seventy five now what is more beautiful about this, these thing happen really really fast how fast? I put thousand here and you will see how quickly this becomes incremented by thousand, thousand num will get incremented one by one thousand times and hence this should become yes! One zero seven five now ready lets double click, double clicked quickly this became one zero seven five that why computers are very fascinating and programming a computer is more fascinating because the power is in your hands to make it compute, it computes really really fast, very very fast correct? Ok. So now let me do something I will set the number to zero ok? This is the part of this code now let me remove this, this doesn't matter, double clicking on it as I told you will show you the existing numbers value but that's there here itself so we don't need to worry let me keep it here so I can see it properly and let me remove this, this is not useful now what will happen if I press this number will be assigned zero and then number will increment one at a time and will become thousand. These things will happen so fast that you will not even notice what's happening. Double click here you see it became thousand correct? Now let us go slowly and see what is happening here? You will be surprised that this is so easy to do, go to control go to wait and tog in wait here now you all know what is going to happen. Right? Set number to zero change number by one wait for one second so you can peacefully see how these changes one by one with the seconds pause in between, all of you I am sure you would have guessed what this is going to do. Let us double click and see what exactly it does, double click! And here its starts one two three four five six seven and so on, you see this it's now incrementing it will go on like that till one thousand I don't have so much time so let me just stop this come back instead of thousand let we make this just twenty and see what happens? So double click on this once again one two three you see this will go on till twenty, four five six seven eight nine ten you see as you can note you are basically creating a counter which changed one number every second so basically this takes twenty seconds. Correct? Twenty!! There you are hip hip hurray!! So we wrote a first neat code correct so this is what loop does, it helps you it gives you the power to repeat something repeatedly for a designated amount of time. Please note one thing that you should do and another thing you shouldn't do, one thing you must do is you must pause and then write your own codes try to play around not just do what exactly we did but do a lot more than what we are showing, play around and use all these things and try to understand what is what? You can always click on, let's say u dint understand what is wait for one second right click on it you will get a help click on it and there you are, you get all the details about that particular command wait specified number of seconds and continuous with next block with the example here correct? Let's close this so you should try doing this that is what you are supposed to do and what you are not supposed to do is simply stick to what we are explaining you should not be doing that. We are using example to explain things so that we will motivate you to explore more stuff so please go ahead and explore a lot more things here before going any further only then can you understand programming really well so let's get back to our original question. I was just going to teach you something that going to come very handy in solving this problem, what problem? The cat is trying to go ahead and come back and then keep going ahead and keeps coming back and then it incrementally keeps moving like this and it get's braver brave with time. Right? How do we programme that using a repeat until loop? How do we avoid writing a huge code for it? Remember? It was going twenty steps at a time twenty first forty next and then back sixty next and then back so let us try doing that using repeat until the repeat lopper ok so I will take this repeat lopper and then put here and ten times I will make him go ahead, how do I do that? Motion move twenty steps, instead of twenty steps I will say twenty steps times one, twenty steps times two twenty steps times three, how do I do that? I mean that sounds little complicated, let me figure this out on scratch. How do I do this? Let me go to data, create a variable called steps, steps is the number of steps that the cat takes and yeah there you are! Steps gets created and initially steps will be twenty and then I am gonna change steps by increments of twenty which means what does this do? See what this does. We discuss this when you double click on this you see this segment of code now is different, this segment of code is different when you double click on this only this gets executed you all know that by this time correct? So when I click on this initially steps gets assigned by twenty ok? Double click steps is assigned by twenty and steps is changed by another twenty which is forty ok? And next if I again continue to say change steps by another twenty it becomes incremented by another twenty again if I say set oh I am sorry not set delete this change steps by twenty it will again get incremented by twenty ah! Well once again I need to gave control wait for a second between stop so that I can see what's happening? Correct? So what will this do? All of you? It's only obvious once I click on this steps get to set to twenty go back see in one second it becomes forty one second sixty, one second eighty and then stops that's precisely what I want this code to execute and give me a output like this correct? This is what I expected. And that's precisely what it doing that right now, correct? Now what if I put this in a repeat loop, what happens? Ten times this entire things get's executed, what get executed? That sounds very complex you see let me just keep it aside this is not using, we are not using this all though we will soon come to this cat go back ok. Let me execute this. What will this give me? Repeat ten times this entire thing, double click let me see twenty forty sixty eight, twenty forty sixty eighty is not visible because I am not putting a waits, wait here let me put that here forty sixty eighty, twenty forty sixty eighty it keeps doing this ten times correct? As simple as that but then what if I want this up to let's say thousand? How do I do this? I initially set steps by twenty and wait for a second and change steps by twenty and then let me remove this things and put a wait here but then I will not set this to twenty here itself see what I am doing slowly and patiently I will set steps to twenty but then repeat this ten times so that it keeps incrementing twenty at a time what does this do? Let's see double click on this you will see twenty forty sixty eighty hundred and so on hundred and twenty hundred and forty hundred and sixty and one eighty and finally two hundred correct? Two twenty oh its goes on its ten times right? So its Last one is actually two twenty we start with twenty so its two twenty, figure this out look at this code and see what it does it simply starts from twenty and ends in two twenty, what is the motive? What exactly am I doing? I have to move this cat incrementally right? Of what use of all these things? Now we are going to use this, we are going to use this and make this cat move the way we want her to move let's go back to this repeat and keep it here and what do I want the cat to do? I want basically a variable sets steps to twenty to come here sets steps to twenty and then what? I want her to move twenty steps ahead ok, instead of twenty I want to put steps here, what do I mean by steps here? I will take steps here and put this here please not what I am doing, is a need thing in scratch, nice facility in scratch. You set steps to twenty and you move twenty steps, first just execute this, lets see what happens ten times click she keeps moving ahead, that's what is expected with the pause of one second what happens now? Pull her back, what happens now? Double click, she moves ten steps, ten steps, ten steps steps is twenty move by twenty steps and then wait for a second ok? Keep doing this I can consider doing the following keep this here and then keep this here so that you set steps to twenty and then move twenty steps and then wait for a second move twenty steps correct? I was assigning steps to twenty here every time that's not required initially if you assign that should be enough, double click ahead twenty twenty twenty twenty twenty so on ten times let's watch her walk twenty twenty and then she stops of ten moments correct? Now what do we want? We want her actually go by twenty steps and come back by twenty steps you see so now what I should do? Is I should go back motion move twenty steps steps ahead and move back a few steps. Please note the steps is a built in thing here I mean By that I mean steps here means the cat should move so many steps we are also calling the variable steps here don't be confused about it right? These two are actually different things here ok. So now she must move back by twenty steps how do we do that? Go to operators there is very nice facility here what is that? I can do multiplications of two things correct? I want to multiply minus one with data and take steps put that here I can multiply steps by minus one what do I mean by that? If steps is twenty steps times minus one is minus twenty so I make her move minus twenty steps so twenty steps ahead and twenty steps behind when I click on this, let me remove all these things these are not required we delete them ok, when I double click on this you see what happens, she keeps going back and coming forward, going back and coming forward you can put a second pause here to make her go ahead come back go ahead come back she will do this ten times come back, up come back twice up come back four five six and so on correct? Ok. So she is doing! we are almost there but what exactly , you remember what exactly we wanted her to do? We wanted her to go ahead twenty steps, come back twenty steps go ahead forty steps next. So so so! What do I do? Come back to data think I should use change steps by twenty ok, people this is where the place where things are getting slightly complicated and I tell you the reasons why it's getting complicated ok it's because as I am going slightly fast off course it is the lecture videos but you have to liberty to pause the video and think about what I am saying, if you don't do that I will be going on fast and fast right? So based on your taste and paste you may want to pause the video and then try yourself how exactly you can make this cat move ahead and come back an incrementally keep doing this so what is this go to set steps by twenty repeat this ten times move steps number of steps that steps is actually twenty so move twenty steps, steps is a same twenty, so move twenty steps wait for a second and move twenty steps in reverse direction minus twenty steps pause and then change steps by twenty means, steps is no more twenty it becomes forty here and then come back and start from here, you see I don't again assign steps twenty here I come back here and then I continue. What is steps now? Is actually forty. So wait for a second you will move for forty wait for a second and then you move back by forty and so on and you will be surprise to see that this is precisely what we wanted our cat to behave like. Double click twenty ahead twenty back, forty ahead forty back, sixty ahead sixty back, eighty ahead eighty back, hundred ahead hundred back one twenty minus one twenty one forty minus one forty one sixty minus one sixty one eighty minus one eighty two hundred minus two hundred now that must tiring right? But the cat doesn't get tired simply because it's a computer programme you can make her move the way you want her to move ok? Now can you play around these numbers here and make her move really fast and many numbers of times I leave it to you.

# SCRATCH ANIMATION 1

So let's do one thing, you can in fact take some other object here not necessarily a cat. How about taking an aeroplane and making it take off, take off exactly it will go for some time on the ground and then take off, go for some time in air and then land let's say, how would you do this? Make it move forward, ok! Chase the angle of the aeroplane gradually may be and move it straight perfect. And then again change the angle, change the angle perfect right and move it straight, straight! I also told you it should also land and then rotate it again, rotate back and then land it down, land it down you must ensure that you don't land like this, gradually come like this and then go ahead ok? Shall we try to see if we can do it or not? Yeah. Ok perfect.

So let us try to build the programme for aeroplane taking off and then landing so first of all what we need do here is, we have cat as the sprite here and we need aeroplane! So how can you do that? In scratch you have many different options of sprite, here you can go to new sprite you can choose any sprite from the library, so I will just do that just click on it and see how many sprites do you have? You can choose any one, any sprite that you want so here since we need aeroplane, I will just click on it you just double click and you will get sprite here on your scratch screen, now we have aeroplane and cat sprite here so we need only aeroplane so I will just delete the cat sprite here just right click on the sprite and select option delete. So here is the aeroplane and I want to reduce it size since it won't fit into the screen of scratch so I just you can even increase the size of the sprite as well as decrease the size of the sprite and now but I want to shrink it so I will just use shrink there are two options here as you can see, one is grow and another is shrink so I will use shrink here just choose it and click on the sprite and yes it is shrinking so here we have the sprite. Yeah! I think this is the perfect size for the aeroplane that it can fit into our screen ok, so I think you know how the aeroplane takes off and then land let us try to build the programme for this, scratch programme for this. So first of all, what do we need to do here is? We have to move the aeroplane on the run way yes! We have to move the aeroplane on the run way so I will just click on the motion category and we have move options so first of all I just click here move ten steps ok so maybe I want to move it twenty steps or fifteen steps say so let us try to do that, it is just moved fifteen steps but since we need aeroplane, we need to show aeroplane running on the run way I will use move with the delay options, how can you do that? So first of all I will just say move five steps and then I will insert a delay and then move ten steps and insert delay and then move ten steps insert delay and this way you can see the aeroplane moving on the run way, I will just show you what is the difference, here if I just enter move fifteen steps here we have it will just move fifteen steps but if I enter move five steps and then insert some delay and then I insert one second delay and after that I insert move ten steps how will it move? Let us see? Here move five steps, move ten steps here you can see the aeroplane moving and it looks good to see the aeroplane moving this way on the run way rather than just having fifteen steps in one go so for that you have to insert a delay, how can you insert delay? With wait option and where if the wait option? Wait option will comes in the control category, in control category you have wait one second ok in this way you can make the aeroplane move on the run way, first of all, we have to move the aeroplane on the run way and that is done, so what is next thing to do? Then we have to rotate the aeroplane, yes by some degrees so here we have turn fifteen degrees so I will just use this option I will just say turn fifteen degrees here since it has to rotate on the left side I will just use this option here you will have to move, you have to turn the aeroplane by fifteen degrees after that I will again like to insert a delay so, I will just insert wait one second after that it should move some steps so I just write move ten steps yes so let us have a look how does, how does it look like here in scratch? First one move five move ten then rotate and then move ten steps yes, it look perfect! So let me try to run it again so that you get the clear idea of, how it is functioning here, first of all move five then ten then rotate then again move ten, yes this looks perfect. This is how aeroplane should take off, yes. So here we have the aeroplane taking off after that the plane was the plane is rotated by fifteen degrees? After that what we can do here is, we can move it by some steps, you can move the aeroplane by say hundred steps for that what you have to do here is? For that I will again like to insert a delay here so let us insert the delay here and wait one second and then I will again go to the motion category and I have to rotate, I have to move the plane by hundred steps, so instead of ten you have to write hundred here ok? Now we are done with rotating and moving the aeroplane in the air ok, once it has reached the position let us try to run this piece of code. It rotates and then it moves ten steps and then it moves hundred steps, so now our aeroplane is in air, now again what should we do, we should rotate it by fifteen degrees, so again we have to rotate the aeroplane by fifteen degrees for that I will again insert a delay here so that it looks kind of real and now I have to rotate it, so last time we rotated it left hand side now we will rotate it to the right hand side. So we will use this option, turn fifteen degrees after that what should we do? We should move the aeroplane by some steps in the air after it has been rotated so just take move ten steps, after that again insert delay, wait one second so I will just run this piece of code so that you can get the clear idea of what is happening over here, yes. So let us try to run it, move five steps move ten steps and rotate then move ten steps and then again hundred and then rotate and then move ten steps, this is how aeroplane is working here. Now we have the aeroplane in the air, it is moving in the air for that to make it look real what you have to do here is, we have to move it by say twenty steps and then wait for some time then move twenty steps and then wait for some time I will insert this code again and again so that it looks real that the plane is moving in the air so let us do that. I will just write here move say like no no no ah this is wrong, here I will just write move twenty steps then wait for one second then move twenty steps again and wait for one second ok I will insert it again move twenty steps and then wait for one second ok once more we have to insert it, move twenty steps and then wait for one second ok so now our aeroplane is moving in the air ok? Let us try to run this piece of code, so that you can get the clear idea of what is happening, move five steps move ten steps then turn and move ten steps and move hundred turn move ten twenty twenty twenty twenty so it has moved twenty steps four times, that means it has moved eighty steps ok, now we have to in order to land the aeroplane we have to again rotate it so, I will just use the option of rotate here, what we have to do here is we have to turn the aeroplane by fifteen degrees again so we have to turn the aeroplane again by fifteen degrees so I will just use it, turn by fifteen degrees and then move ten steps also with that after that you should wait for one second, wait for one second say and then what we have to do here is we have to move the aeroplane down in the downwards direction, so you have to move the aeroplane down, so I will just move the aeroplane by some, by some steps here so I will just write move say fifty steps here, after it has moved fifty steps here then you have to again rotate the aeroplane, after it has moved fifty steps you have to again rotate the aeroplane so what do we need to do with that? For that we have to use the turn option so then as an always inset some delay and then use the turn option you have to turn it by fifteen degrees and then you have to move ten steps again and then wait for some time after it has rotated now it is on the run way again because it is landing, since it is landing then you have to move it by some steps, it is your choice how much steps that you want to move here, I will just use wait I just use move ten steps here and then wait for one second, wait for one second then again I will move it by ten steps so let us try to run this again and see how this aeroplane is taking off and landing so it goes up in the air then it moves there for some time and it rotate and comes back, see we did a mistake here while taking off we moved it by hundred steps but while landing it is only moving by fifty steps, so what we will do, I will again insert move here and this time I will move it by fifty steps so that it looks real and again you have to insert a delay so that it doesn't appear has crashing on the ground so again will run it, double click and run yeah it is going ahead in the air then straight for some time about eighty steps then it has coming down and then again come down and now it is on runway ok, so this is how you can make the aeroplane take off and land in scratch through a scratch programme but you can see that there are many repetitive instructions here, we are using for example here we are rotating it by fifteen degrees then moving it by ten steps and again we are using this code again and again so what we can do here is we can actually group them in a repeat loop. I hope you are already familiar with the loop concept here so what will we do, we will group the repetitive instructions here so that it becomes easy for us to write the programme and the programme becomes short so what we will do here is for the repetitive instructions we just we will just group them in one loop, I will just use the repeat loop here for example here we have move five steps then wait then move then turn then wait move then move by hundred steps and wait so we can't insert a repetitive loop here because here we are moving by ten steps and in the next step we are moving by hundred steps then turn then move, move, move wait move wait here we can insert a repeat loop here because we are moving twenty steps four times so we can insert a repeat loop here and we can repeat it four times this set of instructions move twenty steps and then wait for one second after that we have to move turn then move then move move turn and then move and then we have to move again and again so ah I will just insert a repetitive loop here, a repeat loop here four times in place of this instructions so what will I do is, I will just copy this but we have to do just duplicate it and keep it here so I will just insert a repeat loop here in place of move twenty  and wait one second ok? So what will I do here, I just take this loop and what we have to do, we have to repeat it four times yes and you can now delete this and insert it here so let us now, see how this programme with repetitive loop works. So let us see, it is moving then again in the air then straight one two three four yes perfect then it is coming down and then landing yes!! So as you see this programme is same as this programme but the length is somewhat shorter because we used a repeat loop here, so let us now get back to sir and see what is happening in the other end.

# SCRATCH ANIMATION 2

So you people saw that we could make a cat move. Scratch gives you a whole lot of possibilities in fact, I can make you people move, you want to see how? Yeah. So what we will do is, we will take a picture of Ravi kiran and make him do exercise by writing a piece of code. In fact you will be, you will just give me two poses I will take a picture of these two poses and then I ensure that I make you move around so much it looks like you are exercising. Ok? Get up let's take a picture.

Hello everyone in the video we saw that we took the photo of Ravi's so that we can create a simple scratch programme there he is exercising. So we took two photos of him let me show you, these are the first photo he is standing and the next photo he is waving his hand, his hands are one eighty degree and his legs are stretched. See we are going to use these two images to create simple exercising programme. So we make Ravi jump and exercise using these two images. So let's see this so what I will do this is my, I opened my scratch profile, you can see that there is a sprite ok? Which is the default sprite sprite that is a cat instead of this I need ravi's picture so that I can use it so attaing to we will hide this you can right click on it and hide this hide this sprite and here you can see there is a upload button you can upload your own images and create sprite from there so I will upload the Ravi's image one is this ok? You can see since this image is rotated in left side and i need to rotate it ninety degree ninety degree right foot that I will go to the info you can see there is a I icon I will go there and i will rotate it tats it. Ok. This is done, let me hide it for time being let me upload the next image this is there again this is also rotate it ninety degree left and I need to rotate it ninety degree right i will go to the info rotate it ninety degree this is done. Let me show this ok? So what I am doing them is hiding it showing it hiding it showing it so when I do you can see that it looks like exercising or getting to different different position see so it looks like a animation if I do it little bit fast then it will look like he is actually exercising so we can create such kind of animations through images, this is what actually animators do, they create different still images and then they play it in a very high frame rates so that it looks like the person is moving in the actually moving looks to our eyes that the person is actually moving so this is exactly what we are going to do so I will be using this sprite only so whatever function i am going to put on this sprite only this the second image so what I will do I think you got in tuition what I am going to do. Now I will create a repeat block here ok, what I will do I will go to motion let me check where are the show ok? Show and hide is there so first time we will hide this image ok? And after this I will show the image let me let's try this I am going to do this for ten times ok. It looks like nothing is happening but actually so many things are happening but since there is no delay in the between it looks like he is on still bit it went too fast that it look that he is in single position so what I am going to do, let me first show and then hide, ok so what i am going to do, I am going to put a delay let me see when the delay ok wait after showing wait for one second and after hiding also wait for one second ok, now let's see awesome! Great! So this is all what we are going to do, so it looks like that he actually, he is exercising but we all know that by using the images we did this, he dint do anything so such kind of animation is easily possible in scratch and you can do a lot of things in scratch by set simple, simple techniques which people know so you can create your own animation you can create your own series of images you can have series of images you can have beautiful animation through this. You can download this scratch also for example, I am using this offline you can see this offline scratch 2 editor, so you can download this scratch tools in your pc and you can use it offline also thank you.

Ok, that was so much fun seeing Ravi kiran workout, Ravi kiran I think you should not play this animation for the mornings exercise replacements definitely I will use that only! I don't think you will lose any calories if you keep watching them, you probably gain calories so what you should do is maybe, you should use it as a reflection motivation he looks so energetic when he does the jumping jacks you see, so when you do jumping jacks you might lose out energy after two or three minutes but if you play the animation the energy looks intact may be, it can see as your mirror reflection and you can work out, you should try doing that ok, how about you doing an exercise problem right now? Why don't you do what is called knee rise, a knee rise is you should lift your left knee put it back and then a right knee it will look like stationary jogging ok? One! Another one sis you should try moon walking animation for moon walking, you know what is moon walking? Have you seen Michael Jackson do a moon walk? Moon walk is about walk, it appears as though you are walking forwards but you will actually be moving backwards ok, try taking a couple of pictures and try making it appear as though you are moon walking second Exercise.

# SCRATCH ANIMATION 3

Who said programming is for geeks!? We just now illustrated that the programming is so much fun right? We saw a friend of ours doing jumping jacks we made him do repeatedly and we also saw how we can make a flight land and take off right? So we will see one more fun exercise before we stop our adventure scratch.

Hello guys, hope you would have seen a lot of fun filled activities that you can do using scratch, so you would have seen jumping jacks recently that is nothing but we are using some two different images and we are alternating displaying them in a smart way that it appear to the user that the person is actually jumping, so something like that you can do a lot of cool stuff, sir has given you some exercises I suppose, please do try them also do explore still what all could be done, you use your creativity do a lot of stuffs and this is how you can improve your logical listening and the ability to get things done using a computer. So please do explore scratch to the fullest possible and in this programming screen cast we would see yet another fun activity, we are not going to use this cat in this I hope you had used the cat a lot you had made her do a lot of work so let me say this cat must be hidden, so I will hide her, so in looks category you have an option ‘hide’ let me say hide and say double click on it, she is hidden done! So if not cat what am I going to use? So I am trying to stimulate eagle catching a snake, so this is what I am trying to stimulate, for that we need to check out the sprites for eagle and snake, some sprites are available in the library from scratch or you can also draw your a new sprite as per your wish or if you have an image file something like that you can upload it or you can readily take from your web cam as well. These are the options provided in scratch, I have an image file I am going to upload it because in this library I had checked animals category eagle snake is not provided in this library, so I am going to use image that I had taken from the internet ok, so let me that is available in my system right now so let me go to it and let me download, I had downloaded and kept it so let me take that image, eagle ok and open it ok so I want the eagle to be positioned here so eagle would be flying in the sky right? So this gives an illumination that the eagle in the sky so I had positioned it here so roughly the co ordinates you can say one eighty six, one eighty nine so let me set it so that the eagle comes, if you want to repeat the exercise again and again the eagle comes back to this position and starts flying from this position, let me set the position for the eagle, set ‘x’ to one eighty six ok, and set ‘y’ to one twenty nine or may be slightly above one thirty ok? So given that I have set so eagle has come to this position see one eighty six, one thirty two eagle has come to this position and I wanted to catch a snake, so we wanted a snake that is happy that is happily lying on the ground, the eagle has spotted the snake while flying and so it flies down, it will fly down and come and catch the snake. That is what we wanted, so let us upload the image of the snake, for this the snake is very happy that’s what I said let me show you the image as well, first let me show you the image I guess it’s better if I can show you the image ok, so here in downloads I have it ok, so smile and cry there are actually two images see this is the image of the snake, see the snake is smiling right? So the snake has been smiling and happily lying on the ground so when the eagle catches the snake it shouldn't remain happy, it should cry so when the eagle catches the snake we should make it cry, so but how can we make it cry? Same old trick that we used in jumping jacks we will have another image of a crying snake and we let that particular snake appear when the eagle catches this snake and this snake would be hidden, this is what we are going to do. It will be clear as the programme proceeds, if you don't clearly understand what I am telling now it will be clear don't worry please observe it this is the smiling snake image that I had downloaded and another is a crying snake image ok let me, see the snake has been crying you can see the tears flowing from here. The snake has been crying see the snake is crying alright? So I am going to use these two images, I have to upload the images so let me click this icon to upload it from the file so my first is the smiling snake tats what I want smie.png file let me upload it so this particular snake, snake, I want it here see we had set the position for the eagle that is not visible when we click on the snake so please do understand that scratch treats each of the sprites as independent objects so when we want this particular snake to do something on the event of the eagle we need to use something on the event of the eagle doing something, we need to use the options given under this category events so in this programming screen cast you will learn about the options given here, how to make use of them this will be major take home lesson out of this programming screen cast ok? So this is the position we want our snake to be ok? Let me say this is our snake, this is in position minus one seventy nine minus one twenty nine so let me set it set 'x' so that 'y' do I set the position because in case if you want to repeat the experiment you cannot every time adjust the snake adjust the eagle to ask it to do as you wish, you wanted to get adjusted automatically you want to adjust once and set it once it is set that has to get adjusted automatically that is what you want them to do right? So that is why we are setting it, so 'x' is minus one seventy nine and 'y' is minus one twenty nine so set 'y' to minus one twenty nine so if you don't understand how the coordinates are being set please you can refer here the positions are shown here may be I will this particular portions, this particular place you can see the coordinates. Ok? So please note the coordinates minus one seventy nine and minus one twenty nine and this is the position we want our crying snake also to come so let me upload the crying snake to the name is cry.png this one let me upload it so crying snake has appeared here so let me set the position minus one seventy nine and minus one twenty nine let me set them set 'x' to minus one seventy nine and set 'y' to minus one twenty nine initially I don't want the crying snake to be, see now the smiling snake and crying snake are in the same position so it is imposed one above the other, super imposed so I don't want the crying snake to appear now so let me hide it hide options is available in the looks that you would have seen in the cat right? Let we don't need it so we had hidden it so let me hide it so now let me double click ok so the crying snake is hidden now alright. So now we have to start animating with the eagle so let us say the snake is happily lying on the ground the eagle has spotted the snake it has to fly down and catch the snake, so how would you instruct it to fly down? Move few steps it keeps moving in the straight line it can't hit the snake so it has to move few steps slightly change its direction again move change its direction move change its direction this is how the eagle has to fly, this is how we have to instruct the eagle to fly, so let us do that. So in motion category we have this thing move ten steps for the eagle please note that all these instructions we are giving for the eagle so eagle so you can see the blue box around the eagle which denotes that this particular instruction are given for the eagle sprite ok? So move ten steps then I need to move in this direction so this would be approximately you go thirty backwards in the 'x' axis and ten backwards in the 'y' axis that is the direction I wanted, this is just an approximation let us see how it works in cast if it doesn't works how we wanted we can adjust the parameters any time, this is just approximation an eye approximation that's it so and also note that this may differ with respect to the images, I had taken some images if you have some other image the based on the image dimensions of how much space it takes, based on that this image is a smaller image so it takes less space so for that these parameters may work but for any other image, if you take depends on that image parameters may change so it's better you can, you try once and then you finalise the code that's what I am going to do now to so I am asking it to move ten steps see it moves ten steps see it moving in the backward directions so I need to ask it to come here so I should say to change your x position you move change your x position and y position this is what I am going to say, change your 'x' position by this is your visual approximation I guess minus thirty and change your y position by minus ten I feel this would be better let us check how it goes ok yeah its approaching towards the snake see its approaching towards the snake so I keep clicking for each click it displaces by some amount of distance so I just, I need to use repeat loop so that it approaches here till the snake so let me use a repeat loop control repeat ten times, let me try with ten times so let me again restart. I set the eagle to the initial position now let me repeat it ok, it has come half way so let us give a try for twenty times ok, let us give it a try now. It is in its initial position and let us give a cry, oh it is crossing the snake so let me try nineteen times because for ten it is not very closer to the snake and for twenty it is crossing. so I am slowly decreasing the value from twenty but it is just a hit and try last. I had said it depends totally on the image whatever image you have downloaded based on that the things may change that's why even I am showing you the hit and trial method. Ok so I had set the eagle to the initial position let me execute this ok so it has touched the snake, no! Not fully right? This portion, this portion has to touch this is just the tail part has touched the snake so what should I do now? The 'x' position is perfect 'y' position it has to come slightly down that is the tail portion is touching the snake it has the eagle has to come down so that it can completely touch the snake, so I have to make it come down, to make it come down I should change the 'y' value 'y' position means the 'y' position controls whether it has to go up or down if you give a positive value it goes up for a negative value it goes down, similarly 'x' position is for left and right if an 'x' is given in an negative scale the object will move to the left and for the positive value it will move to the right. This is how the working is so I want it do come down so I should give a negative value of 'y' let me say I will experiment this is purely an experiment so let me experiment change 'y' by minus ten how far it moves, let me see ok, change y by minus ten ok? Let me again restart it is set to its initial position, let me repeat. Ok perfect I guess now it's perfect or shall we move still down we will try, ok let me make it minus twenty ok, it is set to the initial position let me repeat ok perfect it is now it can easily bend down and take the snake that's it means the snake is much reachable to the eagle currently so this position is perfect, so eagle has been moving like this so based on this, this snake has to cry so how will we do that? We have to hide the smiling snake and make the crying snake appear, that is what we are going to do, so but in jumping jacks and here what is the difference is in jumping jacks you had the alternating of pictures after few seconds so it is alternated in a cycle something like that you had, you waited for sometime it has alternated in a cycle but here the appearance of another image depends on some other event so event management is what we are going to do so when the here we have set the kite! Set the eagle ready to fly so let me say it has to pass a message, broadcast a message that is it has to pass a message but it has been it is flying so let me broadcast the message I can create a new message I will say flying so the eagle will broadcast a message that I am flying something like that so it is broadcasting the message that I am flying so given that it broadcasting the message only then the smiling snake has to come. And the crying snake has to be hidden if the message, broadcast message has been received so let us first configure that, it should not remain hidden if you so this it will remain hidden forever we want it to be hidden only till it is flying, till the eagle is flying the snake is unaware of the eagles arrival so it is smiling once the eagle has reached closer to the snake, only then that it starts crying only then the crying snake should appear till that it should be hidden, so when it is flying this crying snake has to hide that is what we wanted so when it is flying? How do we do that? Yeah we have a block when I receive flying so the eagle has been telling right I am flying so when it has been flying the crying snake sprite can be hidden so this is achieved here and the smiling snake has to be shown if it has been flying means when the eagle has been flying in the sky, the smiling snake is not aware of it so it is happy so the smiling snake has to be sure so let me say when I receive flying when the smiling! The message flying is sent the smiling snake has to be shown so let me say show from the loops. So when it starts flying it sends out a message that I am flying so in that stage the snake is unaware of the eagle approaching it has to smile that is why the smiling snake sprite we had set it has, when I receive flying message it should be shown, I have to be shown and for the crying snake it is an opposite, if it is flying then no need for the snake to cry, if it is flying in the air so this has to be hidden so we have set it here ok then ahh when the snake touches the eagle sorry I am sorry! When the eagle touches the snake the snake has to start crying so this snake has to disappear the smiling snake has to disappear and the crying snake has to appear so when it touches, how will these two come to know? If it is a living object they would know that someone is being touched, disturbing us all these they know they would react but here we have some images that’s it we are making the image behave as it is a real object so. how would we get things done here? So for that again same like how flying message have been sent we need to sent out another broadcast message if this particular thing is touching this thing, so for that let me see touch is under sensing yeah! This colour is touching this colour something like that we have an option maybe we can use it if this colour is touching this colour let us use that, if some condition then that block we will use it now, we will use it so if some condition, the condition is if the colour is touching this colour so the colour you can change it, how would you change it is? You click on it and just check which colour you just wanted to appear, just click there you would get that colour see this colour is this the brown appeared here and now this colour should be this one ok green appeared here ok let us bring this, I am fitting here ok, if this colour is, if the brown colour touches this colour then it has to send out another message so for that again we are using the broadcast option but this time the message is not it’s flying, you have to send a new message so may be for the situation I will use ‘trapped’, so if it is sends out a message ‘trapped’ this snake will have to disappear, and the crying snake has to appear so I will configure it here, the smiling snake have to say when I receive ‘trapped’ it has to be hidden, so hide ok so we have configured the smiling snake let us go ahead with the crying snake when I receive trapped its trapped this has to be shown so the crying, snake will cry but still will the eagle leave the snake? No, it will eat the snake, so let us wait for some time till the eagle eats the snake, so let us give a wait, wait for one second, one second is lesser for a living eagle but this is a computer image for this one second is too much so one second would be enough it would eaten the snake so after one second this has eaten the snake the snake that would appear should disappear so again I will give a hide option, a hide option alright so hmmm I hope you are clear here if the crying snake has got means when the message trapped has been sent that is when it will be sent? When the particular colour, this portion of the eagle touches this portion, that is the eagle has touch with the snake so when it has touched the trapped message will be sent and because of the trapped message the smiling snake would disappear and the crying snake would appear, which makes the user fevered due to the eagle now caught the snake the snake is crying. So the crying snake would appear even though the snake cries eagle would not leave it, it would definitely eat that snake so this is why we are given a wait loop, it cries but still the eagle is merciless it eats the snake and after it has been eaten the snake should not appear here that's why we are hiding it. So now here after it has been trapped the eagle will be happy that it has got a food so we need a sound effect there right? So let us have a sound effect, sounds you have an option here let's see if we have something from the sound library something you can record it with your microphone or if you have a sound file you can upload it as well these are the options available here. See, from the library animal I don't have a eagle sound so that is why I had taken an eagle sound that I will be using it, I have a sound file basically I will be using it, you can get the eagle sound from the internet as well you can use it so eagle.mp3 this is my sound file let me open it and show you how the sound is, I will play, this is how the sound of the eagle will be hence it has caught the prey so that we are going to use it here so we have loaded the sound into our project ok once it has sent out a message that it has trapped it, it has to happily give the sound that it has got the food so we have to insert a sound, play sound eagle! Yeah you have an option let us use it here, it will play this sound. So after it has played this sound the crying snake would appear and it will eat the snake and once the snake has been eaten that snake would disappear also the eagle would come back to its original position that is what we wanted to be done, so we have to wait for it to be eaten so once it is eat it has eaten the snake it will rest for some time and then it will fly back right? So let us give the wait loop but slightly more time than the snake, snake had got one second waiting time, that is one second it has eaten and one second let it take rest let me say wait for totally two seconds so there is a wait loop and once the wait is over it will again fly back so all these I have in pieces so this particular thing is nothing but I will create a blocks for it so I will say fly, fly is a block so this is my fly block and this I could say as target, target ok! So this would be my target block and or maybe I could club these two because if it is targeting it will come near the snake and it will stop that will never happens right? It will bounce opens the snake and it will eat it so I can club these two blocks as well so this is my target this is the flying part and this is the targeting of the snake part sp these are the two parts so now let me use these blocks initially the eagle has to been shown let me say show then it starts flying then it aims at the target then once it has eaten the snake it is satisfied it will go away so that I am denoting by height that is, it has gone away, it has gone out of the frame so it has gone somewhere so I am hiding it so basically this is what I am trying to do this is the portion of smiling snake and the crying snake, ok let me check this how it works? Let me double click on this particular block which instructs it, wow! That was so good let me show once again, is the flying too fast? Do you feel that some more delay could be better so that you can feel that it is actually flying, it looks like the eagle is running! So maybe we can insert a delay, this is the computerised image so obviously the speed of the computer is much higher than what! Other living beings are so definitely it would look as it is running so let me insert a delay so that appears as it is flying, one second is too much delay so let me use the delay loop but ok, let me say wait for point one second, that is better I guess zero point one seconds let me wait and now let me again run this, let me double click wow! Now you could see that the eagle was flying just observe one second the eagle is flying it has touched the snake, the snake cry, still it ate the snake and then once it is satisfied the eagle has also gone. See it doesn't wait for me to given description, computers are so fast ok still we will try once again eagle is flying got the snake, snake crying, eaten eagle is flown fine! Wow this is so nice you can try in more creative versions of this as well you can do a lot of things with scratch please do explore a lot this would improve your logical reasoning ability and thanks for watching this screen cast have a nice day! That was again a lot of fun! Did you see how you could pick different pictures and make eagle catch a snake and then go forward right? So you can actually animate, have your own animations, put people over there and then create a story out of it, any more questions in mind? Sir any one can create animations? Absolutely yup, you can create a animation made a video out of it and put it on you tube as well, you should probably try doing that, you have any other challenging exercise in mind? How about this? A bus comes, there are four people are standing, a bus comes picks one person and goes away and same bus comes picks the second person and then goes away third person goes away, fourth person goes away that's it. Each time number of people should reduced, should reduced yeah reduce by one ok? Try doing that and a good challenge would be make a ball bounce in that rectangle region where you animated, you create a ball which is an object and that ball should bounce around and stuff like that so you see there are actually many games written just using scratch, very beautiful games in fact they are addictive you can probably create snakes game, heard of snakes game yeah! Use your arrow keys and the snake moves the things like that a little complicated but it is not so difficult, you probably should try doing it, there are a whole lot of live examples on scratch website itself where in you will be getting code on how exactly a game works and you can see through the code and understand how you can create a variant of them.

# MORE ON SCRATCH

We have now come to the flag end of our first chapter it's time for us to summarise everything that we learnt so far ,we first saw how to make our cat move, how to make her rotate, what was the introduction to programming? And then I told you how to group a bunch of tasks? And there lies the heart of programming, right? we spoke about loops and variables and i told you how you can make the cat recite tables, then we went ahead and showed you a couple of fun examples using our newly learnt programming skills it is now time to see a few more topics in scratch which we will take up in the forth coming chapter.

Hey you can see there are so many featured projects available in scratch, you can always explore them, but before we get into the details of how to design our own scratch game there are many important concepts available in scratch so let us have a look at them.

Let us start with arithmetic operators. What are those? Pretty straight forward pull this, this side ok I have put this here now I will enter let say five here and six here and when I double click on this you see I get eleven, double click eleven. Double click eleven right? if you change this let's say to fifteen and then double click, you will get five plus fifteen which is twenty ok and whatever is the result if you would want you can you can now put let's say ten and eight here, double click you will get two. Ok now look at this, there is something more that you can do, let me delete ten here and let me put this here you see what happened? I repeat I can put this here do you see what's happening here, I can put this here so five plus fifteen is twenty, twenty minus eight is, what is it? Whatever is the answer, you will see that here, it's twelve correct? And I can go on doing this in fact I can take a subtraction thing and then I can divide the answer by two and I can pull this entire thing here like this, so you take this and place it here so it is five plus fifteen is twenty minus eight is twelve by two is six double click and you will see six ok? You can ease this, you can use these arithmetic operators the way you want and also there is an interesting function here, see what this does? Isn't that self explanatory? I don't have to explain this to you correct? It fix a random number from one to ten double click ten, double click six again four, double click one double click six, double click four, double click one see it is very random it doesn't follow any pattern ok don't worry so much about how exactly this is done? That's a very complicated subject all together but you please be aware of this very beautiful facility of scratch, which gives you a number between one to ten picked randomly. So it is not just limited to one to ten as you would have guessed you make this hundred it will pick a number from one to hundred, I just made this hundred right now double click you will get a random number twenty two between one to hundred twenty two, twenty three, fifty five, twenty three, fifty eight, seven, ninety seven, eighty five, eighty four you see there is no particular pattern, you can start from any number up to any number ask for a random number you will see we will using this a whole lot in our explanation throughout our course we will be using this, right? So scratch is that way very interesting and very straight forward and self explanatory that you just see it you could guess what this function is ok, So let me do something very interesting then let me keep this aside may be even delete this, let me do something interesting. Let me consider, I have to pick a random number from one to hundred and put that here you see what I am doing and then pick another random number from one to hundred and put that here, and what is a result? Double click on it one ninety so you pick a random number you pick a another random number, you are adding it double click one forty four so some of two random numbers pick between one to hundred is giving you some random number here and some random number here, is giving you so much ok ninety five, one sixty one, one thirty nine, hundred and so on in fact what you can do is you can make our cute cat recite these numbers, let's see how she can do it? Go to looks, go to say hello keep it here double click on it she will say hello, you take whatever you want, you want her to call out my name? Hi sudarshan! Double click, she say 'hi sudarshan'! What you can also do is delete this, can you guess what i am going to do next pick this and put this here what will she say now, she will tell you a random number the answer basically she less spit out one one two, twenty four, one forty four, seventy three and so on, one not four ok I can play around this and make this ok remove this what was this doing here? You see the pieces here I pick a random number from one to hundred this was not required ten in ten not required, you can put whatever you want here, put whatever you want here and make the cat say, I will just make her say ten plus twenty, what is that? You all know its thirty but let me see whether see says thirty or not, double click here and she says thirty hey! That's right, so it so much fun to keep doing this you know the you can just try whatever you want, go to the operators and do whatever kind of operations you want and then make her say whatever you type right? Isn't that awesome? Right so now it's left for you to explore these things and now move onto the next topic.

We will now make our cat ask your age and say something about you, what is that? What exactly does she ask for and what will she do based on your age? You should wait and see, ok so now, let me teach you something go to operators and you see this join, this is nothing but two strings that you type here will be joined as the one string, you see you type let's say my name here and a space and you type nptel it will show 'sudarshan nptel' ok, whatever you want you can put ok? New space and then Delhi here it will show you newdelhi ok? You put first space second it will show you first and second ok so of what use is this? Let me make the cat do something and then make it clear to you what could be an application of joining these two strings, let me now go to looks and take say and then let me create a variable called name. Ok done so name will contain nothing to begin with as you know it always starts with zero set name to let say your name, my name I will put sudarshan space so name will be put as sudarshan and cat will say call out my name right now, you see what's happening, it will say sudarshan but I want her to say hello sudarshan, how do i do that? Pretty simple. Remove this, Say hello here space and then put a name here it will say hello and whatever is the name said to sudarshan perfect! Now click on this, she will say 'hello sudarshan' is that clear? Perfect. So she now saying 'hello sudarshan' and then I want her to take my age let's say I create a variable called age and I will set age to whatever is my age ok, age is now thirty five and I want her to tell me whether I am an adult or not? Off course I know I am adult, I am eighteen plus but I want the cat to decide whether that I am eighteen plus or not? How do I do that? There is something called if look, an if look which does the following, so if the age is greater than or equal to eighteen how do I do that? Go to operators am being little fast here because I have a felling you are now, you people are now familiarising yourself with the scratch programming and you are way too comfortable with whatever I am doing right? If not you may want to pause and then try it out yourself slowly and then see what is happening here ok? Ok. If the age is greater than seventeen, if age is greater than seventeen means what the age is greater than or equal to eighteen, seventeen is not allowed if the age is greater than seventeen then the cat will say, it says hey you are an adult, you can do anything you want, although this is not true you cannot do anything you want by anything you want I mean anything that is allowed for you to do right? But if the age is less than something it must say oh oh! You are not an adult. How do I do that? Go to control, go to if, go to data take your age, go to operators use this less than if age is less than eighteen which means seventeen sixteen fifteen and so on the cat must say, how do i make her say, cat must say oh oh! You are still a kid! The world is not so open for you ok, now what happens if I execute this let's see, i will click on it first ok, i should put a pause here you see that's very important, control and i make cat wait for let say three seconds and i am going to set my name to sudarshan the cat will say 'hello sudarshan', it will wait for three seconds and age I am going to enter here based on what is my age it will say so and so ok let's see, double click hello sudarshan wait for three seconds hey you are an adult you can do anything you want but if my age where to like say fifteen, I am far from fifteen but assuming that i am fifteen i double click on it, it will now come here, off course say hello sudarshan wait for three seconds my age is fifteen now it will come here, it will see that the my age is greater than seventeen no! Which means this will not get executed my age is less than eighteen yes! Fifteen is less than eighteen so this will get executed let's see. Double click on this hello sudarshan wait for three seconds and then this will happen, ok if you are finding this a little confusing what I will do is I will separate this from this forget this for the time being will not worry about this we will only look at this code, if age is fifteen double click it says oh oh! You are still a kid, if age is ten what will be the answer? Age is less than eighteen off course this should get executed let's see, double click yes! The same thing, if my age was twenty five then I am an adult it should say you are an adult double click yeah! You are an adult, you can do anything you want, now do you understand the importance of this loop? It just checks for a condition and does what you ask the cat to do here ok, now, let me add a small animation to it just for clarity, one second, hey you are an adult and move a few steps forward ok, it must move some twenty steps forward if one is an adult so set age is twenty five which means this must get executed, it must move twenty steps ahead see it says and it moves ok, let me give it a pause so that it is visible to you people. Wait for a second and then move, see it moves it says hey! You are an adult and moves by twenty steps again double click says hey you are an adult and moves twenty steps but if I am ten year old double click it says oh oh you are still a kid world is not so open for you it doesn't move ahead right? Because in my if loop here this gets executed and only this is displayed. Correct? Now if I were to take this and put this here let's say wait for a second and then move twenty steps backwards so if you are not an adult which is by the way the case here given that your age is ten she will move back a little ok, Let me say minus thirty, she obviously! Minus forty also so that she goes back twice the distance double click she goes back correct? Again double click she goes back correct? She says this and then goes back correct? But if it was some, if i was sixty here old she will call me an adult and go forward, if i was five year old, she will say I am not an adult and still a kid and she will go back. I believe you are seeing her move back here, five here let's say forty she will move ahead correct? Perfect! May be, you may want to write this code and then check if it's really working the way it working with me right and you may want to explore how if works with the different example and a small exercise for you all, what do you think the if then else loop does? This is called a loop ok? I told you how if then works, if then is what is given here you must tell me how if then else works? Go on. Before anything let me attach this here and try executing this programme for one last time with my age double click says 'hello sudarshan' and my age is given to be this and it moves forward and says you are an adult and you can do anything you want perfect! So let me move on to the next lesson.

So it was a whole lot of fun learning scratch, I must tell you people that not everyone has this bend for programming in the beginning, it takes some time to seize in your minds and scratch does that to your mind. You will have a clear understanding of how a loop works, we saw how a list works, what is a variable, what is a block? You see, block is something that repeatedly does a kind of instruction that you give and you can use that any where you want, it is actually called a functioning programming. Will see those details very soon, may be you are wondering, Is programming so much fun? In fact it is indeed so much fun but scratch makes it a whole lot of fun so now let's switch our guess and look at python as a programming language trust me if you are very good at scratch you will not face any difficulty learning python. So you mean to say that there are lot of similarities in python and scratch, I would say that they are exactly the same although you will not have this facility of drag and then using repeat loop and things like that, scratch is made for eight year old, python is lot more powerful and hence given that it lot more powerful the way you write a piece of code in python is very different and can be just a little more challenging than how you do it in scratch.

# INTRODUCTION TO ANACONDA

In the last week you got to know what is programming all about, we all saw that programming can indeed be fun. We showed several nice examples on scratch asked you to create your own scratch programmes, I am sure many of you tried doing it. Let us now switch modes and start learning python. Python isn't as easy as scratch but we assure you that we will certainly make is easy and interesting to you. So get your cup of tea pen paper and computers ready let's start.

Hi everyone on this video we will see how to download anaconda. Anaconda is a tool to which you can write your programmes in python and even combine it literally, so why we are suggesting anaconda? Because, anaconda comes with a lot of libraries so you don't have to install those libraries again or those packages again so it is very user friendly for beginners so in this course throughout we will be using anaconda only and with anaconda there comes an ide know as spyder so we will be using spyder, so let's see this.

# INSTALLATION OF ANACONDA

So let me just type on Google anaconda ok, type anaconda python, so it is comes under the snake to and python ok, so you can see the link, this is the link which is anaconda, you can click here so I will just type url ok, huh you can sign up here also that you get news letter and new versions if there release anything you will get a notification also that so just go to the download anaconda and click on it, ok you can see you can download it on windows, you can download it on Mac and even Linux so today I am going to install the anaconda on windows ten so you can easily download it by following the same instructions which I am going to provide you though this video, you can download it on Linux also and Mac also so let me go to the windows version ok, just click on it ok you can see the anaconda 5.2 is there for windows installer, there are two types of version python 3.6 version and python 2.7 version. In this course we will be using python 3 or 3.x I mean 3 point any of the released version, so we will like you to install the python 3.6 version rather than python 2.7 versions even though the difference is very less some syntax literally syntax differences are in 2.7 and 3.6 version but there are some libraries which are compatible in 3.6 version and not compatible in 2.7 version so in this course we would like you to download the 3.6 version just click on this download button, I already downloaded this in my computer so I will just check my, I will go to the download ok, So you can see anaconda 3.5 is already there so I will just double click on it ok so just make it I agree it's up to you what you want, Just me or all the users I am doing for just me ok you can check where you want to install the anaconda, you can make it as a default also as default it goes to the c drive so it's up to you so I will use the default version, I will go there just install it, it will take some time.

# INTRODUCTION TO SPYDER

Ok guys you can see that the installation is completed so I just press next, I don't want Microsoft visual code so I will skip this you can check this read manual how to learn, how to get started with anaconda? So let's see this what they provide. So getting started with anaconda can go through all these things ok? You can check how to start anaconda I mean spyder id on windows and Linux and Mac other operative system and you can learn more about anaconda cloud, so this is done so I will search here go to the search pad and type spyder so just click it. So you can see the spyder this, I opened this spyder IDE, this part is the part where I write the programme and you can see this name, this name is the name of my programme so I can change this for example what I will do, start save as let say and I save it as my first.py ok, ok? This part is the part where you can see the console so you can see here it's written I python console so in this console you can mostly you will get all the outputs related things, this part is the part where you can see the variables and how they are, how and the files another we will discuss. Variable means how they are changing and how? What values they are taking it so we will learn about this in other videos and you should know that in the python console you can it works as the interpreter console so you can do some kind of programming also example if I write here two plus three it will give me five because it interprets here it's not like other languages like C, C++ and java where you have to compile it first then you have to run it, it works as interpreter and line well and interprets, in console you can write those things ok, let me just write a simple command of print here ok and let me show how to run this so I just write print ok and in the columns my first programme ok and you save it, here you can see the run button just click on it yes run it, my first programme. So this is how you do your programming, let me do something some other stuff, this erase it and let me just create a simple variable ok, "a" is equal to ten which means "a" is a just visualise it as a bucket containing lets say ten cookies or a class which contain or a jar which contain ten cookies ok? Let me create another bucket or a jar containing thirteen cookies let's say ok, now I want to create a jar in C I want to add these add a number of cookies of which representing A and which representing B in the C so I want C as the C should contain all the cookies which representing A plus all the cookies representing B so I will just write A and cookies in Bok I will save this now I want to print cookies in C let's see, cookies in C it should come twenty three the best thing about thing is you can revise the thing what are the variables you are created, you can see those variables, what represents in the console also write here a only and see a it will show ten, and b show thirteen and c show twenty three and here you can see all the variables that helped in created a of type integer will see about in other videos, a is of integer type and value it has is ten, b is also integer and maybe is thirteen and c is also integer and value is thirty three so these are the basics stuffs you can print, so you can print here also hi, my name is amit verma ok so print, print it print directly here so this videos was about just installation of anaconda on windows so just follow the same step on Mac if you are using Mac operation system or if you are using Linux operating system then the instructions are same just go to the anaconda cloud download the latest version, make sure you know the architecture for example I take I downloaded the anaconda python

3.6 with 64 bit so you should know the architecture of your operating system, if it is 64 bit you can download 64 and 32 bit both but if it is a 32 bit you have to download the 32 bit version only so make sure you know the architecture and this is pretty much so I just wanted you guys to see how to download the anaconda, we all in all the videos we will be using the spyder ide only, it is very easy you can import, it has a lot of libraries present so you don't have to download some libraries by yourself, if you just download python open source library which comes in the python you have to download it again, So independently you need to download those packages so with anaconda you do not have to do that because it comes with so all those library, most of the library some of the libraries you might have to download and if you have to download those libraries we will tell you, how to download it, so this was it please if you find any problem regarding installation of this anaconda please post it on forum we will help you we will also provide a documentation how to download it on Mac, widows and Linux so yeah thank you.

# PRINTING STATEMENTS IN PYTHON

I have now opened a spyder terminal without before going further let us look at some basic functionalities of this interface off course we told you a whole lot about anaconda installation just now I will just tell you one fact that will be helpful for me to teach you more of python, this is called a console where I can check some commands and this is the place where I type my python code, you will soon get familiarized with these things don't worry if you don't understand completely, what I will do is in the interest of making the video visible to you all I will maximise this you can also do it by going to view and then saying maximise current pane, current pane  means the pin where you all write now, you click on it, it becomes full like this and if you want to get back to its normal state you can go to view and say restore current pane ok, for the time being I will maximise this, maximised it increasing the font and then let's start. We saw what are variables I can say 'a' equals ten, 'b' equals twenty and then say 'c' equals 'a plus b' and then I can print what is in 'c', Its thirty correct 'c' was 'a plus b' and see now when you print what is in 'c' this command print says what is in 'c' display it, if I say show 'c', it will not show anything it will show you some weird stuff like this because python only recognises the word print it doesn't recognise the word show, I cannot simply say what is in 'c'? It will not know what we are talking right? There is a particular way of saying something print 'c' is the way of saying, ok it shows thirty now I can simply say 'c' also it will say thirty but it's a good practice to say print of 'c' always, you will know why very soon. Now if I say 'x' is equal to within quotes sudarshan and I say "print x" just the way I said print 'c', I say print 'x' it will show what ever in 'x', so you must be wondering why should I say print 'x', I can directly type sudarshan here, if your point is to display sudarshan, simply type sudarshan here now note this whenever we are trying to teach you something we will use a very silly example that is in the interest of teaching it is in no way indication of the importance of a command so print 'x' displays sudarshan am just trying to illustrate what print does. I am not telling you the exact usage of the print command, exact usage will come next.

# UNDERSTANDING VARIABLES IN PYTHON

You just now saw what print does, it isn't just this let us see more of it now. Now I have three t a s in my course now my first t a let me call it t one is amit, t one equal amit, when I say t1 it will show amit as you can see ok, and my second t a s name is simran and my third t a s name is vidya. Now I have t1 t2 and t3 now see what I am going to do, no I am going to say print hello t1 inside quotes I am going to put t1 and then I am going to close and guess what will happen, it will say hello t1 why? Whatever you put within in the quotes it will simply show whatever is inside that's it but if I say print hello and then close the quotes and then put a comma and then say t1 close the bracket and then say enter you see what happen, you probably can guess what is happening here I need not explain, it saying hello followed by whatever was told in t1, what was told in t1? Amit was told in t1. So it is simply saying hello amit, if I were to say print hello t2, it will say hello simran, where to say print hello don't worry about what comes here it is a help look up on how print command works, so please ignore this, This is too much for you to understand or even see what is written here so just ignore this thing that is coming in here, so let we say "hello" comma t3 it will say hello vidhya ok, there are my t a s, I am the instructor if I say I equals sudarshan and now if I say print I it will simply print sudarshan, if I say print hello I it will say hello sudarshan ok? If I say print, see I can use up arrow to go to the precious command, I just said up arrow, up arrow there are four arrows you see up, down, left, right? If I press up it will goes to the precious command, previous command. If I again press up arrow it goes to this previous command it is print of I see and then the previous and then the previous I will be using this quite often don't get confused don't be worried how I am able to type something very quickly I am just using this is called the history look up, I am just going and seeing the history so what if I say print hello t1, t2, t3 what will it display? Let us see what it will display. So it says hello amit, simran vidhya because t1 t2 t3 are amit, simran and vidhya. Now let's do something print hello t1, t2, t3and then comma how are you all doing, how are you all doing today? Then I close again it is self explanatory isn't it? You now know what this does, it simply says hello amit simran vidhya how are you all doing today? See programming is all about this, if I now realise that t1 is amit but amit said "sir please display my full name" my full name is amit verma then I will say t1 is equal to amit underscore verma let's say I put it like this ok? Now when I say print t1, it says amit verma now you can wonder what happened to the precious t1 which was actually amit, it is over written by that I mean whatever you assign the latest t1 is amit verma will be the latest value ok? If now I say t1 is equal to let say India, t1 print t1 will give you India, amit verma is now gone. Correct? Ok. So let me now get back amit verma up arrow up arrow up arrow up arrow t1 is equal to amit verma and what I will do is I will display this once again, up arrow up arrow up arrow huh! Hello t1 t2 t3 how are you all doing? Its look at the output here it was hello amit simran vidhya how are you all doing? Now t1 has become amit verma this t1 is now amit verma, so it will say hello amit verma simran and vidhya, simran vidhya how are you all doing today? See hello amit verma simran vidhya how are you all doing today? Correct, now let me enhance slightly, I would want to put a "and" here after t1 t2 I want to put "and" right? when you have three names after two names you have to put and, and then after hello amit verma simran vidhya, I want to put a full stop, so I will put a full stop here before how starts correct? This, look at this command properly, look at this entire command properly let me highlight it for you, look at this entire command, what am I trying to do here, I am simply saying “hello t1 t2 and t3 full stop how are you all doing today?” Perfect it says “hello amit verma simran and vidhya how are you all doing today?” Amazing now you understand how print command works and how you can club variables with print command, let us conclude this discussion with a very small exercise. I say age is equal to thirty enter ok? Print the number you entered was put age and close it will say the number you entered was thirty, now if you change age to some forty and then print it will say entered was forty correct? Now I will say “age” whatever it was, what was it? Print age will tell you what age was, age was forty. Why not thirty? Because you wrote it once again, once you write age equals forty the latest assignment for age will be taken, by assignment is what I mean forty gets stored in this variable called age ok. If I say age equals age plus one this is a little worry some what does one mean by age equal age plus one, python is a particular way of functioning? When someone says age equals age plus one python simply takes the age value and add one to it and assigns the new value to this namely age print age will show you forty one. Correct? Don’t worry so much about what these variables are? You will become very clear with them with time ok? Variable is more like a house where you store some entity right? You store you keep book in a cub board right? a variable is like a cub board and your book is like a value that you assigning see it that way it will become handy to you to call age and the value of age whatever it is right now forty one will be ready for your reference right? so don’t worry even if you didn’t understand if you understood only fifty percent of what I taught still you are through with understanding good amount of python in the forth coming lectures.

# EXECUTING A SEQUENCE OF INSTRUCTIONS IN THE CONSOLE

So we saw what is print command and let us do something, let us use it properly. When I say a equals one and then print a, you get one. When I say a equals a plus one it means you are incrementing a by one so what will happen to a? It will become two. I again say ‘a’ equals ‘a plus one’ and then I say ‘a’ becomes three, ‘a’ equals ‘a plus one’ and then ‘a’ equals to four and so on correct? You now follow what is print command very well ok, so now I can actually say ‘a’ equals ‘a plus’ one and then put a colon and then say print of a these two commands gets executed simultaneously firstly it will as good as executing the command and then executing this command so if I type the same thing and plus enter it will display six, seven, eight, nine, ten, eleven, twelve and so on ok, let me clear the screen by typing clear on now a equals one once again print a which will make it print one, now look at this a equals a plus one print ‘a’ again, I put a colon ok, another this is semi colon not colon sorry and then I say ‘a’ equals ‘a plus one’ once again and I put a semi colon and say print ‘a’ once again, again a semi colon ‘a’ equals ‘a plus one’ then I say print ‘a’ what is this do? It will simply executes all these commands in a sequence enter, two three four why because ‘a’ was already one I incremented it by one it became two and then it printed ‘a’ and so on ok? So what I should do is I should probably print a, a equals one assign that and then print a and then put a semi colon and then increment a and so on now this is a sequence of instructions do you understand that, a equals one is the first thing assign a to one in plain English this what happening, assign ‘a’ assign one to a not a to one I am sorry don’t assign ‘a’ to one assign, you assign one to ‘a’ and then you print what is in a and then again you make a become one more that’s what this means, sir why does we write it as a equal a plus one that’s the convention, you must learn it that way, this is how you can talk to python ok? ‘a’ equals ‘a plus one’ incremented, print ‘a’ now, now ‘a’ will be displayed as two, ‘a’ equals ‘a plus one’ now ‘a’ becomes three now, ‘a’ will be printed as three, a equals ‘a plus one’, ‘a’ becomes four then ‘a’ will be printed as four, perfect let’s execute this and see this is what it happens, it displays hip hip hurray! We are done. It says one, two, three, and four you have learnt one of the important aspects of python, important aspects of programming in general, and programming is all about executing a sequence of instructions to the computer. Telling the computer what to do right? Ok. Perfect. So now you understand how to use print command, what are variables, and how you can type something in a sequence?

# WRITING YOUR FIRST PROGRAM

It is now time to learn a new command, please note that print command just we showed you right now it is used to display something meaning output something to the screen your monitor right? Now let us learn how to take something as input from your keyboard by writing a piece of code.

I have now opened a spyder terminal as you can see here is my what is called the console where you can type your code, here is my place where I can type my code, what do I mean by that. Let me remove whatever is present here, once you become a sufficient programmer you can, you will understand what was present and what I am deleting makes about documenting your code anyway let's not worry about it. Now look at I can now say a equals one, a equals a plus one and then after a equals one I say print a and then after a equals a plus one I will say print a ok now I want this to be executed all at once. This is equivalent to remember we just console we were trying to type something in the previous discussion a equals one print a, this is same as a equals one put a semi colon and then say print a correct? A equals one print a and then a equals a plus one and then print a will simply execute all these things at once this can be given here or the same thing will also be given here, when you give here you must execute this code by going to the menu click on run and then click on run, it will first ask me to save this file, it is just like saving your simple ms word file I will type print dot py, it should always have py extension here like how you are ms word files have doc extension and note pads file have docs txt extension, your python has dot py extension click on save wherever you want to save and now when you go to run and say run it will run you and see the answer one and two is the answer it says running this particular file ok, let me write more, a equals a plus one and then print a ok, we will copy paste this and then repeat this again and again, perfect one this is so much fun I want to see what exactly happens now from here I planned to execute it, so how many times we should print a? Once twice three four five times six times seven times eight times and nine times let's make it ten times now, now what do you expect to be the output? It should have one two three four five six seven eight nine ten. Correct? Let's see if this is what does, go to run and click on run, you will get se perfect one two three four five six seven eight nine ten what did you learn just now? You can write a sequence of using semi colon here I will show you wait, where was that? Here. Like this you can type or you can type it like this or even you can save it for future reference right? You can save this file here is your programme what does a programme do? Assign one to the variable a and print variable a, as I told you variable is more like an entity which stores a value, a now has one but the same a is now incremented by one and then you print a and then you increment a by one print a and so on, so where minimum pre requisites is expected in this course and many of you do not know anything about programming, so all of you who is seeing programming for the first time will find this lessons very self explanatory and people who knows some programming already might be getting bored because we are talking about basics but very soon you will see that we will pace up and cover the whole of python and get into some really cool computing applications so please tolerate our simplicity for the next few lecture videos.

# TAKING INPUTS FROM THE USER

All right so I am gonna teach you people how to take input from the computer, what do we mean by this? I will illustrate with an example. Firstly let us make a full screen then look at this print what is your name? Ok, it shows what is your name? And then I say n equals input what is your name? Same thing I am going to put here. This will also say what is your name? But it will wait for you to type for your name. See what happens I say enter see it is saying what is your name and you see the cursor blinking here, do you see the cursor blinking here right? I will type sudarshan, now it stops. You know what it has done right now, it has taken sudarshan and as assigned sudarshan to n how do you know that? Say print n and there you are sudarshan ok if I say n equals input what your name is? And I type iit underscore Ropar and I say print n, it prints iit Ropar ok? Let me put this to use now I hope, you all understood what we are doing here, correct? So, let me go to view and then restore current pane and I have this ok, so let me now write a code for this. Print, I need not print I directly say n equals input what is your name? And whatever you input I will then say print hello n can you stare at these two lines and tell me what this does? Ok let us run and see that is the best part about programming. If it is hard for you to guess what something will do you can run and see what it does? So if I we may guess it will ask you to type your name and assign that name to n and it will say hello n depending on whatever name you are going to type run and its asking me for your name, what is your name? I will type let's say India, my name is India. Tats it, it says hello India. Correct ok, let's add some more spice to it I will say print hello n how are you n print I hope life is treating n well so what happens, let us save this first run this and see what happens. It says again what is your name? I will type my original name sudarshan, what should it now say hello sudarshan how are you sudarshan I hope, life is treating sudarshan well I am very bad at punctuation so let me correct those things right now. Dot here and how are you sudarshan after that I need to put a question mark hello n and then exclamation may be correct? What is your name as a question mark? Right? Now let me save this and run this. What is your name? I say sudarshan. And it says hello sudarshan exclamation how are you sudarshan question mark I hope life is treating sudarshan well you see what we did, if instead of sudarshan I were to input something else wish let me try now. Let me go to run and then say run its asking me for name I will take some other name ok, I will type ram, what is your name? Ram. How are you ram? I hope life is treating ram well, now isn't that interesting you type something and that is put as a place holder in many places and there is a customised message for that person. Ok? Watch this video couple of times to get a hang off what we are trying to say this is going to be important, we are going to take whole lot of input from users and should understand how that is done.

# DISCOUNT CALCULATION

Imagine I walk into a restaurant or let say shop where there is a discount of ten percent which means if you buy something for hundred rupees you actually got to pay only ninety rupees, if you buy something for hundred and fifty rupees you going to pay only one thirty five rupees so you will get a discount of ten percent. Which means whatever is the price you should multiply by point nine? So now let us write a programme which takes as input the cost and then tells you the price after discount basically it must simply multiply by point nine and then give you the answer. Ok now how do I go about this let me teach you something. Print, what is the cost? Alright? What is the cost?  And you enter something what should that be cost c is cost c equals input I will say what is the cost? This is how it is done right? End a bracket like this and it will wait for you to enter. When I press one fifty it will takes c as one fifty see print c I am sorry, print within bracket c correct? Now I will say print, let's say answer is equal to c times point nine. It is throwing some error you see you must understand this. Whenever we ask for c the c you wrote one fifty but the computer seized it as, it doesn't see it as a number it sees it as what it called string. Now assume your is like your vehicle can only take diesel it cannot take petrol, if you fill petrol you are doing a huge mistake, you probably making your vehicle unusable correct? So this is now trigged as string on string you cannot do any kind of mathematical operations, there is a way to handle this. What you must do, you must convert the string to a number, if this done by the following way. We say d equals hint of c now print d, this will be a number now this you can do answer equals d times point zero point nine and now if you print answer it will have one three five, if you find it difficult what I said let me start from the beginning ok? So let me clear the screen. I say c equals input right input enter the cost of the item and then you will enter something let say two hundred ok now print c is two hundred you cannot do any addition or subtraction on the c now that's how it is, you might be wondering why? Now all these programming languages comes with its own way of handling, you probably you are wondering how difficulties it to ensure that we the people who made python gave us this facility of manipulating c directly why should we do this? But this is the standard. What should we do? We should say d equals convert this to integer that's the command for it int of c a note if you say d equals integer of c it will show weird error like this, name integer is not defined, if you can guess it is just saying that I don't know, what do you mean by integer? When I say d equals int it knows. It converts the value in c on to an integer like value, so d is now assigned the value two hundred, this two hundred is in d and since it's an integer it is ready to take your arithmetic operations so you can do math on it. So what is my math? Answer will be zero point nine times please note you should put a star here that's the multiplication symbol not your typical x you do in school days ok, if you do that it will show an error, you put an star here, star is on your key board it's just on top of the number eight. Point nine times d that the answer now I will print answer it will give you one eighty is the discounted price after removing ten percent from what you input. Now what is our job? Our job right now would be to write a code for this for an entire thing how do we go about? Print hI how are you doing? HI how are you, let us compute the final price after discount we say enter right? What is that? C equals input enter the cost of the item and then you should do d equals int of c then you should finally say answer is equal to zero point nine times d and finally you must say print the cost after discount is close the colon comma say answer and there you are, this will do everything that will wanted to do. First it will print hi how are you etc and then it will say enter the cost, you will enter whatever you enter goes to c and then d will be a convert of c, you are converting the petrol car to the diesel car it's something like that ok so that you can you can write the way you want. Ahh see there is a difference between a diesel engine and a petrol engine you cannot treat one like the other correct? So d equals integer of c, so now d becomes a integer variable you can do all sorts of mathematical operations on it I told you this how python works so you should not ask lot questions there ha ha! Right? So answer is equal to zero point nine times d if you don't understand this is how you compute discount right? Ninety percent of the original value is what you get by ten percent discount if you need a pen and paper to work this out but trust me my math is decent enough that I can tell you that this is the formula. Answer equals point nine times d so then finally I print the cost after discount is the answer let us see what happens to this code, go to run and then run this, this is short cut as well f5 to run it, I am doing it the age old way don't call me a bad programmer there are lot of nice ways in which you can run this code but I don't like using a portal like this I like doing it on the command prompt but to make you understand I am using it the old school way so run. Hi how are you? Let us complete the final price of the discount enter the cost; I bought something for thousand rupees, One thousand rupees so what will be my discount? Ten percent gone should be nine hundred let's see and I press enter whoo! Here you are nine hundred. Do you see the sped with the chit computed nine hundred now that is computers that is programming. You write in such a long way what it should do and it executes in like boom! A fraction of a second and that is what makes programming interest. So what did you just now learn? You learn how to write a piece of code to compute the final discount price right? sky is the limit go ahead see this video again and try to write a code which takes a value and then tells you what will happen if you were to sell this item with fifteen percent profit which means if you take something for hundred rupees and you sell it for hundred and fifteen rupees go ahead write the code and see if it works fine or not?

# MOTIVATION TO IF CONDITION

Let us now see how to instruct your computer to understand your “if conditions” using python. I keep repeating the same thing again and again please don’t feel bugged this is very important, it takes time for you to understand everything right? you might be wondering why is he talking so much about if, why should I worry about traffic signals and how will that be useful in computers. You will soon see as the weeks pass by how will be using this “if condition” the way we keep saying hi to your friends every now and then you see them, you will keep using this if condition to your computer whenever you are programming you would be that frequent is noted.

# A REMINDER ON HOW TO DEAL WITH NUMBERS

So now you know how to take user input in python now let us try to perform some operations on the input and see what happens?

Ok so now let’s start, let me now type print statement print hi how are you? Just like that I am typing this ok, then I am going to take my choice to be equal to input please enter your choice ok, what will happen? Let me run and see. so as I told you, you can go to run and then run and you can always run by this short cut key mentioned here which is f5 on my machine, it can be different on your machine too depending upon what you are using an apple computer it will be different, if you are using a windows machine it will be different and if you are using a Linux machine it will be different. If you don’t understand don’t worry all that you should do is you should see what key board, what key is given here and you should press that that’s one way of executing, clicking on run is another way there is yet another way which is this button here right? Ok. So now let me press this button here. It asks this let say run and it says ho how are you here you see that, let me zoom, hi how are you here and it says please enter your choice. I will write ten so the choice is taken as ten but I don’t get to know what my choice is so what I should do for that is say print c that’s it again run this it will say, hi how are you please enter your choice I will say ten it will print ten. Correct? Now I will say if my, see you know what? Your c will always be a string here ok? What do I mean by that? Let me delete this and spend some time here and tell you what I am trying to say. When I say print c here it will show ten. Let me do one thing, if I say, c plus 2 it throws a error, I told you because c will be c in as a string it is not seen as an number ok, I will show you something more funny when I say c into five ok? It will then tell you its true colour you know c into five will means if c is ten c into five is ten ten ten ten now ha ha ha how funny is that? So c into five according to me is fifty when c is ten. As I told you I am going to repeat this more often than what you can consider as acceptable computers are very dumb they will not understand, I repeat computers are very dumb you must precisely tell them what you want the computer to do? If you simply say c into ten it will print ten ten ten ten times because c is a string according to the computer it sees as a string for that you convert it to the integer by integer I mean a number you must give this command, d is equal to int of c what is this mean? It means c was a string when I entered like this using this command but I am converting that into integer and the resultant int is an integer by integer you mean a number. The resulting number will be stored in d when I say this now see the magic. When I say d into five what do you expect? See it says fifty very good! It’s a nice computer. I mean it listens to what I say but it is bad enough that it is not smart enough that it can figure out what I am trying to say. If I say c into ten it will put ten ten ten ten ten times, only when I convert that to a number will it give me the right answer right, initially you see it threw of an error when I said c plus two that’s because c was a string. If I say d plus two it will say twelve. Correct? Perfect.

# UNDRSTANDING IF CONDITION’S WORKING

So what I will do here is I will go here please enter your choice c is entered I will print c and I will say d equals int of c, which means now d will be, let me remove this, this is not required why should I print c. Directly say d equals int o f c now look at what I am going to do next if d is equal to one colon enter print, you entered one that’s it. And now I will run this you probably have no clue what I wrote here this is doesn’t sound like English, this doesn’t sound like mathematics it doesn’t sound like any language and you are very bugged I know it but stay patience let’s see what this does. Run hi how are you please enter your choice I say one and it says you entered one ok let me figure out what this is doing? I will repeat this once again run enter your choice I will press two and choose nothing. Remember what we did with our scratch programme, yes! You do remember right? Look at this we used set c equals something and then we said if c equals something then do something c equals two c equals three then something. Similarly, similarly I am trying to do it by using python right now. Python is not as beautiful as scratch where we can simply drag and drop stuff here; you need to do some work. What is that work? You should type things that you want, what do you mean by that? Look at this. I say if d is equal to two then print you enter the number two, whatever you put here it will put, if I if I replace this by my name it will display my name now let me run this. Please enter your choice I say one it says you entered one let me run this once again. I say my choice enter your choice I say two it will say you entered the number two correct? Let me do more of this. If d is equal to three then say, what should I say? Print you entered the number three no that’s very boring I will say I am tired, I am tired just like that for fun I am saying something else here so let me run this. But my choice is used to be one it says you entered one, if my choice happens to be two it says you entered the number two and if your choice happens to be three it will say you entered three! No that’s because that’s not what I am saying here as I told you the computers that you are that that you are using is so dumb it will just do same time very faithfully so it will just do what you are asking it to do not more not less so it will actually say ahhh I am tired so now it’s very clear to you how if loop works in python how word of warning, you should type if within brackets only you should type let’s say d is equal to one please note it is not one equal in python it is two equals. You can ask me in scratch you put one equal here why are you sir putting two equals here? My friend English is different from, Hindi is different from, German is different from, Sanskrit is different from, Tamil, Telugu, kannada, you name it all this languages are different, they come with their own style and you must stick to that style otherwise the computer will slap you what do I mean by that? I will tell you what I mean by that. Instead of print if I say print you see and you run here it will say please enter your choice I will say one and you see name prnt is not defined it got confused basically, it will show u a lot of unwanted stuff these scary stuff, you start wondering you will probably close your laptop and run away. All these entries you know, don’t worry about them it quite advanced, computer gets confused how to react so it will throw up all these unwanted text so we should be very careful what to say and what not to say. Let me write it back to print and then click, it saying please enter your choice I enter one it does show you, you entered one.

# REALISING THE IMPORTANCE OF SYNTAX AND INDENTATION

Ok, so coming back two equalities are important and after this there should be a colon, if there is no colon it will behave weirdly look at this I have removed colon right now I am running it, see invalid syntax it is even showing these arrows saying that ahh the syntax invalid syntax. The complicated word don't worry about what this means? It means computer is instead of angry with you for having, made a mistake. It says something is wrong here, what is wrong here? You should put a colon here. Why? Because that's how python works and then you come to the next line, you see you have printed, what, why have I left this space here? Let me not leave this space here and then let me execute. You see expected and indented block ahhh whatever that means if you dint understand don't worry you are not alone it is very difficult to make out what the computer says whenever it is angrily scolding you this is called compilation error and even its also complicated so don't worry all I mean is it is not very happy so that's because python expects you to give this space here and maintain that space everywhere whenever you are inside a if loop like this what do I mean? Let's go back to the scratch and look at this if something is true then you execute this, you see you can even put statements here go to looks and say you put hello here then it will say then you make this one so that its comes here double click its goes to here good morning for five seconds, five seconds over? Yeah over. Now hello for two seconds and then it will stop. You saw what happened? It will come inside and execute these two things whatever is within this yellow if loop similarly in python whatever is indented it will show. What do I mean by this? I will come here and I say "print" so you are doing well I see I will type this now I will execute this ok? Now always it asks for your choice and says one it will come inside and what do you expect it does. It execute these two things this is like in scratch whatever is inside this yellow thing is equivalent to whatever is written after providing this space in multiple lines after it. So when I do this and say enter you entered one so you are doing well I see, it says everything that is here it will not say anything in the next line, it will not say anything in the next line you must note this. Ok so that completes what is a if loop as end always things are not as easy as simply watching what I do and then nodding your head, right? You should try writing this code all by yourself exactly what I did here, you should repeat it; you should also do more than that ok? And please note these things come by practice. In the forth coming weeks we will be using "if" almost every single program will have if some where here or there you will get used to it very soon it's all a matter of time. Thank you all.

# INTRODUCTION TO LOOPS

Observe what I am going to do now, I am going to say print hi how are you ok? And I am going to type this many times, hi how are you ahh I am going to just copy pasting this so let me copy this and then paste this. You all know what is copy pasting right? ok you all have studied ms word note pad and things like that in fact even on your on your cell phones these days you have copy paste facility ok so I can as many times I want so you see I have type this ten times now when I run this code its says ‘hi how are you’, ‘hi how are you’ ten times ah what if I wants this to be displayed hundred times? Remember we discussed something similar to this on scratch right? You recollected right? We introduced you to what is called the looping structures, How to repeat a task many times, how to instruct the computer to repeat a task many times? I am gonna do that right now just observe this ten time if I were to do this hundred times I must copy paste this hundred times that’s a very tds task so python has a nice way to do this. Let me show you how it’s done. For I in range ten what does it mean? Don’t worry I will tell you print ‘hi how is everything’ or maybe I should comma here I should precise ok good hi, ‘how is everything’ and I question mark here, my English is bad, ok so for I range ten print Hi, how is everything? What does this do? This sounds again very weird remember the ‘if’ statement we wrote something and then we put a colon here right similarly we write for I in range ten print hi, how is everything? There with my speed don’t worry let’s see what happens when we execute this and then I explain what exactly this is doing ok? So wow see what happened it was so fast you people didn’t probably observe it. Hi how are you is how I typed it ten times but here hi how is everything? Is displayed ten times how did this happen? Let’s observe. So I is a variable by now you all know what is a variable you have done it in scratch we have been talking about it in python so I becomes a variable in range ten means this in English this statement means create a variable I range ten means start assigning the values zero, one, two, three, four, five, six, seven, eight, nine to I. I repeat for I in range ten means assign zero to I come execute this statement, assign one to I come execute this statement ok assign two to I come execute this statement, three to I execute this statement and so on right? Perfect. So I varies from zero to nine and hi, how is everything? Is displayed ten times. Confusing? Don’t worry as and always practice makes a person perfect let’s practice more of this forward. For let me remove this delete for I in range five print I now this will teach you how for loops works. Ok, let me run this, what is happening? Zero one two three four I variable takes the value zero one two three four, whatever you put here it’s starts from zero and comes five numbers ad one at a time is assign to i. And then that particular value of I gets printed here this is zero one two three four so where to say ten here you guessed it right, it will say zero one, two, three, four, five, six, seven, eight, nine if I say look at this print hi you are in number let say close it comma hi what does it do? Let’s see. huh ah you are in number zero, you are in number one, you are in number two, you are in number three, you are in number four and so on right? Perfect. This makes it very clear to you what we are doing. Ok, let me display the multiples of two, how do I do that? I will delete everything let me start from beginning. For I in range ten colon and then you should put a tab here I should come here and tab is because you cannot start from here, you should always start from here. Ok. So let me type print I will simply say I into two, you know what this does? This will simply display your two tables. Let me click this then and observe what happens here. Run and boom! Zero, two, four, six, eight, ten, twelve, fourteen, sixteen, eighteen but we knew two tables till twenty that's because, I in range ten means zero, one, two, three, four, five, six, seven, eight, nine. The last number will be nine, so nine times two will be eighteen that's why it is stopping at eighteen, if you want it to stop at twenty you should remove this and put eleven here and now look at the magic so its starts from zero and goes till twenty. Correct? Ok Perfect. That's about for loops observe them slowly patiently practice a few problems and you lead through. What we will now do is we will go ahead and discuss more of for loops so it will become very familiar to you ok, it is a this point that you must understand everything of what I have said about for loops ok so take a pause observe the video once again if you want, write a few programmes and let us see more of for loops in the forth coming lectures.

# LOOPS: SUM OF NUMBERS

Ok, let us now use for loop to do some very basic mathematics, are the reason why am doing this basic math with for loop is to teach you how to use for loop ok? Let me ask you this question. What is one plus two? Off course three. What is one plus two plus three? It is six. One plus, two plus, three plus, four plus, five plus, six plus, seven plus, eight plus, nine plus, ten what is it? It is fifty five. Let me use a for loop to do this ok? To add the first few numbers, why are we doing this? Because we are crazy! Ha ha alright this is a usual question people ask why do you make us do all these things? We have a calculator we can add any numbers we want right? Why are you asking us to do all these things by using some complicated programming. Now the point is, do not add these numbers where eventually useless to us right? But to understand how python like programming language can be used to add such numbers up to whatever level right? It is just an exercise problem for you to learn the intricacies of form ok? Let's go ahead and try to write a code for this. Answer is equal to zero. Answer is equal to answer plus one, Answer is equal to answer plus two print answers. What will be the output for this? Let me check. Initially answer will be zero and then answer will become answer plus one, we have discussed this on scratch in detail this sounds like this is the weirdest thing for a beginner programmer right? he will start wondering what do you mean by variable equals variable plus one it doesn't make sense please note this simply means the existing answer what is the value for it ok I am adding one to it and I am adding two to it right? answer equals zero, answer equals answer plus one so answer becomes one and then answer equals answer plus two answer becomes initially it was one it becomes one plus two three so I am printing answer it will be three. A good thing to do is actually print answer in all lines if there with me is the reason teaching you people this, answer and then finally answer ok? Then run it say oh oh oh!! I should not write in quotes you see yeah otherwise it will just display answer correct we have discuss this once again so computers are dumb it will not understand what we are trying to ask them to do so initially answer is zero, answer is zero and then print answer, answer will be zero printed fine, answer equals answer plus one so answer became one and I print the answers so it became one here answer equals answer plus two, answer was one and then I add two to it so it becomes three so answer is displayed as three correct? Now let me go ahead answer equals answer plus three print answer now they should be like zero one three six this is same as adding one plus two plus three you see why? Six that's what is happening here think think think think think think that's the most important part here ok, so answer equals answer plus four print answer what will this be now? This will actually be lets execute and see. zero one three six and ten that's because one plus two plus three plus four is ten, one plus two plus three is six, one plus two is three simply one then nothing zero right? if I keep doing this lets say answer equals answer plus five print answer, answer equals answer plus six print answer, answer equals answer plus seven print answer I am killing time as you can see ok that's because many people wouldn't know how to do these things right? Some of you are very familiar in programming some of you are taking this for the first time right? So it's good to be slow and precise, answer plus nine print answer and finally answer equals answer plus ten and there you are when I execute this, this should be equivalent to what was that, one plus two plus three correct? , one plus two plus three plus four plus five plus six plus seven plus eight plus nine plus ten that we saw already which is fifty five let's see if we get this number ok? Ready steady go! Let's run this programme and see boom! Zero one three .... It is up to forty five, it is not showing me fifty five which should be the answer why is that? Can someone tell me because you are calculating answer but you are not printing it so dumb of me correct! So now let me execute it, I will get the last one to fifty five which is the sum of numbers from one two three up to ten I am not happy with this I want the sum of the first hundred numbers, how do I do this? This is where our for loop will come to our rescue let's see how this is done. Look at this answer equals zero perfect for I in range ten answer equals answer plus I now wait very patiently let me not go up to ten let me now only go up to three and see what happens initially I take zero answer is also zero answer gets added zero plus zero is zero and then I takes one answer is zero, zero plus one is one I takes two answer is one plus two which is three so if I execute this it finally display the answer when I execute this I get three here perfect right! All of you take a pause, if you are new to programming this is the place where you should understand these four lines properly, properly ok, I repeat when you execute this, answer gets incremented one by one zero one and two gets added to answer. Right! if I makes this five you see what will happen, answer will be zero and then zero plus one, zero plus two, zero plus three, zero plus four so on and so for I mean zero plus one is one and then it will be one plus two is three, three plus three will be six, six plus four will be ten right? it is going to be the sum of zero plus one plus two plus three plus four because range of five means zero one two three four lot of details learning is always about braking the details in to small chunks and then observing very carefully right? so let me execute this I get ten perfect. It's for you now to neatly understand what is happening here ok pause the video right now, understand and then come back.

# LOOPS: SUM OF NUMBERS (CONTINUED)

I am sure you are back; you know understand what exactly this code is doing. Let me now give you a mind teaser and then let me indent this. I repeat indent this, initially it was here correct? Now if I do it like this what will happen? U know what will happen. This will now be part of the for loop, it will increment answer by I and then it will print the answer right? Look at this yea zero one three six ten see its printing zero, zero plus one, zero plus one plus two, zero plus one plus two plus three, zero plus one plus two plus three plus four and it so on now magic let me now give more details here print first I numbers when added gives, when added gives answer what does this do now? What will this do now? Anyone? Look at this. Ok let me zoom down so that you can see the entire line so answer equal zero for I in range five answer equals answer plus I print first whatever is in quotes will be just printed like that. I the value of I will be printed, I numbers when added gives the answer let me run this. First zero numbers will added give zero

# LOOPS: MULTIPLICATION TABLES

Alright, so let us do one more exercise from python for loops. I am now going to display the tables ok? So initially, I will say‘t’ equals input what tables would you like to display? And I will take the input that will be ‘t’ and as and always you know I will say ‘t’ equals int of t so that it becomes a number otherwise you know we have seen this already it becomes difficult to handle, it becomes characters, strings otherwise ok? So now print‘t’ simply just like that let’s see what happens here so that we know ‘w’ are going in a right way. I input ten here and it says ten but what I want is the ten tables here you see, how do I do that? Print‘t’ comma and then there must be a multiplication symbol one comma and then is equal to and then the answer t into one right? What’s happening here? Observe, let me run this and see. What tables would you like to display? Ten tables. So ten one times is ten is displaying right? So let me display ten two times is equal to t times two whatever the t is, see what tables would you like to display? Ten ones are ten, ten twos are twenty so again let me run instead of ten if I type eight it will display eight ones are eight, eight twos are sixteen correct? Let me go up to some let say, one more so how do I do that? Print let me go to three x three comma equals comma t times three execute this, this time let me display ten itself, ten ones are ten, ten twos are twenty, ten threes are thirty remember your tables right? now I want this to be displayed ten times as you know for that I may have to use a formula ok, for see how I am going to use it, for I in range ten print ‘t’ comma x and then here one two three is given right, I want that to be I here as I varies from zero to ten I comma equals t times i, see the point is you see a pattern here the pattern is one two three that pattern is replaced by this formula so how do we do that? We do that by replacing this pattern by of for loop here ok? Alright so let me now bring these things ok so let’s see what happens? I run the file it is showing me some error that’s because I haven’t put a comma here let me put a comma here ok, again run what table do you like to display? Ten tables perfect! It starts from zero you see, that because for loops starts from zero. Ten zeros zero, ten one times is ten, ten two times it goes up to nine that’s because, I in range ten means from zero to nine, if you make this eleven and then run ten here and then you will see it goes up to hundred right? Isn’t that fun? You can now display any tables let me now display two tables perfect three tables perfect, perfect perfect! Four. Perfect that was all tables from two to ten right? So this is the power of programming where you need not necessarily display tables from one to ten rather zero to ten you can go on up to any number you want by specifying it here. You have now learnt everything of for loop almost, this much is enough for you to start using for loop, very good.

# INTRODUCTION TO WHILE LOOP

I am now going to give you all a scenario where while loops are seemingly important. Assume I am a doctor I have my own clinic, I must treat my patients and assume I am a popular doctor there are a lot of patients and what I do is, I give them the token number, my receptionist gives them with the token numbers. I start with one two three four five and I go on like this and I treat my patients one after the other. How can I write a program for this? I will now write a program for this and put a monitor outside my clinic though which is in the reception area where people can see which token number is being called right now and that concerned person come inside. Let us see how we can achieve this using a while loop. Ok now look at this I will first write the code and then let us understand what this code does? While c is equal to one print token number n may please come in ok? Perfect. What will this n be? It should be a number starting from one and then it should become two it should become three and so on, so it should first be assign one ok, what is c? I will tell you in a minute let me assign 'c' to be one so now what I will do is, simply I will say input 'c' equals input continue question mark zero score one what do I mean by this? See you know what happens here? 'C' is a number basically ok so I will say int 'c', ok 'c' is a number. When 'c' is zero it will come here and it will see while 'c' equals one when 'c' is zero, 'c' is not equal to one so it will not pass this loop, it will not come down, it will not execute this but when c is one it will execute this now you have practised this enough right now given that I have asked you to practice it, you know what while loop is? While loop simply keep executing whatever is below while loop, whatever is below while loop, while loop keeps executing this until, until this becomes false, I repeat repeat repeat when the condition given here is true where? Whatever is inside this bracket after while when the condition is true it will keep executing this, it will keep executing until this becomes false but the moment this becomes false it will come out of it ok? It's like a bomb detonator right? so it will keep executing this keep executing this until this goes false, once this goes false boom it comes out, ok. Let me type here print thank you thank you this is the end of our day rather today ok? Before anything I will say print hello everyone we are starting, now let's see what happens to this. As I execute this the first line will get executed ok, hello everyone, we are starting, hello everyone we are starting 'n' becomes one, 'c' becomes one, while 'c' is one, absolutely 'c' is indeed one so you come inside print token number 'n' may please come in, token number one may please come in. Continue continue zero or one if you say one here 'c' will be taken as one but then you see the input always gives c as the string you see we have discussed this many times right? So I make it a number ok? For some time you please ignore this command you just by ignore I mean just write it know that you have to write it. With time you will understand what that is? Right! the one big mistake that people do especially teachers teach everything about programming in one shot it doesn't work you have to have this Mac to ignore a few details and grasp only the most important things with time. These details will become easy on your minds provided you keep looking at it again and again right? so don't worry much about this we will revisit this sometime later ok, actually we have discussed this already but don't worry much about it, it is to convert entity c on to an integer ok fine now I come here and I again go here and then see continues ok? Continue one ok enter, oh oh! It is saying token number one may please come in. This should not be the case; it should say token number two may please come in correct? So what should I do? I should ensure that I increment ‘c’ by one ok, increment not ‘c’ I meant I am very sorry I must increment n by one why? So that next time when it comes it says token number two may please come in so let me come here exit this by pressing zero see it exists, In a moment it will be cleat to you, you may again execute this print hello we are starting hello everyone we are starting, token number one may please come in continue zero or one this is for me ok? If I say one by one I mean the one here is the token number one, the one here is a command to my programme to continue or not ok? when I say one it says token number two may please come in, so what I will do as a doctor I will treat my patient once that patient is done he walks out for the next patient I will say I will enter one here enter, token number three may please come in my next patient comes I treat this little kid and I say bye to the kid and the kid goes out and then I again press one as a doctor it says token number four may please come in, the next person comes in, I treat that patient and then I keep doing this, keep doing this, keep doing this so, I am a very hard working medical practicener so I treat some twenty patients or even more I go on I go on I go on you see and finally I arrive at let say thirty six patients and everybody is done. What do I do now? Continue zero or one right? It says continue zero or one I say I am done. Let me press zero enter it comes out .Thank you this is the end of the day, why does it day that? Because when you press zero c becomes equal to this right? And ‘c’ becomes equal to zero and when c is not one if it’s only one it enters when it is not one it will come out, once it comes out it will say thank you, this is the end of our day. And that’s what happened here, as and always I repeat and I will repeat this many times handover the key idea about leaning a programming language, when it comes to a programming language the key idea is repeatedly do the same programme a few times, delete this programme and re do it again delete it and again re do it keep doing this until you are very familiar with what is happening where and then take up more challenges on the same topic and then write your code and there you are, you would become a expert in no time. Right? Take this tip very seriously now close this screen and you try writing this programme all by yourself. How would you make this counter increment and then stop when you ask it to stop by pressing zero ok? Perfect! See you all.

# LISTS PART 1: INTRODUCTION

Hello guys, welcome to yet another programming screen cast, in this programming screen cast we will be teaching you a cool data structure that python provides you namely the lists don't get don't get terrified by the name data structures it is nothing but a way by which you can organise your data, you can arrange your data that is what you call as data structure basically. So list the name is even we use this particular thing list in our day to day life may be for when you have to for shopping you plan a list of items which you must definitely buy, off course we do buy items that we remember at that time when you go to a super market or based on the things that your neighbours in the super market buy off course there are changes in the list but I believe most of us have the habit of taking a smaller list of item that you must definitely buy that, that is they are at most necessities you have also that may be changes as I have said so the lists in the python also very much similar to that, so this is a flexible data structure naming that just now I have said you have a plan of your shopping item that you wants to purchase something but as need comes that may be changes in your purchases right? Something like that as and when there is a need you can add or delete items from the list. That's just a flexible data structure, let us see more of it. So to introduce it I will start off with a sort of a recap that you had in your previous week lectures, so sir had introduced the concept of variables and how you can club variables with print statements I hope you remember that ok, let me give further more hint so that you remember really well. So we had used the different variables for example we had that, me be open a console alright the console is opened now ok so I hope you remember the example that sir has given. We are three TAs in this course t1 is the TA whose name is so double quotes we will use for a string that is for names we use double quotes amit is the first TA simran is the second pa and I am the third TA vidhya ok sir has used the variable I to denote the instructor sudarshan so this is the example sir has used to demonstrate the variables. So let me give you another situation, what if you need to store the details of the students who are taking up the course? So course students the strength may vary so we cannot say like this like here it's fixed we know that for sure there three TAs so we have used these variables t1 t2 t3 so course students the strength may vary just as how I had given an example of shopping list you have a plan of buying some stuff but when you go to the super market you may buy something additional or you may feel that this particular thing is not really of much use to you or you may even some time you may plan something but you may get something better you may get a replacement there is definitely there are changes so something like that even here the strength of the students this is not something that is really well determined so what if it is determined? So for example let us consider a pg class room there may be some twenty students so in that case your solution would be s1 s2 s3 up to s20 maintain some twenty variable. Will be idea you will get as an extension of this concept as we already taught you but now a day's things are changing it's not just the class room teachings that happen every where or if you would see even our nptel course students strength keeps varying there more people may enrol in the beginning and as weeks go by some people drop out too or by all that what I mean is this is highly variable it's never a case that there are twenty students and there will be twenty students till the end that is never a case there may be additions or deletions so something like that even though is very much similar to our shopping list this is highly variable thing so to handle it we need something which is really flexible so for that we will use something called as a list. So let me write it, it's called a list. I hope it is visible to you guys ok? So I have increased the font size list, so that we call structure as list which is a flexible data structure where you can add and delete items really easily that is what you call as a list, so let me let us go ahead with our shopping example or the student example anything as you wish. I hope shopping example is more connected to us because most of us will have the habit of maintain a list of things that you need to definitely buy so let me take that as an example itself so let me say shopping that is the name of my list so these are the things I need to buy so, I wanted to have a list so this is how you create a list please note this, please note this you are using this type of brackets square brackets we call them as we are using this kind of brackets in your keyboard this is near to your alphabet p this key, please see this, this brackets we use to denote list, his is how we are creating a list. So if I say shopping and just the brackets and empty list by name shopping is created. So that is the list is available but now items are presented in the list so that is what is happen happening if you just save this. If you want to insert item say for example I want to buy bread, coffee, sugar these are something I definitely want to buy so this is my shopping list as of now ok so if I say enter the list is created, if I just say a name equal to an you think the square brackets within that I enter the items separated by commas this is the syntax by which you create the list ok? so if I want to know what are in my shopping list I will say as usual this is how you had this for this variables print t1 say amit print t2 gave simran print t3 gave vidhya and so on just as how you had it print command can be used here as well print shopping if I say print shopping see I am getting the list here, bread, coffee, sugar but see I don't wanted with this quotes and that square brackets, commas I don't want it in the some tactical format, I want to write in a format that is friendly for us, how will we write down the shopping list one by one, one below the other right? so I want to print it in that fashion, how will I do that? Simple, will use what we had studied in the previous week our friend loops mainly for loop will be more friendly here, can also try using the other loops I leave it as an exercise you try using the other loops most of the task which can be done by for loop can be run by while loop as well something like that so you just try using while loop as well I will demonstrate using a for loop. So the method is, for the, let me just recap the for loop how we had used it for numbers. For I in range three what does this do? It will count zero one two, if I say this and a colon see the next line appears indented. I will say print I ok, let me say enter see I got zero one two so to deal with numbers you had used a functionality called range dealing with the items of a list is really very simple so let me demonstrate how that is done for item I or item anything for item in the name of the list you have to give here, what is the name of the list? Shopping. Shopping is the name of our list; I will give shopping for item in shopping print item ok. See bread coffee sugar it has displayed the item just one below the other just as how we would write it in a piece of paper, see observe it is very much similar to what you have leant in your previous weeks in fact much easier than that as well for numbers you are using this functionality of range for list it is very simple just use the name of the list for item in shopping this is the intuit name I have given actually you can give any name here this and this you can give any name but this two has to be same you can give any name item is intuit even that it is a shopping bag shopping basket trolley sort of thing given that so things inside it called as items so I had used the name item so this is how you print the items of the list in this fashion one by one alright, this process of checking through each item in the list and printing so let me print the list as such for your understanding once again if I say print shopping just as how I already set print t1 print t2 just we had said like that something like that if I say print shopping it prints in this manner with the square brackets and quotes something that is computer friendly format not a human friendly format to make into human friendly format we are using loops. See what this loop does here for item in shopping, it get into the list called shopping and checks over each of the item, each of the entity present here and it will print that entity the first iteration this is called item is assign the value bread so bread is printed here in the next step coffee is coffee is assign to the variable item so coffee is printed in the next step sugar is assign to the variable item so sugar is printed so this process is called the iteration this is nothing but your iterating over the list that is you are checking through each item that is present in the list, you are checking through each Element that is present in the list so this is how you iterate over the list that is you would check through each and every element in the list. Ok so I have printed all the element in the list what if I want to add another element? I want to add another element let say I would say curd, let me say curd so I should use a functionality there is a functionality available for this it is called as append it is an English word only append is nothing but adding to the end so append functionality would add it to the end of the list so if want to add another item say for example curd I would do it by shopping. Append of this is a bracket so inside the bracket give my item is curd, I want to buy curd as well so I have appended it to the shopping list so we shall try this print shopping we will see what is in the cart? Now see initially we had this as our shopping cart that is shopping list bread coffee sugar due to this command appending another item curd your shopping list has become like this bread, coffee, sugar as before and curd is added to the end of the shopping list ok, this is how I had added an item so, is this the only way by which we can add an item it is only possible that you can keep adding at the end what if you want to change the priorities

# LISTS PART 2: MANIPULATION

What if you wanted to buy something say oil, you wanted to buy oil, definitely oil is not even a single drop of oil is there in your house and you need to buy oil for sure even if coffee is not available sugar is not available you are ok with it but you definitely want oil so you want oil to be pushed to the first part of the list so how would you do that? There is a method available for that as well that is called as insert method that is you insert at the given position, you can specify at position you need to insert the item so let me say bread is needed for me then I want oil so and please note that an indexing starts from here this is what called as indexing so I will demonstrate that first then let us go into this thing of inserting the items at some specific positions ok so if I would say shopping of one this is what you called as an index at this list shopping at index one what is the element present? So index is something like a position holder something like this so by saying one what do you expect do you expect bread would be printed? Right? Let us see what happens, see coffee is taken that is the second item I had asked one its printing second item instead of the first item so let me try zero, shopping of zero let me try see it is now showing the first item so what do you understand from this? Computers count in a different way exactly computers count in a different way the counting starts from zero, human start counting from one we will say one two three four, four items are present the computers will count it as zero one two three. What is the third item? If we ask the human he will say one two three third item sugar, but computer will say zero, one, two, three, third item curd so, I guess you understand the difference between the computers counting and the humans counting it differs by one so if you need the second item you should say two minus one, at the index one so indexing, so this is what you call as indexing, index is nothing like a position holder so this is position zero, position one, position two position three according to the computer so this is how the individual elements can be accessed, individual element can be accessed using indexes indices index using that you can access this elements ok so is there we had printed it using some way already using the for loop so using this indexing mechanism can be printed using our range function which we are used in numbers where is that is another way of printing the elements of a list slightly around about way compare to the simpler way we had done earlier but this is a recap for your range functionality of previous weeks as well so let us see this way of printing each element in the list let us see it for I in range how many elements are there in our list one two there four for computer it is zero one two three range functionality starts counting from zero we want to count till three so what should be the number given inside it should be four because this particular number is when it reach four it will stop counting so it will not consider four so till three only it will consider, that what we too wanted so we will set for I in range four print I will say shopping of I please observe the square bracket are being used here shopping of I ok, see same thing bread coffee sugar and curd this is printed so this is slightly roundabout way simpler way is for item in sorry I have to give the space here it's very there is a change in colour which is not visible for item in the list name is shopping, shopping print item see we got the same thing so this are two different ways by which you can print the items of the list one after the other just like how we write in a piece of paper, for one we just by using a variable inside a list other was using the indices index so when will the index will be highly useful is if you want to extract only just few elements in that case indexes will be highly useful for example I had given you another instance like students information eight in the beginning I have given you an example of students information in that case let us assume that every student has role number or id number whatever you can say so they have arranged in that order based on their id number data has been arranged and stored so in that case if you want to extract the details, if you want to extract the name of the student whose id number is twenty you can use these thing like students of twenty so in our we have four items I want to extract the second item so for second I should say I should say two minus one so index is one shopping of one this extracts the second item so in our in that case I felt student of id minus one or id in case if you start counting if you say in zero position you have something like that name let us say heading name has been that's how we generally write so let me say name then you start writing student one student two and so on so this is generally write right? So let us assume in this way only our list has stored items so at zero index we will have this thing name which sort of a header so s1 student whose id number is one will be at index one, student at id number two will be at index two this is one way by which you can manage the difference between a computer and a human spouting. Just an idea I have given you can try implementing this student information list in any way as you wish so in that cases when you want this particular specific caleman to be accessed in that case the indexing will come highly handy let me say as I now given that I have said about indexing, I will come back to the thing that, I had said already inserting your items at random position that is at a given specified position see for example, I remember that I need oil definitely bread definitely I need next priority is for me is oil coffee may go in a lower priority that is even as coffee is not available I will not be that sad but oil is definitely needed something like that let us assume this is the scenario so, I want bread oil coffee sugar curd in this order I want my list to be so append as I had shown you already if you would see I had used terminology, I have used append so, I have used the up arrows as we have said already up arrow is used for history look up in case you want to go down you can use the down arrows as well up arrow is for going to the precious command if you want to move one down just assume that it is like a history look up is nothing but like a ladder you can climb up and down the ladder to see what you have done in the past ok so I had appended one item let me say I wanted shampoo, I append shampoo so, I have appended it because of it I want to print the shopping item let me print it see shampoo got added at the end so is this the only way by which we can do, can we change the ordering I want it as bread oil coffee and so on I wanted to be like this so how will I insert it for that we have a command called insert let us see it, what this command specifically does this? If you give position and say insert the item it will insert the item into that position so it will shift the other items automatically see how flexible it is, if you are if you had realised something if you are using a variables like i1 i2 i3 for different items and if you want to change the priority just assume how much of work you have to do, i1 i2 you had made coffee and you wanted i2 to be oil and i3 to be coffee if you make i3 equals to coffee then sugar value will be lost so you have to move it to the next next next so how much of shifting you have to do man all this is neatly handed by python data structure mainly the list. So let us see inserting the specified position let me insert shopping. Insert is the functionality insert you need to give the see some help was given index and object is says so its nothing but object is nothing but a position as I said so position and whole depth not position exactly as per humans position that is one less you need to give and then the object whatever you want to insert into your list object is the general terminology python users items object items anything you can say so this is the method by which you do you have to give the position first so I wanted it to be the second position as per humans so it must be index two minus one that is one, one I need to insert item I say oil I want to insert oil in this basket so I had given it, it could have got inserted and let us check that as well so see it got inserted correct oil coffee sugar curd shampoo so this as per my priorities see how handy python lists are there is much more functionalities that are available in this let us see them as and when the need comes also we will give you a hint on whole things that we may be using frequently and as and when the need comes you can check out our video we will explain also you can check out the python documentation there are a lot of functionalities available in this list let us see something which may use frequently so this list specifically has stink items see bread oil coffee all this are distinct items but sometimes some lists may have repetitions' for example consider census data something like that you have list of all ages of people living in a city something like that so there will be definitely repetitions right? you cannot say that if a person is thirty years old is living in the city no other person of thirty years old should live in that city definitely there will be repetitions so in that case you may be interested in finding how many number of people of specified age group is present in that city for that we have a functionality called count let us create that data set from random data set we say ages if the name of our list let me just say twelve, twenty three, thirty four, forty two, fifteen, eighty seven, twelve, sixteen, twenty five, twenty three, eighty two, fifty seven, seven, three, two, three, one, twenty ok. just assume that this is the list of ages of different people so we had created a list we have a data so in this definitely there are few repetitions as you could see twelve, twelve here somewhere else could also be same so in that case I want to see how many people of age twenty five live in this city? So how will I do that? I have a functionality called count that is been provided by python lists it is use that ages is the name of the list dot count is the function inside it you need to give what is the value? What is the object? What is the item for which you want to see, I said the age is twenty five this what I want to check so I will say twenty five see it says one so basically in this list just one time the number twenty five occurs that is one person of age twenty five is there in our list. If I say ages of count of twelve let me say see it shows two because we have twelve here as well as here see we have twice what if we given age which is not present in our list for example let me say seventy I don't think there is a seventy year old person in this list, see it says zero so it will return zero which is the element which you want to check in the list, if you use this functionality it will take this element check prove then this how many number of times this element is present in the list and it will return that answer, if it is not present it will return zero see how neat it is this functionality will be useful in one of our upcoming exercises that is why I have taken the data set where there will be repetitions see how many elements are there in this list? I want to know this. Can you manually count it? One two three four five six oh I am getting very tired shhh its seems too long it's looking like Nile river something like that I don't feel like I want to count it is there a way I can ask python to count and tell me, off course yes! There is the way that's why you could see my tone was not very sorrow full something like that I am joyful because there is a way by which I can count how many number of items are there in this list?  This list is very handy especially in case when we have a lot of data items for example let us say the student data I said list of students in our online course see there are thousands and thousands of students so manually counting over each name and then saying that these many students are present is really difficult checking in our shopping list I have created a really small shopping list just with five items so it was easier to count but see even ages data the sample just I said just it is a  city but this is would probably be in a street, this data could probably belong to a street only. If you take a people of all ages in a city it will be really much longer than this even this we cannot manage manually though real world data may be grow really really big in that case we have to make use of some functionalities of python to make our life simpler so once such functionality helps us find the length of the that is what it called as length the terminology uses length that is how many number of elements are present in this list? We can check that the way is you have to say Len that is the functionality length of the name of the list see ages it says eighteen there are eighteen elements in this items and let us just if you are sort of is it really a eighteen I need to count it, let us test with the something that we know shopping list, shopping list has five items as I have said six oh! Did I had another yeah I added oil yeah fine there are six items see it counts properly even I made a mistakes humans can make a mistake but computer will not make mistakes so this is the functionality that come handy in many situations so please make a note of it this is how you find the length of the lists so in that way where we can use the range functionality and indexing facility and print the elements in a list for example let us see ok so I will look up the history see that time I had just four items in my list, I had just four items in my list ok, see it is printing just four, now I say print shopping, if I just change it here I will not look up further print shopping if I would say print shopping see there are more than five items so if you manually want to give an number it is difficult so in this case you ant python to intelligently alter the number as and when your list grows or shrinks so let us use the length function here so length of shopping, shopping I need to close it ok, see now you got all the items so what it does? It will find the length and it will iterate that many number of times so it will go that many number of steps so this is the list the length is six so it will go and take this element, this element this element and every element will be covered so by using this length functionality and loops you can iterate ever each item in the list this is one way further simpler way is as I said is for item in shopping this is really a simpler way because you may not even think about the range, length and all these things so for each item in shopping as long as there are items in this list keep printing just as how you give the instruction in English with some modification you can do it in python really an easy way ok.

# LISTS PART 3: OPERATIONS

Alright guys, we have list of ages of different people see there are some ages we have had checked using the count functionality let us check once again probably. Ages.count of twelve see it returns two whereas ages.count of seventy it returns zero where as some other values say ‘twenty five’ it returns one twenty three, it returning two see look at the let us print the ages so that we can refer it again print ages see these are the values I cannot verify it manually I am believing but still if I want to verify its taking me some time to check whether the answer written is right or not? Also I couldn’t get an idea of what is the age group of people there living in that city something like that say for example this area has a lot of youth population or this area has a lot of senior citizens something like that some kind of inference I couldn’t male out from this data do you think is there some way if I modify this list you can infer something? Just give it a thought. Yes one simple way, if you would thing out is if the data is sorted that is it is arranged in ascending order or descending order something like that if it is sorted we can get an idea what is an minimum value what is the maximum value or may be examine the mid position to get the average of it something like that you can get some analyses you can give some more idea than having the idea like this if it is sorted that is if it is arranged in some ascending or descending order you can get some more idea so let us see how can we sort it, there are some sorting algorithms you would have heard of techniques like bubble sort quick sort merge sort don’t worry if you haven’t heard of it nothing its nothing never not a problem it’s not a rocket science don’t worry if you haven’t heard of it you can definitely learn it in fact will be pitching some sorting technique in our course as well also python has a functionality which lets you sort the numbers even if you don’t know any of the sorting techniques just say sort it will sort the numbers for you so let us see how sorting can be done. Ages.sort this is the way you will get the sorted list see ages could have been sorted now I want to print the sorted functionality so let me say print ages I obviously use the history look ups using the up arrow ok, see now the ages has been sorted now you can verify count of twenty three was two yeah there are two values twenty three I the previous list since it was distributed some way I couldn’t really check if it is correct or not something like that just not for checking but we can make many analyses the least is one year old kid to the eldest person in the city is eighty seven year old person so here just two senior citizens and till thirty five it can generally consider as youth so yeah more youth is there in this particular factious city that we have considered probably because am an youngster who is typing in the ages I have list of age that is young it all depends so you can make some analyses right? That’s what is the point. Sorting by sorting you can get some analysis will be easier so sorting is one such technique that will make most of your things handy it will make most of your things simple your complicated life would be made simplified with the sorting technique so this I the way how you can sort it in case you want to reverse it what is the method? That is sorted in the reverse sorted fashion sort will generally do it in the ascending order so if you want the descending order ascending order is nothing but the elements if you would see one less than two less than three that is if you go in this way from left you will see the numbers in increasing order like see if you have seven here something less than seven definitely lie to the left of it, it will never come in the right of it. If the numbers are arranged in that way if you take any number all numbers less than that will be in the left side and all greater than that will be on the right side if the numbers are arranged in this fashion then you call numbers then you call numbers are arranged in ascending order and if you want the numbers to be arranged in descending order what is the way? Just think for some time you will get the idea. I hope you got the idea, nothing? Just if you would print this thing in the reverse order it is descending order, descending order is nothing but as I had said you would scan from the left numbers will go in the decreasing trend so eighty seven if I say in the reverse order eighty seven, eighty two fifty seven forty two see if at this point forty two the numbers they are before it that is if you would read it from the left to right just assume that it is left to right in that case you will get the numbers on the left will be greater than it and the numbers from the right will be less than it this is true for all numbers in that sequence this is what we call as a sequence mathematically, in nothing but a list in that list if all numbers are like this then you called as the list is sorted in descending order descending is decreasing order, ascending is increasing order, ascend and descend they say right? For mountain if you are climbing up the mountain they call it as ascend and climbing down the mountain is called the descend right? so that's how the names comes ascending order and the descending order so let us see how can we make it descending order like I had said reversing the list that is by reading in the reverse order you are getting it so can w reverse the list? Yes! We can there is a functionality called reverse let us see there ages,reverse reverse ok it got reverse so let us print it now see as I has said it is in the reverse order eighty seven eighty two fifty seven and so on it is in the reverse order it is now in the descending order so by reversing again back you will get the ascending order so you alternately keep reversing you will get ascending and descending order once you apply sort you will get the ascending order and if you keep reversing you will travel between ascending and descending order ok this ordering of numbers comes handy in many situations and please note this method by which you use built in python functionality to perform this sorting operations that is ordering arranging in a particular order that is what we call as sorting technically sorting operation fine, so let us see some other powerful technique using lists for that let me consider some other data set the student data set that I had said you, let us have some data ok students data set let me just have some fractious data students equal to some names let me say let use doubles quotes let say arun I will say 'rajesh' I will say for example 'Harish' some three girls akanksha, lakshmi, varsha some names I have taken some lists of students are they first sorted? I don't think so I had just randomly entered the names see why do I sort them? Because that's how generally we store right in academic records in an alphabetical order they will be arranged and that is called the attendance order so let me sort them we have just now learnt how to sort apply that technique and then sort so they would have been sorted so let us print students yeah they have been sorted akanksha Harish lakshmI rajesh varsha so these are the students we have in a class just assume that so in that let us assign the role numbers to them let us say as I had said you can have it as zero generally our role numbers will start from one but computers starts from zero to counter that you can either make changes in your programme as you write code in the future or you can intelligently insert some mean values in the zeroth position that is first value that you see is index zero at zeroth index insert some dummy value then you can manage this as one two three so let us do the this way of managing this as one two three by inserting a dummy value in the zeroth position let me say let us insert the course name for example so let me say how would I insert at specific position we had seen using the insert function right? So let me say students. insert at the position zero, zero is the first position see the computers perspective and the humans perspective differ by one so you for first position for humans and zeroth for computer so at zeroth position let me insert the course name for example joc this is our course name I have inserted it tight? Now let me print see I have an inserted it double variable right and now I wanted to extracted a subset of students what do I mean by that? Three are some six students in my data set in my class basically data set is nothing but in my data six students I have considered some six random names I have considered there are some six students I want to extract a subset that is the few of them only I don't want to considered every one of them so how would I do that? This is the special operation called slicing let me write that for you slicing.

# LISTS PART 4: SLICING

See this is also an English terminology you are cutting the slice of it we say right in English something like that from this whole list slice of bread can say from whole bread pack you take some slices of bread something like that you assume this your whole bread packet a list are separate a packet you want few slices of the bread something like that you assume so this is the terminology we call as slicing. So slicing let us see how can that be done so that general syntax how you can slice a list is as follows list name and within the square brackets you have to give start till end plus one please note this syntax especially this end plus one so whatever the number you give same like your range function it will count one less the end point will be counted one less so if you say from role number one to three I want, you should say one to four so that it counts the third person something like that so this is how you should do slicing operations as I said I want one numbers one to four let me say for that what should I say? Students starting position is one here we have to give the index please not that start index this is the sorry here should I be giving the I did again starting index to end index so see this is start index and end index this is computers perspective you should give the numbers in the computers perspective here to manage the human and computers perspective differences I have inserted this dummy value at the zeroth index this is one way by which you can manage or its better I that recommended you guys used to the computers perspective and code it in this way only and upcoming exercises will also be in that way only we will make you familiar with the computers perspective don’t worry if you don’t feel the comfortable at this stage, definitely you will get comfortable as you keep practicing. The key thing here is practice just keep practicing you will find everything is easy for you. Fine ok so I want the role numbers from one to four right? So how would I do that? From one to four is what I wanted and please note here it is end index plus one because same like your range function this functionality the slicing functionality also would count one less than the specified end point so you want to count till four then you should specify it as five see let us see, see this is written one two three four akanksha arun Harish lakshmi this is extracted so we have extract of the slice and please note that it is not necessary that you have to give both these indices the start and end that is optional if I say students and if I just give colon what happens? See entire thing is taken so what does this mean? The default value they have some default values it means if you haven’t specified something what should the computer do? That has also been defined in python. For the default value for the start is the actual starting index that is zero the default end value is the total length minus one see because there are seven elements in the list so it will be zero one two three four five six so seven minus one, length minus one is the default end value zero is the default start value, if you dint give anything if this is also one way by which you can extract out the entire list. This is slightly roundabout but this would actually give you the entire list this is yet another by which you can get the entire list and if you just give the starting value and you let end value as such what happens, let us see, see I am giving the third person and I left the end value blank what happens? See from index three the third person Harish till the end it has taken so whenever you don’t give a value it will look up to the default value and use it so here you haven’t given end so it took the default end value of the length minus one and it has taken so basically in English this translates to start from the third person and go till end this is English translation of this slicing statement. Ok what if you don't give the start value? Simple it will take the same default value zero let us see that too ok am not giving the start value I will say up to five so up to index five it has to stop that means the four persons it will take see it has taken from the beginning, it has stopped before the index five see zero one two three four at lakshmi it has stopped see the end value is always when it reach five it will skip it, it will not consider five so till four it has considered. So I guess you understand the slicing mechanism that is to extract out the subset of a list we use this slicing the start value and the end value, end value please note that you have to give one extra because same like your range function it would stop counting when it reaches this particular given end value, given end value here is five so when it reach index five so it stop counting, it had quit it had counted only up to four so you have to make sure that you give the indices such that extract out the parts that you need. Any if you don't give start value the default start value is zero if you don't give end value default end value is, I am sorry I had said that has length minus one it is length see because we are giving one access right? Till length minus one you wanted to move means you have to give length as the answer. The default end value is let me write it as commands here default start sorry that is a start index is zero and the default end index is nothing but the length of the list I will highlight this statement default start index is zero the default end index is the length of the list these values will be taken only when the user has not specified the value that is see here start is specified, end is not specified so in that case this start value will be taken and since end is not specified it will look up to the default end value and take that value here end is specified start is not specified so it will take up the default start value only when you don't give any value it will take the default value, if you have given some value it will take the values you have given ok. so let us see one more thing students of two to five so it has taken indices two three and four it has just printed that so when you have given the values it will take the values that you gave if you haven't given the values it will take the default values that is what the concept of default values. If you have some experience of c programming something like that you have something called default lock switch case default locks have you heard something you can relate to it, it is very similar to that even if you have haven't heard please don't worry, the concept is if the user has provided something that has to be taken, if nothing is provided take this particular standard value default can be transferred into standard value, so these are the standard values that has been defined this will be taken. Alright I hope you have seen  the difference operation in list these are not only operation that can be done list, there are a lot of them you can refer to the documentation online, I would recommended you guys please try with different kinds of list, different things with numbers names with the floating point values for that you can take the stats calculation for example because if it is fourteen percent tax then when you multiply by zero point one four you will get some decimal  values the floating values we call that in computer science you can get that as well so I would like you guys try different data type list with this we cal as data type that is the data is of type integer or a floating value is nothing but the decimal point is there so real numbers something like that you, it's a string or a word something like that you try a different type of data's, different type of list, different type of operation and only by practice you can realise that lists are really flexible and they come handy keep practising thanks for watching the video happy learning.

# LOOPS AND CONDITIONALS: FIZZBUZZ 01

Hey hey amith, come we will play a game know I will teach everything come ok ok, shubudha and suma come let's play a game come come cheer up come. Ok see the game is called as fiizzbuzz ok, we are six in numbers so the game goes like this, he will start from one and then two and for all multiples of three will tell fizz and then multiples of five will tell buzz and multiples of three and five will be fizzbuzz! Yes. So you got it. Play? Ok ok. Should I start? Yeah. One two fizz four buzz fizz seven eight fizz buzz eleven fizz thirteen fourteen fizzbuzz. Hey this is so easy know so is it? Ok then, we will check from one not one, you start from one not one. One zero one fizz one not three, one not four, buzz, one not six, one not seven, fizz, hey hey!! That is fizzbuzz yesss! Actually now it's getting complicated. Correct! Yeah so complicated, hey this is a nice game I think I should write a piece of code for this; it will be a kick back for me. Why do you think of coding all the time? Go get a life dude, exactly! Hey this is what I enjoy, will definitely write a code for it. Hello, every one today I am here to code for fizzbuzz we saw you guys saw me playing that game and I told that day that I am going to code for it so let's see how we can code write a programme for fizzbuzz. So first I will start my spyder, so assuming that you have all installed anaconda on your system, I am using my Mac you need to start your spyder so go to your terminal and type spyder in even though it is same in windows you can go to start and write spyder in your application and you will get a spyder idle and just click on it and the spyder will open so let's see, so you need to install anaconda first, with anaconda you get spyder idle, without spyder also you can work on your simple terminal or command prompt but we strongly recommend you  to use spyder because it is very easy for you guys. So I am already created a file name here fizzbuzz here, if you will just start a spyder you will get a new file here you can save it by name you want so I saved it as fizz buzz now here I will write all my programme for fizzbuzz. Okay guys let's do this. So this is the place where I write my programme, this is the place where all the outputs and console related things will come so I hope you guys know the interface well anyways you can always Google up and look what are the things here so this is the new Ipython console which starts with the spyder in the Ipython means instructive python so you can visualise things here will see in some other examples in how we can use the Ipython so here the grid thick python is that you can even do the computation on fly, fly means in console you can write things and it will give the output so for example, if I write two plus three, it will just show me 'five' so, it's better you like that if you does the computation on flier that then in c, c++ where you have to create an dot exe file and you can run it on your computer which will show you the output here interactive Ipython console are works like this do the computation on fly. So what the fizzbuzz let me explain the fizzbuzz game again so what we did there suppose a person is saying the number and that number is multiple of three then that person has to say fizz rather than saying that number at loud so suppose if I am saying one two and three comes then I have to say fizz or any multiple of three comes like if six comes then also I have to say fizz or nine comes then also I have to say fizz and if the multiple of five comes then I have to say buzz if ten comes I have to say buzz and if multiples of three and five comes for example the number which is divisible by three and five I have to say fizz buzz for example if fifteen comes I have to say fizz buzz rather than just saying fifteen and a person if says rather than saying all these three, if he says the number then he is out of the game so what we are going to programme, if a number comes which is the multiple of three I have to say fizz if it is a multiple of five I have to say buzz if it is a multiple of both of three and five I have to say fizz buzz for that I need a loop so what I will do I will just print a number from one to fifty and check whether my loops work and then I will just divide that number with all my taking care of all the condition I will write those condition and I will complete my programme for that first I need a loop. Loop the syntax of I assume you know the syntax of loop is very simple and straight for I in range, range is the pre defined function in python where it takes the number and prints the range from zero to that number, even negative values it can take but I am taking positive value so suppose if I type in a range fifty then just print the number, it will gives the value from zero to forty nine so it starts from zero ok zero to forty nine we will get but what I want is print numbers from one to fifty for that I will write for I in range one to fifty one because it is giving always the number minus one values I will write print I that's it we will get numbers from one to fifty ok great now I know how to write a loop so I will just write that loop here I will write for I in range one to fifty one ok now I have a number at every iteration of this loop I will get a number from one to fifty one suppose if I get a number which is the multiple of three I have to print fizz rather than that number once I will write there print if condition, for that I need if condition I have to check whether the number is divisible by three or not if it is divisible by three then I have to print fizz I will write if I mod three is equal to zero which means I is divisible by three print fizz that's it and if it is divisible by five for example the number ten it is divisible by five then I have to print buzz but that I will write if  else I mean if it is not divisible by three but it is divisible by five else if I mod five is equal to zero which means if it is not divisible by three but it is divisible by five print buzz and if it is divisible by both three and five then I have to write fizz buzz for that I have to write a condition if I mod three is equal to zero and because  I need both the condition that is I mod five is equal to zero so if I mod three is equal to zero and I mod five is equal to zero it means then the number is divisible by both three and five and then I have to print, print fizzbuzz that's it we are done so let me click the console here I will just run this programme you can run it from here you can see that hi am getting fizz buzz fizz buzz fizz buzz fizz fizz something like that I don't know which number is giving me fizz which number is giving me buzz and which is giving me fizz buzz for that I need to print that number also so I will write the number, I will associate the number with their corresponding fizz then in order to print two string plus operation for example if I want to concatenate see here amit sudarshan with plus operator I can concatenate these two string it will show me amitsudarshan, hence I want to print I associated with that number means it will show if three  is coming it will show three is equal to fizz, it will show like this three is equal to fizz  so it will show like this three is equal to fizz for that what I will do in the print statement I will first convert the number, number is integer so I have to convert that number in to a string ok for that I will write str of I for convert my I whatever I am getting here fifty one into a string with plus operator I will concatenating this string with the fizz, if it is fizz right? I will copy this and if it is none of this neither fizz nor buzz  nor fizz buzz then I have to write else just print  you can just print I also head start I will show you what string does for example I file a number x is equal to one two three four if I write str of x it will give me a string and it can always be you can save this also and you know this sting is character only so if you write a two get the value at two array is always starts with zero it should show me two ok let's get back to the programme everything is complete now I will just run it and let's see what happens great it's running everything is fine  for one I don't have anything for two nothing three  should print fizz ok I am getting fizz I will give a space here should look nice, better whatever you programme it should look nice ok we will run it again yeah perfect for one I am getting nothing, for two nothing, three is fizz, four is nothing, five is buzz, six is fizz, because it is a multiple of three nine is multiple of three fizz, ten is multiple of five buzz twelve is multiple of three thirteen fourteen nothing, fifteen is fizz ok see this fifteen I am getting fizz but I suppose to get fizz buzz because fifteen is multiple of both three and five so, here is the problem this is the very common problem we get according to me and according to you guys also I suppose that this programme is very the logic is very perfect  I don't think there is any flaws in the logic but still I am getting fizz here rather than fizz buzz I am not getting fizz buzz so what's the problem? So can you guys guess? What did I do wrong, in next video I will explain what is the problem and we will see how to correct it and how to write a right programme for this fizzbuzz? You can post all your problems regarding this video in discussion form you can also discuss there so I encourage you guys to discuss this problem on discussion form will see this in next video.

# LOOPS AND CONDITIONALS: FIZZBUZZ 02

Hey amith! Yes sir. You have some time? Sure sir. Please sit, I stopped you because you look a little worried and you looked a little engrossed in some thought. That's what are you thinking? Yes sir exactly. So I was ready to write a code for a puzzle game I don't know whether you have heard about it or heard of it or not, its very famous puzzle I know not very difficult one but when you try to write a code of it then you get stuck somewhere correct, and it's not only me I tried to Google it and I found that many actually more than eighty percent of the people get stuck there and its very surprising that how it is happening? And why we all of them are getting stuck at that point only? Correct. So you are discourage thinking that probably your programming skills are poor, exactly yeah. Because of which so have you become programming all this while? Yeah yeah I think I am a good programmer. But this particular thing as a bouncer coming that's what you are saying, yeah ok so I also personally feel that fizzbuzz is slightly hyped on the internet, so historically speaking there is some story behind fizz buzz I am sure you know of it if not I will just brief you with it. It is believe that a whole lot of programmer so who claimed that who are good programmers cannot even write a simple code ok? Ok the simple code in codes a good candidate example for a simple codes is this fizzbuzz as you know you describe it right? It's some multiple of three and five that you want to display in a particular way and like that right? Exactly. Let me illustrate what exactly is the problem. Do u know of any tongue twister? Yeah when I was kid I use to know some. Can you tell me some tongue twister of your choice? My friend once told me so I will try. Bity bought some butter, but the butter was better so he bought more butter to make the bitter butter better, that's not the exactly the tongue twister good that you made a mistake, I wanted you to make a mistake, is it impossible for you to get this tongue twister some amount of practice required yeah but without practice its plane impossible for you to get this right? True true. I will give you one another tongue twister. Ok. She sells sea shells in the sea shore we have been narrating this from ages like; can you try do narrating this? Can you repeat it one more time? She sells sea shells in the sea shore she sells sea shells in the sea shore yeah yeah exactly its difficult the point here is that there are certain things that comes by practice certain things that they are very difficult to do it on the first time right? a simple sentence like as good as it gets say that as good as it gets see that's easy, certain things are easy certain things are difficult I some of this in the fizzbuzz is very hyped a programming example and its of discourages people thinking that they are not very good programmers yeah right? with practice I am sure you can do a super fizzbuzz programme too according to me super fizzbuzz you give as input some bunch of numbers and you should display which numbers which multiples to display and which multiples not to display right? Like a variant of visible right? so I think we should not take these things very seriously all that matters here is enormous amount of practice, you keep practicing I am sure fizz buzz is just a very bad candidate example anything for that matter will be a kick off go ok? so my call for all programmers learning especially in our course is that there are going to be questions that are not going to be straight forward that you will find it difficult to code that's to be perfectly fine with a few attempts you will be able to do it, my sincere request for you all is not to switch on the timer then coding right? Fizzbuzz is probably difficult for you but given one hours time do you think you will not be able to solve it, so programming should not be timed according to me. Forget all the competitive programming challenges. Programming according to me is an art it should be treated like an art at least up to the point but you reach that and say you are an expert and then you try competitive programming but until the point you master programming I think you should be slow, you should be deficient and you should not time your programming. So do these experiments take an innocent programmer who doesn't know a whole lot of programming but he is newly introduced to programming some students from let say from joc course and gave them good amount of time and motivate him tell him about this fizz buzz and I am sure he can solve it. Ok in last videos we saw that the programme we wrote a programme but we think that logic is correct but still we are not getting the desired output so we will see out what's problem in the so you can see here in fifteen I am getting fizz I should get fizz buzz but I am not getting the fizzbuzz and according to me the logic is correct still I am getting fizz only, I should get fizz buzz. Actually the problem is that what we are doing is whenever a number is coming first we are trying to find whether that number is multiple of three then if it is not multiple of three then we are checking whether it is a multiple of five and if it is not both the multiple of three and five if it is not multiple of five then we are checking that multiple of three and five then, we are showing fizz buzz ,now see let me debug this, let me go through all the numbers suppose if I am getting number three it will get a three mod three is zero it means that three is divisible by three I will get a fizz. For six also will show me six mod three is zero six is divisible by three if ten comes then ten is not divisible by three and ten mod three is not zero reminder is not zero then it will go to the next condition else ten mod five is zero hence it will print buzz perfect for twelve if twelve comes twelve mod three is zero ok fine it will show fizz let say fifteen when fifteen comes the first condition is I mod fifteen is equal to zero which means fifteen mod three is equal to zero yes fifteen mod three is zero so it will just print fizz and get out of the element that's the problem hence the first the problem is first condition is I mod three its only checking for three divisibility but we have to check for both three and five if it is divisible by three and five both then you should print fizzbuzz and then only you check for three and five hence I will change the conditions here let me re write the code ok so I will write here if I mod three is equal to zero and I will check for both I mod five is equal to zero and print I plus equal to fizz buzz else if it is not divisible by both then I will check if it is divisible by three if I mod three is equal to zero then I will print let me copy this print fizz else if it is divisible by five then print buzz else it means that it is not divisible by three and five just print ok it is always good to write a function for whatever programme you are writing here I just wrote the logic by code here but it is better if I define a function and then call that function I just into function fizz buzz in here I will give the range value let's say python don't take care of the indentation code is huge and you will find difficulties while printing that's why we are using spyder indentation where it automatically it if you are using some other difficult for you guys to do the indentation or you can use the sub line text also but we recommend you to use spyder this is ok I can call this function from the terminal I will just run this I will call the function fizzbuzz and the range I will use this r here again run this we will call the function fizzbuzz and the range let's check one two fizz four buzz fizz nine is fizz twelve is buzz fifteen is fizzbuzz eighteen is fizz twenty is buzz ok you can even write the function you can even call the  function here that contains fizzbuzz and you can give fifty one and I will run this cool so we are done with the fizzbuzz programme you can even modify this programme you can create your own game something like that you can change the numbers in a conventional fizzbuzz game we use three and five you can use some other numbers some other prime numbers ok we strongly recommend you guys to play with ait and if you are getting some new question regarding fizzbuzz and new ideas also please discuss on the discussion form ok it will help other guys also it will create more new ideas for us and for you guys ok we will meet again good luck all the best.

# CROWD COMPUTING – JUST ESTIMATE 01

Sir, can I speak to you for five minutes? Hey hi Ravi, how are you? I am fine, Something is bothering me so much, I am humming this really melodies Hindi song from nineties, Their hero talk's about heroines anklets and it was shot in a south Indian temple I am unable to recall that, Will you please help me with that, So a south Indian temple and a Hindi song, south Indian temple and a Hindi song, You said in the nineties film, Ya nineties, And what was in the lyrics, I don't know hero was praising heroines anklets or I remember it something to do with heroines anklets, Anything in the lyrics some two three words, No, tried goggling it, Sir I tried goggling it but I couldn't find the result appropriate result for that, Ok,So its very difficult to find now, You need know with the lyrics nor do you know can you hum the song? No. How can we find now? Ok so maybe we can think of let asks someone may be who knows both south and north, they should know. You are the best example for that, ha ha you have south Indian friends and north Indian friends and you love Hindi songs ok that reminds me ok I have both south Indian as well as north Indian friends I have loads of them on face book, yeah and maybe we should try posting it there one or the other person will see it and let's see whether he can answer huh? That's a great idea ok super. Oh my god! This is awesome I really want to learn the science behind it let me ask the professor. Ravi saw what just happened with our little exercise, see all that we did was we went to face book and we I add a line on your behalf and it was wonderful to see that is less than ten minutes, ten to eleven minutes I believe, we got the final answer what is most surprising there? Is that the first hit was wrong second hit was also wrong if I am not wrong right? The third hit was "The song" how do you think that this happen? Any guess, see goggle did not help us right? Asking us expert of bollywood may not be of any help, if you simple say south Indian temple nineties bollywood song and finally saying that it has some lyrics about the anklet of the heroine may not land you anywhere, what just happened? I don't know actually I am very curious w couldn't find it in Google and within thirteen minutes we find it in face book, that's right see what just happened is called it's a very popular social computing technique it's about asking people to solve problem and a bunch of people come and solve a problem but then you see we got the answer free of cost we didn't invested any infrastructure that's because on face book you have a whole lot of well wishers who don't mind helping you out correct? I roughly have some five thousand friends on my friend list and out I am sure not all of them would have come forward to help me, some of them came forward out of them only a few people knew something about this song we have many many friend on my friend list who are outside India right they may not know anything about bollywood correct? They off course are of no use there are roughly two thousand three thousand people who probably watch lot of bollywood songs and a part of them tried helped me out and a part of them could help me out with some inputs out of which one was the right answer, right? This is how social computing works. Sir I have a question, do you think this works every time? Or for every one? Yeah that's a good interesting question you need big numbers let's say you have some two hundred friends in your friend list if you try posting this may be ten to fifteen would be interested to talk to you about it and two or three will give you some hits that may not be right, so bigger the number more the merrier seems to be here right? This is precisely how your Wikipedia works this is precisely how your stack exchange works where you go and post a question you will get an answer immediately just the way we got an answer right now, this is how the even quora works right? So imagine imagine if you were asking this question some twenty years back you wouldn't ever get to a solution, yeah. The only possible solution here was face book there are many application where you switch on the application on your mobile it will listen to the music of the song and tell you what the song is but that useless here because you dint know what the song was, you dint even know you couldn't even hum the song right? You just gave some bare minimum details and you got the answer. What is more surprising is that you could not Google it, could you? No. we tried it Googling it and it shows up something else, it is no way close to the answer correct? So there are kind of questions where you won't get answers in Google but with your friends help you can get it in face book, that's right, perfectly. So we can even hypothesise by saying the problems of the world and the problems of our county, can be outsourced and rather crowd sourced and we can ask people to talk about it and we might even get the best possible answers, this is precisely what's goes by the name wisdom of the crowds, where the power of many can beat the power of one super intellectual. That's very interesting.

# CROWD COMPUTING – JUST ESTIMATE 02

The professor just now explained “the power of crowd”, you see crowd is not a bunch of people they are the people who answer my question quora or create articles on Wikipedia which I read how crowd can helpful in many ways. It can help me to find out the number of gems in this jar or help me to estimate or calculate the weight of this bag. You must be wondering how, let me show you. Estimate the number of gems in this jar? One eighty, Two twenty, Four hundred, Around six hundred, Thousand, Five hundred, Three hundred may be, five fifty. It’s around fifteen hundred, two thousand, five hundred, five hundred? Ok I expect above five hundred, two hundred and fifty, fifteen hundred ok, two fifty four hundred, one thousand ten, two hundred only two hundred? Two fifty, two hundred, two hundred, two fifty, three hundred, three fifty, two sixty three, two seventy five, three hundred, two fifty, two twenty, one seventy, hundred, seven hundred and twenty, five hundred, six fifty, three hundred probably, one seven eight one, two twenty, seven hundred, what is this all about? Will let you know say two hundred, seven hundred approx, one seventy, five hundred, ok let’s say one ninety seven, two fifty, around two hundred, hundred and fifty, two seventy, two thousand two thousand? I think two fifty, hundred and twenty five, three fifty, four fifty, around three twenty, four sixty seven, thousand, three hundred, I will get one after that sure, if you guess it right, ok then two fifty, six hundred, five hundred, one twenty, two fifty, one fifty, four hundred, five hundred, not more than three fifty, four hundred, four hundred? Three hundred, two fifty, and what about you sir three hundred ok hey guys guess what happened? There were actually three seventy five gems in the jar and the crowd guessed it right. I know no bodies guess was exact but if we take the mean of all their guesses it is actually coming out to be close to three seventy five wanna know how? Let me show you. As you all know we conducted an experiment which numbers of people were asked about the number of gems or candies in the jar, all though nobody guessed it right but still we got the right answer, how? Let us find out. This kind of experiment was first conducted by Francis Galton in a country fare, ox weight judging it competition was organised in which around eight hundred tickets were issued, every participant were asked about the weight of the ox. The actual weight of the ox was one one nine eight lbs and if we take the median of all the estimates it came out to be one two zero seven lbs as you can see one two zero seven lbs is very much near to one one nine eight lbs this is the wisdom of crowd, this is the part of crowd. So now I would like to explain the reason behind it why does wisdom of crowd prove out to be very useful here? So let us look at this picture we have actual value ok, some people under estimate the actual value of the ox and some people over estimate the actual value of the ox. Ok the under estimation part and the over estimation part it actually cancels out so what is obtained at the end is the actual value. This is the same principal that is playing behind me. For example we have given data comprising of minus three minus two minus one zero one two three here we have median as zero for example zero is the actual value here and some people guessed it one some people guessed it two and some people guessed it three and some people under estimated the value like minus one minus two minus three ok nut if we take the difference of the values of the median here will get minus one, one, minus two, two, minus three and three and then we take the summation of all these values in at the end get the median of the data, this is the same principal that is playing behind the wisdom of crowds and this is the same principal that is playing behind the median of the data ok, the under estimation part actually cancels the over estimation part and the answer that we got from the wisdom of crowd from many people estimate is actually the actual value of the ox. So as I said over estimation minus under estimation will comes out to be zero so the answer that is obtained from the estimates of the crowd is actually the actual value is the true value of the ox. So what we did in our experiment was we asked people about the number of candies or gems in the jar, in our experiment around seventy five people participated the actual number of gems in the jar were three hundred seventy five the median of the data that is obtained is three hundred and the average of all the estimates that is been obtained is three fifty one so as you can see the median is not giving us the right answer here, even the average the mean of all the estimates is giving us the right answer here so according to the recent studies it has been observed that if we take the other aggregate measures then they can also be prove to be useful in these kinds of experiment I repeat recent studies have shown that if we take the other aggregate measures than these aggregate measure can be prove to be more useful in these kinds of experiment. So in these kinds of experiment you have to calculate the number of aggregate measures but aggregate measures are main you have to calculate median mean and other things so here we calculated the average of all the estimates and that came out to be three fifty one and as you know as you can see three fifty one is close to three seventy five as compare to the median that is three hundred so here we calculated the average of all the estimate and that came out to be near to actual number of gems here we didn’t calculate mean as total number of the estimates some of all the estimates divided by total number of the estimates, we didn’t calculated mean by this formula what we did was we calculated ten percent trimmed mean, what is ten percent trimmed mean? Let us find out. In ten percent trimmed mean what you have to do is , you have to remove the ten percent smallest and ten percent largest values the sample that has been obtained by this method you have to calculate the mean of the sample obtained, yes here we are using ten percent trimmed mean you can use five percent, eight percent trimmed mean as per your data actually its depends on the data but on our data that we obtained from seventy five people ten percent trimmed mean give us the right answer gave us the answer that was very near to the actual value, so how can we calculate ten percent trimmed mean? First of all you have to remove the ten percent smallest and ten percent largest values, how can you obtained ten percent smallest and ten percent largest values? In this particular thing you have to sort the data what do I mean by the sorting the data? Sorting means arranging the elements in ascending or descending order so if we sort the data, if we arrange the elements in the ascending order it will be easier to calculate the ten percent smallest and ten percent largest values and then you can easily remove them from the sample, after that you have to calculate the mean of the sample that has been obtained after removing the ten percent smallest and ten percent largest values this is how you can calculate trimmed mean. So as I already said this is the wisdom of crowd, this is the power of crowd, these kinds of experiment explains the wisdom of crowds in the better fashion ok so what is wisdom of crowds? And we look out its definition it is actually the collective opinion of group of individuals, collecting opinion is always better than the expert opinion. Yes collective opinion is always better than the expert opinion; if we take the expert opinion in these kinds of experiments it won't be as accurate as collective opinion, you can do that actually ask the number of gems in the jar from an expert and you can actually ask the number of gems in the jar from crowd. The collective opinion of the crowd is always the better than the expert opinion. There is another example from a day to day life based on wisdom of crowd and that is Wikipedia. Wikipedia is a crowd source here any one can write edit articles as you know anyone can write and edit articles on Wikipedia. There is also an encyclopaedia available online called Britannica, Britannica is curetted by experts and Wikipedia is curetted by people like us by the crowd, if we take the comparative study of Wikipedia and Britannica recent studies recent research has shown that Wikipedia is as good a Britannica, yes this is right, Wikipedia is as good as Britannica isn't it amazing and an encyclopaedia that has been curetted by crowd by the normal people and an encyclopaedia that has been curetted by the experts you can do the comparative analyses you will get to know about it that encyclopaedia that has been curetted by crowd is as good encyclopaedia that has been curetted by the experts. So this is another example of wisdom of crowd.

# CROWD COMPUTING – JUST ESTIMATE 03

So before jumping to the programming part of crowd computing let us try to manually analyse this data, As you know we collected the from seventy five people we have the data from seventy five people, The guesses of seventy five people as you know the actual number of gents in the jar were three seventy five and if u look at the data, Some people over estimated it some people guessed it seven twenty, One zero one zero and fifteen hundred and two thousand two, But some people also underestimated it, Some people guessed it hundred, one fifty, Two hundred, Two fifty, Three hundred, So this is the underestimation part many people underestimated the data and many people over estimated the data, As I said the over estimation part it cancels the under estimation part, So the average that comes out is actually near to the actual value so if we calculate the average of this data let us calculate this basically average from a1 to a75 yes, it is coming out to be four twenty six point three seven but guys recall remember that we don't calculate this mean, this mean is basically some of all the estimates divided by total number of estimates what we do here is we calculate the trimmed mean so I will basically delete this. So how can we calculate the trimmed mean? Trimmed mean is basically you to remove the ten percent smallest and ten percent largest values it depends on you whether you take ten percent or twenty percent but be careful in your choice, here we will be calculating through ten percent trimmed mean for ten percent trimmed mean we have to remove the ten percent smallest and ten percent largest values. How can you do that? First of all you have to sort the data sort the data means arranging the data in ascending order so here first of all what I will do is I will sort the data in ascending order, how can you do that in a spreadsheet? Just go to data and click on sort sheet by column a I will just do that because no other columns here so I will just do that hey w have sorted the data as you know hundred one twenty one fifty one ninety seven you can figure out that this data is sorted in ascending order so now what we have to do is, we have to remove the ten percent smallest and ten percent largest values and we take the ten percent of seventy five it comes out to be seven point five I will basically take the floor value so I will remove the seven smallest and seven largest values let us do that so we remove hundred, one twenty, one fifty, one fifty, one fifty there are five values have been removed so two more are still to go so remove one seventy and one seventy five too so I have removed seven smallest values, now it's time to remove seven largest values let us go till the end and remove the seven largest vales so, I will remove two thousand, one seven eight six, fifteen hundred, fifteen hundred so three more are still to go will remove this one zero one zero, will remove thousand and will remove seven twenty too so after deleting the ten percent smallest and ten percent largest values this sample has been obtained from one eighty to basically to seven hundred so I will like you to calculate the average of this particular sample rather than the previous sample so will calculate it, will just put the formula here again in this spreadsheet its average from sorry average from a8 to a68 exactly so we calculate the average as I said it came out to be three fifty one. It is basically three fifty one point six and the actual number of gems in the jar were three seventy five as you can see that its very very near to the actual value, it is an amazing concept the wisdom of crowd if we it comes into play in many of our day to day experiences and many of the things experience so look here as you know we calculated the trimmed mean and the trimmed mean is come out to be three fifty one point six which is actually very near to three seventy five that is the actual value, isn’t this amazing?

# CROWD COMPUTING – JUST ESTIMATE 04

So let us try to program this particular concept, first of all I will take a list name estimates in which I will store the estimate of seventy five people I have this data here or so, I will just copy paste as you know we have to calculate the trimmed mean for trimmed mean we have to sort the data so I will just sort this particular list by using the functions sort this let us check whether there it has been sorted or not? Length of estimates just prints the values so that we get to know that, estimate list has been sorted. I just run this, as you can see the list has been sorted we can see hundred, one twenty, one fifty, one seventy, one seventy five the sort function is by default sorts the data in ascending order, I repeat the sort function by default sorts the data in ascending function in ascending order so don't link this so now calculate the trimmed value as you know we are calculating the ten percent trimmed mean here so what do we do here? Will calculate the trimmed value as point one of length of estimates but here we can see that it will give us floor value or decimal value. So will type cast to int so that we obtain an integer value out of it. Here point one into seventy five will be seven point five but if will type cast to int it will take the floor value so seven point five into seventy five point one will be seven point five and it will take the floor value so TV will become seven. now what we have to do here is we have to delete the smallest ten percent values and largest ten percent values so first of all let us move the smallest ten percent value it is very easy in python you can just start the list from TV it will remove the first ten percent values if you want to check let us check here for I in range length of estimates we have to print here print estimates of I let us try to check this now yes we got the values here as you know initially it started from hundred but now it is starting from one eighty that means it has moved the smallest ten percent values so now let us try to remove the largest ten percent values here to so it is also very easy here in python we just remove the largest ten percent values for that we have to decrease the length of the estimates we will just take it from length of estimate to minus TV it will remove the largest ten percent values. Now we have to calculate the mean of this particular list that has been obtained by removing the ten percent smallest and ten percent largest values. So I will just what I will do here I will just use the function mean here that has been already defined in python that is mean you have to calculate the mean of estimates but mean function has been defined in library statistics so we have to import that, import that from statistics import mean, so that we can use here, so let us try to run this particular programme and check whether the mean that we calculated in the spreadsheet and the mean that we get here is same or not let us try to do that yes it is coming out to be exactly same, it is three fifty one point five nine and as you know that actual value was three seventy five we have discussed this fact earlier when we were discussing the values when we were discussing the estimates in a spreadsheet so as you can see we are don't with the programme here but in python there is much easier way to calculate the trimmed mean to there is an another way so let us try to do that also here for that you have to import other library that is scipy from scipy you have to import stats import stats. We just have to use the estimate dot sort here we need not use any other estimates because as you can see that we have deleted the values here so we don't need that here I will just remove that, I just comment it so now what we have to do here is we have to calculate the trimmed mean in python there is a function name trimmed mean which will directly calculate the trimmed mean so it is basically stats dot trim underscore mean you have to supply the particular list on which you have to calculate the trimmed mean so it is estimates here now you have to supply the percentage, the percentage of values that you have to remove from the top and bottom of your list so as we know we are using ten percent here so I will write point one here so the function here is first of all you have to import the stats library here from scipy after that after that the function is trimmed underscore mean in trim underscore mean you have to supply the list that you are operating on that is estimates here and then you have to supply the particular percentage of values that you need to remove from the top and the bottom, it is ten percent here so we will right point one here I will just straight way print the n value here let us try to run this again it is coming out to be three fifty one as you can see the values that we get here is three fifty five point one nine same as we got in spreadsheet same as we got in the programme so as you can see that it is very easy to calculate the trimmed mean in python very much easy in spreadsheet and very much easier than the previous programme. I hope this program was useful to you. Have a nice day happy learning.

# CROWD COMPUTING – JUST ESTIMATE 05

Hello everyone, In this programming screen cast I will show you, How can you plot the estimates of the crowd, So that you can to get to know, Which estimates whether median, mean was close to the action value, So how can you plot different numbers on python, Let me show you, For plotting different numbers for plotting any sort of data you have to import one library and that is import matplotlib.pyplot you have to import matplotlib.pyplot I will name it as plt ok, So let me show you how can you plot different numbers I will just write plt.plot and I will pass a list here, So I will pass like one, two, three and four, So let us see our plots first of all you have to save it plot.py. We have a graph here we plot it one two three four, one two three four are plot are plotted on the y axis, Let me show you that here it is one two three four, So in this where you can plot particular data with the help of mat plot lib you must be wondering here we didn't pass any x value, But in python it automatically generates the x values if u dint pass any, I repeat in python it automatically generates the x value if you don't pass any, Now I will show you if you want to pass x value then how will it look like, So I will pass one more list and will be like one four nine and sixteen here one two three four will be the x values and one four nine sixteen will play as the y values, So let us see how the plot comes out to be here, So as you can see we change the y values, So we get one four nine sixteen here the line that has been obtained is not is not straight it is curved and one two three four are playing as the x values here, Ok this is how you can plot in python through mat plot lib there are many other options that are available you can also use many other option I will let you know one option is you can use the labels as x labels and y labels you can actually label the x axis and y axis, Here we will do plt.y label I will write some values this is how you can label your x axis and y axis ,Let us see how this happens see we have label here some values ok since I use the y label it will be shown in y axis you can use also x label if you want to use any label in the x axis ok,So this is how you can use the labels and this is how you can plot through mat plot lib,Next if you this is the line graph if you want plot in other fashion if you want to plot dotes in other fashion let me show you how you can do that for example if we if I want to plot in dotes so let us see how can do that. In plt.plot you can write another argument how do you want to plot the data, So I will write RO here, if I write ro let us see how the plot come out to be, see the plot comes to be in the form of red dots, It is one four nine and sixteen, So this is you can plot in form of dotes, Other options are also available for example if you want to plot in the form of red dashes, So you can write R hyphen hyphen in place of RO now let us run. This is how you can plot in the form of red dashes, I will show you some other option too for example if you want to plot in the form of blue squares you will just write BS will just write BS let us run that, Yes here you have plot in the form of blue squares BS stands for blue squares yes it is one four nine and sixteen and there is one more option that is available that is the form of green triangles you can also do that let us run that let us run that, So here you the data in the form of green triangles let us check that, Here we have one four nine and sixteen, So in this way you can plot in the form of red dashes, blue squares and also in the form green triangles, This is how you can plot the data through mat plot lib in python.

# CROWD COMPUTING – JUST ESTIMATE 06

Here in this programming screen cast I will show you how can you plot the estimates made by the crowd as I said first of all you have to import the mat plot lob library so, let us import that, import matplotlob.pyplot as plt her is the estimates added that has been given to us so I will just plot this array so let us do that plt.plot I will just write the estimate arrays here, let us plot that sorry it is matplotlib so as I said if you don't give the x values python generates the x values on its own so since we don't need any sort of x or y values here we just want to show the estimates here so what I will do here is I will take this estimates array in the x axis and for y axis I will keep it constant I will keep constant y values for all these estimates I repeat I will keep constant y values for all the estimates for that I have to create another array for example another list I will create like y so for I in range length of estimates I will just append any constant value here for example I will take five if so here we need to plot estimates that had been taken in the x axis then we will take y so let us run this programme now, as you can see we have a straight line here but you want to show the estimates these are discreet values not continuous so I will just use ro here or r dash dash I will prefer to use r dash dash, r dash dash in the previous programming screen cast I have explained r dash rash stand for red dash lines so let us run it again so we have the estimates in the form of dash lines red dash lines so now I want to plot the mean and the medium median of all the estimates so that I can get to know that which value is nearer to the actual value so for that as it has been explained in the previous programming screen cast that, that you have to trim the sample you have to trim the ten percent smallest and ten percent largest values so let us do that I just first of all sort the estimes, estimate.sort that will arrange the elements in the ascending order after that I have to decide the trim value that is ten percent in our case so I will just say rim values is equal to point one into length of estimates, length of estimates now we also have to type cast with tool so I will just do that in now let us trim the values so now estimate is equal to estimates trimmed values colon it will trim the values from the estimate list we also have to trim the values again so let us do that it will start from the trimmed values, in trimmed value in our case is seven so it will start from seven and it will go till length of estimates minus TV so I will just write the colon here length of estimates minus TV so I array the same has been obtained is from TV from the trimmed value length of estimate minus TV this has been already been this has already been explained from the previous programming screen cast so we are done with the y array we are also done with the estimates array the trim estimates array so now let us plot that the trimmed estimates array there is some error let us cut this x and y must have same dimension ok, now we have to append according to the new length of estimates please note that I will append according to the new length of estimates please note that the length of estimates has changed so we have to append y according to the new length of estimates so let us do that so we have another graph with is showing the estimates please observe that in the previous graph we had till two thousand but in the new graph we have till seven hundred because we have appended the values we have trimmed the ten percent smallest and ten percent largest values so we are done with plotting that so let me now show you which value is nearer to the actual value, actual value in our case is three seventy five so first of all let me plot that I just write plt.plot plus write three seventy five because we are taking constant values and I will represent it by say green triangle so let us run it again so we have three seventy five here now let me calculate the mean and median of this data for that I will import statistics import statistics so it will be like plt.plot first values value is statistics.mean of the estimates and while I was constant that is why I represent by say red dots let us do that red dots run it again so we have red dot here that is the mean of the estimates and the green dot here that is the actual line let us also plot the median of the data so I will just write plt.plot statistics. Median I just write estimates take y as constant and represent it by blue square yes let us run it again so as you can see that blue square that is the median it is here and the mean that is red dot that is major to the actual value and blue square that is the median it is far away from the actual value so actually it depends on the data you have to calculate many aggregate measures like mean, trim mean and median of the data and whichever aggregate measure is nearer to the actual value you can take that these kinds of experiments work on the aggregate measures I hope this programming screen cast was useful to you guys and have a nice day  happy learning thank you.

# PERMUTATIONS-JUMBLED WORDS 01

Hey, what did you just order I will give a hint ok oh brownie yeah you are right ok you will also do it, yeah! Flavour, yeah! You see what just happened one person asks the other by giving him or her a jumbelled word and says can you tell me the right English word? It is actually very complicated why for the following reason. Look at the word rainbow it is rainbow just seven lettered word and when u jumble it looks something like this are you really one English word which makes sense but if you see all possible combination of this letter seven letters there are not just dozens or hundreds there's actually more than five thousands possible combinations with this seven letter so it's practically impossible for a person to guess what a word could be it involves enormous amount of sort of thinking intelligence and seeing through the letters and then putting it in the right order only one among's the five zero four zero possibilities is actually the right word so similarly we can go on asking a random take a random English word shuffle it and pose it as a challenge right? As an always let us see how to write a computer code for this.

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# PERMUTATIONS-JUMBLED WORDS 02

A person has a word on his mind he shuffles it and asks the other person to guess what that word would be. This is the game, sounds interesting? I said always this being the joy of computing course let us play this the fun way using our programming skills. Let's get started.

Before starting up with the programming let me just give you the summary of what the game is all about, this is nothing but there are two players in the game, player one and player two say initially we will starts up with player one's turn so player two will have some wording in her mind what the person does is he would re arrange the letters of the words, basically he would shuffle the words and write those letters in some random order, in some shuffled order not the original word exactly he would write the shuffled one and present it to person one whose turn is at now so this person is presented with the shuffled word, he is supposed to guess what the actual word would be. If he is able to guess he would get a point and this person would pose a question to the other person and the game goes like this whoever gets the maximum number of points is a winner or it may be draw if both get equal number of points this is how the game will go. So with this kind of strategy let's play this game using out python programming skills. So let's starts with it so we have to play the game so play you have got, we have to play the game so for this there is a warning that could be seen here it says undefined name play so we have said we want to play the game what the thing has to be done we have to define it so let's start defining, define play this is the way by which you would start a function definition so that is play is something that you want to perform this is not something that is already known to the computer so you have to define that this is what is a thing you have to do when I say play so you are giving a definition that's why you have to use def and then the name whatever you wished to give so I start my thing so as he would see second line appear indented indentation matters here so as long as you maintain this level of indentation all this would be considered as the tasks to be done if think play is called so let's start off with what all we need to do so we will have two players may be player one he will have some name let me say player one name player one name let us input the name input for him to input we should ask with a suitable message let me start with the message player one please enter your name so this is for player one same you have to repeat with player two, player two name you have to input here input player two please enter your name so alright we have collected the names of the players so initially both the players will start with zero points so let us assign it the points to player one is zero points to player two is zero again and I said that the turns will be alternated so one player would pose the question and the other will have to answer and the next time it is reverse so there are some turns here instead of the players giving the question we would have the computer question to both the players so the thing is like the computer is the third party which would give the questions of the same difficulty to both the players so there would be no biases also with it that is the idea so we will make the computer pose the questions and both players will get turns alternately just like how it happened in the previous scenarios the turns would be alternated and the question will be post by the computer and answer as to be given by the players. So to keep track of whose turn it is let us have some variable or turn initially let it be zero it can be any number its generally a practice in computer science to start counting from zero so that is why we have given zero alright so this game will go on in definitely until the player one of them has to stop this is the two player game definitely we need two players only then competitive spirit will be there as long as both are willing to play the game would continue in case if one of them wants to stop then the game would come to an end so we need to continue it definitely that is being taken care of this thing while loop and condition, condition I am given as one that means true so while true will always be true and we would end this only when the player wants to stop that we will see later. We will be dealing with that too now what should the person do, he has to get the question from the computer, this is basically the computers task, computers task that was to be done here what is it doing? It has to picked word there would be some picked word computer will pick some word so it will have some dictionary from that dictionary it will pick some words so choose that will be a function we will define later so computer is choosing some word and it is written as picked word so that one word is picked from which dictionary or from you can considers as it has some list of words from that list it is picking some word so then it has to create the question this is the task is to create the question, create the question so question is created by jumbling the pick word jumbling is nothing but we are just rearranging the letters of the word, you are jumbling the picked word so now once the question is ready you can print the question. So far so good now let us come to the players part first is should maybe we should start off with the player one as the first player then the player two we will alternate it to keep alternating the turns we should keep track of the value of turn. So turn is set to zero so let it be the case that at turn zero the player one will be playing at turn one player two will be playing at turn two again player one will play at turn three player two will play it goes on this way so you would see at the even values of turn it is the chance of the player one at the odd values it is the chance of player two so to keep track of even and odd we will use modular operator that is nothing but operator which would divide and return the reminder of so to get to know if it is even or odd we can divide the turn value by two and see the reminder for all even numbers reminder will divided by two is zero and for odd numbers the reminder is one so that fact will be making use of it here and we will code. So as I said we are going to use the Modular operator if turn modulo true turn modulo to this equals to zero single equal to is assignment and double equal to is the condition check up of equalities so this basically what it does is it will check it will check this expression turn one two that is it will take the value of turn divided by two and check the reminder if that reminder is equal to zero then this segment has to be executed so you have to say that this is this person turn so let me print a message saying player one name we know the name we have got the name of the player, player one name this is your turn alright so I have printed this message so I have to get, I have displayed the question here you would have seen here there is a statement print question so computer has to printed the question so that has displayed the question so I have to ask what's on my mind so that computer is asking what's on my mind so that the computer is asking what's on my mind so it has picked some words what is the actual word co it is asking what's on my wind so you have to get the input for this question the player would given answer that as to be taken as an input so we will do this now. Answer is equal to input of what is on my mind so computer has got some word in its mind so that word the player has to guess so that is the question make sense ok this is the question posted by the computer and for this question the user will give an answer which would be inputted and stored in this variable ans now, if the answer given by the user is exactly equal to the picked word if this answer is exactly equal to the picked word then you are supposed to increase his points by one so he gets one point so there will be some points already player one has score during the run of the game initially it is zero but when the loop goes on as an when the loop goes on the points will change for the players may be one player would have guessed lot of words but another one would not have been able to do so there is the chance that a difference between the scores exist so we will retrieve the precious for add one to and make it the new score so player one point is nothing but his previous point plus one this is his new score and you have to update him of his new score print your score is like I can say it is stored in this variable points of player one which is stored and we have fetched the new value and we have displayed in this score after this also if this answer is right then the score will be incremented otherwise you should say that if answer is wrong. Hence print the some suitable message and say the correct answer it will be I would say better luck next time I thought the word that is stored in the variable picked word so I thought this particular word like this you are displaying the message to the user irrespective of what happens here the next has to be have to ask if he wants to continue the game or not? So that is for that let us use another variable let us call it c, c for continue probably c you take the input from the user if he wants to continue or not? So may be continue quit there is two possibilities so let me post a message press one to continue and zero to quit so he has to decide if he wants to continue or quit so this is the message post and for this the user will give an input which will be captured by the variable c and we have defined zero is for quit we have said to the user that zero is for quit so in case if he says zero that is I want to quit then input c is zero that is he wants to quit then you have to thank the user probably you can say you have to thank both of them actually because till some point have time they would have played so you have to thank both of them player one name, player two name you have to thank both of them and you have to display the points they have scored till now like before this particular word has occurred player two would have scored some point as well as player one would have got something you have to display the somebody. So unique the two points value has you have to tag them by greeting with their names and showing them the summary of the game till now so you have to thank them and you end the loop you quit the game, how do you do that? You use the break statement so once this break statement is encountered this break statement once when it is encountered the corresponding while loop would come to an end so that is the game will come to an end. This is the game ok so if this is for player one this particular thing whatever we are done till now is for player one next what should I do is for player two the procedure is just the same as whatever we have for the player one now the changes what we have to do is instead of player one name player one points we would be using the player two information there that's it so let me copy paste everything copied so let me paste it but indentation has to take care of it this has to be done in case if the turn value is odd so for odd turn value it is player two's turns so I will have to do this indentation so please take care of the indentation it is in the same column, column nine and this is in the column nine yeah this is in the same column and this thing has to be indented now everything looks perfect right? so we have indented this as I said this is not player one this is player two let me make the changes it is player two it is player two and thanking message is just the same whenever if anyone wants to quit the game comes to an end so you have to thank both of them for their participation till now as well as show them the summary so thanking part is the same just that updating of points defers from player one and player two because each of them have their own task so you need to have two different variables to keep track of both the right let me save till here ok let me revise once again what I have done this is for we start of with the playing of the game here we will ask for the names of the both the players, player one and player two initialise their points zero and to keep track of alternating the turns we use the variable called turn. Computer will create a question here instead of the usual way, where two people will create a question alternatively so this is just for sake because the with the case humans are playing one person has very good vocabulary and he is picking some very difficult words which is not very easy for an average person to guess, he is permuting that he is shuffling the word in a difficult way that may be the case and the other person may choose the simple words that may create the biased state as you could sense till now so to ensure fairness we will give the task of creating the question to the computer so the computer is doing this just guessing what the jumbled word is, is the task of the players so computer starts here this has to be repeated continuously until players want to quit the game so we are using this particular loop while one, computer is choosing the word from its list of words randomly it will choose some words and it will jumbled the picked word and create a question so it is now after creating the question it is displaying the question so you have to alternate the turns so I am using modular operator which would return the reminder after division so for odd numbers it is player two start  even number its player one's turn for odd numbers reminder would be one when divided by two and for even numbers that is zero divided by two that's that the taken care of here if turn mod two is equal to zero if it is a even turn then it should be the turn of player one because we are starting the turn with zero that is the usual practice in computer science to count from zero so that is why we are doing it, it's not an issue in case if you start from one as well you can in fact start from any value, the thing is you should alternate the turns between player one and player two as long as  you use this mod two as your conditions so this will say this players turn, this is the question has been displayed already what is that the word I have on my mind? That is the question post to the player; the player has to guess his word. So whatever is his he will type the guessed word so if that is the correct guess whatever he has answered is the exact word the computer has picked then the points will be increased by one otherwise there is no increase in points he will intimated of the word the computer has thought and then computer would ask them if you would like to continue the game or quit the game, if they would like to quit you have to greet them you have to thank them for their participation till now and show them the summary of both their scores and you quit the game, you are thanking them you are showing the summary and then you are quitting the game the same goes for player two as well just that updating the points is different see it is the point of the player one is updated and points of player two is updated in case if it is the, so in the turn of the corresponding player his points would be updated in case his answer is correct. That is how this game goes, did you sense any missing gap here , there is a missing gap here also there are some missing definition which we need to do, we would do it in the next videos, in the next video we will see the missing gap and will define and undefined terms now may be choose jumble all these are intuitive to us but the computer may not know what has to be done so that we need to define them and it is intuitive to us define these things, that all we will do it in the next video thanks for watching.

# PERMUTATIONS-JUMBLED WORDS 03

Alright, in the previous video you would have seen the outline of our play method so what is the game so we have seen the outline in the previous video may be we will give you a very quick summary we have two players each have been given zero pints initially and there is a variable turn to alternate their turns the computer would randomly pick the list a word from list of words it has jumbled it and  then post the question based on whose turn it is player is supposed to guess the word the computer has in its mind he has to give his answer if that is exactly matching with what I computer has thought his point will be increased by one else he would be hide of the correct answer there is no change in his points and the next thing is ask the player wants to continue the game or not even if one player says I want to quit the game the game comes to an end because this is a two player game so the game comes to an end so once the player says I want to quit you need to update them of status till now and thank them for their participation and then you quit the game this is what is the outline you have seen till now. I said there is some missing gaps in the precious video I hope you would have sensed the gap let us try fixing it if you would see here the turn value is initialise to zero then if it is even it is player one's turn odd is player two's turn but have I updated the value of turn I haven't so that should be done close the missing gap here observe the indentation it has to be exactly indented with this if turn mod two is equals to zero that is one turn is them you have to increment the value of turn you have to increase the turn is equal to turn plus one so you have incremented please observe the indentation this is in column nine as well as this is in column nine fine so indentation is fine now play method is over so I have what did does is as I had given the outline everything is perfect and while the players turn is over the value of turn is incremented by one so that the next player can take his turn if you would try running it and turn is equal to zero it is the player one turn to play the game then turn values is increased so now the new turn value is one so this is the turn of player two at two against it goes to player one at three it goes to player two so at every value of turn which is an even number player one gets a chance to play and that odd values player two get chance to play this is how the game goes but the programme is not yet complete still he first thing is here we have send choose the word so the computer has to choose the word from the list of words it has so we have define what exactly the computer has to do so will do that here let me say define choose this is something we want the computer to do basically it will have some lists of words so now let me start off with some random words some random English words let me say rainbow computer science, computer science basically then programming this may be some word then mathematics let us move away slightly from academics player this is a word condition this could be a word reverse some word and one two three four five six seven eight so may be let me have two more words let us make it a list of ten words you may have any number of words that's never a problem you can increase it to hundred words thousand words you can have if you have lot of words during the continuous playing of the game it will be fun because there is a lot for the computer there is a lot of challenge for the computer to choose a word then pick any word you can get any word it will be fun so let me say water this is a water and then we would say boat let me say boat these are the words I have just some random words I have given these are the list of words so from this list of words it has to randomly choose a word so for randomly choosing a built in library or random let me import it import random I have imported random library now I can what are all the functionality we find in the library so I would say random. Choice  I say randomly you need to choose from this list. So the default functionality for it is random.choice so in that random library there is a function for choice which will pick some element from your list so words is a list of our words so you have to pick something from list let me say pick equals random.choice this is a pre defined function and then the choice from the list of the words so from this list of words you pick the random word that is what just line means so after the word is picked just return this pick so now if you would see choose you say choose the control would come here it sees the list of words random.choice of words it sees this command so picks random words from this list and returns that value that returned value is taken as the picked word that is assigned a picked word so this is how the picked word things works now so we have faced this and then something which is not defined to the computer is jumbled let us do that now ok I have to define jumbled, jumble or a given word let me say, you have some word is given to you, you have to jumble that word that is nothing but the you are shuffling the word or you are rearranging the letters that’s what you are doing. So for this as well you have something in random library you have some functionality it is called random.sample so you choose some words from this word you choose some letters from this word how many letters you have to post? May be actually I need all these letters so length of this particular word so what do I do this? I need to choose some number of letters from this word. How many numbers of letters? Basically all the letters I need to choose in some random order, I need to choose all the letters in some random orders for example let us take one word water, w a t e r there are five letters and five positions blank positions for our shuffled word I need to choose some letter from these five letters so from the five letter word consider all the five letters but the thing is you pick any one letter in a random order so maybe, I may pick tea, so tea is assigned in the first position, in for second position from the remaining I will pick one letter may be or so on this is how random word would be generated and this would be written as a list but what we need it as, we need it as a word so list is nothing but it would written as for example water let me say t r a w e this is the way it has picked the letters t r a w e will be the five elements in your list. But we want to merge those five elements, those five characters and make it as a single word so we are basically what we want is we need to join those things so for doing this there is a functionality called join so maybe, I would say join these things and for this join functionality we will generally give a separator that is in case you have a space you terminate your join there something like that may be if I would have some words where there are some spaces you terminate there, terminate at a place where you need basically if you would have learnt about string tokenizer  this sounds as an complicated terminology that’s very simple nothing but if you are I will just give a simple example there is some sentence that has been given to you, you need to count how many words are there in the sentence how would you so that?  Count the number of spaces, the space acts as a separator between the words, so counting the number of spaces will give you a hint on how many words are there in the sentence so something like that so space acts as a separator so some separator has to be given based on which joining as to occur that is when as soon as you see the separator you perform this operation, perform the join operation, here I don’t want to see any specific separator may be space or a comma or a pull stop on seeing that only I want to perform this operation I don't want to something like that so let me just give the blank character this blank character.join that is don't see anything just keep joining the letters so t r a v a w e is the order I just randomly said so just keep joining them so just take t again just keep joining r take the next letter a join take the next letter w join take the next letter e join and this is how this particular functionality works. So this is basically separator, in case if I give some other separator what could happen is only one seeing that particular separation that separator this particular operation would be static I don't want that, I want that to be happening irrespective of whatever be the character so I just give a blank character this is a syntax that has been define so we are following it so let me say this is my jumbled word so after every individual character as been joined I get back to the word let me call that word has the jumbled word alright so once jumbled word is time you return this jumbled word, so this jumbled word is nothing but your question, so when you say question jumble of pick word the word is taken the letter of the word are shuffled that is the letters of the word are separated and they are randomly picked and you could get a list, you join the element of the list to get back a single word and you return that single word this is how this functionality works. And something else have we defined everything, just check once, we are getting the name points and turn choose has been just now defined then jumbled we just now have defined and now if we have something here we don't have to defined yeah thank yeah thank here is needs to be defined, fine so let's do that as well let's do that. Define thank, I have passed player one name player two name player one's points and player two's point I have passed these four values so what should I do? I have to print player one name I have to print his name and say your score is, player ones score I have to display and will have to do the same thing for player two as well let me copy paste this sorry let me copy let me copy and paste it I just need to do some modification so that this is player two's name and player two's score has to be and you have to thank them for their participation till now, maybe I would say thanks for playing ok then I will have to print let me just say have a nice day this is purely your wish whatever you need to display is nearly your wish I will just print have a nice day. Sorry I need to give a bracket. Everything looks good let summarise we have define the play functionality we input the names of two players initially have their points at zero there is a variable turn to track of whose turn it is and the turn is alternative that is taken care by this mod two operator the task of creating the question is given to the computer, the computer will choose from its list of words, jumbled the letters of the word then display the question so based on the question and the turn the player will be given a chance to guess what the actual word is. So the guess will be taken and it will be compare whether the answer he has given and the actual word is the same, in case if it is the same his points will be increased by one and it is updated here. He is updated of his new score here, and then in case if his answer is wrong you just intimate the correct answer and don't update the score in here and you will be giving him the chance whether he wants to continue the game or he wants to quit. This is the two player game so if one person wants to quit the game comes to an end in case if they choose to quit you have to greet them thank them show them the summary of the game till now and then you quit the game, each and every functionality for that choosing random word that random library we had used so from the list of words randomly you pick a word and return it so this is what the choose functionality does and jumbling is nothing but, you randomly select the letter from the word you keep it in some order and then you list the merge of those letters in different order what you have, what you have at time you merge you join the letters that is what you are doing here and you return the jumbled words and in thanking this is purely your creativity and this you are updating him of the status till now you are giving the summary of the game he has played till now, so you are giving the summary and you greet him with some nice lines basically this is what you are doing and see there is something that we missed here it's your score you need to display the players true score stat playing this game. Let me run, play this alright, player one let me enter your name let me say abc player two xyz some where it is coding ok this is wrong let me continue this is computer ok score is one, let me continue this is science computer science ok so as you could see the game is very interesting you can go on and on the thing that makes you difficult is when the number of letters increase as you could see here this was difficult for me to guess because there were so many letters here and as the number of letters increase the complexity so you try  playing  by long words in English you try doing it and there may be some variations in the game as well you can limit the number of guesses here I am just telling I am giving one chance if it is correct I will increase the point else I will say you are wrong may be you can have a variant in which you would give the player more chances to play the game he can take some three guesses something like that the creativity part is all in your hands so creativity sky is the limit please do try different variants and do discuss in the discussion form of how you have implemented those variants thanks for watching this video have a nice day.

# THEORY OF EVOLUTION 01

Hey avni, how fascinating it is we have a wall power thing cellular body to this form yeah but this is not a single time process you know it takes millions of years to change into this current form from a single cellular form and a single cell has a collective sticks with changes with time, brilliant yeah and the recent studies have shown that the single segment of dna contains a huge amount of information, nice, I would like to stimulate this evolution process through competition. Let us try. You saw the conversation between avni and simran, avni is trying to explain to simran that there is some sort of randomness involved in evolution of species from one to the other we all know that life started off as amoeba and now we are as Homo sapiens humans. The randomness attached to this so called explanation of what intense evolution can actually be stimulated the exact model is little complicated so what will do is we will try stimulating it at a very different level well yes we can stimulate and see how evolution has happened in fact there is a very popular topic called genetic algorithms just precisely this it learns from evolution or how amoeba became human being and that very perches one can mimic and used that as a sort of technology to come out with brand new ways of accomplishing something. So what we will do is genetic algorithms say is way out of the scope of this course I will try to tell you something which is a very very very tone down version of genetic algorithms in fact biology's and people who actually know genetic algorithms might even complaining by saying what am I teaching you people but I am teaching this in keeping in mind of the brand beginners in programming who may not have any idea of programming, either programming nor biology. But still for its a very interesting topic to just like that discuss and then write a piece of code. So what we will do is we take a huge binary strength let say some twenty digit binary strength all of them are zeros the rule is simple, you pick some digit randomly and make zero for one with a same probability you know how to do this you now are familiar with the random function right you say you import random and then you can use a lot of functions within that you are very familiar with tat, using that you can do this the point is you take some digit in this twenty digit string and in case that digit is a zero you make it one with the small probability keep doing it and eventually you will see all the zeros will become ones. How much time does it take? It's based on the length of the string and the probability which you switch zeros to ones sounds complicated but don't worry the programme is actually very easy once you understand this you can in fact go ahead and study what makes the genetic algorithm but again as a asset out of the scope of our joy of computing course which is at the very foundation basic level but you will have all the interest on earth to go and study and even understand what makes genetic algorithms, so let us now go ahead and see screen cast of how to execute this program.

# THEORY OF EVOLUTION 02

Hey everyone, as you all say I just had a conversation with my friend avni and I am really fascinated by this idea of evolution. Evolution is actually the change in inheritable characteristics over successive generation and how this change happens? This change happens through genes, genes have dna and dna have coding information in the form of beds in the form of zeros and ones and if this information changes if zero becomes one or one become zero it actually needs to evolution so I will like to code this particular fact but before jumping to the programming screen cast of evolution let me give you some prerequisites, first of all we will be looking at the concept of file handling in python how can you read and write files in python? We will be looking at txt file the files with dot text extensions so let us start with that. So here I have a file named file one dot txt you can create your own file. So here is my file I have written hello, how are you in it as you can see hello how are you has been written here I need to read this file and write something in it so how can you do that? How can you realise through python? So first of all you have to open this file so let us do that, we will just write with open your particular file name, my file name is file one dot txt and a signer name to it for example I will add a sign my file and I have to read that file so I will use a function read here I will just print what has been written in the file so I will just write file one dot no you have named it my file dot read let us do that see, it has read the file. We wrote hello how are you and it is the reading the file as hello how are you, so it is working for example now I write, we want to write I am fine in the file so how can you write? I just use the write function here for reading the read function and then the same way I will use the write function here my file dot write for example I want to write I am fine, ok after that please not the fact that you have to close the file so I will just close the file my file dot close, close function is used to close the function here so let us run it there is some error here that is unsupported operation that is not writeable yes this file is not writeable so I will just write in w mode there are different modes in which you can open the files, r is used for reading mode and w is used for writing mode. So I need to write the file so I just use w mode so let us run it again now the exception is now the error is not readable so we need to read and write the file simultaneously so there is another mode that is r plus mode there are various modes in which you can open the file, I will use r plus mode r plus mode opens the file read as well as write please note this fact so I will run it again. So here we have I am fine now if you read the file you have I am fine in this so let me check that, our file was file one dot txt let me open it again. So since we executed it twice we have I am fine I am fine again in this so let me show you by zooming it we have I am fine I am fine twice since we ran it twice, so this is how we can read and write files here so the error we encountered here was that we have to open the file in particular modes so if we want to open the file in reading mode then you can only read. If you can want to open the file in w mode then you can only write and if it is R plus then you can read as well as write. So there are many such modes in which you can open the file let us explore that. For example here we have different modes so here we have different modes we have R mode in which read mode is used, W in which write mode is used to edit and write new information to the file and we also have A mode that is appending mode, you observe this fact when I wrote I am fine in the file hello how are you, it got erased off so if we I opened the file in A mode that is in append mode then hello how are you would stay in the file and I am fine would be appended in the file and next is R plus that we use just use it is use to read and write the files and yes special read and write mode which is used to handle both the actions when working with the file. So there are four modes in which you can open the file first is R that is read mode next is W for write mode third is appending mode in which you can append the data in which you can append means add the data to the file ok by not erasing the precious data and the fourth one is the R plus mode that is the special mode if you want to read and write the file simultaneously. In this way you can do the file handling in python,

# THEORY OF EVOLUTION 03

There is another prerequisite that is required for evolution programming screen cast and that is random library. Random library is basically used to generate random numbers, I will explain you some of the functions in random library, so first of all start with rand in I will just write random dot ranint one to five. So as you can see it has generated an integer from one to five rand int is basically is used to generate a random integer from one to five please note the fact that one to five one and five are includes here it may generate one two three four or five let us run it again and see the randomness involved here now it has generated four again, again it has generated two again four five so as you can see that five is inclusive here that's why it is generating five. Again five four one so basically it has generated every number except three let us run it again nor it has generated three also so random dot ranint is basically used to generate the random integer in the given range, in which the both the range are inclusive I use another function here random dot range please specify the range here I will again specify one to five so let us look at the output of random dot range, sorry its random dot ran range so let us do that this is four one three one two two four three as you can see that it has generated five so ran range basically generated number from one two upper limit minus one it will generate number it will either generate number one two three or four it won't generate five because it generated number from lower limit to upper limit minus one. This is how ran range works. There is another function name random dot random it will generate number from zero to one, it will basically generate decimal numbers from zero to one so let us run it. As you can see that it has generated point four five it will if we run it again it has generated point four zero, point eight three, point five three, point six two, point four seven, point nine two, point six two, point seven three, point eight eight, point eight three, point five two, point zero five four as you can see that every decimal number that has been generated here is from zero to one so random for random is basically used to generate random numbers random decimal numbers from zero to one. There is another thing involved in random dot ranrange let us explore that. And that is for example I want to generate numbers from one to nine so I will write here one comma ten and I want to take a step of two I only want to generate for example odd numbers so I will write two here it will either generate one three five seven or nine let us look at this for example it has generated nine, let us check whether it generates only odd numbers or not one three one three again odd number again odd number odd number odd number odd number again odd number odd number odd number as you can see it is only generating odd numbers so how you can make random dot randrange work like that, you just need to specify the range and also you also need to specify the step that random dot randrange has to take for example here I specify two because if it will generate one then with the step of two it can only generate random odd numbers so if you want to generate random even numbers let us do that too for that I have to start with two then it will generate only even numbers and I will generate till eleven because ten is also odd number even number so let us run it so it has generated eight ten even number even even even even so as you can see that it is only generating even numbers that's how you can make random dot randrange to generate even and odd numbers and many others and many other numbers of your choice this is how it works. There are many functions involved in random library you can always Google it and explore many other functions so just I will give you brief of that. So here we have many functions involved. Will have random dot choices we have random dot set state get state randrange choice shuffle sample, random dot uniform, random dot triangular, random dot beta I will suggest you to go through all these random library functions they are pretty interesting I will give you a demonstration of another function that is random dot choice let us do that, random dot choice basically generates random numbers from given list so I specify the list here for example I specify one two three four so it can only generated numbers from this particular list that I have specified here so it has generated three one four two as you can see it can only generate number from the particular list, you can also specify the list for example I will specify the list is my list is equal to any number you can specify one one, forty five, sixty seven, eighty nine ok so random dot choice you have to generate from my list so let us try executing it. So it has generated eighty nine since we gave only four numbers eleven, forty five, sixty seven, eighty nine it has generated eighty nine run it again it has generated eleven, eleven again forty five, forty five, eighty nine, eighty nine, forty five as you can see that it is generating number from the list that is specified, we have specified the list has eleven, forty five, sixty seven, eighty nine and it is only generating numbers only from that. So I will encourage you to explore more and more about random libraries it has many other instructing functions too.

# THEORY OF EVOLUTION 04

Now, we are done with the prerequisites required for to run screen cast of evolution. So let us start with the programme of evolution. Here we are given a dna file name dna underscore data has explained earlier every evolution process happens through dna and it has some coded information in the form of zeros and ones so we are this particular information zeros and ones so first of all we have to read this file that has the coded information I hope that we have watched the previous programming screen cast in which we explained how to handle files through python. So I will be using that straight away, so I will just write with open my file name is DNA underscore txt DNA underscore data dot txt as my file. So I will just read it, I read it bit by bit and store it in ‘x’ so I just write my file dot read as explained read is used to read the file, read function is used to read the file here I will also make x as a list so that we can use it as a list. We have a list of zeros and ones we need to store it in list so I will just convert it into list. We have ten thousand beds given in that DNA file so I will call it function on this ten thousand bits so I will just write for I in range zero to ten thousand, I will call a function evolve here evolve ‘X’. Now that we are done with reading the file we have also created a list x that stores the DNA information I will just call the function evolve on that particular list and that is ‘x’ so I will just define it, define evolve this is called on the function this is called on the list ‘x’ so I will just write index here I want to generate an index randomly in the list ‘x’ I repeat I want to generate an index randomly in the list ‘x’ so to generate randomly we use as I explained random library so I will just write random dot randint we have to generate  from zero to length of the ‘X’ as explained in random dot randint we have to specify the range and for that range the both limits the upper limit and lower limit they both are inclusive. Then I also need to generate another variable randomly here I will call it ‘P’, I will again use a random int function here random dot randint and I too I need to generate it from one to hundred so algorithm that we are using here is if this particular variable ‘P’ if it will be equal to one then only the change will takes place so I will just write here if ‘P’ is equal to is equal to one then only the change will take place if the bed at index position at ind is equal to is equal to zero then it will be changed to one is equal to is equal to one. Else it will be changed to zero obviously, so I will explain again what we are doing here is we are generating two variables randomly here first is the index of the variable of the particular bed in which change has to take place this change actually depends on the variable ‘P’ which will which will generate randomly if this particular variable ‘P’ will be equal to one then only the change at this particular index ind will takes place if ‘X’ at int is equal to is equal to zero then it will be change to one otherwise it will be changed to zero so first of all it all depends on P. If the P that has been generated randomly here is equal to one then only the particular bed at Ind position in ‘X’ will be changed the change will takes place. This is how we are using we are using the evolve function here. So now we are done with reading the file we are also done with the defining the evolve function here so I will just print here again I will just print ‘X’ again so let me do that let me run it again so there is some error that is list index out of range so I will just look into it, we have ind here and we are generating it from zero to length of ‘X’ so here is the problem we generating till length of the ‘X’ we should generate it from length of ‘X minus one’ since we all know that randrange generates the random value in which the both ints of the range are inclusive the upper limit and the lower limit are inclusive and we have in list the numbers that are generated the indices values are always from zero to length minus one I repeat in list we always have indices values from zero to length minus one ok. If we have given a list of length five then we have indices in list as zero one two three or four we don’t have five here and random dot randed generates random values in which both the ints of range are inclusive so we have to write length as ‘X minus one’ in this particular case the problem will be solved so let us run it again so we have an array here again the array the list that we read from that particular file it has come out here so you can actually print the ‘P’ and ind values and check whether the changes taken palace or not so I will just print the ‘P’ value and see whether the changes are taking part or not so let me print it so it is generating ten thousand times so let me see whether it is whether the value is one at the particular instance  are not so let us look at it we have seventy eight, eighty five, eighty one till now we have a I haven’t encountered one oh here we have one so in this particular case we have one so this particular case the change would have taken place ok you can always try run it and check whether the change is taking place or not we have one again so in this particular case the change will take place otherwise the change will not take place so please note down the fact in mind so I don’t think we have another one here let us check it again so you can try run this particular algorithm this particular programme and see whether the change is taking place or not I hope this programming screen cast is useful to you guys have a nice day happy leaning.

# PRACTICE IS THE KEY

In the last week we presented four cool ideas involving rigorous programming, well it is said that programming is a skill which is best acquired by practice. So keep on practicing, we are there with you in this journey of programming.

# MAGIC SQUARE: HIT AND TRIAL 01

Amit, do you see there are nine squares here, yes sir. So you know how pseudoko works? Yes sir I know. Right! This is very similar to pseudoko but you don't have to know pseudoko also I am just trying to see if you if I can motivate it through something to know already. Here are nine squares one two three four five six seven eight nine can you plug in the nine numbers one two three four five six seven eight and nine here so that each side adds up to the same number and each row and column everything whatever whichever way you add should add up to the same number. Ok you were saying that you want some programming challenge right? Yeah, yeah. I thought I would start giving you something like this. Go ahead try, so I have to choose the numbers from this set that's right. No repetitions no repetitions allowed yes. The sum should be equal to one particular number right? Correct. The rows all the columns, correct. So this you would be surprised to know that this is this is more than two thousand years old. It was invented it was first noted in china and there is very interesting history you can look up on the internet but the best part is three cross three is very interesting as you increase the dimension it gets more interesting I will tell you something a lot more interesting once you solve this go ahead. Firstly do you think it is easy to solve? I think very easy. Very easy? Go ahead. Ok let me use this number yes two six correct so which would be seven eight fifteen adds up to fifteen so if you right some numbers here even that should add up to fifteen go ahead. Six I have used seven I have used two I have used may be you are yeah write something nine then eleven which means should be a four so what happens here two plus nine is eleven plus four is fifteen, two plus six is eight, eight plus seven is fifteen prefect! But then you should put some numbers here so that this adds to fifteen, this adds to fifteen not just that this adds to fifteen, this adds to fifteen oh! I forgot to tell you even the diagonal should add to fifteen ok go ahead. This is I used nine, four. Ok do one thing you go ahead and try I will take a break and come back sure sir, once you are done let me know, ok got it perfect. Hey amit you have done it, I have been watching you yes perfect, perfect, yes it's a four plus three plus eight which is fifteen perfect. You can write fifteen properly here and then fifteen here nine plus five plus one fifteen, nine plus six is fifteen ok remove this and yeah fifteen here fifteen here eight plus one plus six, fifteen seven plus five plus three fifteen, our boy has done it, ok four plus nine plus two fifteen, fifteen look at this nine plus six fifteen, eight plus two plus five plus eight is fifteen perfect. I will ask you another question now this is just one way of doing it, there could be many ways to do this or I don't know this may be the only way can you tell me if there are many ways of doing this, there can be multiples solution also yeah why not, oh my god it took me a lot of time to get this solution also ha ha ha so give me a chalk so what you can possibly do is you can try writing a piece of code as always right? Try to see whatever the total possible ways, how will you go about this? See I solve this in the brute force way trial and error you mean trial and error correct? I change some I put some number ok, check what's the sum is going on right? With iteration I found this correct? For this I can write a code definitely and computer can do it in fraction of seconds, so I will write a code same brute force way. Perfect. But it will give me solutions, all the solutions, try doing it. Definitely. I will tell you something more two things one is one dimension is lower by that I mean, let me look at two cross two board which should be something like this so can you tell me what would be the version of this problem here, for two cross two there were nine squares here so you considered one two three four five six seven eight nine. For four squares what would you do? One two three four I will considered. Can you quickly try if this is possible with one two three four I am going to give you just two minutes time. Go ahead something you will observe one two three four. Ok. Basically one should be there somewhere for sure definitely, ok. Diagonal should be same as this one three may be, do one thing let me write it in bold so you say one here and a three here some should be four this should also be four but you have two and four left this become six, you see you can try all possibilities you will observe and it is impossible to do it in two cross two, exactly  while it is not possible to do it in two cross two, it is possible in three cross three and surprisingly for four cross four it's so difficult that your computer is not enough, ordinary desktop is not enough to find a solution for four cross four guess why? Think about it, write a piece of code and may be you try explaining why it is not possible for four cross four. Sure yeah. Good job.

# MAGIC SQUARE: HIT AND TRIAL 02

Hello everyone, today we are going to write a programme for ‘magic square’. In the video you saw amit was trying to solve magic square of size three cross three and was taking a lot of time because he was trying to solve his solve the magic square using brute force way. Which is possible for three cross three square matrix but not possible for let’s say five cross five and seven cross seven and even nine cross nine. So we will see that how we can write a programme so that we can solve any kind of magic square in an efficient way, or in an easy way. So we know that in three cross three magic square the sum is actually fifteen so let me revise the rules of magic squares. What you have to do, you have to create let say I give you a number ‘N’ you have to create a square of size ‘N’ cross ‘N’ such that some of each row is m is my magic number, some of each row should be ‘M’, sum of each column should be ‘M’ and even the diagonals, sum of the diagonals should be ‘M’. If it happens then you got the magic square, the question is, is it possible to create a magic square for any number N? Let see if I give ‘N’ is equal to three. So my matrix should become three cross three that is it should have exactly nine elements, ‘N’ sum of each row should be some ‘M’ the ‘M’ could be anything. But it should be unique I mean the sum of each row should be ‘M’, each column should be ‘M’ and the diagonal should be ‘M’ so we will see how we can do it, how we can create a programme in python so it can solve any kind of magic square. So will tell you that this, the size of I mean ‘N’ can take any number but it should be odd it should not be even and we will write a programme for that. Before that I will encourage you guys to look at the Wikipedia page of magic square it has all the entries regarding magic square and even history of magic square and even the algorithm to solve the magic square I am going to use this algorithm only. To solve to write a programme in python so before I go writing a programme let me tell you some facts ok see you can see here magic square size three and magic square of size five we know that what is the magic square of size three that is two seven six nine five one four three eight if you sum it then each row is becoming fifteen, six plus seven plus two is fifteen, nine plus five plus one is fifteen and even in diagonals sum both the diagonals sum is fifteen. Here is another magic square of size five you can find the sum of each row and each column it will be sixty five so before we go on writing a programme let me tell you some fact. Fact number one is that for any magic square, for any cross any magic square the magic number is always this which is ‘n’ into n square plus one whole divided by two so let say for three cross three magic square n becomes three so square of three is nine plus one is ten divide by two five into three is fifteen so magic numbers become fifteen and for magic square of size five that is ‘n’ is five, five square is twenty five, twenty six t which is fifty five oh sorry sixty five sorry. Yeah, so we are going to use this fact another thing is that whenever you see the, whenever you see the magic square we will find a pattern here, the pattern is that see the position of one, the position of one is always in the middle a row and last column ok whatever magic square you use, you can even see the magic square of size seven and also you will find in that a middle row. The one is always in the middle row and the last column. So we are going to based on that we are going to use this algorithm we are going to define the algorithm when we are going to use this algorithm to write a program so let’s see this. The step number one is first we have to determine the location of one, so will fix one after that will go on go on to other numbers and will fix their positions and by this will try fitting our numbers and will get our magic square. So first is we know the as I told you in any magic square one is located at the position n by two comma n minus one for any general n cross n matrix square matrix n by two will give me the position of row ok, so for example for three n by two becomes one  I am taking the integer number so zero one first row is zero and second row is one third row is two so one is this position and n minus one is the column position which is two so second column so am I position of one is this which is first row and second column. For magic square of size five becomes five by two which is two second row zero first zero one two second row and the last column that is n minus one which is four fourth column we are assuming that always we always assume here that the matrix starts with zero cross zero, zero comma zero sorry. Ok now after determining the position of one will see where, where two lies now see here the position of two here also there is a pattern for two also you can see two is here when one is here again the two is here so let's say the position of I take this in the position of one which is n by two comma n minus one as 'p' comma 'q' lets say the position of one is 'p' comma 'q' this is 'p' comma 'q' this is p this is q then next number which is two is located at p minus one comma q plus one oh see this one is at two comma four now if you take p minus one which is two minus one is one and q plus one 'q' is here four, four plus one becomes five now five  is not there what is this mean? So let me tell you the second line anytime if the calculated row becomes minus one then make it n minus one and if column position becomes n then make it zero. Here, column position is becoming 'n' which is five, five is n so you have to make it zero so it will come zero. So now see my location is p minus one which was two oh sorry one and 'q' plus one which was zero so two zero oh one zero I am sorry so new location is one zero so I will put my two here so like this I will do for calculate for other number also now after this I will calculate for three again using the same thing now the third step is not the step the third fact is if the calculated position always contains the number suppose calculated position I mean already contains a number then decrement the column position by two and increment the row position by one. We will see this how it happens. Now the last fact is if any time the row position becomes minus one and column position becomes n switch it to zero comma n minus two. So I am going to use these three conditions or facts or algorithm say it is not algorithm because I have denitrify the steps these are not steps these are facts only so I am going to use these facts as a condition to write my programme and based on this I will create my magic square so in next video we will see how you can use these steps to create magic square.

# MAGIC SQUARE: HIT AND TRIAL 03

So in last video, we saw the conditions which we can follow to find our magic square of any size ‘n’, precisely ‘n’ should be odd. Let me help you with an example to solve, to use these steps to find the magic square of size for us three cross three matrix. Now here I have a three, I created a three cross three matrix here ‘x’ represents the elements which I will write as I will follow the algorithm and these are the locations of these elements. For example the matrix starts with zero comma zero which is the location of this element is zero comma zero because this is the first row which is zero is been told you the always the first row and first column will be zero comma zero please follow this convention, location of this element is zero comma one, location of this element is zero comma two similarly location of this element is one comma zero because this is the first row location of this element is one comma one, location of this element in one comma two, location of this element is two comma zero, location of this element is two comma one and location of this element is two comma two. Now let me go to the condition the first step was to find the location of one we know that one is already always located at the middle row and last column so we find that. See in this example the size of the value of ‘n’ is three because this is the three comma three matrix and we want to create the magic square of size three cross three. So yeah so let’s see the first step which is determining the position of one, since ‘n’ is three so the location of one is ‘n’ by two comma ‘n minus one’ which is three by two comma three minus one, three by two becomes one integer division and three minus one becomes two. So position of one is one comma two, so I will put one here awesome, now the next step is to determine the position of two, so for that what we have to do, will take the last determined location, last determined location is one comma two is ‘p’ comma ‘q which becomes ‘p’ this becomes ‘q’ and will find the next position, position of next element as p minus one comma ‘q plus one’ so position of two becomes ‘p minus one’ which is one minus one and q plus one which is two plus one so it becomes zero comma three ok. Now any time, any time is calculated column positions columns position becomes n see here the column position is becoming n, n is three. So make it zero so I will make it, making it zero it becomes zero comma zero. So I will put two at zero comma zero, two I will put, I will put zero comma zero ok great so now the next element is three I have to determine the position of three so I will take last calculated position p comma q this is p this is q and again I will follow the same thing. So the position of three becomes ‘p minus one’ and ‘q plus on’, ‘p minus one’ is zero minus one and ‘q plus one’ is zero plus one it becomes, what does it become? You should write it here, it becomes minus one comma one again I will say the same thing any time the calculated row position becomes minus one here row position becomes minus one make it ‘n minus one’ so ‘n minus one’ means three minus one, so three minus one is two this is as it is two comma one so my position of three is two comma one, let me see where is two comma one, two comma one is here so here I will write three great again position of four same algorithm two one as pq two minus one is one, one plus one is two this I simple one comma two, one comma two I will now see at one comma two one is already present so what we have to do? If the calculated position already contains the number then decrement the column by two and increment the row by one so what I will do, decrement the column, column position was two determine position was two, so decrement the column position by two which becomes two minus two is zero and increment the row by one, one plus one becomes two at two comma zero I will put four great, position of five becomes this is pq two minus one, zero comma one so one comma one I will put five here, position of six one minus one, one plus one two zero comma two so I write zero comma two is six, position of seven becomes what is zero minus one into plus one, now this is little bit tricky. See whenever we saw here see here that zero minus one is becoming minus one will see this condition that any time if calculated row position become minus one then make it n minus one we have to check here both the condition is if position of row become is becoming minus one and the column is becoming n which is three here then you have to follow this condition that is the fourth one, if anytime row position is becoming minus one and, and is very important here column becomes n which is to zero comma n minus two so I will make it to zero comma n minus two three minus two is one so zero comma, at zero comma one I will put seven ok. now next is eight, eight is that so zero comma one is my pq so zero minus one and one plus one so it becomes minus one and two so I will use this condition only that is row is becoming minus one and I will make it n minus one, n minus one is three minus one which is two so it becomes two comma two, so two comma two I will put, eight and the last is nine so which two comma two is my pq so two minus one becomes one and two plus one becomes three so two minus one I don't have any problem with this the row, the column is becoming three so I will wrap it up so it becomes three becomes zero becomes one comma zero so I will get one comma zero. As you see this was the exactly these were the elements of magic square, you can see here I got the magic square you can find the diagonal some of diagonal and row and column it is exactly fifty nine, eight plus one plus six is fifteen, nine plus one plus five is fifteen. So this is the algorithm we are going to use pretty simple. Giving me this algorithm I can definitely say that you can write a programme, you can use the steps and I will encourage you guys to write a programme by your own and you can ask any question in discussion form, in the next video I will try to code this, write a programme using this algorithm and we will create magic square of this size. If you think you can find another pattern and you can write so another algorithm please, please, please post it on discussion form, it will be very good for us and for other students also. Thank you.

# MAGIC SQUARE: HIT AND TRIAL 04

Ok, now we are going to write a programme for matrix square, so what the programme should do? Should take an input n and is the number of elements in magic square since it is a square matrix so the number of elements will become n cross n it will take n and generates the magic square. Now in last video we saw the condition which you can use, we saw the algorithm which we can use to create the magic square, so will use that algorithm to write a programme. So let's start. As I told you it is better to write a whatever programme you are creating whatever logic you are creating in a function so that you can use that function many time rather than just writing in python file its better if you create a function so I will create a function let's def magic square ok, it will take an argument that the numbers of rows or number of columns which is same for a square matrix in magic square. Ok in python I need to define the matrix because the element should be present in the matrix then I can change the elements accordingly so I will define a matrix of size n cross n with initially initial all the elements will be zero all zero so matrix will contain all zero and then as we go as we go down to the algorithm step by step will change the elements from to one from zero to other elements since let's see if there is a three cross three square matrix so elements will be from one to nine. For other it could be different but zero will never come because it will always start from one, so ok let me define the matrix magic square lets see. so for that I need two loops I will write here for I in range three it means it will go from zero to two, zero one two then for j in range again three ok, before that let me define an define a variable named magic square magic square so it is a list I am using it as a matrix inside this first loop I will define a simple list and which I am not keeping anything there which means whenever this loop is, whenever I go inside this loop it will create a new variable, a new list and as soon as this loop gets over it will again go to this loop go inside this and again create a new list l so what I will do inside this j loop ok I will enter the element in l so I will write l dot append, append is to enter the elements in the list so I will append zero so what will happen? See in the first run when the I is zero ok so I become zero so this n is created after this will go inside this loop j loop here j is again becoming zero so for this I am going l dot append zero so first time when j is zero l has element l will contain l will contain zero then it will go back to j, j becomes one j in range only so it will again append zero so l will contain zero comma zero. Again it will go back to loop this j becomes two, two is within the range so again l dot append will run so it will become zero so l will contain zero comma zero comma zero next time j becomes three, three is not in range because three is out of index of range three so this loop will break so it will go to i, now I becomes one and I am not doing anything it so as soon as this before ending the I loop I have to append this l which l I got that is zero comma zero comma zero in this magic square so what I will do after this loop but inside the I loop I will magic square dot append l so what it will do as soon as the j loop gets over the l whatever value I have in l after the j loop I will apply it in the magic square. It will become a first loop similarly when the second run of ith loop again I will create a second row of using the j loop and I will append it in here magic square ok so the magic square will contain a contain the matrix containing zero only zero so I can print this so in order to print this I will just copy this ok you can simply comment this using the yeah you can simply comment this. So what I will do I will print here print magic square I comma j ok let me delete this ok lets run this. So nothing is there because I have to call this function so magic square n is what in range I have to put n because we are making it generalise programme not for three so for n I will put three here and it is showing like this zero, zero, zero, zero, zero, zero, zero, zero and I want it to be in a matrix only so for that in python three what you can do, you can write comma end is equal to like this it will join the I mean print statement always prints one and then by default it keeps new line character till the end so I will put this it will create, it will print the whole everything I this loop in one line so in this I will write magic square three ok so what I am getting here I am getting everything so I just write a print statement here print and I am not print anything great, so what's this print statement does it print everything in one line that's why I am getting everything in one line but after this loop if I just print nothing, nothing means new line character is always there so after this loop after the first loop it will print a new line character and next I will get the next row I mean in next print I will get the next row so this is how so our matrix is good everything is fine perfect I am getting my three cross three matrix now this is the way this is the conventional way of creating the matrix python is a very cool way of creating such things, the best thing about python is that you can write loop inside a list also suppose I can write magic is equal to this is the matrix I want to create at inside this I will write the loop for I in range three it means for I in range three and I will put here zero it means it means for I in range three put zero in this list so it will create, it will create a list of zero three zeros in a list because I in range put three times I have to put zero in this list now same thing I will use like this for I in range put zero and put this list three times, I want this list three times, three times like this so that I can get my magic square for that put this list for j in range three it means that for j in range three put this list inside this list awesome I got my magic square. Great so you can try this I have already written this loop, these loop for creating my simple square matrix of zero elements.

# MAGIC SQUARE: HIT AND TRIAL 05

Ok, the first step was to find the location of one so we know that the location of one is always the middle row and the last column so we I will fix it let me put I is equal to n by two and j is equal to n minus one, you can guess what these are these are the local variable and j are not have been defined but as soon as they use this warnings will go away so I will not care much of this ok I will create, I will create a variable name num which I will initialise to n cross n so that I can use it again, I may not have to calculate it every time I will take count variable initialise to one, count variable is my elements so as soon as I put elements I will increment the count variable so that I can put the next element so first I have to put one then as soon as I put one I will increment the count to two so that I can count again some location ok now I need to follow some condition multiple times, multiple times means what? Until and unless I do not exhaust all the elements, all the possibilities of this matrix which for example three cross three it is nine, it means that I have to check the condition for nine times exactly nine times so I will write a while, while this count is less than equal to num so what this is mean? Until and unless I have put all the all my elements into their location my count will not go to for example three cross three the count is count becomes more than nine that is ten I have to stop here because I have put all my elements to their respectively locations so I don't have to follow the conditions again so that's what this is the condition which I am going to follow the while loop then let see this as I told you the algorithm that this whenever I find the position, the position of one is fixed ok whenever I find a position I have to check two things first is if row is exceeding row is becoming minus then I have to make it zero or n minus one sorry this column is becoming n then I have to make it zero but before that the important thing is if row is becoming minus one and, and, and this is or there is a difference between and, and or, and column becomes n then you have to switch it to zero comma n minus two, this is very important because this is the and condition so it should come first so I will write this so I will write if I becomes minus one which is i's and j becomes n then what is this? I have to follow you write it here condition three condition four yeah ok what is the condition? I have to make j is equal to n minus two, column is becoming n minus two and row is becoming n minus one so j comes n minus two yeah and row becomes zero sorry so I becomes zero this is correct ok else if and condition is not followed it means either of, either this is happening or this is happening let see if j is becoming n column is exceeding so column value is exceeding when this is happening what what you have to do is if column is if column position becomes n then make it zero ok so I will make j equal to zero so or so this is a or, or condition so I don't have to write else here because I have to check both things, for any one of this if after checking this I will check this also not only this or this. I become less than zero then I is equal to n minus one which is this condition is what row is becoming minus one great, now these conditions are over, now what else. One more condition is there if I calculate the location but that location already contains some element then I have to something for that I have to check if my magic square I don't know why I find even though writing magic square ok if magic square location which is I comma j is not equal to zero if it is something else then zero it means it already here something ok then I have to put j is equal to let me check the condition what was it? If the calculated position already contains a number then decrement the column by two and increment the row nu minus one, so will decrement the column by two and j minus two and I is equal to I plus one and after this it might happen that after calculating after doing this the sum of these conditions might follow that either this is happening that is the row is exceeding and column is also row is going to minus one and column is exceeding only column is exceeding only row is exceeding so I have to check that for that I have to go to the go back to the loop now see this in all this condition I did nothing with the column values, column values not incremented so if anything is happening with this, I will check the condition but count will, will not increase so I can do this as many times I as I can so, so for that I need to check this conditions again so for that I will just write continue what does this mean that? After doing this it will skip all the rest of these things which is inside the loop and go back to the while loop and do the things again since the count value is not incremented yet so it means that I am doing the doing everything again without any condition again great so this is done I have checked it else this means what if it doesn't contain any value it means that it is empty it means it contain zero I can put I can safely put my element for which I am calculating the position for that I will write magic square is equal to count value because count is my value and I will put count plus one which means count is equal to count plus one increment the count value which is my next element this is done everything is done now whenever I put an element to its location I have to switch to the second element and I have to follow the p q wala thing so for now whatever location, whatever location I have calculated for this element and take it as p and q and I will follow the p minus one and q minus one q plus one so p minus one means I become I minus one and j becomes j plus one this is it so this was what? Write it here condition one great we are done with this, this is over. Print the elements; do you know how to print the elements? I will copy this just wait one second awesome this is done I will call this function here, function is magic square I will say three for three let me run this ok I am getting an error here, type error list in this is must be an integers or slices not so where it is getting I j ok let see this here I have written I is equal to n by two lets what did I put as n see three by two it is one point five and, and what I am doing using this as indices of my matrix so one point five doesn't mean anything in list it should be exactly integer for that you can use the integer decision that is three two division sign and two equals of course exactly one, so I will put two division signs here that's it let see this tada let me print one more thing here so that I get the value print the sum of each row slash column slash diagonal is what is it? Put plus here STR of n cross sorry n into I need to take this square for square my operator is two stars into n plus one and bracket. Try here m is equal to three into three two one m ok I think you understood what is taken, c I didn't put any multiplication sign here which is a sin, listen to this sum of oh my god sorry I should run now please god awesome, I am getting magic square and my sum let me try this for seven let's see this awesome this is exactly the magic square you can check it and the sum is one seventy five let me try this for five one is great so this works, so this was the magic square programme you can try some other algorithm if you find some pattern this is some basic algorithm which I found on Wikipedia you can check it please make sure whenever you find any difficulty let like sometimes it happen you writing a code but you miss something then dot understand what wrong you are done this is very simple code on few lines sixty one sixty and some add-ons so finding mistake here is not very difficult but if you have a thousand lines of code it will be mess it will be very difficult to find any bugged there, its better you always you can always search for the problems for example this I can search and I can try this here so it will re direct me to the stack overflow, stack overflow is the best website to insist to your problem most of your problem will be ensured here and even if some of the problems are not insured you can ask it and easily get insured here in very less time works like a charm so please make use of it and we are done with the magic square, you can do this programme in many ways this is one of the way so please try finding other way also or you can even use some other parading for example you can use recursion to make this to create this programme. So thank you every one please posts your queries and other things on discussion form.

# LET’S PROGRAM AND PLAY

Mario, angry birds, candy crush, I am sure most of you have played these games. Well now it’s time to develop a new game through programming.

# DOBBLE GAME-SPOT THE SIMILARITY 01

Do you know this game? What game is this? Dobble dobble! Dobble. Do you know how to play this? Yes. Then show me, here is the card, here is the another card can you tell me what is common between this and this? What is common? Tell me. Horse horse, horse horse very good. Very good, now what is common here and here? Can you show me what is common? Ok take your time, ok is only one object common, this and this what is that called? Candles candles that’s right. so the pint is there is only one symbol common in these two cards one and only one and the game is all about one person asking the other person show me what is common between this card and this card ok? So let us play this further. Tatswit you are recording! Yes I am recording yes video is getting recorded so you better be good. Tell me what is common here and here? Ohhh click click click what is common? Ok there is one pink here which is also here what is that? Now you should find it, I am not give you any hints ok ok you be silent ok I am silent. Ok see something is common here what is that? Something is flies! I don’t know what is here! This and this, what is that called? Dragon! A dragon that’s right, very good so as you can see there is precisely one symbol common between these two cards, there are eight symbols here and eight symbols here and precisely one symbol is common and that happens to be a flying dragon so the game is all about we figuring out what is that common object given in any two objects, as you can see its very difficult to figure out what is common here just like that. There are eight objects here eight objects here you see, you should compare every eight objects in this card with every eight objects in this card that happens to be eight into eight sixty four which is of huge space for us to compare and only one of them is the right answer. Say hi fi say hi fi hi fi give me a hi fi ahh.

# DOBBLE GAME-SPOT THE SIMILARITY 02

I think the rules of the game is very clear to you all of you. Simply look at these two cards they have eight symbols each. This is one game which I am very curious to know why our mind cannot spot the exact commonality between two cards, it feels like the moment you spot it you feel how is that all but it takes the long time to spot it. The best part about the game is the way it is being made you see the cards are such that. Pick any two cards there are roughly fifty cards in this deck. Any two cards that you pick will have precisely one symbol in common you not have two symbols in common it is not so happen that no symbols are common exactly one symbol is common and it's for you to find out what's that is? So the rule of the game is very clear. Two people are sitting and one person flashes two cards the other person should spot the common object, so in this case you see this is the common object. Some other two cards you see this is the common object. This is the common object and so on

# DOBBLE GAME-SPOT THE SIMILARITY 03

Well this game appears to be simple but actually it is not, wanna play? Let us try to stimulate the pythonic version of this game. So first of all let me give you a brief introduction of what dobble game is all about. As you know we have a set of cards here each card has some symbols on it, each card has definite number of symbols on it in fact if we pick any two cards we will find the definite number of symbols on it and if you pick any two cards from a lot they will have only one and one symbol in common. You can clearly observe this here that each pair of card will have only one and one symbol in common. So this is what dobble game is all about as we have discussed so how can we realise this through programming? In fact this is much easier in python so will have two list here list one and list two each list will have definite number of symbols and symbols are represented by alphabets so if we have seven alphabets in list one you must have seven alphabets in list two also so we have definite number of symbols and you also need to take care of this fact in mind that each pair of list should have only one and one alphabet in common this is something that we need to check in programming and see how can we play this. For example I have two list here list one and list two, list one comprises of the symbols or alphabets AGTFC and list two comprises of the alphabet TJYPOL so as you can see here only one and one alphabet is common and that is T and they have definite number of alphabets in the list, we have six alphabets in list one as well as six alphabets in list two. This is something that we need to do through python or programming so let us start with our programming screen cast.

# DOBBLE GAME-SPOT THE SIMILARITY 04

Hey, every one as we have discussed we need to have two list here, list one and list two each list will comprise of definite number of symbols and there should be only one and only one symbol in common between these two list as we discussed we will be taking alphabets as symbols here so I will be using a particular string here that is called string dot ascii underscore letters since this is from string library we will be importing string here. Let me print it and show you as you can see here we have lower case and upper case letters in this particular string we have A to Z from we have A to Z in lower case and A to Z in upper case, let me zoom it so that you can see it properly. You see here we have A to Z in lower case letters and A to Z in upper case letters I will be using this particular string for symbols. I will have a symbol list here, since this is the string I just convert this particular string to list symbols is equal to list, string dot ascii underscore letters so we have our symbols here now we have to generate two more lists I will call them card one and card two. I will initialise it also so we will be having only five symbols in each card because I have initialised the list to five elements to five zeros so first of all I need to find the positions of the same symbol in card one as well as in card two, I will have two variables here first is pos one that will be the same position in card one and next one will be pos two that will be the same symbol position in card two so I will have random dot randint it will be from zero to five it will generate random numbers zero one two three same way we can have pos two that will be the same symbol position in card two. So will have random dot randint from zero to five ok I will write here also pos one and pos two are the same symbols position in card one and card two respectively. So we are done with the positions ok so next what will be doing here is will be having a character named same symbol and I will just randomly choose it from our list that is symbols. You need to if you need to choose from the list you can use random dot choice as we are using random we should import random here too. So we are don't with the cards we are done with the same symbol, now we need to append the same symbol in card one as well as in card two ok. so I will have here if pos one is equal to pos two then card two pos one is card two pos two or pos one whatever you want to take you can take since they are equal, will be equal to same symbol same with card one pos one is equal to same symbol if they are not equal then we need to take care of some things here so concentrate we will have card one pos one as same symbol as you discussed earlier and in card two at pos two we will save same symbol but at position one in card two we will have another symbol we will have card one position two will have another symbol so we need to again randomly choose it from random list symbol, random dot choice symbols and similarly for card two at position one will have random dot choice symbols aren't we doing something wrong here because if we are choosing from the same list even card one and card two may have the same symbol so we should remove the same symbol from the list symbols so I just simply write symbols dot remove same symbols. Please note the fact why I am doing this here because we don't want same symbol to repeat in card one and card two so what we will do here we will just remove the same symbol from the list symbols. Even, even we don't want repetitions in the same card; we don't want the same symbols to repeat in similar in same card. So we also need to remove the other symbols that we are choosing for card one and card two so I will just write here symbols dot remove card one pos two and here also symbols dot remove card, card two pos one. So we are done with position one and position two and we are also done with this particular symbols that will be there in card one and card two at pos one and pos two. So what about the remaining position apart from the pos one and pos two, will also be appending some symbols here which are not similar obviously so let us do that. For that I will have a while loop here I will initialise a variable I while I is less than five ok we need to take care of this fact in mind that position one and position two shouldn't be here so while I is not equal to pos one and I is not equal to pos two only then this while loop will execute. So I will have two more characters here alphabet one and alphabet two so I will write random dot choice from symbols as and always we should also remove this alphabets from symbols so I will remove this alphabet. Symbols dot remove alphabet one and for card two will have alphabet two.. That is random dot choice, random dot choice symbols symbols dot remove alphabet two. So we are done with generating alphabet one as well as alphabet two so next we need to append this in card one and card two in card one the ith position append alphabet one in card two it ith position append alphabet two, so we are also done with the remaining position also implement this i, so we are done with pos one and pos two the same symbol as well as the remaining positions. Let me check whether the cards that are desired are the desired cards are showing up or not. So we will just print card one and card two so is to check whether our code is working fine or not. Ok let me check, what is common here is S S ok let me check again there is some error list assignment index out of range card one pos one so let me check that. We need to print pos one and pos two here so as to check whether it is taking the correct pos. It's Four three it is correct its one, one is correct, four three is correct, three one is correct yes we have an error here it is taking one and five also so we will just generate from zero to four, now it's fine so please note the fact that if we will generate random numbers, random integers from zero to five then it can also generate five. For example it can generate from zero to five zero one two three four or five so it will just generate it from zero to five, zero to four because in list we have assignments from zero till four ok so we will generate the random integers from zero to four will write the ranger as zero comma four please note this fact here. So we are done with the cards now since the player is playing this game we should also know we should also let him know that whether he has won or not so I will just write will just input a particular symbol here so ch input ask the user to spot the similar symbol ok if ch is equal to is equal to same symbol then you just write print right else print wrong. So let us try to run this, so we have to spot the similar symbol it is m here yes again run this we have to spot the similar symbol and that is r, again run similar symbol here is it is difficult to find it is though ok so this is how we can play dobble game through python and I may have been wrong before but please note the fact in mind that if we generate random integers from a particular range for example if we write from zero to five then it can also generate five. It can also generate zero that means it can generate zero one two three four five, the both ends are inclusive so we will just write zero four because we need to have list assignments from zero to four ok please note this fact in mind because we were wrong at this and other wise the code is working fine, I hope you enjoyed this programming screen cast happy learning.

# **WHAT IS YOUR DATE OF BIRTH?**

Well guys you know I share my birthday with sachin tendulkar I feel so lucky I think it’s very rare for two people to share the same birth date is that so? Let us find out.

# BIRTHDAY PARADOX-FIND YOUR TWIN 01

Let me take the list of my friends, eight of my friends namely, rishi aeyathi rishi anamika akruthi vidya amit and simran. I go to all these eight people and ask them what is your day of birth by day I mean Monday or Tuesday or Wednesday Thursday Friday Saturday Sunday right? They all will give me their respective answers hoping that they know their day of birth, all of us know in fact. They all will give their answers but one thing is for sure that two people here would be born on the same day when I ask them I noticed that this was the answer and you see that there are two people born on the same day as you can see. This is like stating the obvious you take eight people and you ask them what is your day of birth, what is your day of birth a person can be born in one of these seven days, when you take eight people definitely for sure you will find two people with the same day of birth. Now let me some other experiment let me go on the roads and ask people randomly their dd mm by this I mean their day of birth and month of birth, dd stroke mm, let me ask a few people and see if I find two people who were born on the same dd mm, now if I ready to do this experiment do you think one can guarantee me of a repetition if I go and talk to forty people may be not. Look at the previous example, if I take eight people for sure two people will be born on the same day but if I ask them dd mm there are three hundred and sixty five possibilities, three sixty six possibilities if you include the leap year. Unless I have more than this number assume, I had some four hundred people then I know some two people will have the same dd mm if I take less than three sixty five can I guarantee? May be not. But I will challenge you all that if you go on the road side and stop forty people who are going by and ask them their dd mm in forty or fifty counts you will definitely see two people with the same dd mm you don't believe me come let's go and conduct this excise and check whether what I am saying is right or not?

# BIRTHDAY PARADOX-FIND YOUR TWIN 02

We just now saw the professor talk about birthday paradox. Do you share your birthday with someone else? Let's check in the crowd. Can I please know your birth date and month? Yeah it's eighteenth December, fifteenth July, fifteenth July ok thank you, birthday is fifteenth January, sixteenth November nineteen ninety four, twenty first august, my birthday is on fifteenth of January, my birthday falls on fifteenth June, we have professor abhinav sir with us, sir may I please know your birth date and month? So its eleventh of December, hi mam can I please know your birthday and month? First of December, nineteenth January, yeah its fourteenth august, twenty second April, yeah its fifth January, fifth October, thirty first august, thirty first august thank you, twentieth November, sure its twenty third July, forth Jan, fifteenth July, eighteenth July, eighteenth September, mam twenty July, twenty first December, yeah its twenty sixth November, thirteenth march, February twenty, twenty four august, third august, fourteenth of july, twenty second of September, thirteenth march, sixth December, eighth December, june twenty ninth, twenty three may, twenty seven june, fifteen march, twenty fourth august, twenty first April, its fifteenth august, seventeenth September, eighth April, eleven june nineteen ninety eight, twenty fifth December nineteen ninety seven, first January, tenth july, eighth October, fifth October. Ok thank you.

# BIRTHDAY PARADOX-FIND YOUR TWIN 03

As you know we collected the birth dates of some fifty people and guess what, we don’t need three sixty six people to encounter the collision. Actually we encountered a collision in sample of nine people so let us try to simulate this paradox using a program. Before we get into the details of the programme let me give you a brief introduction of the programme for birthday paradox for birthday paradox we will be generating the birth dates randomly I repeat will be generating the birth date randomly and we will not fetch it from the data. We will only generate the birth dates randomly. So now the question arises how can we generate birth dates randomly so birth date will be generated according to the birth month as you know some month comprise of thirty one days and some month comprise of thirty days so we have January, march, may, July, august, October and December they comprise of thirty one days we have April June September November they comprise of thirty days so birth date will be generating according to the birth months as we are generating only birth date we are not concerned about the birth year. Yes we are not concerned about the birth year we will be only generating the date and the month that’s it. So there is an exception here as you know February has twenty eight days or twenty nine days, if the given year is a leap year then the February has twenty nine days else February has twenty eight days. So how can we determine that the given year is a leap here or not? So our question becomes that. For a leap year most of the years that can be divided by four are leap years you know this fact. But there is an exception here also century years century years I mean like five hundred, six hundred, thousand, sixteen hundred, seventeen hundred these are the century years. Century years are not leap years unless they are divided by four hundred, that means if we have seventeen hundred it is divided by four but it is not a leap year. Because this is a century year ok but if we have sixteen hundred it is divided by four hundred ok that means sixteen hundred is a century year as well as leap year. So I repeat most of the years that can be divided by four are leap years but there is an exception here century years are not leap years unless they are divided by four hundred yes unless they are divided by four hundred so now I will show you the pseudo code of how to determine whether the given year is a leap year or not? So we have first of all we will check if the given year is divisible by four hundred then it is definitely a leap year else if it is divisible by hundred then it is not a leap year. I repeat if it is divisible by hundred then it is not a leap year OK, else if the year is divisible by four then it is a leap year and for rest of the cases the given year is not a leap year. So will go through the pseudo code again, first of all you have to check whether the year is divisible by four hundred or not if it is divisible by four hundred then it is a leap year, else if the year is divisible by hundred that means it is not a leap year else if year is divisible by four then it is definitely a leap year else for rest of the cases the given year is not a leap year. So first of all let me try to code this particular function this particular thing of how to determine whether the given year is a leap year or not? So let us do that. So I will just open anaconda spyder and try to code this. So first of all let me start with how to determine whether the given year is a leap year or not? So I will just generate here randomly or you can also make the user input that particular year, so I will just generate it randomly I hope you know the random library we have already explained it. For example I take the year from ninety three to twenty eighteen ok? As we are using random we should import random here too. Next is we have to write the pseudo code here not the pseudo code the proper code according to the pseudo code. So let us do that I will use if here, so first of all I will say if year percent four is equal to is equal to zero ok and year percent hundred is not equal to zero then I will use or year percent four hundred is equal to is equal to zero, this is according to the pseudo code that I just now explained, I will repeat it again century years the years that divisible by four are leap years but there is an exception here century years are not leap years unless they are divided by four hundred this is the pseudo code that we are using here so after that what I will do is we have generated the year randomly and we will also return the if condition for the year if this conditions satisfies that means you have to print given year is a leap year OK else you have to print given year is not a leap year. So we are done with the programme for leap year so let us try to run it, I will just write leap dot py. Given year is not a leap year so let me try to print the value of year also so that we can get to know whether the given year is a actually leap year or not. Whether our program working fine or not? So let us check that twenty sixteen given year is a leap year? Yes you can see that twenty sixteen is divisible by four and it is also not a century year so it is a leap year let me try to run it again we have twenty eighteen, yes twenty eighteen is not a leap year. Again so let me try to zoom it so that you can also see ok here we have two thousand five is not a leap year again two thousand three it is not a leap year again, twenty sixteen yes this is a leap year again twenty eleven it is not a leap year, two thousand one not a leap year. So this is how we can code the leap year thing so let us move ahead and try to understand the integrities of the programme for birthday paradox. So now that we are done with the leap year code I hope you understood that code so let us move ahead ok, here what we are trying to do here is now we have to write a pseudo code generate the birth dates randomly ok. So as I said different month comprise of different number of days so we have to take care of all the months and we have to generate the days accordingly so we have here if the given month is February and year is a leap year then we have to generate the day randomly from one to twenty nine, if the month is February and the year is leap year then we have to generate the day randomly from one to twenty nine. If the month is February and the year is not the leap year then we have to generate the day randomly from one to twenty eight, yes. We have to generate the day randomly from one to twenty eight then if the month divides by two ok, we have months numbered from one to twelve ok, and if it divides by two that means and the month is less than seven that the month is either april or june ok, then we have to generate the day randomly from one to thirty, january february march april and june ok. So this is it for the month that is less than seven now if the month divides by two and the month is greater than seven, what we are referring here? We are referring here, referring to the month august, october and december then we have to generate the day randomly from one to thirty one ok now if the month doesn't divide by two and it is less than equal to seven here we are referring to january march may or july these months comprise of thirty one days, yes these months comprise of thirty one days. Now the last condition is left that is if the month doesn't divide by two and the month is greater than seven then we have to generate the day randomly from one to thirty, we are referring september and november here, we are referring september and November here, please note the fact that we are not checking the condition for february, since we have checked it at the first place. Since we have checked the condition for the February at the first place we don't need to check it again. So we will start with February because February is an exception here, we have either twenty eight days or twenty nine days in February. After that we are done with February then we can take into count the months that comprise of either thirty days or thirty one days. So I repeat here first of all you have to check whether the month is february and the given year is a leap year or not ok, if the given year is a leap year then you have to generate the day randomly from one to twenty nine other wise you have to generate the day randomly from one to twenty eight, after that you have to take into count the months that comprise of either thirty days or thirty one days. If the month is divisible by two and it is less than seven then you have to generate the day randomly from one to thirty, if the month is divisible by two and the month is greater than seven then you have to generate the day randomly from one to thirty one, if it is not divisible by two and it is less than equal to seven that means we are also referring july here, you have to generate the day randomly from one to thirty one and the last case if it is not divisible by two and it is greater than seven then you have to generate the day randomly from one to thirty. This is how we are going to proceed in the program, you will get to know more about it when we will write a program about birthday paradox.

# BIRTHDAY PARADOX-FIND YOUR TWIN 04

Welcome to the programme screen caste of birthday paradox, first of all let us take the birth dates of some fifty people and try to find out the collisions. Here we will be generating the birth dates of these fifty people randomly in order to generate the birth dates randomly will import the library here that is called random. Let us take the birthday array on list, now as you have to generate the birth dated of fifty people will have a while loop here I is equal to zero while I is less than fifty. First of all let us generate the year randomly, year is equal to random dot randint, randint is the function that is used to generate the integers random, you also need to specify the range in the function randint here I will specify from eighteen ninety five to twenty seventeen, why I particularly choose eighteen ninety five because the oldest person every lived was one twenty two years old so if you subtract one twenty two from twenty seventeen you will get eighteen ninety five, that's why we choose eighteen ninety five I am writing here the oldest person ever lived was one twenty two years old. Now we have to check whether this particular year is leap year or not? Why do we have to check whether this particular year is leap year or not because in order to generate a particular day we have to check that the particular year is a leap year or not because in the month of February in a leap year we have twenty nine days otherwise in the month of February we have twenty eight days so we will also check that the given year is a leap year or not. In order to check whether the given year the randomly generated year is leap year or not we will have if here, if year percent four is equal to is equal to zero and year percent hundred shouldn't be equal to zero or we can just say that year percent four hundred is equal to is equal to zero. Will have a even here leap this is equal to one if the condition satisfies else this is equal to zero. Now let us generate the month randomly, for month we will use the function randint again, month is equal to random dot randint what will be the ranger it will be from one to twelve.  Now we are done with generating year and month randomly. Let us proceed towards day, now let us generate the day randomly. In order to generate the days randomly we will have a nested if else loop here for a nested if else loop will have if month is equal to is equal to two and the given year is the leap year, what do I mean by this? Is this that the month is February and the given year is a leap year so we will generate the day from one to twenty nine that will be random dot randint from one to twenty nine? Else if, if month is equal to is equal to two and leap is equal to is equal to zero, if the month is February but the given year is not a leap year so will have day is equal to random dot randint as one to twenty eight next we have to check please note that the seventh month and the eight month they both have thirty one days so we will have to heart quote this to so will have here the month is equal to is equal to seven or month is equal to is equal to eight will generate the days from one to thirty one next we have to keep the fact in mine if month percent two is not equal to zero and the month is less than seven so we will generate the days from one to thirty one you can easily figure this out, please note the fact you can easily figure this out, why I am writing so because if the given month is in odd numbered month and the month is also less than seven then we can say that we will have to generate the days from one to thirty one. Next condition if it is greater than seven that is the next condition. For that the given month will be a even numbered month,

month two is equal to is equal to zero and month is greater than seven and it is less than twelve too. So we will here day is equal to random dot randint one to thirty one we have specified the condition for twenty eight, twenty nine, thirty one so the rest of the month will obviously comprise of thirty days so we will just have else day is equal to random dot randint from one to thirty this is done. So we have now the birth date of these fifty people plus please note the fact, please not this fact in mind that we don't have it in a specified format, we need to have it in a specified format in order to do that I will generate a, I will import a library here a separate library here that is date time. Next we have to convert it into particular format so I will use the function of date time that date time dot date and I can specify the format here in the round braces will have year month day this is the format I am calling, you can call the any other format but I am calling year month day. Then we have day of year, why do we need to find the day of year? As the professor explained in the lecture that we will only consider the dd mm this is self-explanatory if you want to find out the collisions in the birth date we need to only consider the dd mm of different people so we have the dd mm of different people but we need to generate the day of the year. If you will generate the particular day that the person is born on then we can find out the collisions very easily for example if I am born on second January ninety three that means I am born on second day of the year in the same way we will generate the days of year of these fifty people and try to find out the collisions. In order to get the day of year, we will just use the simple function here please concentrate we have dd that is date in a particular format dot timetuple dot tm y day this is the function that is straight away give you the date of year ok, if you want to code it you can also code it this will be good practice for you but for simplicity sake we will only use the function here. Now we will increment the I variable so that the while loop can proceed and we will also append this particular day of year through the list birthday, appended day of year that is done next sort this particular list when this particular list is ready just sort this. What is sorting? Sorting is basically arranging the elements in ascending or descending order, since we are using the function sort here it will erase the elements in ascending order you can always code it, this will be good exercise for you guys but for simplicity sake we will use the function sort here. Now we are done with the code we just want to print this particular array so that we can get to know the collisions. It will be easy on our mind to find out the collisions if the array sorts it that's why we are sorting the array. Just print birthday I and increment this while loop. I think we are done with the programme and let us try to run this programme. See we have a list of birth dates here particularly the day of year, if you want to find the collision lets have the look at the output. We have fifteen fifteen one collision here we have one hundred four one hundred four the second collision, two not four two not four that is the third collision and here and I don't think we have another collision here so will have three collisions in this list of thirty people, we can always increase the number of people here and will get more collisions. Let us try to run this again if we run this again we have forty eight forty eight one collision seventy two seventy two second collision, one eighty eight one eighty eight third collision I don't think so another two eighty three two righty three forth collision so you can always increase the number of people and try to find out the collisions the collisions will increase and you can re run the loop re run this program again and again you can have different number of collisions too. So this program clearly illustrated the presence of the birthday paradox. I hope this program was useful for you guys and you enjoy it this programming screen cast see you till the next programming screen cast thank you.

# BIRTHDAY PARADOX-FIND YOUR TWIN 05

Now that we are done with the program of birthday paradox, you must remember we used a library name date time minute so let me try to explore this library and show you what else can we do with library date time, date time library can be used to display even today’s date so how can you do that? So you just write date time dot date dot today with round braces so it will display the todays date the current date ok you can even segregate the date in terms of year month and date for that what you need to do is , I will just write for that you just need to write dot strftime in round braces you have to specify what do you want to display, I need to display only year, just write percent y so it is twenty eighteen next I want to display the month you just write B capital B so it has displayed the current month and if I write d it will display the date so it is thirty first. Ok so it can be used to display many other things also so I just go though them first of all if you want to display the week number of the year, week number of the year how can you display that. I will just write so that you also keep a track of what we are doing, week number of the month, how can you do that? You just write date time again dot date dot today dot strftime and in round braces you just write capital W percent capital W. So week number of the month is twenty two. I repeat week number of the month is twenty two. So after this what can we do? After that if you want to display week day of the weak so you need to write, what do you need to write? I will again write week day of the weak again you have to write date time dot date dot today dot strftime and in that you write small w percent small w press enter so week day of the weak is four ok that’s how you can use the day time library there are others thing through which you can also use I will show you one or two examples more first as if you want to display the day of year ok, how can you do that? Print day of year for that again you have to write date time dot date dot today dot strftime and in bracket you write small j so let us press enter, day of year is one fifty one. after that you want to display the day of the month and the day of the week is also possible that is you can also do that so I will just show you how you can do the day of week ok so let us do day of week for that you just need to write first of all write here day of week yes and how can you do that? What do you need to change here? You need to write percent ‘A’ capital ‘A’ that will display day of week. So day of week is Thursday it is actually Thursday today so this is how you can use day time library you can also use it to display the current time, how can you do that? I will just show you. You just write date time again date time dot now so it display the current time, first of all it will display the date and then it will display hours minutes seconds this I show it works date time dot day time dot now it display the current time. I hope this date time library exploration is useful to you guys happy leaning.

# <u>WHAT’S YOUR FAVOURITE MOVIE?</u>

Babumoshai, zindagi lambi nahi, badi honi chahiye this is a dialogue from one of my favourite Hindi movie ‘Anand’, you must be wondering what I am talking? Well this is what our next joy is all about, movies.

# GUESS THE MOVIE NAME 01

I am going to tell you all a nice new game, the game is called guess the movie name. Right so the game is very simple, I will try to write a movie name like this ok? There is this movie with one two three four five six seven letters can you guess the name of the movie? Obviously there are gonna be a lot of movies with this with seven letters so I will give you a hint, it's a Hollywood movie and a very popular one can you guess? It's obviously tough, so what will do, will go step by step you tell me a letter in the movie if the letter is present in the movie I will writer that letter in the right place. E, e is not there in this movie go on sir take L, L, L no it's not there, A yeah A is there as the fourth letter in the movie ok, sir take S, S? S is not there in the movie, T, T is there, sir I titanic yes perfect. Oh ok when I said I, I should actually write I here as well but anyway now you understand the rules of the game correct? So let me play the another game right now ok? By another game I mean another instance of the dame game. Alright? There can be space in between two I will not exclusively write the space here tat's it ok. Ten letters go ahead I, I is not there sit take A, A no it's not there A is common so I thought of taking A, A is not here ok it's not there. E E is there, in only one place O I think you guessed it, the moment you think you have guessed it tell me ok D no D is not there, sir T, T for Tarzan no T is not there S is there S is not there, R R is not there only giving me letters that are not here go on D D is not there, N, N is there, K yeah it's a book phone book no its not phone book this is the book and this is the six letter word sir hint? Hint! It's a very popular Hollywood movie so popular that everyone knows it. Including a two year old may be a three year old two year old ok ok from two year old to sixty year old everyone knows the movie very popular movie dreamt it many a times giving hints is not part of the game just try to keep them interested, S no, you know what no more letters I think you guys should guess what is the name of the movie I am sure all my audients over there who are watching this would have guessed long long back the moment you gave the second word it's so obvious what's the name of the movie sir P is there, P is not there C is there no very very very easy so easy that there are very few movies with this second word book, J J yes J is there jumangee! Jumangee!! No no no no no jungle book! Jungle book ha ha ha wasn't that easy? What was confusing for you people? I never heard of this movie as I guessed it, you haven't heard this movie? No I am not, I have heard but, iit students they don't know what is jungle book movie any ways never mind just kidding ok last last game and let's get started with some discussion about this this game alright? Ok go ahead space is there between the word, no. One word, E no E is not there A, A is there S is not there S is not there, R yes R sis there O, O isn't there, U isn't there aggressively you should guess probably D, D? No. M, M perfect C, C no, E E no again here audients would have guessed T, T perfect MATRIX ha ha ha ha ok so let me ask you people something why was it difficult for you people to guess? You see there are twenty six possibilities per slot even for a three lettered word there are so many possibilities if it was just three letters something like six letters will be twenty six into twenty six into twenty six into twenty six that's a huge possibility space and people generally tend to think of all possibilities because there are so many movies with six letters that you will go on exhausting the list of movies you have watched only one of them will match and they match slowly the idea is to probably use vowels first because vowels will certainly be there once vowels are there you can start with abcde and then go till z right? But the idea of the game is to guess it very quickly now he has the question for you all, can you try writing a piece of code can you programme? And use a let say audients should you start the programme user starts the programme and then there is a random dashes that come and then user inputs a letter, the letter is there it should pop up the letters the way I wrote, if it is not there it should say letter not found, how easy or difficult would this game be? Easy to code, easy to code! How will you start off? What will you do? I will take an array of the number and the size of the array of will be the number if characters present in the movie name I will store the I will also firstly now firstly don't you think we should have list of movie names for the computer, right it should have a list of movies name from where it will pick the movie name uniformly or randomly it should randomly pick, once it picked it should see what are the total number of dashes assume it picks the movie inception I n c e p t i o n inception is nine letters correct? You should put nine dashes and then the user inputs z it should say beep no z not found, if he inputs I you must go to the first one and then I inception and you have two I's correct? It should display two I's and the remaining stuff should be dashes, you can add in some nice variety to the game where you can give them points, you give them ten points if they guess it in one go they get ten points every chance they take you should deduct one point that makes the game interesting right? So can we try writing a piece of code for this? Let's try.

# GUESS THE MOVIE NAME 02

So, how does one go ahead and write a piece of code for this, all you need to do is have a list of movie names, as a python list of movie names which you can modify as per your requirements and then you must randomly put a movie name assign it to a variable and that will be the secret word and the user will start punching in the letters as in, when she punches the letter it should be displayed in this blank spaces. It should be displaced if the letter is there or not, the idea is to guess the movie name with very few attempts.

# GUESS THE MOVIE NAME 03

You would have seen an interesting game in the last video sir has a movie in his mind he just writes as many number of blanks as there are number of letters in that movie name the person for playing with sir has to guess what could be the movie he has in his mind, just filling the blanks is difficult see even you would take movie name with five letters say then even for that you have a lot of possibilities errors so many movies that could be available with five letters so it is difficult for the person on the other side to guess what could be the movie he has in his mind so he is given the flexibility the person can say I assume that this particular letter is present in the movie name so he can guess one letter at a time so he says whether this letter is present or not he asks sir basically whether this particular letter is present or not, smart strategy will be generally to start with vowels because mostly vowels will be present in any name so a e I o u with the vowels if you start you can open lot of blanks you can open up and you can guess the movie name that is the smart strategy generally we use so for example if I say a I want to guess whether a is present in the movie name so in case if a is present the person who has posted the question will write a in those blanks wherever a was present in that movie name in case if it is not present he would say it is not present you come up with the next guess other letter. So, this way by unravelling each and every possibility of letter you try to guess the movie name this is how the game goes, alright? We have just taken few top rated movies from imdb that is the internal movie data base so let us have a list of those movies here first as a part of our programming let us have a list of those movies, movies let me have it as a list the first we had was 'anand' in the second was 'drishyam' and the third was 'nayakan' the forth was 'anbe sivam' so go I have taken the fifth is 'gol maal' the sixth one is 'vikram vedha' so it's 'vikram vedha' the seventh one is black Friday the eighth is 'dangal' ninth is 'manichithrarhazhu' and the tenth one is 'taare zameen par' so may be for each of users because we have to switch between caps and small it would be difficult so let me make everything in small letters that just for our ease, it actually doesn't matter may be you can have all caps or anything as you wish its convenient for me if everything is in small so let me make everything in small letters. I have ten movies I have taken top ten from the imdb, you can take as many movies as you wish or you can include your favourite movie as per your wish this list completely up to you, you can have as many the ones which ever you wish you can have it any ways as you want this list is completely up so we have our list of movies now let's play we will start our game. So this thing if you would see here it says undefined name play so you say I want to play but the computer doesn't know what to be done if you say play so you have to define what is play? So you are defining def play this is the particular syntax to define new functionality if that is if I say play this are the tasks that has to be done, you are defining it. So we will define what has to be the game we will define basically what has to be done if someone says play. So here let me say I have two players as you have seen in the video there were two players and one person was posting the question here that we make it the computer posting the question the list of movies we have given are already the list of movies from this list the computer is posting the question so just that is the change here and we will assume the there are two players to play the game so this is the two players game basically so we will have player one name I want to each will be having a name it will be good if you would refer them by their  name they will feel a personalised thing of playing the personalised play they could feel if you use their name instead of just calling player one player two x y something like that you just use their own name it would be good so I have to input his name so I input it to get the input I should give him a suitable message that please enter your name so only then he can understand what do I actually except in this time and he will give the proper input so I have to give the message input message that is for this particular message, read this message and give your input something like that what could I give? I can give player one please enter sorry enter your name, I have asked his name the same has to be done for player two as well so let me copy paste it I am sorry I will copy paste this let me copy and let me paste here, the only change here is it's player two not the player one again it should be player two and here to we need a different variable that's why there is a warning ps here that is that's the same name there is a warning that you could see here this warning appears because there are two things with the same name p one you had used earlier p one name and the same p one name you had used earlier so if I change it may be this particular thing will be resolved and now this warning is because it is assigned but it is never used but this will be used in the upcoming part of the code but that would be taken care of that would be fixed this particular warning symbol that is appearing here would be fixed as the code proceeds. So let us so let us start off with the next part so for both the players initially we will have zero points so let me initialise that points for player one is zero points for player two is also zero so I have initialised and if you would see we alternate the turns of the players like may be player one gets a chance to guess the movie name the next would go to the next player this would be an interesting variant if you would alternate and see who gets the highest score probably there are other variants may be familiar with this is one of the interesting variant that we will be implementing now alright so keep track of the turn who has to lay next we will have some variable called turn so let me initialise it, let me start the value of turn with zero so why zero nothing special it is general practice in computer science to count form zero you can start from anything basically what has to be taken care of is the turns must be alternated between the two players so player one gets a chance now the next must go to player two the next chance must go to player one the next must go to player two and this chain must be happening so to take care of that you can start from any value only thing is you have to take care of the strict alteration that is the both must get the chance equally that should not be the case that same person get chance again and again that should not be the case so that we have to take care of it that's it. So for that we have a variable turn it's just a general practice of computer science to start counting form zero that why I have initialised zero nothing special you can start from any value, so let me say the players are willing to continue the game or not willing to continue the game so this could be the two states  as the players keep playing some may be it is a two player game so we need both the players to play in case if someone is not willing to continue then the game comes to an end so to keep track of whether they are willing to continue or not so I would say willing I will keep a variable willing initially will have that to true because this is the start off the game so let us keep it true and as long as the players are willing you have to play the game so let me say while willing while willing so as long as they are willing, as long as both the players are willing you continue the game so this is how now here comes the actual game and I say the turn has to be alternated so same here I have started the value of turn with zero you may start with any particular value at the value zero it is the turn of player one at the value one it is the turn of player two at the value two it is the turn of player one again at the value three it is the turn of player two again and this is how it goes as you would observed if the turn value is zero two four six eight and so on it is a turn of player one if it is one three five seven nine and so on it is the turn of player two so every even value of turn is the turn of player one and odd value is the turn of player two, how do we keep track of this? Even and odd values there is an operator called the modulo operator, what it does is, it divides and returns the reminder of division so to check whether the number is odd or even you would divide the number by two and check the reminder, for all even numbers dividing by two would give you the reminder zero for all odd numbers dividing by two will give the reminder one so we will use this concept and check whether the particular value is odd or even so the turn has to be check if it is odd or even so let me check if turn modulo sorry its turn modulo two is equals to equal equal to zero why did I use double equal to because single equal to is assignment basically whatever be the value I don't care just assign this value something like that this is I want to check the equality, I want to retain the value and check it with some standard value so for that some equality check I will apply use double equal to symbol so if this is equal to zero that is even value the turn is even value, if it is even then you divide by two you will get the reminder zero so if that happen if turn turned out to be even number then it should be the turn of player one ok let me write that it is the turn of player one ok so we have to first start off by telling that it is his turn print player ones name you have taken print his name and say that it is your turn player one this is your turn just say him the message alright? So what should we be doing now? We should randomly pick a movie from this list of movies, there is a list of movies with us from that list we randomly pick a movie so let me say that it is picked movie picked movie is nothing but for randomly picking a movie you need there is a built in library called random we need to import it lets import it now import random so this is a built in library that would be used in order to pick something randomly from a given list or from a given range something like that for random selections this particular library is used so for picking randomly from the given list we use the function called choice it is define in this random library so I will be calling it as random dot choice within this I need to pass the list as the argument let me the name of this list is movies so I will pass movies so basically what this does here, from this list movies you pick a random choice and assign that to picked movie variable so you are picking a random movie this is what is done here. Now once I have picked a movie I need to create the question that is basically I need to give blanks that many number of blanks so here I will be stimulating blanks as stars so if the movie contain say for example here in our six letters say dangal is the movie I have picked it contains six letters I need to represent six blanks here I am representing blanks in stars so I need to represent six stars so I should create the question in that format this is my picked movie is dangal for that I need to create a question so I have to create a question here so let me do that let me say, you create a question for the picked movie that is picked movie as some six characters in my case as I have said then six stars must be present and in case if the picked movie for example say it is something like anbe sivam, 'Vikram Veda' or taare zameen par these movies have spaces in between so the spaces may not be encoded by stars, spaces must be retained as spaces just the letters must be encoded as stars so this encoding thing I have to do this thing let me say I have to create the question so this is what is the task I have to do, create a question and once the question is created I need to capture the created question so let me capture it in a variable called question so I had created a question alright so I have once the question is created and captured I need to print it print this question print the question and till now the answer is not yet said that is the person has not yet said the answer so let me say he has not said the answer not said I will have a variable to keep tracking of whether the person as said the answer or not said the answer, so let me say this particular value is true not said is true. Alright so as long as he says the answer now we need to continue here so I will say while not said not said as long as the person as not said the answer continue the game so what should you do? You should ask him to guess a character so let me take the input character let me say it as ch, ch be my character or let me say letter that is better intuitive name, so let me say letter letter input I have to input the character your letter this is the your intuitive message I feel you can modify this message as you wish that’s never an issue the thing is as how the question will be post and the person is ask to guess the whether this particular letter is present in that movie name or not that is what I am doing your letter please enter your letter this is what I want so I will inputting that so when I have to check if that letter is present or not so let me say if that letter is present is present of that letter in the picked movie, a movie was picked right so we have to check whether that letter is present in this picked movie or not so if this is present you have to unlock the blanks wherever the letter was present that part has to be taken care of here, else it’s a simple thing so let me complete that first and lets come back to this part of unlocking else what should I do print his letter whatever he has print he has guessed is not found, this is the message to be print, this is simple so let’s see till now we input the name of the players start with zero points each we have a variable turn to keep track of who has to play now and we have a variable willing to keep track of whether they want to continue the game or they want to quit the game so as long as they are willing we have to play the game we first start with checking the value of turn we first start with checking the value of turn here if that is an even value then it is the turn of player one else so corresponding else let me write here so that it is, it is not skipped let me write the corresponding value it is in column nine so I need to write the corresponding else in column nine it’s in column nine ok hence this is the turn of player two ok so once they’re both the turns are over what should we be doing is we need to increment the value of turn that is this person has played now this will be the turn of next person so I am increasing the value by one, turn is equal to turn plus one this is how I have incremented the value of turn so this is the outline of our play method we have meant completed it just an outline this is a rough idea of how the game will go maybe you pause for sometime look into the outline understand the flow of the game then proceed with it let’s proceed with the missing parts here in the next session of this video thanks for watching till now please watch next videos in order to completely understand how this program will work thank you.

# GUESS THE MOVIE NAME 04

Alright guys, till now we have seen general outline of the game let me just give you a very quick summary of what has happened till now see these are the list of movies we have collected this from imdb top rated Indian movies, when we come to the actual game, when we come to the actual playing part the first thing we do is we are getting the names of the players, this is just to give the personalised feel to the players and we start off with zero points to both the players and we have variable we started off with zero points to both the players as you could see here and we have a variable turn that is use to keep track of whose turn it is that is the turns has to be alternatives, the players will have to get their turns in the alternating sequence for that reason we have a variable turn and we have another variable called willing that says whether the players want to continue the game or they want to quit the game, so this particular thing willing initially I have to set it true because I have to start the game and during the run off the game the players may choose to quit at any time and it is a two player game so it requires both the players to be present in the game, in case if any one says I want to quit I don't want to continue next the game comes to an end and the status till now will be displayed at the end of the game. So, this is how the game goes and this is we had assign we had started the value of turn from zero and this is just a convention in computer science to start counting from zero nothing hard and fast you can do anything as you wish, I had started from zero so the turn zero two four six that is the even numbers turn corresponds to player one that is been captured by this modulo operator and odd number turns will correspond to player two we had given an outline of the game in player one so we had, we had started off with this see when it is the turn we have to display that this is this person turn so I had given a message player one this is your turn something like that I had given a message. I have randomly pick a movie from the list of movies we have and create a question out of it, creating a question is nothing but we would have blanks in the conventional game, here I am encoding the blanks with stars that is, in case the movie has five letters in its name, five stars will be displayed in place of the letters and please note that in case if there are spaces for example if you would see 'thare zameen par' or 'vikram veeda' these movies have spaces in their names, so in case there are spaces the spaces would be left as such, only the letters will be encoded as stars, so in case if I take 'taare zameen par', five stars followed by a space followed by six stars followed by a space followed by three stars. This is how we need to encode it this encoding process let me call it as we are creating the question, so we create the question we take the question and print the question, we show the question to the player and we have a variable to keep track of whether the player has set some answer or not so initially he hasn't said that answer so I said not said is true, so as long as he says the answer repeat this, what we do is, we ask him to guess a letter that could be present in the movie, and if that letter is present we need to unlock those spaces, those star where that letter is present otherwise we have to say the letter is not found, for example see if 'taare zameen par' is the movie that has been picked and if the user wants to know if the letter 'u' is present in the movie name, you could see letter 'u' is not present here so I will say 'u' not found. In case if he types letter 'a' a is present twice here, once here once here so in all these corresponding places I will unlock I will say this is ‘a’ after unlocking it becomes star a, a star, star, star a star, star, star, star a star and spaces would be just spaces there, there is no encoding nothing but in the precious encoding all letters are made stars we would unlock only those places where the letter ‘A’ is present, this unlocking part has to be done otherwise we will just display the message that this particular letter is not found ok ,now let us look at the unlocking part so let me say after unlocking those letters we get the modified question, so I will say this is modified question, let me call this as a modified question to get the modified question what should I do? I should unlock, unlock all that I need is the previous question or the question that comes during this run because as you could see with A unlocked the user may either guess the movie name or maybe he could require more guesses like maybe he want to say I want to unlock more letters may be he want to unlock if letter I is present if letter e is present, he may try for various possibilities so it is not always the case that with just one letter unlocked the user can guess the movie name so it may happen anything so I should take the modified question from the previous run. Modified question from the previous run and the picked movie name I need to have a reference because I need to unlock his choice of letter if it is present. If the letter is present I need to unlock it so, for unlocking it I need to refer to the picked movie name originally and this is his character preference I need to pass so I need to unlock those letters using these parameters, I need to have unlock functionality which requires these parameters so the modified question here and here the difference is this is the question modified based on the previous one see, I will give you an example just take the same thing taare zameen par if this is the movie if the user wants to unlock ‘a’ these will be unlocked and now the new question will be star ‘a’ ‘a’ star, star space star ‘a’ star, star, star, star space star ‘a’ star this will be the new question, this has to be passed and suppose during the second guess the letter he wants to unlock is ‘e’ then ch will carry the value of ‘e’ and to unlock ‘e’ where all e has been present in the movie you need to refer to it, so I am passing the picked movie as well, I need this value as with so this is the modified question based on the previous one so what will be the this particular thing in the first run, so for that let me store here modify question is nothing but your question itself in the first run, in the first run it is all stars, during the subsequent runs the modified question may vary that depends on what are the letters you have already unlocked ok? So you have modified it so modified question is obtained to you so let me now print the modified question, I have to print the modified question. Ok print it modified question now I should say the user has to decide now, if he wants to continue, if he wants to guess the movie name with this or he wants to unlock another character so let me say this is decision so d let me call it d, I will take his decision as a form of am input I can give him an message see press one to guess the movie or two to unlock another character another letter I could say so this is once the unlocked the modified question has been displayed after unlocking the particular letter he has to decide whether he knows the movie name he want to guess it right the way or he wants to unlock another letter so this particular decision is taken care of here. So once the decision has been taken in case he wants to say I want to guess the movie name, if that is we had given if one is for guessing the movie name, if his decision is one then you have to collect an answer from him, answer you have to input it from him, let me say, let me give a colon here for clarity sake let me say here it is, I need to give a message let me say your answer, I am asking him to give his answer, so for this he will give an input answer we need to check if that answer is exactly the same as the picked movie name so, I am checking if answer is exactly equal to picked movie. Ok, in case if that is equal then I would say increase his points by one so already his new points for player one is nothing but his old points plus one, this is his new points so, I have increase his points by one. so I have to let him know that his answer is correct, I have to print that his answer is correct and I will have to make this not said I have used the variable not said based on which game will continue I have to make it false because initially he has not said the answer so I had made it true now given that he has said the answer not said should become false, names I have used so it is easier to understand the code this is generally a better practice to name your variables such that it is easy for anyone to understand the code, the code is not just for you even if you go work in the collaborative environment in a company or anything even you work in teams it may be the case that you would be the one who could and there me be someone who tests your code because it's generally the thing is pretend to cover our mistake that is how human tendency we tends to cover our mistakes, so the person who has written the code if he is given an opportunity to check whether the code is perfect or not he knows what are the mistakes could occur during the, during the coding process he would have tested and he would have got some mistakes, he would try to hide that mistake so the actual quality could not be could not be so in order to get the actual quality what they generally do is they would ask another person there would be another person called the tester, he would be the one who has assigned the responsibility of testing the code, so for him to test the code for him to give inputs such that you can check through how the flow is what are all the calculations being done to check through all that the code must be highly readable and when you work in teams, when your particular portions is being used by someone else or someone else portion you are using and you are depending on it, it is highly important that your code is properly readable and to make it readable one time that we generally give is you give your variable names in a intuitive manners see just as how I have given western modified western picked movie not said all these are English words that speak in itself, you may not say for what purpose you have been using this particular variable name that is quite intuitive people can easily understand just any one knows English will be able to understand the flow of this programme in such a way we are coding as to be done this is a general tip that people use to give when working in teams. So this is the take home message you need to take here alright let's get back not said initially the person has not said the answer so it must may true now given that he has said the answer not said should we make false, not said should be made false. And now I should print him his code, the person name player's one name I have to print here his code your score ok let me say this is points of player one so how much as he score, I should let him know. So this is done till now what has happened we have unlocked the character we have modified the question and we for modifying the question we take the modified question from the previous run, in the very first run the modified question is nothing but the question itself with all letters encoded as stars and spaces left as such, this is the modified question at the first run and subsequent runs, the modified question will be based on what are the letters you have unlocked in the previous runs. So that is why we are using the modified question on both the places, this is the place where generally the beginners would feel it difficult to understand please pause here try to give it a thought, try to take your sheet of paper, you try running it as I had said an example take may be this particular name or black Friday or 'vikram Veda' something like that. Spaces as well as the name is also length here so take something like that, you would get an idea of what all could be the case may be user wants to guess one character he is not able to guess the movie just with one character and because of which in subsequent run what should be the expected modified question you just try run once you pause over here think understand this fully then lets proceed to it, so I would recommend it you understand up to here completely so please give a pause here understand completely and once you are completely thorough please proceed to the next part alright I hope you would have paused and understood the flow till here lets continue and each player must be given a chance whether he wants to continue the game or quit so that will have to ask here. Will say we are asking this along this it has to come so after he has said the answer so this is, this loop runs till he gives the answer after he has said the answer you have to ask him so this consenter column thirteen, ok column thirteen we should go to column thirteen I should ask whether he wants to continue, so for continue let me use c this is for continuing. I am inputting his decision his choice whether he wants to continue or quit the game so let me say press one to continue or zero to quit let me save this and I would see if zero is for quit so if his choice is zero, if his choice to continue or not if it is zero then I should give him a summary, so let me say I have to say print player one his score we have to print so let me print his name first player one name your score ok I have to print his score so which is points of player one, points of player one this is his score now I need to do the same for player two let me I am sorry let me copy paste the same because it's the same just that in place of player one I am going to use player two, I will paste it and I will have to change here see its player twos name and even here I have to change it it is points of player two and now I will just have some nice greeting message would make it more personalise so let me have a greeting message here thanks for playing, have a nice. This is up to you, you can give any number of messages as you wish, this is just to give a personalise feeling to the players that's it, have a nice day. So this is we have treated and now we are not willing to continue so this particular thing this value we had set willing as true that should be set as false, willing should be set as false, will make it as false. This is he is not willing to continue that is the meaning here alright so let us have a very quick summary of what has happened till now, we have the list of movies taken from imdb the top rated movies from imdb we have inputted the players name, initialised both their points as zero have a variable to keep track of turn, that is whose turn it is player one or player two, every even turn is of player one and every odd turn is of player two which is taken care of the modulo operator and this particular variable willing is to take care of whether the player wants to continue the game or he wants to quit it and when the player is willing and it is his turn to say this is his turn we pick a random movie create a question that is encode all the character with stars and spaces are left as spaces only, this is how the encoding as to be done. To create a question, print the question and then this will be the modified question for the first run that will be clear when we see this particular part and we have a variable to keep track of whether the user has said the answer or not said the answer, here the player has said or not said the answer so initially he has not said the answer so not said is true so as long as he did not say the answer continue it, you ask him to guess for a character if the character is present you have to unlock the character else you say the character is not found, that letter is present or not present based on that you take a decision, you say you have to unlock the character or you say that letter is not found and in unlocking process what you do is, you take the modified version in the previous run, in first run it would be all letters encoded with stars, you refer to the actual movie name and unlock all those spaces where this particular character has been present, you unlock it that becomes your modified question print the modified question and once the modified question has been printed the user has two options he can guess the movie directly or he wants to unlock another letter, in case he wants to guess the movie, you say this is the you ask him to answer if that is the picked movie say that it is correct and you increase his points by one and you output him his score and since he has said the answer this particular value of not said should be made as false and yeah here there is a missing case as you could see in case he wants to guess the movie name what if it is wrong? It is not the correct answer. What should we do? That has to be handled so let's say else that comes under else part that is his answer this is in column twenty five and this is also in column twenty five see indentation is important in python that's why we are checking it see column twenty nine yah column twenty nine so in case if his guess is wrong, he wants to say that this is the movie name for example if you could take, for example you can take this two 'drusyam' and 'nayakan' these two are having same seven letters so based on seven letters he would have unlocked some character and he suppose has the feeling for example the letter y is common in both the things suppose he has unlocked the letter y he may feel that he could have answered something but what if that is not the correct answer so we need to inform that it is not the correct answer so please try again, try unlocking some more characters and try finding the movie, that part has to be taken care of, so let us print a message for that, print let me say wrong answer try again, try again ok so he had said that it's a wrong answer so I am just asking him to try again so ok this part is taken care here now alright and in case if letter is not found we say letter is not found and once after he has said the movie name correctly you will intimate him his movie is correct and increase his points intimate his new score and said this particular value of not said to false because this is the variable we used to keep track whether the person has said the value or not said the answer or not as long as he has not said the correct answer this particular value will remain true and once he has said the answer this not said must become false you make it false and once not said becomes false the loop comes to an end so the game this particular turn is over so once it is over, you just ask him if he wants to continue or quit, in case he wants to, he wants to continue let him continue with the next thing in case he wants to quit you have to say this is the summary of the game till now you have to display his scores and give some nice greeting message and set this particular value of willing to false based on which we tracking whether the person wants to continue his game or he wants to quit the game so this particular thing you have to set it to false so this is how the game goes, this is for player one now let's see the same thing will be there for player two as well so let me just simply copy paste it but please make sure that indentation is proper let me copy and I am posting it here and please make sure that the indentation is completely proper. So yeah we may not increment, we have to increment the value of the turn why? Because after this turn it will be the turn of next player in case he says he is not willing if someone says 'I am not willing', the game comes to an end there itself. If there are telling 'I want to continue' then the value of the turn will be incremented based on which the next player will get a turn see if you would see every value of turn if it is even number it will be the turn of player one for odd number it is a turn of player two we need to alternate the turns, that's why we are increasing the turn value by one so initially you will get an even number then an odd number then an even number then an odd number that is player one then player two then player one then player two and so on this is how the turns are being alternated alright this is the outline of the game if you could see this is the outline of the game so maybe you can pause here and just see what all has happen, how is the flow of the game is going on and there are something which has been not yet oh yeah here is the syntax error ok let me verify it I need to use the brackets that's a syntax error. I print the question ok and yeah the same since even I have copy pasted the same error would have occurred so I am correcting this, here I would say there are some warnings occurring here, these names are undefined so let us define all these things in the next part of the video also maybe you can pause this, pause it here you can take a look at the flow of the game, you try to understand how it will goes and then in the next video everything will be clear once we have defined these undefined terminologies as well

# GUESS THE MOVIE NAME 05

Alright guys in the previous video we had seen the outline of the game and the flow of the game how the flow would go this is how the game is played all this we had seen let's give a very quick summary, we had collected a movies from imdb and we had inputted the two players name for personalisation purpose we had started with zero points each we had a variable to keep track of whose turn it is and we had a variable to keep track of whether they are willing to continue the game or not. And in the persons turn you have to display the that is his turn, pick a random movie from this list create the question that is encode all characters except by star print the question then as long as he says the answer you repeat this process you ask him to guess the letter whether it is present in the movie name, if it is present you need to unlock it else say that letter is not found, for unlocking what is that we are doing is, in that movie name if that particular letter is present that particular star is replaced by the actual letter and other letters just remains star this is how it happens, it happens repeatedly sometimes because with one character sometimes the person may able to guess the movie name in that case if he wants to guess the movie name, you get his answer check if it is the correct movie name, if it is correct display the message it is correct and increase his score by one point and say that this is your score and once he has said that it is correct this particular thing comes to an end, the thing of guessing the letter unlocking the letter these thing comes to an end. In case if he is wrong you have to ask him to try again and in case if he wants to unlock another character and he doesn't want to have a guess, you need to repeat the same process again in this particular subsequent round of unlocking what you have to do is, the letter that is already unlocked must remain unlock only that shouldn't be encoded by stars again so the thing that are noted know to the person is what encoded as stars so this is how the game goes and after he has given the answer ask him he wants to continue the game or quit based on that you act accordingly if he wants to quit say that this is the scores and this is the status till now and give him a good greeting message and say that you want to quit the game so just say willing is false where the games comes to an end, in case the person wants to continue what should be done is you should increment the value of turn as you could see here you could increment the value of turn so that the next player gets the chance to play, the thing continue this way you can continue the game on and on as long as not even one of them wants to quit the game, you can play the game on and on so here there are some things that are undefined you could see this one create question is present and unlock this three functionalities are not yet defined so let us define them one by one create question this is functionality that is not defined so let me define it so let me say here let me say I have to define the particular process of creating a question so let me say define this is the way you create a new function, this is creating a question is a function you want to be created so we will create this question based on a given movie there movie will be given here so based on a movie. You need to create a question alright so how do we do this? First let me capture the length of the movies let me say n let the movie contain n characters this included spaces as well please note that this length include spaces as well so for example if you would take this thing if you would take 'golmaal' movie there is three characters under space and four character so it will be counted as eight characters so it is ‘golmaal’ has three characters space and four characters so it will be counted as eight characters so this is how the length of the movie will be counted so I need to calculate the length for this we have a pre defined function length, length of the movie that has been fast raved that is for which movie you want to create a question get its length and for encoding we need to process the individual characters if it is a space you should remain you should keep it as a space only otherwise you should make it a star so we have to process has individual characters so to get individual characters what you do is maybe you can say may be I will say letters, let me say the letters is the list I am creating out of this movie name list of movie, movie is a string that is this as a single word will be passed so that would be decomposed into individual letters by this functionality list so that letters would be a list. We will have now also create a new temporary list may be I will call it as temp that will be used later we will come to know what is the purpose why we are creating this new list because we need to output the encoded format so the encoded format based on individual character I will have it has a list and then I will join it to a single string that is how I am going to do it so let me do that, so we have to hydrate over that is we will move over every character every letter of the movie name so that is present every letter is present in this list letters so I have to move over this for I let me have a variable most over I in the range of how many letters are there? That is captured by this value n which shows the length which calculated the length range of n ok right? if what we should check, if that is space I should just keep it as a space otherwise I should make it a star so let me say that is found in letters if letters at this position I is equal to space then to this temporary array to the output array I have to append that is I have to add at the end, end of the current list I have to add a space else if it is not a space I have to add a, add the same character, I have to add the character star so every letter is encoded by star except this face that is taken care of here if it is a space you append a space else you append a star that’s what you have done now your temporary array contains the encoded form of your temporary list contains the encoded form of your movie which is nothing but your question you call that as your question so you need to return it but your question is a single word not a list so you need to make it a single word for that we have a built in functionality called join, so let’s make use of it and we will join it so let me call it as may be let me say it is a question is nothing but irrespective of any character just keep joining, just keep joining every letter you convert the letter into a string format because string format it is easier you to join so you are converting into string format for every letter x in the temp, so how this works, let me once again say this temp is the array which has, temp is the list which has the stars and spaces with respect to the chosen movie name so say for example if our movie name is ‘golmaal’ then with respect to g the star is appended then with respect to o the star is appended with respected to l the star is appended here is a space so space is appended then star, star, star, star this is how you have encoded and you have it has a individual character for each of the individual character you convert it to a string and keep joining into one composite string that is as one question so this I don’t want it as a list of eight characters but, I want it as one string of eight characters that is how one word of eight characters something like that I want, so that is what I am doing by this join function so I have created this question so let me return the question that’s it create question no words, if you would see here then you would say create question what you do is you calculate the length of the movie you convert it into individual characters the list of individual characters because based on the characters you have to append that is if it is a space you just keep it as a space that's what is taken care of here if it is a space then keep it as the space itself else you have to put a star and then now from the list you merge back and get back the original string, get back as a string that is a question so you get back the question you return that question that is instead of having a list of some characters you merge those single individual characters and make it as a single string and return that single string that is what we are doing here with the help of the join functionality all these are pre defined functionalities alright so create question part is over so sorry this is picked movie sorry for this picked movie, its picked movie given that I have copied pasted even here I am sorry this is picked movie not pick alright ok so now the warning occur at is present define name is present and undefined name unlocked so we will define them slowly so let us define that, define so let me a single functionality is not define is present is present so what are the characters what are the things I have passed one letter and a movie so I have to see if this particular letter is present in the movie name so this is what I have to check in this functionality so let me say I will have a counter, so for this what do I do is my logic is I will count the number of occurrences of the letter and this movie there is a functionality pre defined functionality for strings to count the number of occurrences of characters say for example my name vidya, if I want to count the number of occurrences of letter a in vidya I have a functionality if I pass vidya and a to it I would get back one as the answer because letter a occurs once in string vidya in case if I would give the input as t, letter t is not present in vidya so it would output zero so it would return back the answer as zero because this occurrence of t, that is letter t occurs zero times in the word vidya where as the letter a occurs one time in the word vidya, if it occurs multiple time may be if you could take some may be you take from the same movie itself say 'dangal' letter a occurs two time so it will return two so if the letter is present it will return non zero value that is some positive value if it is not present that function will return a zero value, if the value is zero then the letter is not present that is the logic we are going to use say we are making use of the pre defined functionality of strings ok so I need to get the count so let me get the count so let us say c for count equals to movie dot count that's it this is the functionality count of letter so this particular functionality will count how many times this letter occurs in that movie name so it will count if this value of c is equal to zero that means that letter is not present so is present this what you are asking if the if that is present so given that it is zero it means it is not present so you have to return false return false sorry false else what should be the case? It should return true I guess this is quite intuitive only thing that we are using is this is the pre defined functionality count that counts the number of occurrences of the particular letter in the movie so if the movie doesn't contain that letter then it will return zero which will say that false for which we are returning false that is this letter is not present in that movie name or else we will say it is present so is present functionality has been find so if we could see the warnings are disappear the warnings are disappeared so here only warning that is still there is unlocked so we need to define this, let us do it or maybe you can pause here for some time just see to the flow once, once again and now we have defined this two new functionalities you would, you may even see to the flow and pause it in that particular point where this functionality is being called you trace what is happening how things were done just trace it for clear and understanding so maybe, I would recommend you guys pause here understand the flow completely and then proceed, now comes the unlocking part we will do it so before going to the unlocking part please do make sure that you have understood the flow till here. Alright I hope you guys have paused and understood the flow till here let us get with unlocking part define unlock? For unlocking what are all the things I am having the modified question let me call it question itself here it doesn't matter and then you have a movie name that is the picked movie name you have passed that you have passed then the particular letter alright oh here it is letter than I am sorry for this, this is letter because we are not using the term character, character is the technical term that generally being used in the computer science for letter, letter is a intuitive term for anyone who knows English, so its better if we use letter itself so here also its letter sorry for this confusion ok so far so good, so we have requesting that is the status till now that is how many letters have been unlocked or nothing has been unlocked all letters will be encoded as stars and spaces will remain as spaces if something as been opened then that particular letter will be visible others will be encoded as star and the spaces will remain as spaces so the movie will contain the actual movie name and the letter which the person wants to unlock that is this will in this case note that it will never happen that letter is not present because here itself we are making a change if the letter is present only then we are getting into the uncloaking part if it is not present we just say that it is not present so here that particular thing you need not worry about what if the letter is not present what will I unlock if you need not worry about it the letter will be definitely present. Ok so I need to process character wise so similar to how we did create question part we will break the string into list of characters that is what we will be doing now, we are going to refer to this movie so let me call it as reference, reference list, list of movie ok then I want the question list as well so let me call it as question list or may be ql for short anything but it's not a problem we can have that is question list, the question list is the list of the question, question is nothing but the status till now how many characters are under has been known to the player and how much has been not untracked so this is the question basically here and same like that for output say we will have a temporary list, if you could understand the create question part well this would be easier for you so that's why I had recommended that you guys please do pause and take a look at it in case you feel again you need to take a look at it it's never a problem you can pause here take a look at it you understand that really well then now proceed here alright so, I believe you guys are now very clear with all the parts, so let's proceed similarly, I would find the number of characters in the movie that is determine using the pre defined functionality like number of characters in the movie. So let me iterate in the similar way for I in range of n so for each of that what should I do, if the particular character in this reference list is the actual letter or if it is a space you have to append the space only else you have to append the stars or the characters that is already there as per the scenario so let us do it. Let us first have it if letter in the movie actual movie name if the letter is equal to sorry space or if that letter is equal to the letter that you had asked what should you do? You should append the actual letter to the output; you should add the actual letter to the output. Why do we do that? If it is a space or if it is desired letter in that particular position see for example may be 'anand' is the movie that is been picked and a is the letter they are seeking for at the first position the letter is a which is the desired letter so in that case I should reference that list I should move to that list and take that particular character and append it to my output, so my output are also showing, a is the letter by player is seeking to unlock an a is present in that position so a has to be appended in the output. Else if it is none of them else what should I do? I should check whether this particular question list has star in it in this position or it is not star that mean say for example the same movie anand in the first iteration probably the person has unlocked a and he wants to unlock d now let us just assume he wants to unlock d so what is the thing that noticed, so our letter now we want to unlock is d and the question is a star a star, star and movie is anand this is what is pass to us so the reference list will contain anand in single characters a list of single characters the question will still contain a star a star, star as a list of characters and this is our output list and we have taken the length and it is this reference list at this position is not a space also it is not equal to our letter d so it will come to the else part so you have to check so he wants to unravel d it shouldn't be the case that you should give him star, star, star, star d you should give him a star a star, that should be the output right so you have to check whether the question list has star or some other character that is that character might have been unravel during the previous runs so you have to check it so let me check it if the question list at that particular position contains star, if that is containing star then you append star then you append a star else what you do? Sorry it's a spelling mistake append ok and here it is else you append the actual character sorry, sorry it's again a spelling mistake append the actual character where is the actual character present? It is present actually in both the places in this reference list as well as the question list because during the previous run of unravelling also you would have referred to the reference list and unravel so you can use any of the two question list of I or reference list of I that is ith position of question list or the reference list you can use any of them. Let me use reference list, reference list of I so I have appended it ok so in a similar fashion so now given that I have iterated in this would be done for all the characters all the letter of the movie given that we have completed now we have we would have got the output list as a list of characters we want it as a single string so similar to our previous operation we would do a join functionality so let me say this is my let me say this I will copy, this is what I am basically precisely doing this way I am copying and pasting it so question is already given so let us say this as question new just for clarity sake I am saying this is question new, new question we got and now return the new question so whatever we have done is almost very much similar to what is done in the create question part so if you could understand that there you can definitely understand it what you need to take care is if it is space or if it is required character then you append the required character else you see if it is star that is it has been not unravelled the previous run then you append a star else you append a actual character here you could use reference of I or question list of I as well anything can be used because during the previous runs of unravelling as well you would have referred to the reference list and you would have followed the same procedure so it doesn't matter both will actually be the same value you could use any value and then now we are joining the output list into a single string and we are returning that string this is how unlocking works alright? Now may be you can pause and have a look at the entire code how things go on, how everything works, you just have a look at it, let's run this in the next video.

# GUESS THE MOVIE NAME 06

Alright, in the previous video we have seen the various parts of the code we had seen the play method and the various other sub methods it is using so let us just have a very quick recap of it, we had collected the list of movies from the imdb site we had inputted the two players name and we had started off with zero points each, we had different variables like turn and willing as the name suggest it make sure that turn is alternated the game goes when both are willing to play even if one person is not willing to play game comes to an end and to alternate the turns we are making use of this mode operator and during each turn of the person you need to randomly you pick a movie and create a question how question works here, if you could see create question what it does is it splits it to the individual characters it sees if the individual letters it sees if the letters is a space you just keep it as a space some other letter other than space you encode it as star. So this is how question has been created and once question has been created question is displayed to hum and the player is asked to asked to guess the character guess a letter guess a letter that is present in the movie name so he has to guess the letter, once he has given the letter once you first check the letter is present in the movie name or not so to check if it is present in the movie name we are making use of a built in functionality count that will say how many times this letter has occurred in the particular word so movie is our word in that movie how many times this letter has occurred, if it has not at all occurred it will return zero otherwise it will return a non zero positive value so if it is retuning a zero you say it is false that is it is not present is present while we are playing is false otherwise it is true so that's how we are returning the false and true values correspondingly and if it is not present you say that this letter is not found if it is present you have to unlock  the unlock the letters, so for unlocking what do you do? You do something very similar to that what we are done in create question we open the sting into list of letters you open the string in the list of characters and based on the each character if it is a space or that actual letter which he has wanted to unlock you append that letter into the output string otherwise if it is not unlocked yet so and this is not that letter as well then you have to append star else you have to append letter that has been unlocked till now this is how we have created the output list now we merge the individual characters of the output list into a single string and return that single string so this is how unlocking has been done once unlocking has been done you print the unlock version and based on that the user can take two decisions either he can guess the name of the movie or he can unlock another character if he wants to unlock another character the same process is repeated if he wants to guess the movie name you ask him to answer check if it is correct, if it is correct say that he has said the correct answer and increase his points by one and show him his points and once he has said the correct answer this particular thing of asking for a character unlocking guessing all this come to an end so this is how the movie goes sorry this is how the game goes and this once he had said the answer what should you do is, you should ask him if he wants to continue the game or not if he wants to continue let him play that is then the value of then the if you say he wants to continue then the value of turn is increased by one in order to fascinated the next player to play the game in case he wants to quit then the game totally comes to an end before ending the game you show the status greet them well and then you end the game this is how the play goes ok before starting it running of the code I would like to just say something here if you would see I have typed as with hint these are all some machine independencies may be some operating system independencies in some cases it may happen that you need to type it as or in some system  Linux based system I have used you need to use the function raw input for string and input for the other data types these are some few decencies you can always all ways look it up to what is the problem online and you can fix it please say some independencies on Mac os so I had rectified them this is very trivial just I am type it as string it into if some of you have already familiar with java or heard of java even java works like this takes all input by default as string so we need to convert it  into the type whatever we need here we know that the input is either one two something like that its a number so I am typecasting it as a loop so this is just a few independencies so for a Mac this is not independencies for Mac os for Linux os one independencies I observed was it takes two functions raw input and input so you can look it up depends on may be for windows there may be some independencies you can always look it up in case  if something doesn't work you can always look it up, you can see as to what is an solution, you have lot of solution available online on stack overflow there are many such sites where people generally post their problems and there are many who help with solutions to all these this purely managed by people it's amazing you get good response there and if you know some answers if you contribute that is different feeling that is different kind of feeling proud feeling you would get I would recommend that you guys be a part of such community where you can interact with your fellow programmers and you gain more knowledge as you would generally would have heard pier knowledge is something that is uncomparable to other types of knowledge, even in your college the study time college or school events you could remember you would have done a group study those memories are something those things are something that as no match for it so it is always the case that clear learning has a very good advantages associated with it so one such clear learning clear learning thing for programming is this thing stack overflow or such online platforms you can always look up to it may be you can try reading through solutions already presenting can gain more knowledge and you can also contribute if you know something there  is website for almost everything for hose not from computer science background as well stack exchange has got lot of websites you can see to it may be you would have seen through it already if not also please be a part of such community so where you actively discuss about the current happenings of your field and you stay updated with it so this is I am just telling this so that you guys can be a part of this amazing community where you stay updated with what happenings in your field that's it this thing I had just looked up and I had included it nothing now let's run the program. Ok player one please enter your name ok if he enter let me say abc player two please enter your name let me say xyz ok some characters are there one two three four five six seven eight, eight characters without any spaces so I am not able to guess what is the movie so let me play smartly, it is generally a practice that they you usually unlock vowels you try with all vowels then why this is generally a practice in almost any movie name you see you would have this letter occurrences almost all the time almost all the time so this is general a practice so I will start with a so a is present only once so this is unlocked ok and maybe, I want to unlock another character I would unlock e, e is not found ok let me unlock i, yeah I is found let me unlock another character o, o is not found u, u is not found y not found, yeah I guess now I know the movie name let me guess it 'drishyam' correct? We will see your score one which says no I want to continue or quit let me continue alright oh this is the problem we had to make the small rectification then everything will work good, let us do it very quickly see I have not change the name here its p two its player twos turn I had copy pasted I had missed changing the name so let us do it very quickly so this is how it happens actually you run and by then something unexpected comes then you come to know that oh I skipped it, this is how the programming happens so in case you are making mistakes please never get disheartened all these is learning process failures are the stepping stones for success they say this is very much true in case of programming through lot of mistakes only you come to know what ever is the correct way t two I have to change here and yeah somewhere else yeah here I have to change same lines so ok somewhere else I have to change, do I have to change anywhere else no I guess yeah fine alright let me save let me open a new console letter let me open fresh console and let me run it run ok player one abc player two xyz your turn someone two three four five six seven eight, eight characters let me guess a I am lucky that I am probably getting the same movie let me unlock another I get I yeah I got the same movie let me guess now its drishyam correct? So let me continue xyz your turn one two three four five six seven eight all this is probable actually because we are randomly picking a movie that's it so it may be the case that both the players get the same thing this is highly lesser the probability but if you want to make it still the least, you want to decrease the probability that both the people get the same movie name you increase the number of movies in the list that is what just here we have ten movies because we were not interested in scrolling through the imdb list our aim here is to learn the programming part and not to watch movies so I just had to pick some ten movies from imdb maybe if you would pick more movies there were two fifty movies they were displayed there if you would pick some hundred movies two hundred movies this probability that the both people get the same thing is very rare so maybe I hope next time I will get some other movie yeah I am getting some other movie so let me say with my letter 'a' these are the things so let me unlock another letter 'e' yeah now I am I guess now I am able to guess the movie see with vowels you are able to guess the movie very quickly 'taare zameen par' so wontedly let me say without spaces I want to check if it will works fine wrong answer! See this not excepting the movie name contains spaces so you have to give spaces it doesn't except the other way perfect! So I have to again say some letter so let me say 'r' even that I know that this is the movie name say 'r' ok so it is unlocked so I have to guess the movie name this time properly with spaces I will give 'taare zameen par' ok yeah abc your score is two and I want to continue xyz one two three four five six seven eight oh this xyz must be very lucky person he is getting the same thing this is this occurs because we have a very less number of movies, if you increase the number of movies in the list you will get a better random a better choices every time yeah fine continue again I am getting the same wow ok oh its some different movie alright see I was saying the same movie again and again and I was thinking this is same movie no this is different movie so I am I don't know the answer what would I do? Ok let me give my same answer it will anyways it's a wrong answer let's see yeah I have to try again with some other letter ok a e I o u y alright now I know the movie now let me write 'nayakan' sorry I should say that I want to guess the movie then I will so ok I will write 'nayakan' perfect! Now, I want to quit let me press zero see I am greeted, abc your scores is this, xyz your score is this and I am greeted alright this is how the program output had appeared and this is how the game was played, you would have seen in that and during the run off the game one thing you would have noticed this sometimes one movie got repeated often some movie may get repeated often this is usually a case even when we play manually right? see suppose assume that you are playing same game for some one hour two hour something like that then sometimes it may happen that you forget that you had all already set the movie name you already asked that as a question and you may pose the same question again it happens, the same thing happens with the computer as well, computer how does it does? It has a list of movies it picks a movie name randomly it's purely a probability so we can never guarantee that you will never get a repetition that is in case for example say the movie 'dangal' if it has been picked I never give you a guarantee that again 'dangal' will never be free so that we can never guarantee in this randomly picking method because for this repetition to occur the probability associated the math behind it and the probability is not zero probability it is not a zero probability there is some minimal probability with which this repetition can occur, it is evidently visible because we have only ten movies in our list, in case if you increase the number of movies if you have some hundred movies in the list this probability of repetition is very, very less and you it is not very evident that there is a repetition but there will be repetition the probability of it is non zero so that there may be repetitions may occur there, you can never guarantee that repetitions would never occur so may be for that you can think of some strategies can you think of some other way other than randomly picking from a list can you pick the movies such that repetitions will never occur can you pick the movie from the list from some other method you can brain storm with lot of various methods as well, as well as to implement this, this is not the only way there are many ways by which you can implement the same thing you can come up with the variants in the implementation we have done you can probably make this particular thing we had copy pasted for player one and player two right you can make this particular thing as a sub function and you just make it as a single function or instead of making this many lines so that is your you making this as a single function and reusing the same function when needed you can do that, that is one of the improvisation you can do with this implementation and as good as a lot can be done, this is not the one standard way there are many ways also for the game there are many standard variants may be you can have some constraints like maximum ten guesses, if allowed more than you cannot you cannot allow the person to guess something like that, you can have constraints and may be you can vary the points based on in which guess he gets his right you can vary the points in, you can do anything per imagination sky is the limit you can do a lot of variants you can go on and on you can you can come up with many different implementations you would you would you can even create some brand new game with this as well with the different variants set up as I had said with different incentives schemes with different points and you can have a any number of any type of game you can create with this. So I would recommend that you guys do give some thoughts over it come up with your own variants do discuss that in the discussion form and as I had said in this video itself clear learning is something that is that has no match, so I would recommend that you guys please discuss in the discussion form and faster clear learning you also enrich your knowledge as well so thanks for watching the video till now please do take part in the discussion form please discuss your variants your different strategies of implementation or any idea regarding this you have got you can please do discuss that in the discussion form thanks for watching till now have a nice day.

# DICTIONARIES

Hello guys, welcome to yet another programming screen cast, in this programming screen cast we are going to see of data structure called as the dictionaries. I hope you are familiar with the term data structures because in your precious weeks you have been introduced to data structure called list. This week we will see a data structure called dictionary. So dictionary as the word says it is similar to the English dictionary or any language dictionary for that matter. How is the data stored in the dictionary? You have words and meanings corresponding to it so there are two items in this dictionary words and the meaning something like that even in python dictionaries you have two elements and the relationship between the two is what is model by this data structure called dictionary, technical terminology is key and the value is what they say it is key and the value is what they say, key is nothing but the unique identifier for a given value for example this is the common example which people generally say there may be multiple people sharing a same name in a college in that case to identify a student uniquely the college may assign some id number something like that there is unique id associated with an item that is called as key and the value is nothing but the other details of the item for example assume there are more people by the name amit so if there are multiple persons with name amit they may assign some id number and they may store the person's name. So that the id number would uniquely identify which person is being referred to as something like that, in such applications this dictionary data structure is highly useful, so we will take some other example this student id and the student details is something with generally people say so let us take some other example in this screen cast, I will take the currency conversion concept so here our keys that is our unique things will be our name of the currency and the value will be the conversion factor so I will take Indian rupees as the base so I will take the different currencies dollar euro yen whatever you say I will, dollar is the key and value is nothing but one dollar equals how many Indian rupees? That will be my value this is what I am going to take in this screen cast. So dictionary as you could see is a data structure that models this kind of relationship that there is a unique identifier and value associated with it. So two, two elements relationship what is captured here ok let us see how shall we create a dictionary. Very simple, how did you created a list? You said l equal to this kind of brackets right? This will create a list something like that for creating a dictionary you have to give the name of the dictionary conversion factor let me name the name that itself conversion factor this is a dictionary see this is the curly braces as you could see her, these are the curly braces, curly braces denotes the dictionary, so if say this and press enter a empty dictionary gets created so if I say conversion factor and press enter see an empty dictionary is what is shown as the output, so now what should I do? I should start adding the values that is I should adding the conversion factor for various currency systems, as I said I will be taking the Indian rupees as the standard one to which we want to convert, so that is from this particular currency system we dollar or euro or yen whatever be there from there we want to convert it to the Indian rupees that's what we want to do here, I want to add the value of dollar please note that these are just approximate value I want to add the value dollar so how should I do that? Here goes the syntax please observe, conversion factor this is the name of the dictionary and I need to use the same brackets that I used in list, here is should give the value of my unique key here the value of my unique key is I will say dollar, dollar is my unique key and equal to what is the corresponding value, I assume that it is sixty rupees, one dollar is equal to sixty rupees this is just my assumption so I give the name of the dictionary as you could see here I gave the name of the dictionary this brackets this square brackets within the square brackets I gave my key that is the unique identifier and equal to the corresponding value this is the syntax of how you have to add a new item, so new item got added how would we know? Let us just display the conversion factor see a new item got added dollar colon sixty so dollar is the key corresponding value is sixty it got added, if I want to add further more item I can add it let me do that, euro equals let me say eighty this is just an approximate see I print it now see you got it, dollar colon sixty, euro colon eighty another item got added up so this is how dictionary is worked you have a key which uniquely identifies your value and the associated thing is called as a value. Now let us see how can we view the dictionary? So this is in console I am doing it, but when you do it do it in your programme you generally use the print command just like how you gave print the list something like that you can give here as well print conversion factor which is the name of our dictionary, you can give this see this got printed dollar and then euro and now let us see how can we access the individual items. So I want to know that is the value of euro, how would I do that? I would say conversion factor of the key is euro that is I want to know the value corresponding to euro, see eighty is shown as the output that is corresponding to euro the value stored is eighty so this is how you access this specific value, you retrieve the specific value now if you want to list out all the keys present in your dictionary, you need to say conversion factor which is the name of our dictionary dot keys this is the functionality if I put this see I got the list of all keys that has been enhanced into some special data structure so if you wanted to be there in the list I need to convert it into list, you could see here if I converted into a list I got the output as a list this list the all keys that are present in your dictionary if you want to list the values that is the associated item you want to list it, it is simple instead of keys you put values and as you could see here the values are listed corresponding values are listed in case you want to fetch both keys and values simultaneously that is what are all the items that is present in the dictionary is what you want to see, the dictionary name here it is conversion factor dot items it will list the items in the dictionary see it says dollar comma sixty euro comma this whole thing what you call as tupple that I something but something which is constant as dollars and sixty are associated ones you cannot separate it something like that also if you want to know what are all the functionality that is available conversion factor dot and press the tab that is you get the all functionality whatever is available from this whatever is the item you want to know you can type that for example let me say pop is the functionality I want to know, I want to know what this functionality does? You just put a question mark at the end so it gives the documentation corresponding to it so it says it will remove the specified key and return the corresponding value the key is not found it will raise an error for example here dollar and euro is there now if I say conversion factor of yen I know that there is no such key but still let me show you see it has raised what is called the key error states that this particular key is not present in your dictionary so the key error would be raised in case if the key what you ask is not present otherwise it will remove the key and return you the value associated with it, there are lot of functionalities available as I had said using the tab key you can get to know of all those functionalities, now we shall see updating, this for example I had written dollar is sixty and euro is eighty and I suddenly came to know that the value of dollar has been changed so I want to update it, how would I do that? Please observe it is very similar to insert the thing on your key conversion factor of dollar is equal to the new value, let me say not sixty its sixty five let me say sixty five and enter now let me list the dictionary see it got updated it was sixty initially it got updated to sixty five so this particular statement is same for create as well as update, if such a key is present already it will update the old value and write the new value if it is not present it will insert a new value that is what this statement does. This is the same syntax for both create as well as update so how shall we delete a specific key from the dictionary so let me add one more element conversion factor of yen, this is just my approximation I assume that it is fifty rupees this is my assumption ok so let me enter so now let me check what is in my dictionary so my dictionary contains dollar euro yen now I want to delete specific some values see for example yen is something I had inserted but then I realise that I don't need it so I want to delete it so how would I do it? Observe the syntax d e l is the key word for delete del space name of the dictionary name of the dictionary and inside that you need to give the key so it would search for the value corresponding to the key and delete that key value back this what this functionality would do, key would have got deleted now let me display see it was deleted see if the initial stage when we print it we had yen I had deleted it so if you would see in this updated dictionary there is no specific value corresponding to yen in case I give some key which is not in the dictionary it would throw a key error as you had seen earlier so this is how you handle with dictionary as I had said let us do a small currency conversion so I want the value in rupees, this is thirty Euros I want the value in rupees it is nothing but the value of Euros currently e into whatever is the conversion factor conversion factor corresponding to Euros is what I wanted now check the value of r it's two thousand four hundred so thirty Euros is nothing but two thousand four hundred so in such situations you see a data structure like dictionary is very handy, I would recommend that guys use this tab key and explore whole lot of functions that are available in dictionary keep exploring keep learning happy leaning thank for watching this screen cast have a nice day.

# SPEECH TO TEXT – NO NEED TO WRITE - 01

Hey shubudha! What happened? I see you are not able to sync your lips properly and that's why your expression are not good. Actually the problem is I don't know Punjabi and I don't know the lyrics and its meaning so if you can help me with the lyrics then I can sync my lips according to it, I hope that won't be a much trouble for you. No that is not a trouble for me; I will just write a programme and let you know the lyrics. Oh! Programme that's awesome. Yeah I can write the programme for this, yeah sure then you give me then I will by heart the lyrics ok perfect.

# SPEECH TO TEXT – NO NEED TO WRITE - 02

Hey everyone, as you saw shubudha struggling with the lyrics of the Punjabi song. Here I am to help her through python, through coding. You must be surprise how can we do it. In python there are many state of the art technologies one of them is speech recognition, speech recognition is an apn that has been provided to us by goggle. In this particular library it will just convert the audio file that you supply to text, it will convert the audio file to text so it will be easier for shubudha to by heart the lyrics then and improve on her dancing skills. So first of all how can you install it? Yes you first need to install it, you just use the command sudo pip install speech recognition, you just go to your terminal and type sudo pip install speech recognition it will ask you the password and then it will be downloaded to your pc, so it is very easy to install this particular library that is speech recognition. You just need to type on your terminal sudo pip install speech recognition and then there are many steps involved here to I will just you to create the audio file with the help of your microphone as I did just now. As I am Punjabi, I know Punjabi so I just spoked the words of the song in microphone and recorded the file then I converted that particular file to dot wav extension, yes whatever you record whatever you need to convert to text u need to convert that particular audio file to dot wav extension, I repeat this is very necessary speech recognition only works on dot wav extension files, you must have recorded the files and it maybe in dot wp4 and any other extension but if it is not on wav extension you need to convert it into wav extension so first of all you have to create and audio file with the help of your microphone and then you need to convert it into wav extension. Wav extension your audio file can be convert it into wav extension online, there are many sources that are readily available, you can easily convert your audio file into wav extension so basically what's steps are involved here, first of all you have to install speech recognition, I repeat you have to install speech recognition with this particular command that is pip install speech recognition after that you have to create an audio file with the help of your microphone. Whatever you need to convert it to text and then you have to convert this particular audio file to dot wav extension these are the three steps that are involved here, first of all you need to do these three steps and then you will program. So let us do this three steps and program.

# SPEECH TO TEXT – NO NEED TO WRITE - 03

Hey everyone, I am done with the steps, I first installed speech recognition then recorded to the file through micro phone and then converted the audio file into dot wav extension so these are the steps I have asked you to do I hope you have done with the steps, since you are all done with the steps let us start with the programme. So first of all you need to import speech and recognition that you have installed say it is basically import speech underscore recognition as sr you can use a name sr or you can just go with speech recognition it's your wish. So first of all I create a variable name audio file that I will assign to my audio file so my audio file is sample underscore simran dot wav, speech note the fact speech recognition only recognises dot wav file so you have to convert your audio file to your dot wav then it will basically use audio file as the audio source, you the audio file as the source then you have to initialise the recogniser sr dot recogniser it will basically initialise the recogniser after that what you have to do is with sr dot audio file because you are using audio file here what name has been assigned to your audio file mine is audio underscore file we will use it as source, what you have to do here is audio is equal to r dot record source, so this function will basically read your audio file reads the audio file. So now what we have to do now here is we have to write try print audio file contains r dot recognise underscore goggle audio it will basically print the text, print the text that is there in your audio file, it will convert your spoken words to text, except sr dot unknown value error, you must be wondering what I am doing here, here basically it will raise an exception sr dot unknown value error in case it is not able to understand the audio, you will have to raise the exception here in case your goggle speech recognition module is not able to understand your audio file so I will write here print Google speech recognition could not understand audio, you are also have to write another exception that is except sr dot request error this will be, this will occur if the we are not able to get the results from Google speech recognition so you will write here couldn't get the results from Google speech recognition so I use the word here exception instead of error please note the fact that exception is a type of error, I repeat exception is a type of error .so as you saw we first use the audio file as the source then we initialise the recogniser and then we read the audio file after that will Google speech recognition will basically recognise your audio file and convert it into text and in case it is not able to do that it will raise two exceptions, it can raise two exceptions first of all is if it is not able to understand the audio and second one is if it is not able to get the result from Google speech recognition, so we are done with the programme here let us try to run this programme and check whether I will be able to help shubudha or not, let us try to check that. Yes we have the audio file here; audio file contains that particular song and let me check that, yes! That is right, isn't it amazing! We can just record our audio file and convert it into text so I am very happy I can help shubudha and I think she can by heart the lyrics and improve on her dancing skills I think so. So as you saw speech recognition is a state if the art technology that has been provided to us by the Google through python, so you can use this and convert your audio files to text, it is very very easy through python so thank you I hope this programme is very useful to you and have a nice day happy learning.

# MONTE HALL - 3 DOORS AND A TWIST 01

Hey you are eating something? No nothing. Something there on your mouth nothing, is that chocolate? Yeah, hey you are eating chocolate, yeah show show show this is almost over wait I have one chocolate  I will give you guys would you guys please share that? No, no, no, no I want one chocolate, I won't share with you, I want, no no last time also you took last time also you took no I want no I want, I won't share I will not share please share it please share it hey wait, wait, wait, wait see I have an idea let us play a game ok, if you are lucky, if you win the game you will get the chocolate, are you ok for this deal? Ok I will play the game, anyways I will win I will not share with you we will see we will see, will see will see let's play the game yeah will see. so as I have said I have hidden my chocolates in one of this three bags, you are suppose to guess where would I have hidden, what would you like to choose? Hmmm I will choose this bag, this is your choice among the three bags let me make your life simpler, I will open this bag this is empty so the chocolate must be present in one of this two bags would you like to retain your choice or you would like to swap? No no no you are just tricking me this is my final choice I will stay with this, this is your final choice then I will close this ok now your turn is over I will call shubudha after her turn I will announce the results ok shubudha come as I have said I have one chocolate I have hidden it in one of these three bags you are suppose to guess which bag I would have hidden so which one you would like to choose? Out of three I have to choose one yeah, I will choose this bag, this is your choice yeah among three, yes let me make your life simpler I will open this bag, this is empty so chocolate must be present in one of these two bags would you still like to retain this choice of yours or you would like to swap? Oh this seems to be an interesting strategy let me think, I think I will swap I will swap this will be my final choice. Ok shubudha your turn is over, you had chosen that bag has your final choice, yes I will close this and I will call amit and will see the results ok, amit come ok. You had chosen this bag to be your final choice this contains, this is empty oh god and shubudha you had actually made a swap when I gave you a chance, yes, you initially choose this and you choose this let me open simultaneously, this is your final choice so let us zoom into this, this contains the chocolate and you win the chocolate, this is yours, thank you I won, can you please give me a bite no no no we had decided that we will not waste at least half half no no, I will not share one fourth at least no no no no I will not share. Ok so we saw that the mother is playing trick on the kid, she is trying to force him to swap force him to change the decision, the point is, is it really required what exactly is happening here? I am sure we all are confused right? So let me do one thing in the beginning of the video beginning of this lesson you saw three people playing a game right? The chocolate game and vidya takes them and makes them play a game while she puts a chocolate inside a bag and the point is to find out what is that bag which has the chocolate ok and amit actually loses the game correct? Let me now call amit and ask him this question why exactly he lost the game? Was it out of shear bad luck or was it because his strategy wasn't correct. HI amit, hello sir, I saw the game that you played, it was interesting, I am just wondering if you understood the nit critics of the game, the game is not simple as it appear to you right? so what just happened in the game, vidya tempted you people to play the game to win the chocolate correct? As she was smart, she said she opened up a new bag which did not have a chocolate and tempted you to change; you did not change while shubudha changed right? You lost shubudha won my question is did shubudha get lucky here and did you get unlucky? Or if you are given a second chance should you swap? Or should you not swap? What do you have to say? Sir, I don't think it matters because in the end I have to choose one cup out of three, yeah so why does it matter that I have to swap? My chances are same, looks like it correct yeah, out of three I have to choose one in the end I don't think it make any difference I don't know I think it was just a luck, yeah so you know life is full of deceptions we think we think this is not the right way to do but that will be the right way to do, sometimes we feel that this is the right way to do and we will end up realising that this is not the right way to do here is one such lesson of life where I am going to show to you that swapping has a huge upper hand over not swapping by that I mean when vidya tricks you, you should succumb to the trick, temptation of changing, if you decide to always change your mind, you have higher chances of winning the chocolate shall we see how? Sure. Have this three cups with me one two and three perfect as an example I don't have chocolates I don't have jar so we should adjust with this, I need something to ok correct, this will be the chocolate alright? I am going to hide this some where you are not going to say it right? This is the chocolate three glasses alright? Ok. Turn that side you do not know where the chocolate is ok there are two glasses where there are no chocolate and there is a glass with a chocolate, let say ok so let me clear my table so that it is visible to you right amit? No chocolate one chocolate here start, I have to choose yeah, two glasses do not have chocolate one glass has a chocolate; hmm I will choose middle one, middle one? Yes it doesn't have a chocolate you choose middle one touch the middle one; this was your choice correct? You choose this yeah but then I am going to show you that this doesn't have a chocolate and now I am going to throw this away from the scene now you are left with two choices this or this? If you want you can change your mind, change your mind and choose this or don't change your mind and stick to this what would you like to do? I will not change my mind ok the point is the same right? Yes sir changing mind or not changing the mind doesn't matter right? You are not changing your mind you lost, its luck na? Its luck yeah so what if this had instead of this, this had it you would have won exactly correct, again you got unlucky right? so we will do one thing, ok why don't we repeat this experiment let say some fifty times, ok as long as it is possible for us. Sure you do one thing you stick to only on strategy hmmm you say sir I will not swap come what may, use that strategy some thirty times ok? ok and then the next thirty games will be you will always swap whenever I say whenever I show you an empty jar like this you will say I am sorry whenever you show an empty jar like this you will say sir I choose this but I plan to swap ok change sure, thirty this way thirty that way let see how it goes, ok we will make a note of how many times you win in the previous in the first strategy and then see how many times amit wins in the second strategy and let us decide for others which is the better method. Now we all have a feeling that it doesn't matter what strategy you choose, you win or lose with the same chance, the chances of winning or losing is a same or may be not will see this in detail with this experiment. Isn't this counter institutive? We thought both seemed a like you see amit told me that how does it matter it looks like in both the cases I am choosing something my chances of winning should not change but our quick experiment that I did with amit I was making him turn that side and I was trying to play the game with him he won more number of times in the second case where he was swapping right, he was winning less in the first case where he was not swapping see if we play the game once we can probably say oh I lost because of bad luck we played the game many a times if you observed we played the game twenty times each for this strategy and this strategy somehow the second strategy of swapping seem to have an upper hand, why? So now the question is how do I even experiment and decide that one strategy is indeed better than the other? Computation to our rescue lets see.

# MONTE HALL - 3 DOORS AND A TWIST 02

So first of all let me give you a brief introduction of what Monte hall problem is all about, in Monte hall problem we have three doors two of them comprises of goods and one of them comprises of a price say bmw in this whole programme screen cast we will be using bmw as the price. So in Monte hall problem participant is asked to choose any one of this doors, after he or she had made his or her choice the host ask the participant to change his choice what the host basically does here is he opens a particular door out of these three doors ok now we are left with only two doors one that has been chosen by the participant and one that has been opened yet so now the host ask the participant to change his choice, it depends on the participant to swap or not swap his or her choice now what the participant do here is should he swap or not swap this is the question that we should answer ok so this seems question will be exploring will be answering in the programming screen cast will have a kind of setup here in which we have three doors two of them comprise of goods and one of them comprise of bmw after the choice whether the participant swapped or not swapped we will we will try to explore this fact that what is the optimal strategy here, whether the participant should swap or not swap? Let us try to explore this fact, so let us start with the programming screen cast of Monte hall. As I said we need three doors here so I will be taking a list here namely doors ok this is the list doors, I will be taking another list here called goat door that will keep a track of the door that comprise of goats. After that I will be taking two variables swap and don’t swap this will take care of the fact of number of swap wins and number of don’t swap wins, swap will basically keep a track of number of swap wins and just I will write here number of swap wins and don’t swap will keep a track of number of don’t swap wins don’t swap wins ok now that we are done with it we will generate a random number here that will keep a track of what particular door will comprise of bmw? So since we are given only three doors here we will call them zeroth door, first door and second door so we have to generate a random number here random dot randint from zero to two ok so either any of these three doors will comprise of bmw so I will just write here ‘x’ will comprise ‘xth’ door will comprise of bmw ok now that we are done with it after that I will just write doors x will comprise of bmw yes we are done with it now, now the rest of the doors should comprise of goats now I will just do that here I in range just write zero to three since this will take into count that this is zero one and two so if I here is equal to is equal to ‘x’ then to continue ok, continue what does continue what does continue do here, what purpose does it serve here, if I write continue then it will go to the start of the loop if I write continue it will go to the start of the loop since the ‘xth’ door is already comprise of the bmw we only need to take into the count the doors except ‘x’ so if I equal to is equal to x here it will again go to the start of loop and implement i, because we are not considering x here we are only considering the doors that comprise of goat else what we are going to do here else will just see doors of I it will comprise of goat so you just write goat here, one thing we have to do here we have to append goat door also here goat door dot append and particular index that is I here so we are done with the doors two of them comprises of goat and one of them comprises of bmw after that we have to make the user input its choice so let us do that choice is equal to int enter your choice after that int input you just write int input that will make the user input his or her choice now we have to we also now that user has made his choice now the host will ask the user about swap or not to swap but before that he will open a door, so door open will take a variable here door open I will again generate it randomly here so I will write random dot choice goat door now please note this fact that we can only open the door that comprise of a goat we can only open the door that comprise of a goat so we are only considering goat door here, I hope that you are familiar here with random dot choice we have already explained it so goat door is a list here from which from which it choice from which a door will be will be taken randomly so here we are we have to open the door only comprise of goat so I will write here open a door that comprises of goat. Ok now that we have opened the door after that what we have to do here is we also need to take care of this fact in mind that the choice and the door open should be same here for example if the participant has made a choice of a door that comprises of a goat and we are also opening that particular door please note the fact that this is not allowed here so we need to also take care of this fact that choice and the door open shouldn't be same so I will just have a while loop here while door is equal to is equal to choice I will again and again choose randomly from the list goat door so I will just write door open is equal to random dot choice goat door you just need to write here I will write the comment here door open shouldn't be equal to choice made by the participant ok so we have also done that after that we have to give a choice to the user whether he or she wants to swap or not so you just write swap int input we don't need to write int here you just write input do you want to swap? He or she will answer in y or no or in y or n ok so done that after that you have to apply a if loop here so just write if ch is the choice made by the participant yes then it will be y then what you have to do here is if doors choice is equal to is equal to goat that means he made a choice of door that comprised of goat at initial stage then he will win here print player wins ok and you have to increment swap variable here else what you have to do is else print player lost, I will again explain the if else loop here what we are doing here is if ch is equal to is equal to y that means the player has chosen to swap if he has chosen to swap then we have to we have to take care of the fact that if he has chosen goat in initial stage in initial choice then if he swaps here then obviously he will get a bmw, because already a door comprising of goat has been opened and a door that has goat he has chosen in its initial stage and now he is swapping now if he will swap he will get bmw ok so he is winning through swap so we will increment the swap variable here else if he has chosen bmw at his first place and now he is swapping obviously he is going to lose this game so now we are done with if ch is equal to is equal to y and if he doesn't chose to swap here what will he do here? If he doesn't chose to swap here will write if doors choice is equal to is equal to goat. If his choice was goat initially then and he doesn't chose to swap to then he is bound to lose the game so you will just write print player lost done else if he didn't chose goat at his first place then he will win because he has chosen bmw in at in his initial choice so just write print player wins. Ok and now he has won because he hasn't swap so you have to increment the don't swap variable here so just increment the don't swap is equal to don't swap plus one. After that you have to print this number of swap and don't swap wins, so let us try to run this programme so just save it Monte hall dot, sorry so let us try to run this programme again there is some error here I will just show you it is saying random is not defined because we haven't imported the library random so you just write import random, list index out of range it is giving us an error list index out of range so how can you handle that? Let me think about it, we can easily handle that I think we need to initialised the list here we just initialise it with zero ok since we are only using three doors here multiplied it by three also initialise this particular list here that is goat door so you write zero here and multiplied with by two, so again run it name door is not defined so it is basically doors not door let me check if I have used it in some other place no so let us try to run it again, enter your choice so you have to enter your choice from of your door choice it can be zero one or two so I will enter zero, do you want to swap? I will choose swap, so player wins ok now the player has won because he has swap, again I will run it, enter your choice I will enter two here, do you want to swap? No. Player wins also here so again run enter your choice I will enter two, do you want to swap? Yes I want to swap the player lost because he or she swapped again run one do you want to swap? Yes. Player wins again enter your choice? Two. No, the player lost so how can we keep a track of this how can we find out the what is the optimal strategy here, I am really confused so what I will do here is I will run this programme again and again for example ten times and then try to find out what are the number of swap wins and number of don't swap wins here so that I will just take a variable here so I will just take a variable for example I took j, j is equal to zero and I have while loop here, while j is less than ten I will run ten times for that you have to indent it properly so indent it properly so since we have to run it ten times so I will just use a while loop here for that you have to indent it properly ok, so now this has to go here, this has to go here, this here now it has to be inside the while loop after that the choice thing will also inside the loop so you just write choice here this if loop was inside this if loop so this is nested basically this else is for the above if please indent it properly I will request you otherwise you will get an error after that this else so for the above if ok now that we are done with it. So let us try to run it. Enter your choice, zero. Do you want to swap? Yes. Enter your choice, two. Do you want to swap? Yes. Enter your choice, one. Do you want to swap? No. Two swap, yes. One swap, yes. So it is going on and on I think I didn't I think we didn't increment the j point j variable here, yes you need to increment the j variable here so just write j is equal to j plus one so that it just run for definite number of times ok so I will just do that, ok we have exhibit it now let us try to run it again so that we can get what is optimal strategy here. Enter your choice, zero. Do you want to swap? Yes. Player wins, do you want to swap? No. Do you want enter your choice, two. Do you want to swap? Yes. One Do you want to swap? No. Zero, do you want to swap? Yes. Enter your choice, two Do you want to swap? No. Enter your choice, two. Do you want to swap? Yes. Enter your choice, one. Do you want to swap? Yes. Enter your choice, one. Do you want to swap? Yes. Enter your choice, zero. Do you want to swap? Yes. So here number of swap wins as you can see is five and number of don't swap wins is one so is swapping is the optimal strategy here? Do you really think so? Is swapping the optimal strategy here? Let us try to run this once again and try to find out what is the optimal strategy, so let us try to do that again here zero, Do you want to swap? Yes. One, no, two Do you want to swap? No. One, no. Two, yes. Zero, yes. Two, yes. One, no. Zero, no. One, yes. So here also number of swap wins are greater than number of don't swap wins so I will run it once again so has to get the clear cut idea of what is happening here? Enter your choice, zero. Do you want to swap? Yes. Player lost, one, Do you want to swap? Yes. Two. Do you want to swap? Yes. One, Do you want to swap? Yes. Two, Do you want to swap? Yes. Enter your choice, one. Do you want to swap? Yes. Zero, do you want to swap?

No. Two, no. One, no. Zero, no. So number of swap and number of don't swap wins are equal here so we are getting difference distribution but in every case that we have run here number of swap wins are either greater than number of don't swap wins or they are equal to number of don't swap wins. So we can say that we are not sure of this fact that we will always win if we swap but there is higher probability of you winning if you swap. Yes this seems right from the programme that we are doing here, number of swap wins are always greater than equal to number of don't swap wins so we can say that there is higher probability of you winning if you swap and there is lower probability of winning if you don't swap, so now that we are done with the programme, we will just go through the programme once again so what we are really doing here is first of all you have to make two list, first one is doors and second one is goat door we are taking only three doors here, zeroth door, first door and second door and we have two goat doors here, two of them comprise of goats and one of them comprise of bmw, next what we are doing here is we are taking the track of number of swap wins and number of don't swap wins through these variables after that we will have a while loop, while we are using a while loop here because we need to keep a track of number of swap wins and number of don't swap wins ok, so I am running this programme ten times, you can run it fifty times, hundred times and check what is the optimal strategy? According to me it is number of swap wins it is swapping strategy and it will work in your case too please check it after that and taking the variable 'x' here this particular door except door will comprise of bmw then we are using a for loop here because the rest of the doors would comprise of goats except door comprises of bmw and rest of the doors comprises of goat, now what we are doing here is we are using a new key word here continue why I am using here because except door already comprises of bmw and if I is equal to is equal to 'x' here then we have to continue, we have to bring the control of the programme to the start of the loop because we are not considering 'x' here we are considering list of the doors here and else if I is not equal to 'x' then doors at I will comprise of goat, I also need to append in goat door list because that is keeping a track of doors that is comprise of goats after that you have to enter your choice ok the choice the participant chooses after that door open now the host will open one of the doors ok please note this fact in mind the open door and the choice shouldn't be equal ok? they shouldn't be equal so we are using the while loop here so door open is equal to random choice door since we are already using a goat door list here I will just make a choice from the list goat door ok, after that you have to make the user input whether he or she wants to swap or not ok, you just ask the user do you want to swap y or n if ch is equal to is equal to y then if he has if ch is equal to is equal to y that means he or she wants to swap and he has made a choice of goat at first place and if he has made a choice of goat at first place and now he is choosing to swap then he is bound to win, now he will get a bmw so in this particular case the player wins he has won because he has chosen to swap so we will increment the number of swap wins else he will lose the game after that if he doesn't chose to swap and he has chosen goat in his initial choice then he is bound to lose the game, else if he has chosen bmw and he is not interested to swap that means he is going to win the game, in this particular case number of don't swap wins will increment it will be incremented. Now since we are using while j less than ten we are using j variable here we need to increment j here so we incremented j here after that we are printing number of swap wins and number of don't swap wins. I hope you understood this particular programme on Monte hall and this programming screen cast is useful to you guys thank you happy leaning.

## **ROCK PAPER AND SCISSOR- CHEATING NOT ALLOWED -01**

Rock paper scissor I win, rock paper scissor I win, rock paper scissor draw, rock paper scissor draw hey guys what are you playing? We are playing rock paper scissor ok. What is that? This is scissor, ok this is the paper, this is the rock, rock cuts the scissor, the scissor cuts the paper and paper cuts the rock, the rock cuts the scissor, the scissor cuts the paper and paper cuts the rock did you get it? Can you explain? You will not understand please let us play don't waste our time don't disturb ok please. Rock paper scissor, why are you cheating? Where did I cheat? You changed your sign just now! No no don't lie, you show scissor then you changed to paper! No no you are simply telling, I don't want to play with you. Hey vidya can I talk to you for few minutes? Sure. My friends were playing rock papers scissors game now I understood I goggled it. One of my friends were cheating I want to create the mechanism where no one can cheat, is there any way to do that? Well I remember having studied something like that oh yeah! In nptel there is a course called joy of computing in which actually the professor teaches a mechanism through which we can play rock paper scissors in which no one can cheat, definitely you should take a look at it. Oh yeah! Thanks vidya definitely I will take a look at it.

## ROCK PAPER AND SCISSOR- CHEATING NOT ALLOWED -02

I was a ten old kid when i play a whole lot of games with my dad and dad would invariably win be it arm wrestling or be it running dad was number one or even with indoor games such as chess or tick tack tow anything that involved logic dad would was dad was way faster than me, there is no absolute game where i could win over him and that is when i came across this game called the rock paper scissor where muscle power or logic had no role to play, it was a game which involved sheer luck and i was a happy kid and this game became my the favourite. Let me now tell you the rules of the game called rock paper scissors there are three items here as i told you rock paper and a scissor the point is all these three items are equally powerful and at the same time equally power less every single item is less powerful than some other item and more powerful than some other item for instance a paper is less powerful than a scissor because a scissor can cut though a paper while a paper more powerful than a rock because it can completely cover the rock that's the story with the paper now coming to the rock a rock is more powerful than a scissor as you see a rock can smash the scissor while a rock is less powerful than a paper simply because a paper can cover the rock, lastly a scissor, scissor is more powerful than paper as it can cut the paper into pieces but it is less powerful than a rock as stated already, two people randomly choose an item and they choose one of the three items rock paper or scissor the moment they choose it they declare it by showing the symbol a close fist denotes a rock a palm for a paper and a V symbol for scissor. Once they pick they both decide which item is more powerful than the other and declare who wins as you see there are several possibilities here there is also a possibility of draw please note that is when both people choose the same item, let me give you a few examples assume dad chooses rock and i choose scissors obviously dad wins because a rock can beat a scissor in case dad chooses paper and i chose scissor obviously i won because a scissor cuts the paper and assume dad and me both ended of choosing rock we call it a draw, let me enumerate all the cases. Assume dad chooses rock son also chooses rock it is a draw, dad chooses rock son chooses paper, son wins, dad chooses rock son chooses scissor dad wins, paper and rock dad wins, paper paper obviously a draw, paper and scissors son wins, scissor and rock son wins, scissor and scissor a draw, scissor and paper dad wins these are the only cases you can see, this is a great game to play but with time you will realise that there is room for cheating, how do we patch this up? Is there a way to patch it? The course is about the joy of computing, is there a solution to this problem through programming? Any ideas, here is a straight forward solution this is how the interface will look like after i finish coding, the first player is asked to enter one of the three numbers zero one or two where zero stands for rock, one stand for paper, two stands for scissors so firstly the son comes and enters a number and then goes away from the place, the screen is cleared and then comes dad and he enters his choice of the number zero one two based on what is the item that he has chosen and then he goes away from the place and both of them come together and they observe who won the game, that we both of them are committing to their item nobody can cheat here, the computer finally declares who is the winner well it's very easy to program this but don't you think its cumbersome for dad and son these two players to come inside their room punching their choice and then go out, they should keep doing this a game requires both of them to come punk and go back and then both of them to come together and then see the result, looks like this is not a feasible way of creating a good interface, is there a better way? Let us see. Here is a very cool way to do this, dad and son they both secretly come and chose a placeholder in a ten digit number, what do i mean by this? Assume these  are the ten digit holders and the son says i am going to choose the third digit which means his item is going to be hidden here and dad comes separately and then chooses his placeholder, assume he chooses the seventh digit which means his item will be stored in the seventh digit and then now dad and son both of them sit together, dad doesn't know son's placeholder son doesn't know dad's placeholder they both sit together and they start typing their ten digit numbers by hiding their secrets item in their secret place holder, you see the idea is cool can we make it a little cooler? You see with the lot of intelligence it looks like one person can guess what the other person is choosing it looks like it although i am not so sure, you should tell me whether with some amount of intelligence one player can guess what the other player player secret placeholder is, so let us make this game a lot more complicated. I am going to bring in two new changes to the game so change one, you are not going to assign zero one two for rock paper scissor even that's a secret, a zero for son could be rock but zero for dad could be scissor so they input this information secretly just the way they input their placeholder secretly and the second one would be why take a ten digit number with entries being only zero one two, let us take an ordinary ten digit number each placeholder can be anything between zero one two three four five six seven eight nine, what i will do is, i will put my secret zero one or two in my secret placeholder and you see you can make it further complicated by increasing the number of digits, why necessarily ten digits you can make it fifteen digits or twenty digits right it probably will become difficult for people to enter that bigger number but is till you realise that bigger the number of digits more complicated the game becomes for the other person to guess that is.

## ROCK PAPER AND SCISSOR- CHEATING NOT ALLOWED -03

As you know rock paper scissor is a two player game so first of all let me do this assignments for say player one and player two, for player one we have zero as a rock, one as the paper and two as the scissor, similarly for player two we can do the assignments but let to we have zero as the paper, one as the rock and two say as the scissor. Please note that we are using dictionaries here to do the assignments, now we are done with the assignments you can also make the user input this assignments it's your wish completely your wish now we have to make the user input the number and the secret bit position in the number for that I will a while loop by true , first of all let us make the user input num one that is for player one the choice of player one input this is for player one, player one enter your choice for player two will have num two input player two enter your choice, we also have to make the user input the secret bit position for player one we have bit one int input player one enter the secret bit position for player two we have bit two int input player two enter the secret bit position. Please observe that we are using int here for num one and num two we used only input the default input tag in python is string but we want bit one and bit two to be integers so we will type cast it to int next please see that this while loop will run till infinite times we have to restrict this while loop, this while loop should run till the number of times till user wants so will have one more parameter here say ch and will make the user input this parameter will have ch input will say do you want to continue y is for yes, n is for no. If ch here is n then while break out of this while loop break, now we have now we are done with the assignments as well as input of the required parameters so next step is to quote the rules for rock paper scissor I assume that u have watched all the videos and you know the rules of rock paper scissor but still for the record let us revise for rock paper scissor. In rock paper scissor we have three entities one is the rock, second one is the paper and third one is the scissor, we have rock paper scissor, paper can cover the rock and rock can crush the scissor and scissor can cut the paper please observe that nothing is one important here everything is equal powerful here so let us try to code for rock paper scissor, for rock paper scissor we will have a function rock paper scissor, will pass num one num two and bit one bit two next step would be calculating two more parameters which can be calculated from num one bit one num two bit two, will calculate p one this is for player one, this is basically int num one bit one mode three what is p one here? P one basically is the placeholder at bit position bit one in num one and will mode it with three because the numbers that we are input here that we are going to input here are basically decimal numbers but we want answer in zero one two so will do mode three here similarly for player two we have p two in which will have int num two bit two mode three not that we are done with calculating p one and p two you can code the rules very easily will do if player one p one is equal to is equal to player two p two then this will be a draw, print draw then the next rule else if, if player one p one is equal to is equal to player two rock and player two p two is equal to is equal to say scissor as you know rock can crush the scissor so player two wins here sorry player one wins here, player one wins. Next rule if player one p one is equal to is equal to rock and player two p two is equal to is equal to paper as you know rock can crush the paper cut as you know paper can cover the rock so player two wins here. Next rule player one p one is equal to is equal to paper and player two p two is equal to is equal to scissor as you know scissor can cut the paper player two wins here, print player two wins else if player one p one is equal to is equal to paper and player two p two is equal to is equal to rock as you know paper can cover the rock so player one wins here print player one wins. Next rule else if player one p one is equal to is equal to scissor and player two p two is equal to is equal to rock as you know rock can crush the scissor so player two wins here print player two wins. Next rule else if player one p one is equal to is equal to scissor and player two p two is equal to is equal to paper as you know scissor can cut the paper so player one wins here print player one wins. So now we are done with coding the rules too we are done with the assignments we are done with the input of required parameters and we are done with coding of rules too, now so now let us call this function here will call rock paper scissor and will pass num one num two bit one and bit two so this is done too let us try to run this programme. There is no end of literal here did let us try to run it again so enter your choice, choice is basically number here I will input a three digit number one two three I will input three digit number here too five six seven say secret bit position for player one zero for player two it is one it is a draw, please observe that here we are input number num one and num two are input in the form as a string so they texting will start from zero and go till number of digits minus one I will repeat then texting will start from zero and go till number of digits minus one ok so we will have to input the secret bit position accordingly now let us look at the output we enter one two three and five six seven and the secret bit positions are zero and one, for player one the secret bit position is zero so it will be one, for player two secret bit position is one it is six, if we mode six mode three then it will be zero, for player one we have one for player two we have zero for player one, one is paper and for player two zero is also paper so it results in a draw, if you want to continue you can type y and if you don’t you can type no will have one more example here will do five six seven eight say and two three four let us enter the secret bit position for player one it will be two say and for player two it will be one it again calls for a draw let us have some another example two three four, one six seven eight zero and three it is again draw. One more example seven eight nine, four five six two and one player one wins so you can try it as many number of times as you want. I hope you enjoyed the programming screen cast I hope to see you again in the next programming screen cast thank you, have a nice day.

## ROCK PAPER AND SCISSOR- CHEATING NOT ALLOWED -04

Hey amit, sowjanya yeah hi, you are playing rock paper scissor game na yeah I will show you something read this ok. Ok you wrote a programme yeah, so that is nice, player one assignments, player two assignments enter the position ok enter a five digit number it looks awesome yeah I think we can go with this way and play, can we play? Awesome yeah! Just two minutes enter your assignments yeah yeah yeah what about me? Enter your assignment ok yeah just give me two seconds. Now you guys can play ok! oh it's my turn my turn yeah now I play yeah, yeah oh yes see I won ok, ok let me second turn now I will wait, wait no, no see again I won yes oh god! Thank you so much draw!! Did you like the game? Yeah, yeah yeah no one can cheat. Oh! Yeah.

# SORTING AND SEARCHING – 20 QUESTIONS GAME - 01

Hey amit lets play a game, what game? It's called twenty questions game what!!? You have to keep a number in your mind between say one to ten thousand ok and I will guess it right less than twenty attempts, less than twenty attempts!! Yeah. I don't think it's possible oh it is very easy. Ok the rules goes like this I will ask whether the number is less than equal to or greater than some number so you have to just say yes or no or it is less than or equal to or greater than, you will ask me whether my number is less than, equal to, greater than your some random number yeah yeah. I will say a number, ok let's try this say you keep five and will start from one to ten, one to twenty I will ask whether your number is less than ten, you have to yes then I will ask again some questions sure let's play this, yeah? Yeah I will crack this in less than twenty attempts ready? Will see ok keep some number ok I have, between one to ten thousand? Yeah. Is your number less than five thousand or equal to five thousand or greater than five thousand? Less than five thousand. Less than five thousand, ok so is your number less than two thousand five hundred, equal to or greater than two thousand five hundred? Less than two thousand five hundred. Less than! Cool, now is it less than thousand two fifty equal to two fifty thousand two fifty or greater than thousand two fifty? Less than thousand two fifty, less than ok. Again is it less than equal to greater than six twenty five? Hmmm less than. Awesome, I don't think you will crack in twenty attempts, ah no wait. Is it less than equal to greater than three twelve? Less than three twelve. Awesome, less than equal to greater than one fifty six? Less than. Hmmm now it's getting closer, is it less than equal to greater than seventy eight? Less than. Ok less than equal to greater than thirty nine? Less than thirty nine. Awesome, is it less than equal to greater than nineteen? Less than nineteen. Is it less than equal to greater than nine? Greater than nine, so it's greater than nine, is it less than equal to greater than fourteen? It's exactly fourteen yaar! Yeah wow I cracked in eleven questions see; exactly eleven how is it even possible? What's the logic behind this yaar? That's a good question.

# SORTING AND SEARCHING – 20 QUESTIONS GAME - 02

You saw what just happened we played a beautiful game and even we wrote a piece of code for it. Let me now bring in a small variation to the game, in the variation is the following. I will tell you that I have a number in my mind and that number is less than one lack common guess the number, how this difference is this game from the previous game so, assume you keep the number six hundred in your mind, I know it is less than one lack, right? How do I? How on earth will I guess that it's going to be six hundred? I should ask you this question is it greater than five hundred? Is it greater than ten thousand? Is it greater than twenty thousand? And so on right? is there a smart way of doing this? Think about it. I am going to tell you something which looks like very different from the game I have been discussing a number one to one lack guess what the number is? A variant of what we did just now right. I am going to give you an example which looks very different but these two things are the same, example is that of how would you search for a word in English dictionary? Let's say you want to see the word proclaim p r o c l a I m proclaim how will search for this in the dictionary? You will open the dictionary in the middle, why middle? You do not know whether proclaim is in the first half of the dictionary or the second half of the dictionary, the moment you open the dictionary in the middle you will get words starting from 'n' letter 'n' so you know for sure that first half of the dictionary is all those words is starting from a b c d efghijklm n because you saw the word when you open the dictionary in the middle, so your word proclaim should be in the second half of the dictionary so what you do is you discard the first half and only look at the second half and again go to the centre of it centre of second half will now be let's say 's' words starting from s, now what can you conclude? From 'a' to 'n' was first half of the dictionary which you discarded and now when you take the midpoint of the second half you are seeing 's' which means your word proclaim should be in the first half of this little cut down dictionary, in the second half its in the first half if you know what I mean. Right? Think about it, every time you open the dictionary by half you are discarding one part of it and retaining other part of it, correct? And so on and so forth you finally reached the word proclaim in no time because your dictionary size is reducing by half correct? What is this to do with the problem that we were discussing? Keep a number in your mind from one to one lack and I will guess what the number is. Can use the way I searched through a English dictionary that same method here, think about it, it is indeed possible. What you do is ok; do you have a number in your mind? Is it less than fifty thousand? Or greater than fifty thousand? You see what I am doing, like the dictionary I am trying to see, you see this line, is it on the left side of the, is it the first half of one to one lack or is it in the second half. If a number is in the first half you will say "less than fifty thousand" if your number is in the second half then you will say "it is greater than fifty thousand" and there I am, I know for sure I can now reduce my search space by half and now I will ask you id your number less than twenty five thousand or greater than twenty five thousand. Do you see that this method is exactly the same as searching through an English dictionary? Let us now go ahead and try to write a piece of code for this. Whatever we did just now is popularly called the binary search, the binary search words look complicated but there is nothing here. Binary means cutting into two pieces and search in only one piece and continue doing it your space where you are searching will get halved and very, very, very quickly note the usage of the word very, very, very it is indeed so quick very, very, very quickly you will narrow down to your search key in no time this is called binary search.

## SORTING AND SEARCHING – 20 QUESTIONS GAME - 03

Look at this array with thirty elements, can you spot the umber eleven here? Ah looks very difficult, you should look through every single element from the beginning till the end but what if this array was sorted,  by sorted I mean what if this was in ascending order of the numbers, don’t you think it would have been very easy for you to search for the number eleven.  This is the sorted version of the array, the list; do you see eleven right now? What did you just now do? You used binary search to find eleven, so the moral of the story is if you want to find for an element search for an element in a list, it is easier for you if the list is sorted. You see this is in general true with your life, you will find what you want, if you keep your life organised, if your list is organised in some ascending order if it is in ascending order searching for an element becomes very easy, you see so we have been teaching you how to keep, how to how to search for an element if the list is sorted but we dint teach you how to sort a list, let us now see how can one sort a list.

# SORTING AND SEARCHING – 20 QUESTIONS GAME – 04

HI guys, in last video we saw, we saw that how we can perform binary search on a list of numbers which are sorted. So we saw that the, the list we provided for the function was already sorted which means that numbers where in ascending order one two three four five six seven eight something like that but in real world what happens that whatever you data whatever the data you get is not is not sorted it is actually some random numbers in some random orders and whenever you have to perform such kind of search that let's say binary search and then you have to sort it, so sorting is the algorithm which as a lot of applications so in this program we are going to perform such sorting algorithm known as a bubble sort it is a very easy algorithm I will tell you how to, I will tell you the exact algorithm when you can use the algorithm to write your won program, we will write the program for bubble sort also so let's see this, let me first given an array so let us assume that we have an array of size five and the elements are five one four two eight so what I want? I want to sort the elements of this array it means it should the final array should look like one two five four and eight so bubble sort is like that so in first loop what I will do so I will have two loops exactly so in first iteration what I have to do, I will compare the first two elements and see whether the first element is less than the second element are not, if it is not if the first element is not less than the second element then I will swap their positions so here five is not less than one so I will swap so one should come before five so it will become one five now I will move to the next two elements which are five and four so I will go there I will compare them and see the same thing which is, if five is less than four then it is ok otherwise I will swap them so five is not less than four so I have to swap them so it will become four and five so now I will move to the next two elements which are five two again I will perform the same I will check the same condition so five is not less than two so I will swap them so it become two and five I will move to the next two elements same conditions since five is less than eight so I don't have to swap them so my array my first loop is over, I will go to the second loop and the second iteration in second iteration again I will compare the first two elements so I am doing the same thing so you can find the pattern here, in the first loop inside the first loop I must have another loop in that loop I should do all those things and whenever the second whenever the first iteration of first loop gets over the second instance come I will do the second again do the again perform the same thing so again I will perform compare the first two elements one and four since one is less than four I don't have to swap anything I will move to the second two elements four is not less than two so I have to swap them so two comes before four it becomes two four, I will move to the next two elements four is less than five so I don't have to swap anything now next two elements less than eight I don't have to swap anything so after this my my array is sorted so I need the exactly something like it one two four five eight so now you know the algorithm I give you the simple basic idea of the algorithm I am not writing the pseudo code kind of thing, you will use this idea to write the program it's the very simple program and I think that anyone can write it so we will see in the next video.

# SORTING AND SEARCHING – 20 QUESTIONS GAME - 05

So, let me write the program for this algorithm which is bubble sort so let me write the function def bubble and the parameter will be the array which I want to get sorted ok, first let me compute the length of this array so for length there is a function length of' a' it check it returns the length of any lists which you provide there so length Len of 'a' will return me the length of this array or the list so I will write my first loop which is for I in range of ten which is the length of the array, so first loop is first loop should until the number of elements in the list ok? The second loop should start from the first element but after, after all the swapping see what was happening here, so after all the swapping the maximum element was going to the last position, so in the second pass we, we don't have to we don't have to compare that element or include that element in the comparison so we can safely exclude the last element which we got in the last pass so every time we will exclude one such element in such way will get the sorted array. So the second loop will go for j in range it will start from zero ok but it will go until n minus I minus one, minus one is because of the range issue because we have to go until the minus one part and minus I is that when the ith element when the maximum element gets to its the the last position then we have to exclude that element for that I am putting minus i, so it means that when I is zero it means that it is the first iteration it means there is no maximum element which went to the last position so we have to go to the last element in the second loop but after the after I becomes one it means that one of the maximum element first maximum element went to the last position when we have to exclude it so for that n minus I is required and after this after when I becomes two it means two maximum elements first maximum element and the second most maximum element went to the went to its correct position it means nth position and n minus one position we have to exclude it so that s why this condition is required. Then I just have to check the condition which ever I was talking about if a of j it means the current position is greater than a of next position which is j plus one if such thing happens then just swap it, I will create a temporary variable temp and I will assign it the value of aj in aj I will put a of j plus one and in a of j plus one I will put the temp variable, this is just swapping of two elements that's it that's all my programme should be my sorting is complete let me just create an array 'a' which I gave in the example of which was five comma one comma four comma two comma eight ok this was my array I will call my function bubble and pass this array let me print the elements of this sorted array for I in a that's it I will print 'i', great my program is complete I will just run it cool one two four five eight so my array was five one four two eight got one two five four eight which is one two four five eight which is sorted see you can create your own array and you can run this programme and you will get the sorted array so this is how we perform a sort we perform sorting this is one of the algorithm, this is not the best algorithm that we have there are so many better algorithms for example quick sort, merge sort, heap sort and so other sorting algorithms but this is the basic algorithm which you guys can use, if you if you do not know much about sorting and after you learn more of the programming and algorithm and data structures related things then you can go on to the more advanced sorting algorithms so that your time it takes your function takes less amount of time in order to sort an array thank you.

## SORTING AND SEARCHING – 20 QUESTIONS GAME - 06

You saw how you can sort a list, in fact you can sort it with an in build function python function as well  but we taught you just for completeness sake for interest sake how can you manually sort elements in a list in fact sorting is a huge topic in itself we can go on and on and even give you a course of thirty hours duration only on sorting, there are so many techniques so beautiful so deep that it will take so much time but let us just limit our discussion to one type of sorting technique which we showed you just now, it is left for you to now go and explore more on different techniques with  which you can sort.

## SORTING AND SEARCHING – 20 QUESTIONS GAME - 07

Hello every one, so today we are going to write a program for searching, actually our main objective is to understand the binary search, you saw the play how a person can easily get a search a number in a list of numbers sorted numbers very easily in a less number of steps, will see why binary search works but before that we must understand how to simply search a number in list of list of given numbers for that we will use the linear search it is a conventional simple search will go through each values and try to find our element which we want to find in that list and we will see how many numbers of steps it takes and then we will switch to binary search. So I will start my spyder I will type spyder in my terminal and wait for it, ok I will write my programme in a new file entitle zero I will change the name so what I want first I want first as list of numbers for this programme we will take the numbers in sorted order in ascending order, the numbers will always be this one, two, three, four, five, six something like that I means it should always be sorted I mean the numbers will always been ascending order, first I have to create a list of numbers for that I will write I will write a for I in range let's say I want numbers from one to thousand one thousand one in range if I want a number from let's say one to n then I have to I have to give the argument as n plus one because it always prints up to number less than the given number, I will write here let me save this as linear search ok awesome, for that I need a list so let me write the name as element if I gives the simple bracket to create a list I will use this, so in order to add elements into a list there is a function name append I will just write elements dot see it will show you all the functions which you can use with the list that is the beauty with spyder so it is just tap it will give you the append and I will write here that's it I created the list ok that's it I can just run this to see my list ok I will run the program I will just type element and it will show me the numbers from one to thousand, you can see all the numbers from one to thousand awesome! So we recommend you write all the programmes in function, let me create a function here which I can easily call def linear search and I will give the range here that's 'n' so that the 'n' I can give here also I will give the parameter 'x' I want to find now let's see this. I will give the indentation ok awesome this is done I have created my list of numbers now what I want? I want to search this element 'x' in this list for that I write another loop for I in see now my list is complete I have created the list I can just iterate through the elements of this list which is very easy just I have to write since list is iterative what do I mean by iterative? You can just goggle it you will understand what is iterative, there are some elements in python which are iteratable and some are not iteratable if it is iteratable you can go through it like a loop I will write for I in the list name which is the element ok see here element was created I can for I can go though it here also I can element if I just print it, it will give me the elements so I can iterate through the elements through the elements of element ok for I in element ok now I am iterating over the well use of element I will write its only one if condition, if I is equal equal to my number 'x' which I want to find and I will just print yes! I found my number position str ok since in this example the numbers are sorted from one to n I can just write str I minus one or I can simply I can write str I ok so we have to call a name ok I have to also print that whether found the number or not, if I didn't find any number so for that I will use the flag value let say flag is equal to zero ok, initially the value is zero, if I found this number then I will put the flag value to one after that I write a if condition if flag is equal equal to zero I print number is not found that's it good so my programme is complete now let me just run this and that's it. I will call the function linear search see whenever I type this it will show me that it needs to 'n' and 'x' so n is my range so here I will put thousand one next what number let's say I want to search fifty ok just type it will show me yes I found the number at position fifty because the number has sorted. Ok cool now I want to know how many iteration it took for me to get to this number for that what I will do? I will create a variable count here zero its initial value and whenever I am going through an element here I will do count plus is equal to one I am increasing the value by one and if I don't need to whenever I find a number and the list I don't need to go again go through the other elements of the list so I will just break the loop here because I don't need to go to find go to look at other numbers because I already got my number so here I will break and if flag is zero it will find and I will print the count that's it let me write here number of iterations is equal to plus str because I need to change this into string awesome! This is done let me check. Linear search give me one thousand sorry the number is fifty seven say ok now you can see, yes I found my number at position fifty seven number of iteration it taking is fifty seven because before fifty seven I have to check all the numbers one two three four five six seven eight nine up to then I will go to the fifty seven so you can see that if the number of elements in the list are very huge let's say one million and you want to search a number which is very far end let's say the number I want to search is one million only then it will take one million near one million iteration to find that number which is the very huge, in the game we saw that when we can use the advantage of sorted number whenever the number is elements in an array or list is sorted we can use that extra information for searching a particular number easily that is the correct for binary search so will see how we can use this extra information to search an number in a list of sorted numbers and drastically decrease the number of iteration so that we can in very less number of iteration or to say the very less number of time in less time we can search the number so in next video will see the binary search and will see how we can iterate

# SORTING AND SEARCHING – 20 QUESTIONS GAME - 08

Hello guys, in last video we saw the linear search we saw how inefficient it is to search a very big number in a very big array suppose you have million of numbers in a array and you want to search close to million then it will take around a million of steps to search that number in liner search so it is very inefficient because you do not have that much of memory power or that much of even time to spend on an array of size one million and that too if the array is sorted so here what we are going to see is the very very good algorithm known as binary search whenever you get an array you sort it either in ascending order or in descending order, you are getting some extra information which you can used to search the search any element in an array in very less number of steps that exactly what we are going to see here so in the game, in the game that you saw in the video there were two players one was ravi and amit they were playing a game so ravi has an idea that he can guess a the number which amit has thought in a very less amount of iterations so the game is like this you search you guess a number between one to thousand and then the second person asks second person first computes the midpoint of that array so in one to thousand the midpoint is five hundred then he asks whether this whether the number that you have guessed is equal to five hundred or not if it is five hundred then ok I got the number if it is not five hundred then it is he ask whether it is less than five hundred or it is greater than five hundred so suppose the number you have guessed is less than five hundred then you can see that you can safely discard the other five hundred elements in the array because your number will never go beyond five hundred you have already said it is less than five hundred and the numbers are sorted so you can never go you do not have to search the other part of the array so you can safely discard the numbers from five hundred to thousand so number of elements that you have left now is one to five hundred then again the I will ask I will compute the midpoint of this array which is left that is one to five hundred which is the midpoint is two fifty then you will then I will ask you again whether your number is two fifty if you says yes then ok if you says no then again I will ask whether it is less than two fifty or more than two fifty, if it is more than two fifty then again same logic I will use since it is more than two fifty it I do not have to search it less than to fifty because it will never happen that the element will present in the array between one to two fifty so I can safely discard the numbers from one to two fifty now the arrays which I have left is two fifty to five hundred again I will ask the same question that is I will compute that midpoint of two fifty to five hundred and I will use that midpoint to ask the question and go to the left part of the array and right part of the array this is the logic of binary system in this you are a every each iteration you are whenever you are computing the midpoint you are discarding the half part of the array this way you are size of the array is reducing by two at each iteration so you will exactly use this logic to do the binary search so let me right the definition of the binary search. Def binary search it will take two arguments one is the array let's say array name is 'a' and other is the element that I want to search great, now I will create two variables I will use this variables to position my array, I want to know I will create a variable let's say first position first pos which is zero because array starts with zero so I can safely take zero as my first position and the last position last pos sorry which is the last position of the array, python is an function named it as len through which you can compute the length of an array so suppose if I type Len of a I will get the length of array now it gives the length of array gives the number of elements in the array but we know that list or array whatever you said in python starts with zero so you have to take minus one so let me try this in console suppose my array name is array one and the elements are one comma two comma three comma four comma five ok now if you type length of array one you will get oh there are five elements so you will get five but the position of fifth element is what? Is actually four because the array starts with zero, zero one two three four so suppose you want to retrieve the last element will type array sorry array one and if you type four here you will get the fifth element or you can type array length of this array one ok this length of array one will return the number of elements in array one which is five so number of elements in array one is five if I put minus one here so this this whole thing means it is five which is return by the length of array one minus one which is four so this will print give me the fifth element fourth element which is five so this is exactly what we are going to do here so I will put the last position is length of array minus one ok also I will take a flag value which I will use to see whether I found my element or not ok this warning means that all this variables have been you have created this variables but you are not using it so as, as soon as I use it all this warnings will go ok now I will create a while loop let's see how to do this? And let me write this first and I will explain it how what exactly I am trying to do. So I will type first pos is less than last pos and flag is equal equal to zero what is this mean that? Keep this loop going on until the first position is less than the last position which is this two variables, if first position is always less than last position and flag is equal equal to zero I told you flag you can let me write a comment  here flag means that flag go to zero that element is not been found so it means that you should continue the loop until first position is less than last position and you haven't found the element you are looking for till than you have to continue this loop ok so what I will do first what was the strategy, so first I will compute the midpoint of the array so I will compute mid is equal to you write so I can compute the know to compute the midpoint of an array I will take the first element the position of first element I mean the first position of the array and the last position of the array and I will add it and divide it by two this why I can iteratively I can compute the midpoint of any sub array so I will write first pos plus last pos divided by two now if I write divide by two it will return me a float value, you can check here suppose if I five plus two divide by two it is giving me three point five but I exactly want a integer value so suppose the number is odd it should give a integer value either left or right I don't care much but it should give me a integer position because it is a position if I use this float value then I will not be able to retrieve the element at that position so I need an integer value for that I will do the int there is called as integer division in python that is I will write five plus two, to this division it will give me three this is what I want so I will put this one more division operator here and I am done so I computed the mid now what I will check? I will check whether this element 'x' is present in mid or not I will write if 'x' is equal equal to a of because made is of position if it is there then I am done, I have to check for another thing so I will write I will make the flag is equal to one it means I found the element and I will write print element present at position I can write I can give the position of this mid variable which is I can write str mid you can add plus one also so that you can get the exact position but I am not writing it because I am assuming that you all know that array starts with zero so in that context I am just printing the mid value that's it you can return from here, it means that whenever you found the whenever you get the element which you are looking for you can return it you do not have to other computation I will just return from this, this return will return the function value which I am not returning anything it means it will get out from the function to the main I am calling the function that's it ok now if it is not present at the mid position then I have to check whether it is on the left side or right side, for that I will write else if it is not present at the mid position if this 'x' is less than array of mid which means if it if 'x' is less than mid I can discard the right side of the array which means what? I can safely write that last position is shifted to what? Mid minus one because mid and after mid I can discard because mid is I the first question in the first condition I have checked whether it is present at mid or not, if it is not present at mid then I can discard the mid and all the elements after the mid so I will safely I will reduce the size of my array by positioning the last position to mid minus one this why I am discarding the other elements of the array or if it is not wait sorry if it is not less than the mid it is more than the mid then I can discard the first part of the array which is from zero to mid I can discard for that I will just shift my first position has first pos is equal to mid plus one that's it ok and then the programme is over and if the function doesn't returned anything after all these things it means the array I didn't find the value which I am looking for so I can safely print here after this loop it means that loop is over but still I am not able I didn't find the element I will write the number is not present that's it now in last function the linear search you were printing the number of iterations also how many iterations it takes so here also will I would like to print the number of iteration it is taking to find this value 'x' so I will create a count variable count assign it to zero and at every iteration of while loop I will increase the value now here I have printed the element where it is present I will just print that the number of iterations are which is string of count ok cool, I just need to call this function so in order to call this function here I will create an array so let's say my array name is 'a' and I will enter the elements in this array for I in range one to five hundred let's say a dot append I and I will call this function which is binary search and I have to give the and let's say the number I have to search is seventy so our programme is complete so I will just run this program to see the output. Ok that is showing the number is not present and I exactly did what the logic was present for the binary search and I know that number is there suppose if I type 'a' you can see that seventy which I am looking for is present but still it is showing the element is not present, let me try this for small number lets say number of elements or ten what I want to search is four let me try this, still it is showing that number is not present let me see the numbers one two three four five six seven eight nine ten, and four is there now what is exactly happening is that whenever the condition that I put here while first position is less than last position it might happen that some time the first position and last position will be the one number let say first position and last position are pointing at four at that time still we have one iteration search is required but this condition is will fail that is first position is less than last position it is showing that here the first position is not less than last position it is equal to last position and here it will fill and I will not be able to search this element so to do this I have to make this condition as less then equal to which means when first position is less than equal to last position until then you have to search for the element now you can see if I run this, it will show that number is present at position three zero one two three and the number of iterations is four because four times I have to switch to this, now let’s see this, number of elements will be hundred and one and then I will search for number seventy and run this see the number is present at sixty nine, seventy. Now in linear search what we were looking for, we search for a number the number of elements we put where thousand ok ok and we were searching for the number thousand and it was taking thousand of iteration now here let’s see how many times, how many iterations it will take? I will run this the see the number is present at position nine nine nine and in ten iterations I was able to search this number ten iterations where it was taking thousand iteration in linear search it just takes ten iteration in binary search, suppose I make this one million let’s say ok and I want to search this number let’s say one lack ok see how many iteration it will take in just twenty iteration, in just twenty iteration I was able to find the number, whether as you all know with to find this number it would have taken one lack iteration in linear search so this is the beauty of binary search, you can do actually it takes exactly log of the number of elements in the array that is if the number of elements let’s say sixteen then it will take four iteration and thirty two then five iteration and sixty four six iteration so on because every time you are dividing by two and discarding the rest other part, you are taken only one part at a time this way one half of the array is every time discarded and you are reducing your space by two exactly by two every time by two by two by two by two becomes log of two log of base two so yeah we are finished with the binary search please if you have any questions post it on discussion form or you can if you think you can, this is a very simple programme you think you can create a more efficient more elegant program like this better than this please post it on discussion form it might help other people to learn ok guys thank you.

# <u>Substitution Cipher – The Science Of Secrecy</u>

So this is what our next joy is all about, what happen? Didn’t get it! Now? Yes! Our next choice is about CIPHERS.

# Substitution Cipher

# The science of secrecy 01

This is personally the most exciting idea throughout the course at least according to me, we are going to talk about the science of secrecy. If you know what that means let me illustrate this with an example. Assume Romeo and Juliet they want to communicate with each other secretly how would they go about communicating? Assume Romeo wants to send this long a letter to Juliet then he suspects someone in between, may probably tamper with the letter and then see what is written, he wants only Juliet to read it. In case someone else gets hold of the letter they can of course read it but it should not make sense to them. How would he ensure that he encrypts the letter? A very simple method goes as follows. So this is the letter Romeo wants to write for his sweetheart Juliet as you can see it's a love letter, so what he does he firstly writes the letter and then shifts every single alphabet by five units so which means A becomes F, B becomes G and so on so from A to Z this is what happens to the letters and this particular letter that Romeo want to send to Juliet takes this form and now the letter finally looks something like this and Romeo sends it to his beloved Juliet and Juliet gets this and knows what exactly Romeo has done to his original letter basically they have communicated about this protocol before and what Juliet done has the following as you would have guessed obviously she shifts it back five units which means she makes F A G becomes B and so on and there you are she gets the original letter, you see the shift being five is the secret key here it can be five between Romeo Juliet and laila and majnu it can be fifteen now, what if someone gets to know that the method used by the lovers is simply shifting it by some units and mediator comes and catches hold of the letter communicated by this boy friend to his girl friend can he break his code? So this middle man comes and catches hold of the letter sent by our boy to his girl, he knows that the idea here is shifting every single letter by few units so what he does is try all possible shifts, he first tries one shift and tries to see if the text makes sense or not, so whatever you are seeing right now let us take this three lines text as the letter that the boy writes to his girl and then it is shifted by some units that the middle man doesn't know but he wants to crack it. So he tries assigning A to B, B to C, C to D by that I mean he assumes it to be one shift and tries to see if the text that generated makes sense or not as you can see it is not making sense and he tries a two shift and then a three shift and then a four and then a five and then a six and than a seven and finally when he does eight he sees that the shift is actually making sense completely all the words are English words and hence he concludes that the shift this boy is using happens to be eight and there he is, he has deciphered the encrypted text with his little intelligence so what do you infer from this? Such a method is used between two people to communicate can be broken easily by a mediator a middle man, if he knows provided that the technique used is shifting but he doesn't know by what units it being shifted so what he does? He tries all possible shifts; from shift one to shift twenty five twenty six and so on tight? One of them should give him valid English text once he gets the valid English text he concludes he has decrypted the secret. What we saw just now is called the popularly the Caesar cipher in the literature of cryptography. A very well know technique which is not being used mainly because of its simplicity and the fact that anyone can break it, now I am going to teach you a technique which is a little more complicated rather a lot more complicated than what I explain just now the Caesar cipher I am going to tell something else which is lot more complicated, which appears as though nobody can break this. Here is a lot more complicated way of encrypting a given text and here is how it goes. You take every single English letter and assign a complicated symbol to it and a Romeo writes along a letter for his Juliet and simply takes every single letter, every single symbol and instead of using the English alphabet letters he uses the corresponding symbols that is agreed between Romeo and Juliet now according to what I just now illustrated A stand for this symbol, B stand for this symbol and so on up to Z and Romeo converts his letter his love letter to something as complicated looking as this, now its look like it going to be impossible for anybody to crack this isn't it? Who on earth knows what letter is map to what letter right? This looks like a perfect way of communicating. Or is it, is there a way to break even this cipher? Let us take a break and let me tell you all a nice story, a story that looks completely unrelated to what we are talking but is very related. There was this mom who had ten sons all of different heights and one day the sons trying a prank on their mom they wore masks disguising themselves and they come in front of their mom and then challenge her to identify who is who? All ten sons they appear in front of her like this and say mom identify us, they say that in chorus and mom is perplexed because she cannot see the faces of her sons but then I told you something there are all of different heights, so do you think it will be difficult for the mom to identify who is who? All ten sons are masked and they appear in front of their mom and say mom identify us? Hoping that they are they will make the mom perplexed and that she will not able to identify them, little do they know that mom can identify her sons just by identifying their heights she knows that the shortest is this, the second shortest is this, third is this so on and the tallest who is this and she goes on in no time calling out their names, patting on their back one at a time she says athri, brugu, kuthra, vashista, gowthama, kashyapa, angerasa, rama, bheema, shaama over all ten names right? What exactly did she used to identify them, just their heights and what's the moral? The moral is that despite the fact that sons concealed their identity with their mask, mother used some other parameter. So what's the moral of the story, why the story am it's are encryption decryption theory? Do you see a connection? Well there indeed is a connection and this is the connection. You can try to conceal the letters in the form of symbols but then you see English speaks out aloud, what do I mean by this? English has a very peculiar statistical property, sounds complicated? Nothing at all. It I just I just mean the following, the most frequently occurring word letter in English always happens to be this, isn't this a beautiful idea? Its sounds as though it's impossible for anyone to decode a substitution of the letters of the English alphabet with some random symbols, its look like it's impossible to break this code but you see how English speaks out aloud and anybody can very easily decrypt the text. The seemingly looking seemingly complicated looking technique has a huge loop whole in fact it is as easy as a previous one that we discussed. So this subject is called cryptography where the idea is all about making ciphers and also breaking it, almost every single technique that has come about the history is broken and the ones that is currently in place I believe are just in the status of yet to be broken, so the science of secrecy is all about making and breaking the codes, it is reached than what I illustrated just now I invite you all to read more of it I will give you some references to good books for the same you can go ahead and read more of what makes cryptography.

# Substitution Cipher

# The science of secrecy 02

The professor just now explained a method to share your messages secretly, so let me try to show you how can you share your messages in a secret way. Hello every one before we start doing the programme for substitution cipher I would like to explain the concept in python name string. First of all what is a string? String is used to represent text in python; text may comprise of spaces, numbers and even alphabets. So how can you use string here, for that first of all you need to import a library called string. Now take a variable named x and I would initialised it to a string for example it is hello world, please note the fact that every time you initialise a string you have to close the string in double quotes, you have to close a string in double quotes. So here we have the string called “hello world” and if you want to excess a particular alphabet in the string x, what you can do here is, you can excess the particular alphabet in “hello world” by giving it a particular number. For example if I want to excess ‘H’ in “hello world” I would write print x of zero as you know the indexing start from zero in arrays as well as in strings so if I want to excess the alphabet H in “hello world” what will start from zero so ‘H’ is at zeroth position so I would write print ‘x’ of zero, so here we have h and for example if I want to excess the alphabet ‘O’ here, ‘O’ has two occurrences so if I want to excess the alphabet ‘O’ in the first position that is, how can you do it? ‘H’ is at zeroth position, ‘e’ is at first position, ‘n’ is at second, second l is at third position and that ‘O’ is at fourth position, since ‘O’ is at fourth position I will write x of four and now let us see what is the output, here we have ‘O’ so this is how you can excess the alphabets at particular position in a string ok. There are many other functions available in string for example if I want to excess the alphabets from second position to fifth position, how can you do that? You can actually do it in one go without even using for loop or any kind of loop so I will just say ‘x’ of two colon five and I will just print, it prints llo, this is how you can excess the alphabets in particular range of positions ok now if I want to know if I want to get the length of the string so for that you have a function what you need to do is, you will write Len and then Len and in bracket write the name of the string, here we have ‘x’ so we will write ‘x’ so let us print it so it is eleven, you can clearly see that the length of string ‘x’ is eleven, now if I want to convert the string in upper case there is straight away a function available in python what you can do here is, you just write the name of the string that is ‘x’ and then dot and then you have to write upper, upper is a function available in python used to convert the available string in upper case so now let us see what is the output, for that we should print it, print ‘x’ dot upper yes it converts the string or string “hello world” to upper world to upper case. So we are done with the strings now there is one more function available in python in fact there are many more functions but here I would like to explain you one more for example in “hello world” if I want to replace one alphabet by some other alphabet how can you do that? I will just write ‘x’ dot replace for example I have to replace ‘h’ with the ‘j’ so I will just write, replace ‘h’ in double quotes comma then ‘j’ in double quotes so now the output is “jello world” instead of hello world so this is about strings in python now we will do the programme for substitution cipher and will use strings there.

# Substitution Cipher

# The science of secrecy 03

So let us try to code substitution cipher here, for substitution cipher we have an input file here named ip underscore file, it has some text I need to convert this text in such a way so that it is not recognise by the third person for that I will be using substitution cipher so for that I need a string of alphabets in which I can substitute the letters, python provide with the string named string dot ascii underscore letters I will print it so I will write print string dot ascii underscore letters as you can see we have the letters in lower case as well as in the upper case so I will be substituting this string in such a way so that it is not recognise the text present in our input file is not recognise by the third person I will be using this particular string for that so in order to execute the substitution part I will write import string after that I will initialise a dictionary called dict, I hope that you are already familiar with dictionaries, strings as well as file handlings because will be using all three of them in this particular programme and if you are not familiar with these concepts I would suggest you to go though the previous videos, please go through the previous videos and then watch this programming screen cast. Now we have dictionary named dict is equal to dict is equal to curly braces this is how we initialise a dictionary, for converting this particular string to a substituted format I will write for I in range length string dot ascii underscore letters. In this particular dictionary if any letter is at ith position I will be substituting it by I minus one position, position letter it's your choice by with which letter you want to substitute a particular letter here if I am substituting a letter given letter by its them by the previous letter so we will have here, 'A' will be substituted by 'Z', 'B' would be substituted by 'A', 'C' would be substituted by 'B', 'B' would be substituted by 'C'. This is how our dictionary is going to work, so I will write string dot ascii letters, string dot ascii underscore letters at ith position will be substituted by string dot ascii underscore letters at I minus one position here I used minus one you can even use plus one, plus two, even minus two minus three it's your wish I am substituting it my minus oneth letter so that you can tip your get idea of how our programming is working, it would be easy to observe this particular dictionary so I will just print dictionary here so that so you can get the idea of what is happening over here. So here we have the dictionary as you can see capital 'z' is 'A' substituted by capital 'Z', 'B' is substituted by 'A', 'C' is substituted by 'B', 'D' is substituted by 'C' this is how our substitution is working, so we have all the substitution in this particular dictionary named dict, now we will use this dictionary dict to convert our input file. For that first of all we need to open our input file named ip underscore file so I will write with open ip underscore file dot txt as f then I will write while true I need to read this file by character by character and I need to substitute a particular character according to the dictionary so I will take a variable here named 'C', 'C' is equal to f dot read now I need to check one or two things here first of all I will be checking whether we have encounter end of file or not, for that I will write if not 'c', if the if the file pointer hasn't encounter any character that means it has reached end of file, so in this particular case we will print end of file and it will break out of this file loop and if 'c' if the file pointer has encounter a 'c' which is already present in the dictionary, if ‘C’ in dict in that particular case I will take a string here name data, please initialise the string data here, how can you do that? You can initialise through double quotes so we have initialise our string data, data is equal to dict of ‘C’, if ‘C’ is present in dictionary in that particular case data will be equal to dict of ‘C’ and if ‘C’ is not present in the dictionary then we can’t do anything will just write data is equal to ‘C’ instead of dict of ‘C’, please note the fact that in our dictionary dict only alphabets are present, in lower case as well as upper case and if in our file some if in our file some integers zero to nine or some special symbols are present that will not be substituted so here is an exercise for you, if you encounter a file with some special symbols and integers how can you apply substitution factor on that particular file its an exercise to you, its left to you how can you do that? So here we have the string data which has the substituted file, which has a substituted data so I will just print it. I will just write print data let us try to print this. So now we have the data, but the data is exactly like the input file so there is some error in our programme, here if we are reading the file we need to supply a bullion value here so I will just supply one and try to read it here we have the substituted file so here is the mistake, when we are reading the file when we need to supply some bullion value here one here. So as you can see that it has substituted our input file and this is something we can’t read and we can’t recognise ok so what will I do here is, I will store this output in a file, you can get a clear cut idea of the substitution happening over here so I will take another file here named file is equal to open op underscore file dot txt and I will open this file in writing mode since we need to write it, and what next we need to do here is I need to write this particular string data in our file so I will just write here file dot write under curly braces we will write data ok after we are done with the writing part I will close this file I will write file dot close round braces so now let us try to run this programme again so here we have the substitute text let us open the file output file or op underscore op underscore file as you can see here we have the data in the substituted form this is something we can’t recognise we can’t figure out that this text was actually this and we have substituted it by using substitution cipher so now we are done with the programme, I will go through the program again. First of all there is a string present in python name string dot ascii underscore letters we use this string for the substitution part here we are substituting the letters present in the string by I minus oneth letter, here we have for that we are using dictionary we have initialise the dictionary here in dict after that we are executing the substitution part in this particular part we are writing dict string dot ascii underscore letters at ith position will be substituted by I minus oneth letter this is how we are doing the substitution please note that dict dictionary has dictionaries have keys and value in this particular dictionary keys are the letters present in the string and the values are the substitute letters I repeat keys are the values present in the string and values are the substitute letters, after that we are opening our file our input file and we are reading a character by character for that we need to supply a bullion variable here one as to read the file character by character then we are checking some conditions here first condition we are checking here is whether we have encounter the end of file or not, if we have encountered end of file in that particular case our programme will print end of file and it will break out of this loop and we are checking if the particular character present in the input file is actually present in our dictionary, if it is not present then it will then it will only write this ‘C’ in data otherwise if it is present it will write dict of ‘C’ the substituted letter in the data. At the end we are writing this string data in output in an output file so that we can get the clear cut idea of how the substitution is working and then we are printing the data and finally we are closing the file. So as you can see here we have two files here this is the output file and this is the input file, here the in input file you can read the text but in output file the output file is converted to some other text using the substitution cipher which is not recognised which can’t be recognised by the third person until and unless he gets to know that the substitution cipher is used here so this is how you quote for substitution cipher and convert many more text, I hope this programming screen cast was useful to you guys. Happy learning.

# TIC TAC TOE – DOWN THE MEMORY LANE

Do you remember this? Do you remember this? And this? Yes. And next choice is about TIC TAC TOE.

# <u>TIC TAC TOE – DOWN THE MEMORY LANE 01</u>

That was such a good time we used to play a lot of games in our childhood, I was just going through some of my old books, my rough note book would look like this when you scan it from back. I guess most of you are getting these nostalgic memories on seeing this, this is the famous game called tic tac toe which we all used to play in our childhood we love playing it such a enjoyable it's such a good memories we are re living it this feels so great so in this lecture series I would take you down your memory lane you all shall re live your childhood days, the fun way through computing. That were good old memories we all had almost played this game in our childhood for those who want to re live those memorable days I would start with the rules of the game, the game is something like this, there is a three cross three board which is nothing but board containing three rows and three columns that is a total of nine cells. There are two players who are involved in this game each player is given a symbol, one is given a symbol 'X', one is given a symbol 'O' they are suppose to alternate their turns, during their turns the player is expected to occupy a vacant slot in the board, when they alternate their turns the one who is able to occupy three slots such that the three slots come in one single row or the three single slots come in one single column or they occur along this diagonal or this diagonal or one of the diagonal, the one who is able to do this first is considered the winner, this is how the game goes if no one is able to occupy their positions in such a manner then it is declare a draw no one wins, this is how the game goes, I hope most of you had gone back to your childhood days, if you would observe the rules of the game this is much more than you playing established your victory. This is much more than that the planning the thinking involved requires a lot of envision ability you need to envision what would be your move as well as what would be the move your opponent make given his next turn, you have to envision both, you would play such that, you increase your chances you maximise your chances of winning as well as the you minimise the chances that your opponent wins this is how the intelligent player would play this game, so when both players play intelligently the game would mostly be a draw. This particular strategy where you want to maximise the chances of your winning and minimise the chances of opponents winning is called the min max strategy this is min max strategy which is a very famous game strategy and all such min max games are too thrilling and given that this joy of computing course we would now see how we shall play this game in the fun way using computers.

# <u>Tic Tac Toe – Down The Memory Lane 02</u>

Hello guys, welcome to the programming screen cast of tic tac toe, we had some discussion about the game which we played in our childhood namely the tic tac toe. It has the three cross three board that is nothing but the board consisting of three rows and three columns that is the nine cells, so according to us that is three rows that is row one row two row three, column one column two column three that is how we count but computers count in a different way, let me show you see this is how the index is done in a computer for a board sort of structure this is nothing but matrix basically this is a matrix for a computer that is 2d array that is the two dimensions, there are two dimensions namely the rows and the columns so it's the 2d array this is how the computers represent the board has that is how they interpret the board as see the counting starts from zero, zero one two that is how it counts we count as one two three so please note that if we say second row second column then it is first row first column for the computers so that is whatever is our count you subtract one from it then you will get the count as per the computer perspective this is how we have to we have to take care of the perspectives of humans and computers, first I would like to tell you this regarding the indexing part ok now let's get started with the game, basically let me just give you a brief overview here let me give you see this is three cross three board that is the board containing three rows and three columns so players there are two players in this game each players is given a symbol one is given a symbol 'X' one is given a symbol 'O', they have to initially the board is fully empty we have to place their symbols in one of this vacant positions in the board, this is how the game goes suppose player one places here next time player two cannot occupy the same position he has to occupy the vacant position something like that the game goes and if you would see the players who occupies consecutive three cells along any of the rows or any of these columns or along this diagonals or this diagonal is consider to be the winner. That is the player who is able to first occupy three consecutive cells in this manner is the winner, in case no one is able to do then it is a draw so here as I had discussed in the lecture the game is not just about your moves, it is about your ability to predict how your opponent will play so every player will play keeping in mind that he has to maximise his chances of winning and minimise the chances of opponents winning for example let me just give you overview it is better if you could have a pen and paper draw a board because for indexing purposes I have taken this particular image, but plane board be a beneficial but no problem you please take a paper and draw this kind of grid and follow my instructions and based on that you can understand so this is basically an empty board initially so the first player comes places his symbol here, why this place? Because along this diagonal or rows or columns any way at least one of this ways he has to occupy three consecutive positions only then he can win maximise his chances of winning this particular position one comma one for the computer and second row second column for humans this particular position is intersection of many of the winning possibilities, along this diagonal along this diagonal row this column along four of the winning possibilities there are three columns three rows two diagonals among eight winning possibilities on four of the wining possibilities, this particular cell is present so smart player would start playing this position to maximise his chances of winning. So this will be the first players move so the second player when he gets the move in the next turn he would want to maximise his chances of winning also to block him from the winning that is to minimise his chances he wants to block him from winning so here if you have a ‘X’ so he would like to block one of the four wining possibilities for him so maybe he can place his coin somewhere here so that this row possibility has been block now given that this row possibility has been block in the next turn if the player ‘X’ places his coin here it is a waste so he would place his coin somewhere here so now if you have seen till now here you have ‘X’ here you have ‘O’ here you have ‘X’ so if somehow this is now ‘O’s turn if somehow in the next turn ‘X’ could be placed here ‘X’ would be the winner ‘O’ thinks in that way if I miss this particular location next time ‘X’ will place his coin here and he would be the winner so he would place his coin ‘O’ here to block his winning opportunity I guess now you have understood what the strategy the players would be using they would want to maximise their chance of winning as well as they would like to minimise the chance of the opponent winning if both the players are smart this try dry running some any sequence of run if try dry running just think that both the players are equally smart it would be a draw. In case if at some point of time the player losses the foresightedness he doesn’t think of his opponent in some point of time suppose assume that in this particular instance ‘O’ doesn’t place his coin here he places somewhere here the next turn when it for ‘X’, ‘X’ would place his coin here than thinking of blocking O it is beneficial for him to place his coin and win, so your aim will be both, more priority to your winning at the same time not a minimum priority to defeating your opponent, you have to minimise his chance of winning, you have to block him as much as possible as well as increase your chances of occupying three consecutive cells so that you can win so this is how the game goes as I have recommended if you had taken a blank sheet of paper and dry run how the game would go just play the game without using a computer, just try you would understand how the game is played, what kind of smartness is required from the player for the game and so on so this is how the game goes so let us now start coding now ok so as I had said it is a 2d array the board is the 2d array for the computer so that arrays has been defined in a package called num py let me import it import numpy, I had import it so let me call it as board that is the name easier for me so let me call it as board I would say numpy numpy dot array, array of we need an array of three rows and three columns, we need to give it as a list and within the list each row as to be a separate list so you have blanks initially so for blanks I am representing it as hyphens that it is a blank cell that is what I mean by hyphens here so I have three columns in each row so three hyphens I would be giving so all these three or blanks sorry for this first row has been given now the second row so let me give ok second row is over now the third row ok first column of the third row, second column of the third row then third column of the third row so all three rows has been defined ok now for player one the symbol I allocate is X I had told it is ‘X’ for player two I allocate the symbol O so that is what I had defined here player one symbol player one symbol for symbol I am using ‘X’ player one symbol is ‘X’ player two symbol is ‘O’ so now let me start the game let me say play. As we would have seen till now just saying play serve the purpose because it is something we are defining see you got a morning symbol it is an undefined name so we have to give the proper definition as to what as to be done when you say play so let me define it define play ok give colon here so here start the function definition. So every particular turn, turn has to be alternated the first turn if it is given for ‘X’ second turn must be given for ‘O’ so how many turns would be given totally? It is three cross three grid that is there are nine cells so there will be nine turns so let me say for I will use a for loop because I need to repeat it nine times turn I use intuitive name so that it is easier to understand turn in the range you would have seen this syntax till now this is how you use for loop for a pre defined number of times, I wanted nine times so let me say nine for turn till the range of nine what you do is, it start from zero till eight it will count so zero two four six all these turns should correspond to player ‘X’ that then one three five seven that is even turns is for player x and odd turns is for player ‘O’ so how do we capture this? Using the modulo operator let’s do that. So I will use if turn modulo two is equal to zero that is if it is an even turn you should say it is the turn of ‘X’ let me print ‘X’ turn so this is turn of ‘X’ ok so you should allow him to place his symbol on the board so let me call place player one symbol and then after he places his symbol initial two moves if you would have observed the winning possibility from the fifth move only but still this single loop will be running for all the turns so let us include this check here but it will not be used in the initial stages but in the subsequent stages it would be needed that is as soon as there is a winning move, that is if there is some vacant slot still not that is all nine slots have not been occupied but still there is there was a winning move then the game has to stop so for that let me say you have to if won let me use terminology like this if one if someone this player one symbol this person has won then break that is you should quit the game this is nothing but we are quitting the game else so this is for player ‘X’ we had done else that is the turn is odd turn so all these things we would copy we would do this for player two whose symbol is ‘O’ so I had used ‘O’ player two symbol and if player two symbol this particular thing has won we have to check that is why do we check this? As you would have seen in this dry run during the fifth turn there is a possibility that one player can win that is from fifth turn any turn anyone can win there are good amount of possibilities that there can be a winning move so if that particular move if he has placed it in some position and if that placement is a winning placement then you should quit the game that is why we are using this so far so good this are intuitiveness so I guess you can understand it that is you have to repeat this things nine times that is because of the nine cells present in the board, if for every even number turn this is for ‘X’ and odd numbered is for O allow the player to place his symbol into the into a vacant slot in the board check if that is a winning placement quit this is what we have done till now. Ok as you could see here, see these are undefined names place and won so let me defined that here now ok define place so we have used p one s p two s let me say symbol it is a common terminology player one symbol player two symbol whatever this is be the it is a symbol so I had said that ok so first what should I do? Print the board so since it is a matrix that is I have to display it in the rows and columns format for that we have a predefined functionality to convert it into this is in the list format we need to convert it into rows and columns format for this we have a predefined functionality let us use it numpy dot matrix there is a functionality here numpy dot matrix within in that you have to pass the name of the variable here it is board that is it will take the value of the board in list format convert it into the row and column format and print it that is a purpose of this functionality ok we will print the board then we have to say we have to get the input for in which particular row that is how does it, what does it signify if I is, you are saying that you should allow him to place his coin where would he place? In some particular cell that is identified by the row number and the column number so row get the value input from the user row is an integer so let me type cast it initially its easier now ok, row I should get an input this is actually dependency on Mac system that's it in Linux systems as much as I have used this particular type casting thing is needed these are all some trivial things depends on your operating systems in case you are using some other systems and there is some other requirement you can always Google it and get back the correct method as per your system this I am doing it with respect to the Mac system ok so I had type casted I am taking the input so I should say enter your rowth position enter row what  are the values for rows that is allowed? One or two or three these are the allowed rows ok that is as per the humans perspective I told you counting that starts from zero why why is it that we are giving one two three here because this is as per the human perspective you are familiar with computers so you can understand this indexing mechanism but where as if you would ask this must be designed so that anyone can play the game so for human perspective the counting starts from one that's why we are giving like this then adjust it in the upcoming steps so you get the input and copy paste ok column so I should say enter column column one or two or three so I had said that this is the value you have to input it but what if the person has inputted a row and column that is already occupied or what if by some type of mistake or something he has inputted a value that is out of range of the board we have to handle it, all these we are doing it to ensure that your programme is robots and fault tolerance that is whatever be the fault that may be occurring due to the users input it must be handle your programme, your programme must be able to handle it we should not say that it is your mistake we should be able to handle it as much as possible that is one of the principle in software engineering that you have to make your systems friendly and fault tolerant that is if user commit some mistake even your software may be able to handle it you must be able to display him some polite nice message that this is the reason why this cannot be accepted please try again with the valid thing something like that, that is courteously that is the practice followed generally so let us do that here so let me check if so if it is a valid input I have to say ok you can place it, how would I do that? In case if it is an invalid input I should repeat the process so I need to I need to use while loop here so let me use it while one because it has to turn infinitely till you get the correct input so correct input so if it is a correct input I have to break so what is the correct input? Row the value of row is greater than zero, one or two or three all are greater than zero right? and the value of row is less than four, one two three all are less than four and even column must be same like this column should be greater than zero and column is less than four and that must be a vacant column for vacant what are we using it? We are using a hyphen so I am going to check that one, board of should we use row column? Or row minus one column minus one? it is row minus one and column minus one, why do we use it? As I had shown you the indexing earlier computers start counting from zero where as humans start counting from one this we have taken the input in a human friendly manner but we have to convert it into a computer friendly manner that is why we are deducting one as I had shown you that time second row and second column of humans is nothing but one comma one for computers so you have to subtract from both the indices row as well as column this is how the computers works so if the humans says second row first column it is nothing but two minus one first row and one minus one zero, first row zeroth column for the computer so this is how the computer will inter crypt because why is this different? Because humans start counting from one, computers start counting from zero that is the reason so if this particular thing is blank, blank is given by hyphen, if all these conditions are satisfied that is row and column are within this range one or two or three so it is greater than zero or less than four is nothing but one two or three and column is also the same and if the deserve vacant position then it is a valid input you can break here that is you can let him play yes broken from the loop so corresponding to this here you should let him place board of row minus one and column minus one is nothing but the symbol, whatever the symbol he wants to place this is how the place functionality works please check once again. It is you are displaying the board asking them for input to enter row and column value in case if they enter a faulty input or if that is not a vacant position you will repeat asking for inputs till they enter and if you again and again ask inputs they will get irritated so you should say if this particular this is for checking that whether the input is a valid input, if it is not a valid input you should print saying that invalid input invalid input please enter again some polite message only then they will understand that ok we had made a mistake and that is why you are asking the input again and again otherwise they may think that there is some fault in the system that's why even though I gave an input it is asking again that is what people will think that is why we have to print a message in a way that they can understand easily this is important please note this, this is important so we will ask input as long as it is not valid once it is valid we break from this loop and place that particular players symbol here so see place players one symbol in even number turn player two symbol in odd number turn so whatever is the symbol being pass here that would be placed in that position so this is how it happens so place has been done so please see to the flow till now may be you can pause here for some time and then you can see it and yeah see we had missed one more thing, if player one has won we should say break, player two has won you should say break if no one has won at the end of nine turns that the all the cells have been occupied and still no one has won we should see we should tell that, that has to be included so let us include it if you should say if that is not won if that person has not won player one has not won see not is an operator that is predefined that is one will say if the person has won or not that is it is true or false? If it says true if you are applying not it would become false so it will invert the output of this particular thing, it will invert whatever is the truthfulness into it, that is what not does ok and not player two is also not won, not won of player two symbol if both haven't won then you should print it is a draw ok so why did we do that? Because draw is also possible so that is why we had done this, this ok indentation is important it has to be at this particular position only column five because at the end of all turns only you have to check that is all nine cells have been occupied and still no one has won that means it is a draw, so we are checking it here fine let me change the indentation here too ok so this and this will be in the same level yeah nine and column nine fine ok please pass till here understand the flow of the programme and in the next video we will see how this functionality won, which has whether the person has won or not can be realised.

# TIC TAC TOE-DOWN THE MEMORY LANE 03

Aright guys, in the previous video we had seen the outline of the game we have a play method and we had a place method; play method alternates the turn between the players ‘X’ and ‘O’. Even number turns is for ‘X’ and odd number turns is for ‘O’ and place method checks whether the position is a valid one and a vacant one it is so it allows the players to place it if not it asks for input again and again till he gives the valid position, so this is how game had gone till now so let us see this particular thing that has been undefined won so let us define that method now define won so won we have a same thing symbol as an argument we have been passing symbol so let us see so what should you check? Check rows, you should check for rows if any of the rows has been occupied or you should check for columns, if any of the columns has been occupied in a consecutive manner or check the diagonals, diagonals if any of the diagonals has been occupied so you have to check all these things and return the answer any of these things has been occupied then that person has won. So see check rows check columns check diagonals all these are intuitive to us we have to define it, let us define it one by one. check rows, define check rows we have pass the symbol ok, I need to check each of the three rows so I need to use a variable check this so let me call as row r let me call it as r, r in the range of there are zeros so let me say three, r in the range of three for each row I will have a counter count is initially zero that is we haven’t counter the occurrence of that symbol in the row so count is initially zero now let me iterate through every cell in the row for that let me use c column that is each row has three columns has a cell of three columns right? one cell belonging to each column we have it so we have to iterate over each of these cells for c for cell or column you can take anything for c in the range of three because each row has three cells, ‘c’ is in the range of three so let me call it I will say if board at the r and c index contains the symbol I will increment my counter, count equal to count plus one, why do I use the counter here? Because it may be the case that in some column or in some row ‘X’ ‘X’ and the third thing may be ‘O’, in that case it is not winning placement XXX is a winning placement so I need to count how many times I had seen the symbol ‘X’ so that is why I am using something called as counter, at the end of this loop where we had iterated over each cell I will check the value of the counter, if counter is equal to three because each row has three cells, if all three cells have the same symbol then it is a winning move I should print that it is a winning move so let me print this particular symbol has won let me say this person has won and I will return true here I am returning so this particular functionality comes to an end, in case if the first row doesn’t satisfy it the loop for r will be check once again and for the second loop same check would be done for the third row the same check would be done so once for all the rows this check has been done and none of the places this has been encounter that means there is no winning movement along the rows so in that case this check rows thing here that is winning is not due to occupying three consecutive cells or any row along any row the person has not occupied three consecutive cells so in that case you should say it is not a winning thing so I should return false that is the person has not occupied three consecutive cells along any row that is the meaning here I hope you can understand this check rows functionality otherwise no problem you can pause the video here dry run the code and you will definitely understand it. So check rows is done check columns is pretty much similar so let me just copy paste it copy let me paste ok this is check columns just the name and some indices change because here a column value is fixed rows changed so let me change this as c and r so the rows change that is the column value is fixed rows change that's the only change here otherwise it is pretty much the same I guess you can understand this we would let me show you see in check rows we fix up this row we check each cell we count in check columns we fix up the column in each of the row corresponding cell we count if you don't understand please don't worry dry running is the key here you sit think like a computer just have a index table something like this and dry run it you will definitely understand it, it is very easy for all that it requires is some amount of thinking that's it it's very easy please make sure that you understand till here ok check rows is done check columns is done now check diagonals so yeah I will show here wait ok see the diagonals zero comma zero one comma one two comma two the general format it is of the form I comma I so both the values are same along this diagonal and this diagonal differs so let us three cells so let us manually do it two comma zero one comma one zero comma two so please make a note of these values that's it so for this I comma I general format you can use a loop something similar to what we are use in rows and columns just for that other case we need to use the thing manually let us do it now define check diagonals check diagonals for the symbol check it so for that diagonal let us do it if board of zero or maybe we can even manually do it no problem it is up to you, you can use a loop as well as you can do manually anything zero two is equal to board of one one that is allowed in maths but not in programming so we have to use and operator and we have to say board of one one is equal to board of two comma zero that is one of the diagonals and that particular thing is equal to this corresponding symbol only then this person has won so let me check it you can use any of the three values I am using one comma one that I equal to symbol in that case you say this particular symbol has won and you return true ok next is along the other diagonal here you are returning so in case if this diagonal has been satisfied this particular functionality would come to an end if this diagonal has not been satisfied only then the control would be transferred here you have to check this diagonal, board of zero comma zero is equal to board of one comma one and board of one comma one is equal to board of two comma two and board of one comma one is equal to symbol this can be return using the loop as well I will leave it as an exercise for you guys please do try using a loop to simplify this thing instead of using these many comparisons you can simplify it, alright if that is the case then this is pretty much the same so let me copy paste it copy paste ok in case if both the cases didn't satisfy then it means along the diagonals there is no winning move in that case you have to return false so you had checked rows, checked columns, checked diagonals and see this won functionality is check rows or check columns or check diagonals if in one of the places or operator works in this manner if at least one of this values is true the final answer is true, if along the row or along the columns or along the diagonals there is a winning movement then you would return true that is that person has won, if all three places there is no winning movement only then it will return false that is this person has not won so this is how this functionality works ok let me save it ok so I hope you would have understood the code till now please do pause understand the flow of the code really well and then proceed towards running the code, you can try dry running it as I said dry running is the best way to understand the computers perspective please do understand really well before proceeding further let us run this code and see what happens in the next video.

# TIC TAC TOE – DOWN THE MEMORY LANE 04

Alright guys, in the previous video we had seen how to quote for this game so we had a play method which alternates the turn between the players, even number turns are for ‘X’, odd number turns for ‘O’. During his turn he has to place symbol in a vacant position so here we take care that his input is proper within the range as well as the position is vacant. So based on this, this functionality works and for checking the winning moves we have a functionality one, it checks the rows with their any of the rows three consecutive cells has been occupied by this player or along the columns or along the diagonals so since we are using or operator works in the fashion that if one of the value is true then final answer would be true so if all three ways the player has not occupied only then it will say it is false that is this person has not won, only then it will say something like this, this is how the functionalities works. I hope you had passed, i had asked you to dry runner i hope you had dry run and you have a clear idea of how the code works how the flow is, please do that make sure that you are very very clear in it then you run the code alright? I hope you are now very clear, let us proceed towards running the code. Ok, let me run it see it is extend and i have in a matrix format so it is asking me to enter the row, let as i had said two comma two is the smartest move because you have four winning possibilities in which that particular cell is intersecting so let me say two column is also two, see it has place the cell ‘X’ here let me now you could see initially the board was empty now i said two comma two has to be occupied so it has occupied. So now if i would give again two comma two see its saying invalid input because it is not a vacant position so let me give some other input one comma two so this position has been occupied, so this is ‘X’ turn so let me give some input two comma one this is x turn so for the player ‘O’ this is the optimal placement what if he chooses some other position, why this is an optimal thing because if he leaves this opportunity here in the next turn ‘X’ may win so he shouldn’t leave this so let me say if he didn’t predict the next move what could be and he has chosen something else he wants to maximise his winning chance now this is ‘X’ turn so he may win here, ‘X’ has won, it displays the message that ‘X’ has won so if as you would have as you would have seen it is important that both players are equally smart, try that variant as well, try playing in that way, you would definitely see a draw if both players are equally smart alright guys thanks for watching till now please do try different variant and also for tic tac toe there are many more variants you can browse online please do try a different kind of variant and discuss in the discussion form of what are the variants you have discussed, how did you implement? What are your strategies? What are the new things you came across? Please do discuss everything as clear learning is the best way to learn, please do discuss with your Ta’s and i wish you a happy leaning. Have a nice day.

# TIC TAC TOE – DOWN THE MEMORY LANE 05

Alright, now we have seen one of the implementations of tic tac toe there are actually many such implementations possible, there are many other ways also there are some variants to the game as well you can look up to it and I would recommend you that try implementing some other variant which we haven't discuss and please do discuss the strategies that you take up to implement those variants you explain those variants the strategies you take up please do discuss all those things in the discussion form happy leaning.

# RECURSION

So we have a box here, there is another box inside this box and there is another box inside this box, there is another box inside this box, you must be wondering what I am doing? Then it is called recursion, so let us discuss about it.

# Recursion 01

Sir, may I talk to you for five minutes? Yeah sure. Sir actually I would like to show you this comedy scene. I see that you are always obsessed with movies, when are you going to discuss with me something technical. Sir, actually you are sort of person who actually sees computing in every walk of life, this is something we generally use in computing, acha such a concept is there in this clip so I would like to show you. Sure sure show me. This clip is it? Yeah yes sir. Sir, did you just see what happened here? Isn't this the concept we use in computing namely the recursion? Very much, so that's a good find vidya, this is a very interesting scene so I will do one thing I will try explaining what exactly is happening here huh? Concentrate look, there are several people right? There are some let's say fifteen people, yes sir correct? And then this very person, first person asks a question to his neighbour right? And then he asks that question to his neighbour so this kind of a caste gate continuous you see and goes till the first person and some where here this fellow knows what is the answer, he tells him the answer and the caste gate comes back and hits the source and there you are bingo! You have the answer. Right? So this is exactly sort of how recursion works, you keep passing on the requirement was function inside a function and it goes inside and inside well within the last step and comes back and there you are with the answer that's a very good observation vidya very nice, very creative of you. Thank you sir. Recursion is a very powerful tool in computer science and we are going to illustrate that with a standard example called computing factorial of a number.

# Recursion 02

We all know factorial of a number, it's a very easy thing to compute so fact of five is five into four into three into two into one, we all have been doing it from our school days right? Which is actually one twenty. So what is factorial? Let us take a look at it. Factorial of a number let's say six is six times five times four times three times two times one, do you observe that fact of six is six times fact of five, what do I mean? I mean factorial of a number one can compute by computing factorial of a number one less than that multiplied by the number so fact of n is n times fact of n minus one. So let me thing of how one could write a program to compute factorial of a number. That's going to be very easy, if the input is n I will just write a for loop and then for I equals one to n I will say answer is answer times I initializing answer to one, pretty straight forward right? The best thing about computer programming is that one can do the same thing in several ways. I told you how to compute factorial of a number, I will now tell you how to compute it in a way that is slightly complicated but I am going to introduce to you all a very important programming idea called the recursion. Remember, how I defined factorial of a number? We observed that factorial of n is n times factorial n minus one so how about this, I define fact function and then all that I do is written n times fact of n minus one whenever n is greater than one if it is any if n is equal to one I simply written one and that's what this programme s doing now wait a minute I am I am defining a function and calling a functions name within the same function isn't that weird. There is actually nothing weird about it every programming language gives you this facility, the facility of calling the same function within itself and python also gives you this facility as you can see this programme will simply give you factorial of a number. Try to understand what is happening here, you are computing a function by giving as a input a number and the function is calling another instance of itself. I am sure my terminologies are sounding complicated but you will get used to it as you see more examples on recursion.

# RECURSION 03

Hello guys, I hope you would have seen the concept of factorial of a number, let me just give you a brief over view of what it is. Factorial of a number is nothing but you keep repeatedly multiplying the number starting from one till you reach the given number and what is the answer you get is nothing but you call as a factorial of a number, let me give you an example. Let us take the number four, start with one, next number it is we have not reached four yet so take the next number two multiply it one into two is two, still we have not reached four so take the next number three multiply it two into three is six we have not reached four so take the next number it is four we have reached four so with this multiplication we need to stop so till now we got the six as the answer of multiplication and multiply four to it you get twenty four that is your answer so we say factorial of four is twenty four that is one into two into three into four is twenty four that is what you call as factorial of a number. You start from one and keep multiplying the next numbers until you reach the given number ‘n’ to get the factorial of the number ‘n’, if ‘n’ is the given number, in my case its four n is equal to four, so if the given number is n you start off with one two three and so on till you reach n keep multiplying and what is the final answer you get is what you call as ‘n’ factorial ok so how would we programme it? As you would have guessed from the procedure of calculation I had said we can do it using iteration let us do that so I said factorial, that is the functionality we need so let us define a functionality factorial for we need to find the factorial of given number ‘n’ so let me define n here ok so I need to start from one and go till I reach n so we can use for loop right? But for loop uses the function range to deal with numeric values and as you would have seen in your previous videos and till you practice you had till now range function would take if you ask for range of five it would start by default zero and it will count till four that if you would say five it will count one less than that, that is how it does. But we want it to start form one and we want it to count till n ok so we need to tweak in the range function to get this thing so we can define the start value for the range function if we don’t give a start value till now we had an given the start value for most of our programs so if we don’t give a start value the default start value is taken as zero but we can tweak in we can give some start values as well so let me do that in this program so I need to store the product so let me a variable call product so initially we have one as the product because with one of you multiply anything you will get the same number as the answer so you keep multiplying but from where will you start one into one into two into three into four into up to n this is how this has to been done right, so we have to start our multiplication from one let me define something called as product which carries the value one initially ok, so let me use the loop to iterate over the range that is to move over the range of numbers from one till n so I need to start from one so let me for I in range of I can give a start value as I had said start value one comma stop value, so whatever is the stop value it will stop one less than that, I want to stop at ten so what should I be giving here? N plus one a simple tweak so in this range that is from one to n plus one that is one two three it will count up to n so this is what we wanted we had taken that value as ‘i’ whatever is the new product after taking this ‘i’ will be whatever the product till now multiply with ‘i’ I hope you understand this that is if I want the factorial of three whatever is the product how will I do? One, one into two is two till now my product is two, two into three is six this is how I take the factorial so whatever is the product we had computed till now that product into this particular value of 'i', this is how we compute the factorial. So at the end of the loop that is till you have reached 'n' whatever you keep multiplying, whatever is the product that is your answer so you have to return the product here so this particular functionality will take a number 'n' find its factorial and return the answer that is what it is doing. So let me, I have to use this functionality I have define the functionality now I am going to use this functionality so let me take an input sorry input, input generally takes in terms of strings in Mac so I need to type cast into int this is a Mac dependency that's it, it may differ from operating system to operating system with practice you will get to know how your operating system expects your input syntax to be, based on that you please modify this I am doing this with correspondent to Mac so I am type casting it I am typing it the input let me say enter the number I will get a number but see factorial is defined only for positive numbers for negative numbers factorial is not defined, in mathematically factorial is only defined for positive numbers. If the number zero is given what is the case, if zero factorial has been defined as one that is the mathematical definition that is why see I had defined the product as one, in case if I given zero for I in range one comma one so it will not at all go inside the loop it will directly return the value of one here so this is like a dual advantage sort of thing, zero factorial is one also one is the number with which you multiply any number you will get back the same number so these are the research why we started off with the initial product value with one here. Ok you can start give zero but not any negative number I will enter a positive number so, let me give a clear message enter a positive number I am asking you to enter a positive number if the user still enters the negative number we should be smart enough and we should not allow this input so I should check it, if 'n' is less than zero that is it is a negative number I should display the message that factorial is not defined on negative numbers, on negative numbers see why we are doing this is this is what we call as fault tolerant major if the user has mistake given the some input which is actually incorrect we should not throw at him some random search and terrify him, we should give him a polite message so that he can understand what mistake he has done, he will not repeat it so this I one of the software engineering principle we are following, if 'n' is less than zero you that is it is a negative number say that factorial is not defined on negative numbers else let me say factorial so let me say f for factorial, I will get back the answer, I will call factorial of 'n' so I had called the factorial of 'n' so it will return an answer right that answer be captured in f so now I have to print it. So let me print it print let me say factorial of 'n' is f that is 'n' factorial is 'f' that is what we wanted to be printed. So let me save the code let me run it or maybe, I will give you a brief info, it will take an input it will check if it is a negative number, for negative numbers since mathematically factorial is not defined we display the message that it is not define, if it is a positive number it will start computing the factorial, how factorial is computer is? You start off from the number one and keep multiplying the consecutive numbers till you reach the given number n, at the end of this multiplication whatever is your answer is what you call as the factorial so you return that particular product and you print it here, factorial of n is this you are printing it this is how the working is let us execute the programme let me run it ok, it is asking me to enter a positive number let me show you let me enter a negative number see factorial is not defined on negative number so this particular thing is fine ok so let me run again with the correct input positive input let me give a number six, ok factorial of six is seven two one into two into three into four into five into six is seven twenty that is what is displayed as the answer here fine? So our programme works fine this is not the only way to calculate factorial there is another way as well so you can understand that way for that please do observe the solution try, try enumerating running programme on different inputs, you will find some pattern if you are able to crack the pattern you will understand the alternative way of calculating the factorial. Thanks for watching till now have a nice day.

# RECURSION 04

Alright guys, you had seen iterative version this version where we use the loop to find something is called the iterative version, the iterative version that is we are repeating something again and again this version is what you called as iterative version, so we have seen the iterative version of finding the factorial in the previous programming screen cast, I have told you to observe the solution you could find some patters, I hope you would have found some patterns, using that patterns we will solve the factorial problem here in yet another technique what we call as recursion, so recursion is nothing but a technique in which the  programming languages allow a function to call itself again, it may sound confusing at the beginning but as you go it will be rally very easy so see as I had said if you want to compute the factorial of four, one into two the product is two ok then the next time when you approach three the product till now is two, two into three you compute is six, the product till now is six, six into four is twenty four that is how you got the answer right? and observe that you need four factorial you were making use of the values of three factorial that is one into two into three the values is six is being used to compute four factorial and you need three factorial you were using the values of two factorial so there is a dependency between these values so when you need larger values you are requiring a smaller value to compute the larger value so there were so dependency so the n factorial dependent on n minus one factorial this was the general pattern you could observe from the solutions mathematically also there is yet another definition of this is one definition of n factorial so one into two into three into till n another definition of n factorial is n into n minus one factorial so for n minus one factorial what would you do? N into n minus one into n minus two factorial so ultimately if you keep expanding where would you stop? As I had said zero factorial is one so that is where we stop, if you hit zero you will say I have reached the n so one is the answer here so till now whatever has been multiplied multiply one to it that is nothing but by multiplying one answer won’t change so we have reached the answer, so zero marks the end of this procedure here that is here we are depending on a smaller value for a larger value competition this process stops at the point zero right? so that place where we stop we call as the base case or the anchor case I would write it here probably base case here I will use the green colour because this appears ok so this is the multicolour command basically this appear in green colour it would be easier I guess base case or the anchor case, we can call it this is nothing but point where your recursion, recursion is nothing but your dependency on something of the same type that is factorial of n is a computation it depends on factorial of n minus one, how is that computtered? The same way. Again you apply the same formula n minus one into factorial of n minus two so you are basically doing the same thing but every time you do it again you are doing it on a smaller number so then you multiply one into two into three up to hundred if you multiply up to ten then use it to find eleven then use it to find twelve you are basically break it into smaller things and solving it so when you do that, that particular process where the bigger thing was depending on something similar smaller instance of it a bigger value of n depended on the similar kind of thing of a smaller value right? the similar factorial computation is similar three factorial depended on two factorial, two is a smaller value compare to three so something like it n depends on n minus one, n minus one intern depends on n minus two this chain of dependency how this is being that this process is called as recursion. A function calling itself with a smaller instance is what you call as recursion. So if it keeps calling at some point it has to stop calling here, we have to say that ok we have hit the answer, we have arrived at the answer we have to say that. This particular point what you call as base case or the anchor case point where the recursion stops so this is what you call as the anchor case so let me modify this functionality this particular thing is not being used here this is the iterative version iterative version now let me start off with the recursive version recursive version as I had said that factorial of n can be represented as n into factorial of n minus one this is yet another way of defining the factorial function mathematically, so this particular way we are going to use here so where will it start? Factorial of zero is equal to one what if negative number occurs? They won't occur because here we are taking care of it, if negative numbers have been given as an input we don't even call the functionality we are calling it only on zero or greater than zero only so here we may not worry about handling negative numbers because in the actual calling part we had handled it so we are not worrying here ok this thing we are going to say so first the first step in recursion is always putting the base case base case, here base case was factorial of zero is one so I should say if we have passed the number as n, n is equal to zero I should return the product as such right so let me bring use this statement the product is initially set to one right? this is not needed actually I can even remove this I can directly say return one so I should return one so I had return one that is factorial of zero is one that is why id n is zero I returned one else what should I do? I should return n into n minus one factor n into n minus one factorial so I should call it as factorial of n minus one ok let me give you illustration before we actually run the code, so if I call factorial of three let suppose let start with a smaller value so that it is easier you can extend it for any value even factorial of hundred anything you can extend that's not an issue ok so how it begins is it will first check if n is zero three it is not equal to zero so it will come to the else part it will see three into factorial of two, so value of factorial of three depends on the value of factorial of two, similar instance but the value inside is smaller ok so now this particular thing has to be called ok factorial of two gets called so again see factorial of two so two is passed here n is equal to zero no two is not equal to zero so its comes here it will say two into sorry two into factorial of one ok so now factorial of one gets called ok what is factorial of one gets called, so one is passed here if one equal equal to zero so false come back to else return one into factorial of n minus one so it will compute one into sorry factorial of one minus one is zero ok so now ok factorial of zero, zero is passed here n equal to zero, zero equal to zero its true so it will return one so this will return one so now your calculation will go in the upward direction ok let me show you the upward calculation ok so one is returned so it goes one level up, one level up so it will calculate one into one what is the answer one so that particular value is passed to one level up ok so two into what is the value being passed there one so that is being calculated so two into one is calculated two into one this one there is a passed from one level below for it there is passed from one level below so the computation breaks down into smaller numbers and once you get the number where you get the where you know the answers you trace back till you get the larger answer that is how we are working here ok so two into one is two ok let me now say let me now say so two has been passed here already it was having three, three into two, three into two ok what is the answer? Six so this thing has found it so it will get the answer has six, so this answer will be passed on to the main function value we have called this so when you call factorial of three it will keep splitting the problems into smaller ones until it knows the answer and once you know the answer trace in the backward direction till you get the answer for the larger problem this is how recursion works ok this is the Woking of recursion let us start a new console and let us run it ok? before that I have to save this I will save ok let the console start ok so it has started so let me run the file enter a positive number the same negativity check all these are the same so I am not giving a negative number so let me give back as positive number let me say five, factorial of five is one twenty so in a similar fashion it will break down five will break down to four, four will break down to three, three to two, two to one, one to zero for zero you know the answer now use that answer trace in the backward direction and you will get the answer as one twenty. So this is how the recursive programme work, I hope you have understand the concept here or it's not a difficult thing it's a easier one all that you need is some practice that's it take some examples you try thinking in this fashion, try breaking into smaller pieces then once you get the answer try constructing back the actual answer. So I hope you have been remembering the movie clip that has been shown at the beginning of the video, the person would ask answer for something to a person before him, the thing keeps on propagating till someone who knows the answer, once you get the answer you propagate in the backward direction to the person who has asked the question, in this case what is the factorial of three? That is the question asked by your main function that is question asked by your sir, so you cannot compute it directly so you try splitting it into a smaller instance so that is something analogue us to asking the person before him whether he knows so whether this thing can be calculated it checks even this cannot be calculated so again split this is can be calculated no so again split can this be calculated? Yes, calculate use the answer and propagate in the backward direction, I guess now you understand why we had shown the movie clip and also how the concept of recursion works, this is really a very powerful concept in computer science, it will make it institutive things be easier to code that is something will be intuitive for us we will have in mind that this is how things has to be done. But how you would translate it into a code? It may be difficult if only iteration value for rescue recursion is handy in many situation you would come across as you keep learning some advance stuff even in some stuff what you had learned till now you can have a recursive version for some problems please note that for this we had an iterative version as well as a recursive version so for every recursive version there is an equalent iterative version too, so you can use anything that is convenient for you and in some places recursion is highly intuitive and easy on minds so once such application will see in the next video. Thanks for watching, have a nice day.

# RECURSION 05

Alright guys, hope you had learnt about the powerful technique of recursion, I hope you remember search algorithm that you had studied previously in your previous weeks namely the binary search, let me give you a very quick overview of what is does? It is basically how we search for in a dictionary or in a telephone directory something like that. Why did I take dictionary or telephone directory? Because the words or the contents in the directory are sorted or that is arranged in some ascending order they are arranged in the alphabetical order so that is why we can apply this technique so binary search basically calls for a sorted list, you want a list that is arranged in some specific order, either ascending or descending it calls for list arranged in their order and in that list you want to see whether a particular number is present in the list or not whether a particular item is present in the list or not, so what would you do? You would check the middle of the list, if that is your required item you are done else you check if it is less than your required number or greater than your required number, accordingly you will branch on to the left half of the list or the right half of the list. So this is how your binary search works you have seen a iterative version of binary search in your previous weeks I guess so that is easier but still an easier thing like how I had given you an intuitive version of binary search like you have to scan into the left half and you have to scan into the right half something like that I had given you an intuitive version so such a intuitive thing can be easily translated to code when you use this technique of recursion so let us see how we can use recursion through simplify the process of writing code for binary search. Let me say I will define binary search, binary search define binary search for this I need list or you can call it list l I need a list l, some element x which we want to find, you were processing it by sing the index values, you would have seen that till now you may be familiar but still I am telling you a very quick recap, indexes in computers starts from zero human start counting from one cut computers start counting from zero so the counting starts from zero so if your list has five elements the counting would be done as zero one two three four, zeroth element, first element, second element, third element, fourth element this is how your computer counts so you have to give the starting index and the ending index these are required for your binary search right, so initially your starting index is zero ending index is actual end of your list so that is entire list then as you scan the middle element you keep discarding one half of the list and just keep searching in the other half so that time it will change so these are the Para things you need to do binary search, you need a list you need the element x which you want to search you need a start and end index of the list where you want that is you won't search the entire list all the time, you will search parts of list by discarding the unnecessary half this is how your binary search works so we needed this four parameters ok let me define as I had said recursion requires base case, what is the base case? When we have only one element left in our list, if just one element is left in our list then we have to see that particular element if that is the required element then you say yes required element has been found else you say element is not found ok so for element not found I am going to return value minus one I am going to assume that my list has all positive values and minus is not the value in my list also I am going to return the position of the element, I am not going to return the element even if have negative element I don't mind let's find I am going to return the binary search basically here I am going to return the position in which the element is present at the list, so that is what I am going to return so position starts from zero right? so if the element is not found I will return minus one that is the element is found somewhere outside the list, that's what basically I mean so I will return minus one in case if it is not found so base case is the base case is one element just one element in the list that is the base case so if there is just one element in the list how would you start and end, they would be equal, start and end will be the same value because there is only one element that is the start as well that is the end so I will start using the start and end indices, if start is equal to end I would check if l of start or end you can use anything inter change of it because they both are equal in this case, if that are equal to your required element x you return the start value, that is return that position hence there is just one element say suppose assume you have just one element ten in your list and what you want to search is fifty, fifty is not present in the list and it is having only one element then you have to say element not found so for that I said the code I am going to use encoding I am going to use is minus one so I will return ,minus one, element fifty is present in the minus oneth position meaning that it is not present ok, so I will return minus one in that case ok so this is the base case if the base case is not true then else that is there are more elements in a list, in that case you have to find the middle element and you have to discard the unnecessary half and search through the required half, you have to split the array into halves and search through the required half that is what you have to do right? so divide the divide the array into halves this thing we have to do for that we need to find the mid position, mid position is nothing but start plus end here I have to put up bracket because I need to calculate the sum first then I have to divide by two this is the midpoint right this is intuitive start and so here will be your start and here will be your end the midpoint will be start plus end divide by two this will be your midpoint when you divide there is a possibility that you can get some value that is floating point may be seven by two three point five, what is the meaning for position three point five? Either you have to take it to three or to four so for that we are using a functionality int so we are type casting it to integer because position is a integer we want it in a integer that is first position second position third position this is what we want, we don't want three point five position so will use int so it will have its own conversion it will take it to three point five to three or four maybe we can run here and check int of three point five let me give three so it is taking to floor functionality, floor is nothing but the integer that is closest to it the greatest value of the integer that is closest to it if we take the number line three point five is somewhere here three is here and four is here it is going to the left side so int functionality is going to the left side and taking the greatest integer two one everything is present in the left but the greatest in the left the integer is three so three is returned as the answer. So that will be taken here let me start a new console better because for this program new console ok so this is the midpoint we have computed, we have to check mid element is the required element so if the if l of mid is equal to x that is the case we have done we have found the element return the mid element else I have to check if it is greater or less so I would say l of mid is greater than x means you have a sorted list and the middle element is greater than your required element so where should you search, definitely your element will not be in your right half you can discard the right half and you need to search the left half so your array would now shrink from the starting position to mid minus one so mid position didn't have the element also there is no chance that it will be after mid so it would be before mid so we need to shrink the array up till mid minus one position that is one position before the mid till that we need to shrink the array, that is we will do as return the result of binary search on the same list for the same element x start position is not change see because we are we have to check the left half of the array so the start position is the same but the end position is not the precious end position, previously we have the bigger list we want to split it so the end is mid minus one why mid minus one? Because at mid position we had checked, at mid position didn't contain the element 'x' so an 'x' is the smaller value that is 'x' is l of mid is greater than x meaning that x is lesser than l of mid so 'x' being less than the mid value will be present in the left half so from starting position till one position before mid whatever is the array left search in that array and return the answer if what we mean, see it is really intuitive see this is the left half, this is the left half than using of iterative method recursion is easy you can translate your intuition into code very easily if you use the recursive ideology ok so this is the recursive way to call the left half of the array, call the procedure on the left half of the array search the left half else you have to search the right half, so for the right half end value is the same but starting point is different right? we had searched till mid we didn't find it at the position mid and the value x is greater than the mid value so it would be present from one anywhere from one position after mid up till the actual end point right? so your start value will change or I would say return binary search result on l, x mid plus one and end, the modify starting point is mid plus one till mid position we had checked there was no element that is why we are checking the path after the midpoint, here we are checking the path after the midpoint, here we are checking the path before the midpoint that is the left half and the right half whatever we had it intuitively in our mind we are translating it to code very easily so this is the power of the recursion so it would recursively keep computing and we will get the answer so you can take any example, I had given you an example of how recursion works in the factorial screen cast similar to that you take a list you try applying binary search on it, you try how recursion works, you would have understood how recursion works so this particular search the bigger list depends on the value of searching on the smaller list so whatever the result you get after searching on smaller list will be translated back and will be returned as the result for searching on a bigger list this is how your binary search in the recursive manner so we had defined binary search now we have to used it, right? so let me give a random list you can also change it with your version you ask the user to input how many numbers he want to input, you input that many number of numbers or till he presses some keys you keep getting input you can by now I guess you all are familiar with the various conditional construes using that you can modify the code as you wish, now my motto is to demonstrate binary search so I would just give a smaller list twenty forty five sixty seventy ninety this is my list I have given please note that I have given a list in the sorted order, if it is not sorted you have to apply sorting first then after you get the sorted list then you have pass the sorted list to the binary search, binary search expects the sorted list so that is very important please make a note of it, so this is my list and I will input the x value from the user ok, let me input the value of 'x' from the user and I have to typecast the input in my Mac machine so I am type casting you please follow as per your machine dependency input enter search key search key I would say that is what number you want to search, I would input the search key from the user, that is my 'x' so binary search will return the index and so let me store it as index is equal to binary search on the list l search key ‘x’ starting position is always zero for the bigger list initially we start with the entire list zero and the end position is zero to length minus one this is the five element list so it will be four so let me generalise it so let me say length of l this will give the length of the list that is how many elements are present in a list will be return by this length of the list minus one this is the ending index so you initially start of your search with this particular index this is the suppose my search key is eighty this will be the entire key is passed so it would find the mid value sixty it is righty is greater than sixty so this half is not needed this is neglected and this particular thing will be executed and seventy ninety this will be my new list in this list I will search for my key seventy and ninety this will be taken as the mid value as we have seen that floor value is taken so this value is taken as mid value, eighty is still greater than seventy so it will skip to the right half so here there is only one element ninety so this case would be executed ninety is not equal to eighty so it will return minus five so I will come to know that this thing is not present so something like that it works you can trace through it I hope now you are clear with how recursion works and you are clear with the binary search concept too so you can understand it you all that you need is take a pen and paper and trace through how some example work, that is what is required with that you can easily understand the concept ok so it will return the index where the element is present in the list, ok I have got the index so if I just say the index this is not enough for a normal user for us element sixty is present at the third position if I say two they will say what is this computers doesn’t know even this so we have to translate it into a human friendly format right, so basically what is that we have to do, we have to add one, one is represented as zero, the index in second position is index one, third position is index two so basically whatever is the index it returns add one to it and display it to the user and if it returns minus one, you should not say it is present in the zero you should say that element is not found, print appropriate message so first we will give a check that if index is equal to minus one in that case you should print x value not found ok x value is not found ok else you should print, that is minus one is the not the case that you should print that x value is found at found at position index plus one, computers counting system and the humans counting system differ by one that’s why we are adding one and display ok so this is how you have done, let me save the code I hope you are clear with it we have a list we input a search key this you can even modify it to getting the input for the list element is from the user, you have to sort the list please note that you have to send a sorted list for binary search to occur, so you have to sort that list input a search key then you apply binary search on it how it works? If there is just one element in the list it will check if that element is the required x element x, if it is the case it will return the index else it will return minus one so whenever the element is not found it will return minus one that is how we have encoded it. In case if the list have more than one element what it does is, it will find the mid element and based on the mid element since the array is sorted it will discard one of the halves if the element required is exactly the mid element we are done if it is less than the mid element you have to search the left half so it will discard the right half it will search only the left half if it is greater you have to search only the right half it will discard the left half so for this to understand this really well I would suggest that you take a twenty element list basically and you try tracing it on paper, basically to understand this clearly you should work out a lot on papers and less on computers, computers can do this in fraction of second but humans to understand this strategy it requires some practice so please take a pen and paper take some twenty element list randomly you sort it maybe for that you can use the computers as well because sorting a twenty element list may take some time, so you can use the computer as well you sort it or you take the sorted list of the twenty elements you randomly give some search element you give some element which is present in the list as well as some element which is not present in the list try to understand how the various runs of the programs are and you will really understand the process very easily after practice all that need is needed is practice please do practice practice practice that's it now let us run this program let me run it ok it is asking me to enter a search key let me enter eighty, eighty not found perfect! Ok now let me enter ninety, let me enter ninety, ninety is found at position five perfect. We got it when I entered eighty, eighty not found at this list so it's says eighty not found and when you enter ninety, ninety is found here it is found at position five so it says ninety is found at position five so it works, I would recommend that please you take some pen and paper and work through the various example, work through list with large elements as well here just that I wanted to demonstrate the recursion technique I had taken a smaller list and already sorted list, you try different things unsorted list you sort it then apply binary search at huge list, a list where you get somewhere it is if it is a twenty element list try to find the sixth seventh element try to find the eighteenth element try to find something exactly near middle or near the middle something like that, you try various possibilities you will understand actually what is logic of binary search how it works, all that is needed is practice with pen and paper. Keep practicing thanks for watching have a nice day.

# RECURSION 06

Alright guys, hope you had seen a few examples of our recursion work, this is yet another mathematical function or mathematical sequence I would say which can be easily computed using recursion. So this is what we call as the Fibonacci sequence let me give you some explanations in commands before we start off with the coding. So we are going to see what is called the Fibonacci sequence. I guess you guys are familiar with it, still let me give you a brief overview so the Fibonacci sequence starts with zero and one this is the zeroth Fibonacci sequence same like computers it here the counting starts from zero, zeroth Fibonacci sequence is zeroth Fibonacci number is zero let me write one by one that's better, zeroth Fibonacci number is zero sorry I am sorry for this, zeroth Fibonacci number is zero the first Fibonacci number is one second Fibonacci number is nothing but now you have got two numbers let me put it ok, given that you have got two numbers you add this two numbers and you got the result has zero plus one is one so this is your second Fibonacci number. The third Fibonacci number is nothing but add the last two one plus one two, two is your answer, the fourth Fibonacci number is one plus two three, three is your answer, the fifth Fibonacci number is three plus two five this is your answer and so on and keeps going keeps on going it is a infinite sequence basically. The initial two values are nothing but the seed values they had given, these are the in our recursion terminology this is the base case or the anchor case, this is the base case or the anchor case for these two things we know the value we can directly written the value, you can directly say this is the value, they say I won the first Fibonacci number they can directly say first Fibonacci number is one if someone is asking you the tenth Fibonacci number something like that you don't know you need to calculate it so using zero and one you find two, using one and two you find three, using two and three you find four so do you see a bigger value of the Fibonacci number that is the nth Fibonacci number depends on the value of n minus one Fibonacci number plus n minus two Fibonacci number so it depend on the previous two values right? so here we can apply recursion and we can easily find the nth Fibonacci number so our objective here is to find the nth Fibonacci number, Fibonacci number we can have the iteration of the code too but in this case I would leave the iterative version of the Fibonacci number as an exercise for you guys please do try it, it is not very easy it requires some amount of thinking but it's not herculean task something like it it's not rocket science it is do able so you need to invest some amount of time you invest some amount of thinking, if you do that you can find iterative way how to find the Fibonacci number, so that I would leave it as an exercise, I would demonstrate the recursive way for you. Ok so let us start off I need to define Fibonacci, I would say Fibonacci of n, nth Fibonacci number is what I want to find, I know the zeroth Fibonacci number and first Fibonacci number, zeroth Fibonacci number is zero, first number is one so in this case I need to return if given n is zero I need to return zero, if the given n is one I need to return one so if n is zero or one I need to return back n itself so that is my base case ok that is why I had given a space here as well this is the base case and maybe I will put a dash here this is the base case and this are derived from the base numbers ok so this is the base case so let me first put it, if n is less than two and similar to that the negative value of n Fibonacci is not defined we need to check that in the main part where we are calling this functionality same as factorial how it is defined just for positive numbers and zero, zero and positive numbers Fibonacci is also defined only from zero for negative numbers it is not defined so let me say zero and one right so let me say n is less than two just return back that n, if it is zero return zero if it is one return one so like that n ok else if it is greater than or equal to two from two onwards we are dependent on the previous values so you should return back the Fibonacci of n minus one plus Fibonacci of n minus two maybe you want the second Fibonacci number you should return Fibonacci of two minus one that is Fibonacci of one plus Fibonacci of zero that is what we calculated, Fibonacci of one is one, Fibonacci of zero is zero so that is the first Fibonacci number is one, zeroth Fibonacci number is zero so zero plus one, one that is how we calculated right? so that is what we are doing you take the previous two Fibonacci numbers and add it is what you are doing, if you need the third one what you do is second plus first, first one you know the value so substitute one, Fibonacci number second Fibonacci number you don't know so you calculate second Fibonacci number plus one for getting second Fibonacci number again you call it for second it is first plus zeroth so zeroth plus one is one, one plus one is two so you will get the third Fibonacci number so you can trace it on a paper and you will understand the working so this is how basically Fibonacci numbers work if it is less than two just return back the same value otherwise return back the sum of previous two values right? This is the intuitive way of explaining Fibonacci numbers see we have translated our institution we have translated the intuition into code very easily all thanks to recursion with this powerful technique we can really made this code very very easier but for iterative version you can write an iterative version I don't say that with iterative version it is totally impossible you can write an iterative version but it requires some amount of thinking invested here ok so I have got the answer so I should take an input n is equal to integer input what I wanted so I will type cast for because of my machine dependency I type cast input enter the positive maybe, I can say or non negative that is because zero is also defined right! so I should say non even in factorial please make the correction there, it is non negative because zero factorial is one also defined so enter an non negative number, enter a non negative number is the input is the message I got so and from input I will get still I need to check in case that person has by mistake entered an negative number I shouldn't throw at him some, some results that is incorrect so I should throw him I should tell him the suitable message properly so I will print undefined for negative numbers n is less than zero means that it is a negative number whatever input he has given what is undefined? Fibonacci number is undefined for negative numbers. Ok so I will say Fibonacci numbers are undefined for negative numbers ok so I had given it Fibonacci numbers are undefined for negative numbers else if it is not a negative number than I should print the nth nth Fibonacci nth is not correct because we gives first second I have to put st and nd here properly so that formatting is needed I would modify the message instead smartly Fibonacci number at position sorry it is position n is I should call Fibonacci of n alright? I hope you guys can understand this till now see I take a number input a non negative number zero or positive I agree if it is a negative number I will say it is not it is undefined I print this message else I call this functionality Fibonacci or n ok if I call this functionality Fibonacci or n if that number is less than two for zeroth Fibonacci number is zero first Fibonacci number is one so this is something defined as the base case for us so in that case I will return the value of n otherwise I will compute the sum of previous two Fibonacci numbers right? so this will input the nth Fibonacci number so for first let me verify further let me enter five, if I enter five I should get five as the answer or maybe enter four I should get three as my answer let me verify it first then maybe I will find the other Fibonacci numbers ok let me run the file ok enter an non negative number I will enter so just for verification I enter minus seven Fibonacci numbers are undefined for negative numbers, yeah! I got the proper message ok now let me test it if I enter four, the fourth Fibonacci number is three, I should get three as my answer let me run it, enter a non negative number let me enter three, sorry four, fourth Fibonacci number is three, third Fibonacci number is two fine you can do it. Let me enter four Fibonacci number at position four is three yeah it is working so now let me find out the tenth Fibonacci number let me do tenth Fibonacci number, Fibonacci number at position ten is fifty five maybe you can list the numbers like this and you can verify but its correct so once its working here definitely it will work for many value so using this program you can find the Fibonacci number at the nth position in the sequence, in the Fibonacci sequence what is the number present at the nth position, so here nth is luckily this is one thing in mathematic which is very close to computer science counting starts from zero so nth number is nothing but for humans counting style n plus oneth number pleas have that in mind here counting starts from zero. I would recommended you guys invest some time thinking if you can do this in an iterative fashion it requires some amount of thinking to make it iterative it is possible it can be done but it is not straight forward it requires some amount of thinking so when you invest some time for thinking in the iterative style you would definitely appreciate the technique recursion because it's lets you do something straight away from your intuition whatever you understand from someone explanation you can straight away try staring the code recursion allows that thing, that particular thing is difficult when you do it in the iterative way but still it's not a rocket science it's doable please do invest some amount of time try doing the iterative version and please do discuss in the discussion form of how you tried the iterative version what are the challenges you faced? Or if you can print the sequence Fibonacci sequence if they are asking the nth Fibonacci number, can you print the sequence starting from zero till the nth Fibonacci number, can you try that variation? You can do a lot of things pleased do explore and keep trying more examples so that you understand how recursion actually works. Thanks for watching, have a nice day.

# SNAKES AND LADDERS – NOT ON THE BOARD

As you all know life is full of ups and downs and so is our next game, our next choice is about snakes and ladders

# SNAKES AND LADDERS – NOT ON THE BOARD 01

Six, that will be one two three four five and six. You will get another chance, oh! Six oh oh ok right one, that’s a one you are on thirty seven that perfect, four, one two three and four. So the game that we are playing is called snake and ladders the game again involves a whole a lot of randomness and there is no place for logic or thinking so the idea is very simple, you start with two pawns as you can see, I am red and she is blue we start with one and this is a dice where I roll and then if it’s five I start with keeping I start keeping five so let’s say for example we both are here and I roll this it is two and one two and then she rolls, can you roll bhawana? It’s three and she puts three and whenever we encounter a ladder we climb up and whenever we encounter a snake, a snake bites you see we come down to its tale and then continue you see assuming I climb up like this and finally come here finally come in here just in case I put two from here the snake bites and I may have to come down to twenty three so that’s the rule of the game, the point is anyone who reaches hundred wins the game. So bhawana where were we I think I was eighty one, you were on eighty one, I was on eighty seven ok yah perfect and I should I play maybe I play once more, I think you were about to getting bite by snakes so let’s go, but I was explaining the game I think I should get a second choice. No no I don’t think in this game you will get a second choice hey come on I am not a cheater cock so let’s continue from here one two ok fine I will accept that it was two!

# SNAKES AND LADDERS – NOT ON THE BOARD 02

You saw this game it has very simple rules anybody can follow it and start playing it immediately, so what you do is you roll a dice and move your pawn accordingly assume you are on let's say fifteen number fifteen you roll a dice and you get a five which means your pawn moves sixteen seventeen eighteen nineteen twenty and twenty in case if it's a snake you come down a few units if it's a ladder you going up a few units so this game is very popular all over the world although it originated in India it is now very well known in different parts of the world and there are variants for this game as well so that aside we will now see how we can write a piece of code to stimulate this game,  where two people can simply sit in front of their computer and then play so a word of advice since you people are beginning to code you needn't worry s o much about doing the exact stimulation, putting a board and displaying a snake and ladder may not be required let us do it with playing text. Let's see how this can be done.

# SNAKES AND LADDERS: NOT ON THE BOARD 03

Alright we just now saw favourite game of most of us in childhood snake and ladder being played by two persons, now let us play using a computer programme.

Hello all welcome to the programming screen cast of snake and ladder, in this programming screen cast I will take you down your memory lane where you had played one of the favourite games of I guess almost most of you in your childhood namely the snake and ladders, snake and ladder the board would generally consist of numbers from one to hundred arranged in a ten cross ten grid there would be snakes in some point and ladder in some points ladder would generally take you up the board that is from lower point to a higher point and snakes would generally takes you down the board from higher point to a lower point, this sort of symbolises theme of life that life is full of ups and downs so this would be favourite game for many people because if is somehow close to the real life scenario alright let us now see how we shall play this game in a very cool manner using our python code through our computers. Alright let's get started. Before getting started with the code I will show you the image of a snake and ladder board I had got an image from an internet this is here this is a snake and ladder board as I had said there are numbers from one to hundred in an ten cross ten fashion there are ladders somewhere as well as snakes somewhere I am sure you would have played this game in your childhood this is just journey down the memory lane alright, I would ask you guys to please pause this video at this moment make a note of snakes and ladders in this board various points from where to where is the transition of each ladder. For example this particular ladder starts at twenty one and takes you to eighty two and this particular snakes starts at sixty four and this takes you to thirty six like that you please make a note of this snakes and ladders by pausing the video, this would be of greater interest to us during the later part of our code, please pause the video at this point and make a note of the snakes and ladders alright I hope you would have paused and made a note of the snakes and ladders now let's get started with the coding part let me minimise this image let just started with the coding part when someone plays the game what are the two things you suppose to do, in the computer someone is going to play the game you are suppose to show him the board I call that as show board functionality the next is he is suppose to play the players are supposed to play and the end point of the snake and ladder we have to define, this snake and ladder this board has end point at hundred, hundred is the winning position ,when some player reaches hundred first he is considered as the winner, two players would play and the one who reaches hundred first is consider to have won the game till than the game will proceed in case the player wants to bought in the middle he can totally quit the game if the players continuously proceed hundred is considered as the winning position, this is how we are going to configure our game here so as I said first you are suppose to show the board to the players then you have to ask them to play this is the higher level abstraction details and this is considered to be good programming practice that has to, you have to abstract as much as possible like see a person who wants just an higher level overview he would read this he would understand the entire point is hundred and you are suppose to show the board and play the person is suppose to play this is what the game expects someone to do so this is the higher level overview so this is considered to be a good programming practice to give a higher level overview, your code must be organised such that by reading these overview the person should be able to get the higher level idea of what your code will do and if they want to del deep into the code then you may allow them into del deep into it but for a person who just need high level overview he should not be made to go through the final details unnecessarily this is considered to be a good programming practice for this purpose we are using different functions and another use of functions is reusability if you want to repeatedly do something and if you want to reuse that the if you want to use the same functionality at various places then functions come to your rescue they greatly help in reusability so it is very good programming practice to use functions as much as possible and the higher level overview must be easily obtained for any one so the naming of functions also I recommend that you give such that it is easily understandable by someone it is also the syntactically it is corrective you can say f one f two f three function f, f one f two f three and so on but it doesn't give the overview of what is the function may contains some twenty thirty lines of code or may be more than that as well as an how the requirement is for the code may grow longer and longer that maybe there but to understand what that particular function say f one is doing you have to go through those hundreds of lines of codes and understand this is not good programming practice so to facilitate that a good practice is to name the function accordingly suppose the function is in calculating the matrix multiplication say just for example I am using it matrixes have got nothing to do with this game now I am just giving an example suppose the function is performing matrix multiplication you are suppose to name it as matrix multiply that is a good convention so that just by the name the person who needs a higher level overview can get it without getting deep into the code so this is the good practice please make a note of it and use functions as much as possible because it simplifies your work greatly alright so now let's gets started with it, we have to get deep into the functions, show board and play are the two functionalities and now I need to say what this show board has to do so let me close this image, now I don't think it is needed, let me close this because in show board I am suppose to open the image again, show board functionality this is the functionality, this is a image file so I have to import the image library from the package PIL input image library this is how I have imported the image library so let me now open the image img is the object that captures the image, image dot open here you are suppose to give the file path where the image file is present in my case it is present in the same directory as that of my code so I am just giving the file name in case you have stored your image in a different path please do specify the complete path so that the image shall be opened alright now once the image is opened it has to be shown so I will say image dot show this open show all these are pre defined functions in the image library and as you would see as I had said they had given the names such that you can understand what it does what it does, image dot show there will be a lot of transformations involved in showing from as you know computers deal every in terms of binaries zero and ones from that to a image that is understandable by humans to transform there are a lot of steps involved but all these are abstracted at what we know is, if you say show it will show the image so that is enough for us we need not del deep into the binary level information so in this way your functions must be named open show all these are self explanatory what this particular thing is going is clearly understandable just by this so htis is the show board functionality currently I am not going to code for play so let me command it but this will also be an important step so I have return it and command it and once show board works will go with play, let us we had written the code for show board let us see if the board is being shown then we will go with the next functionality play I will save this file and I will run this see the image is shown, so this is the image that is that has been shown also the board will be shown so this functionality works fine no problem with this  let me close this now and ok the next thing is we have to go with this particular functionality play we will see what are the other functionalities required in play if it requires any other functionalities or any other libraries has to be imported as and when we del deep into we import and we will define new functions we will make the code more readable and if you just give this code probably to your maybe younger sibling maybe ten, twelve years old sibling who barely understands English that person just by seeing the names would be able to understand what are the checks that are being performed how the flow of programming is, everything would be very clear we will make it in such a way we will modularised this program this particular process is called modularisation like you split the you split the single units of tasks into chunks like show board this is one particular task I have made it once chunk and play is one chunk the play consist of some sub chunks we are making the functionalities into chunks so that anyone can get the higher level overview, even test it someone who just understand English with the names they would be able to understand what is the flow of the program, you may check that alright we have done with show board functionality let us go into play functionality

# SNAKES AND LADDERS: NOT ON THE BOARD 04

Alright guys, in the previous video you would have seen the method of show board the functionality I had mentioned here, I had written the code board for functionality so that the board is shown to the player now what is pending is the play functionality, let us see what is needed in play functionality ok I had commander it back then because I want you to show that show board functionality works so now play is require so I will remove the ash so that its now no longer a command and now we want play functionality to find play here as I said in this game we having two players p one and p two let us save them p one will have some name because we want to have it as personalised interface, it should not just be like player one player two he had got a sweet name, we need to refer to him by it by his name then he will be very very happy so we need to refer to him by his name and lets input his name from him lets input his name input is taken by this functionality raw input we are inputting a string so it is taken by raw input, string is nothing but a set of characters names generally consists if characters right so that is what technically we call it as string this is a string so we use raw input functionality to input strings so this is the message that has to be displayed. what should we be displaying? Maybe we shall say player one please enter your name looks good alright the same has to be done to player two as well, instead of typing it again let me copy and paste this and do the modifications wherever needed, it is player two's name and it is player two so the first would be inputting the player one's name and then inputting player two's name so let me command that as well for completion sake input player one name this is input player two name why do we have commands is? To help us understand the code at later point of time. Maybe when you are doing this now maybe you will find it later easier but once you have done sufficient number of exercises and when you want to look back at what you had done previously that time there may be some confusions so to avoid that we have commands where we write in English format that is understandable by us we can look back to the commands and then get to know the syntax that has been used and all that, it would be easier for our revision if needed so that is why we are using commands alright, so initially both are starting at that is they don't have their, they have their dice placed on the floor not on the board so they will roll the dice based on what the number appears they will place it on the snake and ladder board so ground let me denote it by point zero player one has points zero this is nothing but I will command this as well initial points of player one, player one has got point zero alright? Ok the same as to be repeated for player two so let me copy and paste it this is for player two so I will just give a summary of this four lines we input player ones name the same we do it for player two then we have set the initial points of both the players to zero and there will be one variable it will denote whose turn it is, let me initialise it with zero, you can initialise it one as well but in computers generally the counting starts from zero so I have done it with zero but it is not mistake even if you start from one so the game will be played continuously so let me say while one this is nothing but an infinite loop the game would be played continuously so I have put an infinite loop here I guess most of you would have some confusion here that I said when you reach the end points the game ends but here I am saying the game will be played continuously how is that possible? You would have it is not contradicting this is a doubt you may have so it will become clear as the code proceeds there is a check inside whether the end points has been reached or not if end point has been reached then we will break this loop, break this loop is nothing but we come out of the game that is what we call as break this loop this is a infinite loop that will be broken that is the game would end when you reach the end point the check would be perform at a later point alright so turn zero is for player one so I would check turn zero is for player one turn one then I will increase the value of turn by one and I will increment if by one so turn will become one turn one will be for player two turn two will be again for player one turn three will be again for player two so every turn is alternated so if you would observe the pattern here turn zero two four six and so on for player one and turn one three five seven so on are for player two this could be abstracted from this way that to take the value of turn you divide it by two and take the reminder if that is zero that is the turn of player one otherwise it is the turn of player two so to check the reminder we have the operator mod which is denoted by percentage we will be using that now if turn mod two equals equal zero note that single equal to is assignment it is nothing but you are assigning the value as zeros let the value be anything I don’t care, I want it to be zero then you have to use a single equal to here you have to check the equality I don’t want to change the value I want to fetch that value I want to compare it with some standard value in that case you have to use the double equal to symbol please note this point ok if that is the if this is the turn then this is player ones turn player ones turn this is alright this has to be notified to the player so I will print a message player ones name his name has to be mentioned I will say your turn player one this is your turn a message has been put and I had said the player may want to quit the game in the middle as well we have to give him this options till you reach the points hundred I don’t want to I want to quit it even that option will have to be given to him for that matlab this will give more personalised feeling that is why we are giving this option as well that I will take it as his choice so c nothing but ask players approval to continue asks players wish, players choice let me say that as players choice to continue this is what will be doing next let me call this as c, c for choice or whether he wants to continue you can take it anyway I will just call this c or you can name it as choice as well that is up to you I am using c input see that was the function I was used previously the raw input that was a specialised function to input strings nothing but something like name or your address something like that to input that you will use raw input here I will ask one or zero one will denote he wants to continue zero denotes he want to quit the game so I would input his choice I will have to say him press one to continue and zero to quit so I will get his input if his choice is zero if the choice is zero I will have to the scores and I will have to quit the game so let me show the scores the points is stored in pp one and pp two, pp one is the points of player one, pp two is points of player two. First the player ones name has to be mentioned player ones name quote here the points this has to be mentioned and the same thing has to be done for player two as well so I am copying and pasting it player two points of player two this is done, you have to mention the points and you have to say quitting the game quitting the game thanks for playing something like that just a humble message, a message has been displayed and now we have to come out of the loop that is this all these are for the user to give a personalised experience to the user we are using all these print statements but what is the instruction to be given to the computer to get out of the loop is this statement break, break statement will come out of the loop and it will end the game basically that’s what this will do here and otherwise if choice is zero this would be executed if not it would just come here observe the indents, indentation is important here it is in the same level of if so if c equals equals to zero this check would be made if the user hasn't pressed zero that is he doesn't want to quit it would come directly to line number forty here which is in the same indent level here you are suppose to roll a dice let me when you roll a dice you would get some number that let me store it in a variable called dice so to roll a dice to stimulate this what happens is what happens when you roll a dice some random number from one to six something from one to six would come when you roll a dice so I to stimulate this what should I give the instruction to the computer is you pick a random number from one to six to give this instruction I have to use this random package random library has to be imported I will say import random this is the library that is needed since we needed it we imported it here now I will say random dot the functionality is randint this is the functionality randint this requires few to give the end points that is what should be the start value and what should be the end value, the start value is one in our case and end value is six, this particular functionality would generate a random number from one two three four five six this is what this functionality is doing, you generate a random number this is basically stimulation of rolling a dice alright we have generated a random number now you are suppose to display what was showed by the dice, print dice showed dice showed the points that is captured by this variables dice so dice showed this point is printed to the user and now once this is once the dice has been rolled you have to add that many number of points to your already, already the points whatever you have to that you must add this points so let us do that. P one this variable p p one stores the points of player one this is nothing whatever is the pint take it you add the value of dice so that will be your new points and when you are playing the snake and ladder sometimes the it may happen that when you reach some position there may be a snake or they may be a ladder so accordingly your points will increase or decrease the change has to occur so to stimulate that let us say the points one is nothing but you check if there is a ladder at this particular points if you think why do I pass the same parameter here and here is this particular line what it does is it will add the points add the points here I guess the name would clarifies so commands is not needed but why do we pass this here and here I will explain, this is the place where you have landed after rolling the dice now you are suppose to check if that particular position has got a ladder if that has got a ladder your points this particular value of this particular value of pp one the points is no longer valid some higher points you would get so that higher value of points would be stored here so that is why we are using same variable pp one in both the cases I hope it is clear in it will be clear when we deal deep into this particular functionality check ladder this may be a ladder a positive thing which will increase your points or it may happened that it is a snake check if it is a snake of the same the same way check ladder and check snake are two functionalities which we will define later these are the things that we have to do and sometimes it may happen that in some points you are all at say ninety five you are at position ninety five and your dice showed six so when you add here ninety five plus six is one not one but your dice has board has just hundred so what will you do then you will stop at hundred and you would say that you have won so that particular check has to be done here so this is I am sorry for this, this is the same players player ones points has to be changed I am sorry for that and here as I had said you have to check in case during this addition process have you gone beyond the end point so in that case you have to come back to the end point so that check let us make, if the points, points of player one has exceeded end, end point then you are suppose to say you bring it to end point that is ninety five he was present, six came up so he is supposed to go to one not one since the board ended at hundred you stop at hundred that is what we are stimulating here, this is to the functionality here is to check if the player goes beyond the board beyond the board this is being check here alright? Now you are suppose to print him his score as of now you say his name player one name your score your score is this value pp one that is storing the points of player one alright, if he has reached the end points as I have said earlier this is an infinite loop you may wonder that when you have reach the end point you have to quit the game but here we have given an infinite loop now to break this loop we are performing a particular check if you have reached the end, if you have reached the end with this current points that is whether this particular current points is the end point or not here checking if the person has reached the end I am sorry I have used the extra bracket its seems sorry if it's just a space if this if this particular player has reached the end you are suppose to print the message player one name his name you are supposed to give and won player one this person has won the game you are suppose to display it and once you had displayed it you come out of the loop you break that this particular place we are using break because the user wants to quit the game here we are breaking that is here we are quitting the game because one person has reached the end point this is the check for player one this is a player ones turn anything maybe possible right player one may reach the end point player two may reach the end point so we will perform all this checks in player two as well alright let get back to this if turn then else this all this whatever we had done are for player one let us the same as to be repeated for player two so let us copy paste everything copy ok let me paste it this is for player two player two you will modify everything to player two player two sorry this is player one I am sorry for this because here the player wants to quit and we are suppose to display both their points this is player one and player two and now here it is player two is player so his points are added so layer two player two alright player two player two player two player two player two player two layer two player two alright so we are given rough overview of the game let us save this up to here I will give you the summary of what all just happened till now we have inputted the name of two players we have initially set their points to zero and we have some variable called turn which will actually keep track of who has to play now and the turns zero two four six and so on are for player one, one three five seven and so on for player two to keep track of that we are using this particular operator modulo denoted by percentage which will return the reminder after the divisions so we are dividing it by two if the reminder turns out to be zero it is player ones turn otherwise it is player two turn so one thing I guess I skipped here is I skipper here is I should have should have incremented the turn, yeah I should increment the turn right? Once one player has done I should give the chance to the next person so for that I need to increment the turn that I have to do I have done that alright so what in each players turn what are the checks that are being performed let me tell you that, first we say that it is his turn we are asking if he wants to continue or quit if he is quitting we specify the status till now as to how many each person has scored and we are quitting the game, in case he wants to continue we are rolling the dice, dice would give you a number from one to six some random number that number is added to his points and then it is possible that, that particular position there may be a snake or a ladder we will be performing a check there if that is a snake or ladder and appropriately the points would be changed and in case during the process of rolling a dice the points exceeded the end point as I had given an example earlier if the position was ninety five and the dice showed have six addition would give one not one where as your board just as hundred so end point exceeded so in that case we should bring back to end point so this check has been performed here and after that and after performing all the check after performing the normalisation part we are displaying his score and we are checking if that particular position is the end point or not if the person has reached the end point is declared that this person has won and here we quit the game, the quitting here is due to the users choice quitting here is due to reaching of end point that is the difference here the same thing is done for player two as well alright so the higher level overview of what play does is given to you and check snake check ladder reached end these are the three functionalities we have defined here that is we have we have just given the names here what this particular functionality is suppose to do we will define later we will be defining in next few minutes let's see to it. I hope you have understood what all happened till now you may pause for sometime look back to what has happened, you proceed once you got all the pieces till now clear.

# SNAKES AND LADDERS: NOT ON THE BOARD 05

Alright guys, in the precious video we had seen how to display the board as well as higher level overview of the game playing so in that we had used some functionalities like some functionality we had used it like check snake, check ladder we will define that now define check snake or check ladder may we let us start with it check snake check ladder initial view I had asked you guys to make a note of all these points where ladders were there the points where it start the point where it ended I had asked you guys to make a note of it, I hope you would have made a note of it based on that we will be writing this code if the first ladder we saw was at eight so we say check ladder this takes an argument points so some specific points is given we are checking if there is some ladder at that point and if there is a ladder we will return what will be the new points after crossing that ladder that will be returned if the points is equals to eight this is where we have the first ladder then you are suppose to print that you have to notify the user that he has actual stepped on to ladder you print that this is a ladder then you return the new position the ladder at eight actually landed at twenty six so you return twenty six now the next ladder is at twenty one twenty one the thing is the same so let me copy paste it or I will type ladder this ladder lands you at eighty two so I will return eighty two so the same pattern will continue for other ladders too so let me copy paste this block after twenty one the next ladder is at forty three that is taking you to seventy seven the next ladder is at fifty that is taking you to ninety one the next ladder is at fifty four that is taking you to ninety three the next ladder is at sixty two it is taking you to ninety six the next ladder is at sixty six it is taking you to eighty seven the next ladder is at eighty that is taking you to hundred the end point here if you haven’t stepped into any of these points that is once after a dice is rolled we check whether that is a ladder point are not these were the ladder points if the points at which we are making a check by synch match any of these things that means you have not stepped into a ladder so you get beck the points what was added you don’t have any bonus so else this is not a ladder not a ladder so you return the same points that is no addition of points no increment of points nothing is there no change in points just return back what you got this is how check ladder functionality works. Alright! In a similar fashion we can write check snake functionality check snake the first snake you have is at forty four let me check that if points aright this event takes an argument points at this particular point whether there is snake or not if points is equal to the first snake is at forty four if that is equal to forty four you have to print snake you have stepped into a snake you need to note the player so you have to print the snake so snakes will bring down your points so you will come down to twenty two so that you have to return it I will say points is the next snake is at forty six right? print snake print snake and then you return the value where it takes you back it will take you to five so return five the snake at forty six is taking you down to five so you are returning so this is the pattern for all the other points so let me copy paste this ok the next snake is at forty eight that will take you to nine the next snake is at fifty two that will take you to eleven the next snake is at fifty five it is taking you to seven the next is at fifty nine that will take you to seventeen the next is at sixty four it will take you to thirty six the next is at sixty nine that is taking you to thirty three the next is at seventy three it will take you soon to one the next is at eighty three that is taking you to nineteen the next is at ninety two that will take you to fifty one the next is at ninety five it will take you to twenty four the next is at ninety eight it is taking you to twenty eight else these were the points where there were snakes otherwise you have not stepped into a snake not a snake so there is no change in points you just return back the same points this is how check for snake has been made this is nothing this is very simple that I had shown you a board in the very first video when I have asked you to pause the video and make a note of the snake and ladder where it starts where it takes you to based on that I had written this code, you check if it is those special points in that case you display a message whether you have sadder at a snake or ladder and accordingly you modify the points you return the modified points so now I guess you could get it in the play functionality when we were using see we add the points and we check if that points is a ladder if it is a ladder here you will get the modifies points if it is not a ladder you would get back the same point here similarly the check snake functionality also works, the difference is ladder takes you from a lower point to a higher point if you step into a ladder, snake will take you from a higher point to a lower point if you step into a snake that is only difference if you didn't step into a snake or a ladder you would retain the same points alright, so these two functionalities are over so reached end this particular functionality is to be realised let us do that now, realise is nothing but now you del deep into a to check the code nothing but you give the code that is what you call as realise reached end at this particular points you are checking whether you have reached end nothing if the particular points is equal to end point you say true else you say false that is the logic this is the plain thing there is an end point defined you just check whether this given value is equal to that end point if that is the case then you return true true else you return false this is how this works let me save and I am getting an error is it input only I guess the same input will work input I guess it will work yeah oh that is the raw input for other use I am sorry for this confusion this is input, input works alright so we are almost done with the code everything has been defined see the code has been modularised well the first thing we have to do is we have to show the board then we have to play, he next step is we are playing the next step is we are playing, first we show the board then we play that  is the what we are doing this, so this is the highest level of overview so now if you want to know what this particular show board functionality doing is we have an image file we take it open it and show the image that is the that is what this particular functionality show board is doing then the next functionality is the play functionality which is our main functionality here what you do is, you input the names of two players you initially set their points to zero you have some variable call turn which denotes which player has to play now then you ask him if he wants to say that this is this person turn if he wants to continue or quit the same so based on that if he wants to quit, you quit the game there by denoting the status till now or if he wants to continue you roll a dice based on what, what is shown up in the dice you  add his points and then check if he has landed in a snake or a ladder if so you modify his points otherwise you do it you do not do anything you just retain his points so that basically what you do is you add his points and then you check if he has landed into a ladder or into a snake and sometimes it may happen that during the process of rolling a dice you may go out of the board in that case you are supposed to stop at the end of the board itself so this check is done here then you display his score and say if he has won when his points is exactly the end points you just you have to check if he has reached the end point or not, if he has reached the end point you say that he has won this is the higher level overview you have a variable turn denotes who has to play if he wants to quit display the status and quit otherwise roll a dice add points check if has landed into a ladder or a snake and if he has reached the end points you quit the game, this is the higher level overview now the detail functionalities checking ladder checking snake this I had defined based on the board I had shown to you, to check those specific points if that particular point has been reached you display that he had stepped into a ladder or snake appropriately and then you return the points where he will land finally, you return that if he hasn't stepped into it just return back the same points so that there is no change in the points that's how these things works and reached end this functionality what it check is whether the points this person has got is same as the defined end point or not this is the check performed here, this occurs continuously and one of the condition will lead to quitting if the game, one is the player voluntarily wants to quit the game the second condition is someone has reached the end point these are the two constrain these are the two places where the game will quit alright this is how we have coded now you please pause and see till now what has happened or you understand the functionalities, you understand the overview level as well as to the details in depth you proceed from this way itself, these are the major functionalities now let me tell deep into it this way you just consider you have been given a big box you open that box then get a smaller box then you open it then you get a smaller box this is how we are organised a code you understand this well and this once you are very clear with it will proceed with the running of this program alright please pause and understand this very well.

# SNAKES AND LADDERS – NOT ON THE BOARD 06

Alright, till now we have coded the snake and ladder game we have split into various functionalities for easier understanding I hope you had revised it, you had gone through all the functionalities you are clear with what each functionalities is doing, now let us run and see this. The board has been displayed and here you are supposed to enter your name so maybe let me say abc  is my first name, my first player name is abc and second one let him be xyz, abc this is your turn you have to continue or you have to quit? I want to continue the dice showed six and see the initial points was zero abc score is six now because the dice showed six and there was no snake and ladder at that point so the score remain six only now this is xyz turn if you want to continue or quit, I want to continue, dice again showed six so currently both are in the same status, you want to continue yes the dice showed three so I am at nine I want to continue yes both are at the same place, yes I want to continue it's getting interesting I want to continue wow now here the player two has an upper hand this is going interesting you can play on and on the whole day good luck. You have got great pass time now great guys now let me quit this but I am gonna play this tonight, I won't think that I will sleep tonight this is so good I am so happy with this, I am gonna play this I hope you guys too enjoyed this thank you for watching, have a nice day.

# SPIRAL TRAVERSING – LET’S ANIMATE

Well luckily with animation fantasy is your friend said the great film makers Steven Spielberg so let us open the world of animation with our next choice spiral traversing.

# SPIRAL TRAVERSING - LET'S ANIMATE 01

We have been looking at pretty straight forward programming questions, nothing has been very challenging a questions so for. I would now like to give you a question which actually isn't straight forward which requires a whole lot of thinking and of course programming time. So let me pose the question for you all, do you see the animation here? What's happening? You just seeing the white dots spiralling around and going to the centre in a particular way correct? How easy for us to do this? Well! Isn't all it appears though it's pretty straight forward it is not that easy, firstly how can you even do such an animation so let me go step by step and explain to you all how we can actually fill the entire screen with white dotes in this particular fashion, forget what I showed you let us go step by step. Do you now see this four cross four entry of numbers there are numbers one two three four five six seven eight nine ten eleven twelve thirteen fourteen fifteen sixteen and I want the numbers to be displayed in the following way, firstly it should be one two three four and then you must climb down eight twelve sixteen and then towards your left fifteen fourteen thirteen and then up nine five and then right six seven down eleven left ten you should basically traverse this table with these numbers these way alright? What do I mean by that? All I mean is, you should display this numbers in this particular fashion only let me try to be more clear with a nice example so here is the exact question, if the input is four, you create a four cross four table like this in your program don't display it, just create it and each cell corresponds to a row and column isn't it? Look at this first cell it is first row first column I am gonna call it one comma one I will create a list or maybe a tuple whatever is comfortable for you and then use that term one comma one to denote this cell, the second one first row second column this will be one comma two likewise one comma three, one comma four, two comma one, two comma two, two comma three and so on up to four comma four so the input is the dimension of the table if the input was five you should come out with the five cross five table it was six, six comma six if its n in general it's going to be n cross n so given four I create this four cross four tuples one comma one, one comma two so on and then I want to display them in the following order only remember the spiral thing that I showed you, I want it in that order only by that I mean observe one one, one two, one three, one four, two four, three four, four four, four three, four two, four one, three one, two one, two two , two three, three three and three two should be the order in which you must display, you see where I am getting that given a n cross n table you must spiral n like this what is spiralling n you must simply display this addresses of the cells as simple as that, so input is a number some ten and you must create a ten cross ten table and spiral n on that table and display it. Remember we started off with this little graphics animation every simple and straight forward one, looks like the animation of animation that was done some forty years back animation is so sophisticated right now this is far to be regarded an animation but then as a practice exercise I wanted you to all understand doing something as simple as this requires good amount of thinking, I spoke to you about displaying the entries of table right? I am going to use that right now and I am going to show you how we can create this graphics animation using just simple idea of taking n and displaying n cross n cells in a particular way, you now know enough to go ahead and write a quick program which will do the animation, let us see how this is done, amit will now show you how to use your knowledge of displaying the cells in the table and create this graphic animation.

# SPIRAL TRAVERSING – LET’S ANIMATE 02

Hello guys, in this video is little bit different we are going to do some kind of animation here and in order to do this animation will we don’t need any pre requisites we don’t need to know any graphics or any other libraries but two things we need to understand how to store represent matrix in python, second how to print it in different orders so in this video the main crux of the programme is to print the matrix in the spiral order so what do I mean by the spiral order? It means that you will start from the first element is represented let’s say zero, zero comma zero and then you will print the first row then you will go to the last column and print the last column all the elements of the last column then last row in reverse order then first column in reverse order and so on now why is this important? How I am linking it with the animation? So after this program, after we write this program, after we do the traversing of a matrix in spiral order and do the animation with it, you will be able to make something like this. This is exactly the spiral order traversing of a matrix, you can see as I am printing the first row and last column then last row in reverse order first column reverse order and so on, it will go like this in until it it’s the center so in order to perform this animation I don’t need to understand any, any kind of graphics I just need to know how can I print a matrix in spiral order and use that logic to do this in animation so after this video, after we write the program you will be able to know how can we create such kind of animation or such kind of pattern easily so let’s see this in next video.

# SPIRAL TRAVERSING – LET'S ANIMATE 03

Hello guys, let's see this matrix that we have on the screen, it's the four cross four matrix with elements one two three four five sic seven eight nine ten eleven twelve thirteen fourteen fifteen and sixteen so usually how we print, usually in whatever lectures you saw till now in most of them whenever we are using any matrix or printing it or traversing it we usually traverse in row wise manner, what do I mean by the row wise manner? So see this matrix first I go to the first element, my first element is one which is present at zero comma zero so I will go there I will print it, then I will go to the next element, next element is the first row and second column which is here which is the which is two so let's see this I will go to the second element and I will print it. Then I will go to the third element which is present at first row third column which is three then next element is four first row last column now after printing the first row I will shift to the second row first column which is first element of second row which is five here then I will go to the six, seven, eight, second row is over third row nine. Ten, eleven, twelve, third row is over fourth row thirteen, fourteen, fifteen, sixteen see usually we print like this in this program we are not going to do this then we are going to do something else we are going to print the elements of this matrix only let's say in a different order which is spiral order so let's, let's see what do I mean by spiral order. Suppose I have the same matrix I will start with one first element I will print all the elements of the first row which is one, two, three, four then I will not shift to the second row I will just print all the elements of last column which is eight, twelve, sixteen then I will print all the elements all the remaining elements of last row in reverse order which is fifteen, fourteen, thirteen then the first column all the remaining elements in reverse order which is nine and five one is already printed now what is left? In the second row I will print all the elements in the simple order which are left which are six seven I will go to the third row and I will print eleven and I will print after that ten so in this, this is a four cross four matrix then ten is the middle element so this is how we are going to perform and you know if you can print something like this then you just need to add a library which we are going to use a turtle library and with that library rather than just printing the elements will print some kind of dot, coloured dot and it will look like we are animating something so this is what exactly we are going to do so let's do this.

# SPIRAL TRAVERSING – LET'S ANIMATE 04

Hello every one, so in this video we are going to write the program for spiral traversal of the matrix so in the first program I am going to make two programs. In first program will print the matrix in spiral order just what we saw in the last video so will see how we can do this so let's do this, let me create a function name spiral def spiral and what are the parameters of this function? M is the number of rows then n is the number of columns and 'a' is the is my matrix so 'a' is actually a list of list through which I will representing the matrix. Let me define two variables one is 'k' equal to zero and l is equal to zero, let me tell you what these variables are 'k' is actually the starting index of the row which is zero, l is the starting index of the column which is also zero, we know that m is the number of rows in the matrix and n is the number of columns in the matrix so let me write this term, you guys already know that if you want to command multiple lines you have to do with this then you can command multiple lines so let me write this 'k' is index of starting row, l is the index of starting column ok now what we have to do? We have to first print the first row, the first row should be fully printed then we have to shift to the last column and print the all the elements of the last columns so let me write a while loop while see 'k', 'k' is the index of the starting row, starting row index. Why 'k' is less than l? Means that 'k' is when I am going through all the elements of row so I am exhausting all the rows if I am getting over all the rows then I should not print anything because my rows are over so 'k' should be less than m and l should be less than n, what do I mean? L is the starting column index so all the columns are exhausted so when all my rows and end all my columns are exhausted I have to stop the loop, ok while k is less than m and l is less than 'n' first I have to do is since I am starting first I have to do I have to print all the elements of first row so that let me write a loop, for I in range see elements of first row ok I have to exhaust all the elements through all the columns so I have to go through first column index, second column index keeping my row fixed I have to go through the first column second column third column fourth column and so on so my row index is fixed at a just traverse to the column so I will so my range will be l to 'n' to keep this in mind that this 'n' is whenever we type range, range starts from suppose you write in the parameter you give in range is zero comma five so it will go like zero one two three four not up to five ok so hence I am writing 'n' because n is my number of elements number of columns ok so suppose if I give four it will go to an l is zero so it will zero one two three four times but it will not go up to four as four so I can safely write here 'n' ok inside this loop I will print my elements which are print 'a' see my row is fixed ok I will just write 'k' here because the first 'k' index of first row and I so I have to print I have to traverse row all of the columns so I is the column index so row is fixed row is zero and I will print all the elements so the first row. You all know that if I have to print the elements in one line with this space I have to do like this and with apostrophe inverted comma sorry awesome after this when the first row is over I have to go to the second row I have to see I have to go to the second row when the first row is over one two three four I have to go to the second row so second row is important so now here see second row second row but I have to print the elements of last columns eight, twelve, sixteen so in this traversal the my column is fixed where I have to traverse through all the remaining rows see remaining rows are second row, third row and fourth row but the column is fixed column is the last column so I will do this see this before so that we don't forget here let me command this commanding is the best practice ever you guys should command every time whenever you write a code so that you revisit you can understand what you wrote anyone who is trying to understand your program or trying to reinvent your program or use your program can understand what you did so it's better to command so let me write here what we did? We print the first row so printing the first row from the reaming rows after this is over when my elements of first row are printed first ok I have to go to the second row so I will do k plus is equal to one which means that now I am on the second row those row index has been increase k is my row index started from zero but now it is of first I mean second row 'k' is equal to one now I have to print the elements of last column now here I have to traverse route the remaining rows ok and my column should be fixed so let me write this for I in range my column is fixed you should remember so I have to traverse through the rows so which is k to m which is because I have already incremented my k so it will go to the remaining k the last row m is the last row index because m is the number of rows. I will just print here print 'a' I have to traverse through row so 'i' column is fixed which is bought n minus one now why here n minus one because again I have to say this because is the number of columns so suppose my number of columns are four since every time list starts from zero so here it should be suppose number of columns are four number of rows are also four I have to put two in order to be in the last column I have to put three here because arrays list always starts from zero so three array I have to put for that I ma putting n minus one here I put k only because k is already zero so I don't have to take care of minus one here but here I have to take care, again I will write end is equal to grid and my last column is over last column is over means I have to shift my columns to the second last what do I mean by the second last? Since now here n is pointing at let's say if number of rows and number of columns are four, four n was pointing at four so n minus one was three now I have to shift to the last column so second last column so that for that I have to do n minus equal to one let me again command this line printing the last column form the remaining column awesome! This is done now here is some tricky part let's see this now we are on the last second last column a last row second last column so we did this now we have to print this ok see the tricky part here is we have to print it in the reverse order, this is the reverse order sixteen after that fifteen, fourteen, thirteen you have to go in the reverse order for that what will you do, if you have to check the number of rows are not exceeding the row index is not exceeding the number of rows why I am doing it again here see I already checked it in the while loop but it might happen that after this condition the k might become that's a five we should not open here we should not escape the five comes here and not stating this condition that there is a bugging the program so for that I have to put the condition again inside this if it is fine then I will write for I in range ok now what I have to do, I have to start from the second last column I already decremented my n by minus one so I am already on the second last column since n is the number of columns I will start from n minus one where I have to go I have to go to the first column, first column is index at zero now it might happen that since we are doing a generalised program it might occur it might not go to the zero it may for example in this part when I am printing eleven and ten I have to go to ten only I don't have to go to nine after that ten eleven after that ten ok so I have to go I have to go to the last starting column index, starting column index for now it is zero ok for now it is zero but it may not be zero after some time so for that I will write n minus one because again the range, range doesn’t go to the doesn’t include this value, it doesn’t include this value but I need this value for that I need l minus one so that I can go to l is at l since I am going in the reverse direction I am decrementing n minus one, n minus two, n minus three, n minus four up to l I have to go I have to write here that’s minus one minus one is my step so by default step is always one positive one so you always go one two three four five suppose you want to go five to one, five four three two one you have to write here you will start form five and up to zero if you want to go up to one n minus one so let me five four three two one this is my step function ok so I will go up to l minus one and inside I will just print my array ‘a’ n minus one row, row minus one and will start last row because I am in the last row an n minus one because m is the number of rows hence you play that’s it. Great! I am done with the last row ok so I will decrement my rows ok because now I am on the last row I want to go to the second last row so I am here I was here printed all the elements of this row I have to go to the second last row I have to decrement my row index not row index I mean number of rows which is m minus one this is done now next is again let me command this line so that I don’t it printing the last row from remaining rows ok now what is left? I am on the second last row, I have to print the elements in the second last row I have to print ok second last row I have to print the elements of the first column in reverse order again ok we have to do this in the reverse order again so for that again I have to check if l is less than n ok if l is less than n then inside this I will write loop for I in range  m minus one because is already decremented here and I want to start from the second last row second last row to where the first row I have to go until first row so k minus one ok because I have to go until ‘k’, whatever k is so for now k is one thing of the second row because we printed the elements of first row here and we incremented our k to the second row I want to go to the up till second row so that’s why I am writing k minus one here ok and again step function is minus one because I have to decrement ok and here I will write print ‘a’ I of l that’s it awesome this is done I will just increment my l here because again let me tell you l is the l was the starting column index so after I printed this first column all the elements of first column I have to go to the second column once I printed this I have to go to the second column and again do the same thing which I did here I have to do the this thing again here and next thing here and next thing here next thing here nothing is there sorry so this loop will take care of it ok that’s why you wrote this loop let me command this again printing the first column from the remaining column awesome! This is done our program is ready printed everything now let me create my array so let’s array is ‘a’ ok for I in range how many rows I want? I want four rows for j in range ok for columns let me create a index list here l it’s an for now let me I write the elements like one two three four five six seven eight nine ten up to sixteen so I will take the count variable and put it one and here I will just append the count variable and increment it ok so I have the elements so j will take care of a rows  and the elements of four first row let’s say after the first loop of the after the first loop j is over and I append that the list l whatever I got after this loop in the in the in my array ‘a’ in my matrix ‘a’ so ‘a’ dot append l that’s it my ‘a’ is over I have my ‘a’ and I will call the functions spiral, number of let’s say rows are four number of columns are four ‘a’ now let’s see this ok let’s see sorry cool I got my spiral traversal one two three four then eight twelve sixteen fifteen fourteen thirteen nine five six seven eleven ten it is? Yeah. One two three four eight twelve sixteen fourteen thirteen nine five six seven eleven ten cool so this is over you can check this you can write in a different way this is one of the way of writing this program the spiral traversal you can write in a different way also you can create your own logic there is another way which is known as recursion but not we are we will not cover this because it might get little bit complicated but it is cool, you may check depths how we can write a recursive program to print the elements in spiral order so see this and we are going to use this logic spiral traversal in our main program which is creating the pattern.

# SPIRAL TRAVERSING – LET’S ANIMATE 05

Hello every one, now in last video we saw we can traverse some matrix in spiral order now in this video we will see how we can create such animation what we saw in the first video for that we are we are going to use a library named as turtle, if you have installed anaconda the turtle comes with the anaconda, you can check, if it is not there you can simply install with pip just search try searching how to install the turtle with pip and everything is over just a one line one command you can do it just install turtle and you can read more about turtle from here the documentation provided by the turtle library you can see a lot of functions at their lot of functionality they have given let of examples are there you can use the examples to, understand how you can create animation through turtle actually it is used to create patterns and that’s what exactly what we are doing what we show what we showed during the first video so let’s see this so in order to start with, will first create a simple turtle program to show you how it works so let’s see this. First have to import the turtle library so that import turtle that’s it ok after that I have to create a turtle a turtle is a kind of pen which works on the canvas so for that I will create a variable tur lets say and I will create my canvas turtle dot turtle so my turtle has been created now I want to create a simple square box so for that I will write a loop for I in range five I will forward this pen tur turn which I am saying is the pen on my canvas at some pre define starting position so I will just forward it tur dot forward function is there with, with fifty units ok and then after it got provided with fifty units I will turn this pen position within angle of ninety degree so for that I have to use the function dot write it always take the parameter as an angle so I have to take give ninety which means it will turn it with ninety degree and do all this thing again ok that’s it I am done with this I have to write turtle dot done great, let me run this. Cool I got my square see, its working perfectly ok let’s do something else let me create a loop of let’s just say fifteen stances and I will change the angle with let’s say one forty four degree let’s see what happens. Something is wrong let me run this again ok see this, see I created a star so you can using this you can create a lot of beautiful patterns with different colours ok great so we are going to use this library to create our pattern which is spiral traversing and animation depth we saw in the first video we are going to create exactly the same animation with this turtle library. Thank you

# SPIRAL TRAVERSING – LET'S ANIMATE 06

Hello everyone, in last video we saw how we can use basic turtle I mean we can use the turtle library to create basic programs basic pattern programs and basic animations will use this in our will use the turtle in our last program that we did for the spiral traversing to create the pattern that is creating the animation where there were dots printed in spiral order so this is the program which, which I created which I created before traversing the matrix in spiral order, will use this programme to create the animation so let me first import the turtle, import turtle by default whenever a turtle is created the canvas which is created has a default white value the background colour is white I am going to need a background colour of black so I will create a canvas of black background for that turtle dot bg colour that is black ok now I will create a turtle name Seurat to create I have to write turtle dot turtle I created a turtle, whenever I am going to traverse towards the matrix I am going to print a dot for that I need a circle kind of thing so for that I need a width variable I will create a width variable of size let say five and a height variable of size seven and whenever I go to a particular element I will print this thought and I will move forward with some distance so I need that distance so I will write the distance at dot underscore distance which is let's say twenty five ok create, now I created this turtle but I need a initial position for where it should start for that I will set the initial position is dot set position to minus two fifty comma two fifty you can change the position according to your program but I am keeping it minus two fifty comma two fifty it actually starts centre all with always starts from centre so if you put zero comma zero it the position of the turtle will be at the centre of the canvas but I need it at the left most top most left corner so for that I am keeping it the x coordinate as minus two fifty and y coordinate as two fifty while I see this I need the colour of the dot as white because my background is black so for that I will write dot colour which is white great now inside my while loop I will go ok whenever I print first row after printing the first row my after going through the first row my turtle position should be at angle of ninety degrees should move at the angle of ninety degrees so that it can go downwards so in order to go downwards I have to put it downwards I have to put the position of turtle I have to move the position of the turtle within the angle of ninety degree but it should be it should print as the first row as, as usual I mean it should go forward in the right direction it should not print in downward direction from the first row only so from that I will create a variable to check that lets say flag of f as zero ok after printing this ok so I will check here if f is equal equal to one it means I already printed the first row I will move the position of I mean that direction of a turtle at the angle of ninety degree to the right ok and here after printing the first row now I don't need to print the array here just command it and instead of this I will write here ok dot forward with distance which I put as the variable dot distance ok and after I print the first row I need to set my flag value which means if this is ok and after this you always have to turn the position of the turtle at the angle of ninety degree to the right side that's why I created this flag value of the left ok done so first row is over after the first row ok I print the elements of the row I have to print the column I mean in the first iteration I was printing the last all the elements of last column so I have to print in the downwards way for that I have to change the angle of my turtle to downwards it means to the right of ninety degree that is dot right I put ninety degree again here I don't need to print the elements I will just command it and I will write my turtle I will forward my turtle to dot distance, after this again I have to change the direction of my turtle to the ninety degree for that dot right ninety that's it this is also printing I don't need this so I will just command it and instead of printing the element I will just forward my turtle forward with dot distance again same thing after printing the last row I mean the row in reverse order I have to again change the direction so I will do this right ninety here again I will not print this I will forward let me copy this and forward it, distance ok great so now I don't need all of these arrays because I am not printing any of the array nor I need this count value ok I just have to call this let me change the parameters here this is ok let me do this for the value twenty rows and twenty columns ok I am done so I will write turtle dot done lets run this, cool. See I created my spiral traversing so I am getting lines ok this is all what we have to do now little after little tuning up we get created what we get created in the first video what we saw in the first video so this is exactly what you wanted to do you can, you can take this program change whatever order you want to change this is the main program actually, you can fill the colours, you can change the shape you can you can do something else ok but this is the main program ok thank you.

# SPIRAL TRAVERSING – LET'S ANIMATE 07

So in this program what I was doing, I was doing two things first I created a list of colours let me zoom it ok so first I created a list of colours a list colour white, yellow, brown, red, blue, green, orange, pink. Violet, grey, cyan you can give lot of colours you can check the top line, second what I was doing rather than printing the line if you just forward the, the position of the turtle it will just print a line, you can create a dot also so there is a function in order to create a dot which is known as turtle name dot dot and it will create the dot with, with the way you want to create it ok so it will create a small dot the colour we the colour we want to create so in this I am at random on the random basis I am writing different colours ok so that's that I am doing here and for that first you have to remove your lining pen not to remove lining pen you have to write a call off function which is known as penup which will penup means that it will not it will not create a simple line rather than when I will call the function dot function it will create a dot and go forward at the time of forwarding it will not print anything so penup will not after putting penup it will not print anything whenever you print when will you call a dot function it will print a dot there and move forward that's it, that's what I was doing exactly there so and every time what I was doing I was calling a random function so you can see here call randint from zero to ten, zero to ten means within these, these our spoiling colour from this list and using that colour every time so that the colour changes randomly, this is the pronoun for the pattern, it is the same pronoun which we which we which we wrote in the last video only little bit tunes up so you can create your own program you can change this so I will recommend you guys to please change the program and create your own pattern using this parallel order only you can different things also you can one more thing you can do is you can rather than going from the first element which is present at zero comma zero and going to the middle element you can start from the middle element and you go you can go in the reverse parallel order to the first element. So this program hopes like this so yeah it will show a spiral order like this with dot, you can even create some other shapes rather than dot you can see the turtle documentation you can create different shapes actually, you can create your own shape also and I am not doing anything I am just randomly changing the colours by picking up the random colour form the list of random colours which are defined as a variable and I just put it on the same program which I wrote in the last video so that's it so why to do this? Because it's just fun that's it nothing else so please post your questions and whatever problems you are facing in this regarding this video in the discussion form, we would like to help you guys to do this program, to learn interactive programming with python and please if you please I would like you guys to change this program to make something else, if you have some new ideas please please post it on discussion form. Thank you.

# **GPS - TRACK THE ROUTE**

Hey guys, suppose I asked you to go to a remote place in your city so what would be the first thing that will come to your mind? Off course that would be gps, so our next choice is about gps.

# GPS: TRACK THE ROUTE 01

Hello friends, welcome to yet another joy in our course or you must be aware of gps the global positioning system it is something it will let you share your location on whatsapp or track your route using goggle maps. Gps is actually a network of satellites that help tracking the geographical locations of an individual, keeping the technical stuff aside let me give you a nice motivation about this joy. Assuming this story some time back one of my friends owning a pizza delivery company calls me, he has complaints of late deliveries from is customers and he installs a gps tracker on the delivery bikes hoping that this would solve his problems but instead of giving the location name gps tracker gives raw information in the form of latitude and longitude information in order to help my friend I need to write a code that helps him track the route of his delivery boys but can we do it using python? Of course yes, but how can we do it? Well we have amit to our rescue, let’s find out.

# GPS - TRACK THE ROUTE 02

HI guys, in this video we will see given a set of routes how we can plot the route on Google maps so for this we are going to use two languages these are gm plot and csv library what you have to do exactly, what's the problem? Let me explain. So you are given a set of plots ok and is you have to plot these routes on Google map so how you can do this? So suppose this is my file named as route dot csv and these are the coordinates, coordinates means the first number which is shown near latitude second number is shown in the longitude so it is separated by a comma so that's why this file name is route dot csv, csv means dot csv file format is comma separated by this so whatever the values are there separated by comma you can open these file in spread sheet also or in any simple text editor so we will recommend you guys to use blind text or you can use any simple text editor so suppose I have list of coordinated here so in the video you saw that there is a guy and he was wondering around on bicycle and the manager what he did, he put a gps system in his bike so that he can he can see the routes where he is going and whether he is delivering in the correct location or not or he is wondering off any where so suppose he get the information in this file dot csv file and these are the locations where he were where the person went know how to plot this because these number doesn't make me sense I don't know what to do with this I don't know where he went exactly so there is a way we can plot this so in this video we will see how we can plot this on Google maps and how we can see what the routes are been followed by this values so let's tuned on and let's see in next video.

# GPS: TRACK THE ROUTE 03

Before we start plotting the routes let me go through this route dot csv file one more time and will see how we can pass this file using our csv library that I said in the last video, you know this route dot csv file has well need the coordinates in separated by commas it mean that first value is the latitude value and the second value separated by comma is the longitude value. The first step is to pass this data how to how to extract this information from this file in python? So let's see this. So I will create a new, new python file name gps dot py and I will import this csv library, you can read more about csv library on goggle, there is a nice documentation here you can read all the functions and the data provided by the library please have a look at it and I will be using very simple passing method to pass the csv file so let's see this. I will import the library first import csv ok, now my csv file name which I want to pass the name of the file is route dot csv so I will open that file with open route in commas route dot csv in green board because I don't have to write anything as f would say you can put here an variable name here ok after opening it I will use a radio function so I will create a variable name reader and I will use the csv reader function to pass this f, what what! What it does, it will create a iterator through this file ok so that I can iterate over it and I can read the values one by one as the list so, so now my reader variables has on the, on the data of the csv file I have been iterative so I can iterate over it so what will you do? I will iterate over it using the loop for let's say row in reader ok now row will give me row will contain list so each item of this list will be a value separated by comma so first will be the first value and second will be the second value after the comma third will be the third value after the comma like this in string so let me print this so what I will do? I will store the first value in lat that is equal to row of zero ok because zeroth element of this list contain the first value and long is equal to row of one because the first elements contains the second value that is the longitude I just print it print lat print long or I will just create a dummy print so that I get a new line character that's it let me run this, cool! so you can see all the latitude longitude values has been extracted and printed ok so suppose what I do I will try to sum up, let me print the some of the latitude longitude just in some so what I will do here in the last print statement I will print lat plus long ok it should print me the sum of this values so let me run this cool so you can see the sum of latitude longitude ok so here is the problem it's not exactly the sum what you are getting is a concatenation of two strings so as I said this, these are not the float values these are string values so what are you getting is the concatenation of the string values because the plus what in python if you do a plus in two strings it will concatenate the strings you are getting a concatenated value so in order to get the sum of this latitude longitude you have to change it you have to type cast that is we say type casting in c so it means you just change the string value into float value so for that you just need a function float and inside this so what it will do it will just change the string value into float now let's run this now if I run this now you can see I am getting this sum one not six point three four zero seven so now I know how to pass it a simple passing a head of this csv file this is all required for this for program. In next video you will see how, how to plot this latitude and longitude value.

# GPS – TRACK THE ROUTE 04

So now we know how to pass the csv file or using our csv library plot this on Google map for this we need a different library name as gm plot so let me just import this gm plot here to see that whether it is present on this anaconda cloud or not so I will put from gm plot import gm plot let's see, you can see that it's written that error module not found error no module named gmplot means that this module is not present in the anaconda cloud it might happen that you looking for some new module which  is not on anaconda most of the time all the modules and libraries are present in the anaconda that's why we recommended anaconda but there some library some new libraries which are still pending and which are not present on anaconda cloud but you can download it. One thing is that you can list all the libraries which are present on the anaconda, to do that you can just type on your terminal conda list will list all the libraries which are downloaded by the conda which are present on the conda cloud so you can see all these libraries which are present in your conda cloud ok, you can see that gmplot is not there so I am going to download it, in order to download anything from conda you can search each search it through conda the command is conda search gmplot, now it will search this gmplot named conda cloud in its cloud whether it is present in any of its channel if it is present then it will download, it might happen that may the library which we are looking for may not be present you can see that package not found error thrown from conda this library is not present in any of these channels in that case you have you can download the package through pip, so I will download pip install gmplot, now pip will install this gmplot and it will it will be automatically taken by the anaconda. Ok it says that successfully installed gmplot one point two point 0 so I will see this, let me try this command again you can see that it has been imported so it means that I can use this library now ok great let me delete this ok so I will import my csv library import csv next I will import this gmplot from gmplot import gmplot ok before I go any further you can read more about gmplot on you can Google it you can read more about gmplot on get there is a get on library on gmplot on the radio material is diploitation is provided how to place the map? How to create polygon? How to create the marker and how to draw it, how to see it, all these things are present depend in this look. Ok so let's see this. So I will import the csv file, csv library and gmplot library next what I will do I will place mine map on a particular location for that there is a function so let me create a variable gmap is equal to, you can see here also that how to place your map to particular coordinate to a location if you want to place the map to a particular location you need to write the latitude longitude value of that location and you can use the Google map plotter functional gmplot see you have to put the latitude longitude and the zoom resolution here the exactly same thing I am going to do here gmplot dot Google map plotter ok and you can see when I show the argument centre latitude, center longitude and zoom here I will put my location as twenty eight dot six eight nine one six nine this is one random location I put or you can change it accordingly whatever you want to put there three two triple four eight and let the zoom resolution will be seventeen so I will the gamp contains the map location where I want to plot the, second thing is I want to create the icon, the icon which I am going to use to plot my routes so I can just you just copy this line here ok you do not have to carry much, you can create your own icon here but I am going to use this Google icon you can just copy this line and you will get the icon ok I have the icon I have the map now I will read the I will pass my csv file so that I can plot I know how to pass it with open I will open my route dot csv file dot csv ok as f I will create my reader iterator using the csv dot reader function I will pass my file. Let me create a variable name k using this k how will I use this? So let me pass this reader iterator, iterate through it for row in reader ok so row will give me list of all the values which are present in this csv file so I need to there are two values latitude and longitude ok and store it in latitude and longitude, so lat is equal to row zero since this is a string value I need to have the float value I will convert it into float same thing which we did in the last video the same I will store the longitude value is float of  row awesome! Have the latitude and longitude values float values now I am going to use the 'k' as a dummy variable how, if k is equal is equal to zero it means the first time this is the first coordinator I left then what I will do gmap dot I will use the marker, marker function I put the marker function put the marker to a particular latitude to particular coordinate that latitude longitude, put the marker at this lat and long value which this is the first lat and long value with colour, so let me put the colours yellow ok and I will make a 'k' is equal to one which means that this was the first and after this whatever is coming you can do something else if you want so else if it is not the first coordinate which means the 'k' is also 'k' value is not zero means first value is already 'k' and you already put yellow mark at there so whatever is coming after the first is you can you create the I am creating the marker as particular latitude and long with colour blue ok this is done since the last latitude longitude value the lat and long at the last after this iteration will contain the last latitude longitude value again I would like to place a read marker there what I will do put a gmap dot marker lat long of colour red that's it, at the last location I will put red colour marker so we are done I just have to draw this map gmap dot draw with html file that's in my may dot html do it, I am going to run this ok so I created a my map dot html so let me see go to my folder it say let me open this with my chrome and you can see so this is since it is yellow it is starting from which means this is the first location ok this is the first location only that delivery guy started and then went till this particular streets when some where some where these are the blue coordinates given by the gps device and he going somewhere and you can see these are the locations he went these are the blue locations there there these are the red location this is the last location. So this is how you can easily plot the coordinate if you have the coordinated you can plot it on Google map using the python it's very easy though I didn't use any hard core programming there many libraries through which you can do simple things that's why python is so popular you can do a lot of things with few lines of code because there are lots of library and the community is very strong. So thank you very much

# TUPLES: PYTHON DATA STRUCTURE

Hello guys, welcome to programming screen cast, in this programming screen cast we would be seeing about a data structure that python offers you, please don't get terrified by the term data structure it is nothing it's just way which you can arrange your data. That is what you call as data structure in simple terms so python offers a lot of data structures we had seen some of them in the previous weeks we had seen list you have been extensively using lists and you will be also using in future and dictionaries we had seen you had dictionaries in rock, paper, scissors game I guess so will see yet another data structure call the tuples. So the data structure name is tuple, maybe I guess now you can see it the font size I have increased the data structure name is tuple, we call it as a tuple so what is basically a tuple will see right? So initially we will start with creating a tuple, how can we create a tuple? So let me say I want to store the ice cream flavours which I like let us say ice cream flavours let me say, ice cream flavours equal to it's a name of my tuple that's a variable name so I will say to create a tuple what you need to use is the normal paralysis the normal brackets open and close bracket so within this you need to give the items of the tuple, one maybe I can say vanilla then I can say chocolate then butterscotch then strawberry maybe I will say strawberry and so on you can give any number of flavours so alright so let me press enter alright my tuple got created now if I say ice cream flavours see I got this tuple vanilla, chocolate, butterscotch, strawberry whatever I had added so you can use this way or you can print as how had been using print ice cream flavours I think I was just pressing the tab key because of which it got auto fill and please note that I am using the console I am explain using the console if you feel that the screen doesn't appear familiar just that I have maximised the console that's it I will press enter see I got it as a tuple ok so what if I want to print it one by one, how can I do that? Simple just like how you use loops in lists or dictionaries something like that say something like that you can use loop ok let us see how we can print the elements one by one simple just like how you have some you have used loops in lists or dictionaries previous data structure whatever data structures you had seen you can use a loop to print each element individually as you wish so let me use a for loop for I will say flavours flavour in ice cream flavours that is my variable name start my loop I will say print sorry its print I have to say flavour so I have used an intuitive name you can use any variable name here for I j anything you can use even though this is tuple of ice cream flavours I am using a variable name flavour we got it one by one vanilla, chocolate, butterscotch, strawberry alright so we had printed the elements in the tuple and now in case if you want to access the individual elements how would you do? That is I just want to print chocolate let us say how would you do? Let us see chocolate is the second element in my list just like in my tuple so let just like how I did in my list I will use the index indexing mechanism so let me say print what is this tuple name ice cream flavours of the index I have to give its my second thing so let me give two letters see what happens butterscotch see the third one gets printed so if I say two the third one in the list got printed so what does that mean? So the indexing mechanism the python uses is different so that is it starts counting differently its starts counting from zero just as how it did in list you should start counting from zero this is index zero vanilla is at index zero chocolate is at index one butterscotch is at index two strawberry is at index three this is how it counts we start counting from one and but the computer starts counting from zero so if he want the second item we should give index one then we can get the second item as per our view, as per our perceptivity second item but for computer first item that is because its start counting form zero so let us print it print ice cream flavours of one so that is the second item actually if I print see I got chocolate which is the second one actually in the list alright so this is how you access an item simple nothing but you give the name of the tuple ice cream flavours you use this square brackets and give the index at which you are decided item is present say let me say three strawberry got printed let me say I will let it be a typo I will say thirty see tuple index out of range it throws an error in case such an item that is index number thirty I have instead of printing three let in case I printed I typed as thirty that a typo error given that there is no index thirty in the list it says index out of range that is there is a range of the index value for this tuple that is zero one two three these are the allowed values this thirty is out of this range is what the error means so it throws an error right so let us see we have let us just summarise very quickly. We had created a tuple ice cream flavours and we had access the individual element to create a tuple what you did? You just used the open parenthesis and you just inserted the item and you had just access the elements so in case I want to change something for example I don't want to have butterscotch instead I would like to add black current flavour so I want to modify the element or I want to add something newly so how would I do that? Let me see let me say tuple name is ice cream flavours just as how you did in lists the update functionality let us try the same way because we are accessing the elements in a similar way as that of the list using indexes so let us try updating as well the same fashion as that in list so let us see ice cream flavours of it was in index two right butterscotch so let me remove that and I want to make it as black current black current clack current I made it as I don't need an underscore here is a string so basically ok black current so let me press enter see its throwing an error tuple object does not support item assignment its telling something ok let me try adding it as new item so it was having zero one two three right I want to add it at index four I will say index four is black current here I am trying to add it initially I tried modifying it was not allowing me for what so ever reasons let me try adding it the same error is being thrown again probably I have to delete that initial value and again reassign the value probably that is how tuples work let me experiment let me say delete the same key word that we used in dictionaries delete del ice cream flavours of two let me say tuple object doesn't support item deletion oh even this doesn't work so let me ask for help delete question mark that will give me a hint delete not found what happened? Omg! Delete ice cream flavours let me check let me ask for help tuple empty the tuple if I ask for a question it says it will empty the tuple it says I cannot delete an individual item I cannot modify an item I cannot add an item I cannot delete and individual item all that I can do is I can delete the tuple completely I can delete the entire tuple so basically tuples are sort of data structures which is fixed once of you fix something it remains as that as like that forever that is what they technically call as immutable so those who have some knowledge of biology you would have heard term called mutation that is nothing but some changes in your genes that is what they call as mutation so the same terminology they have been using here as well we call tuples are immutable because you cannot make changes as like you had made in list or dictionaries if you use a tuples so tuples once if you fix something it is fixed forever now if I say delete ice cream flavours the entire tuple would get deleted so let me check if I now call ice cream flavours and I call it says name 'ice cream flavours' is not defined so what are the options that is supported is you can create a tuple, you can access the elements of a tuple and you can delete it as a whole you cannot add anything modify anything delete a single item nothing of sort can be done so it is sort of a foxed data structure alright it seems very restrictive right so is there any other option? Is there any other thing that can be done in tuple so let me create a toy tuple for example create a toy tuple let me just use some numbers one two three four given that I have deleted that ice cream flavours tuple so I am just creating another tuple for demonstrating what else can be done so just like the length functionality of the list can be used I will demonstrate length of toy will say four so the length functionality can be used in case I want to see the count I can use the count functionality as well as if you want to retrieve the specific index where the element position is where the particular element is present you can see that toy dot count I want to count the number of times then number two occurs it occurs one time so it says two the toy of count is one let me change the tuple once for you let me making it as one two, three ,four, five, one, two, three, four, one two, three, four, five so I had made a new tuple basically see initially I had toy is equal to one two three four and here I have this you can ask me you said tuples are immutable you cannot change them but how can you change this, so what happens is that thing what I had created already this gets destroyed and something new gets created that's what happens so you, if you changing or if you try modifying in this way the old one get destroyed so whatever you once seek that would remain fixed you cannot change it alright so let me show that for you, length of toy initially it was four now if I say it's says nine so the earlier copy was completely destroyed so now I can show you the functionality count  toy dot count let me say how many number of times the number two occurs, it occurs two times number four occurs two times number five once so count is the function that is how many number of times this particular item is occurring in the tuple so in a tuple you can give numbers initially in ice cream favour example we gave strings you can give it anything even the floating point value you can give anything we will see some applications where the different types of values are given in tuples so count is one functionality and other functionality you can have is index so let me say toy dot index of five let me say five is present at index eight and if you say, say for example two it is present at index one it says but see two was present at two places so the index functionality returns the very first index from the left where the particular item is present, that is what the index functionality does, so these are some functionality that you can list, length, count index that can be used in tuples also but something you can do in list which cannot do in tuple is you are modifying an element, adding a new element or deleting the particular element those things are not allowed so tuples are sort of once you defined its fixed so such a restricted data structure where can it be used. I had post the question and probably you would have thought the same I guess so why do they even create this? Is it even useful? Of course yes it is used in many cases once such example I wills say, say for example you are dealing with rainbow, rainbow is a tuple you can say the colours in rainbow violet, vibgyor right! Indigo, blue, green, yellow, orange, red so once you have defined this as the rainbow no one can further add or delete or undo anything about this so this is we want to the rainbow consist of these seven colours its fixed so that you don't want anyone to make any changes in such scenario you can make use of tuples, in one scenario where this would be useful this rainbow even is a toy example you can consider as I just showed you to illustrate something that it is fixed in the nature you cannot change no now can change this for such scenario you can use tuples for one thing where in computer science they use tuples is in you can say in image processing so in images the each individual unit of an image is called as a pixel this is basically like human body contains cells, cell is the basic unit of the human body something like that, the basic unit of the image is called a pixel so each pixel, if you would as you would have studied in your school physics each colour can be represented in terms of the three basic colours namely red green and blue every colour any colour can be represented in terms of this that is by proportions of red green and blue by taking ten percent red forty percent blue and the rest green something like that by some proportions of red green and blue you can denote any colour so every pixel will have a colour in an image so in case if that is a image of a grass or some natural grass the top portion will be the sky it will have the blue value which will have dominant and the bottom portion will be the grass which will have the green value dominant in case you have sand there will be yellow it will be represented in terms of some proportions of red green and blue as you can see every pixel has variation in colours each colour can be represented in proportions of red green and blue so basically a pixel probably would be defined like this pixel one there maybe millions of pixels in an image depends on your camera resolutions where you would have seen that five mp camera, eight mp camera, thirteen mp camera you heard these days right, there may be different pixels it depends on the camera resolution as there are number of pixels increases the clarity of the image also increases so the pixel one contains some colour say yellow colour so it may be something like point seven sorry its zero point seven comma zero point one comma zero point two something like that some proportions of red green and blue rgb is the order in which they will represent the pixels so in that case they internally represent each pixels in terms of tuples so you don’t want to change this that is this should be the proportion in this image you don’t want to change it that time they will say this is the particular pixel configuration that is this is the colour that is associated with this pixel so in such applications in computer science tuples come very handy alright you can think of many such applications where once you fix something you don’t want to change it in such a scenario if you encounter some such scenario you can consider using this data structures tuples alright guys thanks for watching the programming screen cast have a nice day.

# LOTTERY SIMULATION: PROFIT OR LOSS

Have you ever tried your luck in gambling? Well, you shouldn't let us find out in our next joy named lottery simulation.

# <u>LOTTERY SIMULATION: PROFIT OR LOSS 01</u>

What's the matter? You look worried, yeah I have to share something with you. Ok I have this brother of mine who gambles a whole lot and I have tried convincing him several times not to do I mean not to continue in that path but he never listens to me and he feels he is earning the whole lot of them but I force him that he lose everything one day that is really bothering me. So what kind of gambling is that? Ok, let me explain and that is a system of ten units and he bets on one unit if he wins then he will get the all nine units but if he loses then its, its little less but that is you invest hundred rupees, yes and if you win you get nine hundred if you lose you lose the hundred rupees, yeah precisely actually deceptive yeah you feel as though you can make a lot of money yeah that's why he is feeling he is not loosing but I force that he is, you tried counselling him? Yeah I tried several times I have told them but he never gives an ear to it, ok. So what else can be done? That's what I am thinking I tried to see I am trying to seek your help for it, can is it possible that we can simulate this through a computer? Ok so you want to see whether, whether you can simulate the process of wining or loosing yeah in a game of gambling yeah and you want to see whether one can become rich or one undergo very losses, yes precisely I you see he is educated so if you help me with this I can assure to him that see you are not in the right path and this is what actually happening, I think this is possible but this is popularly called the gamblers rule in problem probability but that aside you want to see if we can simulate the enter process of gambling and try to figure out if someone can even win or not, of course we will win a few times, we will lose a lot of times let's say we play here for one year's time three sixty five days we would like to see whether we make a lot of money or not right? yes very good so what I will do, I will try to help you out with the simulation I will even give you the plots, ok on what happens if you keep playing for the next few days, months, years ok and the inference is starling enough to make your educated brother stop gambling yeah let's try our best though I cannot guarantee that he will stop gambling because gambling is a disease its very tough to people to come out of it but with proper counselling and making people understand the math behind it they might get convince that they will go back upraise oh! If you do that with the computer I will be really grateful should be possible let's try, I cannot guarantee you but I can tell you that once can be hopeful because with proper counselling you can ensure that you can make any educated person understand gambling don't work.

# LOTTERY SIMULATION: PROFIT OR LOSS 02

Hello guys, welcome to programming screen cast of lottery stimulation so let me first introduce that is lottery simulation let me just give you a brief overview, I hope you have seen previous video where the procedure of lottery simulation lottery was explained, I will just give a brief overview so basically there are some sort of coupons was available and you have to bet on one particular coupon and there draw coupon randomly from the available list of coupon and if your betted coupon and the coupon that appears matches you will win for playing this game you are supposed to place some amount and if you win you get something much more than what you have paid, so this is how they try to tempt you to play the lottery so let me explain it, let me take a very quick scenario and explain it so let me assume that there are some ten coupons so basically each coupon contain a digit a number from one to ten so one two three like that up to ten, one number is present in each coupon like that there are coupons that coupon contains the digits from one to ten you are supposed to bet something say I say the number seven will appear if you draw now I will say that so now after listening to my bet seven they will draw coupon randomly, if that randomly drawn coupon contains the number seven I win otherwise I lose, to play this game I have to pay say some hundred rupees if I win that if I get lucky and I got the same number if I win I will get say thousand rupees ten folds see I am getting back ten folds or even if you say nine hundred rupees anything I am getting much more than what I am paying so I am really tempted to play this game so let us see what happens if I play this game, if I get lucky I will get much more than what I paid, if I get unlucky I will lose hundred rupees which is not very big deal, let us try what happens let us try simulating it so to simulate this we will use something called random so because you are randomly picking one coupon out of the available coupons right so we are I am going to use package called random so let us say I will say import random I hope you are you have used this package sometime so basically to deal with something that happens with some probabilistic thing randomness we use this package random so I will I will so say I have a bet so bet is something I have a number so let me say I have to get the input of the bet, the bet is the number I have to get a input, input as you know would generally take everything in terms of strings so I have to type cast it I have to change it into integer type so let me do that int of input so I have to ask for what is your bet? So I will say your bet ok so I got the bet from the user, once I have got the bet now I have to stimulate the lucky draw, let me say lucky draw so what is this lucky draw? I will take a coupon randomly it will contain a number from one to ten so to stimulate that I will use the function from random package it is random dot randint what is this functionality done? What is this do? You are supposed to pass two parameters ‘a’ and ‘b’ as it is shown so a number form a to b between that in that range some number will be there so I want from one to ten so one to ten here both one and ten both the end points are also included any number can be picked from one two three four five six seven eight nine ten, one of these ten numbers could be picked even I have to bet something from one to ten so let me make this statement more clear, your bet from one to ten let me make this more clearer so I have inputted a value I have asked for what is my bet as you could see here I have asked for his betting so let me find so I have inputted what is the number on which I want to bet and now I am stimulating this lucky draw so here I will be getting another number and if so for that before that I need to have something called the account right? Where do I deposit the money that I am winning so let me say I have some account initially there is nothing I got it from the organisers of the game so it is zero and now if I get lucky that is my betted number as well as the one I got in the lucky dry they both turn out to be equal so this double equal to symbol is what I am using it for equality check if you use a single equal to it is assignment so it will change the value sign but I want to check the equality I don’t want to change the values so I am using the double equal to symbol that is equal to the lucky draw then to my accounts what happens? Whatever was there earlier along with that amount say nine hundred rupees gets added and to play this game how much have I spent? Hundred rupees so I have to deduct hundred because I have spent hundred from my hand and I have added some nine hundred rupees because I have won in this place so nine hundred got added and I had spent hundred rupees so the net gain is eight hundred rupees I am getting into my account if I get lucky what if I didn’t get lucky? Nothing I just lose my hundred rupees else if I didn’t get lucky else what happens I will lose hundred rupees so let me say account is nothing but I have to deduct hundred from my previous earnings from those earnings I am spending so I have to deduct it ok so at the end of the game what is there in my account and print ok see I just playing once here I get an input I bet on some number and some lucky draw happens, some number is drawn at random and if that is equal to what I have bet I win nine hundred rupees else I don’t win anything but to play this game I have to pay hundred rupees that is why I am deducting hundred in both the cases so at the end of the game what is that I will have in my account that is after playing this game what do I have, let us see that now let me save this and now let me maximise the console so here is where you get the output right so let me maximise this portion and now let me run this run, run it ok your bet from one to ten it is asking me let me say I bet on seven minus hundred see so I had lost why have I lost actually? I don’t know, I want to see what happened in my lucky draw case so let me print it and modify this programs slightly let me print the lucky draw print the lucky draw whatever is the value in lucky draw I want to print it as well as whatever is in my account I will print it let me save the program and now let me maximise this portion maximise here and now let me run it again its asking me for a bet so let me now bet on five it turned out to be four and so I lose ok let me try again I run again this time let me bet on two it turned out to be ten again I lose I am losing continuously ok it’s all luck so let me try simulating this multiple times and see what happens, let us see this in the next video.

# LOTTERY SIMULATION: PROFIT OR LOSS 03

Hello guys, in the last video you had seen what happens when you try betting on what would appear as a result of a lottery draw we got a lot of losses but we has just run only once right what if we play multiple times there is a possibility that we get lucky and we win a lot of price let us see that so let me assume in the first go if I play this game for one week and here if you see one week is seven days so seven times I have to get the input its slightly time consuming and also irritating on the users and if you think seven times you are entering the input so let me simulate this is  just the simulation right so this is we need not accurately do what we do in the real world so let us make it even this as a random integer that is how we have a lucky draw something like that we will have a random pick let me copy this and use the same thing here for betting thing also because every time while giving the input I didn't have anything in my mind so I was just randomly giving some input right so let why do I want to get the input, let me pick something randomly so let me pick random so I will use the same thing so see bet I have random integer and I lucky draw I have something so now I have to print it very clearly initially I just had lucky draw as a random, randomly pick numbers so I had just printed the value but now given that both the values are randomly pick I have to print it properly so let me print it let me say this is my bet turned out to be bet turned out to be this value that is stored in the variable bet and now the lucky draw lucky draw this turned out to be the value stored in this variable lucky underscore draw ok so I will print this too and my account has zero and every time I will check it I will see if my bet and the lucky draw is equivalent if it is equal I win a lot of price otherwise I lose so let me here even here I have to say amount in your account let me say amount in your game account game account is nothing but it is stored in the variable account may be let me maximise this part so that it's very clear ok so I have modified slightly if you could see I will just summarise the modifications so bet I am just choosing random integer which is my bet which is simulating my bet and this is another random integer which simulates the lucky draw that is whatever the organiser is picking from the collections of coupons that is the lucky draw and I will print the values whatever has been randomly picked then I have my account initially it is zero but as you keep playing see because we have to repeat this thing as I have said let us see what happens if we play  this game for a week so we have to repeat this thing so if I put zero here account equal to zero here every time account will be initialised to zero and you cannot really see whether you actually win or lose so actually when I putted in the loop this must be out of the loop so let me cut this line from here cut and let me put it here initially my account has nothing the game account has nothing and I start my game here and I pick a bet and the organiser picks a lucky draw and if they are both are equivalent I will win some amount if not I don't win any amount but since I am paying hundred rupees to play this game once I am subtracting hundred in both the cases and at the end of the game I will say this is the amount I have in my account currently so given the this is just seven times we are going to repeat it I said we are going to simulate it over a week alright so week has seven days so let me going to simulate it just for seven days seven times so I can print the amount at every instance but if we increase the number of iterations we may get messy that you print the amount at every instance so that time what you do, what is the final amount after those many number of games have been played we will just print the final one that's it ok so I have to repeat this for some seven times so let me use a far loop any loop can be used for is slightly convenient for us that’s why we are using it I in range seven because we want to repeat it seven times and see for loop requires indentation so let me select this entire block so I want this things to be indented so I have indented I have used a tap key I have indented everything so fine so I am repeating this seven times right so my programme is done I will just quickly summarise and we will run and see what happens so initially there is zero in my game account there is nothing in my game account I am doing this I am playing this game for seven days that is for a week I am playing this game each day what I do I bet for one number and the organiser will draw will draw a number randomly from his set of coupons that is what we call as a lucky draw, if my betting and the lucky draw that the organiser has taken if they both are equal I win a prize money if they both are equal I win a prize money otherwise I don’t win anything but to play this game I have to pay hundred rupees that is why I am deducting hundred from both the cases and after each game I am printing what is the amount in my game account so this is what the programming is doing I have saved it now let me maximise this portion value will get the output and let me run it run the file see we have got the output maybe I will maximise this pane that will be clearer so let me maximise this pane see initially the bet was three and lucky draw was six so I lost here so my account has minus hundred because I had paid hundred rupees initially and I didn’t win anything so I had minus hundred and see next time bet and my lucky draw is  equal so I got lucky I got an amount next time I lost so some loss is there next time I had lost so again there is a loss again a loss again a loss again a loss you could see there are some more losses compare to gain but still due to the ratio of the amount we could see there is some profit net profit I have out of the game this is we have just simulated the playing of the game for one week let me just try running again so run again I am getting some amount oh I am getting still more amount I am having more wins again this time I am losing this time I am winning so wins and loss it’s all completely random right so anything can happen so this is I have simulated if I play the game for one week next we will see what happens if we play the game for one month we will see that in the next video.

# LOTTERY SIMULATION: PROFIT OR LOSS 04

Alright guys, in the previous videos you had seen a simulation of lottery game if we play once and the next we saw if we play it for one week continuously now let us see if we play it for one months. On an average I take different months have different number of days on an average let me take thirty days, so for thirty days I am playing it, so just a simple change in place of seven I will make it thirty an thirty outputs it difficult to analyse so let me not print everything as I had done I hope you understood how things work the same thing it will work but given that the number of iteration the number of times we are repeating is increasing so output may look slightly messier so let me command these lines so I am commanding so I may need not print every time what happens I will just what I will do I will just print what is the amount at the end of the thirty times right? I don't even want to print after every iterations so let me bring this out of the loop I will remove this indentation let me bring this out of the loops now this is out of the loop I have simulated this for thirty times I may win or I may lose something may happen but let me see what is the amount after I have played for thirty times that is after I have played for one month completely what is the amount whether I have a net profit or a net loss let us see so let me save the file now let me this is where you get your output right so let me let me maximise this and ok let me maximise this better and now let me run the file, amount in your account minus three hundred so here I have an net loss this time this month so if I repeat this time I won this time I have a huge loss compare to minus three hundred minus two thousand hundred is really huge loss now I won I won something but the winning is less I won I have a loss I have a loss there is lot of alteration of winning and losing then we cannot actually predict we win or loss but it seems the losses if we occur is we incur is very huge so let us try increasing the number of iterations that is we tried for one month right for thirty iterations let us try it for one year what happens if we play this game for one year and let us see that in the next video

# LOTTERY SIMULATION: PROFIT OR LOSS 05

Alright guys, so till now you had seen the lottery simulation game for if we play this game for once, one week, one month we had seen now let us expand and let us see what happens if we play this game for one year. Nothing just a simple change in the program in place of thirty, thirty days in a month right so I play each day so I had simulated thirty, in place of thirty what if I play for one year so one year it may have three sixty five or three sixty six days depends on if it is a leap year or not let me consider the current year two thousand eighteen it's a non leap year so it will have three sixty five days so I will put three sixty five you can even try for three sixty six I don't think much change in result will be there just one iteration one more time you will play extra I don't think much change in results will be there let us plot let us look at whether we could win or we losing let us see  so let me maximise this pin now let me run the file oh it's a heavy loss I am losing some twelve thousand rupees oh my god seven thousand rupees I am losing loss loss win loss loss thirty thousand still heavy loss loss loss win loss you could see I have more loss compare to win so if you keep playing for one year I was able to find there is more loss compare to win so I am having the net amount but still you have a idea that there is a loss but we want to visualise what happens so let us see the visualisation part in the next video.

# LOTTERY SIMULATION: PROFIT OR LOSS 06

Alright guys, so you had seen what happens if you play this game lottery game for one year continuously so let us visualise what happens rather than seeing the net profit or a net loss let us visualise it. So let me say I will by visualisation I mean let us plot a graph basically graph with x-axis and y-axis, x-axis be on which day what do you win or what do you lose, on day one what is the amount you have on your account, on day two what is the amount you have in your account let us plot that and let us observe the trends so to plot will have a package that is amt plot lib I hope you had used the same package I guess it is in your crowd computing experiment the jelly beans experiment I had I hope you had used this package let's just reuse the same package will import it, import matplotlib dot pyplot the name is very big and even that we need to use it lot of times difficult for us to type this big name so let us use a shorter name and alias name to use that you will have key word as you have to say as and use your shorter name so let me say plt so instead of telling matplotlib dot polyglot I will be telling plt this is an instruction this is what this means so I have imported it and now I need to plot it right so I need x-axis and y-axis so I need two lists 'x' denotes the 'x' axis quantity and let y denote the 'y' axis quantity so on being of each day I have to increase one value in my 'x' axis at day one day two day three something like that so at the beginning of the loop I will add that to my 'x' list 'x' dot append is the functionality to add some element into your list so what should I have? It is the first day means I have to add one second day two third day three something like that but you're for loop is starts counting from zero that you want given that this is a visualisation for humans we need the counting to start from one so how can we tackle it just add one to it and append it so if I is zero I want one, I is one I want two so I basically want the values of I plus one let me append it I plus one ok so I have the day has started I bet something organiser draw something if it is equal I win some amount else I don't win anything but to play this game given by that I am paying hundred rupees I am deducting hundred from each instance and after the game what should I do? I have to whatever the amount I have in my game account I have to append this to my 'y' axis that is I have to plot the corresponding thing in my 'y' axis so let me append that to the y axis list so that is y dot append I have to add it to the list the account, account has some amount right so let me say I have to append this value I have done that at the end of three sixty five days what will be there in this x list the corresponding day count day one day two day three up to day three sixty five and in y axis the list have whatever the amount your account contain the particular day that will be available and I want to plot it this is the net profit or net loss you have and after that you have to plot it so what should I say plt dot plot this is the function plot it I can say x and y I want to plot these two things and I want to show the plot so I will say plt dot show plt dot show so it will plot the values with taking the days in x axis we had set and the amount in y axis and it will generate a plot and show us the plot this is what these two lines are doing, let us observe visually what happens with these days let us save the program now let me maximise this side let me maximise the pane maximise the current pane and now let me run the file see so there are sometimes you have increase in your account sometimes you have a money but there is a huge fall if you could see, see this is really a very huge fall so you as you keep playing you encounter more a loss if you could see this if you analyse the fall is very huge so that is what makes it a that is why they say lottery don't work ok this is just once let us try again say see even here you could see there is a loss at the end loss and every time you just observe the y axis the amount positive values are very less negatives are very high especially observe this particular graph just zero to thousand is the maximum you had gain but loss some six thousand something some days you has even had such a heavy loss so that is why the take home message from this is lottery doesn't work it amount they say you pay hundred if you win nine hundred nine times of what you pay this is a tempting statement but if you analyse the interfaces its not that you will win they are actually cheating you tempting you to lose a lot so it mostly the case that you will lose that is the take home message from this programming screen cast we had simulated and we had seen and for those who are interested the next avenue would be there is some probability involved here you could look into the map of it why is that happening? Why is that you lose a lot of time? So if you would analyse the probability of winning and probability of loosing you can get some insides for those who are interested you can check the math behind it why is it that most of the times we lose, I run once more this time I won I run once more this time there is a loss again loss, loss see there is a lot of loss and win I had run some probably six times just once I had won that is by one I say I have a net profit most of the times net profit or loss what I gain out of the game is a loss why this happens? You can Del deep into the probability aspects mathematical aspects that can be done as an extension to what you have learn in the programming screen cast this you had simulated and shown that it will be a loss why that happens that is being explained by the mathematical part, you can look into it alright guys thanks for watching this programming screen cast have a nice day.

# IMAGE PROCESSING: ENHANCE YOUR IMAGES

What is the first thing that you do after clicking pictures? Of course apply filters in order to enhance them, so let us now shift gears and explore a new technique to enhance images through our next joy image processing.

# IMAGE PROCESSING: ENHANCE YOUR IMAGES 01

We see CCTV's and surveillance cameras almost everywhere these days so do you have any clue what exactly they do? They record a whole lot of photo frames and that becomes a video and they have to store all these video transactions across several days so storage can be a problem and hence the video quality is very very less, if you want if you want to store one weeks happenings in your hard disc then the quality will be very less, the quality is high you can only store one days worth transactions so the point is images captured by the surveillance cameras are generally very very low in quality so low that whenever there is a crime it might even be difficult for you to detect by surveillance cameras some of the minute details but you can indeed use your expertise in programming to take those video frames analysing video can be difficult so let us limit our discussion to photos so assume you capture a picture and the picture is of low quality so low that you have no clue what exactly are the minute details in the picture for example you are probably seeing me if you take a picture of mine you probably wouldn't know what exactly is this symbol here on my shirt right? can you use programming to enhance an image so much so that the lesser details get enhanced and are visible as you can see such a enhancement would be a great use for forensic experts especially in a situations where there is crime and you have captured some pictures let's say through surveillance cameras and the qualities very low. How do you enhance them? Let us see a couple of situations and let us tell you we are gonna tell you how exactly to enhance the images so much so that picture make much much better sense then what it did previously.

# IMAGE PROCESSING: ENHANCE YOUR IMAGES 02

Hello all, welcome to programming screen cast of image processing let us explore some cool ideas in image processing through this screen cast before taking you to the programming part let me show you the images using which we are going to do image processing. This is the first image as you could see it looks like some mirror image all that we can make out from this image is just that some news paper clipping some photos have been pasted on the wall and what is written in the newspaper? It's not so clear, this being mirror image it's not so clear for person to understand what is being written here so let us apply some image processing technique, transform this image into a form that all can understand and let us try to see what is there. This is our first image for image processing that we are going to do now, next image is this, this image has been taken from a crime location as said in lectures then this whole exercise of image processing is ended helping find out the truth, this thing is image of bullet shot, a few bullet shots can be seen but there are many feature in this image which are not clear to us we will apply some image enhancement techniques image processing techniques some enhancement technique will enhance this image so that we shall understand about the minute features present in this image, who knows maybe that may serve as evidence let see ok not we start off with the programming component the first shown is the image of the mirror image that needs to be flipped so we can understand what is the written in those news paper clipping that has been pasted on to the wall let us see that so that exercise we are doing now the technique we are going to apply is flipping in some term it is flipping the image that is our first image processing task we are going to do, for doing this we need to downloads the necessary package let us do that we need the image package for pil library let me import that from pil import image, done importing alright I had shown you this image, this image is obtained dot png the name is obtain dot png lets open this image, it is done by this way image equals to image dot open here you are suppose to give the file path my image file is in the same directory as that of my python script so I am just giving the file name in case your image file is in different path you need to specify the entire path please note that the correct path has to be specified else if you just specify the name the file would be searched in the default directory where your script is running please note this point. File name is obtained dot png run with it we have opened this images this part is opening the image we are done with it images in general computers processes them in terms of matrices so that image is opened the corresponding matrix format is taken and capture by this object img the matrix format is capture here and flipping the image is the terminology that we are using here to actually convert the mirror image into actual image we are using the terminology for the thing that is easy on our minds but the technical terminology that they uses transposing as I said that images are generally transferred in terms of matrices so matrices I hope al would have come across them in your high school level mathematics in matrices there is some special operation called transposing that is nothing but your making your all rows as columns and columns as rows that is what you call it as transposing. So transposing operations your are going to apply to this matrix which will actually give you back your actual image from your mirror image that is what we are going to apply now. How will we do that? Let’s see. this is our matrix object of the transposed image so let me call this transpose img this is nothing but image img dot transpose this is the pre defined function transpose of this is the pre defined parameter image dot flip left right this is the pre defied parameter what this line does is? It takes the matrix that is captured by img of object transposes that matrix and this new matrix is stored in transpose img object that object captures the new matrix so now we have got the actual image in the matrix representation that is understandable by the computers but now what we need to do is? We need to convert it into a form that is understandable by humans, let’s do that. The next step is we have to convert it into a human understandable format and save it to a file, to a file in a human understandable format that is the next step we need to do alright so this is done like this transposed image, you take the matrix object which you need to actually save it into human readable format to take this object and call save function on it this is also a pre defined function and it takes an argument here same like how you had done in open you are supposed to specify the path where you want your output file to be saved, here I want the file to be saved in the same directory so I just giving the desired file name. In case you want it to be saved in the different path please do specify the entire path correctly and give the desired file name let me give the desired file name. I will call this corrected because that was a mirror image which was not very clear to us this has been corrected so that it is clear to us now corrected dot png corrected dot png is the file format I want the same as input file format its fine with me so I have given that in case you need may be you can try changing the file format as you required jpg or dmp whatever you can give tat ok so now where everything is done just print a message this is for us to understand that the output file is ready just say that done flipping, done flipping ok everything is done let me run this file I have given command as well for each line of the code so in case you think that you need to understand the code please do pause the video, read through that understand the code then you may proceed with running of the file, let me run this file. Ok I got the output as done flipping let me check the output file I had named it as corrected dot png let me check it corrected dot png yeah I got this yeah this is the file see now it is clear from us to understand what is being written some news paper clipping doctor accused of killing wife something like this yeah some news paper clippings have been there so now I can understand what is being written here unlike the previous image which was not clear to us you can observe side by side this is not very clear this image is not very clear to us where as after applying the transposing operation we got an image that is very clear for us to understand what is being conveyed here so this is how transposing operation has helped us underwhelm something which is which was previously not clear and now that clear with the help of coding computing well among two operations that I have said one operation is done let us go with the second operation I don’t want this thing so let me command it I am done with the first operation so let me command that now the second operation is nothing but image enhancement there are many image enhancement techniques available we are going to use one special technique called as histogram equalisation specifically adaptive histogram equalisation we call it as clahe this is just the technical name all that you can take is that is an image enhancement technique we are going to use it so that I had shown you that image previously, yeah this is the image there are some features in this image which are not very clear to us that could be taken clearance we apply this technique enhance this image so that the portions that are not clearly visible will be clearly visible who knows this is a image taken from this is the image of bullet shot this is taken from the crime location there maybe some features which may be evidences which open up new facts anything can be possible right! Let's explore alright. For doing this we need to import our package cv2, cv2 is the package we need we have imported and now I need to same like that I need to read the image, image img equals to cv2 dot image read is the parameter I am supposed to give similar to the previous one suppose to give the file path here, since my image file path is there in the same directory I am just giving the file name in case your image file is in a different place you need to specify the path completely file name was crime dot png let me read this image, so image has been read so we are suppose to now apply the technique CLAHE image enhancement using clahe that is contrast limited adaptive histogram equalisation it is clahe in case you are interested to learn more about it there are so many good articles available online which reveal you everything right from the math underlying in the concept everything's beautifully explained you can see it, here our aim is to arrival the mystery so we are not getting it into the technical part of it this can be looked up on any time it's there are lot of resources available straight forward you can do it so let us apply the technique to our rescue to unravel new facts who knows anything may unravel let's see. I have to first prepare for applying clahe technique, preparation for clahe this is the step that has to be done this is the equalisation technique histogram equalisation technique this is an enhancement technique to enhance the image some preparatory thing has to be done that is what I am doing now, I am creating an object that will how the properties of enhancement cv2 dot create clahe this is the preparatory face we are done with it now this enhancement technique works well if the images is in the black and white format all that we need is to unravel the facts so let us convert it to black and white formats get to know if there are some new facts and based on that if needed we can even convert it back to colour and we can do it that is our need basis we can do it anytime so let us convert it to nothing but a gray scale image so convert it into gray scale image this is the gray scale image we have to convert the image into the gray scale value so that this enhancement technique shall be really really effectively applied which converted gray image is equal to cv2 dot cvtcolour of to pass this image object then colour underscore BGR2GRAY this has been converted into a gray scale image cvt sorry cv this all these are pre defined functionality in cv2 package so just observe everything has been pre defined so it is with very few lines of code anyone can enhance any image unravel any facts from it this is so simple now a day's ok we are done with making it into a gray scale image now let is ok we shall apply the enhancement process apply enhancement that is the next step we have to do enhanced image is clahe dot apply of gray scale image we are suppose to give the gray scale image to the CLAHE algorithm gray scale image has been passed and after enhancement all these as I have said image processing all this would be delt in terms of matrices img is an matrix corresponding to your original image that has been converted to gray scale and the corresponding matrix is captured by gray Img and that has been applied and some transformation would occur so the image is enhanced and this particular matrix is captured by e and h image that is enhanced image this is the short form I am using it this is captured by this particular object so having said that enhancement is done it is in a format that is understandable by computers now the next step is to make it human understandable let's do that and the output has to be saved in a file save it to a file this is the next step we are supposed to do will do that cv2 dot im right image right here u s always we are supposed to give the path of the file but I want the output file also to be written on to the same directory so I am just giving the desired name, in case you want it in a different path please do specify the complete path I would like to call this as enhance dot png I am ok with the same format in case you want to change the format you can try for changing that too and this image must be an image corresponding to this particular matrix this particular matrix enhanced image the enhanced one from matrix representation which is understandable by computers to representation that is understandable by humans as to be written so we have to pass this as well and this particular file be named as enhanced dot png that is the command we are done with it so once everything is saved to notify us let us print done enhancing I would recommend that please pause the video here take a look at the code I have written with comments please do look at the code understand this once you have understood this completely lets go with running of the code ok now let's run the code yeah so it is done, done enhancing it shows so let us search for this file enhanced dot png enhanced dot png let us open this ok this is the given file which was not very clear now just observe this is enhanced version of the image the cracks on the wall starting from that every minute detail can be observed it seems some marking is over here even that could be observed alright now we have enhanced this image few more details that are not clearly visible in that original image are now revealed after this enhancement we shall analyse this and this may open up new facts that were earlier un realised thank you for watching this screen cast have a nice day.

# IMAGE PROCESSING: ENHANCE YOUR IMAGES 03

You see many a times when we take our own picture we realise we actually appear better then what we are that is because of the inbuilt image processing applications on your cell phone or on your camera so this is a very bid industry as you know look at instagram it takes your picture and shows you ten different ways in which you can sort of enhanced right? So just now we showed you a couple of examples of how you can take an image see it as a matrix and perform operations on it and get a brand new picture out of it right? This was just the beginning it's a big ocean out there you can actually explore to whatever extent you want.

# ANAGRAMS

List slit, lift filt, eat tea, what is common among all these? Let us find out in our forth coming joy named anagrams.

# ANAGRAMS 01

Do you know what is Hooke’s law? Well Hooke’s law states that force needed to extend or compresses spring by some distance is proportional to that distance it was discovered by Robert hook. Well in earlier time’s scientist were more concern about protecting their secrets than sharing new knowledge with the world. Robert Hooke first reported Hooke’s law has this which unscrambled to this which means force proportional to stretch, you must be wondering why are we discussing all these, well look at these two words these are called anagrams, anagram is a word or a phrase formed by rearranging the letters of another word, let me give you some more examples of anagram. For example listen silent, tar rat, dusty study, inch chin, brag grab, stressed desserts so now let us move towards the programming screen cast of anagrams.

# ANAGRAMS 02

Before we move to the programming screen cast of anagrams I like to tell you what are ascii values, ascii is basically A S C I I which stands for American standard code for information interchange, ascii values basically represent character encoding. What do you mean by character encoding here? For example, ASCII value of A is sixty five that means a character represented by some integer value, here integer sixty five represents character A. Similarly we have ascii values for rest of the characters too I will show you, here we have the ASCII table you can see here that the capital A is represented by sixty five, capital B is represented by sixty six, capital c sixty seven, capital d sixty eight which goes till capital z ninety so similarly we have character encoding for this smaller for the lower case letters too for example small a is represented by ninety seven, small b is represented by ninety eight which goes till one twenty two and z is represented by one twenty two similarly for the special symbols also we have the ASCII values for examples exclamation mark is represented by thirty three then hash value is thirty five then dollar sign is represented by thirty six percent sign is represented by thirty seven similarly we have ASCII values for plus minus even round braces so you can see and you can refer this tables for ASCII values and we will be using these ASCII values for the implementation for our program related to anagrams so let us move towards the programming screen cast of anagrams.

# ANAGRAMS 03

Welcome to the programming screen cast of anagrams, I hope you know what are ASCII values, I have explained in the previous video if you haven't watch the previous video please go through the previous video ASCII values basically represent character encoding each character is represented by some integer value ASCII basically stands for American standard code for information interchange this has already been exchanged in the previous video so please go through it now the questions comes how can we find the ascii values in python? How can we find the ASCII values of characters in python? So we have a pre defined function here called ord, ord we have a pre defined function here will be using this to find the ascii value of a particular character so I will just show you for example I need to find the ascii value of capital A so I will just write capital A we have sixty five as I already told you the ascii value of capital a is sixty five of b is sixty six, of c is sixty seven so we will just cross check whether it is giving us the right ascii values so here it is sixty six then capital c we have sixty seven now I will check the value of capital z which is basically ninety, yes it is ninety now we can also check the ascii values of lower case letters so I will just write small a here it is ninety seven that is correct then small b it is ninety eight small c ninety nine and let us also check the ascii value of small z and it is one twenty two yes we have the ascii values here this is how you can find the ascii value of a particular character as I already explained we also have the ascii values for some special symbols like sign of exclamation, @, percent, hash tag so we can also check so let us check for the sign of exclamation it is thirty three you can cross check in the table that we have shown here in the previous video you can cross check with the particular table now we have let us check for @ it is sixty four and for percent sign it is thirty seven for dollar it is thirty six for hash tag it is thirty five so this is how you can find the ascii values of a particular character and some special symbols too so I will advise you to cross check with the table that we have given in the previous video so now let us use this let us use this function to implement anagrams so as you know we have two strings and we need to find whether these two strings are anagram or not? So I will take two strings here I will take str1 and I will input enter the first string I will also take the another string str2 and I will write the input enter the second string second string as you all know by default it takes string as the input we don't need to type cast is as we do in the case of integer or decimal values whenever you have to take string as an input you don't need to type cast it you just write the input function input and in the round basis whatever message you want to display so here we are done we have the str1 and str2 we will compare this str1 and compare the str2 and find out whether these two strings are anagrams or not so let us think about it if we have to use the ascii values how can we use it some of the ascii values use of characters present in the anagrams should be same please think about it the some of the ascii values of the characters present in anagrams should be same because both the strings have same set of characters if two strings are anagrams then they should have the same set of characters for example I showed you one example cat and rat both of these strings comprise of three letters three characters R A and T so if we if I sum these values of these both strings the if this if I sum the ascii value of these both strings the answer should be same of the answer is same then these two strings are anagrams otherwise they are not so I will just now I will just write the code to how to sum the ascii values of the characters present in the string so will take count one here I will initialise it to zero I will take the variable I here we will use while loop here I will run my loop till the length of string one since we are trying to sum of ascii values of str1 so I will take this while loop take this length of str1 and what should I write here? I should write here count of one is equal to count of one plus I will use the ord function here str1 of I so we have the count value here now the count one will store the sum of ascii values of the characters present in str1 I should increment the I value here so that it traverses whole of str1 then I will take count two also and this is zero similarly I will apply while loop here too in order to find the sum of ascii values of str2 I will write length of str2 I will write count two is equal to count two plus ord function str2 of I should also increment the I value here so now I think we are done with generating the some of ascii values of both the strings now I will write if count in one is equal to is equal to count two then I have to print these are anagrams anagrams else what do I need to print here I need to print these are not anagrams I will write these are not anagrams so I think we are done with the program what did we do here? We took two strings then we stored the some of the ascii values of the characters present in the both strings in count one and in count two respectively then we compare count one and count two if they are equal then these must be anagrams otherwise these are not anagrams so I will just run this so I will write anagrams dot py I should enter the first string for example I enter listen then I should enter silent as you know listen and silent these both make anagrams but it is saying that these are not anagrams I think we did a mistake here yes we haven’t initialise I again please look at this we haven’t initialise I again we incremented I here so we need to initialise I again because we are using the same variable in both the loops so it was a mistake here so please rectify it we need to initialise I here again so let me run it again I will write silent and listen, let us check whether it put yes these are anagrams these are right so let us let me run it again I will write some example which are not anagrams so I will write for example wet I will write rat these are not anagrams so our programme is giving us the right output so let me also show you what a count one and count two are storing here so you get to know that these are actually the same values count one here and I also print count two here so let us run again so let us take the strings which are anagrams so I will write rat then I will write tar so see these both count one is also three twenty seven count two is also three twenty seven so both are anagrams this should be same and our program is giving us the output these are anagrams since these we already know that these are anagrams so it is giving us the right output so this is how you can code for anagrams I will go through the program again. First of all we should take two strings str1 and str2 and our motive is to figure out whether these two strings are anagrams or not so I will just take one count that is the count one it will store the ascii values, sum of the ascii values of the characters present in the str1 I initialise the I is equal to zero then I had then I will have the while loop which will run till length of str1 and then we are storing the sum in count one and we are using the ord function to find the ascii values it will store the sum of the ascii values of the characters present in the str1 and then we are incrementing I here and I am printing the some of the ascii values here of str1 then we are initialising another variable that is count two which will store the sum of ascii values of the characters of str2 and then we also have I is equal to zero why do we need to do this because we have already incremented in the previous loop and we are using the same and we are using the same variable here so we need to initialise it again so I initialise it again so I wrote I is equal to zero then I again had a while loop which will count the number the sum of ascii values present in str2 and then I printed these count two value and then I checked if count one is equal to is equal to count two if these both are equal then these are anagrams otherwise these are not anagrams so I hope you understood the logic behind it, what is the logic behind it? If two strings are anagrams then the sum of the ascii values of the characters present in the both the strings should be same this is self evident as I showed you we have different examples here silent listen tar rat and in silent and listen the set of characters present in the str1 and str2 they are same the ascii values should also be same so that's why we took the sum of all these ascii values and checked whether these two are equal if these are equal then the strings are anagrams otherwise they are not anagrams well there are many other ways in which with which you can find out whether two strings are anagrams or not! We like you to discuss on discussion form the other ways of how to find, how to figure out whether two strings are anagrams or not in the fourth coming screen cast I will be telling you one more way through which you can figure out whether two strings are anagrams are not. Thank you.

# ANAGRAMS 04

Welcome again to the programming screen cast of anagrams in this particular programming screen cast I will be discussing another way to figure out whether two strings are anagrams or not. In the previous programming screen cast we have already discussed a way to figure out whether two strings are anagrams or not, in this particular programming screen cast I will be discussing another way this particular method requires the use of sorted function. So what is sorted function? Sorted basically sorts any sequence list for example list and returns the sorted list so I will show you how can you use the sorted function for example I have a list here named x of integer values and the values present in this is not sorted for example they are four three one two now if I write print sorted of x see it printed the sorted value and it printed the sorted values in ascending order so sorted function basically returns the sorted values in ascending order I repeat sorted function by default sorts the value present in the list in ascending order so id I want to sort the list in descending order what should I do? Well we have a way here, how can you do that? For example we have sorted x and this will pass another parameter named reverse and in this I will write true so let us check whether it works or not yes it works, so if I write reverse is equal to true then it will sort the values in ascending order so I repeat sorted functions sorts the value in ascending order and if I write reverse is equals to true it will sort the values in descending order this is how you can use the sorted function, you can also use the sorted function on different data types for example we can use the sorted function on character values too for example I have ‘x’ as the list of characters as q for example w e r t and y now if I write print sorted of ‘x’ press enter so as you see it has sorted the values in alphabetical order so first of all we have e then q r t w and y so if we have list of characters then sort it functions sorts the character in alphabetical order and you can also use the string in this sorted function let us find out how can you use that for example x is a string here and we have python x is a string which has the value python here if I write again print sorted of x and what should it print, yes it has sorted the characters of python in alphabetical order first of all h then n then n o p t and y so now you see you can use a integer list you can use a character list and you can also use the string and use it in sorted function to sort the values present in the particular list or string and I hope that you are already aware of dictionaries we can also use sorted to sort the dictionary so how can you do that, I will take the dictionary here I will again name it as ‘x’ so I will write for example q, q is the key here and one is the value then I will take w, w is the key here and two is the value and then I have e here for example and its value is three and let’s take one more value I will just write four here and then I will take t here its value is zero so we have a invalid syntax here let us find out where is the problem, so the problem is we didn’t use the curly braces yes in dictionary we should use the curly braces and in list we use the square the rectangle braces so we should use the curly braces here and then if I write print sorted or ‘x’ see it is sorted the values in alphabetical order again first of all we have e then q then t and w so it is sorting the dictionary values on the basis of keys yes it is sorting the dictionary values on the basis of keys please note this fact now I have another example here for example I have a list of strings example I have ccccc bb then a then ddd and I want to sort this list on the basic of the length of this particular string yes I want to sort this list on the basis of length of particular list so the output should be first of all since a is the list since a is the string in the smallest length so first of all a should be printed then bb should be printed then ddd then ccccc should be printed so how can you do that? This is fairly easy, you just need to write print sorted capital l and the key should be length just write length here and I press enter here I have the desired output I sorted this list of string on the basis of length of the particular string so first of all I have a here then bb then ddd then ccccc so this is how you can sort this on the basis of length of the particular string so I hope now sorted function is clear to you guys now I will use the sorted function in our program in our program to figure out whether two strings are anagrams or not so first of all we should take two strings I will take them str1 and str2 I will just write input enter the first string then I should write str2 is equal to input enter the second string now you can figure out what purpose would sorted function would serve what purpose would sorted function serve here so we can sort this to list str1 and str2 and if the sorted version of these two list is same if the sorted version of these two list is same then you can say that str1 and str2 are anagrams otherwise they are not what should I write here? If sorted of str1 is equal to is equal to sorted of str2 then I should print these are anagrams else I should print these are not anagrams so let us run this program I just save it as anagrams two dot py enter the first string I will enter listen, enter the second string I will enter silent these are anagrams so our output is correct if I run it again enter the first string hello enter the second string thyui any random string these are not anagrams so our output is correct so I will also show you whether the sorted version of these two list are same so I will just print sorted str1 and then print sorted str2 so now let us try to run this again so I will take another anagram example I will take tar and I will take rat sorry I typed it as r a y so these are not anagrams but you see the sorted version are not even same first of all they are a r t then it is a r y now let us check on the list that our anagrams so I will write rat I will write tar so as you can see the sorted version of these two lists are same so first of all let us a r t for rat and for tar it is a r t so for sorted version so these two list is same that’s why these are anagram and we can also take the example of two lists of not anagrams so I will just write another list for example how and wow how and wow these are not anagrams since the sorted version of these two list is not same so you can now figure out how you can now figure out how to check whether two strings are anagrams are not and I will just go through the program again first of all you need to input two strings that is fairly simple and then you need to check whether the sorted version of these two list are same, if they are same they are anagrams if they are not same these are not anagrams so this program of anagrams was fairly easy you just need to apply a function and there you are you get the answer so we have discussed two ways whether two strings are anagrams or not there are I think many other ways through which you can figure out I like you to discuss on discussion form and bring out some most nice ideas to check whether two strings are anagrams or not. I hope this programming screen cast was useful to you guys if you have any problem any doubt please refer the discussion form we are there to help. Thank you.

# FACEBOOK SENTIMENT ANALYSIS

Apart from sleeping, working and eating what else do we do in our day to day life? Well my answer would be Face booking. So let us try to explore FaceBook with our next joy FaceBook sentiment analysis.

# FACEBOOK SENTIMENT ANALYSIS 01

# FACEBOOK SENTIMENT ANALYSIS 02

You saw what happened? People tend to be putting a whole lot of their life status on facebook in fact someone who is very active on facebook can actually we revealing more than what he suppose to reveal, in fact just by looking at his status updates you can see when he is on a high when he is on a general low in mood. Facebook is such a happening place but sometimes I suspect are people active in their real life as they are on facebook? What comes to your mind when I say analyse a given data? What is data to begin with? So how many times do I smile in a day? How many straights of walk do I take every day? What is my salt intake on an average per day? Sugar intake, how much junk though I eat, how many hours do I sleep? Don't you think if I get this data for one full year and then analyse it, put exactly can one infer, we probably can infer how healthy a life style we are living right? This is what you mean by data analytics, analysing the given data. If something as simple and straight forward as number of walking straights you take per day, say something about your fitness level. What about the data you update on facebook? Your overall activity on facebook, do you think we can infer something from an active facebook user, the fact is it's not something we can infer a whole lot of things. Assume I take all the data that you have shared on facebook from the past one month, status one, status two, status three, status four up to let's say some hundred status messages that you have shared from the past one month by observing this what can I infer? For example if my status was something like life isn't fair as I expected it to be, don't you think I am sounding slightly negative? Right, second example assume my status were to be something like this, I am proud of my country, I made it sound positive but irrespective of the tone in which one might talk the very sentence I am proud of my country gives a positive connotation  right? Now let us look at the third example, et your goals right! so its a very suggestive one liner neither positive nor negative its neutral correct? Do you see the sentiment of the first sentence being negative, second being positive the third being neutral as and always what are we doing? Programming of course. I told you three sentences and even told you and you realised that it was positive, negative and neutral. Is there any way we can write a piece of code which does the same without any human intervention what if we can use computer programs to predict what is the sentiment of a given sentence? Of what use will that be? U know where I am getting that I told you I take one month's data of a person which comprises of hundred status message, yeah he has updated hundred status messages I take all of them s one s two s three up to s hundred and then I look at the sentiment of these sentences assume eighty on hundred where negative which means this person is undergoing some trauma in his life or isn't in general happy person on the contrary what if ninety messages out of hundred where full of joy it means this person if too look out for a spouse you should marry this person right? a very happy partner will spread sort of happiness and happiness as you know is contagious right, what amount of this hundred status messages are positive how many of them are negative? And how many of them are neutral? Don't you think if we ask and answer this question we can say something about this person himself, I am interested in doing the following exercises, I want to take all the status messages I have ever updated on face book ever since I was part of the facebook, let's say from two thousand eight I think I am there on facebook, it's been more than ten years. So what I would like to d o is download all my status updates and try to analyse all of them by looking at the sentiments of individual status updates and see which are the months I was more happy then unhappy and months where I was unhappier than happier, times when I was neutral and so on, you see you can ask a lot of questions but hurdle here is how do I download my data which is ten years old, once downloaded how do I measure the sentiments of my status updates, this in fact is very easy to do today, in fact you can do it for your won facebook data as well and we are going to illustrate it for you. Point number one, we will tell you how to download your facebook data, point number two python to our rescue we will tell you how to use the exact python api to analyse your face book data

# FACEBOOK SENTIMENT ANALYSIS 03

Welcome all to the programming screen cast of facebook sentiment analysis, I will be going step by step and explain you how to do facebook sentiment analysis, we will be giving you background of sentiment analysis and others things too. So let us start with the programming screen cast. So first of all what do you have to do in here is first of all you have to download the facebook data and store it in excel file I will show you how can you download the facebook data, for that you must login to your facebook account, so here is my facebook account I have already logged in here first of all you need to download your facebook data, how can you do that? For that you must go to the settings in your profile as you are saying I will go to the settings here I have your facebook information and as you can see I have a tab on the left pane your facebook information so clink on that here is my facebook information here I have download your information, you can download your copy of your information to keep or to transfer to another service so click on that view as you all can see I have my facebook account information here I have the information In the form of posts, photos and videos, comments, likes and reactions, friends etc etc so first of all you need to create a file here you can see this button create file click on this, as you can see here it is written your file is being processed we let you know when its complete so it will take some time to create the file and then you can download your file yes so I have already downloaded my facebook information, I will I will show you here I have my facebook and my user name and you can see I have all the information in here, I have about you, I have friends, my followers, messages, pages, payment history everything is stored here so when you will create file here and download your file you will be your information will be downloaded in the formation of these folders yes, you will be given one zip folder and in that folder you will be given this information, you will be given the information about your friends about your groups, likes, reactions, messages everything will be there so here is our information, what you need to do next is, you need to store this information in an excel file yes so I will just show you for example I have folder here name post so I will just click here, here I have my post other people post to my timeline and my own post too so I have open this page so here is my post it is basically your post that means only my posts are shown here these are all the posts that I have posted on my timeline, so what you need to do here is, you need to store this post in an excel file for that I have created an excel file for example here all the post I have stored in an excel file so you need to do the same thing you need to just copy your post from this html file and you need to paste this these post in an excel file ok I repeat how can you download your facebook data, first of all you need to go to the settings in your facebook profile in settings you will be given facebook information you need to click on the facebook information tab on the left pane after going there you will be asked to download your information here first of all you need to create file, the file on the folder will be created and you will be given a folder like this, you will be given a folder like this here you will be having each and every information related to your facebook profile, information regarding your friends, your post network information so this is how you can download your facebook data for our programme you need to save this information in an excel file so you pick and y folder for example I picked post and then I open your post dot html which came like this and I stored these post in an excel file, I just copied these posts and stored it in it in an excel file as you can see these are all the posts that I have stored here ok so you just save these post in an excel file and save this file you will be using this file for our programme, this is how you download your facebook information and store it in facebook, store it in excel file so first and the foremost step is you need to download your facebook data and store it in an excel file, then what you need to do is you need to import the pandas library, it what purpose does it serve, it provides easy to use data structure for data analysis, you will get to know how what are this data structures eventually in our programming screen cast? Then you need to import the nltk library it is a library which is used to process human language, yes it is a library which is used to process human language what does nltk do here? It provides sentiment analysis of human data, so what is sentiment analysis? Sentiment analysis involves working out whether a piece of text is positive, negative or neutral, for example if I ask you how are you? This is a neutral sentence and if I say I am very very very happy today this Is a positive sentence and if I say I don’t know where my life is going and I am very sad so this is a negative sentence so sentiment analysis involves working out a piece of text and figuring out what whether it is positive negative or neutral so this is sentiment analysis so what will be you doing here is the file that you have stored, the post that you have stored in an excel file will be analysing them will be analysing them on the basis of sentiment analysis will be analysing your post whether your post are negative, positive or neutral so this is facebook sentiment analysis. Then we will use Vader, what is Vader? Vader is valence aware dictionary and sentiment reasoned, yes Vader, it is used for sentiment analysis but it not only tells you whether a piece of text is positive negative or neutral it also takes into account the intensity of the sentiment yes it also takes into the account the intensity of the sentiment what do I mean by that? For example I said I am very very happy today, that it will also tell me that this sentence is eighty percent positive or seventy percent positive it will also tell me this the intensity of my sentiment ok so will be using vader here, vader stands for valence aware dictionary and sentiment reasoned it takes into account the intensity of the sentiment too you need to download the vader lexicon, what is the lexicon? Lexicon acts as a dictionary here, lexicon acts as a dictionary here and then you need to convert your excel sheet that you have created to data frame with the help of pandas, what is a data frame? A data frame is a two dimensional structure in the form of a table so pandas provides you with the data frame facility as I said pandas provides you easy to use data structure and data frame is one of them, will be using them in one of our program. So this is the step by step guide to our program. How can we do facebook sentiment analysis? I know some of the things are not clear here when we will do programming screen cast everything thing will become clear here so I just revise what we have done till now, first of all we need to download your facebook data, you need to login to your facebook account and then you need to go to settings and download your facebook information there we need to store your information In an excel file that I have already explained then you need to import the pandas library it basically provide you easy to use data structure for data analysis then you need to import nltk library nltk library will help you to process the human language it will basically help you to analyse the human data it will help you in sentimental analysis of human data, sentiment analysis involves working out whether a piece of text is a positive negative or neutral I have already given you a example of a positive neutral and negative sentences then you will not be using only nltk will be using vader too vader also takes into account the intensity of the sentiment, yes then you need to download the vader lexicon, lexicon acts as a dictionary here, you also need to convert your excel sheet that you have created to a data frame, what is a data frame? Data frame is basically two dimensional structures in the form of table which is provided by panda's library so this is it now will be proceeding to the programming screen cast in the next video. Thank you.

# FACEBOOK SENTIMENT ANALYSIS 04

So now let us start with the programming screen cast of facebook sentiment analysis I hope you have watched the previous videos, if you haven't I suggest you to watch the previous videos and start with the program of facebook sentiment analysis and I think you all know what is sentiment analysis now so what is basically sentiment analysis? Sentiment analysis helps you to figure out whether the piece of text is positive negative or neutral yes facebook sentiment analysis will help you realise the sentiment analysis of your facebook data so first of all you need to download your facebook data this tips have already been explained in the previous video so you first download your facebook data store it in a excel file, you download your facebook data and store it in a excel file will be using this excel file in this program so as I already explained you first need to import the pandas library so I will just write import pandas as pd then you also need to import the nltk library which will help you in sentiment analysis, I will write import nltk after that you need to import the vader library so what do you need to do here is, you write from nltk dot sentiment dot vader import sentiment intensity analyser so you need to import these three libraries pandas, nltk and from vader you need to import sentiment intensity analyser then I asked you to download the vader lexicon you can download directly from here I will just write the command please pay attention I will write nltk dot downloader dot download and in bracket you need to write vader underscore lexicon, lexicon basically acts as a dictionary here so you need to also download that for downloading will lexicon you have a command nltk dot downloader dot download and in package you need to download, you need to write vader underscore lexicon hen I will load my file here what my file name is data underscore file dot excel sx this is my file name in the previous videos like in speech recognition you had a doubt that where do we need to store the file which we are going to use in the program for example in speech recognition we used audio file all of you had a doubt that where do we need to store that file, actually you can store that file anywhere in your system bottom line is you need to provide proper path to your file yes, you need to provide proper path to your file for example here my data underscore file is in download so I will give a proper path here the path here is basically users simran setia that is my user name then in that I have the download folder and in that I have my data file ok so I hope that is  clear to all of you, you need to provide a proper path ok you can store your file any where nut you need to provide proper path in your program so I have loaded my file that is done it is basically xlsx so I have imported the library I have also downloaded the vader underscore lexicon and then I have also loaded my data file here and next what you need to do is, you need to read from the excel file so I just write excel is equal to pd I will use pandas library to read my excel file I will write excel file and in bracket where is y excel file it has been loaded in the variable file so what this is doing? This is reading from the read from excel I repeat  pandas library will help you to it is basically pd dot pandas library will help you to read from the excel file so will write pd because u have imported pandas as pd I will write pd dot excel fie and in bracket I write my file variable name that is file here ok so I have read my excel file with the help of pandas library as I already said pandas provide you easy to use data structure so will be passing this excel file converting this excel file to data frame, what is a data frame? As explained, data frame is a two dimensional structure in a form of a table so data frame will help you to analyse your data easily, as you know pandas library provide you with the facility you can convert your excel file to some other data structure which are easy to analyse yes so will be converting our excel file to data frames so I just write dfs is equal to xl dot parse and I will write xl dot sheet underscore names zero here we are storing our information in the first column so I will write zero here I will just show you for example in the first column only I have stored all my post information you can see here, I have stored all the information in the first column so I will write zero here so I will write hash what is this statement doing? Parsing the excel sheet to data frame, parsing the excel sheet to data frame next what do you have to do here is you need to update your data frames as per your requirement for example here I have some blank rows here as you can see I have some blank rows here I need to delete this blank rows from my data frames so I will just write dfs is equal to list dfs and in square bracket I will write timeline because that is the name of my column ok and then I also print the all dfs so what it is basically doing here is it is removing the row from the data frame I will write it removes the blank rows from the data frame. So we will just run our program here and see how data frames are coming so I will just run so I first need to store my file I will store facebook dot py so as you can see I have my data in the form of data frames whatever has been written in the excel file is here because I have converted that excel file to data frames so we have the data frames here now we have the data frames next step would be analysing them so we will do sentiment analysis on this data frames so what will I write here, first of all I initialise my sentiment intensity analyser I will write sid is equal to sentiment intensity analyzer then as you can see in my excel file I have this information too that when my post was posted for example the particular time and date is already been is also been return so I need not analyse this information the particular time information so what will I do here is whenever my whenever this sentiment analyser would encounter this the timeline information the time information it won't analyse it, it will only analyse my post ok I repeat here for example I have here Monday march twenty seven twenty seventeen so this has nothing to do with sentiment analysis so I will just remove this I will not remove here from the excel file I will just hardcoded in my program so that whenever my program encounters this particular statement this time information it skips that information and it doesn't do sentiment analysis on that information so I will just write here I will store it in str1 so what is the information it is basically utc plus zero five thirty so it may be different for your for excel file so you need to check in your excel file if there is such kind of information you can skip that information because this has nothing to do with the sentiment analysis so now I will write for data in dfs a is equal to data dot find str1 this particular command will str1 in my excel file in my data frames and if it, it will basically return a Boolean value here for example if it returns minus one here that means it hasn't found that str1 in that particular a for example here I have the data here I have the data and if it encounter this particular line Monday march twenty seven twenty seventeen and it has encounter this str1 also here so it won't return minus one here because it has found that utc plus five thirty here but it hasn't found that utc plus five thirty here that means it is an information that needs to be analysed that we need to analyse the that we need to analyse this particular information we need to perform sentiment analysis on this particular information so if it is hasn't found this str1 that means we need to analyse that particular statement so I will just write ss is equal to sid dot polarity underscore scores data whatever data it has encounter now, now it need to be analyse we needs to perform sentiment analysis on it so it will print the data here also print the data and then for k in ss write print k and then ss of k it will basically print whether the basic information is positive negative or neutral and it will also print the intensity of the sentiment too I will just run this file then you will get to know what I am trying to say here so I will just run it let us just run it let us just run this so super we have all the information here we have all the timeline information here so for example my first post was this row rohith Sharma wants to read the murder of roger whatever it is and then it has given the polarity also that it is point two five one negative it is point seven four nine neutral and it is zero positive and the it is also given some compound information compound information is aggregate of all the sentiment for example it is negative also neutral also but the compound information says that it is point minus point six nine that means it is negative similarly in the fourth coming post too it has return that is zero negative it is point seven two one neutral and it is point two seven nine positive and it is also given the compound information since the negative value here is zero and here is positive so the compound information the compound sentiment is also positive here so it is analyse each and every information given in the excel file and it is return that whether it is negative neutral positive and also the compound sentiment that we can figure out from the from this information so I think now everything is clear to you guys now I will revise the each step here also so first of all what do you need to do here is you need to import the libraries basically you need to import three libraries here you need to import pandas you need to import nltk and you also need to import vader here so I have imported these three libraries then I have downloaded the vader  underscore lexicon you also need to download that and then I need to load the file here since I have already explained you need to provide proper path to your file you can store your file anywhere in your system then I read from the excel file after reading from the excel file I will pass the excel file to data frames yes data frames is the facility that has been provided to us by the pandas library, data frames are basically two dimensional structure in that form of a table it makes the analysis easy and then I have removed the blank rows that at present in my excel file so this is how you do it after that I have print I have printed my data frames that each and every data frame is printed here after that I have initialised my sentiment intensity analyzer and then I have explained you guys that I have in my excel file I have also the time information the time at which my post was posted on my timeline so I need not analyze this information this has no relevance so whenever my program encounters this particular str that is the time information and then it skips that particular statement and goes to the next statement so for that I have taken a variable a here whenever in my data str1 is found and if str1 is found it needs to skip that for example that means if its values is minus one it hasn’t found that str1 yes I repeat if its value is minus one it hasn’t found that str one so now I need to analyze that particular statement if A value is minus one that means it hasn’t found that particular statement that particular time information so I will just analyze that statement for that I need to write sid dot polarity underscore scores and in bracket you need to supply data then I have printed that data and then for ‘k’ and ss I have printed the particular ‘k’, ‘k’ will denote here whether it is negative neutral or positive and ss k will tell you the intensity of the particular sentiments ‘k’ here is the sentiment ss of ‘k’ here is the intensity of the sentiment I hope that this programming screen cast was useful to you guys and you have understood it well and if you have any doubts please post on the discussion form we are there to help you. Thank you, thank you so much.

# NATURAL LANGUAGE PROCESSING: AUTHOR STYLOMETRY

Guess what? Today is my birthday and I got this pack of letters written anonymously by my friends, will now I have to figure out who has written which letter? Can we write a code for this? Well let us find out in our next joy natural language processing.

# NATURAL LANGUAGE PROCESSING: AUTHOR STYLOMETRY 01

Observe the conversation happening between these two teachers. One teacher is getting her feed back by the students and the other teacher is just listening to whatever this teaching is saying. Let us see the conversation now.

# NATURAL LANGUAGE PROCESSING: AUTHOR STYLOMETRY 02

Hey anamika! Hey how are you? So how is this semester been? Semester has been great, that’s also reflected in the favour of forms that I received from the students which are graduating, oh nice and there is one regular form that I like to show you for two reasons ok mostly they are and second thing is that I have a feeling who would have written this, wait a second they are anonymous feedback right? Yeah so how do you know who is written it? Yeah yeah they are anonymous but from the style which they write I am doing that I have checked their answer sheets from the beginning I have a feeling of the style that they use so this sheet you see there lot of E usage and lot of adjective so I have a feeling who would have written this. Oh wow.

# NATURAL LANGUAGE PROCESSING: AUTHOR STYLOMETRY 03

You see what just happened? One teacher by name anamika is talking to another teacher varsha about her feedback as a teacher, students have given her some feedback and in that feedback particularly anamika is impressed with one such students feedback who has given her a whole lot of complements and most importantly anamika realises that all though the feedback is anonymous she has feeling who might have written that feedback, how? She understands some sort of way it is written the sentence the grammar and the style of writing in general even that she is a teacher of a classroom she probably knows how her students write she is trying to figure out who this student could be who has given her complements. Is it really true that we can look at the style of writing and then conclude that this is the person who has written this letter, let us do a small experiment? We have this huge repository called the project Gutenberg all you can do there is download a whole lot of books that are available in a text format free of cost without out any none of these things are copyrighted material, you can get this material and then analyse the data with your programming skills. Now I would like to take a look at some author's complete work and try to figure out if we can see some signatures in their writing.

# NATURAL LANGUAGE PROCESSING: AUTHOR STYLOMETRY 04

Hello guys, welcome to stylometry with python. What exactly stylometry is? Stylometry is the quantity study literacy style. It is based on the observation that authors tend to write in relatively consistent, recognizable and unique ways. For example some people write in short sentences while others prefer long blocks of text consisting of many clauses. Everyone has their own unique vocabulary sometimes rich sometimes limited no to people use semi colons dashes and other forms of punctuations in the exact same way. Scholars are use stylometry as a tool to study variety of question for example considerable amount of research has study the differences between the ways in which men and women write, isn't it cool? So in this tutorial will see stylometric analysis to set of English language text using nltk library of python let's see this.

# NATURAL LANGUAGE PROCESSING: AUTHOR STYLOMETRY 05

So what's the idea? Scholar T C Mendenhall in news paper said that an author stylic signature could be formed by counting how often he or she uses words of different lengths. For example if you pick a novel and count word lengths in a five thousand words segment and then plot a graph of the word length distribution. The curve would look pretty much the same no matter what part of the novel we had print, so isn't it cool? Will use this idea, will count the number of words of different lengths and then plot it, with this plot will try to understand the author's style of writing. This analysis is very basic one there are many advanced way of doing such classifications but since it is a very basic course will just motivate you with the idea so let's see this.

# NATURAL LANGUAGE PROCESSING: AUTHOR STYLOMETRY 06

Hello guys, see in this video we will prepare the data set. So as the instructor said we will go to the project Gutenberg, you can see the link here. If you go to this web page you will see there are many E books which you can download for free as a text version so we are going to use these E books for our analysis so for you what we have done we have created an Archive which contains all the which contains selected papers which we are going to use for our analysis. So you need to download an unzip the Archive or some papers containing eighty five documents that we will use for our analysis, this eighty five documents have been extracted from this project Gutenberg E book version. When you unzip the Archive it will create a directory called data, so you will have a folder name data, this will be our working directory so what do you have to do? You have to whatever python code you are going to create just create it in this data folder, we have the data will be using nltk library which comes with the anaconda cloud so you can just try importing nltk can your python console and you can see that it is working. Make sure you download all the collections of nltk using nltk download so before we can proceed with the analysis we need to load the files containing all the eighty five papers into convenient data structures ok, the first step is to assign each of the eighty five papers to the proper set since all the files have extend a format we can easily assign each paper to its corresponding authors for this we will use the dictionary data structure so the idea with the dictionary data structure is comes with the key and the value payer so with each key you can store, store its value and you can easily access this particular value using the key so if some of you know the basic data structure then you must know hashing also, in hashing what we do we create a hash of the particular data and whenever we want to extend the data we use that hash value to extend the data so the dictionary data contains the key value payers in Archive key would be the authors name and the list of paper numbers will be the values associated with these keys so let's see the data world.

# NATURAL LANGUAGE PROCESSING: AUTHOR STYLOMETRY 07

So, I am on this web page you can click on the link given in the description to download the data so, if you click you will get to this page, you can just click on the download button to download the data, I have already downloaded it, you will get a archive version of the data just click on it so to unzip it and you will get a folder name data so I have this folder name data I will just open it and I can see all the texts so name of the files are federal lists underscore one two three four and so on, these are the files which have been written by some of the authors and our key idea is to understand their style of writing so that we can get some ideas of how we can distinguish different authors using their style of writing. So our analysis we will use these files as input and we will plot the word length distribution and through that we will try to make some sense that whether different people write in a different way or not so these are key idea and these are input file please download it, see it once and let’s start with the code so we are splitting the codes into five categories so these categories are if these files for example are we are going to split categorise these files into five categories, first there are some fifty one papers known to have been written by Alexander Hamilton and there are in these files there are fourteen papers to have been written by James Madison, four of the five papers are known to have written by john j and three papers were co-written by Hamilton and Madison and there are twelve papers disputed between Hamilton and Madison which we don’t know who exactly was the author of these twelve papers so our idea is to understand the style of writing of these papers and then try to analyse this twelve papers whether they were written by Hamilton or Madison so we will use our key idea which we discussed previous videos that we will try to plot the word distribution I mean word length distribution and with this we will try to see whether whether it is really that people people have different way of writing people have different way of writing papers such that they use different length of words differently so in next video will try to write a code for this and will plot the graph of this length word distribution will try to analyse the data. Thank you.

# NATURAL LANGUAGE PROCESSING: AUTHOR STYLOMETRY 08

So, I will start with the code, I will start with by preparing the data so first I will prepare a dictionary name paper let me first import the nltk so nltk working or not let me check first so I will write in the ipython console import nltk so it's you can see that it's not throwing error it means it has been imported successfully before you start your coding you need to download all the constrain all the collections of nltk so for that I will write nltk dot download this function will open new ppi through which you can download all the collections of nltk this required for this stylometry analysis so I will just download everything so up to we download we will start with our dictionary thing so I will write take papers as the dictionary so the first author is I will name is Madison, Madison and the key is Madison and the values will be list of papers which he wrote so I will that is ten, fourteen, thirty seven, thirty eight, thirty nine, forty, forty one, forty two, forty three, forty four, forty five, forty six, forty seven and forty eight so these are the file names if you see the files the name is the first name is same for all the files federal list but after underscore there is the numbering so I am defining these numbers as a list here ok so for Madison he wrote these as I told in a last video that Madison wrote around fourteen papers so these are the fourteen papers which he wrote and then we will create the dictionary for Hamilton so I will put a comma and then I will write Hamilton and Hamilton wrote around fifty two papers so I will write those fifty two numbers which are present there one, six, seven, eight, nine so I have return the list for Hamilton after that after Hamilton comes the Jay so I will create the key and value payer so jay and the values will be jay wrote four papers so two, three, four and five and after jay we have shared papers of Madison and Hamilton so I will just write shared and the values and the list of values are value list are eighteen, nineteen and twenty so three papers they wrote so collaboratively and there are disputed papers which we, we are not sure of who exactly wrote this papers so I will just write this disputed sorry disputed and the value will be a list of some papers which are forty nine, fifty, fifty one, fifty two, fifty three, fifty four, fifty five, fifty six, fifty seven, fifty eight, sixty two and sixty three so these are the papers which are disputed and we really don't know who exactly wrote so we are doing to analyse for these papers exactly will see that which who who can write this paper by analysing the word length graph of all the authors which you have stated before the disputed papers. Ok so our data is the dictionary is created now I will create I will create a definition of reading files so what the idea is, let me write the definition name read files ok this will taken take one argument name as file name and what we are going to do, we create another dictionary and this dictionary what it will contain. It will contain the key value as the authors name and the key will be the authors name the value will be whatever he wrote will keep all the data in a particular list as a string so the key will be the authors name and the value will be all the data, whatever he wrote so all the data as a single string so let me create a list of name string. So strings as list ok I will create my loop as for file and file name, the file name is my argument. With open I will open my file as so my file name is federal list and with an underscore now will have the curly braces so that I can put the file name so what, what exactly is this? It is the key value which the corresponding value for each key which for each author. So I will put it here and dot txt so I will created the string for the file name so that I can open the exactly file exact file open of particular author. I will open it as f and I will append string which the strings are the list which I created the name of the list is strings dot append and what I am going to append? The file which is which I opened as f, I will read this f dot read and whatever it is reading I will just append it into this string list ok and I will just I will just join it and return it so I will just and what I am going to return is slash n dot join the string that's it. So what I am going to do with this function, each time I will for each author I will send the file name file name is the list of the which contain the numbers and through that these numbers I will read the file which is present in the data folder for the corresponding author and I will get all the data for particular author, read the string and put it in a single list. Ok what I am going to do with the single list will see in the next video. Thank you.

# <u>NATURAL LANGUAGE PROCESSING: AUTHOR STYLOMETRY 09</u>

So, you can see here I have downloaded all the collections for nltk you can see its written whatever I needed to download everything has been installed here and will go to our program let me just zoom it we can see let me ok  yeah everything is visible now I define what my function is exactly and what this paper dictionary is doing, I will just save it just run it again so everything looks perfect by running it I have created the papers dictionary so by if I just type in my let me just zoom this one ipython console here you can see the ipython console. If I type papers you can see the dictionary Madison contains a list of all his papers Hamilton contains list of all his fifty two papers jay contains four papers, shared are three and these are twelve disputed papers so our dictionary is ok let me clear it ok so next what you have to do, you have to call this function so that we create the actual dictionary that we wanted to create our main dictionary is dictionary key value will be the authors name and the list I mean the value the key will be the authors name and the value will be the string of words I mean all the sentence all the words in a single string so that we can analyse it easily. So that what we are going to do we will create a dictionary out of authors so let me write here let me give the name of the dictionary here is federal list by author ok so this is the dictionary and let me write the loop for author comma files in papers dot items ok so let me just try this once what exactly it does so let me let's see the let see my ipython console. I will just try this console for I comma j and I already created this papers dictionary so papers dot I items ok I will just print I and k ok you can see it is printing all the values Madison and in his list, Hamilton and his list and jay and his list shared all the list, the list disputed and the list so its work perfectly I wanted this only so I will go to the next line what I am going to do so authors will give me so what I is? I will give me the name of the author and j is giving me the list so author here the variable author will give me the name of the author and files, files here will give me the list so files contain list author contains authors name so what I am going to do the dictionary which the entire dictionary which is just created I will use this federal list by author and key is what? Key is the authors name here it is author, key is author so I will create a author through which I am looking through and what will be the value, value is the list. So what I will do? What I am going to do? I am going to call this function now read files and I will I am going to call this function by giving the list as a parameter so the file name will be my will be the list so files what it will do? The list will go here ok then all the numbers, numbers in this contain numbers so for all the numbers it will read all the files for that particular author and then contacting it all the words strings whatever it is into a single list. Ok and that list I for a particular author I will use it to analyse. What I am what my idea is? I will use that list I will count all the words of a particular length and I will plot the distribution that's it ok so you will now you know what I am exactly doing why I am calling this function here, so this will give me the list of all the words for a particular author so my dictionary will contain everything here ok I am getting some errors so what is the problem is I put here file name rather than file ok lets run this cool its working so I will just try to print this so that everything looks perfect so for author in papers print this my federal list by author the author author is my variable here I will just print something some sliced hundred character so that lets see everything is working or not?  I am getting something it means the dictionary is perfect. I will just delete this. Ok my dictionary has been created, now the only thing I have to do is on this dictionary I have the data I have created my I have done my pre processing I will do the analysis I will count the word and their length and I will plot the distribution and I will see it whether there is difference or not. Let’s see this in next video. Thank you.

# NATUTRAL LANGUAGE PROCESSING: AUTHOR STYLOMETRY 10

So, everything is ready we have created the dictionary federal list by authors which contain all the data where key s the authors name and the value is the authors data. Now what we have to do, we have to create words distribution word length distribution exactly so will count the word with the particular length will count them I will create a graph out of it. So let me create a tuple name author authors and the first one is Hamilton second is Madison third is disputed then we have jay and the shared documents ok my tuple is there for authors, I will create two dictionaries author tokens ok let me explain this dictionaries let me create first and length distribution so what I am going to do I go through all these authors and whatever dictionary I created in this dictionary the key value is the authors name even see all this authors name are the key values here I will take the I will take the strings the data contains everything and I will remove all the punctuated things other letters only the words will be there I will take only the words because what we want is the word and their length so first I will do I will create a loop for author in authors is my authors list yeah  tokens is equal to I will create a token tokens is equal to nltk dot there is a method for it word tokenize so what it does? It takes a string and create tokens out of it I mean each word will be each string each token tokenize so everything for example punctuated letters words  everything will be tokenized, tokenize what I will take the data, so data is present in this dictionary federal list by author and I am taking for each author this import nltk ok so this is done lets will after will filter out the punctuations for that there is an nltk library for it so I will take the author tokens ok so the dictionary I created ok and the key will be key value will be the author and now here is the tricky part I too remove all the punctuations here so I will do it in a very small code so let me let's see this first for token token in tokens you know what tokens is here ok if any here it is c dot alpha this is a function for c in token so what I am doing for each token in tokens I am checking if it is and alpha alphabet it is made of all the characters only or not any other special characters or numbers something if it is only alphabets then I am I will create a list of it otherwise I will discard it. So for token for token in tokens put the token there so put that tokens in this list that's it so this line is nothing else you can do it in other way also I just wanted to put it in a single line so for token and tokens if that it's an alpha correct for c in token whatever token I am getting here is a string so for c in token means all the words in this string if it is a made up of only alphabets I will put this token inside the list otherwise I will discard it so then I got the words out of the for each authors now I just have to calculate the length of it so token length which I created token lengths ok token lengths will be again let me just write it again for token in author tokens of what authors, for token in author tokens what token put what length of it. So I got the I am iterating over all the words which I got for each authors and I am just getting the list of lengths that's it great I got the list after that I want to get the length distribution of author so I will create a list out of it length distributions authors ok and I call nltk dot frequency distribution this is the function name and I will pass the token lengths and I will just plot length distributions of author let just plot it so maximum length I will put as fifteen up t fifteen words the length of the fifteen words sorry title is equal to author, author means what even it will plot it will show for which author this plot is so this dictionary is length distribution yeah that's right it was throwing the warning error now its fine let's let me just make it little bit big and just run it ok I am getting some error str object has not attribute alpha sorry just should be is alpha because, I am checking whether it is an alphabet or not. Ok let me sorry let me run it again ok you can see the plots here you can see there are plots for Hamilton Madison disputed and jay and shared so there are exactly five plots, there are three authors Hamilton, Madison and jay the shared plot is for documents which were shared by Madison and Hamilton and disputed are those which we don't know. Now see this see the plot it looks exactly almost same but it's not exactly if you see the plot of jay the samples are the word count I mean they are there are two words two length words there are twelve hundred words around twelve hundred words around of length two and so on it goes distribution goes like this for jay it's definitely not what this disputed plot is it means that whatever the documents are there in the disputed are definitely not written by jay you can see there the beginning and the tale is definitely different for jay and disputed but if you see and also if you see the shared one its look similar to Hamilton more as compare to the Madison one but if you see the disputed one you cannot you cannot say who exactly wrote the disputed one either it is Madison or Hamilton but you can definitely say that jay didn't write you can see the plots are different the beginning part is little bit similar to the Hamilton one you can see the beginning of this, this plot of disputed but the tale is more or less similar to Madison or even Hamilton so we cannot even generalise with this but this is a very fun way of seeing what exactly is going on or is it any way related with this at least we were able to say that this jay definitely not didn't write these disputed. This analysis was given when it was given it was it didn't account for the authors way of writing things, how they choose words what kind of words they choose they are different there are so many research on this and still not idea so people are still working on this stylometry and now they now there are so many tools through which you can analyse the authors signature and you can predict or you categorised authors documents by analysing their styles. So it has, so this what the program we did was just to see just to motivate you this idea of stylometry you can exactly actually given a document you can do some analysis and get the authors name if you if you have previous data of it. So this is exactly known as data analysis, now a days we have a lot of data you analyse the data and you whenever you get a new data you try to infer from the past data whether this new data is something or not this was the very very main example of doing it because this is the only this was the quickest and simple way of doing it. If you go little further it will become very complex so since this is the very basic course we dint want you to go through the complex one so we did this basic analysis to understand the data at least and we can see the plots and we can at least infer something through this. There are so many methods of doing stylometry, you can go to the web and you can just type stylometry and you will get a lot of lot of way please see this nltk library this is this is a very good library for analysing text, documents actually it is known as nlpk analysis so please see this libraries if you have any problem post it on discussion form. Thank you.

# INTRODUCTION TO NETWORKX 01

Hello everyone, in this video we will see the network analysis part using the python library known as networkx so our next joy is six degree of separation it is basically analysing the social network. Before we dive into the world of network I would like you to see this cool library of python known as networkx which is used to analyse visualise networks and even generate them so will see a very basic example of a network, how to create them? How to see them and how to visualise them? In this video I will use network to create simple graphs and then visualise them so let us see this. So let me show you a graph so suppose you can see in my screen that there is a graph of three nodes, nodes these circles are known as nodes and these lines are the edges between them all you can say interactions between them. So for example if you see these nodes as people an edges is friendship among them then see it is a social network kind of thing so for example person one is a friend with person two, person two is a friend with person three, person three is a friend with a person one so this is a you can visualise it as a social interaction graph where people are node and interactions are edges between them so you can see in this graph for example there are three nodes and three edges. If some of you know little bit of graph thing then you must know that this is a complete graph between three nodes if you create a complete graph of three nodes you will have three nodes and three edges. What is a complete graph? Complete graph is nothing where when you have an n nodes if any two pair if you pick there should be an edge between them so I will start my networkx program by creating this graph this three node graph so let's see this so I will import my library networkx library using the command import networkx as nx will be my alias here ok I have imported my networkx now I want to create a graph. To creating a graph the command is let's say my graph name will be g and nx dot graph that's it you can see here dropdown list its already showing the graph ok so this will create a graph now what I have to do my graph is empty, it is created but it doesn't have anything it doesn't have any node it doesn't have any edges so what I will do, I will add some node to add an node in a graph there is an command which is known as g, g is my graph g dot add you can see it will come add node my dropdown list so g dot add node is my command now I will add my node number that's it I will just add the name of my node here it is one so I will add one so it means that after this command one as a node has been added to my graph, graph is g so let me see my graph one has been added now I will add two same thing g dot add node in second node two so, node two has been added node one and node two has been added, now I will add node three so g dot add node and three awesome so all the nodes have been added what else now my graph contains three nodes one two and three but there are no interactions, I mean there are no edges between them I have to define I have to tell the networkx to add the edges between the nodes which I want so I want edge between one and two, two and three and one and three. How to do it? Let's see this. So the command is very simple, like adding node I can add edges so g dot you can predict what I am going to write add edge ok and what is my edge? It should be between you can see its written their arguments ad edge u of edge v of edge it means that from where it is starting, from where it is ending. See it is a undirected edge undirected it means that there is no direction from u to v, you can consider v as u, u as v it doesn't matter so I will add from one to two one comma two ok one node has been one edge has been added sorry second is two to three I will add g dot add edge two comma three ok and my last edge is one comma three so I will add g dot add edge one comma three ok so according to me my graph has been created I don't have anything to see here so what I will do I will just print my graph so how to print? I will just add print not graph my edges sorry nodes g dot nodes that's it lets see what happens I will just run it you can see my nodes are one to three it has been shown here ok you can see the nodes are one comma two comma three ok great but still I ma not able to see my let me try to print the edges ok let's see what it does ok my edges also being printed this is edges from one to two, one to three, two to three see this is a very small graph so I can I can using this data I can see whether what kind of graph I have bit suppose the graph contains hundreds of nodes then some two hundred edges then what will you do, you cannot just by watching the data you can see what kind of graph it is and you can visualise it so what can you do networkx provides a visualisation of graph using the mat plot lib library so that is too also very simple I will just import the mat dot lib library so import mat plot lib dot pyplot as my alias plt ok I imported mat plot lib library now this graph which I have created the g I will draw it as nx dot it will show and plt dot show that's it let me just clear my clear my ipython console here ok I will just run my program to see what it does. Ok it is saying that draw is missing the required positional argument g, I didn't draw what I am I didn't say what I am going to draw g is my I need to draw g so I will put g here ok I don't like this fifty ipython console so I will clear it ok I will run it again awesome you can see my graph this is exactly what I wanted to make ok this is my graph of three nodes and the interactions are ok great ok now suppose I told you that this is a graph of three nodes and this is a complete graph of three nodes what if I want to create the graph a complete graph of ten nodes maybe twenty nodes I will have to add this nodes and edges again when there is a way which you know the first thing that will come in your mind is creating a lop through loop you can add off course you can do it. There is an another way also one way is to create a list of nodes, let me show you how, rather than adding this nodes like this whatever I can do I will create a list let's say l and I put the nodes here that's one comma two comma three ok I don't need all these thing I will just delete and what I will do rather than adding just command g dot add node there is one more command which is g dot add nodes underscore from you can see from is written and you can give the nodes for adding it can be any kind of container which can be a list so I will add l here ok to I added same nodes one two three so that I can check whether it is showing me the same kind of thing or not I will just run this program awesome I am getting the same output so you can do something like that. Now there is one more beautiful thing with the networkx it comes with all kinds of graphs pre defined graphs, suppose I want to create a complete graph of let's say what I said before that I want to create a complete graph of twenty nodes I don't need all these things we just delete all this things and I will just add nx dot complete you can see complete graph is there, complete graph of how many nodes I want, so let add ten nodes ok and save it as g let us just delete this awesome so g will be my graph and I create nx dot complete graph of ten nodes one two three four five six seven eight nine ten there are ten nodes and it is a complete graph you can see you can change the number of nodes let's say I want twenty nodes so I will just run this again save it and run it you see complete graph of twenty nodes great so there is a lot of things which you can do with networkx this is just I wanted to show how can visualise simple simple graph so suppose I want to create a there are so many layouts for networkx you can see you can just type networkx generators, generators are the graph generators there are so many ways through which you can create awesome graphs so suppose I want to create a random graph, what is a random graph exactly? So suppose there are hundred nodes in a graph what I will do, I will pick two nodes and then I will put an edge between these two nodes with some probability so what I will do I will take every two nodes and I will add an edge with probability let's say zero point five and I will do with this I will do this step with all pair of nodes so with this algorithm I will create a graph, this graph is known as random graph. So what is the speciality of this random graph? Is that in such kind of graph if you pick any node you will see that the degree of that node will be the average degree will be near to the average degree of the graph so you can learn about this graph this is known as adosh graph or random graph you can just search on goggle and read about this so I can create this graph through networkx the command is gnp underscore random underscore graph so I can just type here so I will just type here gnp underscore random underscore graph and it will be random these are the number of nodes as you can see here gnp random graph n is the number of node, p is the probability but which I am putting an edge between an two vertices or nodes so I can put here let say twenty graph and let's say I was telling you there zero point five is the probability ok so let's see what happens. I will run it again you can see this is random graph so you can see all the nodes have almost equal kind of degrees or you can say it is if you pick any node here the degree is almost equal to the average degree of the whole graph so I can change the probability here I can increase it I can decrease it so I can make it zero point two and twenty nodes you can see the graph here there is one isolated node which is not this graph is not connected means you cannot reach to any node by traversing the edges with networkx you can do lot of thing just go to this networkx page and you can see a lot of things here I just wanted to make sure that you understand the visualisation part and little very little very basic network which we are going to use in our social network analysis or you can say network analysis part which is our next joy so in next video we will see some more of this we are going to use a very good visualisation tool which is known as gaphi and will see how we can visualise a graph using that tool and how you can generate beautiful graphs. Thank you.

# INTRODUCTION TO NETWORKX 02

Hello guys, in last video we saw small example of networkx how we can create smaller graphs and how we can visualise them using our mat plot lib library. So in this video we will see some more of networkx and then will see how to generate graphs in a new format as a file which is known as gexf format and then we will use a new tool know as gephi two to see and visualise the graph. There are so many things which you can do with the gephi tool so let's see this. So again I will, let me clear my ipython console ok let me clear let me make a basic graph random graph using gexf format sorry networkx so I will write my command to create a random graph nx dot gnp random graph its coming here and I will just put my numbers of nodes and the probability with which I am putting edge between two nodes so let me put let me put some more number of nodes so I can see something, let me put number of nodes as fifty and probability be zero point three ok great. Let me just see this graph once so mat dot lib I have imported ok so I will put nx dot draw my sorry I will I will save this graph as G I will save it ok g is equal to values and I will draw this graph. After that I will show this graph great let me see this let me run this ok you can see its very dense because number of nodes are more and the probability is ok point thirty percent with thirty percent probability and adding at just it not a complete graph it looks like a complete graph but it is a random graph ok let me do some, some more things with this graph so let me come to my let me go to my ipython console ok let me make it side by side so that it is visible ok what I am going to do? I am going to just print the number of nodes here sorry not the number of nodes the nodes so this is g dot nodes ok these are my nodes zero one two three four five six seven eight nine two eleven twelve up to forty nine its starts with zero ok let me if you it's coming as a container so let me print it print g dot nodes ok now it is a list so these are the this is the list of all nodes which are present tin this graph same way I will print the edges so print g dot edges ok it is printed the edge list it means which node is connected to which node for example you can see here zero is connected to three, zero is connected to five, zero is connected to eight, zero is connected to ten, zero is connected to so on and something like that so now I can do a lot of things here you can see the edge list here suppose I want to know the degree of each node so I can just print it here. Else print g dot degree of let's say zero show me thirteen zero has thirteen degree. So something like that I can do with this graph I can analyse a lot of things ok you can see you can go to the documentation I can see a lot of things here there are lot of functions lot of algorithms lot of graph type just explore it. What I am going to do, I will generate one more different type of graph here which is known as scale free graphs which was given by barabasi I am going to generate it. You do not have to know what exactly it is I am just I just wanted a new type of graph which is different from a random graph that's why I am generating it so that I can show you that difference. So I will just generate this let it be here so I will generate this scale free graph so it is nx dot the command is barabasi Albert graph barabasi underscore Albert graph ok it takes two arguments one is n number of nodes and second is m, m means each node is each node coming with two edges so how it works? In starting there are very few nodes for example there are two nodes and then at each iteration a new node comes and whenever the new node comes it comes with two edges and these two edges get connected to those node which has higher degree so the probability of getting connected to already existing node is higher for that node whose degree is very high. So this why this process this process happens and at the end the node which is having higher degree at the end will have more degree so and there will be having few nodes of having high degree and very high number of nodes having very low degree so this is known as scale free graph. So I will so I have put this random sorry again I will put this fifty here and m has two and each node is coming with two new edges ok and I will see what else is different from this mode ok I will just run this you can see its exact it's so deferent from this gnp I mean the random graph. Now why is it? You can see there are few nodes one two these nodes which are closer to the centre having high degree and these nodes which are at the periphery are having less some has low as one this is the connected graph you can see all nodes are connected but very few nodes are having high degree and very large nodes are having low degree so this is known as professional attachment new nodes when they come they prefer to attach to old nodes which are having higher degree. So it is known as the in real world examples most of the many nodes are not exactly random most of the networks are not exactly random they are actually skill free the people have claimed like that. So what are our aim is now, this graph what we have generated will see this graph in a new format will write this graph in a file and the format of that file will be gexf so to do that what I will do, ok so I want have to write this graph I created the graph so after that I will just write a statement to, to write a graph in gexf the command is nx dot write underscore gexf you can see it is a gexf is my format the first argument will be my graph which is g and the second will be the path or the file name if you are not giving the path the file will be saved in your current directory. I will just put the name as analysis one dot gexf that's it ok great I will run this again the network has been created and you can see the if I go to my directory you can see this file has been created. If I just open it let me just open it with my sublime text, you can see that this format is there to represent this graph some this is some xml format. If you if you try to see this you can see that they are nodes, nodes with some id and with some label and there are edges with some id and source from where to where it is. Ok and so this is how we this is some xml format so my graph has been created in gexf format why I am doing it? This is gexf format I am going to import in a new tool known as gephi, gephi is an application through which you can visualise graph easily and you can do a lot of different kind of analysis in a in one click so I have already downloaded gephi in my computer it's very easy just goggle gephi the first link gephi download the first link you get go there and just download it nothing else is required. Maybe java one point eight is one point seven or one point eight is required that you can download easily nothing else is there, just download it, I have downloaded it, if you require if you facing any problem with gephi just let us know or put it on discussion form will show you. So I am going to skip that download part of gephi I will just open gephi in my in my system and I will see this graph I will analyse this graph or visualise this graph so this is my gephi then your version point o point nine so I will just go to file and open my graph which I just created on through my networkx which is which is barabasi Albert graph. So this is my graph ok analysis one I will just open it showing how many number of nodes are there ninety number of edges are ninety six is asking me what kind of graph it is, it is a undirected graph I will just put ok we can see the graph is there I will just zoom it using my scrollbar ok I am on windows on Mac its things are different and even though also little bit things are different so you can see this is my graph ok now let me show you a very unique thing here, you can put any node and you can change the directions of the nodes very easily here and you can see how many to which node it is exactly getting attached to, this t is the level if I put this it will show me the node level so this is my node number one there are some plug-ins which I will show you what exactly it is but just let just don't worry about what I am doing ok, that's it I will just unlabeled it so I can see the nodes. Now what I want I told you that in this graph in this particular graph which is scale free there are few nodes which has higher degree and there are lot of nodes which are having lower degree so what I am going to do here is you can see this button follow cursor this is my button this showing the node colour and this is size so what I am going to do I will change the colour according to the degree how will I do it? I will go partition not ranking I will do the ranking choose the attribute as degree and I will change the colour here as let's say which colour is easily distinguishable let say this ok and I will apply you can see these are the dark nodes let me just zoom it the dark nodes are those which are having higher degree, you can see there are so many nodes which are light, now still it is not very much visible or comfortable so what I am going to do I will change the size of the nodes based on again degree, min size I will put let say ten here and max size I will put as twenty alright see what happens. See let me make it twenty and make it forty its clearly visible now see, you can see exactly very few nodes which are very big and a lot of nodes are smaller smaller smaller ok such kind of easy visualisation if you want to just analyse your network through eyes and see what exactly is going on and you can do such kind of thing on there are so many things which you can do. One more thing I will show here you can see follow my cursor and you can see there are so many layouts what is this? I will just put this and I will just run it you can see the nodes are re assigned in a circular way this way you can see a lot of things and you can analyse graph very easily I will just increase the size of nodes sorry edges it is visible, now see the right pane on the right side there are so many things which you can do for example I want to see the average degree I will run it, it will show me the degree distribution. If you see the degree distribution here see the points here you see that there are few nodes which are having high degree they is one node exactly two or one node sorry having eighteen degree there are so many nodes of degree two there are twenty four nodes of degree two so this is what scale free network it follows something like this or it is known as power law this is known as power law. You can read more about this what else you can do there you can run this network diameter and in one click and for undirected you in one click you will see the diameter here so the diameter is four. There are so many things which you can do you can run the page rank modularity so I am not doing it because we don't understand it properly I mean its very basic I wanted to show a very basic thing, how to create a graph in networkx and visualise it using your gexf format you can even using this data laboratory you can import or export this graph in a spread sheet or see comma separated value we have talked about this comma separated values in gps fun activity. You can import it comma separated value file and then visualise it you can then preview your network through this you can see a very beautiful network as been created here. Where the edges are curved you can save it save your network as pdf and you can do lot of things here so this I just wanted to give you a quick introduction to this very beautiful tool known as gephi g e p h I and you can just download it you can see in next in next joy which is network analysis actually six degree of separation you will see what exactly it is. Will see you can you see the importance of network in our life and I just wanted to motivate you with this visualisation tool how sometime it is very important to see things and before just getting to the coding and writing functions to analyse it just sometime you need to watch things closely and then see what exactly is going on, so you can see this is extremely good and do a lot of things with this tool so please if you have any problems if you are facing any problem with this with that gephi download please post it on form we can put a quick video to download gephi, how to download gephi, how to install gephi. So thank you.

## SIX DEGREES OF SEPERATION: MEET YOUR FAVOURITES

Have you ever tried reaching your favourite celebrities! Well in today's world it is very much possible. Let us find out in our next joy six degrees of separation.

# SIX DEGREES OF SEPERATION: MEET YOUR FAVOURITES 01

Hey shubudha what's up? Hello sir I didn't see. So I see that you keep watching a whole lot of videos. Sir, I was feeling very sleepy so I thought I will watch one or two song and then I will get back to my work. That's fine but what peculiar is you keep watching ranbir kaapor's videos all the time. Yes, I am very fascinated by his work I like his acting and I really want to meet him given a chance but I don't know I will be able to meet him it's like, why not? Do you know that it's today not very difficult to go and catch hold of a person let say if you want to even meet the president of the United States. Do you know that you know someone knows someone who knows the president? Sir really I can meet ranbir kappor? Or, any other person who I want to meet. Ok does it sounds surprising if I tell you that you probably know ranbir, no through the common friend's common friend, which means ranbir friend's friend's friend is probably shubudha. It is you, will it happen? That's why so I did it is a very nice science that goes behind that so I will explain it to you. Its gonna take quite some time for me to explain the science behind it, I think I should not be disturbing people here why don't you walk outside, yeah sure. So shubudha you off course of know obama right? Yeah I know but I don't know him personally, that's right. So what if I tell you that you actually know obama through someone through someone through someone is that believable? Of course not, how will I know obama through someone, sounds like it farfetched, yeah. So this is a fact. I know a good friend from Germany and her dad has worked with obama and he knows obama really well ok and obama also knows him very well, so what just happened? I know this friend from Germany and off course she knows her dad really well. I know her very well she knows her dad very well and her dad knows obama very well and you know me very well, so now you see you know me one, I know this friend from Germany two and friend off course know her dad very well three and the dad knows obama very well four, through four people you are connected to obama. That's surprising yeah what is more surprising is probably you think this is a co incidence, yeah because this might be a co incidence because what if this is not true for any two people in the world. Correct, do you observe something? You know obama through three people and every friend of yours; they know obama through four people, four people. You see they know you, you know me, I know yes and so on correct? So similarly it is believed rather it is known experimentally that any two people in the world are separated on an average of six people, any two people in fact. Really? But how can we prove it? Yeah! So my point was this you were talking about ranbir kapoor right? yeah I don't think it is very difficult for you to find someone who might know someone who knows ranbir kapoor really well, so it's all about knowing how to reach ranbir kapoor, the fact is you know someone who might know someone who knows ranbir kapoor really well, well but it's difficult to find, I am only talking about the fact that there exists a path between you and ranbir kapoor through a couple of friends. How to find that? It is a difficult question to answer but a very well known question in sociology is the fact that any two people are connected by on an average of let's say six hobs in fact within India any two people I believe should be connected in less than two or three hobs so it should be difficult for you to meet ranbir kapoor as they say where there is a will there is a way, if you attempt whole heartedly you might meet your dream star.

Maybe I can find that path which is difficult to reach ranbir kapoor that’s right so it’s going to be whole lot of hard work but the science of it is that you can always find such a path, I am sure that you still don’t believe it, yeah slightly yeah so I am not still convinced, very convinced so what I will do is now take you to the desktop and show you by writing the program, I will assume the population of the world is so much and I will show you if this is really true, if we assume some random friendships that any two people are actually connected with very few hobs.

# SIX DEGREES OF SEPERATION: MEET YOUR FAVOURITES 02

Hello guys, welcome to programming screen cast of  'six degrees of separation' so before going to the actual programming screen cast let us see some few pre requisites that is required. The first pre requisite that is required is you must know what is the graph data structure? By graph I don't mean this kind of a graph which you would have been doing in your school physics, chemistry something like that, you take some quantity on your x axis and you take some quantity on your y axis and you observe as you vary this quantity what is the variation in the other quantity, so such thing is what you call as a graph in till your school days, this is not what we are going to say as a graph, we are going to say something like this a set of few points and lines as graph so basically graph is nothing but group of some points and lines that lie between the points that is what we call as a graph and technically we call the points as nodes and lines as edges this is the technical terminology so to illustrate better let me take an example, let us consider the railway network of the city of the country railway network of the country so the nodes or the points let us get used to the terminologies nodes and edges, the nodes in this network is nothing but the cities or the stations and the edges are laid if there is a direct train between the two cities at least one direct train must be there only then we will draw an edge this is how we will construct this railway network. So between any two cities if you take there may be an edge or they may not be an edge depends on there is a direct train or not so for example see two cities Chennai and bengaluru there is an edge between these two cities because there is some direct train between these two cities where as if you see these two cities Chennai and Guwahati or guwahati and Bengaluru there is no edge because there were no direct trains and see this graph is actually dynamic in real scenario because trains maybe introduced every now and then as well as few trains maybe not operated because of various reasons so this graph keeps on changing that I what I been made and dynamic edges maybe newly added or deleted anything may happen so this is just earn example that there existed a direct train then you will put an edge if there is no direct train you don't put an edge that is how it is and if there is an edge you can say that I can reach this node to this node that is from Chennai I can reach bengaluru and from bengaluru I can reach Chennai that is what is what is what is captured if there is an edge but it doesn't mean that if there is no edge we cannot reach this at all so let us see, see there is no edge between Chennai and guwahati but still you can have another city in between Kolkata and you can reach guwahati through Kolkata that is you take a train from Chennai to Kolkata and from Kolkata you can take a train to guwahati this is what you call as a path so path is basically the way which you can reach from one node to another that is what we technically call as a path so there is no edge between Chennai and guwahati but there exist a path so it is not the case that if there is no edge there is no path there may be a path even if there is no direct rain see from Chennai you cannot directly reach guwahati but via Kolkata you can reach guwahati that is the meaning and it is not all the case that there will be just one path there may be multiple paths well as you could see here so if there is a edge you can say that you can definitely reach from this city to this city and if there is no edge that is not the case that there is no way to reach there may be a way to reach even if there is no edge between the two cities that is you have some cities as an intermediate destination intermediate point through which you can reach your decided destination that is from Chennai to Bangalore there is a direct edge you can directly reach from Chennai to guwahati there is no edge but still there is a path via Kolkata to reach guwahati from Chennai as well as from guwahati to Chennai as well the same path can be followed and as you would see most of the railway network is by direct functional that is you have trains from both the directions, if there is a train from Chennai to bengaluru there is obviously a train from bengaluru to Chennai that's how mostly it works that's why it is undirected graph which we call it as undirected that is directions doesn't matter from Chennai to bengaluru or bengaluru to Chennai both path exist so this is what we call as an undirected graph and we would always be interested in finding the shortest path there may be multiple paths that may be present let me show you, see this graph there may be multiple paths that may be present from Chennai to guwahati see one path is Chennai to Kolkata, Kolkata to guwahati other is Chennai to bengaluru, bengaluru to Kolkata, Kolkata to guwahati there are two paths in this smaller example where may be multiple paths may be present in that case you have multiple options we always go with the shortest path by shortest I mean minimal number of switches of trains in this example there maybe some cases where we have something called weighted graph that is the edges will be having weights so basically you can think of it has a distance between two cities or the time the train takes to travel you can consider anything as your weight so that would be captured here as weight so if for example Chennai to bengaluru let us say the distance is four hundred kilometres this edge would have a label four hundred so something like that the distances would be here as a weight and we would like to find the one the path where the total weights that is the sum of individual weights is minimal that is if you some of all the weights Chennai to bengaluru, bengaluru to Kolkata, Kolkata to guwahati as well as Chennai to Kolkata, Kolkata to guwahati you will find the sum and wherever the answer is minimal you will take that path that is what you call as a shortest path in a weighted graph but here we don't consider weights I just say the number of trains I need to switch that in a undirected case that is what I am considering here in this case the shortest path is from Chennai to Kolkata, Kolkata to guwahati just I need to take two trains but if here I take this path then I have to take three trains Chennai to bengaluru, bengaluru to Kolkata, Kolkata to guwahati so I will have to take three trains so this is not a shorter path in this scenario the shortest path will be generally preferred and whether there will be a path always between any two pair of nodes that is a next question that is of our interest and in some cases there may exist a path in any pair of nodes you take in some cases it may not exist. When in case if a graph is such that between any two nodes you take in the graph there always exist at least one path then you call it as a connected graph see if at least one path exists between any two nodes then you call that graph as connected graph. So out railway network is almost the connected graph because from any city in India you can reach to any other city although you may not have direct trains you definitely have a path that is mostly as much as I know the capital cities are mostly well connected so you can reach from any place to the capital city of the state in from that city you can reach to any city of your of your interest so I fell the railway network of India is connected that's a connected graph so this is now just one place where graphs are used, graphs can be used in many places even people may involved in the graphs that is one such graph were people are involved is something which we all regularly use I guess most of all used like daily routine facebook so facebook network is a network which involves people so as I said the nodes are people an edge is put if the two people are friends with each other in that case you will put the edge so this is how the facebook network is constructed this is how the facebook analyses the data, the facebook would store the details of your friendships and analyse it will give you recommendations right, you may know this person, you may want to send a friend request to this person such recommendations all these analysis are done for that data has to be stored so the for storing this kind of data oh who is friends with whom facebook uses the graph data structure that I sit denoted people by nodes and edges are put if the people are friends with each other and you can download the facebook data set from this particular link I had already downloaded so I will not be downloading again so the thing that you have to download is there is a text file called facebook combined so basically they have a different versions I guess the discretion is also available in the link you can read through it the thing is they had collected data at various time intervals the various geographical location and they had combine everything into a single text file that is what we have it as facebook combined dot txt just download that particular thing and one more thing that I want to tell is graph is something that I had shown you if you would see this is a pictorial representation of a graph we have nodes and edges this diagrammatical representation is easier on our minds but how do we store this in a computer? We cannot feed in this figure and ask the computer to understand this, we need to store it in some specialised format, there are some formats by which you can store the graph data so there are many formats actually but I will be discussing just one format that would be used in our facebook data that is how we get our face book data and I am just going to discuss that format. Let us see, see the format is called edge list representation so just consider a sample graph there are nodes one two three four and this is how the edges are present there is an edge between one to two, there is an edge between two and three, there is an edge between three and four. So you just I had said one two two three three four one space two, two space three, three space four this is how you store it so this is what you call as an edge list. Edge list is one of the representation for graph, there are many other representations you can look it up as well as what are the other representations there are some advantages and disadvantages associated with each representations you can take a look at it as an advanced material in graphs.

# SIX DEGREES OF SEPERATION: MEET YOUR FAVOURITES 03

So, as I had said I have downloaded the facebook data see edge list format as I had explained as been used here there is an edge between person zero and person one person zero and person two and so on see the persons have been anonymous that is if you don't know the name and other details about the person they have been associated some dummy id and that id is used to identify the persons that is the way you identify the nodes in the network so it's an anonymous data you can download from the link that I had shown in the power point and it will be available in the description as well you can check out that and you can download this data. Now given that we have seen our data let us get to programming alright to deal with graphs we have package in python called networkx let me import it import network and given that the name is linked here let me use a shorter alias name as n xi will call nx so this is how the package that you used to analyse the graphs so the as I had shown the data was in edge list format so I had to read that data so for that the functionality is nx dot read underscore edge list see the names are also intitutive you can understand what it does and see you are getting the arguments you have to give the path of the file in my case the file is in the same path as that of my programme so I am just giving the name of the file, if your file is in different path you need to specify the complete path. So the name of the file is facebook underscore combine dot txt that is the name of the file alright may be I will maximise this pane so that its easier for you so I have maximised this pane so you have read the edge list so as a result of reading the edge list a graph object would be created that is sort of graph structure would be created and we need to capture the created structure so we will capture it by the variable G, G for graph basically so will capture it this way alright so what I am going to do is, I am going to compare the shortest path length by length I mean the number of edges you have to pass in a if you take the shortest path, that is in our example from Chennai to Kolkata, Kolkata to guwahati was the shortest path that is there we are two edges Chennai to Kolkata and Kolkata to guwahati so that two is the length of the shortest path is what I say. So like that I am going to find the length of each pair of nodes the shortest path that is the shortest path of the each pair of nodes, is what I am going to find. So for that I need to take the list of all the nodes right? let me capture that in a variable N, N for nodes basically let me say is equal to list of the nodes is what I want so let me type cast to the list and I will append the nodes of the graphs by telling G dot nodes this is how you get the nodes of the graphs and you are converting it into a list format so you are getting the list of the nodes, so that is what you are doing as well as the intention that I am having here is I am going to find the length of the shortest path between each pair of nodes each pair of nodes possible so between node number zero and one, zero and two, zero and three, one and two, one and three, one and four and so on so on each pair of nodes possible, I am going to find length of the shortest path and I am going to observe something so, let me do that I will for that I need to store the length of the shortest path in some list to observe something so let me store it, let me say shortest path list shortest path list so shortest path length list let me say I need a list and I am having an empty list here and now as I said I am going to check for each pair of nodes right so I need to run a loop and iterate over all these possible nodes so, I will run a loop for u, u is a node let us say for u in nodes we have capture the list of nodes in the variable N u in N as well as we need tyres like this u would capture one of the node we need to capture another node so that we find the length of the shortest path between that tyre so we will find another will take another node for v in N so what I am going to do is I am going to find the length of the shortest path but it should not be found for all the nodes because if both the nodes are the same then it doesn't make sense right I want to go from Kolkata to Kolkata it doesn't make sense right so these two nodes must be different so only if they are different you should go for finding the shortest path length so I am going to give the check here u s not equal to v that is the nodes are different in that case you have to find the length of the shortest path let me call it l for length so for that we have a functionality in networkx nx dot shortest path length see you have a functionality here shortest path length G source target this is the format so you have to pass the graph object G and your source node is u target node is v between these two nodes you want to find the length of the shortest path and let us print it print let us say shortest path between u and v is of length l we have captured the length on variable l so will display that l here as well as I had said I am going to append each of the shortest path calculator onto this particular list shortest path length list so I will append that shortest path length list dot append the value of l so till now what I have done? I have read the edge list format graph data and I have captured it into a graph object first I had imported the network nx library then I had used it to read the edge list format data and I have imported it into a graph object then I capture the list of all the nodes from the edge list that is the edges where given from that we have to capture the list of all the nodes that we had done and we are planning to find the shortest path length between all possible pair of nodes and do some analysis on it. That is our aim for that will have list, shortest path length list which will store the list of all the shortest path lengths, all possible pair's shortest path length it will be stored here and so we need to find the length of all possible pair shortest path so that for that we are doing we are iterating over the list of nodes we take a node u we take another node v if they both are different we go for computing the shortest path length and we print the length of the shortest path and we append it to our list, so by the end of this loop all pair shortest path is calculated and you have that stored in this particular list. So now from this I want do find the minimum value that is the least value of the shortest path between the two pair of nodes that I can say I can say the minimum value of the shortest path let me say minimum shortest path length is nothing but I want to find the minimum value from this list, this list will contain the lengths of all pairs shortest path so I want to find the minimum value so let me find the minimum shortest path length list from this list I want to find the minimum value just like minimum I want to find the maximum let me copy and paste it and modify I will copy I will paste this I will say maximum so for this I have to use the functionality max so min is the functionality that will return the minimum element in a list max is a functionality that will return the maximum element in the list so maximum of the all the shortest path length would be returned here and the minimum value would be returned here so given that we have found the minimum and maximum what next mathematically? Average right? you want to find the average shortest path length so to find the average we need package called numpy let us import it import numpy this particular package has most of the mathematical operators that we generally use average operator is also defined in this particular package so I have imported numpy and let me find the average so average shortest path length is nothing but in numpy package you have average function see you could get it here average and from this list shortest path length list from this list you want to find the average alright you have found it now the next job is you want to print all these values so let us print minimum shortest path length ok that is stored in the variable minimum shortest path length alright just like this we want to print the maximum as well as average that whatever we have computed we want to print the values so let me copy, copy and paste here and let me make the modification, minimum and maximum and next we want is maximum and that particular thing is stored in the variable max spl so we will have it here and the next thing is we want to find the average. Average shortest path length average shortest path length alright so we have done we have done with our program so maybe we will just I will save it and we will have a quick overview to analyse the graph data set we need networkx package we have imported it and see we are reading it from the edge list format we have downloaded we read the edge list format and we captured the graph object and from that graph we are extracting the list of all the nodes and then we want to find the shortest path length between each possible pair of nodes in the graph. So that is our intension for that to store all the lengths and analyse later we are having a list which stores all possible shortest path length. So to find shortest path length between all possible pair of the nodes we have loops two for loops running you pick a node u, you pick another node v and if they are different you find their shortest path print it and append it to the list that is what you are doing so at the end of this loops what happens is between every possible pair of nodes whatever the length of the shortest path it would be stored in this particular list. So now we are analysing this list, we are finding the minimum of all possible values that is all the values of all shortest path lengths. We are finding the minimum then maximum then the average to find average we need library called numpy so we have imported numpy in numpy the functionality of average has been defined and we are importing it. So given that we have computed all these values we will be printing it that is what we are doing and by six degrees of separations what we mean is the average shortest path length on an average you can reach from anyone to anyone on the earth if you take the shortest path by maximum six that is the average shortest path length would be maximum six this is what we mean by six degrees of separations, so let me give you an intuition for why this is true. So just assume that you have hundred friends in your facebook so we generally tend to make friends with people who are like us right so those friends may also have further hundred friends so the next level if you consider your friends friends that is hundred into hundred and the next level hundred into hundred into hundred see if you go just one level below the number of people you met is increasing exponentially so if you trace just sic levels you can definitely cover the entire population of the world. This is the intuition behind six degrees of separation concept ok I hope you are clear with the concept, you are clear with the terminologies of graph data structure how to deal with it, how to code it, I hope you are clear with it. Maybe you can pause here think for a while and you can then proceed with running the code and this is not just only functionality available here actually there are a lot of functionalities available here maybe let me show you. Nx dot I will just press a tab sorry I am sorry nx dot nx I will show you the other functionalities nx dot see a lot of functionalities available here and you can go to the console as you know by typing in the functionality and giving the question mark at the end you would get the documentation of what that functionality is all about. If you have some idea of graph theory the math behind it, if you have some idea you can understand it really well but still even if you don't have it it's never a problem all this is nothing but the intuitive thing translated into hardcore mathematics that's it its really very easy you can pick up that so you can I would recommend that you guys explore this networkx library really well you understand the various functionalities, you see that what all can be done you will realise that it has whole lot of things which is otherwise difficult if this is not present. If you are asked to do this all pairs shortest path to find the shortest path there is an algorithm that runs for some the mathematics behind it everything it runs for some pages in some standard text books if you see. so we have this particular functionalities that finds the shortest paths that makes our life easier if we have to implement this from the scratch it would take a lot of time. So networkx really simplifies a tasks of dealing with graph so you can analyse the parameters of the graph without having to worry about the minute details that is advantage of using networkx. Alright please do explore it ok let me save and run it so before that let me restore the pane alright so here we have the output will be getting the output in the console so let me run the file. See the shortest path is being computed between the different pair of nodes. Are you getting somewhere more than six the average will convert to six it will take a lot of time because see there are so many nodes see three thousand nine hundred some nodes are there so there are around four thousand nodes I guess, you can check out that in the description of the data set as well, how many nodes and How many edges are present is given there, even that there are so many nodes present it would take some time to compute but you can see what is the minimum maximum average also do explore the other parameters and please do discuss in the discussion form of your explorations. Thanks for watching this video, have a nice day.

# <u>AREA CALCULATION: DONT MEASURE</u>

Do you know the area of your home state? Can you calculate it? Well! You can. Let us find out in our next joy area calculation.

# AREA CALCULATION: DON'T MEASURE 01

Sir, I must say that I am a proud Punjabi, you know why? Punjab has become so popular despite being such a small state. See that's right it is a highlight of our country for a simple reason that you take any actor that matter in Bollywood, you take any outdoor sports you will see good number of Punjab's ok you go on enumerating good actors of our country good number of them will be from Punjab, I have been observing from this from decades look at the hockey team of India, cricket team of india Punjab is dominant, not dominant but there are good number of Punjabis in cricket team and hockey team and you name it. Right? I also wonder the same Punjab being such a small place it is generated a lot of celebrities so far right? So what was you are wondering given the size of Punjab how is it doing so well? Exactly I think it makes just less than one twentieths of India size. I doubt that so Punjab appears to be small it is not one twentieths even lesser than that a lot lesser than that, how do we even calculate this? OK that's a good question so given that we are talking about computing, let me ask you one question if I give you a following fact the size of india area of india is three million square kilometres roughly approximately ok and I will give you the map of India in that I will give you Punjab as well, ok is there any way you can compute the area of Punjab? Just by knowing these two facts fact number one, area of india is three million square kilometres and you are given a raw map of india with these two things can you compute? Sir I don't think so this doesn't make a difference, you think you think data insufficient that you probably can't even go close to computing what could be the area of Punjab right? So let me give you a fascinating idea in computer science called estimation, so in places where you cannot accurately compute something you can estimate rather approximate alright so let me tell you how exactly that is done. Let me use the board right now look at this. You see the you see a square here right and let me write down let me write a bigger one let me write a big square rectangle whatever anything is fine and it is easy for you to calculate the area of this. Right? It is very easy. Once you calculate the area of this don't you think it's easy to compute the area of this, why is that? What do you think is the area of this part simran? It's just the half of the, correct! It is pretty obvious; this will be half of the entire area of this rectangle correct? Yeah so far so good now what if I try to ask this question forget this region, what if I were to give you a random region given that you know what is the area of the rectangle can you compute the area of this region? How would you compute? This will be difficult to compute, ok for a for for someone who doesn't know the idea that I am gonna say, they might even think this is quite impossible question to solve but here is a very nice way of solving this problem. What you do is assume you go there simran OK you needn't go just I am asking you to imagine that you go that side and then take a you have played a dart game yeah yeah where you throw arrows at a target yes yes, you go there and imagine this to be a dart put a dart here if possible and then keep shooting the arrows, the arrows will fall inside the squares it must fall inside the squares, now what you do if you estimate those arrows that fall inside this region ok and those that fall outside this region that will give you a good estimate of what is the area of this region given this region so what I am trying to say, out india map here and try to close your eyes and try to put a dart on the india map and see how many times your random darts fall on Punjab and how many times it doesn't and that will give you a good estimate. So let me illustrate this to you I am closing my eyes and I am trying to put a dot here randomly you see my eyes are closed ok so on and so forth I keep putting dots ensuring that I am putting inside the square if I am putting outside don’t worry so on and so forth now let’s see so there are some points outside some points inside based on what is the area of this region you will have proportionate amount of dots inside and proportionate amount of dots outside, so here is a challenge for you, can you try simulating this on your computer? Can you try to see if you can write a program computes the area of random looking region like this, I can surely try good let’s see.

# AREA CALCULATION: DON’T MEASURE 02

Hello everyone, as you all know I got to know an interesting method to calculate the area of Punjab and let us try to verify this method now. As you see you are given a map of india here in which Punjab region is clearly highlighted so I will just go by the method as sir explained so first of all I will start placing the dots in the concerned region randomly so let me do that so I will just placing the dots one two three four yes in this way as you can see I have placed many dots in this region so now let us try to calculate the number of dots present in the Punjab region as well as present in the india region so let me count that one two three four so you see I have around two dots in the Punjab region three dots are out of the concern region which we won’t count and there are eighteen count that are present in the india region ok and I would like to tell you a fact here that for calculating the area of Punjab I have been given the are of india yes for calculating the area of Punjab I have given the area of india so let us go ny the method I will just write number of dots placed in the Punjab region divided by number of dots placed in the india region which will be equal to area of Punjab divided by area of india so now let us try to calculate it, it will be two divided by eighteen into three two eight seven zero zero zero, two divided by eighteen into three two eight seven zero zero zero which will be around three six five triple two which will be around three six five triple two as you can see this is far from area of Punjab, area of Punjab is around five zero three six two square kilometres which is not right so what do you have to say here? Is this method correct? Yes this method is fairly correct but what we have to do here is we have to increase the number of iterations we have to increase the number of dots that are placed randomly here so what we will do here? I will start placing the dots again in the concerned region but now I will more iterations I will place more dots here so let me do that one two three four five so as you see I have placed many dots here so now let me calculate the number of dots here. Let us do that one two three four five six seven eight there are eight dots in the Punjab region and outside Punjab I around placed five hundred ten dots in total so there will be around five zero two dots outside Punjab region so we have eight divided by five zero two into three two eight seven triple zero which calculates to five two five nine two wow this is clearly a very nice method as you can see this is very very close to area of Punjab so as you see as you observed you have to increase the number of iterations you have to place more dots in the concerned region in order to get the accurate area, in order to get accurate area I hope that now this method is clear to you now we will move to the programming screen cast of this particular method thank you.

# AREA CALCULATION: DONT MEASURE 03

So, first of all let me give you brief idea of what images are, so an image consists of matrix of pixels. A pixel is basically a colour and a colour id defined by mainly three attributes. I repeat an image is basically a matrix it is a matrix of pixels and a pixel is basically a colour. A colour is defined by mainly three attributes. And the attributes are amount of red present in the colour, amount of blue present in the colour and amount of green present in the colour. Yes the RGB values they define a colour so whatever colour you choose they would be having some amount of red, green and blue in them. And that’s what makes a new colour. I repeat every colour is defined by three attributes, amount of red, amount of green and amount of blue present in that particular colour there are only three primary colours and those are red blue and green and if you mix them in some particular quantities and that makes a new colour so this is what RGB values are all about and every colour is associated with some RGB values. So as you see there are some examples here RGB value of white is basically two fifty five, two fifty five, two fifty five. That means it have two fifty five amount of red, two fifty five amount of green and two fifty five amount of blue in it and RGB value of black is zero, zero, zero. So it has zero amount of red, zero amount of green and zero amount of blue in it. So now let us try to implement a program based on this so that you can get the clear cut idea of how to represent an image and how to assign unique RGB values to an image so let us do that. So first of all I will just show you how can you represent an image here so we will have numpy library here first I will do import numpy as np after that I should also import pil yes I should also import pil and hope that you are already aware of pil and you have seen the previous lectures on image processing so I will just write from pil import image, after that what should we do I will create an image of a particular RGB value I repeat I create an image of a particular RGB value to define the dimensions of the image I will take width and height of the image so I will just write width is equal to for example five and height is equal to four so next what do you have to do, as I said image is basically a matrix so we will take an array here so I will just write array is equal to np dot zeroes in bracket what should I write? I should write the particular dimensions I will write height comma width comma three and after that I will write one more argument here and that is dtype is equal to np dot uint eight so let me explain you what does this function do. So np dot zeros would make an array of the given size of the given dimensions so we have as the first dimension is height here and second dimension is width here and that is five and three and each pixel as you know contains three byte values for RGB so we will write three here so the first dimension is height, second dimension is width and three represent here the byte values for RGB as we know we have each by dedicated to each of the colour so we will have three byte values for RGB ok so one byte for red, one byte for green and one byte for blue. So that’s why we wrote three here and then there another parameter here which is called dtype, what is dtype? Dtype basically represents the data type assigned here, the data type assigned here is uint that is unsigned int the data dtype represent the data type and what is the data type here? That is uint which is called unsigned int. So now we are done with the dimensions of the image and we have also made an array of the image so what now we have to do here is we have to make an image out of this array so I will just write a variable here for example img is equal to image image dot form array form array which should be under parameter that should be passed here would be array. After that I should save this image, image dot save and I should give the name to the image for example will be test dot png ok so we are done with that. So now let me try to build this image and see what happens here so now as you know I will just run this file let us try to find out what kind of image has been formed here so I will just write image one dot py save it so there is some problem here let me check that. We have imported image here and capital I image is not being recognised here that is module pil dot image has no attribute form array sorry it shouldn't be form array it should be it should be from array please note the mistake here, it should be from array not form array so I will just run this again. So let me now test that image so it is test dot png as you can see there is black sort of square here so what can you infer from this image? Please note the fact that you only initialise the array with zeros, yes you only initialise the array with zeros and the colour corresponding to RGB values as zeros is black as I have already explained the RGB value of black is zero zero zero and you are only give and you are only given zero zero zero here so if I will make an image out of this array with only zeros that will only be consisting of black colour. So as you can see here according to the width and the height we have an image here and it only comprises of black colour. So as you see corresponding to give an RGB value we construct an image. Let me give you one more example of that I will take one more array for example I took array one and now I will take some other RGB values I will just write np dot zeros and I have here and I will also change and I will also change the width of this image width and height of this image I will just write hundred comma two hundred comma three and the data type would be same I will write dtype is equal to np dot uint that is unsigned int eight so that is done. So now I will assign some unique RGB values to this array so I will just write array one colon comma colon hundred I hope you are aware of this fact of list slicing in python we have already explained you in the previous videos so I will assign some unique RGB values as I said so I will assign two fifty five comma one twenty eight comma zero this is basically orange colour I will write here this is basically orange colour. And this is for the left side and for the right hand side I will write array one I will write colon comma hundred colon as we are talking about the right side so it should be hundred colon so I will write here zero zero comma two fifty five and this is basically blue colour so I will write blue colour here. And again what we will do here we will make an image out of this array so I will write img is equal to image dot from array form array from array now this time the array would be the array one and you should save this image image dot save and give another name to this image so I will just write test one dot png. So I think now our code is complete and now let us try to run this. Now let me check test one dot png so as you can see it as orange colour to the left hand side and blue colour to the right hand side so this is truly amazing we can construct images out of the RGB values so I will just repeat what we did here. First of all we need to import the numpy library as well as image library from pil package yes you have to import numpy image after doing that you need to construct an array, as I said image is basically a matrix so you need to construct a matrix, you need to construct an array here from that array you can construct an image here so I first of all I just took an array of zeros and I supplied dimensions here we supplied with an height here and three what is three denotes here? As we know each byte here would represent RGB values so there will be three bytes here so first byte will represent r second byte will g that is green and the third byte will represent the blue factor here and dtype what is dtype? Dtype is the data type which is unsigned here after that we have constructed the array, now we need to construct an image out of this array so there is a direct function here in python and that is from array you will just write image dot from array and you need to supply the array as an parameter. So I supplied the array parameter here after that we need to save that image and as you know only zeros as RGB values corresponding on to black colour so we only had a black square here, small black square here because we gave the dimensions as a very less dimensions the width and height are very small that's why we had a very small square here small square here so after that what we did was we took another array, array one in array one we supplied some dimensions some new dimensions hundred two hundred and three which I have already explained and then we supplied the data type, data type is as usual unsigned after that what we did was after we change the values of this array for changing the values we use list slicing, in the left part of the array we had orange colour and on the right part of the array we had blue colour after that again we need to construct an image out of this array so we used the function from array so we used from array image dot from array and we supplied array one here after that we save the image and here in this particular image we had orange on the left hand side, blue at the right hand side I hope this programming screen cast is clear to you and if you have any doubts we are there to help you please post on discussion form. Thank you.

# <u>AREA CALCULATION: DONT MEASURE 04</u>

Welcome again as you have seen that we constructed an image from given RGB values now we will do the other way around suppose we are given an image and we need to find the RGB values of the dominant colours here so how can we do it? Well that is fairly easy so here will be taking our previous image only our test one dot png images png image in which we had orange colour at the left hand side and blue colour at the right hand side so I will just write the code in python on how to calculate the RGB values of a given image. So let us start with that. So first of all you need to import the image library so I will just do that from pil import image from pil import image after that what I will do here is I will open my image so I will just have im is equal to image dot open our image name is test one dot png so I will just write test one dot ong after that I will convert my image to RGB so how can you do that? You just covert you just write some variable here suppose I wrote RGB underscore im is equal to im dot convert so this will convert my im image to RGB values to matrix here so this is done now I need to calculate the particular RGB values so I will just write r comma g comma b is equal to RGB underscore im dot get pixel so get pixel is basically a function as I already explained that each cell in the matrix of an image represent a pixel and each pixel represented by a colour so they get pixel here only fetch the correct RGB values from the particular image they get pixel function will fetch the RGB values from the matrix of a given image so the given image here is test one dot png we already know the RGB values so let us check that whether it is right or not so I will just print the RGB values here I will write print r comma g comma b so let us try to run this. Yes! It is giving us the right values we had two fifty five, one twenty eight zero as the orange colour and zero, zero two fifty five as the blue colour so here we have the correct RGB values of the orange colour we have two fifty five, one twenty eight and zero now if I change the x and y coordinates here we can also get the blue colour so I will just write till hundred it was orange colour after that it was blue colour. So I will just write one fifty here so that we can get to know whether it is giving the correct RGB values of blue colour also so I will just run it again yes it is giving us the correct RGB values of blue colour too so we have zero, zero and two fifty five here so I will explain you again what I have done so far first of all we import image library from pil package after that we need to open that particular image so I will just wrote the function image dot open and in argument you need to pass the you need to pass the image name so we pass test one dot png after that you need to convert your image to RGB values to RGB matrix so you need to write im that is the variable name im dot convert and in bracket you need to pass RGB after that you need to print the particular RGB values so I stored these values in this three variables r for red, g for green and b for blue so I just wrote RGB is equal to RGB underscore im that is the converted image dot get pixel. The pixel is basically colour here so it would comprise of particular attributes that is RGB so after that you need to pass the x and y values here the x and y values would fetch a particular pixel and print its RGB value so then I wrote print RGB so as to check whether it is giving us the right Value or not so in this way in python you can find the RGB value of a image. This is the way which you can do by implementing in python with while there is an easy way too, there are many websites that are present on the internet in which you just have to upload your image and they will just give you the RGB value of the dominant colour present in your image. I will just show you one of the websites in fact there are many you can refer many so I will just show you one as an example here. So as you see I have uploaded an images our images test one dot png and on the left hand side we have orange on right hand side we have blue so if I need to know the RGB values of the particular colour I will just over my mouse here on that particular colour and I get the particular RGB value these are basically called colour pickers so we have on orange we have the RGB values as two fifty five, one twenty eight, zero as we discussed earlier and on the blue one we have zero, zero two fifty five yes our program is giving us the same output as this so if you don't want the pythonic way of calculating the RGB values of the image you can go by this method too but I will suggest you to write a code for this and then you will get to know what are the RGB values so I hope that this method is clear to you, clear to all of you we will be implementing this method we will be using this method in our forth coming programming screen cast relate to area calculation so if you have any doubts please, please, please post on discussion form. Thank you.

# AREA CALCULATION: DONT MEASURE 05

Hello all, welcome to the programming screen cast of area calculation so first of all let me explain you how are we going to calculate the area of Punjab given area of India and how can we apply the method explained in the previous video here in python. So as you can see we have been given an image here in this image we have map of India in which the Punjab region has different colour and the rest of the region of India has different colour so please think about it how can we implement this using the method explain in the precious videos, now you are aware of this fact that you can find the RGB value of the image I hope that you are aware of this fact that you can find the RGB values of the image using python as well as the colour picker application any where available anywhere on the internet so as you see we have two different colours here so if I start putting random dots here I can find out which dots are placed in the Punjab region and which dots are placed in the India region, how can you find that? If the RGB value of the particular c and y coordinate is different is corresponding to this greyish part that means it is it lies in the Punjab region as well as if the dots that are placed here correspond to the black region the blackish region that means it lies in the India region so you can calculate the number of dots placed in the Punjab region as well as in the India region so I repeat here how can you do that, first of all you know how to calculate the RGB values of the particular image, so in this particular image we will be having two RGB values one corresponding to the greyish part and one corresponding to the blackish part so what will we do here, we will start putting random dots here will start taking the random x and y values here the corresponding x and y values if they have the RGB values that correspond to the grey colour that means they are present in the Punjab region and if the x and y values that are present in the blackish region and we can calculate the RGB value of that particular region and if it corresponds to the black region that means they are present in the India region so what we need to calculate here is, first of all we need to calculate the RGB value of these two colours of this grey colour and of this black colour and then we need to select the points select the x and y coordinates randomly and corresponding to these x and y coordinates we can calculate numbers of dots placed in the Punjab region and number of dots placed in the India region. So first of all let us calculate the RGB value, so I will be using the easy method here I will be using the colour picker application here as you see I have uploaded the image here so let me check the RGB value of the black region corresponds to sixty, sixty one, sixty and of this grey region corresponds to eighty, eighty one, eighty one so of this black region the RGB value is sixty, sixty one, sixty and of this grey region it is eighty, eighty one, eighty one so using these RGB values let us implement this in python so let me start with that so first of all what you need to do here is you need to import the scipy library here I will write import scipy dot misc I will be explaining you why are we importing this library and as an always we will be importing the image library from pil package so just write import image from pil after that you need to import numpy too I will just write import numpy as np and you also need to import the random so here I will just write import random so we have imported the required libraries now first step that we need to do here is we need to read this image so I will be introducing the new method here using the scipy dot misc library you can import you can read the images so I will just take a variable here img is equal to scipy dot misc dot read and our image name is map zero one dot png don’t worry we will be applying you with this image we will be posting you this image so you just need to read this image after that you need to keep the track of the count of number of dots present in the Punjab region as well as the India region so I will be taking two variables here first of all it is count underscore Punjab is equal to zero and then count underscore IN that is India that is equal to zero so next what we need to do here is we also need to keep a track of count here you will be getting to know why are we taking count here now I will take a while loop and run it many number of times so that we can get the exact area for example I run it ten thousand number of times as I have already explained in the previous programming screen cast in the previous screen cast that we need to have many iterations here that's why I took ten thousand let us see if this gives the right area or not, after that I need to select x and y variables randomly here what I need to do here is? I need to select x and y variables randomly here for selecting the x and y variables you need to know what is what are the dimensions of the image, yes you need to know what are the dimensions of the image so I will just go to the image and go to the image and try to find out what are its property so we have our image here map zero one dot png so I will just double click here and try to find out what are its properties so we have the image dimensions here that is two four eight one into two seven three six so we have the image dimensions here as two four eight one into two seven three six so according to that we will said the range for randomly selecting x and y values so let us write them in the code so I will just write random dot randint and we have zero comma two seven three five here and y is equal to random dot randint zero comma two four eight zero here two four eight one was their length and two seven three six was the breadth and in python we have the x and y dimensions reversed for example here we had the length has two four eight one but here the x value will the x part will be as two seven three five because it is reversed in python x is taken as the depth one and y is taken as the length one I hope this is clear to you so if this is not clear to you we commonly take x as the horizontal value and y as the vertical value in python it is reversed in case of images so x acts as a y value here x acts as a vertical thing here and y acts as a horizontal thing here so please keep this fact in mind while doing while processes image in python. So we have the x values the y values as we read this image, image scipy dot misc we also have the z dimension here while the z dimension is of no use to us because our image is 2d and it is not 3d so we will just take z is equal to zero here. After that we already know the RGB value of the grey region and of the black region so I will just write if img x y z is equal to is equal to sixty then it corresponds to the India region so I will just increase the counter of India here that is count is equal to count underscore n plus one and I also increase the count here count is equal to count plus one else what do you need to do here is else if the RGB value corresponds to the Punjab region the grey region what we will write here x y z is equal to equal to righty here, so then we need to increase the count of Punjab region so you just write count underscore pun is equal to count underscore pun plus one and you need to also increase the count value, count is equal to count plus one so now we have number of dots placed in the India region number of dots placed in the Punjab region now what you have to do here is, you need to calculate the area of Punjab so we will go by our method I will write area of Punjab is equal to count of Punjab region divided by count of India region into the exact area of India so I will be taking the exact area here and that is three two eight seven three two eight seven two six three, three two eight seven two six three and I will now print this particular variable area underscore Punjab so now let me try to run this and see whether this is giving the right answer or not and if it is not giving the right answer so will have to check whether increase in number of iterations will help this method or not so I will just run it, we need to save it so I will just write ar dot py let me try to run it so there is some problem here let me try to check it is not read it is basically I am read it is not read basically I am read please note this fact here so I will just try to run it here again so we have the area of Punjab as four eight zero two three but actually the area of Punjab is five zero three six two so there is some error here so let me try to increase the iterations so I will increase the iterations to one lack so let me try to run it again it will take some time because the iterations have been increased so now we have five four zero nine six the error here is less earlier we had around four thousand error now we have around two thousand error so if you keep on running this particular program you will get some value near to the area of Punjab. Ok, since it is a randomise method this can't give you the exact amount the exact area but this will give you the area that is very very close to the area of Punjab. So we will just run it again let me run it again it is now four nine eight eight nine run it again it is five two three six four oh my god this is very very close to the area of Punjab it is somewhat two thousand it is somewhat two thousand away from the actual value you can check that online, what is the area of Punjab and you can figure out and this is very very close it has only one thousand difference the area of Punjab is basically five zero three six two so, it is very very close to this particular value so you must be having the question that why are we getting the different values every time please note the fact that we are calculating we are taking x and y as random values so we don't know the ratio count of Punjab divided by count of India the, we don't know what this ratio will give us so if this ratio is accurate then only the area of Punjab will be accurate so we need to increase the number of iterations so has we get more and more x and y coordinates and we get a clear picture we get the correct value of area of Punjab ok so since this is a randomised method the answer the solution the value would be different for different times for different number of runs so as you keep on running this algorithm you will get different different answers but only but one thing is sure about it that this will give you area that will be very very close to the actual area of Punjab the error would be very very less. I hope this programming screen cast was useful to you guys I will be coming up with another method of how can we calculate this the logic is somewhat same but will be telling you one another method of how can you calculate the area of Punjab if you are given area of India. Thank you.

# AREA CALCULATION: DONT MEASURE 06

Hello everyone, welcome again to the programming screen cast of area calculation. This particular programming screen cast I will be discussing another way of how to calculate area of Punjab using randomised method so I hope you have watched the previous videos and you know how to calculate the RGB values of the given image. One method is related to colour picker application and you just go to any online website and you find the RGB values which we used in the previous programming screen cast. But in this programming screen cast what would be doing would be using the automatic method the pythonic method of finding the RGB value of a particular image so I will be using this particular method. This method has already been explained in one of the previous programming screen cast so I hope that you have gone through all those videos so let me start with this programming screen cast so as I said I will be using the pythonic way to calculate the RGB values of a given image. So as an always let us import the image library from pil package so I will just write from pil import image. After that I also need to import random since we need to randomly select x and y values here after that I will open the image I write im as the variable im is equal to image dot open and as and parameter I should pass the image file our image file name is map hyphen zero one dot png so we have our image in the variable im, now in order to calculate the RGB values I need to convert this image to RGB matrix so what should we right here? As explained already I write some variable here RGB underscore im is equal to im dot convert and that should and you should pass RGB as a parameter here so that is done after that we will be using the method used in the previous programming screen cast I will take count of India is equal to zero, count of Punjab is equal to zero and we should also take a count here count is equal to zero now I need to have many number of iterations so I will take while count is less than equal to one lack as we did in the previous programming screen cast now what should we do here? We should select x and y values randomly here, in order to select the x and y values randomly here you should know the dimensions of the image. In the previous programming screen cast I showed you how can you know, how can you get to know the dimensions of the image you just go to the particular image and you right clock, right click on the image and go to the properties of the image, in properties of the image you will find the dimensions of the image. So we already have the dimensions of the image so I will write x is equal to random dot randint and in this you should write zero comma two four eight zero and you should also select you should also select the y value randomly so I will write random dot randint zero two seven three five, two seven three five so this is done and will take z as zero after that you need to store this RGB values in the variables RGB corresponding to the particular x and y coordinates so I will just write RGB underscore im that is the our that is your image converted to RGB matrix dot get pixel get pixel and in this I will write x and y, x comma y corresponding to x and y coordinate it will give us the RGB values and now we know the RGB values of the grey part as well as the black part so I will just write r is equal to equal to sixty here if it is sixty here then I should increase the counter of India, count underscore in equal to count underscore in plus one I should also increase the count variable here count is equal to count plus one else if r is equal to is equal to eighty here what should we write? You should increase the count of Punjab so you write count underscore Punjab is equal to count underscore Punjab plus one. You should also increase the count here count is equal to count plus one after the you need to calculate the area of Punjab according to our previous randomised method so you just write area underscore Punjab here is equal to count underscore Punjab divided by count underscore IN that is India and you should multiply with area of India, so we have the precise area of India that is three two eight seven two six three so you should write three two eight seven two six three and now we need to print area of Punjab. So now let us try to run this program I will write area one dot py module run has oh sorry it should be int here so we have the area of Punjab is five three two one three the actual area is around five zero three six two I guess please check on the internet and it is somewhat closer to area of Punjab let me run it again we have four nine five eight five this is closer ever five one four eight two is very much closer so you can always increase the number of iterations and the accuracy will improve with increase in number of iterations so I hope that this programming screen cast is clear to you all I will just explain you this program again so as and always first of all you need to import the image library you need to import the random library after that you need to open the image the concerned image will be providing you with this image don't worry after that you need to convert this to RGB values after that what I initialise three count here, count of India, count of Punjab and simple count here that will keep a track of number of iterations so we are taking one lack iterations here you can always increase the number of iterations. We have the dimensions here dimensions of the image we have already explain how can you get to know the dimension of the image so I will pick x and y randomly in this particular range so I pick x from zero to two four eight zero and y randomly from zero to two seven three five. As already explain you can calculate the RGB value from that particular image using this particular method you get pixel so need to take that image RGB image and corresponding to x and y value it will give you RGB value we have already you already know that the RGB value of the blackish part there is sixty and around sixty and for the Punjab region it is the greyish region is eighty, so r is equal to is equal to sixty you need to increment the count of India and if it is eighty then you need to increment the count of Punjab after that you are just applying the method the randomised method explained in the very first video so we will just take the count of Punjab divide it by the count of India into the precise area of India that is three two eight seven two six three and then you can print the area of Punjab. I hope the previous videos and this video is are useful to you all, you have understood them all I just go through the videos first of all what we did was, we proposed a randomised method to calculate the area of Punjab after that we showed you how can you construct an image from for the given RGB values then we then we did programming screen cast other way around, first of all we had the RGB values and constructed an image now we had an image and we need it to find out the particular RGB value. We also showed you how can you do that. After that we proposed two methods of calculating the area of Punjab using the randomised method in pythonic way for this particular thing we had an image of which had the blackish region as the India part and greyish region as the Punjab part so what you can do here is we already know the RGB values of these two colours, using these RGB values we generated randomise random x and y values here according to this x and y values we calculated the number of dots that are placed in the Punjab region as well as number of dots that are placed in India region randomly. After calculating these two particular parameters we already given the area of India so we had the formula here which was explained in the very first video we did count of Punjab divided the count of India into precise area of India and that gave us the area of Punjab. So I hope this method is clear to you all please refer JOC wiki if you have any doubts. Thank you.

# FLAMES 01

I frankly remember playing a game during my school days. So what we will do is, we would write down two names and try to cancel out common letters and then try to find out the remaining letters and then count them. Once you have a count, once you note this down what we would do is, we would write down this word flames f l a m e s and we would go ahead and we would try cancelling out the letters after counting the given number and whatever remains is the relationship between these two people. For example let us take names of two people amit and amit likes deepika the celebrity right? so what we can do is we can try to see what, what will their relationship lead to right this is a very childhood game that we all have played at least I had played my classmates would be a whole lot of this so a and a gets cancel the amit ‘a’ and deepika ‘a’ and ‘m’ is not in common! ‘I’ and ‘i’ gets cancelled, ‘t’ is not in common so the number here would be m t d e e p k which is one two three four five six and seven, now what we do, we start counting from ‘f’, ‘f’ is one ‘l’ is two, ‘a’ is three, ‘m’ is four, ‘e’ is five and ‘s’ is six so one two three four five six and seven moment seven comes that part is struck out and again I will start with ‘l’ one two three four five six seven and ‘a’ gets struck out, one two three four five six seven ‘s’ goes away, one two three four five six seven ‘l’ goes away ‘m’ is one ‘e’ is two ‘m’ is three ‘e’ is four, five six seven ‘m’ goes away and ‘e’ remains so the fun conclusion would amit and deepika have the relationship enemies, ‘e ‘for enemies so similarly ‘f’ stands for friend, ‘l’ stands for love, ‘a’ stands for affection, ‘m’ stands for marriage that amit will be marrying the celebrity deepika and e means they are enemies which unfortunately turns out to be the answer, ‘s’ is the celebrity is a sister so as I said it is a fun game people would have played this but the reason why we choose this as a example is this represent a very famous mathematics question called the Josephus problem ok if you are interested you can take a look at it but what interest us in joy of computing is to write a piece of code for this and what would we do, a piece of code that will ask for the guys name and then the celebrity girls name and then will tell you what is the relationship between them using this fun algorithm

# FLAMES 02

Hello guys, welcome to programming screen cast on flames. So flames is nothing but a game that is a very interesting game that most of you would have played in your childhood I guess or if not played you would have seen someone playing or you would have seen some where the game is being played some experience would be there. So we are going to relive those childhood days through this joy. So before starting with the programming part let me take you back to that memorable days by explaining what this actually game is, what this game flames is, so we will start off with it. So in flames what are these things f l a m e s it's not just a single word, all these have some expansion sort of thing so f stands for friends, l stands for love, a stand for affection. M stands for marriage, e stands for enemy, s stands for siblings so this is what we abbreviate and call as flames so basically in this game if you could see, see some two names are given and mostly it would be the case that one is a boy and one is a girl it you can that makes sense basically for flames if you see the expansion will have something like love, marriage all those are generally between people of different genders right. so it's generally case that one of your name is of a male and other one is of female so two names are given and you are suppose to predict a sort of just say what could be the relationship that could exist between these two people. For example ajay and priya are the two names we have, what is the relationship between these two? This what you are going to predict. So what we generally do is, we would strike out the common letters between the two names and then will start counting from here, how many letters are left out based on that will start counting and will cross out each and every letter and slowly will proceed till we just have one letter left out in this word flames and that particular letter is the result is what is displayed. This is how it goes maybe let me go slowly. The first thing is I said you should cross out the common letter right? you should cross out the common letters so as you could see there are two names you need to cross out the common letters, we will explore this part of crossing out the common letters see you take a single name ajay, you have two letters here and here that is common that is here is also a letter 'a' here is also a letter 'a', you cannot cross out these two this crossing out is not allowed, what is allowed is you have a 'a' here and you have a 'a' in the other person's name so you can cross out these two letters and this is allowed, so once you have crossed out the names now that remain is j and p r I y, that is the names that remain. So as I said you can remove 'a' or you cross out 'a' and what you have as remaining you compare those two names see as I had said you are comparing these two names and here also you have a letter in common 'y' so you have to cross out this letter 'y' I am sorry this has to be 'y' so remove 'y' and now compare these left out strings J A and P R I you have to compare it. J A and P R I if you compare there is no match there is nothing common between these two strings so in this point of time you count how many letters are remaining, one two three four five, five letters are remaining so here the answer we get is five. Now you have the word flames, you start counting from the left count five because we got five in the previous so count five, one two three four five see so E is where we are stopping at so you have to cross out that E so E enemy is not the relationship you have to cross it out and the next round you should not take it as F L A M S instead you should start from this point S start at S and then you add of the left hand side part, whatever is in the right side you have to take it first then you have to add the left side as it is, that is how you have to take it for the next round so the next round now I get as F L A M now I have to repeat it with this let me do that ok as F L A M one two three four five I am at M so M is crossed out so after you cross out M whatever is in the right there is nothing in the right so leave it then whatever is in the left S F L A this is in the left so you have to check it with S F L A now see we have four letters remaining but we need to go in counts of five so where will we go for the fifth letter, noting lust come back to the first letter and start counting again this is how you do, this is what we call as modular counting that is what we are going to do now go in the clockwise direction sorry anti clockwise direction sorry anticlockwise ok so S F L A so you have to start counting five letters from the left and go back in the anticlockwise direction I am sorry this is anticlockwise S F L A one two three four again come back here five so you are at S so you cross out S so whatever is in the right F L A whatever is in the left nothing so now you have to compare F L A let us do that F L A same like that you have to count five and you have to go out in the anticlockwise direction please note this its anticlockwise because we are going this way we are going in this way so it is anticlockwise so one two three four five we are at L so L is not the relationship so you cross this out you cross out this L, L is not the relationship you cross it out now what is remaining? In the right side you have A you write it then on the left side you have F write it, so you are left out with A and F now we have to repeat it with this so let us do that A and F will count five steps one two three four five so at A we are standing so we have to cross out A and now we are left with just one letter F and this is when we have to declare the result is friends so ajay and priya were friends is what we got. This is how we have to play this game. So what is the thing that is slightly challenging here is you have to do the modular counting and you have to start counting from the part where you left that is that time we had stopped at E so we have to start our string next time with S so S F L A M we have to take, we could not take F L A M S so that is where is the tricky part here so we have to try coding this thing that is what we are going to do in this exercise. We will see that in the next video.

# <u>FLAMES 03</u>

Alright guys, so in the previous video you had seen what is this game of flames. So it is basically you have two names and you have to cross out the common letters across the two names that is within the name if there is letter repeats you cannot cross it out, if it repeats that is if you have one letter at name one and other letter at name two you can cross out the two letters like that you keep crossing the common letters and at the end how many letters are left out you have a count of it and you go in steps of that many number of counts along the word flames from the left and if it exceeds the word length you again come back to the first letter in the anticlockwise direction and you proceed it, this is how you play flames right so we had run through an example we had taken two names ajay and priya and we had run through and we found that at the end of our game the letter that remains is F which denotes friends so ajay and priya are friends is what we got as a result. So something like that we have to play the game so for that will do the programming part now, before going into the actual programming part I would like to give you some introduction to some pre requisites that is needed to code this joy flames so let us go ahead with the pre requisites. So the string is nothing but the list of characters as you could see string is what the technical term we give for words, name is nothing but words right so name consists of group of characters so like that we have a group of characters, group of letters, letters is technically called as characters and a group of characters is what we call as string technically so when dealing with strings we need to know of some functionality of strings and which is available in the python package namely string there is a package name string so let me import that first import so let me say let me start so the first thing we have to do is import this package named string let me do that import string this is the name of the package, yes we have imported this package string, if you feel the name is lengthier you can use as and give us smaller name s g something like that, that's your wish from this string ok I am doing this, so let me say those string I have s is some string strings the notation is we initialising with double quotes so let us say the name of the person is raj, raj is the name of the person so if this is a name of the person I have a mix of capital letter that is the upper case character we call them as the capital letters and the small letters we call them as lower case characters we have a mix of upper case and lower case. If I want to convert it to all in upper case or in all in lower case I have a functionality with a corresponding name let us see that, so first let me see for lower case that is all small letters is what we call as lower case, lower case is nothing but small letters, I want to convert into completely small letters so I would do it as print I want to print it in that format I will print it I will print it see s dot lower is the functionality we have this functionality s dot lower, if I print see I am getting the name in all small letters same if I want it for all caps what I will do is print s dot as you would have guessed its upper cast the capital letters are technically called as upper case characters so we will use the functionality upper, upper ok this is the functionality so let me press enter see RAJ its printed in all caps the upper case letters so this is how you get the complete lower case or complete upper case of a mixed string or even if you give in all small letters if you still apply s dot lower there is no problem it will return the same string there is what this functionality does is it will check if the given input string has all small letters if yes it will return the same string otherwise it will convert all letters to smaller letters and return that converted string similar is that for upper function, it will check if all the input characters that is the input string has some characters if all those characters are in upper case that is the capital letters or not if everything is in capital letters already it will return the same input otherwise it will convert the letters into the corresponding capital letters that is the upper case letters and return the converted string so this is how these two functionalities work. So upper and lower these functionalities for it to work properly to import this package string and the next thing is we were actually looking at individual character level right so we were checking this particular letter ajay A the first letter whether is it present in the next name priya it was present so we cancelled the next letter j we will compare, it was not present next letter A we will compare it is not present, so something like that we will be doing, right so we will compare letters by letters so when we have this has a whole string we cannot do that comparison so we need to convert it into list of characters list of individual letters that is what we need to do it, that will be required in our program so we will see how can this be done so let me say string list is nothing but list of the string s the variable is s so I am passing it ok so it was converted into a list so let me see what is present so let me say print string list ok so it contains the individual letters R A and J so it is the a single string is converted into list of three characters right so this list function precisely does that it will take the individual character make it an element and convert it convert a single string into list of individual characters that is what this functionality does. Another thing you can do with this strings is you can find and replace I hope you would have heard about this terminology this is most popular if you are a movie watcher as well if you watch in movies some movies have comedy scene based on that I hope you remembering three idiots is one of them it has been remade in into some other languages as well so in that they will find a particular word and replace it with some other word and they will make the hilarious situation out of it something like that we can do replace operation here so we will find a character and replace it with something else so let us say I don't want to print J I want to print some other letter in place of it so how should I do that, so I would say print I want to print the result S dot replace is the name of the functionality replace of as I said I don't want J so I will write J here J with let me say I want to print it with hash so I will say hash so if I say it is printed as R A hash in place of J hash is substituted so this is replace function precisely does that it will search for this particular string here and it will substitute this string in place of it, this is a source string this is a target, this is the source this is the target it will search for the source and it will replace it with the target its not necessary that you need to give single letter you can give sub strings as well so let me say let me modify it I will say AJ I can give a substring a part of the string is called what you call as substring in substring what is mandatory is the letters must be in same order if I say AJ and if I say if I press enter see R and the part AJ gets converted into this hash and here this is case sensitive that means if it is in a small letter this has to be in a small letter otherwise it won't agree if you say if I say RA and say enter see hash J is the output so this particular substring RA is searched and in place of that you are substituting hash it happens now I will check with smaller ra small ra I will check see you are not getting something like this you are getting raj completely because it is case sensitive that is whatever is the it is a smaller letter or a capital letter that it is upper case or lower case letter depends on that it will find and substitute so this replace functionality is case sensitive we call it as case sensitive by case sensitive we mean if it is a capital letter you must say capital letter if it is small letter you must say small letter it won't match ok it's the same letter just that, that is in caps and this is in small it doesn't perform that kind of a match that is what we call as case sensitive and most of the passwords are case sensitive if you have been using it in your mail or with your net banking anywhere the passwords are case sensitive so in that case if you a capital A it is different from a small a this is what we call as case sensitive, alright this is how this replace functionality works and even you can give something so let me say I will I will give this capital RA and I can give star hash I can give something like that as well so even here you can give a substring in place of this substring you substitute this substring that can also be given it is telling us star hash and J or maybe I can do this way as well I can say just A see what happens let us see in case of A star hash has been substituted so this is nothing but you find this source whatever is the source in place of that substring you substitute this substring that is what this replace functionality does, I hope you are clear with this why would you use replace functionality in the code will see. alright the next thing that is needed is slicing of a list, I hope you have used it already in a some of your previous weeks but still let us have a recap because we will be using tI extensively here the slicing place a major part here so we will see a recap of it so let us see we had a list right sl this is a list you had R A J in case I want to extract only the specific part of the list, this is hte smaller list so let me take the larger list let me say L equal to I will take letter because we are going to deal with strings so will take letters only a b c d e this has to be a comma f ok so let us consider this list containing six elements this is my list L containing six elements a b c d e f so if I want to extract some portion of this list only and display that is what we call as slice in your taking the slice out of the entire list is what you call as slicing so how would you do that L within the indices brackets square brackets you should give the start index from the let me say index one up till index four let me say let me see what happens ok as you could see I had given index one to index four so what is the output, it starts from index one as you know the list indexing starts from zero so A is at index zero, B is at index one, C is at index two and so on so index one the letter found is B so B is printed index two C is printed index three D is printed index four E is not printed so this is how the slicing functionality works, the end index is ignored so this means when you reach the index four you stop the process you don't print that you don't included that is what this means so this portion of the list is sliced and taken so this is how slicing works and another thing is this is not mandatory to give both the indices the start index and the end index. That is it is optional so if I say list of I don't give a start index I just give a end index let us see what happens ok this is the output see you are getting A B C D just before the fourth index so if you don't give a start index the default value that is taken is zero, zero is taken as the default value and from zero till four minus one three the third index everything gets printed everything gets added to the output, aright what if I don't give the end index? I will say two I don't give an end index what happens from index two see A is at index zero, B is at index one, C is at index two from index two it prints the entire list so the default value of the end index if you don't give is taken as the length of the list so a recap or summary of what we have seen till now. Slicing is nothing but you cut the required portion of the list you just take the part of the complete list is what you call as the slicing you are taking a slicing of a list that is why you call it as slicing, give the square brackets and within that you need to give start index colon end index plus one please note that it is end index plus one because the last index won't be counted into the output it will ignore at the index it will stop at that index so you have to give end index plus one and it is not mandatory you to give both start index and end index it is optional in case you don't give the start index the default value is taken as zero if you don't give the end index the default value is taken as the length of the list so what if I don't give both let me do that what if I do this? The entire list gets printed because the default values are taken so this works basically like L of zero to length of the list is six here so zero to six it will start at index zero print A, print B, print C, print D, print E, print F which is at index five and index six when the counter reaches six it will stop so it will stop here and it will print the entire list so this is how it will works if you give the value it will take that particular index the way you have to give is start index and end index and plus one please note this plus one fact end index plus one and if you don't give the index it will take the default values, the default value for start index is zero the default value for end index is length of the list, ok guys I guessed you have a good revision of the list slicing which you had studied some weeks back as well as you saw the new functionalities of upper, lower, replace in strings please do explore these functionalities also there are some more functionalities that are available please do explore them and get more familiarised with these features with this command also yes one more command that you will be using in this we will see of this  is the list L and I want to get the index of an element I know the element I want to get the index that can be done by using the function index let us see that. Print L dot index is the function you need to pass the element here the element let me say is D, I want to know the index of the element D so if I print I am getting three, the index of the element is three. In case if there are multiple places this element D is present if your list is A B C D, A B C D, A B C D E F just assume if that is your list then the first occurrence of the value that is what is returned as the output here the index function return the first occurrence of this element in the given list, the index function returns the first occurrence of the given element in the given list right because how this functionality works so what all you have seen, we have seen the upper functionality which converts the strings into completely upper case letters, lower functionality which converts the strings into completely lower case letters replace functionality searches for a sub string and substitutes with another sub string then we have seen list slicing with start index and end index plus one then if you don't give the start index or end index what is the default values, how it works we have seen and we have seen the functionality of index which returns you the first occurrence of a given element in the list so this is what we have seen  now this are the pre requisites that are needed to code flames so with this maybe you can start thinking and you can start trying the code flames so from this develop a logic and you check it with what we are telling in the next video that would be the best thing that's the best way you can develop programming logic and you can develop by yourself I would recommend that and please do explore what are the other features available here and by this is by exploring and practicing, you become a better programmer and so keep practising is what I would say ok guys, thanks for watching this. Will see how to implement flames in the next video.

# FLAMES 04

Alright guys, in the previous videos you had seen what is this game of flames, how is it played? We had taken some example and then we had traced through every step of the game how is it played then in the next video we had seen some pre requisites that is needed to quote flames one was using the functionality of string like replace, upper, lower etc and the other one was you have to know how to slice some part of the list so we are going to deal with list so we are going to convert a string into list of characters and that is how we are going to process it. So you have been given some insides on how you slice a list and how you deal with parts of list and all that's what was given to you and index functionality was given to you that will be of great use for us, so let us now go ahead with the programming part so for that before starting of that I have recommended you that you analyse the game you understand the previous two videos understand the procedure of the game and then you use the pre requisites that has thought to you in the second video and based on these two you try coding flames and then in case you have done something you can verify what you have done with what we are telling and in case if it differs what I am going to say now is not the only way which you can do you can do with multiple ways so that may be some differences with which what you have tried and what I have been showing there maybe some differences so in that case please do discuss your strategy in the discussion form, so will be very happy to help you out on your approach and how it will work will it work in all cases that's will help you out so you please go ahead and try it yourself then you come back and check here so just use this video as the verification for what you had tried. So don't completely, blindly follow this video and code line by line you try yourself and come back and use it as a guide. This video is a just a guidance for you and how you can code flames, this is just one way by which you can do, there are many ways this is not the only way there can be multiple ways so you try out the different ways and please do discuss on the discussion form. Ok so let's starts off with this as I have said first thing is we need to let me say I had imported string and now I have to input the name of the two persons so let me say person one p one I will get the name so to get the input I have to use the functionality input function input input and I have to give the message here I will say enter first person name let me call it as first person name I will get the input alright there may be a case that the person may use combination of capital letters and small letters as I have said the replace functionality and all the functionality so strings are case sensitive that is if you give capital letter it will strictly look for capital letter, it won't consider the corresponding small letter version so it is case sensitive so it is better that we convert this input into a common format by using either of the upper or lower functions so I had demonstrated both of the things in the previous videos you can use whatever you wish so this is to ensure that the comparison are not skipped for example we had taken the names ajay and priya assume that they are using capital letters for the first letter of their name and rest they are using in small letters so if he will give an input as capital A j small j small y and the second input will be capital P small r small I small y small a so when you compare it will say capital A which is the first letter of ajay and small letter which is the last letter of priya it will say they are different but where as we want them to be considered as same so it is better that we convert them into a common format so that is why we are going to convert it into lower case let us say lower case so I will convert it into lower case so my new input will be my new the name of the person my new name will be p one dot lower that is a functionality I am going to convert it into lower case so I had said this is just to ensure that the input is in the uniform format so this is how you converted into lower case and full name I had just taken the first names in the example that I have considered in case you are typing in your full names there may be the case that you may give your first name give a space then give a sir name for example ajay Sharma if that is the name ajay space Sharma they may input this space shouldn’t cause a problem anywhere so will trim out all the spaces from the name what if the first person has the lengthier name if it is ajay Sharma and the second person doesn’t have the sir name it just priya this space should not be counted as the extra character in the future runs and that shouldn’t cause the problems so will trim off all the spaces from the names so that will be what we will be doing here for that we will use the replace functionality that we had seen, so will do that the p one is now p one dot replace what do I want to replace the space with empty character nothing the space must be trimmed that’s what I want, so I am replacing it this is what I have done this is just to make sure that the space doesn’t cause problem in the final computation so we had done it for the person one the same goes for person two so let me copy and paste it and do some modifications its person two here its person two here even its person two its person two person two so what have we done till now? We have got the name, we have converted everything into lower case so that the miss matches due to one being in the upper case and one being in the lower case doesn’t occur we are converted everything into uniformly into lower case you can even do it for upper case and other thing what we have done it we have trimmed the spaces so whatever wherever we have the spaces we have cut off the spaces and we have made it into a just list of characters we just sequence of characters is what we have now. So we have the sequence of characters I have said so we have to convert it into list so let me say l one for list one let me say l one is nothing but I have to convert it into list so how will we do that using the list function list of the first person name p one so you have the list of characters similarly we wanted for l two the list of for p two the list of all the character in p two the person two name so now we have the list what should we do? We should remove the matching remove the matching letter from this list one and list two you have to remove this matching letter so this is what you want to do now so it’s not just one letter it is multiple letters so this has to run inside the loop just give a pause here and think how can you do it, this requires some amount of logical thinking to get it perfect so pause here for some time and think will see how this functionality can be implemented in the next video.

# FLAMES 05

Alright guys, so in the previous videos you had seen how to play the game of flames we had raised it with an example then we say some pre requisites I had asked you to give a try to writing a code then this videos we have to take it as a guide lines just a guide lines that's it this is not the only way by which you can do, you can do it in multiple ways so please discuss if you have another way in the discussion form and so in the previous I had started off by asking for input of two persons names, converting them into uniform format of all small letters I had trimmed off the spaces and now we have to see how to remove the matching letter that is the functionality we want to see. So let me say I will define this functionality here define remove matching letter right? That was the functionality right? so let me define it here so it had taken two parameters list one and list two it had taken two parameters so let me use it so what should I do? I have to take a letter of the first list take a letter check if the letter is present anywhere in the second list if so there is a match and you cross out otherwise you proceed with the next letter that's what you are doing right? so for that what should we do we need to run loops so how many loops? Two loops, because we have to match for each letter at which position in the second list you have the occurrence of that letter so we need to have two loops so let us do that. So let me start my first loop for I in range so basically you are actually going through each and every index and checking the element right so I have to find the I have to check through till you reach the length of the string so range you have to give the end value plus one right basically so I want to go till the end of the string basically end of the string is nothing but length minus one that is the index of the nth so length minus one nothing but length we have to stop at this I guess you are familiar with this just that why we are giving it as length I was giving an explanation that's it length of this list you have to pass that is for that many number of iterations that is for that many number of letters you want to make a check right? so iterate it over the first list same like that you iterate it over the second list for j in range length of the list l to k so what should you check? If the letters are matching so how would you check it? You have to use by indexing right so by name of the lists followed by square brackets within the square brackets you need to give the index so we will do the check now if the list one at the position I at the index I and the list two at index j contains the same letters that is check by equal to operator double equal to operator which checks the equality if they both are equal then you capture that letter that is your common letter you have to capture it common letter that is why I have used the letter c, c for common, common letter you can use l one of I or l two of j because both are equal right both the vales are equal so you can use anything here and now you have to remove the elements from the list so that's what we will cross out in a paper we will cross out but in computer how do you cross out so crossing out is nothing but removing it from the list that is what we will be doing it. So let me remove l one dot remove is the functionality remove this element from the list and l two dot remove element same element c alright so now we will have to find once the letter is removed whatever is the string you need to have it and you need to do it for both the you have two strings modified right so let me concatenate it you will understand why soon let me concatenate the two strings two modified strings let me make a new list l by saying l one plus I am concatenating another list I will use a dummy character say star then I concatenate this list l two, why I am creating this concatenate list, you will come to know very soon this is just one way this is not the only way you can do it in any way this is my way of doing that’s it. Its not the, the way of doing it just one way of doing it ok so this is my list and now here I have found a match at some letter so I had removed that letter and I have a new strings and in between the new strings I am using the star as the marker sort of a thing so this person first person name has ended and this is the name of the second person so I am using a marker right? so I will have to return that I will have to return that I have found a matching letter so remove matching letter is the functionality right so I have found the matching letter I have removed it so you have to say that you have to return that value to the call function so I will return a list say inside that list I will pass this concatenated list and I will say true I can pass any values right I can pass numbers, floating point values, characters, bullion values anything I can pass right so I will say I have proved that I have matching letter I have removed it and this is the list you get after removing the matching letter so you are concatenating it so you will run this loop for all the things right the very first time when you find a match you will remove the letter and return this that’s it in case you run through the loop completely and you still didn’t find the match what happens? It will come to the end of the loop right at this point so in that case you didn’t find the matching letter, in that case what should you do? You should return I will say same like that you generate this list this has to be done because this is done inside the loop right in case you came out of the loop this would never be executed and you need to do that so let me copy this line because I want to repeat it again so let me say I have created a concatenated list I will return the concatenated list and I will say false this is the concatenated list but in this two items I haven’t found the matching character that is why you are returning false so this is how you do so basically you run through the loop you check for each letter whether this letter is present in the next list anywhere if it is present you remove that matching letter and you return the concatenated list and return true, true stating that there was a matching letter and it was removed if there is no matching letter just you do the same concatenation process return the concatenated list and say false, false stating that there was no matching letters ok so this is how it has been run so we have been returning it so when we have returned it we have returned list right so we need to capture a list so let me say return list is nothing but we are capturing it and we need to do it continuously for some till three is no matching letter so how will I come to know that there is matching letter or there is no matching letter how will we come to know that? So we need to have a flag sort of thing this is what we call as flags n programming that will say whether to continue to proceed with something or you should stop sort of a flag, flag is an indicator for some directions sort of thing right so that indicates that something is present here flag indicate something like that here our flag will indicate whether there is some matching letters still present or should we still proceed or no that’s it so there is no matching letter you start counting with the flames part that is the second part of the game, first part is we have to cross out all the matching letters, second part is we have to go in the anticlockwise direction of counting to determine what is the relationship between two people in flames in the word flames right so we will be doing it so the first part is we are removing in the matching letter so the flag will say whether we can proceed towards second part or no we cannot proceed so let me use a flag for that let me say proceed, let me use the same name so that it is easier on mains proceed equal to true initially we have to proceed we have to the first run we must check for matching letters and then only we can we cannot directly start counting right we should check for matching letters and then only we can start counting so the first run it is always true and as long as proceed is true you keep checking for the matching letters I will have to indent it now ok as long it is it says proceed you keep checking so the returned list contains two information one is the concatenated list and the other is whether you can proceed or not, that is one letter has been removed and you can proceed or there is no matching letters there is no need to proceed it returns whether we have to proceed further or not right? so it is returning two information so we have to extract that out from this returned list so concatenated list is what I want to make I want to extract out from the returned list so that is present as the first element of the list, the first element of our list is the concatenated list so I am removing it return list return list of zero, zero is the index by which you access the first element right remember that array the list indexing starts from zero the counting starts from zero please remember this ok so I have taken the concatenated list now the second part of the list is whether should I proceed or not? So it will say I have to proceed is nothing or not so proceed is nothing but is present in the concatenated list second position that is index one so I have taken the concatenated list and proceed and in this concatenated list concatenated list consist of the two names the concatenated and there is a star in between right see that's how we have concatenated we have taken the name we had removed that common letter taken that name appended a star then we had taken the second name right that's how we have it so the star marks is a marker that the first string has completed and after this whatever comes is the second string right so star is a marker so we have to find where is this particular index where this start is present so where is the star present we have to find and at that place we have to slice this list so this is where the list slicing what we had seen come into play so will be slicing the list here so let me say star index star index is nothing but concatenated list on this concatenated list what is the index of the element star that's what I am going to check this is what I am going to call as star index and now I am going to extract out the lists again so l one is the name of the first person in the list format that is present before star right so you have to start from the first index and when you encounter the start you should stop you should not include the star you should include everything before star so what should be the end index? Whatever is the place where star is present right? Please give a minute time here think why am I doing this you will understand if you give some time that's where the logical thinking I don't want the star the functions behaves in such a way that whatever I give as an end index that will be ignored so as I had said I have to give a start index and end index plus one so if you just give end index it will just ignore the end index as well so I have to give plus one so I don't want star I want star to be ignored that's why I will give star index as my end index start index I don't need to give I wanted from the beginning so the default value let it be taken and I will just give the end index so I will do it from concatenate the list we have to perform the slicing operation no need to give the start index because I want to start it from the beginning the default value and I have to ignore the value of the star this index where the star index this point the star is present I want to ignore that point so I will give that as my end index star index is where I want to end so whatever is there still you reach star index whatever is there that will be capture in this list l one ok same like that what should be there in l two, anything after star till the end so till the end I wanted so that's the default value will be taken for end index so I need not give the value of end index but start index I should give, start index wherever I give its starts from there. So should I give star index as my start index? No because if I give star index as my start index it will say star p r a I y I just want p r I y I just want the name I don't want star here so I should say star index plus one the next place you should start my next slicing so I should say star index plus one is the starting index here star index plus one till the end, end value is taken as the default value is taken so till the end will be considered so two slices have been made. So this is why we had concatenated the list like this is the name we wanted and this is the name we wanted this is just one way or maybe you can pass it as list of three values as well you can think of it, you can pass it as l one comma l two comma true or l one comma l two comma false you can do it that way as well that is why even another way so as I have said there is no just single way by which you can do, you can do it by multiple ways so this is just one way of doing it, now just re through this loop so you had taken the list of characters, you had set some flag like proceed right, you had set it through true initially because you want to check if there is matching letters are not first instance you cannot take a risk of directly going to counting you should check for matching letters so I said proceed with the checking of matching so remove the matching letter so it will loop through the lists, it will loop through both the lists if it find a matching letter it will remove that from the list it will concatenate with the star in between, return the concatenated list and return true if it has found the matching letter and return false otherwise there is no if there is no matching letter it will return false so from this you will understand whether to proceed with next iteration that is we have found with the first iteration ajay and priya let us take the same example A and A it is return so should I continue with checking for another matching letter or can I continue with counting that is return by this proceed so that is extracted the concatenated list from that list we have extracted out and we have extracted the flag proceed so if it says true that is it has found one matching letter there is a possibility that there is another matching letter so we have to proceed it again if it was not represent if there was no matching letter present if you again check it again obviously you won't get matching letter right? so that time you can stop so that is captured in this flag proceed and this is your concatenated list and star was the separating marker in our concatenated list so based on the star index I had sliced the list into two portions and I have my list l one and l two ready for my next iteration I have found my one matching letter and trying to find another matching letter if possible so I will repeat till all matching letters are found. If there is a position where there is no matching letter in that time I will say that ok this is when I have to stop there is no matching letter I have to stop so this is the part one of the game crossing out the matching letters next comes counting in a circular fashion and predicting the relationship in flames. We will see that in the next part of the video.

# FLAMES 06

Alright guys, so in the previous part you had seen how we can remove the matching letters and check if we have to proceed with finding matching letters again or we can proceed with the counting of the letters. So we had run it using a loop so I hope you went through this thing you could understand this nippet please spend some time here its just a matter of time if you just spend some time you can easily understand what this is, what are the strategy we are using you can understand it, so ok by here we have we have taken two list we have obtained two list and there is no matching letters between these two lists so now what should we do? We should go with counting, how many letters are left out now? So now there is no matching letters, now we have reached at that position J A and P R I there is no more matching letters now we should start counting. So how will we do that? Will do that will take a variable say count just for counting we are counting how many letters are left out so let me say count it is nothing but length of the list l one and the length of the list l two right? Am I right? because J A is the list that is left out to you P R I is the list that is left out to you, the length of the first list is two, the length of the second list is three two plus three is five this is what you want to compute right? so that’s what you are doing, so we should declare the result now after going through the flames letters and we should declare the results so let me say that is result let me make use of a list result the first element is flames F, f for friends so let me say friends the next is love l for love f l a, a for affection m, m for marriage, e for enemy and s for siblings right? so this is my result array so among along this I have to if it is five I have to do it one two three four five I am at enemy so I should cut of it I should take the right part then I should take the left part concatenate it that will be my new result and there I have to proceed that’s how I have done right, so if you remember you would check through the very first screen cast video of this flames you would understand, you can check through it if you have confused here that’s how we generally do will stop at a place will cross out that word and then will take the right part then will take the left part that’s how will do that’s the trick here, had it been just remove I could have done something like that I would say I can start form friends that’s very easier but the trick here is you have to if you had stuck at enemy you have to start from siblings you cannot start from friends that’s where the knack here is so there you have to do it and so let me start coding so let me say where should I stop at this so I have to counting I have to keep counting this way one two three four five in that case what if some other names are given? Some ten letters are in common there are just sic flames f l a m e s there are just six items there are ten letters in common, what will you do? You will have to again start your counting of seven from here right from friends you will that is one two three four five six then seven will friends will again be counted as the seven l is eight, a is nine that’s how you will count it right, and ten is m that is your fourth element so ten mode six is your four that’s your fourth element you want that is as per the humans perception so modularity, modulo operator is that’s what we are using here that is we want to again come back to the starting point and count it this is what we call as modular counting, you keep counting if you reach the end you come back to the starting point and resume your counting that’s what you will be calling it as modular counting and in our computation it is the fourth element but as per the computer indexing it is one less right computer indexing starts from zero, we start counting from one so it’s one less so we have to take care of it so my resulting index where should I make a split so I am standing at enemy I should make a split the right part and left part I should concatenate it right that's what I have to do right so where should I make a split I should find that index let me call it as split index its easier if I give such a name so split index is nothing but I have to find the count modulo operator I have to use because if I exhaust the list I am coming back to the starting point and counting right so I have to use the modular operator this is called modular counting so I have to count it modulo length of the list my list is result I will find the length that is six count modulo six that is the human index and it differs from computer index by one so I have to subtract one so let me compute this first and from this value I have to subtract one. So by brackets I mean first compute this value count modulo length of the result and then from that resulting value subtract that's why I am giving the bracket so then you put minus one ok this is my split index if my split index is what are the values it can take? It can take, can it take negative values? just think for some time, yes, it can take negative value because of there are six letters in common just count f l a m e s one, two, three, four, five, six at siblings you will be standing you cut off that part and to the right you have nothing so what will you say the count was six, six mode six it is zero minus one it will say minus one so it will say minus one which is out of the range of the list right so it can take negative values so if it takes positive values you have you process it in a different way that is if it is struck at middle point it will take a positive value, if it is struck at some of the ends it will take negative value so in that case that is if I am struck at siblings there is no right half right so I should concatenate just the left half so in that case I should process it differently if it is a positive and a negative value so first let us see how to process if it is a positive value then we will see for negative value, let us see if split index is a positive value by positive value what do I mean, it is greater than or equal to zero if that is a positive value what should I do, I have a right half and then I have a left half, right half is nothing but I should start at this, I have to slice the result, result list let's say again here list slicing come into play so I have to slice the right half so this is my split index I have to start at the next point and go till the end so I should say split index split index plus one I have to start at this index and go till the end. Just think for sometime why I am giving this index, because we are on the right half and the next is left half, I want to consider everything except this except this split indexes so what should I give as a end index? Split index because the end index is by default ignore by the slicing functionality, I want to ignore this thing so I am giving that deliberately so I don't need to give the start index because I want it from the beginning that's the default value will be taken so I want to give the end index as this split index. So at this split index I want to take the left half because right half and left half is available so how should you get the new result? Right half followed by the left half right this will be your new result this is the case you have to do in case of this so the thing you have to do in case you, you are struck up at a point somewhere in the middle then you will get a positive value of split index and you can do this in case you are struck at this point you are getting the negative value of split index in that case what should you do? You should take the left half, so left half is nothing but that would appear only when you are struck at the last value, when you are stuck at the last value, your split index is the last value so in case if there are six common letters you will be stuck at siblings there is no right half that's when you will get an negative index. So in that case what should you do you should just take the right half that is wherever you are stuck that split index must be ignored and the other things may be taken right, so whatever is the length of the result that particular point that must be the last element must be ignored the other element must be taken whatever is your result the last element at that point must be ignored and other must be taken aright we will do that ok?. so how do we do it? The last element has to be ignored the others has to be taken so I will do the slicing appropriately I will say result is nothing but whatever is in the initial half I don't care I just want to cut half the last element so the last element must be ignored if I just say length of the result what happens, the entire list would be taken but I don't want the last element so I should ignore the length minus one that particular thing has to be taken care of, why do we give minus one? Because I want to ignore the last element so I will give length of my result minus one so the last element is ignored and you got the result right? so this is has to be repeated right? this will be the run for one iteration like that you have to repeat it till your result array is just result array is being sliced right so at every iteration one of the element f l a m e s one of them getting cut so the result array size reduces by one at every instance when the result array becomes of size one you have to stop and say whatever is left out is my result right? so you have to repeat this process so I will use a loop here I will do as long as the length of my result array is greater than one, as long as there are more than one elements you keep counting as long as there are more than one elements you keep counting and you split the array into left and right halves do accordingly you do that, if you get just exactly one element just stop at that position so now when this loop comes to an end what happens? There is just we have put it in a loop so that it gets repeated as long as there is just one element left out in the array and the loop comes to an end what happens? There is just one element present in the array that is your result. So what should I print? I should print, what is there in my result, result it is a list right so it will be printed with square brackets I want to print it without square brackets and there is just one element present that I know so how would I say, result of zero just the first element whatever is the element just print that answer that's what we are doing here ok. ok I ma sorry here is an invalid syntax, I have to give the name of the array right result so I have done that ok so any other place ok yeah this is where I have taken and fine I guess everything is fine ok so now let me run it before that let me restore this view ok here is the console so here we have return the code maybe you can revise here you can take a pause here and revise what happens now will run it, let us run the code yes run alright now let us run the file, let me see it is asking for person one name let me say ajay, person two name it is let me say priya see it has returned friend as the result so like this you can input many names and yes you will it will compute the flames as per the game and you will get the result please do try this as well as if you have some other strategies of doing it please try that too and discuss with us on the discuss form, we will be very happy to help you with the different ways of solving the same thing, its actually interesting the game looks very simple but when you try coding there are different things to be considered like modular counting and you have to consider the right half then the left half, you cannot consider from the first and you cannot consider common letters along the same name it should consider it among the different name, it's all like there are some intercases which we can we are doing it very easily by if we do it manually but when you are writing a code there comes a lot of logical thinking so you should practice more of such things and also please do discuss if you have played some other games n your childhood and you try coding it and discuss with us what are you have tried and what all you have played or if you want to try for something also please discuss in the discussion form will try helping you out with whatever you want to code so literally you can code anything all that is need is some logical thinking that’s it. So you explore keep exploring a lot and this is how you can learn keep exploring, keep practicing, alright. Thanks for watching, have a nice day.

# DATA COMPRESSION 01

So, imagine you try creating a website for yourself and in that website you have a lot of images and it takes forever to download this images when someone opens your website so what you do is option one you consider removing the images of course that's not a good option given that you want to keep some images. A second option would be to reduce the size of these images by size I mean the bites that the picture takes. So how do you do that? Given an image how can one try reducing the size of the image, probably you will lose the image quality or you might you may not be able to do it for all you know right? Is there a way to do it? Is what we are gonna tell you, we will teach you a very naive very straight forward way of compressing an image. Of course there will be compromise in the quality but then what matters the most is the fascinating idea behind compression of image size by using the very straight forward naive technique.

# DATA COMPRESSION 02

HI guys, in this video we will see a basic illustration of numpy library of python so what numpy is? Numpy is actually is numerical python stands for numerical python so in our precious videos in our previous joys we have seen some part of numpy we have been using numpy a lot of times and to do some tasks to perform some tasks but in this video I will show you what else you can do, how basic operation you can perform in numpy very easily. So what exactly numpy is useful mainly it is useful to perform some matrix operations in python and to do it easily you are using use mostly numpy library so array what say matrix mostly matrix we use numpy and there are lot of lot of matrix if you know linear algebra and matrix and you must know that the application of matrixes are huge in mostly in image processing and even machine learning and ai matrix the applications are a lot mostly if you know matrices and their operations and you can definitely easily cope up with the ai elements and machine learning so even in deep learning so for that we use numpy library a lot suppose some of you wanted to want to machine learning or ai or deep learning whatever it is then you need to understand this library numpy library because you will be using this library a lot there. I will give you very very very basic operations in numpy library which you can perform. So let's see this I will just import the library or let me just oh sorry yeah let me increase this pane in most of the videos I keep both of my pane because I like keeping this but you can when you are working on your programming part or coding part you can zoom out this part, zoom in this part or exclude this part vice versa so let's see this let me import the numpy library so I will write here import numpy as my alias will be np here you can put here it's the convention people mostly put np so you can put np here so you can put x also so looks nice here great. So I will create an array of how do I create array, I have been creating a array of elements before in python as a list for examples see this console here let me ok should be fifty fifty percent ok so let's see this suppose I create a list of name l in my elements are one two three so I will make a list, list is created like this one comma two comma three ok and I will just run it and if I see the elements of this list I tell one two three so same thing I will do but it will create an array what's the difference? What you can do? When you get an array as array just see it is an object when you get when you create an array of let's say elements of one two three you can perform a lot of operations through numpy library which has been return this in this numpy library so you can perform all those operations a lot lot lot of operations there so will make an array of let's say all these three elements one two three first so let's see a is equal to np dot array it will show you here ok np dot array and what are my elements so let's say one comma two comma three so the convention is you pass the list there, the list of array as an argument so np dot array will take an argument an argument will be my list great you just print this let's see what happens, print a let me just run this we will see one two three exactly what I was expecting here I am getting here nothing different I got an array great now let me do some other thing here with this array I have the array just print something let me just write it and you will understand what is it exactly. Type of a ok see showing class numpy dot nd array n dimensional array this is one dimensional it is known as one dimensional array let me print something else one dimensional why because just one row and there are three columns if you see this as a matrix there is only one row this is one two three and there are three columns, first column second column and third column so one row there are three elements so that's why is the one dimensional array ok let me write some more functions let say a dot shape ok let me see this three comma nothing three comma nothing means it has one row and it has three columns, three columns one row great I can print the elements how am I supposed to get the elements same as the list a dot zero let say a one and a two it will give the elements as one two three so exactly same as the list how am I suppose to, how I receive the elements of a list using the indexes same analogy is there ok great I can change the elements also exactly if I want to change the elements of the index let say one and make it let say six and just run it and I can see the array one six three same analyst nothing, great what if I want to create a two dimensional array? Same thing let me just delete all these things I will change array as the two dimensional array so how I put two dimensional list? One list inside that there will be two lists. List number one will be first row ok so first row let's say one comma two comma three ok comma second row is a list which is four comma five comma six that's it let me run it to see yeah, it's running everything is fine you can see a as if you run it array one two three four five six this is first row second row as an array and if you just print the shape of b not the shape of you shape of b shape b dot shape but you will see these sorry what did I put! It's not b sorry sorry ok a dot shape to see two comma three ok so there are two rows two rows and three columns great awesome there are many ways through which you can create arrays and this is this is one way suppose I want to create suppose I want to create an array of all zeros one thing I will just put a list zero, zero, zero, zero, zero I can through loop I can create a list of zeros and I will pass the list that list or I will just directly use that whatever it is but through numpy you can with one line of code you can create such array easily. Suppose I want to create an array of zeros so I will just a is equal to nm dot said zeros and I will just give the dimension as array of matrix of two cross two zero ok I just run it I just see a see zero zero zero zero great or if I want to create an array of one's  np dot ones let me give the dimension as two comma two I will just run it I will see b give me one one one one great ok or if I want to put some particular number in all the at all the places let's say so I can just put np dot full ok and let's say I will give the dimension of array again two comma two and what is that constant let's say I want to give six that's it I will run it I will see c six six six six awesome great or if you want random numbers also that too is possible will just put np dot random dot random so you know random dot random put one zero two one you want from if you want from if you want integer then you can put integer values rand int so I will just give the dimension as two comma two and if you see d see some random values from zero two one between zero to one are coming in this array.

# DATA COMPRESSION 03

So, we saw some basic functionality on num pad in this video we will see some more of these and after that we will start the next tutorial. So let's see this let me import it again import the numpy library again so I will write import, import numpy as np so ok so let me create an array so np is equal to what was the command np dot array and then I will give the list which is let say one comma two. Now, I want to I want to know the data type of this data type of this list or this another numbers what is it? I will just print I will just print as x dot dtype that's it if you run see int thirty two that is thirty two bit int ok suppose I change the array as let's say this as one point o and this is two point o and now if you run this you see a float sixty four bit so this is a floating point value that's why data type is showing is float sixty four great. Suppose you want to be force it to be something else you saw that it was showing int thirty two ok and if I want it to be it t be a sixty four suppose I want more values there which I know can be represented in sixty four bit only I need more numbers more numbers means more digits I mean more binary value I need so I can represent a big number. Suppose I want sixty four bit number to be represented so I need to force my data type to be a sixty four bit so what will I do I will create an array np dot array ok I will give the sorry I will give the list which I say one comma two and I will force it as dtype is equal to np dot int sixty four ok ok and now if I type print x dot dtype you see now it showing sixty four now it is allocated sixty four bit to represent these values, now you can if number which is as big as which can only be represented by sixty four bit binary numbers then you can represent it to before that you would have try to represented to would have been showing some overflow value or some other value. So you can force the number to represent in sixty four bit or even thirty two bit. Great now the best part of numpy is that why we use in post video that I told you that the main reason of using numpy is mostly to do some operations with array and matrix mostly matrix so matrix operations are very very easy in numpy and in most of the machine learning aI based technologies mostly it is matrix operations only. If you see an image it is basically nothing bunch of pixels represented in a matrix form so suppose you see a you see an image of let say five one two cross five one two and the height is five one two and the width is five one two if you read the image in a computer you see a matrix of actually you see a matrix three dimension matrix actually n cross p whatever so three dimensional matrix will be there and each with the size of one two cross five one two why three dimensional matrix because it contains rgb values red green and blue. And each cell of the matrix will contain a value which is between zero to two fifty five representing what range of red it is, what range of blue it is and what range of green it is then mixing all this three you get particular pixel value and then image is displayed in computer, this is the in general way of representing an image or in simple form see a grey scale image grey scale image means zero to two fifty five each pixel will have only value from zero to two fifty five, zero means black and two fifty five means white s whatever number if you are getting will be between black and white only. So there will be only one matrix one matrix you can imagine of let say five one two cross five one two and each cell will contain value from zero to two fifty five only so it's a matrix of the computer sees it as a matrix only and if you know matrix operations easily you can do a lot of things with that with that image that's what image processing people do they see it is in matrix and then they do lot of things with that then enhancement another processing is very feasible with images so let me do a little bit of math with this numpy library so that we get comfortable with the array operations and matrix operations. Ok let's see this I have imported numpy I will create one array let's say x as np dot array ok and let's say two dimensional list will be there so first is one comma two and second is again three comma four ok and I want it to be float so I will just force it using float not float no dot float that is sixty four bit and another array sorry another array of np dot again two dimensional array so first row is let's say five comma six and second row is let's say seven comma eight and again I am forcing it to be float just like that no reason for this just doing it. Float sixty four sixty four bit awesome. Let me see the error first here ok sorry ok so this is awesome let me just run it and I will do all the operations here on my console so its gets easy for me to follow let's see this. I will just bring x plus y x plus y let me see what it does, is it correct can you see whether the x plus y ok or not? If you see that it has done the element by element addition so the if you see the first matrix the two dimensional the first row was one two and the second row was three four. Thus the second matrix the first row was five six and second row was seven eight if you do the element by element addition element by element means what first row first column that is one comma one is one here and for the second matrix one comma one is five, five plus one gets is six so first element is six, since it is float six point is coming, if I just print x then see y ok same goes for the second element which is which is what eight six plus two is eight and third element is ten seven plus three and the twelve great or you can do what you can just print here what np dot add you can just pass x comma y which is which are matrices will show this in image z. Now let me do some other operations let me subtract this two matrices subtraction means element by element subtraction so what will you write one thing I can write is just print x minus y and it will give me the is it correct? Five minus one is x minus y minus four six minus two four oh my god see minus four minus four it gives the subtraction in minus four minus four great or I can just np dot rather than writing add I can just subtract x comma y and it will give me the same result. What else can you think any arithmetic operations next is of course multiplication so I can just write print x multiplied by y and it will give me the element by element multiplication five into one is five, six into two is twelve, seven into three is twenty one, eight four is thirty two. So eight four is thirty two so I am getting or I can write print again np dot multiply sorry multiply x comma y same result or I can even divide the arrays matrices that is x divide by y will give me x divide by y, you can check the result so this is very easy these I am just showing the basic operations you can do lot of other stuffs in using numpy library so suppose if I want to print np dot sqrt which is square root of let's say x only, it will give the square root of all the elements one is one this is what four sorry two square root of two is one point four one, then square root of three, square root of four or same thing I can do it for y also it will give me the element value, there are lot of things you can do with this. So this is just to show one thing is that in programming assignments somewhere we ask how to I think in week three or week four maybe I don't remember, where you have to print the transpose of the matrix nothing else you just have to transport it and print it it's very easy in numpy so let me just delete this part I don't want this float value to be hanging on this whole integer to make it very clear let me just run it and see my x ok one two three four is there, now if I want to print the transpose of this matrix x is very easy in numpy I will just print x dot capital t one three four two that's it you can see one three will become first column will come first row one three two four so if it is small matrix three cross three or even four cross four it is very easy to do this with the way you have been doing right now suppose the matrix size is let's say thousand cross thousand literally very time consuming to write the code actually see even in this numpy library in the library behind the code same thing is going on what we are doing it not like it is magic just doing it, the same amount of time it take to do it but it is easy for us to call this and do some operations that's it nothing else the time actually the same. So, to just reduce the time of the programmer this kind of libraries are build and that's why python is very popular because actually the time program takes is way higher than other programming languages let's say c and c plus plus so python takes more than one and a half let's say one point if a program takes x amount of time, python will take definitely one point five x or may be more than that amount of time because python is just for the programmers to make for programmers to make their life simple that's it and c is to mostly most of the systems are designed in c plus plus so that they are very efficient and they run very fast. Python is for basically to for the mostly the researchers are doing using python to to do their research and see how their things are working. That's it. So these libraries are to make the, to make our life very easy and I very less amount of time we can code. Ok there are lot of things let's say if I want to sum all the elements of this matrix I will just write let me write here print np dot sum and sum of what sum of x it will just print and show me ten four plus two plus three is ten yeah or if I just want to print the sum of first row the sum of first row is just if I want to print the sum of first row I will just np dot sum of tutorial came up x comma x is I will put zero means its sum of all the sorry sum of first column ok so this is the first column this is the second column so column wise it will show first column is three plus one which is four and second column is four plus two which is six so column wise it will give you the result or if I put it x is one then it will show you the row wise so one plus first row is one two so one plus two is three, second row is three four, three plus four is seven so there are lot of other operations which you can do with numpy. I just wanted to give you the how matrix are illustrated in numpy and how we can do lot of operations using numpy on matrixes. So we will be using little bit of numpy library in our next video so this is the very quick tutorial for numpy, you can go through the documents of numpy just goggle it, and many people have written a lot of tutorial on numpy you can just check it, it is very easy to understand and even if you face any problem with the numpy you can just post it on the discussion form. In next video will do something using this numpy on images, Thank you.

# DATA COMPRESSION 04

HI people, so in last video we saw the basics of numpy library and how we can do different operations on matrices and arrays using the python library of numpy. Now in this video I will show you a simple a very naive method to compress an image and how easy to do with the numpy libraries and image libraries now the idea is to motivate you people to see the compression and to understand how exactly image compression another kind all other compression happens and this is the very naive way of compressing so this the compression which I am going to show you in this video today will be lossy compression lossy means we are going to lose some data but still things will be image will be visible and you will be able to see the details of the image but you will definitely use some data though so that’s why this kind of compression lose data after compressing is known as lossy compression. Where as in there are some compression techniques where you don’t lose any data but the size is reduced those kind of compression are known as loss less compression so in this video we will show a very very basic method which you will appreciate and you will see how we can compress an image, so in last video as you have seen that we have told you that image is nothing but bunch of numbers in representing in matrix so whenever you load an image in python or any programming language its nothing it’s just an matrix with some rows and columns so whatever is the height of image or width of image each point of image is pixel, pixel is nothing but value from zero to two fifty five if it is an rgb in rgb scale that it will be a three dimensional matrix but let us assume that we are using some images where the image is a grey scale it means only one matrix and each cell I mean each cell of the matrix will have the values from zero to two fifty five, zero to two fifty five means that if it is close to zero it means it is close to black, and if it is close to two fifty five means it is close to white, in this way you can represent an image in an grey scale, grey scale means there won’t be any colour will be like black and white image but mostly everything you can visualise through it. So let’s see this will take an grey scale image grey scale image and then will try to compress it so I have an image here you can see you search it on Google also lena so it’s a grey scale image grey scale you can see it is a black and white image so whatever if you see if you load it in computer and through python any python library that’s a pil and see this, this is nothing but just a matrix and so the size of the image is five one two cross five one two it means there are five one two row and five one two columns so it means five one two cross five one two pixels are there and what are those pixels? It’s nothing it’s just value from zero to two fifty five so that’s why it si a grey scale image. Ok so what we are going to do? We are going to compress this image now, you all know that zero to two fifty five means it is a value zero to two fifty five means it is value between zero to two to the power of eight exactly two to the power of eight is two fifty six exactly it is a value from zero to two to the power of eight minus one ok so what we are going to do is we are going to map this values of zero to two fifty five to just zero to eight that’s it that’s our compression. So what does it mean exactly? So whatever values you are getting, you will somehow map it eight bit so zero to two fifty five means the values are of eight bit if you do the math then you will understand that eight bit means that you need eight bits to represent a value from zero to two fifty five ok just use your pen and paper you will understand what I am saying you need eight bit integer or whatever you are using to represent value from zero to two fifty five but you want to represent value from zero to eight you just need three bit. So what we are going to do? We are going to map this zero to two fifty five values of the, of this matrix ,of this image to eight bit image to three bit image which means zero to right value that's it that will be our compression definitely we are going to lose some data you will not be able to see the image very clearly but something if you are intension is to do something else with the image or just to see the lines of the image you can compress the images and you can do the analysis in other way you let's see your motive is not to see the image clearly but to do some other kind of analysis and you need very less size image then you can compress such image in an eight bit image to three bit image you can do your analysis so that's what we are going to do so its that I am just using the three bit image you can compress it to a five bit also eight bit you can compress it to five bit or any number kind of thing so let's see this. So I am going to import numpy library import numpy ok and then I will take the image library of image function of pil library so from pil import image great ok I will just load my image which is you can see its lena dot jpg so I have imported the image function this is not image it is actually image function open and then I will then jpg awesome I will just load this image in a new variable lets say pixel map im dot load let me just run it so see it ok yes running if you just type here im dot in my console im dot show show me the image great so everything is perfect here till now ok just as I told you I just want to see this images as the matrix so just using my numpy library I will import this image as an array ok so numpy dot as array is a function too just load this image I will again image dot open and I will pass the image lena dot jpg ok let me just run this let me just see this I. So you can see this is just a matrix so there are I mean dot, dot, dot it means these columns are missing they are not showing the column so you can see there are some numbers bunch of numbers one thirty five, one thirty seven, one thirty eight so it is like these numbers are between zero to two fifty five any number can be according to the pixel brightness and the colour pixels are there ok so what our motive is? Our motive is to note down this numbers this numbers are taking for example one thirty seven you need eight bit you need eight bit to represent these numbers ok suppose this one thirty seven somehow I map it to let say six ok then I need just three bits to represent six I don't need eight bit so for those such way I will be able to for each pixel I will take only three bit rather than eight bit so my overall size of the image will be reduced that's my motive and so let me see this if I select it and see it is written forty one forty two point one kb or if I go to the properties you can see forty two point one kb so forty two point one kilo bites is used in this image. I am going to reduce the size of the image ok awesome let me just remove this line I just wanted to see this line as a matrix this image as a matrix ok I will create one more object of this image same size so that I can whatever mutations whatever things I am going to do on this image I will save it in save those changes in another image so I can see that how that image will be the compressed image and I can relate the size of that new image. I will create a new image of same size so let me write it as img is equal to image dot new and I will take the im dot mode, im dote mode means what kind of im is this object of my original image of other jpg im dot mode means what kind of image, is it a colour image? Or is it a grey scale image? So this im dot mode will send me the type of image, first arguments is the type of image and the second argument is the size of image so I am using im dot size I can pass the size of my original image so I want to create the new image object with same type of image type of means I want grey scale image new grey scale image and of same size so this is a new image blank image kind of see it as a blank image which doesn't have we have not define the pixel kind of pixel it has. What values exactly it has we haven't defined so we are going to defined it according to our mapping ok so yeah this is this is what I wanted to do and I want to note this image and let's say I wrote it their pixel map so pixel new I will write here and img for load ok so my new dummy image has been created now I told you what I am going to do is let me just command some the following lines and I will write what I want to do. Ok so what I want to do from now, I told you that the pixels here are taking eight bit exactly so I want to map this eight bit into three bit so how to do this? Eight means there are two to the power of eight it can contains up to two to the power of eight values it means zero to two fifty five it can have if I want to map it into three bit means it should contain values from zero to eight because two to the power of three is eight only so how am I going to map it? Just divide it if you divide two to the power of eight by two to the power of three you will get two to the power of five ok, it means what is two to the power of five? Two to the power of five is thirty two it means ever thirty for every thirty two gap I will have one number it means if I get a number from zero to thirty one which is a gap of thirty two exactly I will just put zero there if I am getting thirty two to let's say sixty three I will just put one because I am mapping for one bucket of numbers from zero to thirty one so where in this matrix where ever I am getting a number which is between zero to let's say this is twenty, twenty one, twenty four, twenty one, twenty two, twenty six which these all numbers are between zero to thirty one so I will just put zero here other than putting zero to thirty one I will just put zero here because zero will take only I can represent zero in a very three bit value but these are taking more than three bits twenty  eight twenty six so I will just put zero so I can reduce the space ok and same way if I am getting let's say a values between thirty two to sixty three if not here or let's say sixty four to ninety five so let's see all these values are between sixty four to ninety five I will just put two here because two will take only two only three bit number I can represent three bit I can represent to but these values are I am taking with I am representing with eight bit number so that's how I am going to do so let me write it here in the comment section ok. so that you see what exactly I am doing so I will write here what I am going to map so zero to thirty one will be mapped as zero zero, thirty two to sixty three will be mapped as one and let me just delete this so that its everything will visible here sixty four to ninety five you just I am just making a gap of thirty two only is to ok ninety six to one twenty seven will be three, one twenty eight to one fifty nine will be four, one sixty to one ninety one will be five, one ninety two to two two three will be six and two two four to two fifty five will be seven ok great ok this is what exactly I am going to do this see this and according to this I am going to create my new image.

# DATA COMPRESSION 05

Ok, just see this numbers and according to this I am going to write my program let me just ok, let it be there let me just keep some space so that it is visible here ok great awesome so what I am going to do I will loop through the old loop through my old image I will see the values one by one see where it lies in this in these blocks, block number zero block number one two three four up to seven where the numbers are lying and accordingly I will change I will put the pixels in my this new image which is which I have created correspondingly I mean wherever I am getting the number in this image in my old image in that corresponding place of this new image which has created I will put what I am going to assign here so suppose I get a number at let's say four comma three comma two is one thirty eight one thirty eight lies between one twenty eight to one fifty nine so I will put four at three comma two in this new matrix so this will be my new image so I will write it down I will write the loop for I in range of image dot size, size is two sorry get the size of image zeroth is the row number of ok and for j in range image dot size of I sorry one ok so row and column I am going through each row and of each column and then I will check if the old image is which one the pixel map this is the old image which I have loaded in pixel map so if pixel map ok of I comma j is greater than equal to zero I am checking for the first law if if if the number which I am iterating through ok is greater than equal to zero and ok let me ok let me write this pixel map I comma j and it is greater than zero and it is less than equal to thirty one it means this this particular cell value lies between zero to thirty one if it is the case then you write pixel name pixel new I j as I comma j as what will put zero ok let me just we are going to use this line again and again so let me just copy this ok so I will just put rather than if else if else if, if it is not in the first block the value is not lying between zero to thirty one, if it is lying between thirty two to sixty three then put what one copy this because it is our else if part I will put here else if it is not even in the second block so if it is lying between sixty four to ninety five, ninety five is our third block I will put two ok and if it is not there then if it is between ninety six to one twenty seven, one twenty seven, ninety six to one twenty seven is my fourth block so I will put three ok and if it is not there but it is between one twenty eight to one fifty nine then I will put four they're ok it an if it is between if it is not there but in between one sixty to one ninety one then I will put five there and if it is not there but in between one ninety two to two two three two two three I will put six there sorry six there and if it is not there and at last thing which I have to check is if it is between two two four to two fifty five it must be there that's it my work is done here so I have created this new image I have mapped all my blocks to my original to whatever I wanted to so that I can reduce the size ok now what I am going to do I will just save this image with the command image dot save let me write the name as lena is equal to dot jpg ok I will see the matrix also here I saw the matrix as where did I write oh I removed it ok let me write it again so j let me write it that I will write it j numpy dot as array and image dot I don't need to open it I can just I already open it there it's ok lena is equal to dot jpg so I will see this new image as an array and how so my program is complete so let me just run it great so I will saved it let me just check this j you can see all these values have been mapped accordingly if you see this array let me show you one thirty five was a one thirty five was lying between one twenty eight to one fifty nine hence the four is written so all these values all these values if you see it is lying between one twenty eight to one fifty nine that's why I have mapped it with four so the mapping has been done and I have save the image as the lena dot jpg so I will go to my folder and I will see the image, you can see it its looking black but it's not exactly black I will open it and you can see that some part its actually visible the image is exactly same if we look at the image here you can see image is exactly the same image if you see the boundary if you see the face eyes everything is same its just that we are trying to represent eight bit image into three bit image that's why so much we have lost we have lost a lot so it's looking approximately everything is looking black but still you can see the image lining all the edges all the other things are exactly the same so this is compression and if you see the size, size of first image was forty two kb this is just four kb four point eight six kb it is reduced a lot lot ok so this is one type of compression which you can achieve through this just simple naive method there are lot of lot of tons of method for image compression and it is it still a hot topic people have still working on this compression technique. If you see the this image format for example jpg these are also compression techniques and how to represent an image in an very in a very efficient way so that it takes less time but the quality is not compromised so this we just wanted to give you a motivation give you a motivation of image compression and compression in general that's why we shoed this example. You can check out different algorithms for compression and you can even discuss about that those algorithms in the discussion form so that we can we can give you inputs and also we can learn about that. So this was just to motivate you guys how you can how it is simple to see these things through the first principle and achieve such kind of things so it looks like that compression is what exactly is going on the compression when we do the zip and rar of some files and then we see that the size of file reduce and we just copy it or we just send it on email but we don't understand how to what exactly is going on. So these simple methods are actually used in the background to compress those files so image compression is a very hot topic and there are very many ways through which you can compress the image and actually people see the way people study the way we pursue image on based on those studies they do the compression and it is actually it has been very successful venture through this. So please discuss about this technique in discussion form and ask question if you have any. Thank you.

# BROWSER AUTOMATION: WATSAPP USING PYTHON 01

So, we are coming to the flag end of the course and I need to note something, we have tried sowing roughly twenty four saplings so far as different joys in computing and this saplings will grow on to trees by that I mean we have only introduced the concepts the naive way, in fact for every single concept that  we have taught there is an advanced version you to look into it and of course we will help you out if u can ask us for more details, now going ahead with the next joy you all know how watsapp works, ever wondered how to automate it? This very process is called the browser automation so let’s see how you can automate watsapp messages by learning this concept called browser automation.

# BROWSER AUTOMATION: WATSAPP USING PYTHON 02

Hello all, welcome to yet another programming screen cast of joy of computing. This programming screen cast is about browser automation. So first of all let me tell you what is browser automation? As the name suggest I need to automate my browser, if I need to automate my browser I should use browser automation. This is very much possible in python and this is a very easy task to do in python. Suppose I open a website and I click on some links, whatever I do there I need to automate that so in order to automate all the things that I do on a website I need to write a script for it and this script will use python will use the browser automation library of python so whatever I am doing manually here I need to automate it, how can I do that? Well python your rescue will be using a library of python called selenium yes I repeat the library is called selenium so first of all let us install selenium library so please open your terminal window and write pep install selenium so I will be doing that I will just write pep install selenium s e l e n I u m ok so I will just click enter since I have already install selenium library it is showing the output as requirement already satisfied so I will request all of you to install it I have already installed it that’s why it is saying requirement already satisfied so first you need to do here is to install selenium library by using the command pep install selenium now that we are done with installing the selenium library there is one more requirement you need to install here so there are different web drivers available for different browsers for example if you are using goggle chrome there is a separate web driver, if you are using safari there is a separate web driver and if you are using Firefox there is separate web driver so what you need to do here is you just go to Google and you type installation installing web driver for chrome I will be using chrome here so I will use the chrome driver here so the web driver for Google chrome is called chrome drivers so I will just write chrome so there are different links here so go to the download section we have the chrome driver here I will just click here please download the latest version of the chrome driver and it is two point four two as if I click here it will download the concerned web driver for chrome since I have already installed it will not install it, I have shown you the way how can you install a web driver for chrome similarly you can install web driver for other browsers. If you are using safari you can download that for safari too and if you are using Firefox you can download for Firefox too. I just show you so you just need to go to Google and you now need to type install web driver for safari I will say safari so in the download section you have the different web driver for safari too so you can download the safari web driver too if you are using safari so please keep this fact in mind there are different web drivers for different web browsers so you need to install the respective web driver for the web browser that you are using. So now we have now you are done with the installation process first of all we install the selenium library that is used for the browser automation in python and then we also installed the web driver which is also user browser automation so I request all of you to install these two things before proceeding to the programming screen cast.

# BROWSER AUTOMATION: WATSAPP USING PYTHON 03

Welcome again to the programming screen cast of browser automation, I hope you have gone through the previous videos and you have installed the two requirements required for browser automation. First of all you need to install the selenium library and second thing you need to install respective web drivers for your web browsers so if you have these two let us go ahead with the programming screen cast of the browser automation. In this particular screen cast I will be giving you limbs of what browser automation is all about and how can selenium library would help us in doing that. So as and always I will import the selenium library I will just write from selenium I need to import web driver import web driver sorry it is web driver so if it doesn’t show any error that means your selenium library has installed successfully, if it doesn’t show any error that means your selenium library has been installed successfully and if it does show an error that means you need to install you need to check whether your selenium library has been installed or not. So in my case it is not showing any error so that means the selenium library has installed successfully after that I need to initialise the web my driver here so I will take a variable here I will just write browser, browser is just a variable name, you can take any other variable name here since I have imported the web driver I will write web driver dot chrome and in bracket I need to supply the path of my chrome driver I repeat I need to supply the path of my chrome driver here in my case I have installed the chrome driver in downloads so I will just go to downloads and click get info here so here I have the path of chrome driver so I will just copy this and paste here the name of the chrome driver is chrome driver hyphen two so I will just write chrome driver hyphen two I will now click enter here see this is working, the moment I initialise the web driver it is opening the google chrome web browser so it depends on your web driver, whatever web driver you have installed if you initialise that particular web driver it will open your respective web browser in my case it was chrome driver so it is opening the google chrome window here so now we are done with the initialising the web driver, web driver is basically used to open the web browser so I use the web driver to open my google chrome window so next thing what you need to do here is suppose I want to open a particular web site in my google chrome web browser so how can you do that? You will just write browser dot get and in brackets you write the name of that particular website so here I will be just opening the website of selenium library so I just write https colon slash www dot selenium HQ dot org so let us see whether it works or not? So as you can see the selenium the selenium website has been opened there in the goggle chrome web browser window I will show you side by side so you can get the clear idea of how this is working. So here I have the goggle chrome window and on the right hand side I have my ipython console so I will close it again and I will show you how this is working, here I have the Google chrome window here I have the ipython console I will I will execute this statement again since we have already close the window it is not working so I will again initialise my browser window here I will click ok now the it has open now let me open this particular webpage I will click enter so you see the selenium hq dot org has been opened here, you see I didn’t do anything here I didn’t write these particular address in the address bar of my web browser but it has opened automatically I have written a script here in my ipython console so what you need to do here is first of all you need to initialise your web driver after that you can open any web page using the address of that particular web page and this is browser automation and this is called browser automation so now we are done with it, I will be giving you one or two examples of how can you automate your browser using your selenium library now that we have opened the selenium web page suppose I want to click on this link name download how can you do that using browser automation, this is very much possible in python let me show you how can you do that. So I will just go t my ipython console and I will write variable name here it is elem is equal to browser dot find element underscore by link text by underscore link underscore text and the link text here is downloads so I will supply the link name here is download it should be same as the link in the web page and if it is in upper case you should write it in upper case if it is in lower case you should write it in lower case so I will just click enter here so elem stores the particular length to download and now if I write elem dot click it should click on the download link so I will just click enter, you see the download link has been opened here so this is very much possible in python and very much easy to automate your browser so this is how you can open a particular link on your web page you just used to use the method find element by link text and using that you can click on any particular link and you can you can automate your browser. So we are done with this now if you need to write something in this search bar how can you do that? Well this is also very much possible in python. For that you need to double click on your web page and you need to select the inspect option here. Double click on the web page and you need to select the inspect option here I will just do that till open a window like this which has elements, consoles, sources, network well you need not go into the details I will show you here in this you need to know the id of this particular search bar. How can you do that? As you click on this particular statement it will show you the it will show you the concerned area, here we have the body ok so there I will click here and it will show me the particular elements present on the web page if I keep on clicking on this statements so I need to go to the search bar I will search the search bar here where it is present I think it is present here, so I will just click the down arrow after that here it is header I need to search the header only because the search is in header bar I will just click here, here I have the id no I need to search the edit page no I need to search this search selenium yes it has come to search selenium and here you can see it is showing that label for q and here I need to write in this particular text box its id is q as you can see here input type is equal to text id is equal to q and if I take my mouse there it is showing me that this particular id corresponds to this particular text box as you can see this particular id is corresponding to this particular text box so the id is q so I will just go to the my ipython console and I will write search that is again an variable name search is equal to browser dot now I need to find the element by id so I will write find underscore element underscore by underscore id and the id is here q simply q so I will just press enter here so in search I have the text box and to search I need to send some key so I will just write search dot send keys whatever I need to send there off course I need to search the download so I will just write download it send underscore keys download ok let me click enter here let us see whether it is working or not, see it has the download thing written here in the search box I will just close it as you can see it has the download text written here in search box I didn’t do anything I just wrote a script and automated by browser and as you can see browser automation is a very easy thing to do in python and it is very very interesting too, you can do a lot of things using browser automation, this is just step of the ice berg so now we are done with that. So I will just go one by one what we did in this programming screen cast. First of all we imported the web driver from selenium after that I initialise the browser variable here in which I supplied the web driver path after that what we did we opened a particular web page using the browser automation using selenium library after that we opened the link using find element by link text after that what we did at last we search we search for the text box in the in the inspect element. How can you do that? I will show you again that it becomes clear to you, what we need to do? You need to double click on your web page after that you need to select the inspect option here inspect option it will show you a window here and if you keep if you over your mouse it will also show what this particular element corresponds to this particular element in web page for example in this particular case the body it corresponds to the all of web page and if I click the down arrow here I have the various element I need to search for the search text box I click on div id is equal to container here after that I have different elements first of all it is header then it is m body then it is footer as the search box is in header I just click that down arrow of that header after that I need to go to the particular text box so what I do here is I click on this id is equal to search box after that as you can see you keep on clicking on you keep on clicking on the down arrows and you will get this search box this is the label for the search box and then if I click here input type is equal to text and id is equal to q here it is showing the particular text box so this is concerned text box that we are looking for so its id is q so we have used the method find element by id and I have stored this particular element in search variable after that I clicked after that I sent keys to this particular text box I send the text download and as you can see it has written that particular text in the search text box so this is how you can automate your browser and I will advise you to explore more in browser automation as this is a very interesting topic. So in the next programming screen cast I will tell you how can automate your watsapp using browser automation.

# BROWSER AUTOMATION: WATSAPP USING PYTHON 04

Hello all, in the previous programming screen cast we saw some interesting things you can do with browser automation, this is clearly very interesting topic to do in python. Here in this particular programming screen cast as I said earlier I will be teaching you how can you automate your watsapp using browser automation. And for that what you need to do here is as and always you need this selenium library as well as you need the web driver here too will be using the web watsapp to automate your watsapp so what you can do with watsapp automation? For example you need to send hundred messages to your friends or to your any other particular group you can do that using python you don't need to write the message again and again, you can just use, a for loop and your friends using watsapp using browser automation, well I will advise you not to do that, I will be giving you glimpse of how can you do that using python. So first of all let me write let me write from selenium, selenium import web driver as we did in the last programming screen cast. After that you have to import two or three things here so I will just write from selenium dot web driver dot support dot uI import web driver wait after that you need to import from selenium dot web driver dot support. Import expected underscore conditions as ec next you need to import from selenium dot web driver dot common dot keys import keys after that again one more thing from selenium dot web driver dot common dot by import by next you need to import the time library so that now we are done with importing the concerned libraries so after that I will just write I need to initialise my driver so I will just write driver here I am using the variable driver in the previous programming screen cast we used browser here I am using driver in order to show you can use any sort of variable name so I will just write is equal to web driver dot chrome and as explained in the previous programming screen cast you need to supply the path to your web driver here I have the web driver in my downloads folder so I will just write the path here. It's in users simran setia downloads and in downloads I have the chrome driver chrome driver hyphen two. Then I need to open the watsapp window so I will just write driver dot get and you need to supply the web address here so I will just write https colon slash web dot watsapp dot com I will also write wait is equal to wait web driver wait is equal to web driver wait driver it will wait for some six hundred time units and first of all we will run this and see how the watsapp appears magically through this and yes your only watsapp account will appear in it. How can you do that? Let me see, let me run it. I will just write watsapp using python dot py so it's basically capital w please look at it this, this is capital w not small w so I will run it again it cannot import keys, keys is again capital k I am really sorry for that as you can see the web drover is doing its job it is opening the concerned web browser. So let me open it awesome as you can see you have the watsapp window right here in your chrome browser you can also see there is a QR code here so I would request all of you to go to your settings in your watsapp and then click watsapp web slash desktop. If you click here you will get an option to scan QR code please click on the scan QR code and scan the QR concerned QR code here. I just did that and as you can see you will have my watsapp window here, as you can see you have my watsapp window here and what next we need to do here is I need to send some hundred messages to my friend so how can you do that using browser automation? And this is very much possible using selenium so what you need to do here is first of all you need to supply a target you need to supply a target frame whom you want to send these messages. So I will just write target is equal to first of all you need to write the single quotes and then in double quotes you need to supply the name of your friend so I need to send messages to Ravi kiran so I will just write Ravi kiran and then you need to write the string the message that you have to send to your friend so I will just write message sent using python, this is the message that you need to send so I will just write the comments here enter your friends name enter your friends name here the message you need to send to your friend you need to send to your friend. So please take care of these two things according to your watsapp after that what you need to do here is as in the previous programming screen cast we located the download link we also located the search text box using the methods like find element by id or find element by length text. In this same way in your watsapp window what you need to do, you need to locate the particular message box the particular message of your target friend here I just wrote ravi kiran so here I have the ravi kiran I have ravi kiran message bow I need to locate this particular message box using this inspect to so how can you do that? You just double click here and click on the inspect option after that you can see you have the different elements and different elements stock here now I need to locate ravi kiran here how can you do that? As and always we will start with the body here we have the body of this particular web page then click the down arrow and then click again as you can see there is no particular thing for this message box here we have the different message box under the span tack I repeat here we have the different message box in this span tack if you have anything in this span than how can you locate it I will just show you how can you locate the particular message tag or how can you locate a particular element if it is under span tag instead of div tag please note this fact in the previous programming screen cast located elements that were under dif tab but here we have the different elements located in this spanded so how can you do that? This is very much simple in the case of span, so how can you do this using python so I will just write variable here is s underscore arg is equal to inverted commas slash slash span and in span you write contains at title, title is an attribute here what title are we looking for? We are looking for the target so I will just write target I will just close the inverted commas here and I will write target, target plus space inverted commas then round braces then square braces so we are done with it after that you need to write target is equal to wait dot until ex dot presence as you can see you have the presence of element located yes you need to use this presence of element located element underscore located so how you need to locate the element! Using x arg and x path so I will write by dot x path comma x underscore arg that has been supplied here then you need to close the braces when this is done you need to click on that particular target, target dot click this function has been alternate used in the previous programming screen cast target dot click now we have this target is equal to wait dot until ec dot presence of element located. So I will just check it then there should be two braces not one so what does this function do? This function locates for the particular target which we have supplied in x underscore arg and it will keep on locate till it doesn’t find it so there are is a function for it that is wait dot until we will be using this function to locate that particular target. Now it will click on that particular target so let us run this programme again and see whether it is clicking on that particular friends message box or not so it will again give me a QR code so I need to scan this QR code again so as my watsapp appears so as you can see it is not clicking on that particular target it may be we did some mistake here, here we have some mistake target is equal to when it takes two positional arguments but three were given so let me check where is the problem we need to supply one extra brace here one extra brace here so let me run it again. I need to scan the QR code again in order to check whether it is clicking in the supply target or not here we have the watsapp as you can see it has clicked on that particular message box ravi kiran's message box as you can see on the screen so it is working the code till here is working what next we need to do here, after that now I have clicked the target and the next thing is to send the messages to those to that target, for sending messages I need to locate the text box for sending the messages I need to locate the text box and in this particular thing how can we locate the text box? You have various methods like finding the element by id or finding the element by link text in this particular thing we will be using find element by class yes how can you find element by class? Again go to your watsapp window and click on inspect. How can you inspect it? So I will be using that particular method there is it there we have the body then I will be clicking here I need to locate the message box, so how can you locate that? Here I will click here where is the message box, yes I need this particular window after that click the down arrow again no I don't need it no, where is the message box? Yes in footer we have the message box so click on this down arrow after this, no I need to locate this particular message box and then this so as you can see it is under class underscore one p lpp underscore p llp as you can see here div tab index is equal to minus one class is this. So I will be using this particular class to look at the text box so how can you do that? I will name it as input box sorry I will write I will name it as input underscore box is equal to driver dot find element by class by class name underscore name and you need to supply the class name here so it is underscore one plpp underscore one plpp this is how you can locate any particular element using the method find element by class name. So next is I need to send messages so I will just type for I in range fifty let me annoy me friend by sending fifty messages and then I will write input dot not send keys and in that I should write this string and after writing this string I should also press enter so I will write keys dot enter keys dot enter so I think we are done with the code so let me run it and check whether it is working or not. So we will have the watsapp window here again, let me scan the QR code again now that I have scanned my watsapp window as appeared as you can see I am able to send fifty messaged to my friend ravi Kiran and I am sure he will be very annoyed with me what I am doing here, anyways I will explain it so what cool method I have discovered to annoy him, so in this way you can send various messages many messages to your friend and annoy them I would recommend you not to do that but it's your wish so this is how you can automate your browser and you can make your browser do many things using selenium and by just writing this script so now that we are done with it now I will go through the program again. First of all you need to import all these libraries here and I have already mentioned that web driver is a web driver is the library that you need to import from selenium after that you need to supply the path of your web driver in my case it was download so I just supplied the path of downloads and chrome driver too this is how you can do it. Next you need to supply the web address here as we wanted to automate our watsapp so we used web dot watsapp then you supplied the wait parameter if you want your web driver to wait in case in case of slow internet connection the web browser doesn't responds sometimes so this will help in tacking that from tackling that scenario the wait parameter here. After that you need to supply your friends name and then you need to supply the message that you need to send to your friend. Next you need to locate the message box of your friend in order to locate the message box of your friend you need to locate the span tag here because it comes under the span tag, span tag is basically collections of elements so you need to locate the particular message box using the attribute title. After locating that message box you will supply the wait parameter here again so your program will wait until the presence of that particular element is found or not and then you need to click on that particular message box and then you need to click on that target then you need to locate the input box that is box two for that you will be using diff tag for using the diff tag we have shown the various methods like find element by class name, find element by link text and also find the element by id in this particular thing we use the class name in this particular thing the class name is underscore one elpt it may be different for you guys so I have shown you how can you locate a particular element using the inspect tool you can do that. After that one what number of messaged you want to send you can have a for loop and then you need to send keys to that text  box first of all the message you need to send and one thing more you need to press enter after you write the message so I just write keys dot enter I just wrote keys dot enter that's how you can press enter in browser automation so as you can see me did this and my friend is actually annoyed with this because I send this fifty messages to him so as you can see we send this fifty messages by using browser automation. I hope that this programming screen cast was clear to all of you, if you have any problem please post on discussion form will be happy to help you thank you.

# FUN WITH CALENDAR 01

Our next joy is what next goes by the name calendar arithmetic. Some of us are very bad with calendars you know what I mean. It’s very difficult to note what day it is? What time it is? And assuming you are calling a friend who is staying abroad you find it very difficult to figure out what time zone is he in and what could be the time there and is it right to call at this time or not correct? So our next joy is going to be exploring what one means by calendar arithmetic.

# FUN WITH CALENDAR 02

Hello guys, welcome to this programming screen cast on fun with calendars so in this joy will be answering some questions regarding calendars and time that is date and time will be answering some questions regarding calendar and time that is sometimes that may even turn to be tricky as well so let me give an example to say why do I say it's tricky. For example assume that suppose your spouse asks you a question, do you remember the day on which I am born? See she is very intelligent she is not asking you the date because most of the people what they do to is to remember that they would set their passwords to date of birth of their spouse something like that they do so even spouses are becoming intelligent they don't ask for the date they are born, they are asking the day on which they were born. If such a question is post to you what will you do? You as a life partner you are suppose to remember it that's what they expect from you but as a human it is difficult for us to remember this and also in India if you see it's not just about you and your spouse it's about very big family comprising of your family members your spouse family members your children and so on so that a and this is a very big list if you see an Indian family scenario. So its humanly impossible for us to remember days and even come this involves this many people, so remembering dates itself difficult in that case if they specifically ask what is the day, what will you do. Is there a easy way? Can python help you relief from such situations, is there an easy way? Let's explore that in this joy. This is not the only question we are going to explore, we will be exploring some more cool questions as well so we will see them in detail as we go by so before going I will say what are the libraries that we will be using in this joy that makes our life simpler the libraries that I will be using are you can see the list here its calendar, date time, pytz tz stands for time zone so this is the hint I am giving so just guess what would I be having in my mind of what task can we doing with python and time zone given the word time zone what can we be doing let's keep this suspense for now, I will reveal it slowly and before getting it into the actual task we will explore these libraries will see some basic tasks that may help you in understanding the task that we planned to do with this python libraries. You will be seeing pre requisites that are required for you to do the task needed in this joy in the next part of the video.

# FUN WITH CALENDAR 03

Alright guys, I just post you a question that you would have faced in your in your real life or you would have heard of someone saying that I face such a situation. So in this joy we are going to tell you away by which you can quickly get relied from such tricky situations before getting into that actually as I had said we will be seeing the pre requisites that is needed so the libraries that I have said so for we will go with them one by one so the first thing is I want to current time, so how would I do that? So I would say I need the current time that’s my first that’s the first thing I am going to do current time is what I am going to do now. So for getting the current time we will be using this package date time as you had seen in the previous video the package is from that the date time package has the feature by which we can get the current time so we will be importing it date time is a bigger package from date time from that bigger package we will be importing a smaller subset of it even that is named as date time so get used to this also date time is a lengthier name so let me abbreviate it as dt just for our convenience sake so this is the way I have imported day time is a bigger package it has sub packages if you just want to work with dates you have something called date if you just want to work with time you have something called time and you have a lot of things or maybe you can think of it may be I will say let me say import day time let me say I have imported so if I say day time dot and I press a tab see I have plenty of options that is available time time, time zones so many options are available this is the name package that we have so ok I don’t need this date time actually I want the sub set day time this is the super set I want the sub set so let me do this again so that this particular thing that is imported now I want to display the current time how would I do that? Simple dt dot now that’s it see this is the current time two thousand eighteen ten four sixteen ten thirty three and this is the mille seconds count so that is fourth October two thousand eighteen the time is sixteen minutes it is given in twenty four hour format evening four ten thirty three seconds and this many number of mille seconds so this gives in this precession and let us see what happens if we print it, so if I say print day time dot now if I print it let us say see it is given in a lightly organised format compared to giving in a individual value in a tupple format as you could see it returns a sort of tupple format now it is giving in a organised format and you could see this is reverse we generally we write it as four ten two thousand eighteen and if you have used some software like excel and all you are supposed to give the month first followed by the date, it should be ten four it should be ten four not four ten if you use such packages it is giving entirely in the reverse format why is this so? This is the format generally followed in the data bases when you store data value in data bases it will first store the year then the month then the date so you would have heard that computers generally are comfortable processing from right to left right so if you see right to left it is the human friendly format so probably this is the reason why data bases use this storage representation so the same representation is adopted here as well so first comes the year then the month then the date it’s just the reverse of what humans are doing generally and time comes this way sixteen hours that is evening four o clock eleven minutes eleven seconds and these many numbers of Millie seconds that’s the time so now this is how we got the current time so this is nothing but the system time whatever was the time here in my system this is the same time being fixed that is when this instruction was executed this was the this was the exact time of the clock in my system that’s being fetched here.

# FUN WITH CALENDAR 04

Alright guys, in the previous video we had seen how we can fetch the current system time may be will have a very quick recap of that too because we will be seeing and extension of it basically from so let me give a recap from date time that is a super set import the sub set date time given the name is big let me abbreviate it as dt and to get the time what did I do I said print dt dot now what is the name now is displayed, fourth October two thousand eighteen sixteen twenty two four twenty two seventeen minutes these many Millie seconds this is the time. Alright so if I want to know the time in other locations say Singapore, America or whatever I want to know the time so what should I do? So they follow something called an time zone so as you know the standard time that is calculated by the green witch meridian in your school geography you would have studied using the green witch meridian that zero degree longitude is taken as the standard and time is calculated and our Indian time is gmt that is gmt green witch mean time that is the standard timer standard time that is considered and from that gmt our Indian time is heard by five hours and thirty minutes gmt plus five thirty is our time zone like that there are different time zones so I want to know what is the time zone of in that time zone of Singapore. So what should I do? Here I need to work with time zones right so I need to import something called a time zone I need to use something called a time zone for that I need to import the package pytz import py py for python and tz for time zone so I had imported it and now I need to create the time zone object this is how python works, you can know more of what is an object and of all if you had studied course called object oriented programming you would have an idea if not also you will be learning it that's very simple it's not a rocket science you can quickly learn it just know that something like time zone special thing something if you want to work with time zone you need to create some special object this object you are going to create it now. So how would you create it? Simple pytz dot time zone what is the time zone you want to know that you have to give it as an argument here, you have to give it as argument I said Singapore so let me say Singapore. Singapore is the this thing I want to know so I have created a time zone object that is I want to know the time in this particular time zone of Singapore whatever is the time zone followed that is time is followed here I want to know this particular time so I have created an object and now simple just I have been printing right printing the time right with this I need to pass this object time zone so if I pass this object time as per Singapore it is displayed it is eighteen it is evening six o clock fifty five minutes thirteen seconds these many Millie seconds so this is plus zero eight is nothing but gmt plus eight hours, hours is Indian time is gmt plus five thirty right so two and half hours Singapore is ahead of India so gmt plus eight hours so this is how the time is displayed so you can give up specific time zone and know it and so how would I know what I just know the country I don't know the time zone what if that is the case, you can always look it up like they have name the time zones differently our Indian time zones is named as Asia Kolkata so they had given some time zones they had just given the name for the country for some time zones they had given some prominent cities in that country as the identifier name something like that this is the identifier Singapore is not the this is not the time in Singapore in and around Singapore whatever is the countries that follow this time zone this time there so this is the identifier for the time zone region Singapore is an identifier for some regions they have given the country name for some they have given the prominent city name in the country and for some they have separated with continent followed by the name of the city or country or whatever that's how they had given they are different time zones they are defined in python so you have to give that specific value here what if I don't know the time zones that are present in python. How will I get the list of all time zones present in python simple even that is defined already, simple you just need to type pytz dot all underscore time zone time zones just type it see you got a list you got a list of all time zones see oh my god there are so many I am not able to understand where it starts there are so many so ok let me do a thing let me see how many are present there is a list right so we want to know how many are present there so I will use the functionality length so let me use it let me say length of this list five ninety one five hundred times of five hundred plus time zones are supported here in python so may be there may be some variations if you have because in my Linux system I remember getting some five seventy range answer in windows version there is five ninety one may be there is an update software or whatever be that when you are trying there may be a variation if there is an update you will see an different number but currently these many time zones are being supported in python so these are the different time zones so here you can look it up as to what is the time zone that you see pacific, Portugal, Poland, Singapore, turkey you have different things so you can look it up as through what this is, what is the time zone that you want to know so green which is the standard time so Greenwich is our midpoint and all these are behind gmt and all these are ahead of gmt so will get something like that oh no there are arranged in ascending orders I am sorry for this I am sorry for the info if that is the case then India should have come there given that I didn't spot India realise that they have arranged that in alphabetical order so the alphabetical order they have arranged the names see Asia Kolkata is our Indian time zone you could see it here Asia Kolkata this is our Indian time zone, Asia kualalumpur is for Malaysia Kuwait you have a lot KarachI Pakistan, Kabul, Jerusalem you have a lot you can see through the list there are some five hundred plus time zones so this is how you will be working with the time zone basically you will create something called as time zone object and pass that object to your day time dot now functionality you will be passing it, to get the time as per that time zone so in that case what was the output let us see. Just revisit it once we had created an object and now we want to print it, what is an output? Its gives the date time and it says what is the difference from gmt plus zero right means its eight hours ahead of gmt so that's what is known from here that's what is the output that is earthling just that you need to create an time zone object and just pass it through the now functionality this is how you deal with different time zones. Alright guys the question that I have addressed initially the suspense and is given a date how would you find which day it falls on that still pending right we will see that in the next part of the video.

# FUN WITH CALENDAR 05

Alright guys, so in the previous videos I had post you a question and I have we had seen some we have been slowly proceeding to answer that question. So we have seen how can we get the current time from the system also we have seen how to work with different time zones and how to get the current time as per that time zones. So we had seen all those in the previous videos. So now will address that question given a date how would you know on what date of the week it falls, so that is done by using a library calendar let me import it import calendar if you need you can use a shorter name using as key word you can abbreviate it you needed I just use it as the same as it is calendar so there is a functionality here as you would see, calendar dot see so many functionalities come if you would just give a question mark I guess by this time you would be familiar with this so I have been skipping that part that if you need a help of what this particular command is just type the command and put a question mark at the end you will get to know what is that particular command so isleap, leap days so many search functionalities are there see you have a lot of functionalities alright week day this used to be of some use to me so let me ask what this functionality does. Let me ask see the signature is calendar dot week day year month day year, month, day from nineteen seventy you need to pass some years and in month between one to twelve date between one to thirty one you need to pass it and it returns the week days zero to six which denoted Monday to Sunday it is zero is for Monday, one is for Tuesday, two is for Wednesday, three is for Thursday, four is for Friday, six is for Saturday sorry, five is for Saturday, six is for Sunday so just enumerated from zero to six there are seven numbers there are seven days just write it from Sunday to Monday this is how they have mapped it and it will give the corresponding index of the days, so this is how they have implemented this functionality so let us experiment with it calendar dot week day we need to give the year let me say the same year two thousand two thousand eighteen it is some month so let me say July some day someday let me say sixth July what is the day on sixth July? It’s says four, four is nothing but is zero is Monday, one is Tuesday, three is Wednesday, four is Thursday so this day sixth July two thousand eighteen was a Thursday it says same like this you can change the different dates and time and you can see it and if you could see it says zero to six is this it may be trickier it may be really urgent so we need to do something for this right so we will do that in our programming screen cast. How can we do that? And also there is some more intercases that’s need to be looked into will see that in the upcoming part if videos.

# FUN WITH CALENDAR: 06

Alright guys, so in the previous videos we had seen some functionality this library will offer that will be helpful for us in achieving our task so that we will be doing two tasks in this joy. So let's go ahead with the tasks one, task one is given a date, can you tell what day of the week it falls on? Is it a Monday or Tuesday or a Wednesday or whatever what is the day of the week that particular day falls on? Input is a date output is you have to tell what day it falls on. That is what we want to program we want to model it. So what is the library that we will be using, we had seen in the pre requisites as well we have the library calendar and we have a functionality week day that actually returns you the day on which the particular date falls so before this we need to see how to get the day because humans have different format of inputting dates I can say three Jan two thousand seventeen three one two thousand seventeen in excel it is one three two thousand seventeen there are different formats so how to take the input format that is something we need to agree upon as well as there are some more things so let us see that ok. So this is the procedure first thing is you need to input a valid date so what do I mean by a valid date? If I say forty seven two thousand eighteen it's an invalid date, none of the months have the date called forty probably I wanted to type four and it's a typo it had become forty in such cases we have to handle it so that has to be taken care of as well as if I say thirty one nine two thousand seventeen, is it a valid date? No because nine is the month September it has use thirty days so few months have thirty days few have thirty one days as well as any month the date will never exceed thirty one that thing has to be taken care of also just not just month place a role here year place a role here if it is a leap year there are three sixty six days in that year otherwise you have three sixty five days so that extra days is accounted in the month of February so if it is a leap year February may have twenty nine days and otherwise you will have twenty eight days so you cannot give a bound that thirty or thirty one if I say thirty February that's not a valid date right so I get reminded of comedy that comes person says that I will repay my loan by thirtieth of February something like that such things if a person is giving such a input to your computer if your program is given such an input your program must be robot enough handle all these things so will have to check if the inputted date is the valid date otherwise you need to keep getting the inputs as long as he doesn't give the valid date don't stop getting input o say that it is an invalid same keep getting inputs, if you get a valid date stop and use the built in functionality and get the week day also another problem that we had seen it returns the index of the day zero to six it's difficult n human minds to say what day this is we are comfortable with Monday Tuesday representation not with zero to six representation. So we need to do some inter conversion here as well even that as to be taken care of. So we will see one by one what is the thing So first we will see about the valid dates, will see ok so as I had said this month of February is a special month if it is a leap year you have twenty nine days else you have twenty eight days else and what about the other months? Ok so the month of February is a special month you have to decide how many number of days are present based on the year if it is a leap year or not a leap year right so how would you determine if it is a leap year or not a leap year. I hope you are familiar with this procedure you had seen in your birthday paradox joy so still let us revisit it. How would you detect a leap year? If it is a century year, what do I mean by century year? It is a beginning of a century beginning of a new century something like that how would you detect in simple words it ends with two zeros. The year sixteen hundred, seventeen hundred, eighteen hundred two thousand all these are century years. If it is a century year first thing you have to check is it is a century year, if it is a century year you should check for divisibility by four hundred, if it is divisible by four hundred then you can say that's a leap year. For example nineteen hundred it's a century year so nineteen hundred and four hundred check the divisibility, nineteen hundred is not exactly divisible by four hundred hence nineteen hundred is not a leap year where as if you say two thousand it's a century year so check for four hundred divisibility two thousand is exactly divisible by four hundred hence two thousand is a leap year, if it is not a leap year you should check divisibility by four so basically given a year the procedure is you need to check if it is a century year or not a century year. If it is a century year check for divisibility by four hundred otherwise check for divisibility by four if the divisibility is not satisfied say that it is not a leap year if it is satisfied say leap year. This is a procedure by which we get to know a year is a leap year or not. So this is about February and it is the number of days is determined if it is a leap year or not based on this factor this is determined so we have seen how to determine a leap year or not so the other months other than February what is the pattern? Somehow thirty days somehow thirty one days will see that alright other months we can divide into two halves so months till July if you would see the months January, march, may, July all these months have thirty one days they are odd numbered months January is one march is three may is five they have thirty one days, the other months February we have considered earlier so consider February here  consider April june the other two months these months are even number April is number four june is month number six the month these months are even numbered months they have thirty days this is the pattern that is observed till july so from august there is a change in pattern. What is it? It's simply a reversal what do I mean by that? That is this so from august the odd numbered months September November ninth month and eleventh month November have thirty days and the even numbered months from august, august is eight right so from august eight October ten and December twelve these months have thirty one days so this is the pattern from august so there is a change from august till july the pattern is reversed odd numbered months have thirty one days and even numbered months had as you could see here till july this is a pattern odd numbered months have thirty one days and even numbered months have thirty days. From august there is a reversal as you could see odd numbered months have thirty days and even numbered months have thirty one days. So based on these factors we should check if a given date is a valid date or not. So based on this factors we have to check it and use the library functions and display what is the date. What is a day what is the day of the week given a date that's what we will be doing in the programming, we will see that in the next part.

# FUN WITH CALENDAR: 07

Alright guys, so in the previous videos you had seen the different functionalities being provided by the libraries there are pre requisites then you saw the procedure by which we can calculate we can find the day of the week given a date so as I had said if we get an input from an user we need to be careful with something extra facts like there may be a typo or may be thing that is out of range basically that’s not an invalid thing may be enter you need to take care of all those things we had seen we had seen an algorithm as well sort of an algorithm the procedure we had seen so will go ahead and start coding will gets our hands dirty with coding so let’s start. So the first thing is something that is easier is you need to get year as you have seen in the documentation in our pre requisites videos python supports from the year nineteen seventy so user enters year before that so let’s get started with the coding let’s get started with the coding. Python supports from the year nineteen seventy right so the user enters the year before that you should say that please enter something from nineteen seventy some message like that and ask him to enter it so we will do that part first we need to get the input of that year so let me say year year is nothing but I need to get an input, input generally it would be in a string format that’s how python works so we need to type cast it to integer input I will have to say enter the year let me say nineteen seventy to the current year two thousand eighteen enter any year let me say that I will have to check he has entered something like this only if he has enter some other year before nineteen seventy that is something less than nineteen seventy he has entered we would say it is an invalid year so I will have a check here if year is less than nineteen seventy I should say print let me say have to say enter a year in the range nineteen seventy to the current year two thousand eighteen so I have to save this and again he have to give an input right so again I need to take it and this must keep going till he enters an valid year so I need to put this in a loop let me put while one I need this has to be continuously done but while one is continuous loop there is no wareness nothing but indication for true so as long it is true I am just giving the condition as true so it will pass the loop and it will keep running so when should I stop in case he has given so while one is nothing but an infinite loop as you could see that is it will keep asking for inputs and keep checking if it is an invalid input yes its fine you go ahead and asks for input again if it is valid input again you ask for input is it something fair right no right we should not do that so in that case we have to break we have to come out of the loop we have to quit. So that’s called break else that means the year is something from nineteen seventy that is nineteen seventy or above the value is something like that then you near out of the loop is what I mean. I hope you are understanding this part you get an input check if it is before year nineteen seventy in that case you ask him ask the user to enter again if not come out of the loop you do it inside an infinite loop and once then user enters an valid year to break to come out of the loop that’s how the working is. So this is for year and another thing is month a similar pattern will follow for month as well so let me just copy paste this block and I will do the modifications there, let me copy let me paste so here this is month, month enter month, month will take the value from one to twelve one to twelve right January is the first month and December is the twelth month it takes the value and now the checking ahs to be month is less than twelve and what if someone enters minus three is it a valid month? No right it has to be a positive value so you have to check for the positive value as well and month is greater than zero greater than zero is positive and it should be its not less than twelve its less than equal to because the twelth month exit right so it should be less than equal to twelve. If that is the case it’s the valid input so I will remove this I will change it so if the month is within twelve it is a positive value and if it is within twelve then it’s a valid input so I break out of the loop else I will have to say print what shall I say enter a valid month from one to twelve so let me say that I will say enter a valid month from one to twelve. So that is how I got an input from month so I hope you are understanding this snippet its basically you are inputting month checking if it is within the bounce that is it should be a positive value or maximum allowed value is twelve, if it is within the bounce just say ok you are done getting the input otherwise you say that it is the invalid month you have to enter some valid value this is the range of the value and you continue asking for inputs so that why we are using an infinite loop. Once the input is a valid input you break out of the loop right, so this is how you have to get the input for month year month as well as date. For checking date you should check if it is a leap year or not also you should based on that you have to check what is the month that is being inputted and how many dates how many days are allowed in that month. So you should check that and you should get the input right, so for that first thing we have to do is we have to check for leap year. So let us define a function for it, leap let me say leap is nothing but check for leap year check for leap year. What is the input? The given year so I have to check for leap year then same like this I have to have to get the input for date basically so I have to get date let me say date enter the date and for date we don’t have a fixed range something like that so let me remove this part but still you should check if it is a valid date or not so let me say I will use a function here as well I will use a function check valid date so what are the factors that determines if it is a valid date or not? The entered value of date then the month the entered value of date the month and the year and the factor that it is a leap year or not. All these factors determine if it is a valid date or not, so based on that determine if it is a valid date if it is a valid date you break that is you done taking the inputs otherwise you have to say that you have to enter a valid date. Enter a valid date this has to be the message that has to be printed. I hope you are understanding the flow so let me just give you a very quick summary if what we have done till now. The first thing is we are taking year as an input python supports from nineteen seventy so if someone enters an year before that we will have to say that enter an year in this range will have to say that from nineteen seventy to two thousand eighteen the current year enter some year or maybe even if he enters some other ear its fine but ok let me say let me modify this message as well because you can enter anything after that. So let me say from nineteen seventy that’s a very nice message from nineteen seventy two is not because currently it is two thousand eighteen if you do it after five years you will have to change people will ask for some other so it’s all define anything before nineteen seventy is not defined from starting from nineteen seventy this is a nice message so anything before nineteen seventy is not defined after nineteen seventy that is from nineteen seventy and after that you can give any year as an input. So we are getting an input for a year then we are doing it for a month we are checking if it is within the bounce one to twelve and as well as we have to check we have to get the input for date and for getting the date and checking if it is a valid date we have various factors what is the entered date what is the month that has been entered previously what is the year that has been and if that year is a leap year or not, so first we have to check if it is a leap year then we have to use all these factors to determine if the date is a valid date and once everything is valid you are ready to use this inputs into your built in library functions. So here as you could see these things, check leap, check valid date these are undefined functionalities. We have to define them will do that in the next part of the video.

# FUN WITH CALENDAR: 08

Alright guys, in the previous video we have started off our coding we had define some functionalities we have just use the names check leap, check valid date will start defining them one by one. so will do with check leap functionality first so let me define it define check leap? It takes an year let me say it takes an year as an input now if it is an a century year you have to check the divisibility by four hundred otherwise you have to check the divisibility by four that is the procedure right so let's check that first. Century year as I had said the last two digits are zeros so that is they are divisible by hundred that is the check we are doing. If the year is divisible by hundred year mod hundred is equal to zero that is the case then it is the century year and hence we have to check for divisibility by four hundred divisibility by four hundred if this is equal to zero that is it is divisible by four hundred then you just say return true, true that it is a leap year that is it's a century year and it divisible by four hundred hence it is a leap year so you return true else that is the check for divisibility by four hundred fails in that cast it is not a leap year so you should return false aright and now this is the procedure if this is a century year. If it is not a century year so that comes under the else part if this particular condition divisibility by hundred fails what if they input is two thousand four it's not divisible by hundred so this will fail so here we have to go ahead with checking for divisibility by four so will do that if year mod is equal to zero then you return true that is it is a leap year else if it is not divisible by four for example an input like two thousand seven would fall under this category return false that is not a leap year. So the logic is if it is century year you should check for divisibility by four hundred otherwise you should check for divisibility by four. So let us trace through some inputs. If it is thousand eight hundred that is eighteen hundred is the year at is pass to it so eighteen hundred mod zero is mod hundred is zero eighteen hundred is a century year so you check for divisibility by four hundred eighteen hundred mod four hundred is not equal to zero so its comes under else part and it is false that is eighteen hundred is not a leap year yes it is true. It works fine, what is the input is two thousand, two thousand mod hundred is zero yes ok its comes inside two thousand mod four hundred is zero yes it returns true it says two thousand is a leap year which is true. What if the input is two thousand sixteen? Two thousand sixteen mod hundred is not equal to zero so it comes into this else part it checks for divisibility by four two thousand sixteen mod four is equal to zero so it says true yes two thousand sixteen is a leap year is what it says which is true. If you are input is two thousand nine two thousand nine mod zero sorry two thousand nine mod hundred is not equal to zero so it comes into the else part it checks for divisibility by four two thousand nine mod four is not equal to zero so it comes into the else part and returns false. If says two thousand nine is not a leap year, yes it is true. So for all possible inputs this thing works fine so we have correctly coded the mechanism by which we can check if a given year is a leap year or not? So we are done with check leap see the warning symbol has vanished there is just one warning symbol for check valid date it involves this four factors so we will see that in the next part of the video.

# FUN WITH CALENDAR: 09

Alright guys, so in the previous videos we had seen how to check if a given year if a given year is a leap year or not, we had seen how we can do that as you could see if that year is a century year you should check fir divisibility by four hundred else you should check for divisibility by four. So we have realise this leap year check right. So what is pending we have to check? If it is a valid date or not. So for checking for this, what are the parameters let us define let us we had already used the name so let us define it. So let us say define check valid date that was the name I had used so what are the parameters it had taken? It had taken the date it had then taken let us see it had taken date month year and the factor that it is a leap year or not a leap year, date, month, year and the factor that it is a leap year or not a leap year. Even here you can perform if leap year check is needed but given that we had perform we will directly pass in the factor that can be alternative way by which you can code it. So I am checking if it is a valid date I am beginning with my check so the thing is, first thing I have to check is the special month February it depends on it is a leap year or not. So I should say the first check is if this is a leap year the leap year factor that is this functionality has return true or false right this factor is true if the factor l which stands for leap year is true then I should first check if the month is February so I should check if the entered month is equal to two it stands for February that is the case you check what is the date that has been entered that is a leap enter and it is a February if it is less than thirty that is up to twenty nine is allowed right so I am saying if it is less than thirty you say you return true that is that is a leap year month is February and the date entered is less than thirty so its a valid date so you return true say that its a valid date. If this is not true else this is not a valid date so return false sorry false f has to be in caps for this we had done check for February month. Alright the first check that we have done now is for the month of February in case if it is a leap year, if the month is not February what is the case? So that comes here column number nine so the same column number nine see python indentation is important so here if I say else that is the input month is not February so what is the pattern? We had seen till july there was a pattern from august there is a change in pattern so then that we have to capture it. Let us first categorised that into month is before august or after august will categorised into if the month is before august or after august so august is the eight month right so let us see if month is less than eight if that is the case if that is a month before august that is till july till july what was the case odd numbered months had thirty one days even numbered months had thirty days so we will capture that we will check if that is a odd numbered month how will we check? Month mod two is equal to one if it is a odd number in that case you should check if the date must be maximum allowed is thirty one so let’s see date is less than thirty two so in that case we will say yes it is a valid date return true. Right see it is not the February month the special month its some other month only so we have to categorised at as before august of after august that is from august will have to categorised it so if it is before august if it is less than eight the month is less than eight august is the eight month if it is less than that it is before august check if it is an odd numbered month or even numbered month. If it is a odd numbered month maximum allowed date is thirty one so you say if date is less than thirty two it’s a valid date return that it’s a valid date else you return false stating that it is an invalid date so this is the pattern for odd numbered months thirty one days are there. If it is a even numbered month what happens? This test will fail so here you should say else so this denoted the even numbered months in even numbered months the limit is thirty days so the same thing I will copy paste it I will use the same thing just that there are thirty days that is allowed so I have to change this limit as thirty one so if it is less than thirty one that is still thirty one will be allowed otherwise it will not be allowed this has to be there also here what if someone enters a negative value for month we had check the negative value so we check that as well so let me check it here check valid date is the check I am performing. So let me say date is greater than zero it has to be a positive value and this thing both the thing if it is satisfy you say it's a valid date otherwise say it is a invalid date. That's it the thing I had missed it in the previous video so I am including it please make a note of it. So ok will get back to this so what we had done till now? We check if it is a leap year, if it is a month of February then we allowed dates till twenty nine so we had set this if the date is less than thirty as the limit, in that case it is a valid date else it's an invalid date. If it is not February we have two partitions of months if it is before august the odd numbers month have thirty one days, even numbered months have thirty days that we have capturing here. This is the pattern for before august so what if it is after august? So here I should give a else part then right so its column number thirteen so let me give an else part here column number thirteen else so this is the months the month is from august and above month is eight and above for this part same thing just that the pattern reverses for even numbered months you have thirty one days and odd numbered months have thirty days so let me copy paste it and let me make a very small modification with that it's done if that is a even numbered months so this has to be zero so I hope you are understanding this part so given that you are in the eleventh week I assume that by now you are very familiar with this and you understand it fast that's why I am going slightly fast but nevertheless if you feel that you need to slow down you can pause the video think over it you check the if else conditions you take an input run through the flow and you will understand it. Also this is not the only way by which we can write the code there may be many ways and when you have these many numbers of if else there are multiple ways we can viable the conditions. You try for alternate ways as well and discuss them on discussion form. So this all this checks we have done if it is leap year, if it is not a leap year the same things will hold but the only change is February will have twenty eight days, all the other months the thing is same right so let me just copy paste it let me copy let me copy till here copied ok so I should put that right so the leap year check I had done it in column five so let me go back and do it here in column five I will get back to column five ok so here I should give else and say I will paste ok so I had pasted it so the check is very similar that's why I have copy pasted it but may be you can give a pause and you can thin over and you will understand the flow. May be one thing I will do is I will maximise this part I will explain ok this thing I have maximised see if you could see have saved it you could see this, the first check you are doing is if it is the leap year or not. If it a leap year and the month is February maximum allowed date is twenty nine so I had set this limit if all these are satisfied you say that it is a valid date else you say it's a invalid date so you return false in case of valid you have to return true. If it is not February the partitioning is till July the add number months have thirty one days and even numbered months have thirty days. After july the pattern reverses that's what we are capturing here if month is less than eight, eight is august right so if it is less than eight if it is before august this is a pattern if this is a odd numbered months you allow the maximum value as thirty one else you say it’s an invalid date otherwise you allow the maximum value to be thirty that is if it is not a odd numbered month you allow it to be thirty this is till july. So after july the pattern is for the even numbered months allow up to thirty one for odd numbered months allow up to thirty this is the pattern. All this we have done if it is a leap year, leap year February has twenty nine days if it is a non leap year the only change is February has twenty eight days so I have to make a change here date is less than twenty nine till twenty eight is allowed. You say it’s true else you say its false and rest all things follows as such so maybe you can rewire this and write in another ways you can perform the February check at the later point of time as well you can rewire it in multiple ways given that there are so many branching you can rewire the conditions as you wish. Try different rewiring and do discuss in the discussion form its easy but you need to give some thought, you need to give some time here that’s it just pause the video here think over it take a pen and paper you run through the flow of this code just run through it try rewiring the conditions as I had said discuss them in the discussion form with all this it will be very clear it will be very crystal clear to you. So this is how we are checking if it is a valid date alright so we had seen how to check a valid date will see the remaining part after getting the input what are we supposed to do, Will see that in the next part.

# FUN WITH CALENDAR: 10

Alright guys, in the previous videos you had seen how we can get the input and check if it is a valid input and then once the input is valid will proceed with how to check for the weak day that is the day of the week on which that day particular falls. So we had inputted the year anything from and after nineteen seventy is allowed and we had inputted the month and allowed range of values is from one to twelve and we had inputted the date. Valid date factor varies based on if it is a leap year the month the date, date month year based on all these factors your fact that the date is valid or not varies. We had seen how the variation is all these we had seen in the previous videos also I had explained in the slides as well that was we had coded from the slides you can try coding it in your own and you can see how the similarities and differences are between your approach and my approach exist and you can discuss all this in discussion form just that given that there are so many constrains it may appear that its very complex or something but nothing it’s just that my way of wiring the conditions is like this there can be multiple ways you try wiring by yourself you will understand that is very easy alright so by now you have got the input of by year month and date that is needed for our built in function that built in function is found in the library calendar so let me import that first. So that I can use it that is my ultimate purpose of this exercise, I will import calendar right so I had imported calendar so now I will be using it so this functionality week day presenting calendar is what will return as what is the day of the week given a date right? so we had seen how the signature of it we had seen how to use in a pre requisites videos will revisit it now so one thing that we have noticed it, it returns the index it says zero for Monday one for Tuesday and so on till six for Sunday this is how it returns your input it returns the output in this fashion. Zero for Monday to six for Sunday, zero for Monday, one for Tuesday, two for Wednesday and so on till six for Sunday This is how the calendar dot week day the week day functionality that is present in the calendar library works but that is difficult for humans to interpret so we need to convert that as well so first thing what we have to do is we have to get that index whatever it is returning so let me say day index day index is return by this thing calendar dot week day calendar dot week day of see the arguments of year comma month comma day, day is nothing but the date year comma month comma date ok so we had inputted it we had send the inputs for this functionality to work so it will work it will return an index which will say which day it is zero is for Monday one is for Tuesday so on till six is for Sunday but that is not easy for humans right so we will have to get the day from this day index so let me say day I have to get it from day index I will say get day I have to get day form day index day index right see there is a warning symbol which says that undefined that just said the name I haven’t defined what this has to do so let us define it. So let us just define here its fine define get day let me define it so what is the list of days let me say list of days the encoding is its starts from Monday zero is for Monday right so let me say Monday Tuesday, Wednesday, Thursday, Friday, Saturday and Sunday this is the order in which encodes right Monday is encoded as zero, Tuesday is encoded as one and Wednesday is encoded as two and so on till Sunday is encoded as six. So I have created a list of days basically so I guess you would understand you could figure out what will be my next step, I will just return ok for this I need to have the day index right so day index I need to have the day index so I will just return the corresponding value from this list given by day index. List of days of day index this is the most simplest definition I guess you are able to understand it the encoding is zero is for Monday one is for Tuesday and so on till six is for Sunday this I show the lists have their indexing mechanism so we are creating the list of days and whatever is the index that is being passed we will pass the corresponding element in that index as the output that is if the if the passed index is three then zero one two three its zero one two three its Thursday, Thursday will be taken as the output so we will get the name of the day here so that is obtained now we have to print the answer, let me print I will print a very good message I will say date this is how humans do it right by slash will separate it date month then we have another slash and year this is our human friendly format and I am printing in the human friendly format only I will say it falls on this particular day that is captured in the variable day so this falls on this particular day this is how the working is simple will just give a very quick summary and then will go ahead with running of this code see I am getting an input of year I am getting an year I am inputting year if it is after or from nineteen seventy I agree that input and I am inputting month that should be between the range one to twelve and I am inputting date and based on the factors if it is a leap year what is the month that is being enter? I determine if the entered date is a valid date or not then based on this I have I am inputting some date month year right so based on this things I am now passing it to the built in functionality calendar dot week day in calendar library you have the functionality week day we are passing this and it is returning a index it’s not retuning the name of the day its returning the index so for getting the name of the day I am using the list mechanism and I am getting day and after getting the day I am printing the this particular date month year falls on this day. This is how the working is so let me save it and let me run ok so it’s asking me to enter a year so let me say two thousand let me enter a month let me say four April let me enter a date let me say twenty five ok twenty five four two thousand falls on Tuesday I got an output. So like this you can check for any dates once your programs is ready running is see as you could see maximum it would have taken half a second that’s it its all the delay by which you are giving your input that’s it. If you hardcore the input if it is to be red from the code itself it will be really within in the fractions of seconds so this is really really very fast ok by which you can know the day of the week given a date alright guys so we had seen one task will see another task that we can do using these cool python libraries in the next part of the video.

# FUN WITH CALENDAR: 11

Alright guys, so that was a very good exercise given a date you can quickly say on what day this particular date falls on so that is really very cool functionality that python provides and now we will see another task so this is the task if the current time is known as per your time zone can you say what is the time in other time zone? Based on different time zones I have to know the current time that is I can get the system time from the command as I had discussed in the pre requisite video for all the time zones can I get the name. So to get the current time as we had used the date time library we will be using that and to work with the time zones we will be using this library pytz tz for time zone py is for python pytz will be getting the current time and will display the time as per the different time zone as an output so that is what we will be doing in our task. Will see that in the next video.

# FUN WITH CALENDAR: 12

Alright guys, so in this joy fun with calendars we had seen some cool tasks that can be done using some library provided by python we had seen them in the pre requisites also we had seen a task of given a date can you predict which day of the week that date falls on we have seen that so the next task we are going to see is you have to fetch the current time not only in the time zone that is followed in your current location but as per the time zones across the world that is what are all the time zones that is present I need to get that time as per all those time zones that's what I want to do. So let me do that so as we had seen to get the current time we need to use the date time library so there is a super set as well as the sub set as I have discussed in the pre requisites video we just want to work with the sub set so let's import it. From the super set date time library import the sub set date time library as dt because the name is lengthier so I am using a short name as dt so that this is to get the current time and you are going to work with time zones as well so let me import the library pytz py for python and tz for time zone so let me import this as well. So the first thing I have to do is I have to get the list of all the time zones. So let me call that as time zones time zones is nothing but it is given by this built in functionality in pytz library just say pytz dot all underscore time zones so if you just say that you will get the list of all time zones here so now simple just iterating through the list. We have the list of all time zones and we know given time zone we can fetch the current time as per the time zone so we are going to iterate through this list and then we are going to display the time corresponding to each time zone in this list. So let us just iterate it lets say for I in range length of this list the name of the list is time zones just iterate over this list so my current zones is nothing but at the ith element right so I am picking the current zone, I am iterating over the list I am picking the current element where I am standing I am picking the current position element so I have picked the current zone so now I should create a time zone instance time zone object a special object to work with time zones as we have seen in the pre requisite video we will be creating a special object so let me say tz time zone the convenient representation pytz the library name dot time zone is the functionality and what is the time zone that I need it is given by the variable zone this particular time zone is what I need so I have created an object and now I can print the current time easily right so I will print current time at zone what is a zone that is given by the variable zone is the current time how will we get it? From this date time library dt dot now is the functionality and to this functionality I have to pass this time zone object tz so I have passed it and so I got the time zone now right so that's it just we can get the lists of all the time zones from this library pytz py for python and tz for time zone can get all the time zones and we are iterating over this list creating a time zone object special object by which you can work at that time zone irrespective of whatever is your current location a special time zone object we are be creating and we are passing the two functionality that fetch the current time so that it fetches the time as per that time zone and then we are printing the fetched time that's what we are doing just simply iterating over the list. I hope from the pre requisites video if you are very clear and from the slides that I have discussed given the task it's very easy you can easily connect the doubts and you can come up with this code this is really very easy see the code is just some eight lines probably another there are some spaces in between so some six seven lines of code that's it that's what we are using but what is the magic it's doing? You will understand if you get the output. So let me save this let me run this code see I am getting the output oh my god oh so fast it’s done see some five hundred plus time zones we have seen everything everything is outputted just in fraction of seconds that’s the magic so you can know the time at any place in fraction of seconds by using your python programming skills. Alright guys I hope you enjoyed this series on fun with calendars, thanks for watching. Have a nice day.

# PAGE RANK: HOW DOES GOOGLE WORK? 01

We are now going to discus one of the most celebrated ideas in computer science. I will say it is one of the most celebrated ideas heavily applied idea in whole of mankind, I get to say this because simplicities of the idea yet it has tremendous applications you will soon get to know what I am going to talk about but before which let us start our lesson with a small activity. The activity comprises of a bunch of students trying to talk to each other and then they make a note of who all they were impressed with for example, a person will go and talk to twenty others and make a note of list of people that she was impressed with and every single person does the same. All twenty people do this, write down the names of people there are impressed with and they give it to us. Now we create what is called the impression network, who likes whom? It looks like this on twenty nodes and now we ask this question. This impression is more like voting I like him, I like here based on this can we pick a leader amongst this twenty people. First question, how do you pick a leader? Second question, why is it important for us to even ask this question, why is this the most celebrated idea of all times? You will soon get to know after watching this activity.

# PAGE RANK: HOW DOES GOOGLE WORK? 02

So now you just saw a group of people were just asked to choose their leader, each of each one of them was asked to select around five people to whom they wanted to be their leader and now have the leader of the group here, congratulations virender you have been selected as the leader of the group. Thank you so much it would be a proud privilege for me, how did you convince these many people to vote for you? Actually convincing them is not a big task what I feel personally, yeah the thing is that first of all I introduced myself that what type of problems they are facing actually whether they are from research side, whether they are from graduation side that what type of problem they are facing in their hostel life and their carrier life so by asking all this and giving some good suggestions so I think I have been connecting them very properly. Yeah to that reason yeah so due to them actually I feel that they choose me as maximum as possible, oh good so can you guess how many votes did u get? Definitely many, ok you got many votes but let me tell you one thing it’s not only the number of votes that made you the leader but it also the people who voted for you. I didn’t get you, it’s the number of votes that server as the deciding factor in every election but how this? Let me explain you page rank algorithm. So take an example suppose you and your friend are applying for a promotion both of you are working in a company so many of your colleagues have refer to your friend for the promotion but your manager and group has referred you who has high chances to de promote it? I think me, ok now you see the number of people who are refereeing doesn’t matter it also matters who is referring to you so this is the page rank algorithm and a Google also works on this very simple algorithm. That’s really cool, yeah. So in Google we have a underline web graph where the nodes are the wrap pages ok and the links are the hyper links, meaning one page is referring to another page through a hyper link we will connect it through the edged ok and then Google page rank algorithm to find out the top most. That’s really cool yeah I really like this thing yeah. So congratulation again thank you so much and we have a small gift for you, thank you so much winning the election.

# PAGE RANK: HOW DOES GOOGLE WORK? 03

You know what I am going to do with this data. I now have a what is called a network and my network look like this do you see the nodes here and the edges, look at this arrow from this node to this node it just represent that this person found this person impressive. If A finds B impressive B may not find A impressive correct? Now my question is how so you find the most impressive person here in this network and obvious answer that we will have is ok at that person who has a lot of arrows coming in by that we mean many people found him impressive a number of people finding someone impressive should not be an indication of how impressive a person is. For example assume hundred friends of mine says I am a nice person that is different from the prime minister of India saying “I am a nice person” in one of his speeches. Right there is a huge difference so what do you infer? It is not how many people who elect who vote for you it’s about who are those people who vote for you, maybe we should consider that right? So now on this network I would like to ask this question who is the most impressive person? And I am going to give you a technique to find it. You may not even see why exactly we are using this technique to find the most impressive person but with time it will be very clear. Here goes my algorithm, here goes my technique what I will do is on this graph that you are seeing the network I will start from some node and take what is called a drunkards’ walk also called the random walk the point is you start from the node let’s say this node you have so many lines going from this node forget about what this network represents I am going to play around this network. I am going to jump to one of the lines and land at the other end of this line the point whatever you are seeing I am going to land there but I have so many choices to take I will pick one of them randomly then I jump there once I jump there I mark one here saying that I have seen it already and from here I will jump to another node again from here there are many lines going out I will choice one of them randomly and jump there and call it I have visited it once I keep doing this I keep doing this if I visited node more than once I will make a note of it I will call it I visited this node twice and so on and finally I will do this lets say a million times may if I am not happy I will do it a couple of million times or even more than that and after doing this a million times I will see what is the points collected by individual nodes. And I will look at that node which has collected the maximum number of points and I will declare that node as the winner. Why do I do this? What does this signify? You will get to know very soon.

# PAGE RANK: HOW DOES GOOGLE WORK? 04

Now we had a bunch of people and we ask them to take part in an activity and then we drew a network from them and did what is called Radom walking on the network I asked you to do it right will show you how to do it right now but one should do this random walk and then see the points accumulated per node and then look at that node with the highest point and call that fellow a leader. Now you know what? This is precisely how the Google search engine works and this algorithm is called the page rank algorithm. So what is the connection? Look at this, look at this network and take a random walk on it. Which what are those nodes which will have highest number of visits, it’s going to be those nodes which are easily reachable when you are taking random walks. How do you reach a node? It should be easy to reach their neighbours. How can you reach their neighbours? It should be easy to reach their neighbours neighbours the most popular person’s neighbours neighbours so if there is a easy in coming way to a node then that node have a very high weight so intuitively speaking a person is important if important people point to him. A node will get a lot of weight if it is adjacent to a few nodes and these few nodes are all important, by important we mean they are adjacent to a lot more nodes, a node is adjacent to many nodes and these nodes are adjacent to many nodes and so on. Correct? So we are just giving you a very toned down version of page rank because it’s a first course in programming we don’t want to bother you with the lot of details but if you are interested you can take a look at our course on NPTEL called the SOCIAL NETWORKS course there we describe in detail one entire week we discuss on how page rank works. So getting back we have now told you how to take a random walk on a network increment the points individually on the nodes and look at the top nodes thus obtained they are the most important nodes now how does Google work based on this idea. It’s very simple what Google does is as you can see in this animation it takes a website and on that website as you know given a website it will have a lot of hyper links to another website by hyper links we mean a place where you go and you click and it takes you to another page you see that’s called the hyper link ok so I will take all possible web pages in the world that one can think of and I will put an arrow from a web page to another web page if there is a link from this web page to this web page and I will create this huge network by the way this is how the network of internet looks like where a line between two node where nodes represent that sides and the line between two represent that there is a link from this side to that side this was the snapshot of the internet long long back off course it has grown a bigger right now it looks even more complicated right now but just for completeness sake we are flashing the network of the internet this is called the world wide web network so now here what Google does is it takes all these pages of the world and runs a random walk on it. It’s a huge network you see it takes a long time to run random walk off course with sophisticated computers you can do this. A huge network on billions of node you run random walk and you make an note of how many times you visit a node as you are taking random walk, as we are talking, as we are listening to this video Google actually is doing this. It is taking all the websites of the world and then taking random walks on it. Ok you would have heard of crawlers this is precisely what I mean by crawlers. There is a program which crawls the web like this and makes a note of how many times I visited the what node and ranks all the nodes based on the visits to the node ok when you type the word Mysore city right it will look at all those websites which has the word Mysore and then it will realise that there are ten thousand websites that has Mysore in it but it will show you the top few sites that have accumulated top points due to this random walk. You may want to get that straight let me repeat so when type Mysore there are ten thousand hits ten thousand sites which has Mysore in it but then Google will fetch the top few sites to you how does it know what are the top few sites, top few sites based on the points accumulated because of random walk that's it this single idea has lead to a five hundred billion dollar industry today. Google has it stand tall mainly started of this algorithm which till date is the best possible searching technology off course there are several other parameters in search engine users but the basic heart the nucleus of the idea is whatever we are whatever we taught you so far. Now let us go head and look at the integrities of how to take a random walk and then how to crawl on a network.

# PAGE RANK: HOW DOES GOOGLE WORK? 05

Hello all, welcome to the programming screen cast of page rank I hope it is now clear to all of you what is page rank. Page rank is basically an algorithm that is used to wrap notes in a given graph. In this particular programming screen cast I will be telling you how can you find out page rank from random walk method using random walk method so first of all let me tell you what is random walk method. As the name suggests when you have to traverse the graph randomly it’s called the random walk. You pick any node from the graph randomly and look for its out links and traverse one of its out links randomly. Then you look for the out links of the present node and then traverse one of this out links randomly. In this way you traverse the whole graph if it is connected obviously so I will be giving you an example and it will become clear to you what is random walk so as you can see we have been given a graph here in which we have ABCDEF and G as the set of nodes. So first of all I will initialise the counter of each of the nodes so I have initialise the counter of each node as zero then I will pick a node randomly from the given set of nodes so I picked A here and I incremented its counter two then I look for neighbours of A, what are the neighbours of A? We have B we have F and we have G we have to pick a node randomly from the neighbours of A so we will pick the node randomly and we have selected F and we have implemented its counter two then we will look for the neighbours of F. What are the neighbours of F? We have G, we have E so we will pick a node randomly from the set of neighbours of F and increment its counter too we have picked E and we have incremented its counted too then what are the neighbours of E we have A, we have C, we have D now we have to pick a node randomly from this particular set, what is the set here? Set is A C and D will pick a node randomly and we have picked A and incremented its counter so now we have A two B zero C zero D zero E one F one G zero so what do you think here? Is this the final page rank? The values that we have mentioned here, no we have to keep iterating it this random walk method. We have keep re iterating it as many numbers of times as we can and at the end our page rank values will be same as the actual page rank values. so next we will look for the neighbours of A, so what are the neighbours of A we have again G we have F and we have B so we will pick a node randomly from this set so we have picked B and we have incremented it counter too then what are the neighbours of B we have only C as the neighbour as B so we will traverse C and increment its counter so we have traverse C and we have incremented its counter too then as you can see C is sink care what do I mean by sink? C has a sink care it has no outgoing edges so what we can do here? If we encounter sink in graph we will pick a node randomly from the given set of nodes so we will pick a node randomly from the given set of nodes. What is the set here ABCDEFG you can pick any node from the nodes of the graph so next we picked D here so D we picked D and we incremented its counter too then we will look for the neighbours for D. What are the neighbours? We only have C has neighbours of D so we will pick C and we increment its counter we picked C and we incremented its counter then we again came to C and it is a sink it has no outgoing edges so next we need to pick a node randomly from the set ABCDEFG so we picked F and we incremented its counter then we will look for the neighbour of F so what are the neighbours of F? We have E and we have G so next we picked G as the node and we incremented its counter too so in this way you keep on traversing the graph for many number of iterations and at the end the values we have shown here will reflect the page rank values yes so how can you be sure of this method? Well will be simulating it will be implementing this and will be showing you that this random walk method actually reflects the right page rank values so let us move towards the programming screen cast of this page rank implementation and you will be getting the clear idea of how this method works in practice. Thank you.

# PAGE RANK: HOW DOES GOOGLE WORK? 06

Before we move to the programming screen cast of page rank I will be telling you how can you draw graph using networkx. I hope you are familiar with the networkx library we have simulated in the joy sixth degrees of separation so if you haven't gone through the videos of networkx I will recommend you to go through the videos of networkx first. So first of all let me show you how can you draw graph using networkx. As and always we will be importing the concerned library we will be importing networkx here so I will write import networkx as nx next I will showing you carious graphs that are available in networkx so first of all is the barbell graph so I will write nx dot barbell graph and in this you need to supply two parameters apply four and two we will be getting to know why I am supplying these two parameters here then I will write nx I should initialise the g variable here as this so that I can draw it barbell underscore graph and the two parameters now I will draw this graph nx dot draw G as you can see we have the two communities here we have two communities as which comprises of four nodes so in this particular community we have four nodes and in this particular community we have four nodes so next if I write four and three then what happens see, see we have four nodes here four nodes here and in between we have three nodes in this particular graph we wrote two and we had two nodes in between the two communities, so what does the first parameter represents here, it represents the number of nodes in the particular community and the second parameter represents the number of nodes in between the two communities so we will draw it again so you will get the clear idea of what is happening here so this time I will draw I will write five and three and then draw it so five nodes here five nodes here and in between we have three nodes and this is how barbell graphs works. Next I will be telling you some more graphs next will be drawing a complete graph so I will write G is equal to nx dot complete underscore graph and in this particular thing you need to supply what are the number of nodes so I will just supply four here let me draw it nx dot draw and G so I have a complete graph and it comprises of four nodes so those who don't know what is the complete graph, in complete graph we have each node that is connected to its n minus one node for example if I pick this particular node it should be connected to rest of the three nodes so here I pick this node and it is connected to rest of the three nodes so basically it is connected to the rest of the node except the node itself that graph is called complete graph. So next basically it has every possible edge that can be present in the graph except the parallel edges and the self loads next we need to do I will show you how can you draw a cycle graph. I will write nx dot cycle graph and in this also I need to supply what are the what is the number of nodes here so I will write five let me show you next you write nx dot draw G. So we have a cycle of five nodes so what next there is another graph available in networkx that is ladder graph so I will write nx dot ladder underscore graph and you supply one parameter here and let us see what does this parameter do and nx dot draw capital G so we supplied five and we have five nodes on each of its parallel edges we have five nodes here and five nodes here in this way you can draw a ladder graph next is path graph so I will write G is equal to nx dot path underscore graph and supply the number of nodes present in this path graph I will say six nx dot draw capital G so you have a simple path comprising of six nodes then I will draw a star graph so what we will do? Will do nx dot star underscore graph and then I supplied a parameter here let us find out what does this parameter do. And then let me draw this particular graph so since we supplied five here we have five nodes as the sink nodes I hope you know the what is sink, it doesn’t have any outgoing edge so there is a hub node at this centre of this star graph and we have five nodes as the outer sinks we as the outer nodes we have one two three four and five as the outer nodes and we have one sink so whatever parameter you supply here that becomes the outer nodes and there is one hub node so in this particular graph there will be n plus one there will be parameter plus one number of nodes so if I supply four here then there will be five nodes in the graph one will be hub node and the four node and there will be four outer nodes so let me draw it again so we have four outer nodes and we have one hub node here so what next we need to do here is, next is we can draw wheel graph also so I will just write G is equal to nx dot wheel underscore graph and you supply one parameter here again so you will draw this graph so we have wheel sort if thing it is basically four node graph and it looks like a complete graph also so it is a complete graph so we will do it five nodes and let us see what happens so in this particular thing we also have five nodes one two three four and five and we have one two three four five six seven and eight number of edges here so you can see that it basically doesn’t correspond to a complete graph it has a central node and it is connected to each of its nodes so let me do it for six graph or six node graph. So here also we have a central node and it is connected to its n minus one node so this is how wheel graph works then we also have some random graphs present networkx so let me show you one instance of random graph and will be using this random graph for our implementation in page rank so I will just write g is equal to gnp underscore random underscore graph and here I have first parameter as the total number of nodes for example I wrote five and then there is a parameter that is the probability of edge creation next parameter is the probability of edge creation I just wrote point five so with probability point five it will make an edge between two nodes and with probability point five it will not make edge between two nodes so the probability is equal. So next sorry I didn’t write nx here that’s why it is showing an error. So next I need to draw this particular graph so you can see we have five nodes here and it has drawn their edges randomly so let me draw it again it will show different graph because this is a random graph and it is making edges randomly so let me draw it again see now we have a completely different graph. So we will be using this particular graph for our implementation of random walk method in page rank. Thank you.

# PAGE RANK: HOW DOES GOOGLE WORK? 07

Welcome again, all of you we have now come to the implementation part of random walk method so let us start. First of all as and always we will be importing some libraries so first of all let me import networkx import networkx as nx then will be importing random library we will also be importing matplotlib I order to visualise the graph so I will write import matplotlib dot pyplot as plt so I think now we are done with the importing libraries. So as and our previous programming screen cast I showed you how can you draw random graphs using networkx, so will be drawing that particular graph here so I will just write g is equal to nx dot gnp underscore random underscore graph and first parameter is number of nodes so I will take ten nodes here and I will also take the second parameter the second parameter is the probability of edge creation so I will take it has point five and then we also need a directed graph in page rank yes in very first video I explained that we need a directed graph for page rank so I will just write directed is equal to true since we need a directed graph. So next what you need to do, I will draw it and show you whether we are getting a directed graph or not. So nx dot draw g and plt dot show so let me run it. I will write random underscore walk dot py. As you can see we have a directed graph here we have a directed graph here we can see the arrows here but what I want more is I also want the labels of nodes to be visible so I just write with underscore labels is equal to true so that I can also get the labels on the particular node so let me run it again. So now we have labels visible also we have zero one till nine we have zero one till nine since we only passed ten has a parameter here. So ten is the total number of nodes here, so now we are done with drawing the graph here next what we need to do? As with the random walk method we will pick a node randomly from this set of nodes from zero to nine we will pick a node randomly so I will just take a variable here I will say x is equal to random dot choice. I hope that you know what random dot choice do it will pick a element randomly from a list so I need to pass the list here so I will write I for I in range what is the range here? G dot number of nodes, g dot number of nodes so this is the range here so we will pick a node randomly so I will just write a comment here x is the random source node since we have picked x here we also need to increment it counter but how will we store the counter? We will store the counter in a dictionary so what we need to do here? We need to initialise the dictionary here, I will use dict underscore counter is equal to curly braces this represent dictionary here. I also initialise the values of dictionary to zero so I will write for I in range g dot number number of nodes dict underscore counter I is equal to zero. Here what you need to do, g dot number of nodes in dict counter is equal to zero here. Since this is the method you need to pass you need to pass the round braces too. So I also need to implement the counter of source node so what is the source node here? We have x here dict counter x is equal to dict counter x plus one so we have incremented the counter too we have initialise the counter of each node as zero too. So for next what we need to do here, we need to look for the neighbours of x and then we need to traverse all of the graph this way. So I will be taking many numbers of iterations here so I will have a loop that will represent the iterations here so I will start with example for this graph and then what we need to do, we need to look for the neighbours of x. So I will just write I will take a list here list underscore n is equal to list of g dot neighbours of x. Now we need to check we also need to check here whether x is the sink or not here. If x is a sink then we need to select a node randomly from the given sets of nodes from the given set of nodes present in the graph otherwise we need to otherwise we need to select a node randomly from the neighbours of x. I repeat if the given node is the same we need to select a node randomly from the nodes present in the graph otherwise we need to select a node randomly from the list of neighbours of x. So I will just write if length of list underscore n is equal to is equal to zero here then that means this is the sink here it has no neighbours. So I will write then x will be random dot choice again we have to select the neighbours we have to select x from g dot number of nodes. I write I for I in range g dot number of nodes. So we are also done with it, we have handled the case if x is the sink else what we will do? Will select a node randomly from the list of n so will say x is equal to random dot choice list of n list underscore n. Then we also increment the counter of the particular node so I will write dict of counter of x is equal to dict of counter x plus one. I also increment the counter here too dict of counter of x is equal to dict of counter of x plus one so I think we have ended all the cases, I will just write some comments here if x is a sink choose a node randomly from neighbours of x so I repeat this loop this particular loop first of all we are looking for the neighbours of the source node, our source node is x here if it has no source node that means if it is a sink then we will choose a node randomly from the given set of nodes of present in the graph. Otherwise we will choose a node randomly from the list of neighbours of given node ok so we have handled the cases here and we have also incremented the counter of our particular node if that particular node is traversed in the random walk method so now we are done with it so you must be thinking how can you verify that this particular random walk method reflects the exact page rank values well we have networkx to our rescue, we have a built in function in networkx that tells us what are the page rank values so I will be using that particular function and we will be verifying that the values returned by a networkx function and a values returned by our random walk method are same or not? So let us do that, what is the method used here? Page rank so I will just use p is equal to nx dot page rank this is the method that will be using and we will be passing our g graph here. So nx dot page rank is the method that will tell us the exact page rank values and I will print p here and I will also print the dict of counter here so I have to verify whether this values same or not. So I think we are done with it, done with the program let me run it. So we have the values here we have zero as the first node, yes zero then one then two then three four five let me see we have we haven’t sorted this values, we must sort this values so that we can get the clear idea the ranking of node given by the random walk method as well as the page rank method are same or not. This will be difficult for us for looking at these values so I will just sort this values here so I will just write, I will sort this dictionaries the dictionary p as well as the dictionary dict count. How can you sort the dictionaries based on the values it is very much do able in python so I will take two list two dictionaries here first one sorted p that is equal to sorted I will write p dot items and key is equal to item operator dot item getter one if oyu pass one here it will sort based on the values if you pass zero here and it will sort on the basis of the items on the basis of keys as we know in dict we have a key we have a value so it is your choice whether you want to sort the dictionary on the basis of the keys or on the basis of values since we have to sort the dictionary based on the page rank values so we will be passing one here and then again I will write sorted random walk is equal to sorted here we have dict counter as the dictionary dict counter dot items and we also will write key is equal to operator item getter one. Here I will print sorted p and here I will print sorted underscore rw sorted underscore rw since we are using the operator library here we need to import operator two so I will just write import operator I hope we are now done with it and let me try to run it. We have the values here we have zero here and we have two here we have one here we have zero here so as you can see these values are not matching with the exact values so what you can do here is? We can always increase the number of iterations let me increase the number of iterations suppose I increased it to ten thousand let me run it again let me see we have two here by then we have eight here then we have one here we have one and eight here, here it is two eight one nine here it is two one eight nine and there are some discrepancies in these values also so let me again increase the counter suppose I increase it one lack, let me run it again so we have four here four here one here one here then again we have eight six here we have six eight here then we have two here then we have seven here three five zero nine three five zero nine so these values are matching to the exact values but there is still a discrepancy we have encountered that is here we have eight six here we have six eight so I will again increase the counter let me increase one zero here again so let me run it again so we have five nine eight we have five nine eight then four then six then three then one then zero seven two perfect! We have exact page rank values so you can see random walk method is effective to find the page rank values but you have to re iterate it many number of times so it depends on the size of the graph too you have to re iterate it again and again in order to get the exact page rank values the correct page rank values. So I hope that this method was correct to you guys I will explain you this method again. So first of all what we did we took a random graph a gnp random graph to start with and then we choose a node randomly we choose the source node randomly from the given set of nodes where we called it as x then we also initialise the counter of each node to zero and we incremented the counter of x since this is the source node and then we took a for loop in which we will re iterate many number of times this random walk method. First of all I took n in which we have the neighbours of x, list of n represents the neighbour of x and then we checked whether it is a sink or not? If it is a sink then we need to choose a node randomly from the nodes of the graph otherwise we need to choose a node randomly from the neighbours of x and side by side we are also incrementing the counter of the particular node if we traverse that particular node so increment the counter here of this node x and here also. So every time the node x is changing and this particular loop is iterating for many number of times and then we took the page rank method given by networkx so as to check whether page rank values given by the random walk method and the exact page rank values are same or not. And we sorted these two dictionaries based on the values so that we can get a clear idea what is happening here then we printing these two dictionaries so in this way you can find page rank using the random walk method. So my only suggestion is to iterate this method many number of times only then you can get the correct page rank values. I hope this programming screen cast was useful to you all, Happy learning.

# PAGE RANK: HOW DOES GOOGLE WORK? 08

Hello guys, welcome to this programming screen cast on page rank using points distribution method so before getting into the points distribution method actually we will revise it some concepts from the previous weeks as well as we will see some pre requisites that would help you understand the points distribution method better. So will get started first will go with the revision so we had seen a data structure called graphs when we had seen the concept of six degree of separation using the facebook data set we had analysed that between two people what is the distance you can reach we had analysed there we had taken a data structure called graph through which the facebook the friendship network is modelled so we had seen about it. So while explaining about graphs I had taken railway networks as an example so here is a network that we had taken so the graph is nothing but the set of points and lines points are technically called as nodes and lines are technically called as edges. So in this railway network we had the cities of the railway stations the stations as nodes and there is an edge there is a line between the two nodes if there is a direct train from this city to that city. That’s how we had taken we had modelled and we had created the network created the graph so this is graph so nodes and edges are the terminologies I would like you guys to recall and another thing we had seen is neighbours, neighbours are nothing but what are all the nodes a particular node is connected to using an edge that’s what we call as a neighbour. For example if you take this node Kolkata the neighbours are guwahati, bengaluru and Chennai the node Chennai has two neighbours bengaluru and Kolkata so this is what you call as neighbours and whatever you have this network is an undirected graph that is there is no direction the edges are by directional we assume that is if there is a train from Chennai to bengaluru there is a another train from bengaluru to Chennai as that is an assumption we have that’s how our Indian railway network in major cities is that’s an assumption we had made but its not the case always that this kind of representation is enough to represent any data sometimes you may need to consider the directions as well. So we will see an example of such networks in this lecture as a new concept so such graphs are what we call as the directed graphs so here is a directed graph where directions of edges that is from which node to which node is the connection present this direction information also important in directed graph an example form real world that we can take is as I have given in this slide its email communication network. So here the nodes are different email id’s the different people of their email id’s and there is an edge from one node to another node if this email id has sent a mail to another one that is see in this network there are let us take a very small group there are three email id’s we have taken A B and C and generally we keep them anonymously that’s why we take toy example A B and C lets say there are some three nodes so the from this mail there is a mail that is sent to this mail id A to B there is an edge so person A has sent a mail to person B but it’s not necessary that person B should reply so that’s why we don’t have the back edge an edge from B to A whereas the mail he has sent to C, C has replied so if you have sent mail form this person to that person you have a directed edge direction denotes who has received the mail so from A B has received so there is an arrow towards B this is how email network communication network can be constructed. So the point to note here is you have directed edges the direction of edges is very important in this directed graphs and some other example that we can considered is maybe from the academics, the academy year I can say fightation network we have something called fightation network where the nodes are the research paper and if one paper is sighting or it is refereeing to another research paper there will be a directed edge and as you could understand its not necessary that edge has to be bI directional say for example you may sight a paper that has been written in say two thousand five it is not necessary that, that paper authors should sight you back in some other papers or in that same paper so directed edge right so you understand the concept of directed edges this is from academy year and another example of social network just like facebook something involving people that could be something like you could have people as nodes and if this person likes another person, if a person A likes person B there is a directed edge from A to B so this person has expressed a liking for this person this could be used in for example matrimonial sites you can used it you can make analysis of matrimonial sites and you can suggests better recommendation and there are lot of applications which can be easily solved if you model the data as a graph especially as a directed graph this is one thing an example of social networks may be another example you want to give I may give from the marketing domain the marketing domain we have something called as supply chain network so basically the different companies are the nodes here and if this company supplies some materials to this company there is a directed edge so basically let us see this company B in that case company B is acquiring some raw materials from company A and it is making some products and it is selling those products it is supplying those products to company C something like this. So for example if B lets say battery if this company is manufacturing batteries the raw materials needed the electrolytes the lithium rods all those such raw materials it will acquire from different companies so they will be the incoming edges so it is acquiring the raw materials from different companies and once it has manufacture the batteries, the batteries may be of different requirement one you used it for your television remote the one for your clock the one for your car everything the batteries are different so this may manufacture batteries using this raw materials and the manufacture products are sold to different other companies so the car batteries should be sold to the major car companies and the clock batteries to the clock manufacturers and so on so this is how you can form a network using the different companies and the transaction they have such a network is called the supply chain network this would be a directed graph and I don't think there will be a bi directional edge in supply chain network although I am not marketing expert this is just a little knowledge that I have but may be if you feel that there are some examples where there can be bi directional edge in a supply chain network also please do let us know in the discussion form. Like this yeah the motto here is to make you understand that there would be directed edged and what is the what directed edges signify in different networks we had taken one example of email communication network it can be watsapp communication someone has messaged you on watsapp there is a directed edge from that person to you, if you reply back you will get a back edge, if you don't reply you won't get back edge so that can be watsapp communication network, mail communication network anything a communication network in general or in academy you have sightation network and in marketing domain we have the supply chain network in case of social network you can consider it sort of matrimonial networks like which person has contacted which person based on that whether the person has replied back to his profile something like that. You can construct it as a network and you can perform analytics. Once you have modelled your data in the form of a graph data structure analytics is very easy further so that’s why there are many applications where this graph data structure the directed graph especially comes very useful so will get into some terminologies of directed graphs as well so the edges we have right so we will have other names as well nodes as I had said as points and edges are lines is the layman terms other technical names can also be present that is nodes can also be called as vertices and edges can also be called as links I am just telling because in lectures we may use those terms interchangeably or may be if you look up to some material online some research papers or some text books you look up any where these two terms may be changed interchangeably nodes are vertices and edges are links they may be used interchangeably and let us see some terminologies so links are the edges in case of undirected graph they are bi directional so we just say they are the edges but here its directed graph so the direction matters so we will say incoming links and outgoing links incoming edged and outgoing links outgoing edges so for node in this graph if you considered for node A the incoming edges from C to A that’s why we have a tupple C comma A and the outgoing edge is from A, A to B and from A to C like this you can say the outgoing links and incoming links so the representation is source comma target, source is the starting point and target is the destination or the end point of that edge, an edge is resent between two nodes the starting points are called the source that will be given as the first item here and the end point of the edge is the target that is given as the second item of the tupple so like this edges will be denoted by tupples. This is the representation regarding directed graphs alright so I have started off with directed graphs I would like that you guys think of any other applications where this directed graphs can come handy that would be the thing I would expect you guys as a result of watching this video, you must thinking that direction and you will see further pre requisites in the upcoming parts.

# PAGE RANK: HOW DOES GOOGLE WORK? 09

Alright guys, so in the previous videos you had been to introduced to the new data structure of the direct graph that is variant of the graph data structure called the directed graph where the direction of edges matter so let us get into the networkx the package for graphs and let us see what are the support it provides to deal with directed graph data structure. So let's get into the python syntax and other cases let's get started. So as the first step the package that supports the graph data structure operations is networkx that will be importing will do that, import networkx as nx so this is just a short name given that the networkx is a lengthier name so I am using a shorter name nx so I have imported it. So how did you create a undirected graph? If you want to create a new undirected graph, you would say let me say in directed graph so let me say u, u is nothing but you create a nx dot graph this would create a undirected graph right this was used to create a undirected graph in case you want to create a directed graph how would we do that? Let's see let me say G is my graph I will create a nx dot DI graph is the command see observe that d is in caps and g is in caps di graphs is another terminology use to say directed graph. Directed graph di graph is a sort of acronym if you consider it as. DI graph d and g has to be in caps please observe this DI graph so DI graph directed graph object has been created so if you would want to know what are the nodes present in the graph that's let us see g dot nodes, nodes see its empty node view is abstract data type it is using but we are comfortable with the list representation so given that it is using a different representation in python two if you are following the output of g dot nodes will be a list data type but in python three they have a different representation so we need to convert this into list representation so that it is easier for our programming purposes. So currently it is empty so there are no nodes so let us add a few nodes something like this G dot add nodes from add nodes from you have a thing so you need to pass a list as a parameter so a list where you enter the list id's. Ok for example abcd whatever you want you want to give it as a list let me use numbers itself for the nodes. I see I want to create some three nodes zero one and two so I can say zero one and two what if I want to do for hundred nodes? Thousand nodes? If I want to increase, is it possible for me to manually enter all the numbers, no right so in that case what we can do is we can use for loop here and we can do it to demonstrate that even for this case as well I will use a for all and demonstrate for I in range how many nodes you wanted you just need to give that. Let me say I want some five nodes so I have given five so this is there so I want to add node number I for I in range five that is you iterate over the number line till you reach five so from zero it starts zero one two three four it will stop so all those whatever is the values that I has been taking I had said zero one two three four you append that to the list that is what this particular line means. This is one line way by which you can add a collections of nodes if you want to name them in the numbered fashion only so you can added it this way so please observe this g dot add nodes from you need to pass a list given that I want to number it sequentially I can use a for loop and get it done instead of enumerating those numbers manually this is how you have added the nodes now let me see what are the nodes now. See as I have said zero one two three four has been added so as I have said this representation the tupple sort of representation has not very friendly for us. List is friendly for us we can iterate over the list easily so we will convert it into a list. List of G dot nodes so yes see it has been converted into a list format. Zero one two three four has been converted list format so what are the edges present in the graph. G dot edges you would see out edge view it says so in an undirected graph you can just say edges but in a directed graph if you just say edges it by default is taking out edge that is the outgoing edges so we need to mention in edges and out edges G dot out edges there are no outgoing edges and G dot in edges for incoming edges there are no incoming edges and as this is the different data type we wanted to be in a list format so we will be typecasting it into list bit given that currently we don't have any edges so let us add a few edges and again will do it. So let me say G dot add edge you need to give the source and the target source is the starting point of your edge and target is the end pint of your edge. So if I say one comma two and edge it is a directed edge from node one to node two will be created, will be added into a network. So now if I see G dot edges so one comma two is the only edge I have so if I say in edges it's also one comma two in case of out edges its one comma two so basically it doesn't matter if you just say edges where this in edges and out edges terminology matters is if you give a particular node and say what are the incoming edges to this node. Or what are the outgoing edges from this node in that case this thing actually matters so let me say G dot out edges of let me say one let me save this see one comma two where as if I give G dot in edges of one its empty because there is no incoming edge to node one currently this is how this works so let me add few more edges and let us see add edge from zero comma three let us say let us say two comma three, let us say that this be a bidirectional edge three comma two as well let us add three comma four, four comma one please note that the order matters. You are giving the source the starting point as the first element and the target as the end point as the second element this order matters. Please note this as like in undirected graph you cannot give in any order please note four comma doesn't mean that one comma four is present that is if there is an edge form node four to node one node one to node four there is an edge that doesn't, you cannot guarantee that there will be an edge. So this is the specially of the directed graph as you could see now if I say as I have said list will converted into list given that it is easier for us G dot out edges of any node say for example let me say node two so this is an outgoing edge, let me say node three these are the outgoing edges, so from node three there are two outgoing edges one towards node two and one towards node four there are two outgoing edges so you have a list of tupples an edge is denoted by a tupple. List of tupples this is how you have the output of this particular function out edges. Please note this, this will be used in our programming screen cast, this functionality out edges we will be using it also if you want to find the incoming edges you can say in edges these are the incoming edges from zero you have incoming edge towards three and from two you have an incoming edge towards three these are the incoming edges so this is how you can see the outgoing edges and incoming edges. We convert it into list format because that is easier for us. Alright guys so we have seen some pre requisites some functionalities that are provided in networkx package to work with directed graph and what is so important about directed graph is that you have to again and again keep in mind is edges are directed an edge from node A to node B doesn't mean that you have path from node B to node A as well. That is direct edge from node B to node A as well exist this is not guarantee that is if there is an edge from node A to node B there is no guarantee that you will have an edge from node B to node A this is key point from directed graph please do keep this in mind and will see about the algorithm of points distribution method how do we do that and what is the procedure to be followed will see that in the upcoming parts.

# PAGE RANK: HOW DOES GOOGLE WORK? 10

Alright guys, so in the previous videos we had seen about what is called the directed graphs and few terminologies and we have seen how we can use networkx package to work with directed graph so now in this lecture will get started with what is the points distribution method as the title says what is that actually is? What are we going to do? For this example let us take the same graph the same very simple graph with just three nodes for demonstration purposes while programming may be we take complex network as well. We can do that also will take a simple graph for demonstration purpose just this graph having three nodes will be taking this for demonstration and while programming we van have the complex network as well we can take that as an example. So let's gets started so before starting I would just assume ask you to assume just ask you to assume that this is a network social network as you can say of people liking each other. So you can considered this as probably I have collected a data about your family and friends so this are your family members or friends the nodes are the people and this person likes this person the person A likes the person B in that case will have a directed edge from A to B this is how the network as been constructed. If you by if you would assume this way it would be easier for you if I explain the points distribution method so assume that this is our network a simple network, simple network having just three nodes just assume the network is there is a node between two people if person A likes person B there is a directed edge from person A to B so for demonstration purpose will be taking this simple network containing just three nodes while we are actually programming we can construct a complex network as well having many nodes and edges you can construct any network and you can run your algorithm on in. For demonstration purpose will take a very simple network so just assume that these nodes are people and there is a directed edge from person A to person B if person A likes person B something like that. So will gets started with the points distribution method. So initially all the nodes are given equal points so points you can also consider it as some gift items or gold coins anything some item is given to all the three people something equally has been given and they have been ask to play a sort of a game something like that the rule of this game is whatever you are having you should share it equally with all your out edges so we had collected the data of who all you like right so out edge is who all this A likes person B and person C person B likes just person C, person C just likes person A so this is the data right so you are suppose to share whatever you have to equally to all you like so that is the rule so you have to play the game. So initially they have equal number of points but due to this exchange of the points the next iteration the points may not be equal to all the three, the one thing is guarantee is the sum of points will be the same the initially let us say we have given some x points to each person so three x is the sum right? So the same three x the sum three x would be maintain throughout but the individual values each person is having will differ. So let us see, let us see node A node A has got two neighbours, by neighbours I mean those people to whom there is a outward edge that is person A lies these two people, person B and C so there are these are the neighbours. So given that there are two people you should divided into one by two since there are two people that's why it's one by two if there are three people it will be one by three if there are k people it will be one by k. You should partition your total points into that many fragments and you should be giving it too all those people, that is the rule for node A let us see for node B is having you should give entirely to node C one by one so it’s one so you should give entirely to node C node and node C if you considered he just has one neighbours he just likes one person node A so whatever he is having he should give entirely to node A so I guess do you sense something that we have started with equal points each but even after just one exchange in this passion obeying this rule there is more points will be acquired by node C given that node A will also give something to him node C node B will also give something to him even that two  people are giving he will acquire more something like that one node will acquire more one node will acquire less but the sum of the points will be the same that is the if you add up the points with A B and C initially whatever was the sum we started with equal number of points let us say x points to node A node B node C the sum is the addition of all these three values are three x same three x will be maintain after all this iteration like this you have to repeat this process for multiple number of times. So this is how you have to give so now a day’s giving to someone as well as getting to someone right so will see the getting details now. So node A will get something so that will be his new points initially if I give a equal number of points let us say hundred number of points for simplicity they were given hundred points each node A hundred node B had hundred and node C has got hundred so three hundred points total in the system so node each node has hundred points initially so following that rule they had to share their points along with their neighbour who the node when have to give it to his neighbours as well as get some points from someone who likes him, for example let’s say node B he like node C so he would give his points to node C as well as node A likes node B right so he will also give so node B will be getting something as well so each node will be giving something as well as getting something. So we will see what if gets that’s what I mean by new points initially everything was equal following this exchange rule they had exchanged their points and what is the new points. Node A will get everything from node C right so whatever was the old points of C that will entirely come to A that is the new points then for node B he will get half from A because A likes B and C so he has to as per the rule he has to share it equally with both the people right so he will get half of it so do you see that? Initially there was hundred points now even that he is getting just half of it now B will be settled with fifty points after this one exchange. Same like that for node C the new points is he will get half from node A as well as from B entirely he will be getting. So this is how the game has to played game where initially all the nodes were given equal points say hundred points and the points has to be exchanged between the parties following this rule that how many ever outgoing edges you have, how many ever points you have you should share it equally among all the people. So that is the rule following that they should play the game but if you keep playing this game if you keep doing it so after one iteration as I have said we can say iteration one exchanged one round after one round as we had said node A had some hundred points, node B had some fifty points as you could see, node C will be having fifty points if our math you can work out and you will get to know that so this is the status after one exchange. Now with this state there are again following the same rule and node A had hundred points so he will divide into fifty fifty node B had fifty he will give that fifty entirely they follow the same rule and they play this what happens if you play this for say some thousand iterations or ten thousand iterations if you keep doing this what happens? Let us see that in the next part.

# PAGE RANK: HOW DOES GOOGLE WORK? 11

Alright guys, so in the previous videos you were introduced to the concept of directed graphs. You have seen how you can work with directed graph using your networkx package as well as I had introduced you to this game of points distribution. So we had seen the concepts. So let us see as I had said one snap you had done the calculation what happens if we keep repeating this? So let us first try simulating this in excel that is the spread sheet we will try simulating it because we are doing it for some thousand times manually if you do there is a possibility that after some iterations you may get some fractions as well so you may have some errors if you do it manually so let us use this spread sheet software and let us see what happens. Let me open new sheet, I am opening the new spread sheet so let us say we are starting with ok let us say we are starting with hundred points per a node we have the nodes a b and c let us use the same columns for ease of understanding we start with hundred points so I have given hundred points to all three of them and I have asked them to exchange based on the rule and after exchange what happens we had seen. What will be the new points that will be accumulated we have seen let us refresh once new points of A is nothing but old points of C just it will retain that so what is the new value here this is the time one let us say at time t is equal to one this is the values they are initially equal values all hundred points each so at time two what happens? What are the old points of C is what A will have so let me say equal to C one old points and so what for B what is the new points of B is? Half of the old points of A so I will say equal to old points of A is A one here divided by two, half that’s why I am dividing it ok so for C what is the new points? Half of old points of A plus old points of B because C is liked by both this people so it will get more points so half of old points of A and old points of B we have to add them up. Let us fill that formula half of old points of A I will say A one by two plus what is it whole points of B so its B one ok so as I have said it is hundred, fifty and one fifty so this calculation we have done it. What happens if we repeat it for further iterations? So in excel if you drag it the formula will get copied automatically it will get adjusted that is nice feature available in this software so will drag it and see what happens see these are the values after some ten iterations so it is hundred, fifty, one fifty so that one fifty came here half of it comes here so some kind of jugluary happens and that every point as I have said the sum will be constant. The sum is equal to let me give the space so tha you can understand this is the points and this is the statistics we are calculating let us see we are calculating the sum equal to A one plus B one plus C one ok this is the sum so I need to copy the same formula here as well see the individual points the distribution differs but the sum of the three values remains constant so this is something similar to your law of conservation of energy in your physics, energy can neither be created nor be destroyed it will keep getting transferred from one form to another form right so something like that the points are neither created nor destroyed that keeps getting transferred from one party to other party in the system that’s what is happening in this game so we have done this for ten iterations right so let us further extend it let say for some twenty iterations what happens? So some values have occurred so let u stake this let us even see the sum see the sum remains constant with irrespective of whatever is the distribution here among the three items the sum remains constant factor twenty some sort of convergence that is occurring that is by convergence I mean between the previous one and the next one there is no much difference or it exactly remains the same that is what I mean by convergence let us see what happens when do we get convergence let us now fast forward and let us I mean we had check by ten iterations right first we check for the first ten then we check for the twenty and we keep on checking like this know so let us check for lot of iterations so let us some ok so I have checked for some another fifteen iterations so it appears to converge but here its one nineteen point nine nine and one twenty point something fifty nine point nine nine nine one twenty it appear to converge. Ok let us further extend let us see what happens ok. Let us further extent let us see what happens. Let me say I am extending till one forty see here there is an exact convergence one twenty, one twenty, sixty, sixty, one twenty one twenty and it will remains the same irrespective here after how many ever times you repeat it remains the same let us see where this first convergence has occurred so at the sixty sixth iterations it was one twenty sixty six one twenty sort of a stable state has been reached, sixty seven the same sixty eight the same after that how many ever you repeat the same points distribution is preserved and of you want to see the sum let us extend and see let us extend and see here let us extend and see the sum alright so till here see the sum of three hundred from the beginning till the end and after sum the exchange of points keep happening to see we initially began hundred hundred hundred equally the three hundred points were equally distributed to the three parties they were asked to exchange as per the rules so the points after one exchange changed after another exchange there is further change in points in so that keep kept on shuffling the distribution kept on changing but irrespective of distribution sum always remain constant so you can see that the property that has to be observed and this convergence factor is used in your page rank algorithm so we had simulated this using the spread sheet so let us now get into the programming part and simulate it using our python programming skills. This will be done in the next part.

# PAGE RANK: HOW DOES GOOGLE WORK? 12

Alright guys, so in the previous videos you had seen what are the directed graph? How can you work using networkx packages in python and how can you deal with directed graph in networkx also we have seen points distribution method, what is the procedure in which you play and how would you distribute your points, how would you share your points with your neighbours also we have seen a simulation in excel spread sheet that if you keep repeating for multiple iterations the values converge that is the difference between this iterations and previous operations will be either insignificant or there is no different at all that difference is highly negligible in that case we say the values converge. So we will now see simulation using python so will get started. The first step what you have to do is, you need to create a directed graph so let's do that. So how would you create a directed graph? For that we need networkx package let us import that first import networkx as nx so we have imported it so let us now go ahead with creating the directed graph, how would we do that? G is the graph object is equal to nx dot DI graph DI graph this is how you create a directed graph. So we have created it this is the empty directed graph so the next step is you should add the nodes so let us do it G dot add nodes from yes so will have to pass the list so let us say we need some ten nodes we have taken a some small example of three node graph in a demonstration now let us take a slightly bigger graph ten node graph may be you can say hundred node, thousand node graph you can just keep extending the same thing to larger number of nodes with the complex graph you can keep doing that. So let me now give the demo with the ten node graph so I am going to insert ten node from zero one two three till nine so using a for loop single line way as we have seen in the pre requisites portion of this lecture series we will be doing that so it is nothing but it just a I for I in range how many number of nodes you wanted you just need to give, I am going to do with the ten node graph so let me do it ten oh ten nodes from zero till nine had been added to the graph so the nodes have created so now we need to add some edges so let me say add edges so I need to add edges and get back to new graph with edges that is rently we just have nodes and no edges so I need to add edges and get new structure as graph G let me say G would be my new graph I have to add edges that's what I have to do. I have created a graph object I have added the nodes now I have to add the edges so undefined name add edges it says that means I have used this name but I have to define what this actually is so let me go ahead and do that let me say define add edges so first add edges what do you need? Between two nodes there will be an edge right so we need to get the list of we need to access the nodes frequently so will take the list of nodes so let me say nodes is equal to list of G dot nodes so we got the list of nodes now I need a source node let me say S for source for S from nodes and I need to pick a target node let me say T for target from nodes list I have picked it also I should see source and target must be two different nodes if it is a same node it will be a it will be like telling I like myself every one like them self so capturing this information is actually waste of space something like that so will not do that generally that's what we call as loop in graph theoretical terms so we don't prefer loops generally that's what we call as a simple graph we are going to construct a simple graph where there is no loops so we need to see that if these are two different nodes so I am going to perform a check here if s if not equal to p only then I will think if I have to add and edge or not so if I just like that keep adding an edge what happens this loop will run for all possible paths and what you get is a complete graph I don't want a complete graph I want graph to be randomly generated so randomly generating I have to go ahead with this package random so let me import it import random so I have imported it so I am going to randomly add an edge so how could I randomly add an edge so I can simulate it with a coin toss so I may say that I just pick two nodes two different nodes if they are two different nodes what I will do? I have picked a node s I have picked a node t source and target so s and t I have picked two nodes if they are two different nodes what they have to do, I will toss a coin if I get a head I will put an edge otherwise I won't put an edge if a toss an if you toss an coin what is the probability? What is the chance that you get the head? One by two right because head or tail right two possibilities one of the possibility is head which is your decided thing so one by two this is zero point five is the probability of your head so I am going to simulate the coin task if I get a head I will put an edge otherwise I won't put an edge so let me say I am going to toss a coin that is nothing but I am going to generate a random number that's what I am going to simulate so let me say r be my random number going to generate random dot random what does this do? It will generated a random number in the range zero and one so let me take zero point five that is the probability for my head right so I will say if this random number r is within zero point five I consider that I have got a head otherwise I will say I have got a tail this is how I am going to simulate the coin task process so if this r is within point five less than or equal to zero point five what do I do? I consider it as a head, so if there is a head yes I have to add an edge right so I will add an edge g got add an edge from source to target so I have added an edge so I will do this for all possible pair of nodes and whatever is the resulting graph I am returning the resulting graph I will return graph G after adding the edges based on coin toss so coin toss is a random process if you toss a coin you get a head or you get a tail so based on that if you draw an edge you will get an random graph every time so you can experiment what happens instead of having the faced structure you get a different structure every time and you experiment what happens whether there is convergence with or any structure you an experiment it and there will be convergence you need to run for sufficient iterations may be ten node graph so I suppose maximum some ten thousand iterations definitely there will be convergence so why do I say this concretely because there are some mathematical results which say that there will be convergence in any structure so just a intuition that I could give you is just a hint is same probability is a hint you can del deep into it if you are interested there are mathematical results there are proofs which states that definitely there will be convergence if you keep repeating this procedure so you can simulate it so for example I had taken as hundred hundred hundred each right initially all nodes where having hundred points that is how I had taken so I can instead of giving hundred points if there are some k people I can give one by k each the sum turns out to be one so this is one thing in probability right the individual probability is may be anything but the sum of all the probability will be sum up to one right so you consider it like that and you can think of it has probabilistic model and you can prove that its slightly rigorous the mathematical part but its not very difficult actually you can understand it all that you need to know is some amount of probability if you are good with probability you can look up to what the mathematical proof is and you will get to know that definitely whatever be the graph structure whatever be the initial configuration initial we are starting with equal even if you start with unequal configuration it will converge probably the number of iteration it takes to converge may differ but it will definitely converge so that is the mathematical result so you can experimentally verify this with different random graph you run it once you will get a graph you can verify whether there is convergence similarly you can run it again again again you can run it in you will get different graph every time you can verify whether there is convergence as per that mathematical result there should be convergence let us see what happens ok. so we have added the edges we have added the nodes we have added the edges now what should you do we should visualise the graph visualise the graph to plot the graph and visualise it what do I need? I need this package matplotlib so let me import it import matplotlib dot plyplot as plt shorter form so let me say matplotlib pyplot I have imported so what should I do? We should first draw the network so I will say nx dot draw the graph G and anything have a labels, labels are nothing but the entries of the node that is this is node one this is node two this is node five so I need the labels so I will say with labels equal to true I will save this so you will get the graph draw with the labels and now I have to show the drawn graph I will say plt dot show ok done so what we have done till now we have created a directed graph we have added a few nodes added some edges we have added edges randomly simulating a coin task and we are going to visualise the graph that’s what we are done till now ok lets go ahead run this and see what happens. Let me run it see some graph got generated some network is there if I run some other one more time I will get another network another time get another network some other network some other network so every time you are getting a random network so on this you can run your distribution method and you can check whether there is convergence. As I have said there is a mathematical result its says there will be convergence irrespective of whatever is your initial distribution initially however you distribute the points and as well as how whatever is the structure for your graph irrespective of that you will have convergence there is a mathematical result let us verify that in the upcoming parts of the video. That’s why we are generating random graph so if you run this iteration you will have a different output some other iteration you may have different output so we are basically experimenting it so for different structures how is the behaviour we are basically experimenting that’s what we will be going to do. So will do this in the upcoming parts of the lecture.

# PAGE RANK: HOW DOES GOOGLE WORK? 13

Alright guys, so in the previous videos we had seen what is a directed graph how can we work using networkx and what is point distribution method we had simulated that in an excel sheet with an small network with a small graph with just three nodes we juts simulated now we are proceeding towards checking what happens in a complex network having lots of nodes ad edges so the first step we had created directed graph we have added a few nodes and edges were added randomly simulating a coin task now we had visualise the graph as well this is what we done till now now will get started with the points distribution procedure so let me get started the first step is you have to assign initial point right initial points that is a first step that you have to do let so do it so let me say assign points let me call and once I assign the points I need to have a list of points right so this node zero has got these many points node one got these many points and so on I need a list of points so let me say let me call that as points so I will assign points so to assign points I need to assign it to for each node eight so I need the graph. From the graph I will keep each node and I will assign points to the nodes so I need this and you could see the warning undefined name so we need to define this is so let us define it here define assign points and we use the graph G and so we have assign the points so first as I have said for each node we need to assign it so let me first get the list of nodes nothing but list of G dot nodes so let me get the list of nodes and now I have to assign it assign the points so I need a array points let me say P I need a list so for each in nodes so for each node I have to append how many points? We have started with hundred initially right so let us do that same. Append hundred and once all the nodes has been assign the points you return this array P so we had assign the points so what have we done? We take the graph G and assign points to each nodes and we get back the points so we take the graph G we take the list of all nodes for each nodes we are assigning hundred points initially that's what I am doing I am assigning hundred points hundred points I am assigning initially this is not the hardern part through that you have to assign hundred only you can experiment with different initial value this is the seed value you can say, you can experiment with different seed values as well we I have said you can model it as probability you can say one by number of nodes you initialise it you can see what happens you can try different things so let in this demonstration let us take hundred as the seed value. We have hundred nodes we can have different seed values with different seed values the final convergence values would be different but one thing would be the same for example we had taken in our demonstration we had done with our excel sheet we had taken hundred hundred hundred each for each node so three hundred at the end it convergence state it was one twenty sixty one twenty that I remember it may be something like this instead of having hundred initially I had taken ten the values may be different if I had taken twenty five the values may be different but one thing that is same is if you take the fraction one twenty divided by the total three hundred and sixty divided by three hundred, one twenty divided by three hundred that is point four point two point eight this is same something like that even if you take twenty five twenty five twenty five at the end if you take it the individual point divided by seventy five the total you will get the same thing point four, point two, point four that distribution is maintain so that is why I had given a hint you can modulate as a probability problem c the final sum is one if you model it if you getting the distribution is constant so you can modulate as a probability problem and you can mathematically you can go rigorously and prove that definitely there will be convergence irrespective of whatever be the structure of the graph whatever be the initial stage you start with. It’s not necessary that you have to start with equal, equally hundred each that’s even not necessary you can even start with ten points for one node five hundred points for another node three hundred pints for some other node you can start with the any configuration. At the end you will reach the stable state, stable state there was one twenty sixty given that we started with hundred hundred hundred or in terms of scale down to one version that is point four point two point four that was the stability state. Something like that every graph has got a stability state that is the mathematical result so ok let us come back we have assigned the points and now the next step is we should keep distributing keep distributing the points so if you keep distributing the points you would get some final points let me call it as final points the converged points you can call it as the converged points you will be getting you will get the final points by keep on distributing distributing the points along the nodes of graph G so ok again I got a warning undefined name so I need to define it so let me define let me copy and define it here let me define it define this and let me say so I have to distribute the points among the nodes for that again I need the list of nodes let me say nodes is nothing but the list of G dot nodes I need to get the list of nodes I have got it now I have to kick so we have started with the definition of keep distributing functionality so I have to start keep on distributing as I said keep distributing so it is an infinite loop so let me start an infinite loop so if it is an infinite loop when will we stop it’s our wish right so excel even when we stimulate if I wanted I can drag it till one lack iterations you can drag it as you wish so I am going to give the control to the user when he decides to stop that will be the time he will stop. So I am going to give the control so till then it will be an infinite loop so what should I do? I will get I have to distribute the points to the nodes of the graphs from this intial points configuration initially we have equal points that’s the configuration we are starting with and we have to take that configuration and distribute as per our rule ok so if you distribute the points you will get a new set of points in the next iteration. So let me call that as new points it will be my new points so once I get my new points I need to print my new points what are my new points? I have printed so once this new points have been obtained you have to take this as your initial configuration and repeat the procedure the next procedure. How would I do that? I have to say my points will be nothing but my new points so basically what you will do? You get some initial configuration you distribute it as per your procedure after distribution you get some new configuration the new points oh! Take that as your base configuration I am repeating in iteration two iteration one values as your base values in iteration three you will take iterations two values as your base value right? so that’s what we are going to do so that’s why we are doing this assignment now you have to decide whether to stop or not so let me say stop I will say is nothing but I am going to get an input from the user let me say I will take some key from the key board press hash to stop or any other key to continue, you can have any key I have just taken hash randomly you can have percentage symbols but you can have anything you can have anything that’s not hard and fast thing you can have anything. So I say you have to press hash as long as user presses hash they have to continue if you want to continue you have to press some other key. If you want to stop you press hash that is the rule I am defining for the user so I will get the input now given that user has given the input I need to process it if these stop value is equal to hash then what should I do? I should break; I should come out of the loop alright so once I have broken so user wanted to stop. Whatever is the value of new points that will be my final points so I need to return it so I will return new points right once I come out of the loop whatever is the values of new points I will return it back so I keep distributing it so what have we done here? We have assign equal points initially now plus another it has to be equal we had done equal you can try a different variable as well as I have said. You assign randomly some values then you see you can use a random generator again for example from one to five hundred you pick a random value and assign that as point to the node and you can try different things. We have started with the equal configuration. And after that after you have assign you keep distributing the points so in keep distributing what are we doing we have to we have to distribute the points and as a result of this distribution you will get a new configuration you will print that configuration and take that configuration as your base and repeat that you repeat that as long as the user doesn't want to stop as long as the user wants keep repeating it, if he wants to stop he will stop in that place you return whatever is the new points whatever is the last configuration you have got so you return that ok. So here we have undefined name distribution points, yes we have to define it we see distribution point what is the procedure to distribute? Will have to define it, we will do it in the next part.

# PAGE RANK: HOW DOES GOOGLE WORK? 14

Alright guys, so in the previous videos we had seen what is a directed graph how can we work in networkx and then we had seen what is points distribution method, we had simulated it with small network with just three nodes in excel sheet now we had started with the programming part. We had created a network we had visualised it we are creating a random network simulating coin tosses, we had visualised that network then we had initialised the points equally some seed values we had initialised all the nodes we are getting equal points first at the first stage then we have to keep distributing under keep distributing we need to now define how to distribute the points actually so we will do that now. So let me define it define distribute points on a given graph on a given point configuration so let me distribute so as and always we need the list of nodes to operate so let me get the list of nodes list of G dot nodes alright so before starting initially every node has given something to its neighbours so it will have nothing so let me say new points are initially zero right so I need to assign tit so let me say or I in range length of nodes that is how many number of nodes are present length of nodes in n o d e s ok so for that we need a list as well right so let me say new points so let me create a list new points let me say new points new points to append zero why zero? Because the node has given everything to his neighbours and it has nothing. That is the state then we will that’s how we had even introduced in the slides right so the node will give its to neighbours that’s it the next stage we considered what is it getting from? If everyone is giving, someone has to get so what is that it’s getting is the next consideration so first the giving part so this is the giving part so we have to keep appending the zero value because it has given everything it has nothing so this is your new point so now let us say for n, n for node let me say n in nodes I take a node n and say out for out edges so we have to see the out edges and we have to share it with the outward edges right so I am taking the list of out edges list of G dot out edges and you need to give the node id it is n basically so I gave it so how will this appear? List of tuppples an edges represented by a tupples source comma target something like that list of tupples we will get so if this is an empty list what happens, it need not share anything with anyone it will just keep getting whatever it was having previously so we will handle that case here if length of this list out is equal to zero meaning that it has no outgoing edges then it will just keep getting the same points what it had got in the previous iteration right so I should say the new points for this particular node n is nothing bit new points of that node n previously plus whatever is the points it had in the previous iteration previous value of points whatever was the initial configuration will again retain so it will it had got some for example say it had got some ten points again it will get some ten points it may not share with any one even that there is no outgoing edges if there are outgoing edges you need to share, if there are no outgoing edges it will again get some ten points, if ten iterations again ten points it keeps accumulating so that accusation thing we have handled here right? ok so if this is not the case its having some outgoing edges then it has to share some points so let me say share it has to share how much? How should it share? It has to share equally among all of its neighbours, how many neighbours are present? That is given by length of this out so how many outgoing edges? It has to share it equally so whatever is the points it has been having you have to divide it equally among all the outgoing edges so that is the procedure so we have got the value of share and so this is the value of share so this much it has to give the whole it has to give, it has to give to the target nodes in the outgoing edges list right so that we will see so for the items in this outgoing list. What is formats of the items in the outgoing list? Source comma target right so this is of the format source comma target that is how the items in the outgoing list are for each of this in outgoing list your source will be the same list n right target differs so we will take the target value and we will work on it so the new points of target target is nothing but whatever it has earlier plus what is the share it is receiving from this node n so why do we having this new points plus share why do we have this? Because for example we had one node where it will half from I guess from node C half from node A and fully from node B so it had different shares so when you run the loop with node A you would have given half when you again run it with node B what happens is, if you just say it is equal to share the previous share whatever it has got from node A that value will get destroy. We need to retain that as well as this that is why we are adding it ok so this is how the new points calculated ok once everything is done what you do? You return your new points right so this is how the distributions occurs just let us see once this part so we get the list of nodes we have new points initially it is zero given that we have all given whatever they have nothing so given that they have given something has to be someone has to be there to get so each node will get some share as well so we are going to see the getting part now. What happens? We are taking a node n and then we are seeing the outgoing edges so to the outgoing edges only it would have shared so the outgoing edges the target nodes will get the share will get the new points if there is no outgoing edge what happens is? The nodes will keep accumulating the same points in this iterations as well, so that is why I am adding the same points again no if that is not the case there are outgoing edges it has to share it so it will equally share the points it ahs with all the neighbours so this share has to be given to the neighbours so it is of the form source and target that is how you get the list of edges so to all the target nodes you have to increase this share you have to give this share to the target nodes that is what we are doing the target nodes will get the value of share along with that whatever it has got and why we are doing this addition because a node may get different share from different nodes to account that we are doing this addition so once you have computed it for all the nodes you can reform the list of new points. So that is how single time distribution occurs so this is distributing just for one iteration. So in this keep distributing function you are repeatedly calling the distribution functionality and you keep repeating the process again and again till the user stops alright I hope you are now clear with the procedure may be you can take a pause and you can think what we have done in the previous parts what we have simulated in the excel sheets, what is the procedure we have explained, you think you give a pause you will understand it really well. So ok let me clear the screen here actually I will do it fresh ok so this screen is clear here so let me run it let me run so I have got some values initially it was hundred now I have got something hundred and thirty five, hundred and three nine, ninety eight something I have got it so it’s asking me if I have to stop or I have to continue, I want to continue so let me press some other key I prefer enter key it has continued continue continue continue continue continue continue like keep continuing after some time did you see here the values here as well as here there is no much difference I guess it has almost converged so there is convergence so I can up here lets press hash yeah it has stopped so as you could see we had some graph structure you can see some random graph structure is here so I had done distribution once and this is a result of distribution once and it is asking me if I want to continue or stop I had continued for lot of iterations and at the end I found there is convergence right? see the values here its one thirty six point eight nine eight five, one thirty six point eight nine eight five there is no much difference so it has started converging so once it converges after that whatever be the number of times you repeat it the same values will remain so given that I had hit the convergence I stop that's what I have done now so this is how we have checked the values of points after repeatedly distributing it now we have to run the nodes based on the points it has accumulated so for example forty five is the minimum value so this node gets the least rank and one thirty six is the highest value so this node gets the highest rank something like that so I have to rank the nodes so basically you can considered as marks so initially I am giving hundred marks to people and there are playing the game and distributing it and at the end whoever has got maximum marks whoever got minimum marks I have to sort the nodes I have to compare the result with the actual page rank procedure that is implemented in networkx I have to compare that result and so that we will be doing in the next part of the video.

# PAGE RANK: HOW DOES GOOGLE WORK? 15

Alright guys, in the previous videos we had seen what is a directed graph. how can we work with networkx and directed graph and what is points distribution methods and how do we, we have simulated it with excel sheet and there was for a small network and now we had started off with the programming part complex network having lots of nodes and edges. We had created a random network we had visualised it we had initially given equal points to all the nodes and we had started the game and we had asked the nodes to distribute the points based on their outgoing edges based on structure they keep distributing and we decide when to stop this game. Just because in excel we have that flexibility we can drag the calculations as long as we want so something like that same like that I want to give the control to the user that's why we had done it something like that but the user will decide when to stop, whenever the user stops just it will just say it is the final values something we had got it right so now what should we do once you get the final values you should wrap the nodes based on those values and next step would be compare it with the actual page rank values that is provided by the built in functionality in netwotkx we will do that so after they are done distributing we got the final points so I have to what should I do rank by points right I have to rank the nodes by points so let me say rank by points let me call it so to rank the nodes by points what do I need? I want to final points so I want this so I have to rank the nodes by points. Ok as you could see it undefined name see it is the warning message that is given so later define it let u define it ok define this thing and let me see how shall we start with it so we have decide this functionality rank by points so first thing is if you just like that we have points is nothing but list right so if we sort the list its simple and you just rank it that's what you are saying but if you sort the list you will lose the information as to this point belongs to which node so here we need to keep track of two information which node has got how many points it's not just about how many points it's about which node has got how many points so we have to convert this list structure into a dictionary structure we have to convert the list structure into a dictionary structure and this dictionary structure will be applying the sort functionality that is defined on dictionaries and this will be using to display the rank of the nodes based o n the points they have accumulated alright so as I said I need a dictionary let me say d so I have to have the node the points are actually arranged in the same order know zero point is the zeroth element of this list node ones point is the first element of the list something like that, that's how it has been arrange so let me it's just enough for loop one for a loop over I in range length of points so whatever is the index that is the node ideas as well in our case length of final points basically final points or may be even we can call it as a points as well so there it was final points and even here you can call it as points so for I in range length of points that is I am going to iterate over the list I am going to convert it into a dictionary, dictionary with key I the list index is what I am going to make it as key so that is same as node id right so I am going to make it as a key is nothing the corresponding value of points at that index. So I have created the dictionary now I should sort this dictionary to sort the dictionary we have built in functionality sorted you need to pass the items of the dictionary and here dictionary has too things keys and value we need to sorted based on the value part right so for that you need to say key equal to lambda so key here is nothing but based on what value you need to do, what is the factor based on which you need to sort. That is what you call as key and lambda is built in functionality in python basically it like for example alphabetical order ascending order attendance order or whatever you can call it as how would you do it? Based on the first letter you will sort the names A comes on the beginning then comes B then comes C so on. What if you have different demand? You want to sort it based on the last letter, so in that case you have to make use of custom function that you are actually going to write based on that function you have to sort on your requirement something like that if you have some other factor that you are defining and based on that factor you want sorting to happen you will use this kind of a syntax. And lambda because I am not going to define a new requirement that is not present in python this sort by value is a common requirement so python has defined it so it will say the upper f let me call it f nothing but at f at index one so key is at index zero and value is at index one so this lambda functionality sort of converts it into a tupple format and it will take the value at index one and it will sort based on it so this is the defined functionality that has been defined in python itself in case you want it an new requirement here you can say key equal to my function and based on that function it will sort so what is the factor I should considered for sorting that is what is denoted by this key ok so my factor is nothing but I wanted to be based on the first index value that is based on the value of points I want the sorting to take place the relationship between the node id and the points is also preferred and sorting also occurs so this is the specialty so if once it is sorted it will return a tupple of list of tupples basically so I need to print it so I will print this value so even that I have sorted and printed so the ranking part is done. So what should I do now? Once I have ranked let me compare it with the default networkx function I am going to use that and I am going to compare that results. So the default function of networkx is nx dot page rank and you have to pass the graph and it will return a dictionary basically. So that's why we had also used a dictionary right we had why dictionary? Because you need reserve you have to actually preserve the fact that this node has this many points you have to preserve that fact. So let u say this is the result of your page rank result and dictionary nx dot page rank do G just you need to give the value of page rank these things maximum number of iterations how many iterations you want you can customise it you can do it as per your demands but even you don't give there are some default values they would be taken and it would be calculated so we got the resultant that the dictionary and same like this I have to sort and print the sorted values. Right? So let me copy paste this lines so let me just copy and paste it so I will copy given that I have copied let me paste the line here and instead of B it is the name of my dictionary is result so I have taken a dictionary and based on the values I want the sorting to occur that is based on the points accumulated I want the sorting to occur so just a quick recap and then we proceed with running. So we had seen till distribution of points the output even is displayed here so before proceeding further I will clear the screen and I will run so the output is as well displayed here it gets converged to some specific distribution we have seen at till there and once we have got the convergence state we have to rank the node this is the points for the node zero this is for node one, node two and so on so we have to rank the nodes based on it just keep this is the list if you just apply sort on list the fact that this node had this points is lost that's why we will convert it into dictionary as you could see here we are converting the list into dictionary preserving the fact that this node has this many points and we are applying this built in functionality sorted on dictionary and this is the functionality that take care of sorting based on a given parameter based on the pints value we want to sort it so we are using that built in functionality just that to preserve the part of his node has these many points you are using the this many data structure you are converting the list into dictionary. Also we have done this for our method points distribution method and once it is done we are going to compare it with the default networkx functionality page rank that will also return the dictionary and we are applying the same sorting mechanism and we are displaying the sorted result we will see how is the result we get after the points distribution method and what is that built in result that is obtained will see how it is same or different or what is the thing that is the node that is ranked as number one here it is the same built in as well let us see all those things before running let me clear the screen and let me run for you let me run now so as I have got the some network some distribution will continue continue continue continue continue I keep continuing I keep continuing yes I guess I got a convergence see because I was saying these two values are failure there is convergence there is convergence here let me press hash to stop ok I have got convergence here ok so this is the value we have got based on the points distribution method node seven is the least rank node and node six is the highest rank node node seven and node six yeah same we are getting we are getting different values here because as I have said take a different seed value the final value will be different but as I have said if you take the this particular value by sum of all these values if you find the distribution will be same that is here if you take this particular value twelve point something by three hundred if you take the particular fraction if you take the fraction that will be same for both the things you can verify it this built in functionality is using some other methodology also I hope you have seen or you will be seeing another method of random walk to get the page rank values so it is using random walk and some other methodologies and some other seed values it is built in functionality is doing so the values may differ but the ranking if you see node seven is the least rank node node six is the highest rank node that I preserved not only that the ordering is exactly preserved as you could see, seven four zero eight seven four zero eight ok two nine one three five six two nine one three five six wow this is exactly matching so good it is but our method of point distribution the built in page rank method exactly matches as I say whatever be the network structure there is a convergence state there is a convergence configuration this is mathematical proof that you can take it as an advance material from this lecture. You can look it up some amount of probability is the pre requisite for the mathematical part you can look it up as an advance material from the its very required if you are interested and if you are key person and math oriented person highly enthuastic with maths you can look it up requisites are probability theory that’s it you can look it up and see to it, it will always converge there is a vigorous mathematical proof for it. Alright guys I hope you enjoyed this. Activity on page rank using point’s distribution method, thanks for watching have a nice day.

# PAGE RANK: HOW DOES GOOGLE WORK? 16

Hello all, welcome again to yet another programming screen cast on page rank as you all saw in the previous videos we performed an activity in this particular activity people were asked to choose their leader so now we have their data with us we have the data in the form of edges as you all know we need a directed graph in order to calculate page rank so will be creating a directed graph out of this data. First of all let me explain this data, we have this data in the form of edges. If we have one two that is there is an edge from one to two, if we have one space three that means we have an edge form one to three in the same way we have the data in the form of edge list. You can see the rest of the edges so now out task is to make a graph out of this edges so how can you make a graph out of this using python where this is also very much possible and very very very easy in python so let me show you how can you create a graph out of this data. As an always you need to import networkx first so I will just import networkx as nx then you also need to import matplotlib library so I will just write matplotlib dot pylot as plt after that you need to write just one function in order to read the data, yes in order to read the data you just need to write a function of networkx what that function is? I will just write G is equal to nx dot read underscore edge list and in this particular bracket you need to pass the path of your data file so I have stored the data in the txt file in a dot txt file so I will just write the path of this file I will write it’s in users simran setia list of my file name is page underscore rank dot txt apart from the apart from the path that you need to supply here you also need to write r here apart from the path that you need to supply you also need to supply two more parameters here first is create underscore using it describes whether you directed graph simple graph so since we need the directed graph I will write nx dot DI graph and after this I need to node type node type is int here so I will just write node type is equal to int since we have anonymised the nodes here we had the data in the form of the names of the different people but we have anonimised the data now we have the detain the form of numbers, in the form of numbers we have one two three till around twenty five so we have read the graph from the txt file now we just need to draw this graph how can you draw the graph simple nx dot draw capital G and you also need to pass with labels is equal to true since we need to show the labels I just write with labels is equal to true and then we just write plt dot show so I think this is done let me run this program. So as you can see we have a graph here the graph that has been read from the txt file ok so this is fairly simple in python you can easily read the data and draw a graph out of that data so the functions that you need to use here is G is equal read underscore edge list what you need to write here is you have to initialised graph variable G is equal to nx dot read edge list first of all you need to supply the path of your particular txt file after that you need to supply whether you want to draw a directed graph or simple graph then you also need to supply the node type here and we had nodes in the form of one two three so we supply node type is equal to int here so when you run it you can easily see your graph so as you can see this is simply fair in python you can try that and I will now your task is you run page rank algorithm on this particular graph and tell us who was the leader of the group. So I hope that this programming screen cast was useful to all of you thank you.

# COLLATZ CONJECTURE

So now you people have seen may questions many answers a lot of programming and now I am going to teach you one of the smallest programs ever taught to you in this course but one of the most complicated of all course in fact the world still doesn’t have an answer for this two line code it’s a smile while condition and a if condition that’s it. You will see that the code gets done in less than quarter page but the world still doesn’t know an answer to it. So what’s the question? The question is very simple, input any number n if the number n is even make it into half if the number is ten make it five ok if it’s even that is, if it’s even make it by two if it’s odd let say if it was eleven if it’s odd shoot that number up by three and plus one which is if an its eleven make it three into eleven thirty three plus one thirty four simple if it’s even reduce it by half, if it’s odd increase it by three times the number and add one to it that is the way we start with n equals seven so seven becomes into three twenty one plus one twenty two, twenty two is a is an even number so you should half it so twenty two becomes eleven, eleven becomes think about it into three plus one thirty four I will be slightly fast now, thirty four becomes seventeen why? You are halving it ok seventeen becomes three n plus one three times seventeen fifty one plus one fifty two, fifty two becomes twenty six twenty six becomes thirteen, thirteen becomes thirty nine plus one forty, forty becomes twenty, twenty is even it becomes ten, ten is even becomes five, five is odd becomes sixteen, sixteen becomes eight and then eight becomes four, four becomes two, two is even so half of that is one, one is odd so into three plus one is four again we saw four already so whenever you get four it becomes two, two becomes one and one again becomes four you see this is an infinite loop so what we do is we say while the number is greater than one you do this, do what? If the number is even make it become half the number is odd make it become three times that number plus one so seven had how many numbers? Count one two three four five six seven eight nine ten eleven twelve thirteen fourteen fifteen sixteen seventeen eighteen nineteen twenty steps right seven to twenty steps let us take some number big number like two fifty six how many steps it will take let us count. Two fifty six is even half of that is one twenty eight, one twenty eight is even half of that is sixty four even again thirty two even again sixteen wow even once again eight even again four even again two and that then becomes one, one is odd so one becomes four into three plus one four and four becomes two and two becomes one and you enter you stop here you see a small number like seven took a such a long number of steps a number of two fifty six how many steps did it take? Let me count one two three four five six seven eight nine steps as supposed to more than fifteen steps for seven correct so smaller the umber doesn’t mean smaller steps, bigger the number doesn’t mean bigger steps in fact this piece of code is so not understood given an n when will it stop? We still don’t know right this is famously called the three n plus one problem Google for it there is a lot of references for this three n plus one problem also called the Collatz conjecture in fact we don’t even know whether this will ever stop for some input it might keep going on and on it may never hit four two one four two one in both the cases you saw it four two one but it needed four two one you see Google for it read lot about it and know when someone comes and gives you long program and says he has done something very complicated give that person simple code like this and ask him to analyse what is happening that will be a big goggle for him will not really converge to a nice answer on what exactly happening here ok that was fun looking at question that is smallest possessive we have given you very very very big programs that work and very very small program like this that even the world doesn’t know how to answer so programming world is very huge very interesting very fascinating sky is the limit I hope you people have enjoyed the lectures given to you so for so we are ending this course with a quick code on collatz conjecture.

# COLLATZ CONJECTURE 02

HI guys, in this video will see the programming part of collatz conjecture this is the last screen cast of week twelve and last screen cast of me also so it has been a nice journey let's see the last program it's a very simple program of collatz conjecture. Let me review the conjecture once again so what you have to do so given a number n you have to every time you have to check whether if n is even number then you divide it by two that means that n becomes n by two and if it is an odd number then you have to do n is equal to three n plus one. so you repeat it every time until and unless until n becomes one and the point you have to check whether is it possible to make any number one using these methods so this was the conjecture actually we can see it through a program whether can we do it with a any number or not so I will create a definition in python and you can supply any number as an argument to this function and you can check whether we can achieve one or not. So let's see this so I told you I will create a definition so let me write it here ok ok so my definition name will be check numb an argument is number numb let say ok I will check how many iterations I want so for that I will keep a variable iterations so as initially as one now what I have to check every time I have to do two operations one is if n is an even number then I have to reduce it divide it by two and if it is a odd number then I have to make it three n plus one until it becomes one so my check some should be check should be let me put the while loop while numb is not equal to one until it doesn't become one I have to do this operations every time ok so until numb doesn't become one is not equal to one I will check first if numb is a numb mode two is equal equal to zero which means that I am checking whether it fully divides this number fully divides by two or not if it fully divides then it means it is even number so what I have to do if it is an even number then numb my number becomes number I can write like this also numb divide by two and I just make it as an integer because you know when you divide a number it becomes a floating point number it doesn't matter here but still ok and yeah since it is an even number it will divide by two and I will just make my iteration plus one iterations is plus equal to one and if it is not an even number then what I have to do numb is equal to numb into three plus one isn't it and again I will I will increase my iterations value by one ok ok after this it means the loop has been executed successfully after that I will just print my number what is my numb after all these iterations and what are my how many iterations it took to become one off course it will print one just to take check if numb is becoming one or not and this iteration will tell me who many iteration it took to finally reach one of my definition is complete so I will just run it check numb let say twelve great let's just run it. Ok you can see the output one and ten it becomes one it is becoming one after ten iterations let me put twenty let's see it is becoming one after fight iterations see you can check various numbers and see whether yeah it is becoming one after eleven iterations so this is it this was the program there is nothing much to do it was the very basic program to show you how the conjecture behave actually you can read about the conjecture online and so this program was just to show you their limitation of computing also it has been a nice journey with you guys in joy of computing course we hope we will meet again in some other course lets see this thank you every one. Have a nice day.

# CONCLUSION

It is time for us to give our final concluding remarks for our course we have studied more than thirty awesome ideas in computing and we have shown you how to code it from scratch. We started off from a very basic tool called the scratch tool we taught you how to program without even going to a proper programming standard programming environment just by drag and drop technique you try developing some applications some straight forward applications with that we learnt how looping structure works, how variables works, how if then works and then we graduated to python we saw some thirty cool ideas implemented all of them so the computing aspects of this the science aspect of it I need to tell you all one important. Programming is one subject where there is no starting point and there is no ending point you just get in your swim and you keep swimming and keep learning better and better skills now you observed that programming pursue involves some standard understanding some standard raised ways of talking ways to your computer but then the way in which you talk the logic that you use is a different kind of skills correct so you observed that some longest programming exercises required you to think very logically right let’s say for example if you are asked to add two matrices that’s going to be quite easy but we are asked to multiply two matrices you see that’s going to be non trevion. So it is not just about python or c or c plus plus java or any other programming language it’s also about this knack to use the programming language properly. A good analogy would be you know good English but are you a good writer, let’s say let’s go one step further and say if you are a good writer can you be a good poet? You see all these things require a different set of skills and some sort of a unique talent which is not actually born it can be made. So single most important tip I can give you people is continue to program all the thirty plus joys that we formulated where the joys that we thought would be fun programming you probably not see a single text book which covers all these things in fact many topics you may not even see any web reference to it. So there is no beginning there is no end you should just start programming as much as possible. Now given that we have we have completed this course successfully I would like to give you a tip on what’s next? There are certain aspects of programming certain aspects of computing that we did not cover because that was not in our motive, our motive was inspire you to get into programming and do a good deal of it now let me enumerate a few things that you can look at even that you have the knack and experience of going one step ahead and read something advanced. There is this notion of what you called the object oriented programming approach where people realise that programming the day from top to bottom simply writing your code may not always work. We must use a very object oriented approach by that we mean it is not just variables at the low end level you see a bunch of variables as an entity this is based understood when you start getting into this topic called oops which was mainly popularised from the point when c plus plus was developed which was a pre curser which was C, C plus plus had an advantage from C that you could use the type of programming that was easy on the programmers mind so what you can now do is to go ahead and understand how to program using object oriented approach in python. Second point we covered many APIs but APIs are actually limitless when it comes to python so whenever you want to do something you will find an API for it there are several standard APIs most of it actually we are covered in our course but there are still a lot more you can explore which you should that's my second tip to you. A third possible point of improvement could be one can learn python for web programming that was not covered in the course but you now have all the skill set to go ahead and understand how web programming can be done using python. Fourth tip would be if you are a scientist a programming language like python will be indispensable again there are several APIs for a particular type of science you can explore it and use python for scientific computing and fifth tip which is in fact the most important one is when it comes to programming your programming requirements are generally not one or two pages long just the way our joys were in our course they were all very short codes if you develop a big application it might run through several thousands of lines, developed together by several people so what we did not cover is how to manage a big project in python. Anyway that's my fifth tip you may want to look at it. It's pretty easy now that you know most of what entails python. Ok let us now address what is it one can do after learning a course like joy of computing as I said if you are a scientist you can try to implement your knowledge and expertises in python in solving your scientific problems let's say you are fun loving person who would like to develop some unique applications as part of your Btech project or whatever you are studying or even if you are a graduate and you want to develop an application which you are very passionate about seeing the final product you can go ahead and develop an applications of your choice. Another important tip would be the world is converging towards highly collaborative environments you can think of joining one of the open source projects on Github and contribute to the software development. They will be developing a package together collaboratively dozens of them will be sitting and writing the code you can also give in your little piece there. And finally yes of course you will have an added advantage over others when you say you have tried answering several random questions using python and hence you are fit to be a software developer right now please note to be a good software engineer developer all you need is good amount of confidence and good amount of logical thinking ability and we strongly believe that we have given this things to you. Sky is the limit please go explore and as and always this course might have a sequel will keep you all posted there is a possibility that will come out with joc two point o where will talk about a little advanced topics.

PH: (044) 2257 5905(08)
nptel.ac.in
swayam.gov.in